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Short-Term Disaggregated Load Forecasting Using a Hybrid Fuzzy ARTMAP and K-means Clustering Model

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Abstract

Accurate short-term load forecasting at disaggregated levels is critical for energy management in microgrids and institutional environments, yet it remains a challenge due to high consumption variability and limited contextual information. This paper proposes a hybrid model that combines Fuzzy ARTMAP neural networks with K-means clustering to improve hourly load forecasting using real data from a university microgrid. The methodology includes key preprocessing steps such as filtering low-load records, removing holidays, interpolating missing values, and applying cyclic encoding to standardize the data into 96 time intervals per day (15-min resolution). For each prediction, the average load profile of the five most recent weekdays is computed and compared to cluster centroids to identify the most similar group, which is then used to train the neural network. Results demonstrate consistent improvements in MAPE, RMSE, and MAE compared to the non-clustered baseline. The model showed robustness to non-stationary behavior and atypical patterns, even when relying solely on timestamp and load data. The proposed strategy outperformed conventional approaches and proved suitable for complex, data-limited environments.

Keywords: short-term load forecasting; disaggregated demand; Fuzzy ARTMAP; K-means clustering; microgrids

1. Introduction

Short-term electricity demand forecasting is an essential activity for the planning, operation, and efficient management of power systems. With the advancement of measurement technologies and the increasing variability of consumption patterns, especially in low-aggregation environments, the challenges associated with modeling and estimating electrical load have become more complex and uncertain [1]. These challenges are intensified by the nonlinear dynamics of load profiles and their sensitivity to exogenous factors, such as weather conditions and user behavior [2,3].

Disaggregated systems, such as microgrids or institutional environments, present significant consumption variations due to irregular occupancy, intermittent equipment operation, and the influence of academic or administrative calendars [4]. Such characteristics reduce the predictability of load profiles and limit the smoothing effect typical of more aggregated systems [5]. Therefore, it is necessary to adopt models capable of handling noisy data, abrupt changes, and irregular seasonal behaviors.

Among the methods applied to short-term load forecasting in low-aggregation environments, Artificial Neural Networks (ANNs) stand out due to their effectiveness in

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forecasting nonlinear time series, especially in this context. To improve their performance, these networks can be combined with clustering techniques, such as the K-means algorithm, which allows for their specialization on subsets of days with similar consumption patterns, increasing forecasting performance [5].

Building on this idea of segmentation, prior clustering has also been explored in a three-stage architecture composed of self-organizing maps, K-means clustering, and a multilayer perceptron ANN. This model was successfully validated for short-term forecasting in institutional microgrids. The model demonstrated the ability to adapt to operational variations and seasonal patterns characteristic of intermittently used environments [4].

Another approach with lower computational complexity is the Pattern Sequence-based Forecasting (PSF) method. This technique also performs initial clustering of load profiles using K-means and identifies historical consumption sequences similar to the target day's pattern. Future load is then estimated based on the average of the subsequent loads, making it suitable for environments with cyclical and recurring behavior [6].

To improve data segmentation beyond traditional methods, the use of simulated annealing as a deterministic clustering technique has also been proposed, applied prior to neural network training. This strategy proved effective in separating heterogeneous load patterns, allowing greater model adherence to consumption profile characteristics [7].

In a different direction, other neural network architectures have been explored for this task. The application of Radial Basis Function Neural Networks in systems with high variability, such as real-time energy markets, has shown good performance in generalizing noisy and complex time series, reinforcing the viability of alternative architectures in challenging operational scenarios [8].

Despite these advances, a critical gap remains: many models are validated under idealized conditions that do not reflect real-world operational constraints. First, they often rely on *ex-post* data from the forecast day itself, a condition that is unfeasible in practical *ex-ante* forecasting scenarios where only past information is available [9]. Second, they frequently struggle with the operational irregularities of institutional settings—such as strikes, holidays, and academic calendar variations—which disrupt conventional time-series patterns and challenge model generalization.

This paper proposes a hybrid approach for hourly load forecasting at a disaggregated level. The methodology integrates Fuzzy ARTMAP neural networks with the K-means clustering technique, and is validated with real consumption data from a Brazilian public university. The proposed model was designed to estimate the demand for any day present in the database, considering only the hourly order of daily measurements, without requiring continuity between consecutive days.

The choice of the K-means algorithm aims to handle the high variability of consumption data. Instead of trying to teach a neural network to predict a consumption profile for a generic day, days with similar load profiles are grouped. This segmentation creates more homogeneous data subsets, which reduces noise and the complexity of the task for the neural network, resulting in more efficient training and specialized models for each type of consumption profile.

The selection of Fuzzy ARTMAP as the neural network for forecasting is also strategic. Unlike other architectures, this network is an incremental learning model, making it capable of learning new patterns without suffering from catastrophic forgetting [10]. Moreover, its Fuzzy component makes it resistant to outliers, allowing the model to generalize well even with imprecise data or unexpected peaks. The combination of these two techniques allows the approach to segment complexity (K-means) before applying a robust and adaptable model (Fuzzy ARTMAP).

The main novelty of this work lies in a framework that integrates clustering with incremental learning to improve forecasting in variable and non-stationary environments. While traditional models struggle to generalize from a single heterogeneous dataset, the proposed structure enhances forecast quality by allowing specialized networks to learn from stable, homogeneous patterns. This approach maintains the ability to update incrementally with new data—an intrinsic feature of Fuzzy ARTMAP—and ensures scalability for continuous data streams. It thus provides a practical solution for common forecasting challenges, including limited historical data, unmeasured variables, and high variability in user behavior.

The effectiveness of the proposed methodology, with and without the clustering step, is compared. For a more comprehensive validation, the approach's performance will be benchmarked against other forecasting algorithms, including Multilayer Perceptron (MLP) and Decision Trees. This comparative analysis highlights the advantages of the hybrid approach for the unique challenges of disaggregated load forecasting in institutional environments.

2. Materials and Methods

This section details the data used and the methodological procedures for the construction, training, and validation of the proposed forecasting model.

2.1. Data Description

This study utilizes a real-world dataset of active energy measurements from a single building block at a Brazilian university. Data were collected from instruments connected directly to the building's exclusive transformer, ensuring a truly disaggregated load profile free from the influence of external consumption sources.

Measurements were taken continuously from 24 April 2012 to 22 August 2013. The dataset features high temporal granularity, originally distributed in intervals close to 15 min, although with slight variations in time stamps, which were subsequently regularized during the preprocessing stage.

The analyzed period includes atypical operational events, such as an academic strike, holidays, and low-occupancy phases, resulting in irregular consumption patterns and abrupt variations. The time series also presents noise and anomalous values, attributed to acquisition failures or actual disturbances in the electrical system. These characteristics create a challenging yet ideal test case for evaluating the robustness of forecasting models in non-stationary environments.

2.2. Data Preprocessing

The data preprocessing stage was designed to standardize the time series and filter out non-representative operational patterns. The process involved regularizing time records, cyclically encoding the time variable, excluding non-representative observations, and filling in missing data.

Although the raw data had a nominal 15-min granularity, variations in timestamps were observed due to imperfections in the acquisition system. Therefore, the data was temporally regularized, aligning load values to a strict 15-min grid. This step is important because models like Fuzzy ARTMAP rely on pattern similarity; irregular measurement intervals could lead to poor clusters. Regularization eliminated this bias, ensuring that daily consumption profiles were compared at equivalent time points. This preserved signal quality and allowed the network to be trained with more reliable data.

To preserve the cyclic nature of the time variable, trigonometric encoding was used. The total number of minutes from 00:00 was calculated for each record, and Equations (1) and (2) were applied to encode the time variable:

$$\text{Hour_sine}_i = \sin\left(\frac{2\pi \cdot \text{Minutes}_i}{1440}\right) \quad (1)$$

$$\text{Hour_cosine}_i = \cos\left(\frac{2\pi \cdot \text{Minutes}_i}{1440}\right) \quad (2)$$

After adjusting the time series to regular 15-min intervals, each full day contained 96 load records. Based on this standardization, all incomplete days were removed. Additionally, samples corresponding to national and municipal holidays, as well as days with total consumption below 5000 Wh, were excluded. This deliberate filtering focuses the model on representative operational patterns, preventing the noise from atypical days (such as holidays or periods of minimal activity) from degrading its forecasting performance on typical working days. In total, these filtering steps resulted in the removal of 138 days from the original dataset.

To fill occasional gaps, simple linear interpolation was used. The estimate for each missing value was obtained by the arithmetic mean of the two valid records immediately before and after the missing point. This technique is effective in preserving the local continuity of the series without introducing artificial oscillations.

2.3. K-Means Clustering

High variability in load profiles from disaggregated environments, such as institutional buildings, presents a challenge for forecasting models, especially in hourly forecasting contexts at a disaggregated level and with a limited set of input variables. The absence of complementary information—such as temperature, occupancy, or operational context—combined with the exclusive availability of date, time, and load, increases the complexity of the problem and demands additional segmentation strategies.

In this scenario, unsupervised clustering algorithms have proven effective in organizing highly variable load time series, allowing for the reduction of internal heterogeneity and the specialization of predictive models in more homogeneous subsets [4,5]. K-means was chosen due to its simplicity and its successful application in systems with high variability.

According to Hernández et al. [4], K-means can be incorporated into hybrid architectures based on neural networks and self-organizing maps, segmenting daily profiles with similar behaviors in institutional microgrids. This approach demonstrated significant gains in terms of accuracy and adaptation to seasonal and intermittent patterns, being particularly relevant for applications in educational environments.

Similarly, Panapakidis [5] argues that the prior grouping of daily profiles by similarity favors the construction of specialized models for different types of days, contributing to improved hourly forecasting in systems with high operational variability.

From a technical standpoint, the algorithm performs partitioning based on the minimization of Euclidean distance between input vectors and centroids, iteratively adjusting the groupings until convergence is reached. Each centroid represents an average behavior pattern of the group, which favors the structuring of data into more cohesive and interpretable sets [4].

In this study, K-means was used to group daily load profiles by analyzing each day's consumption vector (the 96 15-min data points). Since K-means is sensitive to data scale, each weekday subset (e.g., all Mondays) was standardized using the Z-score before being processed by the algorithm. The mean and standard deviation parameters were calculated from the data used for model training.

For the technical implementation, the K-means++ initialization method was used to select initial centroids, as it promotes a more efficient and stable convergence by choosing initial points that are far from each other. The algorithm used the Euclidean distance metric, which is particularly suitable for time series data represented as vectors.

The number of clusters was fixed at three ($k = 3$). While internal evaluation metrics like the Elbow Method are often used to find a statistically optimal number of clusters, such an approach was not aligned with the primary objective here. The main goal was to create operationally meaningful groups based on similar load behaviors, ensuring each subset had a sufficient number of days for effective model training. Therefore, this fixed value was a pragmatic choice to guarantee that each specialized Fuzzy ARTMAP network could be trained on a robust data sample, preventing the instability and poor generalization associated with sparse clusters.

This clustering step prepares the dataset for the Fuzzy ARTMAP network, allowing it to be trained with more homogeneous information. This approach maintains forecasting consistency even when daily data are unstable.

2.4. Proposed Hybrid Modeling for Load Forecasting

The forecasting strategy is based on a hybrid approach that integrates K-means clustering with the Fuzzy ARTMAP neural network. Fuzzy ARTMAP is a supervised learning model with incremental adaptation capabilities, composed of two Fuzzy ART modules, ART_a and ART_b, which process the input and output vectors. These modules are interconnected by an inter-ART structure that manages the match tracking mechanism, ensuring learning stability. The network's incremental learning feature eliminates the need to reprocess the entire historical dataset with each new input, which makes the architecture scalable and well-suited for environments with evolving load profiles [10–12]. (For a detailed description of the Fuzzy ARTMAP network's architecture and learning mechanisms, see Appendix A).

The model was structured to operate with segmentation by weekday, from Monday to Friday. This approach aims to capture recurring consumption patterns while acknowledging that even within the same group, there is significant variability across days. This allows for a systematic evaluation of the predictor's efficiency, considering the unique behavioral patterns of each daily profile. Weekends were excluded from the analysis due to their atypical and unrepresentative consumption profiles, as low campus occupancy on these days results in extremely low and irregular consumption.

To handle the high variability of load profiles and the absence of complementary information, such as temperature or occupancy data, the hybrid approach uses the output of K-means clustering (detailed in Section 2.3) to specialize the forecast. To predict a specific day, the average load curve of the five most recent weekdays ($l = 5$) is calculated and compared to the obtained centroids; this value was chosen to provide a stable balance between capturing recent consumption trends and smoothing out single-day anomalies. Let $x_{d,t}$ be the load value on day d at time instant t. The average load at time t for these l days is given by Equation (3), and the complete average load curve vector is given by Equation (4):

$$\bar{x}_t = \frac{1}{l} \sum_{d=1}^{l} x_{d,t}, \quad t = 1, 2, \ldots, 96 \tag{3}$$

$$\bar{X} = (\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_{96}) \tag{4}$$

The overall methodology is summarized in the graphical representation presented in Figure 1. The complete pipeline can be broken down into two primary phases: an offline training phase and an online forecasting phase.

1. Offline Training Phase (performed once for each weekday):
    a. Data Segmentation: The historical dataset is first segregated by weekday (e.g., all Mondays).

b. Standardization: Each daily load profile within the subset is standardized using a Z-score.
c. Clustering: The K-means algorithm is applied to group the standardized profiles into k distinct clusters, each representing a typical consumption pattern.
d. Model Specialization: A separate Fuzzy ARTMAP network is trained exclusively on the data from each cluster, creating a pool of k specialized forecasting models.

2. Online Forecasting Phase (performed for each target day):
a. Reference Curve Generation: A reference load curve is calculated by averaging the consumption profiles of the five most recent weekdays.
b. Cluster Selection: This reference curve is compared to the centroids of the k trained clusters, and the cluster with the highest similarity (i.e., minimum Euclidean distance) is selected.
c. Model Activation: The specialized Fuzzy ARTMAP model corresponding to the selected cluster is activated for the forecasting task.
d. Recursive Forecasting: The activated model generates the 24-h load forecast for the target day using the recursive mechanism.

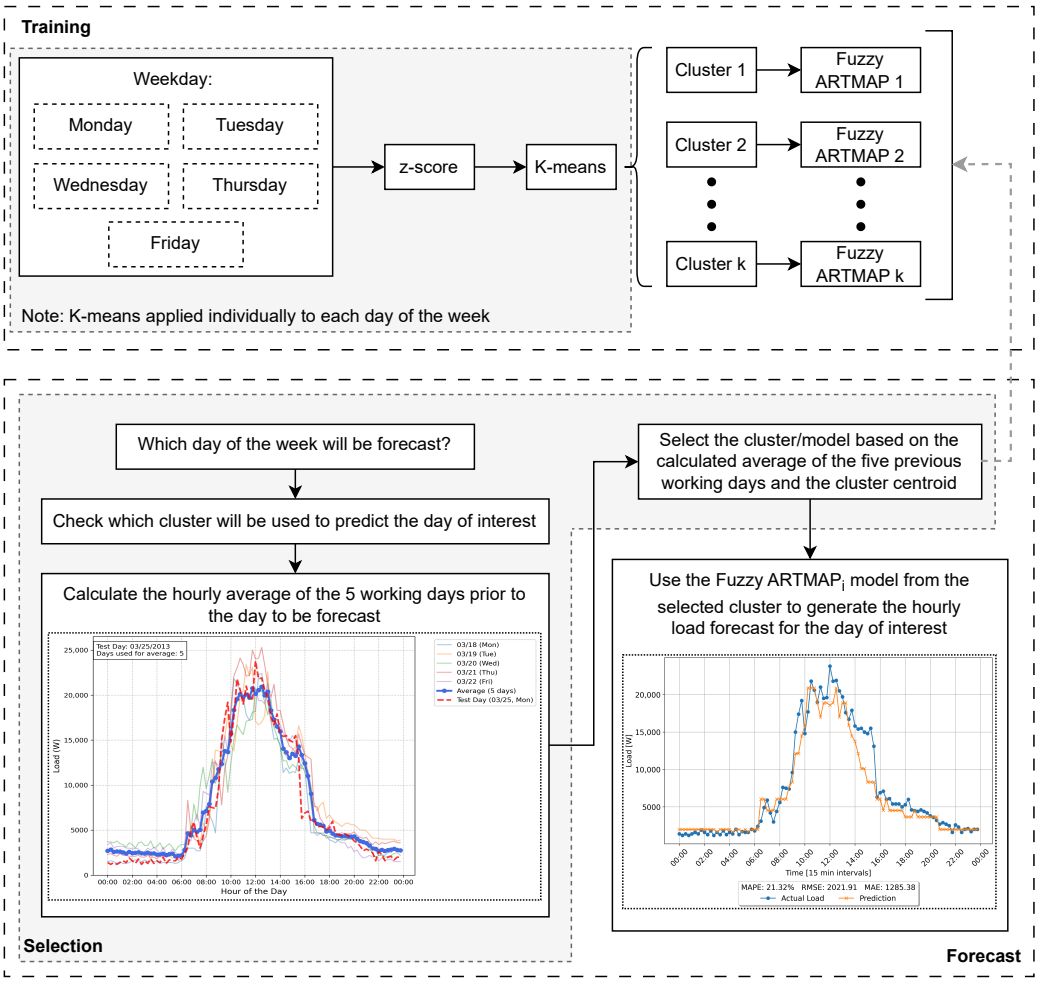


Figure 1. Graphical representation of the hybrid load forecasting methodology.

The recursive mechanism mentioned above is a fundamental aspect of the model's *ex-ante* design. It utilizes a sliding window of three time steps, $[x_{h-3}, x_{h-2}, x_{h-1}]$, to predict the load at step h. To generate the forecast for an entire day, the process begins by initializing the window with the last three known consumption values from the previous day. After the

first prediction is made (e.g., for 00:00), the mechanism becomes fully recursive: this newly predicted value is fed back into the sliding window as an input to predict the next time step (e.g., 00:15). This iterative process continues for all 96 intervals of the day, ensuring the forecast is generated without any access to future consumption data. In addition to the load values, the network input at each step also includes the hour encoded using sine and cosine, as described in Section 2.2.

Formally, the recursive forecasting process can be expressed as follows:

$$\hat{x}_h = f(x_{h-3}, x_{h-2}, x_{h-1}, \text{Hour_sine}_h, \text{Hour_cosine}_h), \quad (5)$$

where $\hat{x}_h$ denotes the predicted load at time step h, and Hour_sine_h and Hour_cosine_h correspond to the trigonometric encodings of the time variable at that step.

2.5. Experimental Design and Evaluation

The effectiveness of the proposed approach was evaluated through a comparative analysis across three distinct modeling scenarios. (1) The proposed hybrid model, consisting of the complete K-means and Fuzzy ARTMAP architecture, was evaluated. (2) An ablation baseline was established to isolate the specific contribution of the clustering component. For this, a standalone Fuzzy ARTMAP network was trained without the initial clustering, using data from 15 randomly selected days of the same weekday as the target day, ensuring the day to be predicted was not in the training set. (3) A benchmark comparison was conducted to contextualize the model's performance against two classic algorithms: a Multilayer Perceptron (MLP) and a Decision Tree. The MLP was chosen because it is a widely recognized benchmark and represents a different neural network paradigm from Fuzzy ARTMAP. The Decision Tree was selected as a representative of rule-based, non-parametric methods. This comparison against diverse, established models provides a more comprehensive validation of the proposed methodology.

The choice of using 15 days for training in the no-clustering scenario aims to simulate a practical setting with a limited historical data record while ensuring a sufficient sample volume for training. The Fuzzy ARTMAP network is particularly well-suited for this task. It can generalize patterns even with a small amount of data due to its incremental learning capability. This feature lets it learn new information without "forgetting" old patterns. By using random days, the approach forces the model to learn more general weekday patterns. It prevents the model from relying on short-term temporal dependencies. This is important because the network's fuzzy component also makes it resistant to outliers and noise, allowing it to perform well even with imprecise or unexpected data points.

The overall validation process was conducted iteratively for all weekdays. For each weekday subset, the model was trained on all days but one, which was used for forecasting. This procedure was repeated until each day in the subset had been forecasted individually. This approach allows for the model's performance to be evaluated in scenarios with limited historical series and without the requirement of temporal continuity between days.

To ensure optimal performance and prevent overfitting, the hyperparameters for all machine learning models—Fuzzy ARTMAP, MLP, and Decision Tree—were systematically tuned using a grid search methodology. Grid search is a technique that exhaustively searches a specified subset of the hyperparameter space for a model. It evaluates every possible combination of values from a predefined grid of parameters to identify the one that yields the best performance on a validation set [13]. This optimization was performed independently for each weekday subset (e.g., all Mondays) to ensure a fair comparison, where each model was fine-tuned for the specific data characteristics of that weekday. The set of parameters that yielded the lowest mean error on a validation set was then fixed and used for all subsequent forecasts within that specific weekday group. This

approach ensures that the models generalize across the subset rather than specializing on individual days.

The parameter grids for each model were defined as follows:

1. Fuzzy ARTMAP:
   - Input module vigilance parameter (ρ_a): values between 0.90 and 0.99 with a step of 0.01;
   - Output module vigilance parameter (ρ_b): values between 0.92 and 0.999 with a step of 0.001;
   - Inter-ART module vigilance parameter (ρ_{ab}): fixed value of 0.8;
   - Learning rate (β): discrete values {0.1, 0.2, 0.5, 0.8, 1.0};
   - Choice parameter (α): fixed value of 0.005.
2. Multilayer Perceptron (MLP):
   - Hidden layer sizes: Varied architectures, including single layers (e.g., (5), (10), (20)) and multiple layers (e.g., (3, 8), (5, 25), (3, 5, 10)).
   - Activation function: relu and tanh.
   - Solver: Optimization algorithms Adam and L-BFGS.
   - Regularization term: 0.0001 and 0.001.
   - Learning rate schedule: constant and adaptive.
3. Decision Tree:
   - Maximum depth: Unlimited, 5, 10, and 20.
   - Minimum samples for split: 2, 5, and 10.
   - Minimum samples per leaf: 1, 2, and 4.
   - Number of features for split: All, 'sqrt', and 'log2'.

The hyperparameter configuration that resulted in the lowest mean absolute percentage error (MAPE) on a validation set was selected as the optimal one for each experiment. Model performance was evaluated based on three statistical metrics widely used to assess forecasting performance [14]:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{6}$$

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{7}$$

$$\text{MAPE} = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right| \times 100 \tag{8}$$

In the above equations, y_i represents the actual load value and $\hat{y}_i$ the predicted value at time i. The metrics are calculated based on the 96 points of the daily load curve.

The model's implementation was carried out in Python 3.10, in a Jupyter Notebook (version 7.0.8) environment. The Fuzzy ARTMAP neural network was custom-implemented for this study, and libraries such as Scikit-learn (version 1.6.1) were also used for data preprocessing and K-means clustering [15]. Simulations were performed on an Intel(R) Core(TM) i7-8700 @3.20GHz (12 cores) processor with 16 GB of RAM. The source code is available upon request for academic reproduction and validation purposes.

3. Results

This section presents the results from applying the hybrid model described in Section 2.4 to the task of short-term hourly load forecasting. The objective is to assess

the predictive performance of the model in estimating the complete 24-h consumption profile with a 15-min resolution.

The effectiveness of the proposed approach was evaluated by comparing the model's performance in three scenarios: the hybrid model with K-means, a Fuzzy ARTMAP baseline without clustering, and a comparison with other classic machine learning models like MLP and Decision Trees. Forecasts were executed for all weekdays (from Monday to Friday) present in the dataset.

3.1. Comparative Performance Analysis

To provide an overview of the models' performance, Table 1 presents the mean and median of the error metrics for all weekdays.

Table 1. Comparative performance (mean and median) of the models by weekday.

Weekday	Hybrid		Fuzzy ARTMAP		Decision Tree		MLP	
	Mean	**Median**	**Mean**	**Median**	**Mean**	**Median**	**Mean**	**Median**
MAPE (%)								
Monday	24.22	22.73	34.03	32.57	33.57	32.84	35.51	34.99
Tuesday	26.44	24.89	29.34	27.94	31.50	30.42	33.84	31.31
Wednesday	28.20	27.55	31.28	29.77	32.81	31.24	35.03	33.26
Thursday	28.08	27.47	30.03	29.61	32.33	30.19	31.59	27.03
Friday	26.55	27.33	32.79	31.05	32.22	30.28	31.09	31.30
MAE (W)								
Monday	1530.61	1376.15	2269.69	2250.97	2295.88	2187.65	2476.82	2458.08
Tuesday	1820.47	1673.75	2040.26	2002.92	2247.45	2235.85	2412.20	2447.79
Wednesday	1736.17	1679.46	2214.76	1996.67	2315.96	2266.96	2474.29	2258.13
Thursday	1792.97	1803.86	2093.23	1990.83	2293.93	2294.65	2380.36	2249.10
Friday	1803.98	1612.37	2097.42	1939.79	2251.74	2332.42	2479.43	2365.73
RMSE (W)								
Monday	2352.88	2067.23	2776.45	2628.51	2968.10	2819.62	2894.21	2734.62
Tuesday	2778.15	2493.19	2847.52	2798.61	2882.02	2907.52	3033.71	2953.63
Wednesday	2565.48	2584.02	3259.43	2840.45	2926.32	2871.61	3105.03	2775.81
Thursday	2671.39	2695.13	2838.01	2774.24	2993.46	2810.65	2957.30	2937.06
Friday	2632.69	2533.02	2816.07	2770.97	3038.25	2806.18	3040.39	2765.12

The hybrid model consistently outperformed the other models in all metrics and for all weekdays. The prior clustering, by selecting more homogeneous training samples, contributed significantly to reducing the error, showing notable average gains.

3.2. Statistical Analysis (Mann–Whitney U Test)

The following table presents the results of the Mann–Whitney U test, comparing the performance of the hybrid model against the baseline models (Fuzzy ARTMAP without clustering, Decision Tree, and MLP) for each weekday. The test was performed for all error metrics. A p-value below 0.05 indicates a statistically significant difference between the models.

Table 2 shows that the differences between the hybrid model and the other models were statistically significant for all three metrics. This confirms that the gains observed with clustering are not random and represent a real improvement in model performance.

Table 2. Mann–Whitney U test results (*p*-values) for the hybrid model versus other models.

Weekday	Model Comparison	MAPE (*p*)	MAE (*p*)	RMSE (*p*)
Monday	Hybrid vs. Fuzzy ARTMAP	0.0000	0.0000	0.0000
	Hybrid vs. Decision Tree	0.0015	0.0021	0.0011
	Hybrid vs. MLP	0.0000	0.0337	0.0406
Tuesday	Hybrid vs. Fuzzy ARTMAP	0.0000	0.0004	0.0031
	Hybrid vs. Decision Tree	0.0000	0.0000	0.0000
	Hybrid vs. MLP	0.0002	0.0000	0.0000
Wednesday	Hybrid vs. Fuzzy ARTMAP	0.0489	0.0005	0.0006
	Hybrid vs. Decision Tree	0.0000	0.0000	0.0000
	Hybrid vs. MLP	0.0028	0.0000	0.0000
Thursday	Hybrid vs. Fuzzy ARTMAP	0.0019	0.0039	0.0008
	Hybrid vs. Decision Tree	0.0000	0.0000	0.0000
	Hybrid vs. MLP	0.0072	0.0000	0.0000
Friday	Hybrid vs. Fuzzy ARTMAP	0.0000	0.0010	0.0043
	Hybrid vs. Decision Tree	0.0000	0.0000	0.0000
	Hybrid vs. MLP	0.0047	0.0000	0.0000

3.3. Analysis of Clusters and Load Behavior

The hybrid approach uses similarity-based clustering to optimize network training. The strategy of using the average of the five preceding weekdays to identify the most similar cluster proved to be effective, as this average tends to capture the recent behavior pattern of energy consumption. This is visible in Figure 2, which compares the load curve of a test day (25 March 2013) with the load curves of the five preceding weekdays. The average curve (red dashed line) closely follows the general behavior of the test day (blue dashed line), justifying its use as a reference for clustering.

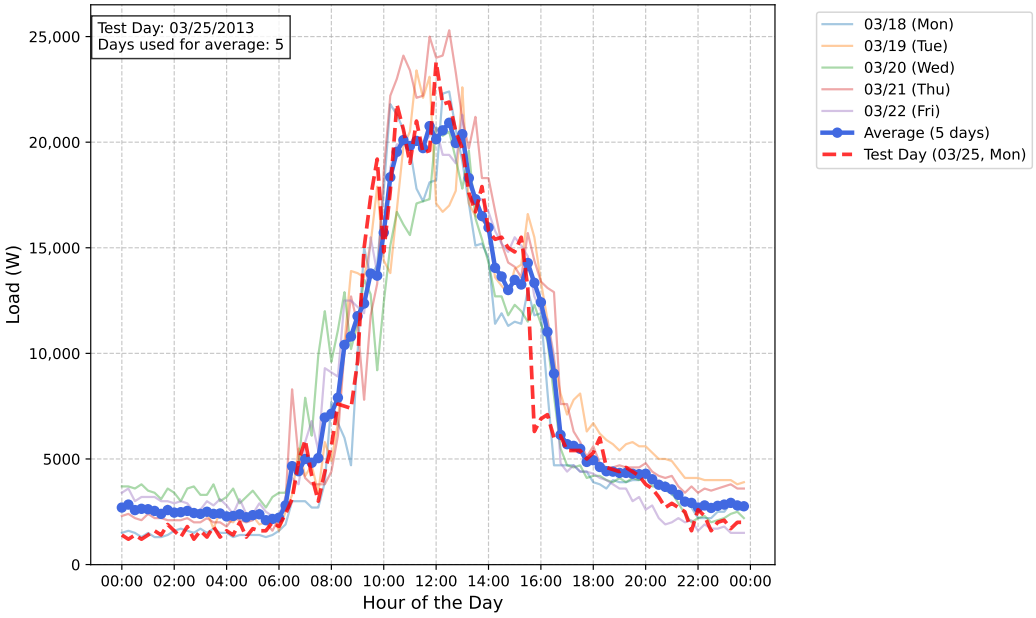


Figure 2. Load curves of previous weekdays, average and test day (25 March 2013).

To reinforce the effectiveness of clustering, Figure 3, a heatmap, and Figure 4, a line graph, illustrate the average load profiles of the three clusters identified by the K-means algorithm for Mondays. This specific weekday was selected for detailed visualization as it typically exhibits the most pronounced consumption variability, making the clustering

effect particularly evident. Cluster 1, for example, represents a significantly higher consumption pattern, with sharp peaks during business hours, especially between 9:00 a.m. and 2:00 p.m. In contrast, Cluster 2 presents a consistently lower load profile throughout the day. The clarity with which K-means separates these behaviors demonstrates that clustering organizes data so the Fuzzy ARTMAP network can learn from more homogeneous and consistent consumption profiles. The ability to identify and use these distinct patterns is a key factor in the superior performance of the hybrid model.

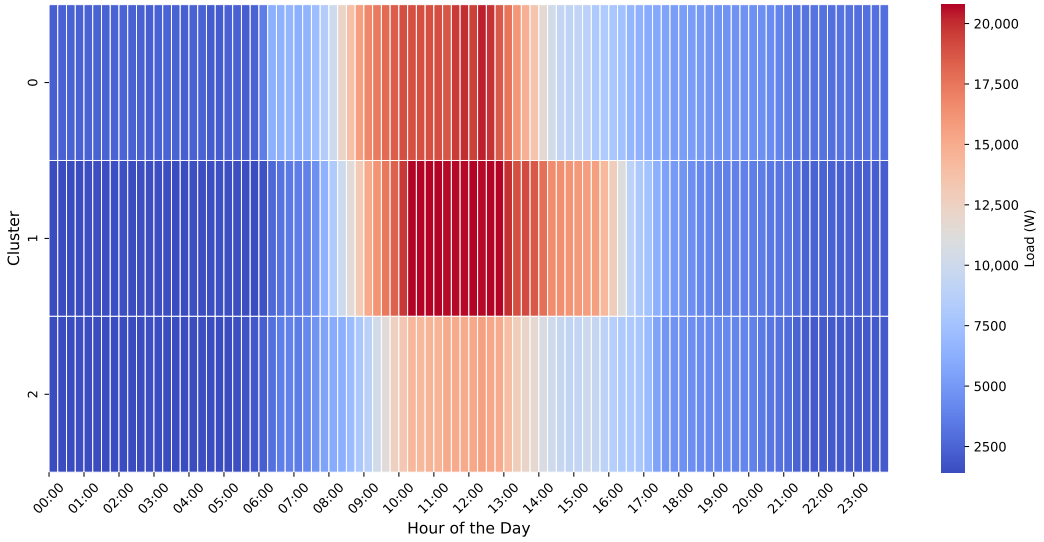


Figure 3. Heatmap of load behavior in clusters for Mondays.

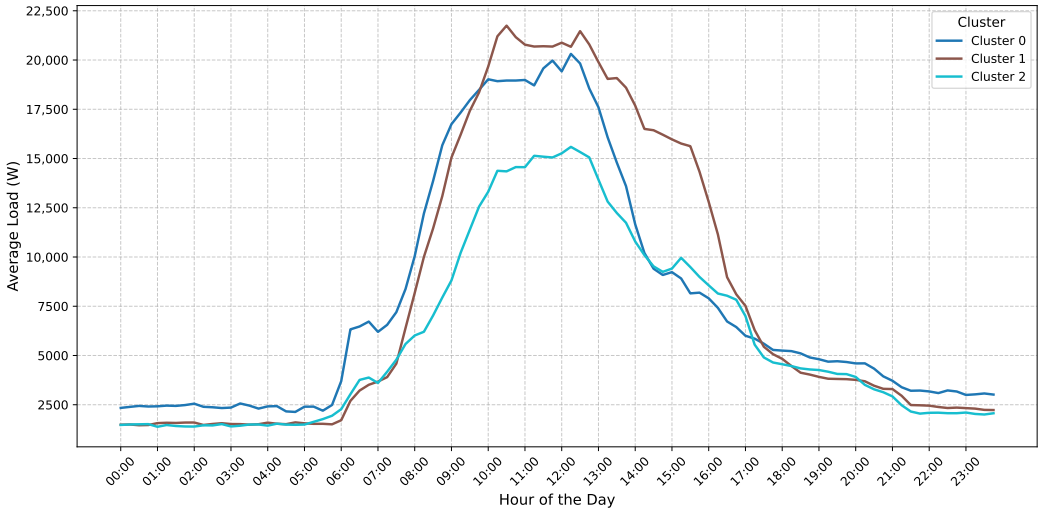


Figure 4. Average load behavior in clusters for Mondays.

3.4. Performance Analysis and Qualitative Comparison

The following tables (Tables 3 and 4) detail the performance metrics for a sample of days, offering a qualitative comparison between the hybrid model and the Fuzzy ARTMAP approach. We do not present detailed results for the other models (MLP and Decision Tree) here, as their performance was consistently inferior, as already demonstrated in the overall statistical analysis and in Tables 1 and 2. This focused approach allows for a more direct and relevant validation of the contribution of clustering to the Hybrid Model.

Table 3. Forecasting results without clustering, with error metrics by date and weekday.

Weekday	Forecasted Day	MAPE (%)	RMSE (W)	MAE (W)
Monday	04 June 2012	39.93	2948.06	2246.72
	02 July 2012	35.74	2828.97	2116.87
	27 August 2012	51.25	4898.25	3362.92
	10 September 2012	24.65	2363.79	1457.81
	03 June 2013	33.82	2495.07	1728.23
Tuesday	17 July 2012	28.94	3288.02	2249.38
	27 November 2012	22.04	1745.00	1378.21
	12 March 2013	34.38	5546.29	3610.01
	14 May 2013	22.06	1574.44	1094.79
	09 July 2013	20.95	1975.82	1338.65
Wednesday	23 May 2012	39.04	4617.05	3301.02
	30 May 2012	38.26	5657.40	3943.84
	27 June 2012	49.60	3581.87	2939.55
	16 January 2013	26.70	2840.45	1917.83
	07 August 2013	21.83	2181.61	1432.81
Thursday	19 July 2012	24.84	2552.45	1681.48
	30 August 2012	30.18	4707.09	3298.65
	24 January 2013	34.26	3311.84	2461.92
	21 March 2013	26.42	3148.23	2288.22
	23 May 2013	23.48	1864.36	1145.37
Friday	04 May 2012	31.64	4076.96	2865.26
	29 June 2012	32.49	3545.73	2770.92
	27 July 2012	30.87	2828.96	2047.80
	01 February 2013	21.56	3753.42	2446.43
	01 March 2013	33.10	4872.23	3444.79

Table 4. Forecasting results using clustering, with error metrics by date and weekday.

Weekday	Test Date	MAPE (%)	RMSE (W)	MAE (W)
Monday	04 June 2012	13.38	1224.33	819.13
	02 July 2012	14.41	2561.90	1485.13
	27 August 2012	12.67	1140.37	764.38
	10 September 2012	14.99	1966.08	1201.25
	03 June 2013	20.37	2346.90	1376.15
Tuesday	17 July 2012	11.81	1132.66	768.39
	27 November 2012	17.12	1775.90	1217.00
	12 March 2013	15.94	2053.29	1228.61
	14 May 2013	16.44	1335.55	901.25
	09 July 2013	14.28	1070.11	770.94
Wednesday	23 May 2012	19.98	3598.44	2153.60
	30 May 2012	14.70	2859.40	1692.77
	27 June 2012	16.92	1972.91	1330.75
	16 January 2013	20.27	1884.29	1238.74
	07 August 2013	20.75	2213.70	1486.98
Thursday	19 July 2012	17.60	2353.28	1435.50
	30 August 2012	18.67	3145.25	1729.06
	24 January 2013	16.38	1804.37	990.95
	21 March 2013	17.01	1944.79	1241.15
	23 May 2013	15.93	1774.39	987.11
Friday	04 May 2012	14.78	2425.45	1439.66
	29 June 2012	17.31	3117.66	1849.57
	27 July 2012	14.78	1618.82	1041.61
	01 February 2013	18.86	2534.06	1575.77
	01 March 2013	20.90	2724.67	1600.83

Results Without Clustering

According to the non-clustered approach described in Section 2.5, all working days were individually forecasted using training based on random subsets of the same weekday. The results illustrate the challenges of forecasting in disaggregated environments, where the model's performance can vary significantly across different weekdays.

The model's performance varied significantly across the weekdays. The average MAPE values were higher on Mondays and Wednesdays, suggesting more complex consumption dynamics on those days. This behavior may be related to the resumption of activities after the weekend and the diversity of institutional routines.

Since the non-clustered approach does not perform prior similarity filtering, the presence of atypical or noisy profiles in the training set directly impacts the error metrics. This highlights the limitations of this strategy in environments with high behavioral variability. The data in Table 3 are only illustrative examples of forecasted days, as forecasting was performed for all available weekdays from Monday to Friday, as described in the methodology.

3.5. Results with Clustering

The hybrid approach optimizes the Fuzzy ARTMAP network's training by using a preliminary clustering step. This method allows the model to learn from more homogeneous data subsets.

For each forecast, the model uses the average load curve of the five preceding weekdays to identify the most similar cluster. The Fuzzy ARTMAP network is then trained exclusively with data from that specific cluster. This process ensures that the model learns from the most relevant data, removing the test day from the training set to maintain causality.

Table 4 provides the MAPE, RMSE, and MAE values from sample test days. The data is illustrative, as the model was executed for all available weekdays to validate its robustness in disaggregated contexts.

Overall, a significant reduction in error metrics is observed compared to the non-clustered approach, especially on days with greater variability in load patterns. This result reinforces the contribution of similarity-based segmentation to model specialization and improved forecasting reliability in institutional environments with high behavioral oscillation.

3.6. General Comparison Between Approaches

This section provides a comparative analysis between forecasts performed with and without K-means clustering. We focus on the hybrid approach and the base Fuzzy ARTMAP model, as these two models consistently outperformed other methods like MLP and Decision Trees. The initial performance analysis and the Mann–Whitney U statistical test confirmed their superiority across all metrics and weekdays, justifying this focused comparison (Sections 3.1 and 3.2).

Table 5 shows the percentage gains from applying clustering for each metric (MAPE, RMSE, and MAE). The gain was calculated based on the relative error reduction compared to the non-clustered approach, using the results from Tables 3 and 4. The goal is to highlight the practical benefit of the segmentation strategy on forecasting effectiveness.

Overall, applying clustering resulted in a significant reduction in error metrics, especially on dates with higher variability in load patterns. This reinforces the contribution of similarity-based segmentation to model specialization and to improving forecast reliability in dynamic institutional environments.

The strategy of dynamically identifying the most representative cluster—based on the average of the five preceding weekdays—proved effective for selecting more homogeneous subsets. This facilitated network learning with consistent patterns and mitigated the impact of outliers.

In addition to the quantitative improvement, the clustered approach also showed greater regularity, with reduced error dispersion. This stability indicates a better generalization capacity for the network, even in operational scenarios subject to unpredictable variations, such as in microgrids.

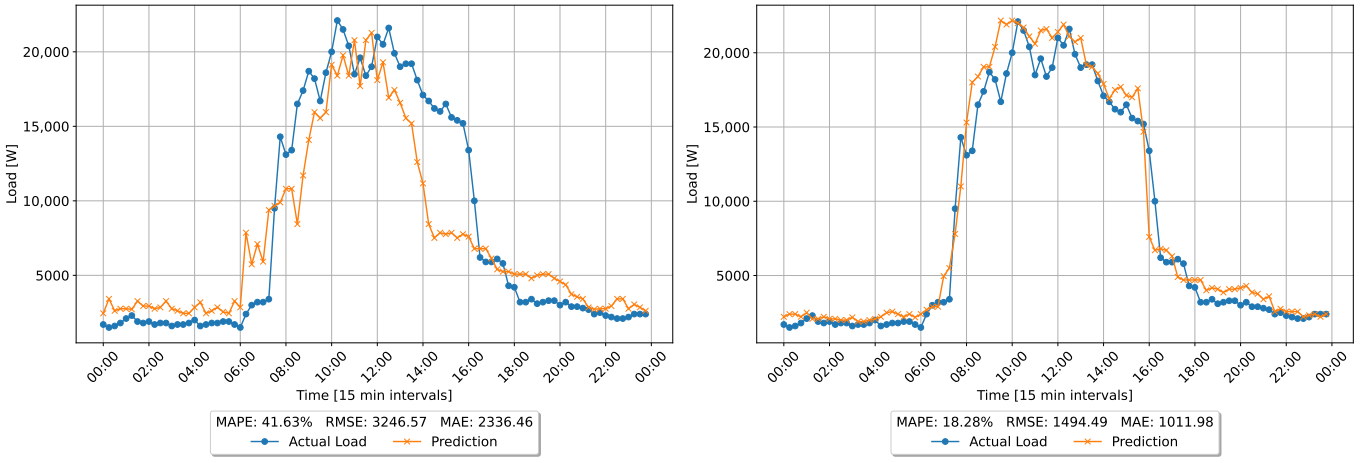
Table 5. Percentage gain per metric (MAPE, RMSE, and MAE) with clustering applied.

Subset	Forecasted Date	MAPE Gain (%)	RMSE Gain (%)	MAE Gain (%)
Monday	04 June 2012	66.49	58.47	63.54
	02 July 2012	59.68	9.44	29.84
	27 August 2012	75.28	76.72	77.27
	10 September 2012	39.19	16.83	17.60
	03 June 2013	39.77	5.94	20.37
Tuesday	17 July 2012	59.17	65.55	65.83
	27 November 2012	22.29	−1.78	11.70
	12 March 2013	53.65	62.98	65.97
	14 May 2013	25.48	15.17	17.68
	09 July 2013	31.84	45.83	42.40
Wednesday	23 May 2012	48.83	22.07	34.74
	30 May 2012	61.56	49.47	57.09
	27 June 2012	65.88	44.88	54.74
	16 January 2013	24.09	33.66	35.42
	07 August 2013	4.95	−1.47	−3.78
Thursday	19 July 2012	29.14	7.81	14.61
	30 August 2012	38.13	33.20	47.58
	24 January 2013	52.17	45.52	59.74
	21 March 2013	35.61	38.23	45.76
	23 May 2013	32.16	4.82	13.82
Friday	04 May 2012	53.29	40.53	49.76
	29 June 2012	46.74	12.07	33.26
	27 July 2012	52.12	42.78	49.14
	01 February 2013	12.53	32.51	35.58
	01 March 2013	36.88	44.07	53.54

Figures 5–7 visually illustrate the difference between forecasts generated by the two strategies, comparing the real profiles with the estimates obtained with and without clustering. It is clear that by grouping data by similarity, the model can reproduce the structure of the load curves more accurately.

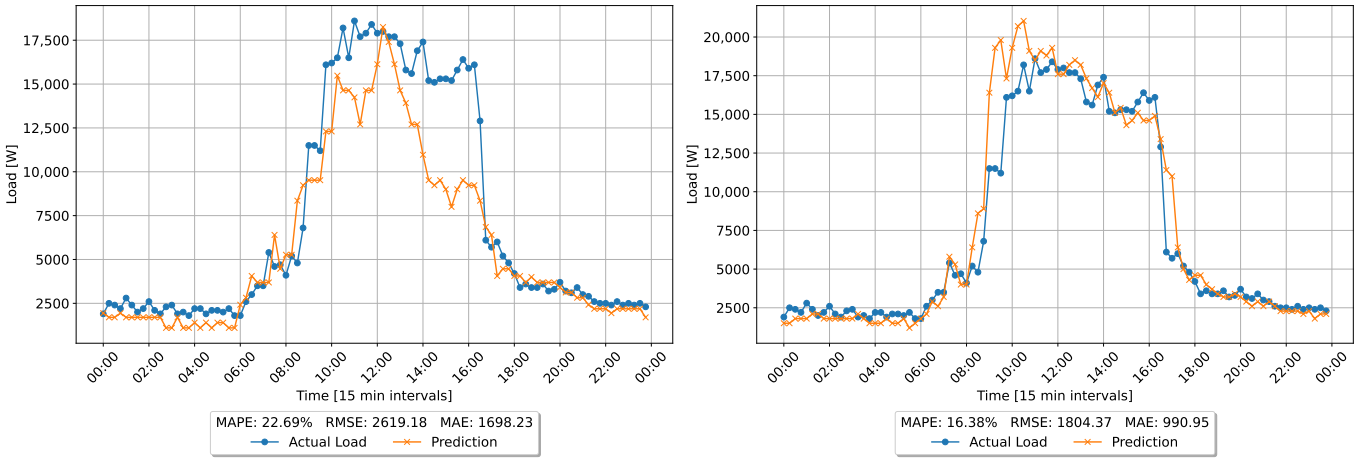
This figure (Figure 5) highlights a key finding: the hybrid model's ability to better capture the load curve's shape explains its quantitative gains. This is particularly true for the peak load, which is a significant period for energy management.

As shown in Figures 6 and 7, the hybrid model's ability to capture load curve trends more accurately is clear across different dates. The clustered approach consistently aligns better with the actual load, showing its reliability in managing the daily variations in energy consumption.



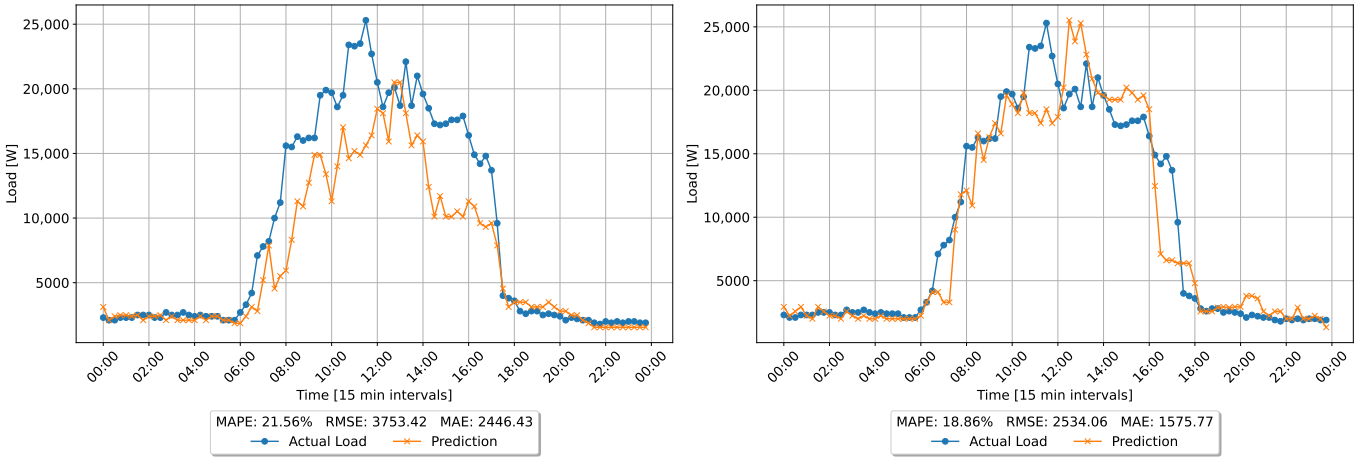
(**a**) Forecast without clustering. (**b**) Forecast with clustering.

Figure 5. Comparative analysis of load curves for 17 December 2012, Monday.



(**a**) Forecast without clustering. (**b**) Forecast with clustering.

Figure 6. Comparative analysis of load curves for 24 January 2013, Thursday.



(**a**) Forecast without clustering. (**b**) Forecast with clustering.

Figure 7. Comparative analysis of load curves for 1 February 2013, Friday.

4. Discussion

The results show that our hybrid model, which combines K-means clustering with a Fuzzy ARTMAP neural network, is an effective tool for forecasting energy consumption in environments with constantly changing patterns. Analysis across all weekdays, using multiple error metrics, reveals that the model not only surpasses the standalone Fuzzy ARTMAP network but also performs better than traditional approaches like Decision Trees and MLPs.

The model's success lies in how clustering organizes the training process. By separating days into groups with similar consumption patterns, the Fuzzy ARTMAP network learns from a cleaner, more focused dataset. Table 1 shows this improvement through the reduction in MAPE, MAE, and RMSE values. Statistical analysis using the Mann–Whitney U test (Table 2) confirms this improvement is not coincidental. Even in the few cases where statistical significance was less pronounced, such as for the MAPE on Wednesday ($p = 0.0489$), the value is still within the confidence limit, which demonstrates the model's consistency.

A key design decision was to use the average consumption of the five previous weekdays to select the right cluster for the forecast. This method worked well because recent behavior is a great indicator of what will happen next. Figure 2 makes this clear, showing how the average curve closely approximates the actual curve of the test day. It is a simple, practical solution that allows the model to adapt to new trends without needing complex configurations.

Figures 3 and 4 show why this works so well. The images reveal distinctly different consumption patterns for Mondays: some days have high, well-defined consumption peaks, while others have lower, more stable consumption. Clustering successfully separates these "types" of days, allowing the forecasting model to specialize in each one. Figures 5–7 show the result of this specialization: the hybrid model's forecasts follow the shape and peaks of the actual consumption much more faithfully than the model without clustering.

This work builds on previous research but focuses on a specific challenge: forecasting load in a real-world environment using only historical consumption data, without extra data like weather forecasts. The results show that even without this additional information, intelligently organizing the data brings great benefits. The model's ability to handle the natural instability of this data suggests it could be useful for other forecasting problems involving complex time series, such as predicting solar power generation or demand in different types of buildings. It could even be applied to other regression tasks.

It is important to acknowledge some limitations of this study. The analysis relied solely on historical consumption, and adding information like temperature, academic calendars, or building occupancy data could make the forecasts even better. The dataset itself had flaws, such as periods of strikes and missing measurements, which, although corrected with interpolation, may have added some noise to the process. The choice of three clusters was also a practical decision to ensure that each group had enough samples for effective training; a longer data history could allow for the use of more clusters, possibly refining the forecast further. Finally, although the model was tested on all available weekdays, its performance in other locations, such as different universities or commercial buildings, still needs to be verified.

In the future, we plan to explore using smoothing techniques to filter noise from the data before applying clustering. Testing the model with external data and in other locations will be essential to confirm its practical utility in energy management. The Fuzzy ARTMAP network's ability to learn continuously and its good tolerance for noise are features that align perfectly with this approach, allowing the model to adapt to new patterns without needing to be completely retrained.

5. Conclusions

This work proposed a hybrid model based on the integration of the Fuzzy ARTMAP neural network and K-means clustering, applied to hourly electricity load forecasting at a disaggregated level. The developed methodology stood out for its ability to dynamically identify the group most similar to the consumption profile of a given day, based on the average of the five most recent weekdays. As a result, it was possible to train the network exclusively with more homogeneous data aligned with the expected behavior, thereby improving the model's accuracy.

It is important to highlight that the good performance obtained is not solely due to the application of clustering but also to a set of essential data preprocessing steps performed prior to modeling. These steps included the removal of load records below 5000 W, the exclusion of national and municipal holidays, the interpolation of gaps in series with missing data, and the use of cyclic encoding to represent the time variable, standardizing the curves into series with 96 daily records (15 min intervals). These actions were crucial to ensure consistency and comparability among the profiles used.

The results showed significant improvements across the MAPE, RMSE, and MAE metrics, outperforming not only the non-clustered approach but also other established models like MLP and Decision Trees. The statistical significance of these improvements was confirmed by the Mann–Whitney U test, validating the superiority of the hybrid architecture. Furthermore, the model exhibited robustness in the face of behavioral fluctuations typical of institutional microgrids, even when operating without external variables such as temperature or occupancy. The adopted structure proved promising for real environments with incomplete or difficult-to-standardize data, contributing methodologically to the literature in the field.

However, the study presented significant limitations. The available dataset contained only load and timestamp (date and time) information, without additional explanatory variables. The analyzed period coincided with moments of academic instability (such as strikes), which affected consumption patterns, making each day potentially atypical even within the same weekday subset. The dataset also presented failures and missing data, requiring interpolation techniques for gap filling. Moreover, the intrinsic challenge of working with disaggregated loads, characterized by high noise and irregular behavior, imposes additional barriers to modeling. Nevertheless, the MAPE values observed are considered satisfactory given the complexity of the scenario.

For future developments, it is suggested to integrate smoothing techniques into the preprocessing pipeline, with the goal of reducing high-frequency noise and improving signal continuity. Additionally, applying the model to different institutional environments and with varying levels of aggregation may help assess its generalization capacity and expand its practical applicability.

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Abbreviations

The following abbreviations are used in this manuscript:

RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error
ART	Adaptive Resonance Theory

Appendix A. Fuzzy ARTMAP

The Fuzzy ARTMAP network is a supervised learning model that integrates principles from Adaptive Resonance Theory (ART) with fuzzy operators. Developed by Carpenter et al. [11], its architecture allows for incremental learning and categorical stability, even in noisy and non-stationary time series.

Its structure is composed of two ART Fuzzy modules, denoted ART_a and ART_b, responsible for processing input vectors and desired outputs, respectively. The connection between the modules is mediated by the intermediate *inter*-ART layer, which dynamically adjusts category associations through the match tracking mechanism [12].

The network's operation is regulated by three main parameters: the choice parameter $\alpha > 0$, the learning rate $\beta \in [0, 1]$, and the vigilance parameter $\rho \in [0, 1]$, which can be set separately for the modules ρ_a, ρ_b, and the cross-connection ρ_{ab}.

The training process begins with the normalization of the input vectors $a = [a_1, a_2, ..., a_M]$ and output vectors $b = [b_1, b_2, ..., b_M]$, as given by the following equation:

$$\overline{I_a} = \frac{a}{|a|}, \qquad \overline{I_b} = \frac{b}{|b|} \tag{A1}$$

Then, fuzzy complement coding is applied, as described in Equation (2):

$$I^a = [a, \overline{a}], \qquad I^b = [b, \overline{b}] \tag{A2}$$

The choice function for the neurons in layer F_2 of each module is given by the following:

$$T_j^a = \frac{|I^a \wedge w_j^a|}{\alpha + |w_j^a|}, \qquad T_k^b = \frac{|I^b \wedge w_k^b|}{\alpha + |w_k^b|} \tag{A3}$$

The winning neurons are then determined by the following:

$$J = \arg\max_j T_j^a, \qquad K = \arg\max_k T_k^b \tag{A4}$$

The resonance condition is verified by the following expressions:

$$\frac{|I^a \wedge w_J^a|}{|I^a|} \geq \rho_a, \qquad \frac{|I^b \wedge w_K^b|}{|I^b|} \geq \rho_b \tag{A5}$$

When satisfied, the synaptic weights are updated as follows:

$$w_J^{a(\text{new})} = \beta(I^a \wedge w_J^{a(\text{old})}) + (1 - \beta) w_J^{a(\text{old})} \tag{A6}$$

$$w_K^{b(\text{new})} = \beta(I^b \wedge w_K^{b(\text{old})}) + (1 - \beta) w_K^{b(\text{old})} \tag{A7}$$

In the *inter*-ART layer, the association weights are defined by the following:

$$w_{J,K}^{ab} = 1, \quad w_{j,k}^{ab} = 0 \quad \text{for } j \neq J \text{ or } k \neq K \tag{A8}$$

The network performs this process iteratively for each presented input–output pair, enabling incremental storage of learned patterns. This characteristic eliminates the need to reprocess the entire historical base with each new input, which favors the scalability of the architecture [11].

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Analysis and Optimization of Fuzzy ARTMAP Parameters for Multinodal Electric Load Forecasting

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Abstract

Accurate electrical load forecasting is fundamental to the efficient operation of energy systems and plays a decisive role in both generation planning and the prevention of supply interruptions. Anticipating demand with precision enables energy generation and distribution to be adjusted effectively, reducing risks for both industrial and residential consumers. However, forecasting is challenged by climatic variations, demographic changes, and evolving consumption patterns, which limit the effectiveness of traditional approaches. Advanced machine learning techniques such as artificial neural networks have demonstrated potential to address these challenges, although their performance depends strongly on hyperparameter optimization. This study applies a multinodal forecasting methodology based on the Fuzzy ARTMAP network to predict short-term electricity demand at nine substations in New Zealand. The method involves an exhaustive search for network parameters, particularly the vigilance parameters ρ_a and ρ_b and the learning rate β, which are critical to model performance. The input data were extended with statistical measures—maximum, minimum, mean, and standard deviation—to evaluate their contribution to forecast accuracy. The results showed that the standard deviation provided the most consistent improvements among the windowing techniques, reducing the Mean Absolute Percentage Error (MAPE) in most substations. Parameter analysis further indicated that specific combinations such as ρ_a and β strongly influence category formation within the network, and consequently the precision of the forecasts.

Keywords: adaptive resonance theory; artificial neural networks; short-term forecasting; electricity system distribution; machine learning; multinodal load forecasting

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1. Introduction

For electricity companies, providing accurate load forecasts is crucial to ensuring that the grid meets demand efficiently and reliably. Accurate forecasts make it possible to optimize supply while maintaining a continuous, stable, and economically viable service that avoids system interruptions and overloads. However, utilities face growing challenges in achieving higher accuracy, particularly with the increasing complexity of the grid due to the integration of intermittent renewable sources and varying consumption patterns [1].

Electricity load forecasting can be classified into four time horizons: very short term (VSLF), covering fractions of a second up to fifteen minutes; short term (STLF), covering hours to days; medium term (MTLF), covering weeks to months; and long term (LTLF), involving annual projections [1,2]. Traditionally, most forecasting approaches focus on overall system demand, which is essential for operational functions such as economic dispatch, safety assessment, and maintenance scheduling [3].

However, with the increasing complexity of power systems, multinodal forecasting that analyzes individual loads in substations, consumption zones, bars, or specific nodes has become equally important [3,4]. While global forecasting helps with the systemic balance between supply and demand, the multinodal approach is indispensable for static and dynamic system analysis such as load flow and voltage stability assessment [4]. Despite the challenges inherent in its greater complexity, this approach enables more effective infrastructure management on a regional scale [4,5]. Both global and multinodal forecasting, however, face challenges arising from the nonlinear and nonstationary nature of load data, which are influenced by factors such as extreme weather conditions, seasonality, and socioeconomic events.

Faced with these complexities, traditional linear models such as ARIMA and linear regression often prove inadequate and unable to capture the required dynamic and nonlinear relations [6,7]. To address these limitations, a wide range of advanced data-driven methods, particularly those based on machine learning and Artificial Neural Networks (ANN), have emerged as superior alternatives for modeling these intricate relationships while offering greater precision in real-world scenarios.

A study by [8] concluded that while traditional statistical and machine learning methods have their strengths, deep learning techniques are more suitable for the dynamic and nonlinear characteristics of electricity load data. Studies on the Greek electricity system have highlighted that even simple feedforward ANNs can achieve a decent level of predictive performance, though their results heavily depend on the careful selection and quality assurance of the input data [9].

The literature reveals a variety of methods for predicting electrical loads, including both classic and innovative approaches. For instance, ref. [10] obtained promising results by combining Autoregressive Integrated Moving Average (ARIMA) with ANNs for short-term electrical load forecasting, while [11] conducted a comprehensive evaluation of various techniques based on artificial intelligence and time series analysis. Ref. [12] demonstrated the importance of feature extraction and influencing factor analysis, showing that data mining improved results when combined with regression approaches such as Random Forest.

In recent years, more sophisticated approaches have been developed. Ref. [13] proposed a Bidirectional Long Short-Term Memory (BiLSTM) architecture that reduced Mean Absolute Error (MAE) by 25% and 46% compared to unidirectional Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) in 24-h forecasts. Similarly, ref. [14] further advanced the use of BiLSTM by incorporating stochastic weight averaging to improve generalization in substation-level forecasting.

Ref. [15] integrated a deep neural network with time series analysis and feature selection, achieving an MAPE of 3.78% and RMSE of 432.4, which outperformed ARIMA, Random Forest, and Support Vector Regression. Ref. [16] employed Temporal Convolutional Networks (TCNs) within a hierarchical framework that was resilient against missing or erroneous data, achieving reliable short-term forecasts for residential loads. At the distribution level, ref. [17] introduced a lean forecasting approach combining clustering, neural networks, and Monte Carlo methods, while [18] proposed a step-by-step framework that explicitly incorporates electric vehicle charging stations into load forecasting.

To further overcome existing limitations, a wide range of sophisticated and hybrid models has been explored. For example, ref. [19] highlighted that while conventional models such as Autoregressive Integrated Moving Average (ARIMA) have their place, more complex approaches like Fuzzy Time Series (FTS) outperform them due to their ability to handle nonlinear and uncertain data at low computational cost. Other hybrid approaches show similar promise: Ref. [20] proposed a method for multi-energy loads using a Radial Basis Function–AutoRegressive with eXogenous inputs (RBF-ARX) model for deterministic components and a Gaussian Mixture Model (GMM) for stochastic ones, improving forecast quality by up to 50%. Likewise, ref. [21] combined an Autoencoder (AE) for dimensionality reduction with a Radial Basis Function Neural Network (RBFNN) for load forecasting, demonstrating better accuracy and efficiency compared to a standalone RBFNN.

Additionally, ref. [22] compared the performance of ANN and Autoregressive with eXogenous inputs (ARX) models for net demand forecasting in distribution substations, concluding that the ANN model was better for direct forecasting due to the nonlinear nature of demand. Hybrid models have proven valuable in other domains as well, as seen in the work of [23], who used a Particle Filtering–ARIMA (PF-ARIMA) fusion model to predict battery useful life, achieving a 70% improvement over other methods by handling data specificities such as capacity regeneration.

A particularly relevant advance was presented by [24] with the Fuzzy-ARTMAP (FAM-ANN) continuous learning network, specifically developed for smart grids. This innovative solution proved capable of incorporating new data without requiring complete retraining, achieving a MAPE of ≈2% (60% better than the static version) and maintaining a consistent performance of 2.32% MAPE over 12 months, thereby combining precision, computational efficiency, and remarkable adaptability to dynamic scenarios.

Recently, multinodal forecasts have gained special prominence for their ability to integrate global planning with detailed local analysis. In this scenario, contributions include that of [25], who proposed a single-stage forecasting method. This approach simultaneously considers scenarios with and without the load contribution factor, offering a more efficient solution compared to conventional two-stage methods.

While traditional approaches require first predicting the global load and then predicting the local loads, the proposed method significantly simplifies computational complexity by performing these steps in an integrated manner while also demonstrating improvements in the precision of the results. Another notable advance was presented by [26], who introduced a methodology for forecasting consumption at all system nodes. Their method was based on the direct determination of specific rating coefficients for each voltage level, providing a comprehensive approach that encompasses everything from substations to supply zones and other critical components of the energy infrastructure. These advances in multinodal techniques pave the way for practical applications in real systems, such as the one presented in this paper.

However, it is crucial to note that the performance of these models depends fundamentally on the proper selection and optimization of their hyperparameters, such as learning rates and the number of neurons. An inadequate configuration can lead to problems of underfitting or overfitting, compromising the generalization of the model [27,28]. In this context, parameter optimization has proven to be a crucial strategy for improving the precision of load forecasting models. Techniques such as grid search, evolutionary algorithms, and Bayesian optimization make it possible to efficiently explore the parameter space in order to identify configurations that maximize predictive performance [29]. This importance is corroborated by work such as that of [30], which showed how hyperparameter optimization is decisive in reducing errors in short-term forecasting models for urban smart grids. Similarly, ref. [31] obtained significant performance gains by applying optimization

techniques to a fuzzy ARTMAP neural network for forecasting at disaggregated levels, combining it with singular spectrum analysis for noise treatment.

Despite the advances achieved with machine learning, deep learning, and hybrid approaches, several challenges remain. Many models, including deep learning architectures such as LSTM and transformer models, are highly sensitive to hyperparameter configurations, prone to overfitting, and often require large datasets and long training times. In multinodal forecasting, where the complexity is higher and data availability may be limited, systematic strategies for parameter optimization and data enrichment are still underexplored. These gaps open space for approaches such as Fuzzy ARTMAP, derived from Adaptive Resonance Theory (ART), which offers resistance to outliers, incremental learning without catastrophic forgetting, reduced dependence on large datasets, and fast convergence [32]. These characteristics make ARTMAP a suitable alternative for scenarios that demand adaptability, accuracy, and efficiency.

This study applies and expands the methodology proposed by [33] for short-term (24 h ahead) multinodal electricity load forecasting, using data from nine substations of a system in New Zealand. The approach focuses on implementing the fuzzy ARTMAP neural network [34], emphasizing an exhaustive search for optimal parameters, detailed analysis of parameter effects, and enrichment of input data with descriptive statistics (maximum, minimum, mean, and standard deviation) calculated from the participation factors in the periods $t-1$, $t-2$, and $t-3$. Its performance is compared with well-known benchmark models, including Multilayer Perceptron (MLP), Random Forest, and Support Vector Regression (SVR), to evaluate the method's effectiveness and investigate how network configurations and input variables influence the accuracy of short-term electricity demand forecasts.

The remainder of this article is organized as follows: Section 2 outlines the key contributions of this study to the field of electrical load forecasting; Section 3 describes the methodology, including the dataset, the fundamentals of the Fuzzy ARTMAP network, the strategies for global and multinodal load forecasting, the evaluation criteria, and the windowing techniques applied; Section 4 presents and discusses the results, offering a comparative analysis of the performance of different techniques and a detailed investigation of the effect of network parameters on the forecasts; finally, Section 5 summarizes the study's conclusions and suggests future research directions.

2. Contributions

This study offers contributions to the field of electricity load forecasting, focusing on the application and optimization of the Fuzzy ARTMAP network in a multinodal context. The main contributions are as follows:

- Systematic Parameter Optimization: We present a rigorous approach based on an exhaustive search for critical Fuzzy ARTMAP network parameters (ρ_a, ρ_b, and β). This goes beyond a simple application of the model, providing a detailed analysis of how parameter interactions influence network behavior, cluster formation, and forecast accuracy.
- Statistical Feature Validation for Data Enrichment: The enrichment of input data with statistical features such as the standard deviation is shown to improve forecasting performance across multiple substations. This provides a new and effective strategy for preprocessing time series data.
- Detailed Multinodal Forecasting Analysis: A comprehensive assessment of forecasting performance at a disaggregated level across nine individual substations is provided. The multinodal approach offers insight into local complexities of power networks that are often overlooked in global forecasting studies.

- Discussion of Model Robustness and Limitations: Key limitations of the proposed method are analyzed, including its sensitivity to parameter configurations and the challenge of applying a single optimized parameter set to large-scale systems.

3. Load Forecasting

This section details the methodology used for multinodal electricity load forecasting. It starts with a description of the dataset and then presents the Fuzzy ARTMAP neural network. This section also explains how global and multinodal forecasts are made, including the mathematical formulas for the participation factor. Finally, the criteria for comparing the results are presented.

3.1. Dataset

The database was supplied by the Electricity Commission of New Zealand and collected from the Centralized Dataset (CDS) from January 2007 to March 2009. It contains active power values measured every half an hour at each substation along with indicators for the day of the month, month, year, day of the week, holidays, time, and load sample value. The substations studied are in the Waikato region, located in the northern part of New Zealand's North Island. The substations considered were Kopu, Waikino, Waihou, Hamilton 11, Hamilton 33, Cambridge, Te Awamutu, Hinuera, and Kinleith.

For the training and validation stages of the Fuzzy ARTMAP network, a specific subset of this data was used covering the period from 8 December 2007 to 7 January 2008, totaling 1488 samples. The hyperparameter optimization for all models was performed on data from 8 December 2007 to 6 January 2008, with the objective of minimizing the error on the final day, 7 January 2008. The forecast period consisted of 48 load values (half-hourly) for a 24-h day (8 January 2008).

Figure 1 highlights various substations, distribution lines, and power stations on the map of the power transmission system in the north of New Zealand's North Island.

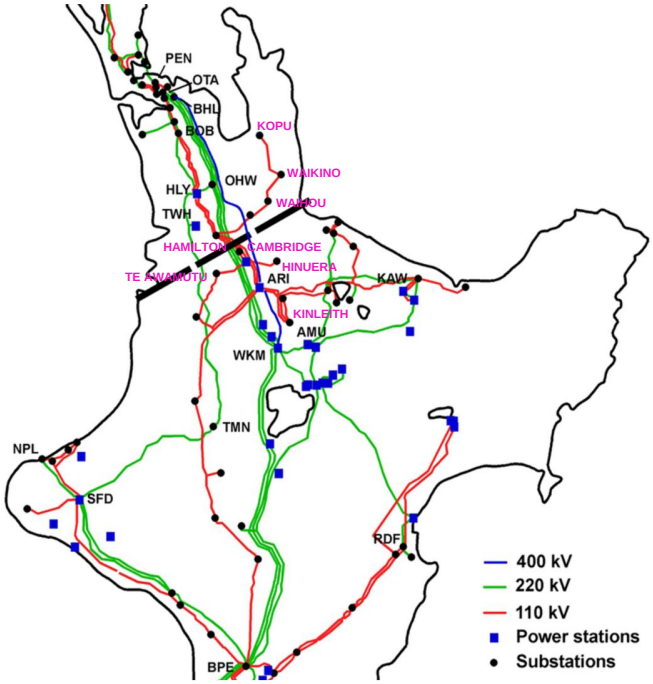


Figure 1. Map of the power transmission network in the north of New Zealand's North Island (adapted from [35]).

3.2. Fuzzy ARTMAP

The Fuzzy ARTMAP supervised learning network [34] consists of two Adaptive Resonance Theory modules, ART_a and ART_b. These modules perform calculations based on fuzzy set theory [36], specifically using the AND ($\wedge$) as defined by (1).

$$(x \wedge y)_i = \min\{x_i, y_i\} \tag{1}$$

Figure 2 illustrates the ART modules of the network. The input data and desired outputs are entered into the ART_a and ART_b modules, respectively; the inter-ART associative memory module is responsible for checking the correspondence between the input patterns and the expected output categories. A critical component of the network is the Match-Tracking used by the inter-ART module, which dynamically adjusts the vigilance parameter to achieve an optimal balance. This minimizes the prediction error while maximizing the model's overall ability to handle a variety of input patterns while at the same time producing accurate results even for unseen data. Prior to training, the weight matrices associated with the ART_a, ART_b, and inter-ART modules are initialized with values of 1, indicating that no categories are active in the network. During the training process, as the network is exposed to the input data, resonance occurs between the input patterns and the predicted output categories, leading to the activation of the corresponding categories in the network. In turn, this triggers the updating of these weight matrices to capture the relationships learned between the input patterns and the output categories. Algorithm 1 shows the pseudocode for the Fuzzy ARTMAP network.

The input data for the global forecasting model is composed of a set of 11 binary bits related to the exogenous data plus three global load values from previous time steps, resulting in a 14-dimensional input space. For each substation, the input of the Fuzzy ARTMAP network (ART_a) is composed of the global load at the current time (h) and the participation factors from three past time steps, resulting in a four-dimensional input space. It should be noted that the raw data are not preprocessed or normalized externally, as the network performs L1 normalization internally.

The Fuzzy ARTMAP network's performance is fundamentally influenced by the vigilance parameters ρ_a and ρ_b and the learning rate β. In this study, we perform an extensive hyperparameter search on these critical parameters, while other parameters such as ρ_{ab}, α, and ϵ are kept fixed. The search ranges were defined as follows: ρ_a from 0.90 to 0.99 (increment of 0.01), ρ_b from 0.980 to 0.999 (increment of 0.001), and β from 0.02 to 1.00 (increment of 0.01).

The vigilance parameter (ρ) acts as a similarity constraint for category formation. A higher ρ value enforces stricter similarity, leading to the creation of more categories (clusters). This is particularly relevant for ρ_b, which clusters the network's output. A large number of clusters results in a narrower range of values per cluster; conversely, a low ρ_b generates fewer clusters, grouping a wider range of output values. This can lead to a "discretization" or "straight-line" effect on the forecast curve, as diverse output values are mapped to a single oversimplified category. The α parameter, or learning rate, is a constant that indicates the extent to which new patterns influence a category's weight update. The β parameter, or training rate, controls the speed of learning, with a value of 1 implying faster training.

The inter-ART vigilance parameter ρ_{ab} acts as a match-tracking mechanism between the ART_a (input) and ART_b (output) modules. Its primary function is to ensure consistency between the input pattern and the expected output category. Because its role is to measure the compatibility between categories rather than to fine-tune the forecast precision itself,

we considered its optimization to be non-critical for this study, and consequently kept its value fixed.

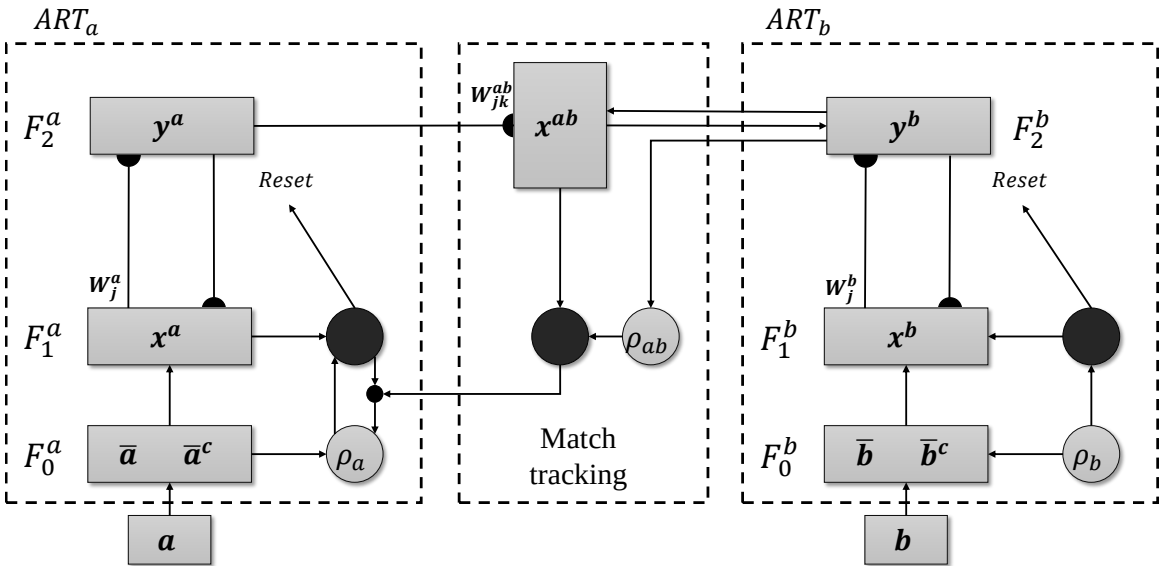


Figure 2. Fuzzy ARTMAP architecture.

Algorithm 1 Pseudocode for training the Fuzzy ARTMAP neural network.

Input: a: Input Data;
b: Desired output;
α: Learning Rate ($\alpha > 0$);
β: Training Parameter ($0 < \beta \leq 1$);
ρ_a: Vigilance Parameter Module A ($0 < \rho_a \leq 1$);
ρ_b: Vigilance Parameter Module B ($0 < \rho_b \leq 1$);
ρ_{ab}: Vigilance Parameter Inter-ART Module ($0 < \rho_{ab} \leq 1$);
ϵ: Increase.
Output: W^{ab}

procedure
  $\boldsymbol{W}^a = 1 \qquad \boldsymbol{W}^b = 1 \qquad \boldsymbol{W}^{ab} = 1$
  $\overline{\boldsymbol{a}} = \dfrac{\boldsymbol{a}}{\left|\sum_i^M a_i\right|} \qquad \overline{\boldsymbol{b}} = \dfrac{\boldsymbol{b}}{\left|\sum_i^M b_i\right|}$
  $\overline{a}_i^c = 1 - \overline{a}_i \qquad \boldsymbol{I}_a = [\overline{\boldsymbol{a}} \quad \overline{\boldsymbol{a}}^c] \qquad \overline{b}_i^c = 1 - \overline{b}_i \qquad \boldsymbol{I}_b = \left[\overline{\boldsymbol{b}} \quad \overline{\boldsymbol{b}}^c\right]$
  while Data for training exists **do**
    for all clusters W_k^b **do**
      $T_k^b = \dfrac{|\boldsymbol{I}_b \wedge \boldsymbol{W}_k^b|}{\alpha + |\boldsymbol{W}_k^b|}$
    $flag = True$
    while flag = True **do**
      $K = \max\{T_k^b : k = 1, ..., N_b\}$
      if $\dfrac{|\boldsymbol{I}_b \wedge \boldsymbol{W}_K^b|}{|\boldsymbol{I}_b|} \geq \rho_b$ **then**
        $\boldsymbol{W}_K^b = \beta(\boldsymbol{I}_b \wedge \boldsymbol{W}_K^b) + (1 - \beta) \cdot \boldsymbol{W}_K^b$
        $y^b = \begin{cases} 0, & k \neq K, \\ 1, & k = K. \end{cases}$
        $flag = False$

Algorithm 1 *Cont.*

else
  $T_J^b = 0$
for all clusters W_k^a **do**
  $T_j^a = \dfrac{|\boldsymbol{I}_a \wedge \boldsymbol{W}_j^a|}{\alpha + |\boldsymbol{W}_j^a|}$
$flag = True$
while flag = True **do**
  $J = \max\{T_j^a : j = 1, ..., N_a\}$
  if $\dfrac{|\boldsymbol{I}_a \wedge \boldsymbol{W}_J^a|}{|\boldsymbol{I}_a|} \geq \rho_a$ **then**
    if $\dfrac{|\boldsymbol{y}^b \wedge \boldsymbol{W}_J^{ab}|}{|\boldsymbol{y}^b|} \geq \rho_{ab}$ **then**
      $\boldsymbol{W}_J^a = \beta(\boldsymbol{I}_a \wedge \boldsymbol{W}_J^a) + (1 - \beta) \cdot \boldsymbol{W}_J^a$
      $W_{JK}^{ab} = \begin{cases} 0, & j \neq J, k \neq K, \\ 1, & j = J, k = K. \end{cases}$
      $flag = False$
    else
      $T_J^a = 0$
      $\rho_a = \dfrac{|\boldsymbol{I}_a \wedge \boldsymbol{W}_J^a|}{|\boldsymbol{I}_a|} + \epsilon$
  $T_J^a = 0$

3.3. Global Load Forecasting

To forecast the multinodal load, it is necessary to obtain the global (forecast) load; in turn, this is used to calculate the participation factor provided by Equation (2), which is fundamental to multinodal forecasting. Global forecasting is carried out by a system called the global load forecasting system, which is based on the Fuzzy ARTMAP neural network [33]. The input data set of the ART$_a$ module in this system contains exogenous data such as time of day, day of the week, month, holidays, daylight savings time, and global load (CG) at three previous time instants ($t-1$, $t-2$, and $t-3$). This representation of the global load includes the "sliding window" technique proposed by [37] and aims to identify the behavior of the load profile. The forecasting result consists of the CG at time t, which is provided by the output of the ART$_b$ module of the Fuzzy ARTMAP network. The exogenous variables are described in detail in Table 1, and are composed of 11 binary bits related to the exogenous data plus the three global load values from previous time steps, resulting in a 14-dimensional input space. Figure 3 details the global load forecasting strategy with the Fuzzy ARTMAP network.

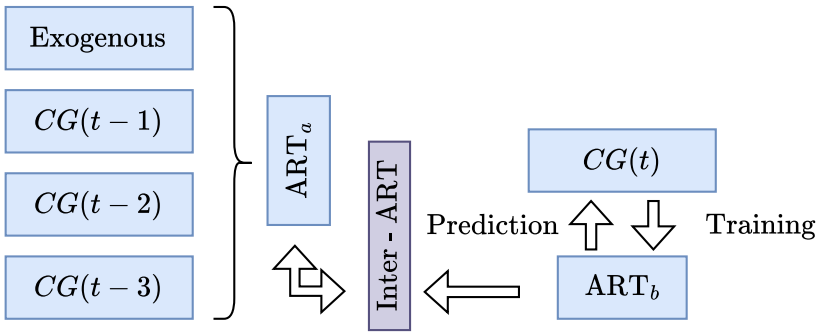


Figure 3. Strategy for global load forecasting with FAM-ANN.

Table 1. Encoding of exogenous variables.

Variable	Bits	Encoding
Day of the Week	3 bits	001—Monday 010—Tuesday 011—Wednesday 100—Thursday 101—Friday 110—Saturday 111—Sunday
Time of Day	6 bits	[000001] to [110000] (representing 48 half-hour values)
Holiday	1 bit	1—holiday 0—not a holiday
Daylight Saving Time	1 bit	1—daylight saving time 0—not in daylight saving time

3.4. Multinodal Load Forecasting

The multinodal load forecasting system (*PCM*) is based on the Fuzzy ARTMAP neural network. Forecasting in this system is carried out in modules, where each module contains a neural network responsible for predicting the load at individual locations such as substations, transformers, and feeders. To perform substation load forecasting, the multinodal load forecasting system uses the global participation factor (*FPG*), defined as the percentage of each substation's load within the total system load, as specified in (2) [37,38]:

$$FPG_j(t) = \frac{CM_j(t)}{CG(t)} \tag{2}$$

in which:

- $CM_j(t)$ is the multinodal load associated with substation j at time t.
- $CG(t)$ is the global load, which corresponds to the addition of the loads of each substation at time t.

For each substation, the load forecast is obtained by the multinodal forecast. The input dataset for the Fuzzy ARTMAP network for each substation j consists of the global load at time t, $CG(t)$, and the participation factors of substation j at times $t-1$, $t-2$, and $t-3$. The network output in the *PCM* system corresponds to the predicted participation factor at time t. The flowchart of the multinodal forecasting system is shown in Figure 4.

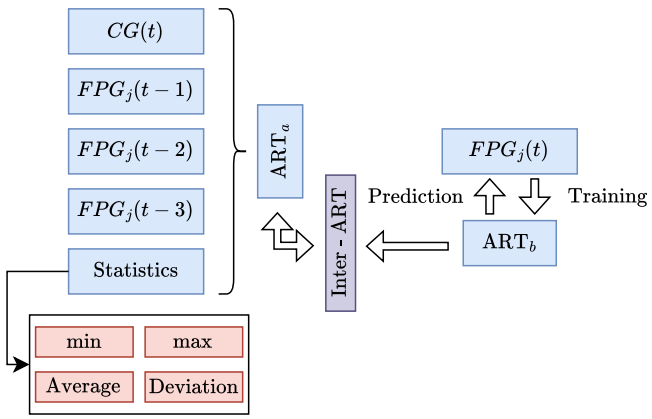


Figure 4. Multinodal load forecasting system.

Within this system, statistical features derived from the global participation factors (FPG_j) at times $t-1$, $t-2$, and $t-3$ were evaluated as potential input extensions to the network. The considered statistics included the mean, standard deviation, maximum, and minimum values, which were assessed individually in order to determine their impact on forecasting performance.

These statistical features are part of a feature engineering strategy. Feature engineering aims to transform raw data into informative input variables that help the model to capture relevant patterns. By deriving statistics from the recent participation factors, each feature provides a distinct perspective on the substation's recent load behavior, such as its central tendency, variability, or extreme values [39–41]. Evaluating these features separately allows the most effective input to be identified for each substation and forecasting horizon.

From a theoretical perspective, each derived feature adds a new dimension to the input space. For example, using only the mean of the past three FPG_j values extends the original three-dimensional lag space by one dimension. This dimensionality increase provides the Fuzzy ARTMAP network with richer information, enhancing its ability to separate patterns corresponding to different load behaviors and improving generalization [42,43].

The mean is obtained as described by (3):

$$\overline{FPG}_j = \frac{1}{N}\sum_{i=1}^{N} FPG_j(t-i) \tag{3}$$

in which:

- $N = 3$ (considering the values at $t-1$, $t-2$, and $t-3$).
- $FPG_j(t-i)$ is the global participation factor of substation j at time $t-i$.

The standard deviation, which measures the variability of the data around the mean, is calculated by (4).

$$s_{FPG_j} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(FPG_j(t-i) - \overline{FPG}_j\right)^2} \tag{4}$$

The maximum and minimum values are determined by Equations (5) and (6).

$$FPG_j^{\max} = \max\left(FPG_j(t-1), FPG_j(t-2), FPG_j(t-3)\right) \tag{5}$$

$$FPG_j^{\min} = \min\left(FPG_j(t-1), FPG_j(t-2), FPG_j(t-3)\right) \tag{6}$$

By evaluating each statistic individually, the most informative feature can be selected for each substation, thereby optimizing the network's predictive performance. Empirical studies indicate that networks using feature-augmented inputs outperform those trained solely on raw lagged values, particularly in capturing trends, variability, and extreme events.

After the participation factor has been predicted at time t, the substation load $PCM_j(t)$ is obtained by multiplying the predicted participation factor $\widehat{FPG}_j(t)$ by the predicted global load $\widehat{CG}(t)$ (7) [33,37]:

$$PCM_j(t) = \widehat{FPG}_j(t) \times \widehat{CG}_j(t). \tag{7}$$

3.5. Benchmark Models and Hyperparameter Optimization

To validate the effectiveness of our proposed Fuzzy ARTMAP methodology, we compared its performance with three widely used machine learning models implemented in the Scikit-learn library in Python [44]: Multi-layer Perceptron (MLP), Random Forest, and Support Vector Regression (SVR). For a fair comparison, each model went through a

hyperparameter optimization process to find its best configuration, using the same training and validation period as Fuzzy ARTMAP. The optimization used a grid search, with MAPE as the main criterion for selecting the best model.

These models were chosen because they represent different approaches to time series forecasting. MLP is a type of artificial neural network that captures nonlinear relationships by learning patterns across multiple layers of neurons, making it a stronger alternative to linear models [45,46]. Random Forest combines many decision trees built from random subsets of the data, which lowers the risk of overfitting and produces more stable predictions [47]. SVR, an extension of Support Vector Machines, models nonlinear relationships through kernels, is less sensitive to outliers, and balances model complexity with generalization by defining a margin of tolerance around the regression function [48].

The hyperparameter search spaces for each model were defined as follows:

- MLP
  - Hidden layer sizes: (5), (10), (5, 5), (10, 5), (5, 10), (20), (20, 15), (20, 30), (100), (200), (100, 50)
  - Maximum number of iterations: 1500
  - Activation function: 'logistic', 'tanh', 'relu'
  - Optimization method (solver):
    - 'LBFGS' (Limited-memory Broyden–Fletcher–Goldfarb–Shanno)
    - 'SGD' (Stochastic Gradient Descent)
    - 'ADAM' (Adaptive Moment Estimation)
- Random Forest
  - Number of trees (estimators): 50, 100, 200
  - Maximum depth: 2, 5, 8, 10, 15, 20, 25, 30, None
  - Minimum samples required to split: 2, 5, 10
  - Splitting criterion: 'squared_error', 'absolute_error', 'friedman_mse', 'poisson'
- SVR
  - Kernel type: 'RBF', 'poly', 'sigmoid'
  - Regularization parameter (C): 1, 10, 100
  - Tolerance margin (epsilon): 0.01, 0.1, 0.5

3.6. Criteria for Comparing Results

The Mean Absolute Percentage Error (MAPE) is a metric that is commonly used to assess the precision of forecasting models. It is calculated by taking the mean of the absolute percentage errors between the predicted values and the actual values. The MAPE formula is provided by (8):

$$\text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{FPG_t - \widehat{FPG_t}}{FPG_t}\right| \times 100 \quad (8)$$

in which:

- FPG_t represents the actual value of the participation factor at time t.
- $\widehat{FPG_t}$ represents the predicted value of the participation factor at time t.
- n is the number of hours, with a reading every half hour.

MAPE is widely used due to its interpretability; it expresses the error as a percentage, which makes it easier to understand the model's performance [49]. Smaller MAPE values indicate more precise forecasts. The use of MAPE is particularly well suited for this study, as it provides a normalized error measure across different substations and load profiles. This allows for a direct comparison of forecasting quality between substations with varying

load levels, since the metric is not biased by the scale of the load. Lower MAPE values mean better forecasts.

4. Load Forecast Result

This section presents and discusses the results obtained from the multinodal electricity load forecasting. It begins with a detailed analysis of the MAPE values for the nine substations, comparing the performance of different windowing techniques, followed by an in-depth look at the behavior of the Fuzzy ARTMAP parameters and their impact on forecasting precision.

4.1. Results

The results presented in Table 2 provide a detailed analysis of the MAPE values for nine substations, using windowing methods that consider the maximum, minimum, mean, and standard deviation values of the samples calculated considering the participation factor at times $t-1$, $t-2$, and $t-3$.

Table 2. Results by substation.

Substation	MAPE (%)			
	Maximum	**Minimum**	**Mean**	**Standard Deviation**
Kopu	10.35	9.68	9.64	4.79
Waikino	6.93	5.24	7.11	5.00
Waihou	3.96	3.99	3.68	3.72
Hamilton 11	4.05	3.79	3.68	4.25
Hamilton 33	3.57	3.35	3.21	3.46
Cambridge	3.16	2.72	3.08	3.07
Te Awamutu	4.34	3.07	3.13	3.5
Hinuera	3.39	3.44	3.47	3.60
Kinleith	5.40	5.40	5.40	5.15

Analysis of the results reveals different behavior between the substations. The Kopu substation showed relatively stable performance in terms of MAPE, with values of 10.35% (maximum), 9.68% (minimum), and 9.64% (mean). On the other hand, the standard deviation introduced a significant improvement, reducing the MAPE to 4.79%. This indicates a considerable improvement in the forecast. Figure 5 shows the load curve, where it can be seen that the predicted values follow the actual load well throughout the day. The range around the forecast curve is relatively narrow, indicating that the variability within the cluster of the forecast values is low.

Waikino, on the other hand, revealed a trend of continuous improvement in forecasts when using different windowing techniques. The MAPE showed a significant reduction when using the minimum value (5.24%) compared to the maximum value (6.93%) and the mean (7.11%). In addition, using the standard deviation resulted in an MAPE of 5.00%, the lowest value observed for Waikino. Figure 6 shows the forecast and actual load curves as well as the range obtained by the standard deviation within the cluster of forecasts. This range is slightly wider than that observed in Kopu, suggesting greater variability in the predicted values. However, the predictions still align well with the actual load, demonstrating the model's effectiveness even in the face of this variability.

For Waihou, the MAPE results varied from 3.99% (minimum) to 3.68% (mean). This variation suggests that the different windowing techniques do not have a significant impact on the precision of the forecasts for this substation. The mean showed the best result, with an MAPE of 3.68%. The forecast curve in the Figure 7 closely follows the actual load

and the range around the forecast is relatively narrow, indicating that the model captures load fluctuations well and has low variability in the forecast clusters.

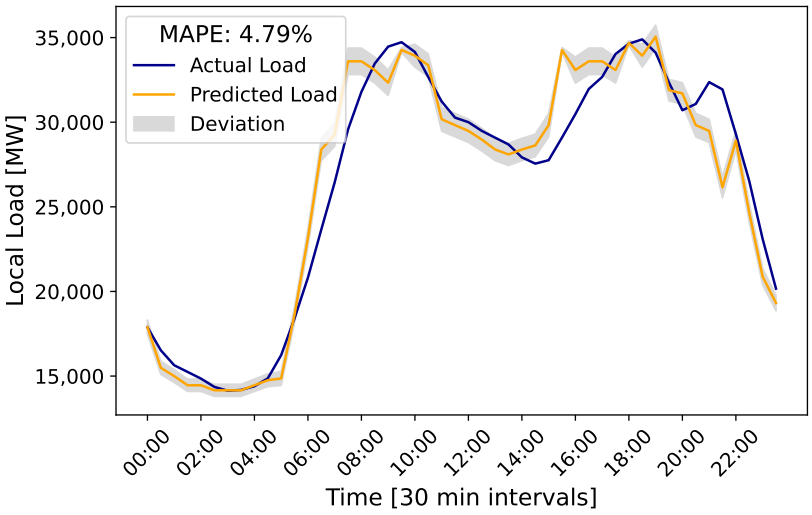


Figure 5. Actual and forecast values for Kopu substation.

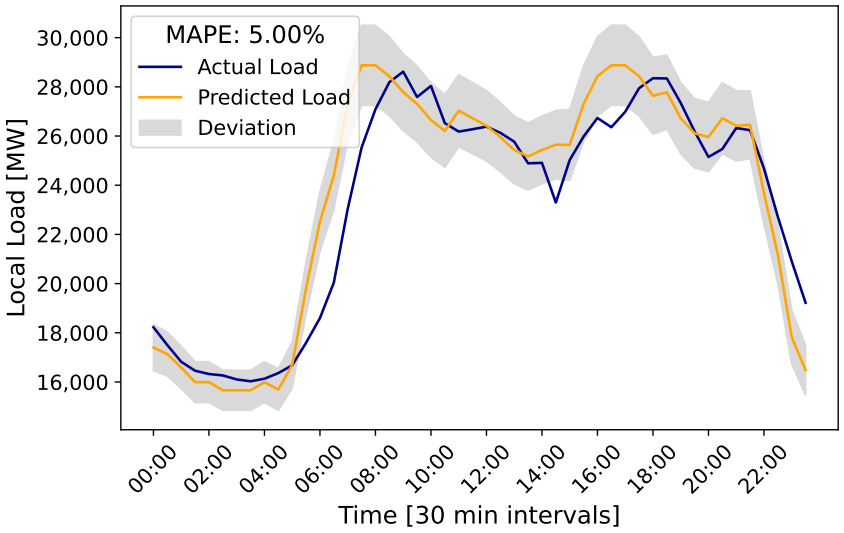


Figure 6. Actual and forecast values for Waikino substation.

Hamilton 11 showed a moderate variation in MAPE values, with a maximum difference of 0.57% between the different methods. The best result was obtained using the mean (3.68%), followed by the minimum (3.79%). The moderate variation suggests that all the windowing techniques offer relatively accurate prediction, although the mean was able to better capture the trend of the data in this substation. As shown in Figure 8, the confidence interval around the forecasts is narrow, indicating that the forecast is quite accurate and that the variability in the forecasts is low.

For Hamilton 33, all MAPE values were lower than 4.00%, with the best result obtained by the mean (3.21%). The corresponding prediction curve can be seen in Figure 9. The consistency of the results suggests that windowing techniques are highly effective for this substation. In particular, inclusion of the mean seems to capture important characteristics that contribute to more precise predictions.

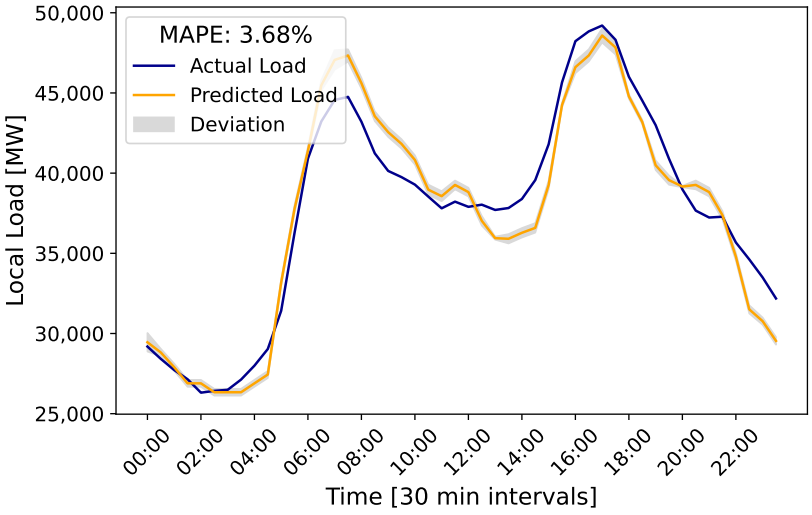


Figure 7. Actual and forecast values for Waihou substation.

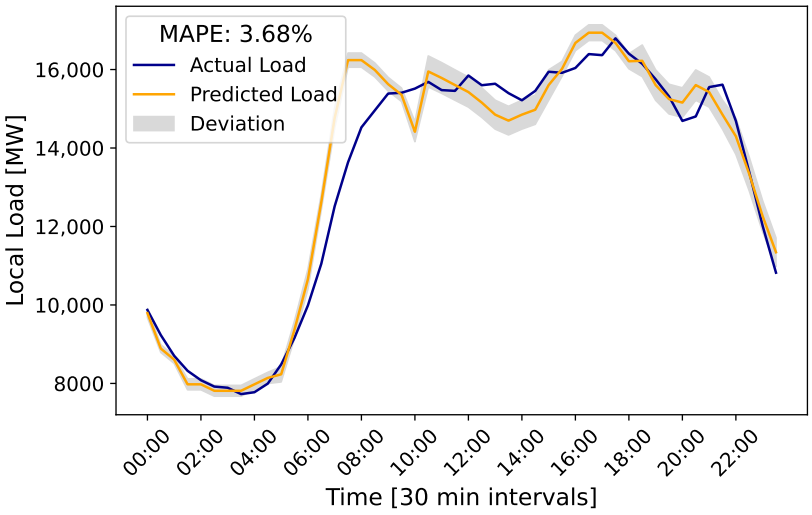


Figure 8. Actual and forecast values for Hamilton 11 substation.

Cambridge showed a smooth and constant variation in MAPE values, from 3.16% (maximum) to 2.72% (minimum), with the minimum providing the best result, as shown in Figure 10. It is worth noting that the minimum in Cambridge was not only the best result among all the substations but also had the lowest MAPE among the four techniques used, showing uniform and accurate performance. The forecast curve followed the actual curve well, with a slight divergence observed between 5:30 and 8:30. The interval around the forecast is quite narrow as a result of the low variability of the clusters.

Te Awamutu showed MAPE values of less than 4.5%, with the best result obtained with the minimum value (3.07%) and the highest with the maximum (4.34%). The forecast curve shown in Figure 11 shows predicted behavior very close to the real values. In addition, the forecast showed no variability within the clusters, suggesting that for this substation there were no significant differences between the load values within the clusters.

Hinuera maintained a similar variation in MAPE values, between 3.39% and 3.60%, showing consistent performance across the different windowing techniques. This performance reveals that all the techniques offer relatively accurate predictions, with the standard deviation being slightly more effective in this substation. The range around the prediction is slightly wider, as shown in Figure 12, indicating more variability in the predicted values; however, the predictions still follow the actual load well.

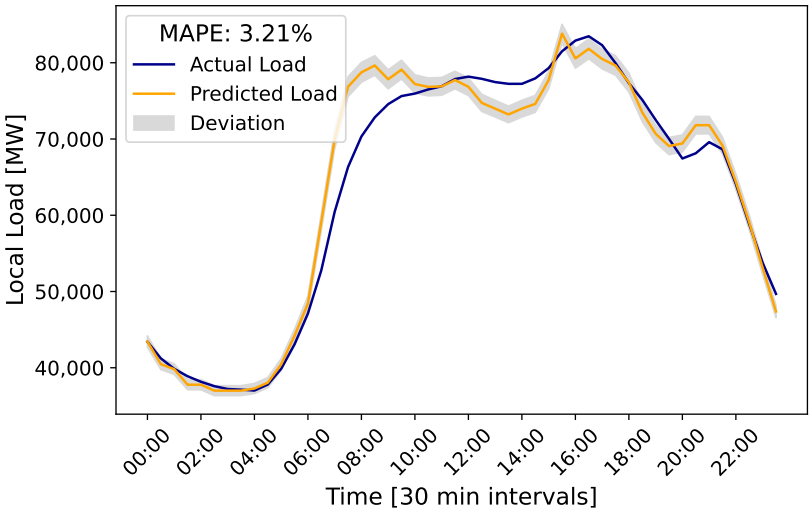


Figure 9. Actual and forecast values for Hamilton 33 substation.

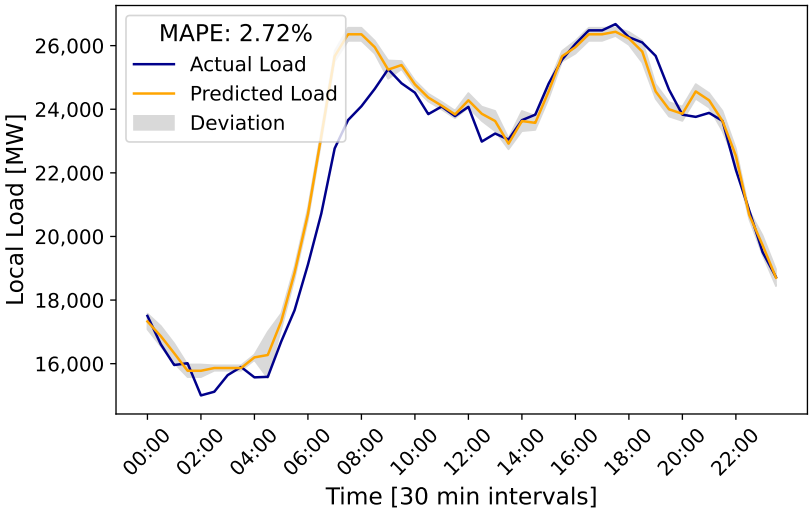


Figure 10. Actual and forecast values for the Cambridge substation.

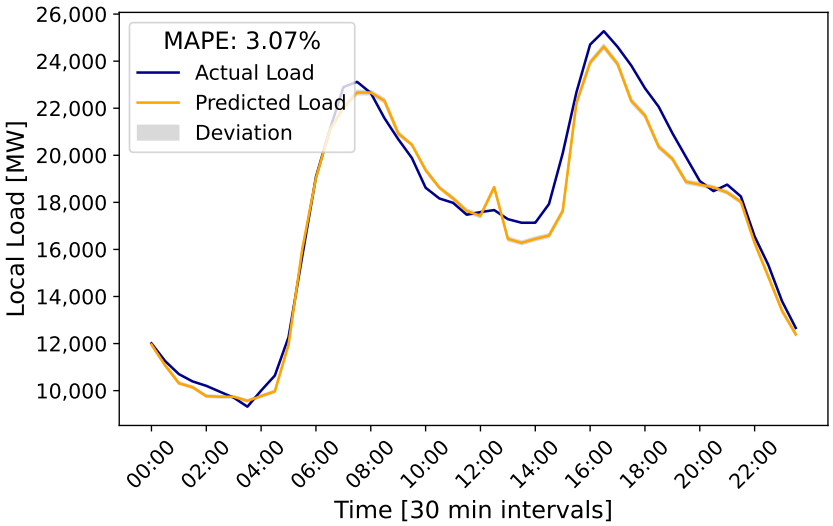


Figure 11. Actual and forecast figures for the Te Awamutu substation.

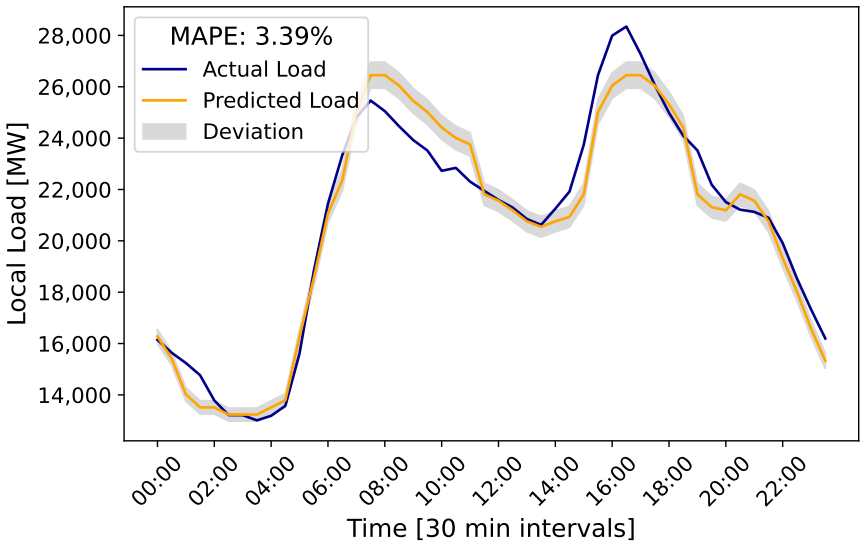


Figure 12. Actual and forecast figures for the Hinuera substation.

Finally, Kinleith showed an improvement in MAPE with the standard deviation (5.15%), standing out from the other techniques that kept MAPE constant at 5.40%. The prediction curve can be seen in Figure 13. It shows greater variability in the interval around the prediction curve, indicating that there was a greater variety of values within the clusters.

In general, substations such as Kopu, Waikino and Kinleith stand out in the forecasts when using the standard deviation. On the other hand, Waihou, Hamilton 11, and Hamilton 33 showed better results when using the minimum value. Other substations such as Cambridge, Te Awamutu, and Hinuera also showed improvements with different strategies. The standard deviation and minimum value proved to be effective techniques in most cases, with each offering its own advantages for predicting electrical load. The combination of parameters that produced the best results can be analyzed in Table 3.

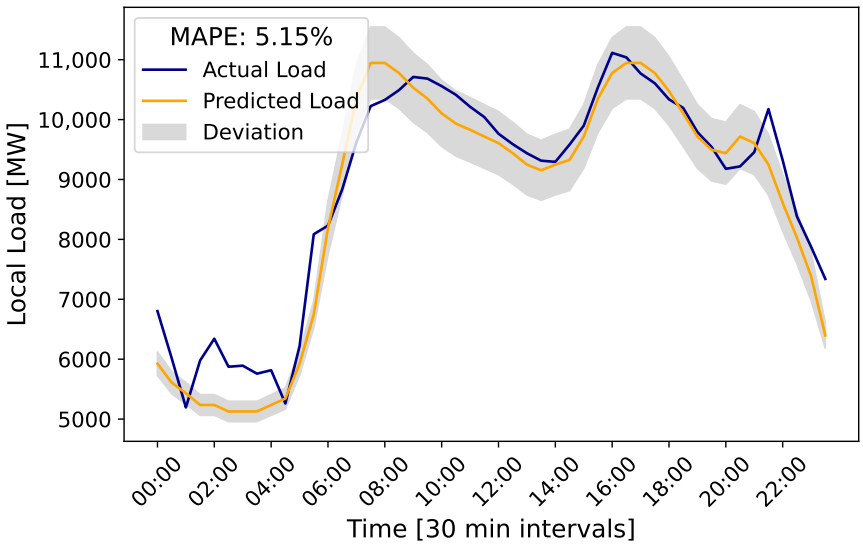


Figure 13. Actual and forecast values for the Kinleith substation.

When analyzing the table with the configuration of parameters that obtained the best results for each substation, we observed a relationship between these parameters and those substations that showed the greatest variability in the intervals around the prediction curve. The Kopu, Waikino, Hamilton 11, and Kinleith substations showed the greatest variations.

This is directly related to the ρ_b parameter, which controls the required similarity for a given piece of data to belong to the cluster. Lower values of ρ_b relax this restriction, allowing slightly different values to be included. On the other hand, values closer to 1 restrict similarity so that relatively close data does not fit into the cluster, which can result in the creation of a new cluster to house it. The Kinleith substation, although it had a wider range with ρ_b close to 1, had high variability due to the β parameter responsible for the learning rate being close to zero. This reduced the creation of new categories, increasing the variability within the existing clusters.

Table 3. Setting parameters for the best results.

Substation	Statistics	Parameters
Kopu	Standard Deviation	$\rho_a = 0.91$, $\beta = 0.96$, $\rho_b = 0.991$
Waikino	Standard Deviation	$\rho_a = 0.95$, $\beta = 0.40$, $\rho_b = 0.996$
Waihou	Mean	$\rho_a = 0.99$, $\beta = 0.92$, $\rho_b = 0.996$
Hamilton 11	Mean	$\rho_a = 0.91$, $\beta = 0.40$, $\rho_b = 0.998$
Hamilton 33	Mean	$\rho_a = 0.91$, $\beta = 0.98$, $\rho_b = 0.997$
Cambridge	Minimum	$\rho_a = 0.95$, $\beta = 0.78$, $\rho_b = 0.999$
Te Awamutu	Minimum	$\rho_a = 0.99$, $\beta = 1.00$, $\rho_b = 0.999$
Hinuera	Maximum	$\rho_a = 0.91$, $\beta = 0.44$, $\rho_b = 0.994$
Kinleith	Standard Deviation	$\rho_a = 0.97$, $\beta = 0.02$, $\rho_b = 0.999$
Fixed parameters: $\rho_{ab} = 1.00$, $\alpha = 0.001$, $\epsilon = 0.0005$		

It can be seen that the Waihou, Hamilton 33, Cambridge, and Te Awamutu substations have high values for ρ_a, ρ_b, and β, which is directly related to greater variability and standard deviation within the clusters formed by the prediction labels that represent the actual load. These parameters indicate greater rigidity in the formation of these clusters, ensuring that only values with high similarity are grouped together. On the other hand, substations such as Waikino, Hamilton 11, and Kinleith have lower ρ_a and β values and show greater variability, as reflected in larger shaded areas around the prediction curves.

The results show notable differences in forecasting performance across the nine substations, reflecting the unique characteristics of each substation's load profile. Substations such as Kopu and Kinleith benefited most from including the standard deviation as an input, suggesting more volatile or irregular load patterns; in contrast, Waihou and Hamilton 33 performed best when the mean was used, indicating more stable and predictable consumption. This demonstrates that the Fuzzy ARTMAP model adapts to the most informative statistical feature for each case, thereby maximizing prediction accuracy. A single forecasting strategy does not fit all substations in a multinodal system; tailoring the approach to each substation's characteristics is necessary to achieve the best results.

4.2. Computational Performance and Method Practicality

The computational performance of the Fuzzy ARTMAP method was evaluated during the forecasting phase to assess its practical applicability. Figure 14 shows the prediction time per sample (measured in microseconds, µs) for each substation and each statistical windowing technique.

The results indicate that the proposed method is highly efficient. For most substations and windowing approaches, the prediction time per sample remains below 1 µs. The mean, maximum, and minimum techniques consistently show the lowest computational cost across all substations.

The standard deviation approach, which often yields the most accurate forecasts, requires slightly more computation. Notable increases in prediction time occur at the Waihou and Te Awamutu substations, reaching approximately 4 µs and 2 µs per sample,

respectively. Despite these peaks, the times remain in the microsecond range, supporting real-time application.

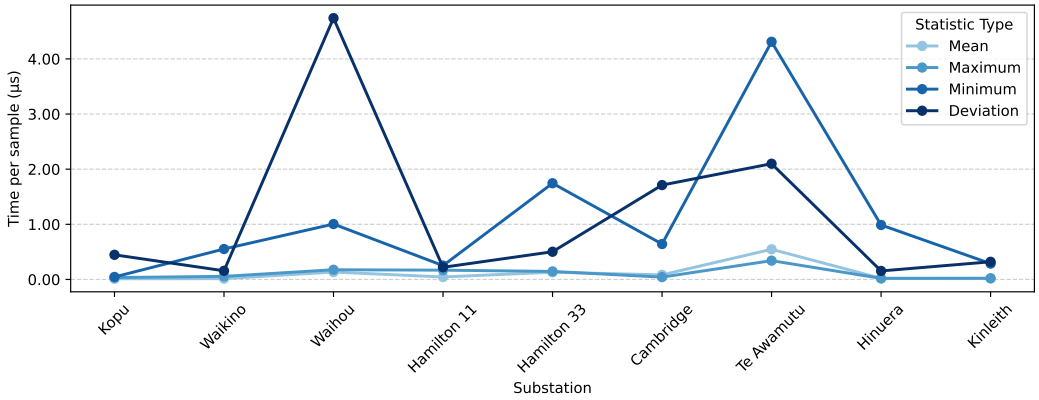


Figure 14. Prediction time per sample for all substations and windowing techniques.

This efficiency highlights an advantage of the Fuzzy ARTMAP architecture. Unlike complex deep learning models, which demand extended training and specialized hardware, this method provides fast predictions with minimal computational resources. The combination of rapid training and prediction times makes the Fuzzy ARTMAP model suitable for scalable real-time load forecasting in dynamic smart grid environments.

4.3. Analysis of Parameters

Analysis of the parameters of the Fuzzy ARTMAP network applied to predicting electrical loads in substations is crucial to optimizing its performance, since it is an architecture that is sensitive to changes in parameters. To adjust the parameters, β was varied from 0.02 to 1.00 with an increment of 0.02, ρ_a of the ART$_a$ module from 0.91 to 0.99 with an increment of 0.02, and ρ_b of the ART$_b$ module from 0.98 to 0.999 with an increment of 0.001. This analysis was applied to all substations, taking into account the expansion of the input data dimension with the maximum, minimum, mean, and standard deviation values of the load participation factor at times $t-1$, $t-2$, and $t-3$. The discussion on the behavior of the parameters is reported for the Kopu substation; similar behavior was observed for the other substations, especially in relation to the parameters ρ_a, ρ_b, β, and the number of categories in the ART$_a$ and ART$_b$ modules of Fuzzy ARTMAP.

Figure 15 illustrates the three-dimensional relationship between the parameters ρ_a, β, and the number of categories (neurons) created in module A of the Fuzzy ARTMAP network. In Figure 15a, it can be observed that high values of ρ_a combined with high values of β result in a greater number of categories. This combination of parameters makes the network more prone to creating multiple categories due to the greater restriction imposed by ρ_a, which prevents new data from being allocated to existing clusters, necessitating the creation of new categories to accommodate them whenever the similarity does not reach the stipulated threshold (ρ). In contrast, although the β parameter allows for fast learning without many training cycles when $\beta = 1$, it prevents the old information contained in the clusters from remaining. Instead, this information is replaced by the current information, leading to the creation of more categories. Similar behavior is observed for Figure 15b–d, which illustrates the same relationship with the insertion of the minimum, mean, and standard deviation of the three previous loads.

Figure 16 follows a similar structure to Figure 15 but focuses on module B of the Fuzzy ARTMAP network. In Figure 16a, it can be observed that high values of ρ_b combined with high values of β favor the creation of a greater number of categories, similar to what was

observed in Figure 16a, but generating a smaller number of clusters because these clusters are responsible for housing the network's output information.

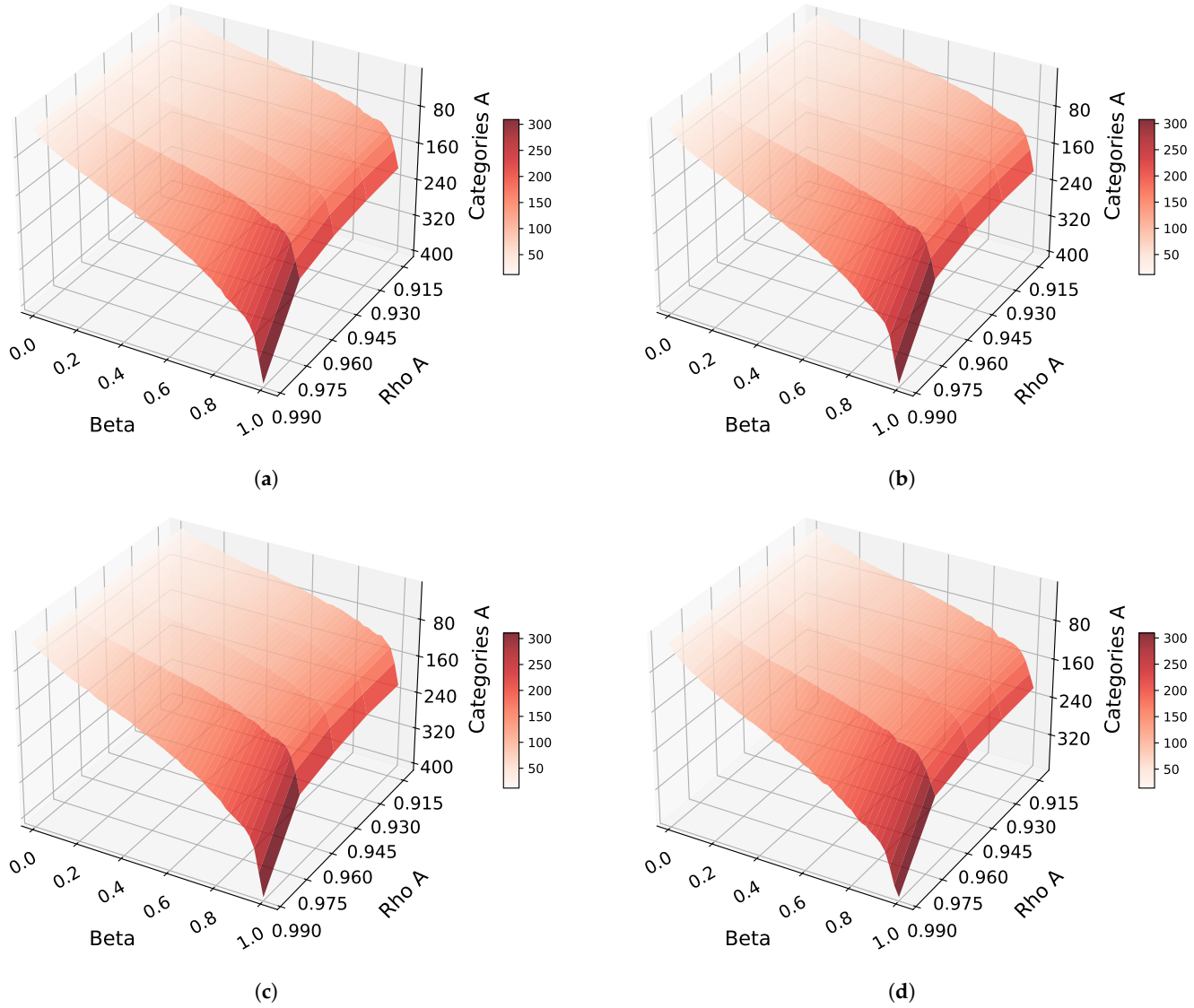


Figure 15. Three-dimensional visualization of the relation between the parameters ρ_a, β, and number of categories in module A: (**a**) maximum, (**b**) minimum, (**c**) mean, and (**d**) standard deviation.

The interaction between the parameters ρ_b, β, and the Mean Absolute Percentage Error (MAPE) can be seen in Figure 17 when inserting the maximum, minimum, mean, and standard deviation of the three previous loads into the input data. This makes it easier to understand which values of β and ρ_b produced the best results.

In Figure 17a, the lowest MAPE values are observed for combinations of high ρ_b values, while β shows more flexibility, oscillating within the $[0, 1]$ range. This relationship between low MAPE values and high ρ_b values indicates that the network needs to create a considerable number of categories in module B during training in order to obtain good results. This allows the network to predict results more accurately when it comes to forecasting. However, the value of β does not seem to have a significant impact on the result, unlike its influence on the creation of clusters reported in Figure 16.

Similar behavior is observed when the other statistics are used to expand the input dimension of the data, as shown in Figure 17b–d. In addition, it can be seen that for Figure 17d there is a greater fluctuation of the MAPE for the walls, causing more irregularities on the surface compared to the others. This unevenness indicates that the model's

response is more unstable when the standard deviation is used as one of the inputs, reflecting greater sensitivity to variations in the data. The more uneven surface also reveals that small adjustments to the ρ_b and β parameters can lead to significant changes in MAPE.

Despite this irregularity, the standard deviation produced the best results for most substations, as shown in Table 2. This is because the standard deviation captures the variability of the data more effectively, allowing the model to better adjust to fluctuations and make more precise predictions. This highlights the importance of selecting the appropriate statistical feature for each unique dataset, which is a key contribution of this work.

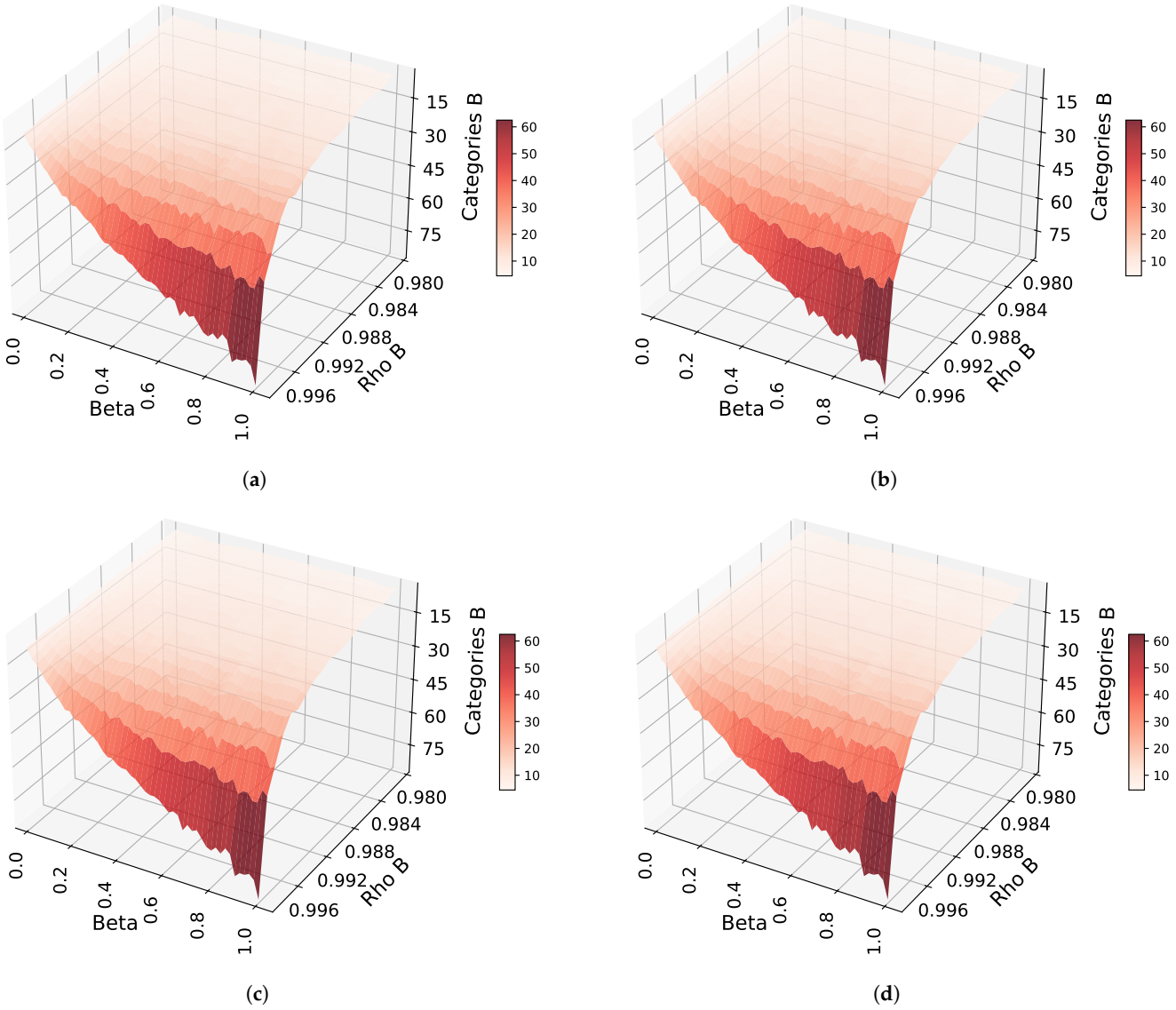


Figure 16. Three-dimensional visualization of the relation between the parameters ρ_b, β, and number of categories in module B: (**a**) maximum, (**b**) minimum, (**c**) mean, and (**d**) standard deviation.

Figure 18 explores the relationship between ρ_a, β, and MAPE. Analyzing Figure 18a,b,d, it can be seen that the highest error rates occur in regions where ρ_a and β are high; in other words, a low constraint in module A (low ρ_a) combined with a high learning rate (high β) can lead to greater imprecision in predictions.

However, a different behavior is observed when using the standard deviation to scale the input data, as shown in Figure 18d, where it can be seen that the lowest prediction errors are achieved when ρ_a is medium or low and β oscillates throughout the range, with values close to 0.6 obtaining the lowest MAPE.

Finally, Figure 19 shows the relationship between the parameters ρ_a, ρ_b, and MAPE. Figure 19a–c illustrates that the highest error rates occur when both ρ_a and ρ_b are low. In other words, relaxing the similarity of modules A and B can lead to greater inaccuracy in predictions. On the other hand, Figure 19d reveals that by including the standard deviation, the behavior of ρ_b is modified. Higher values of ρ_b tend to reduce the model's performance, which means that a greater restriction on ρ_b (resulting in a greater number of categories) compromises the precision of the forecasts when the standard deviation is taken into account.

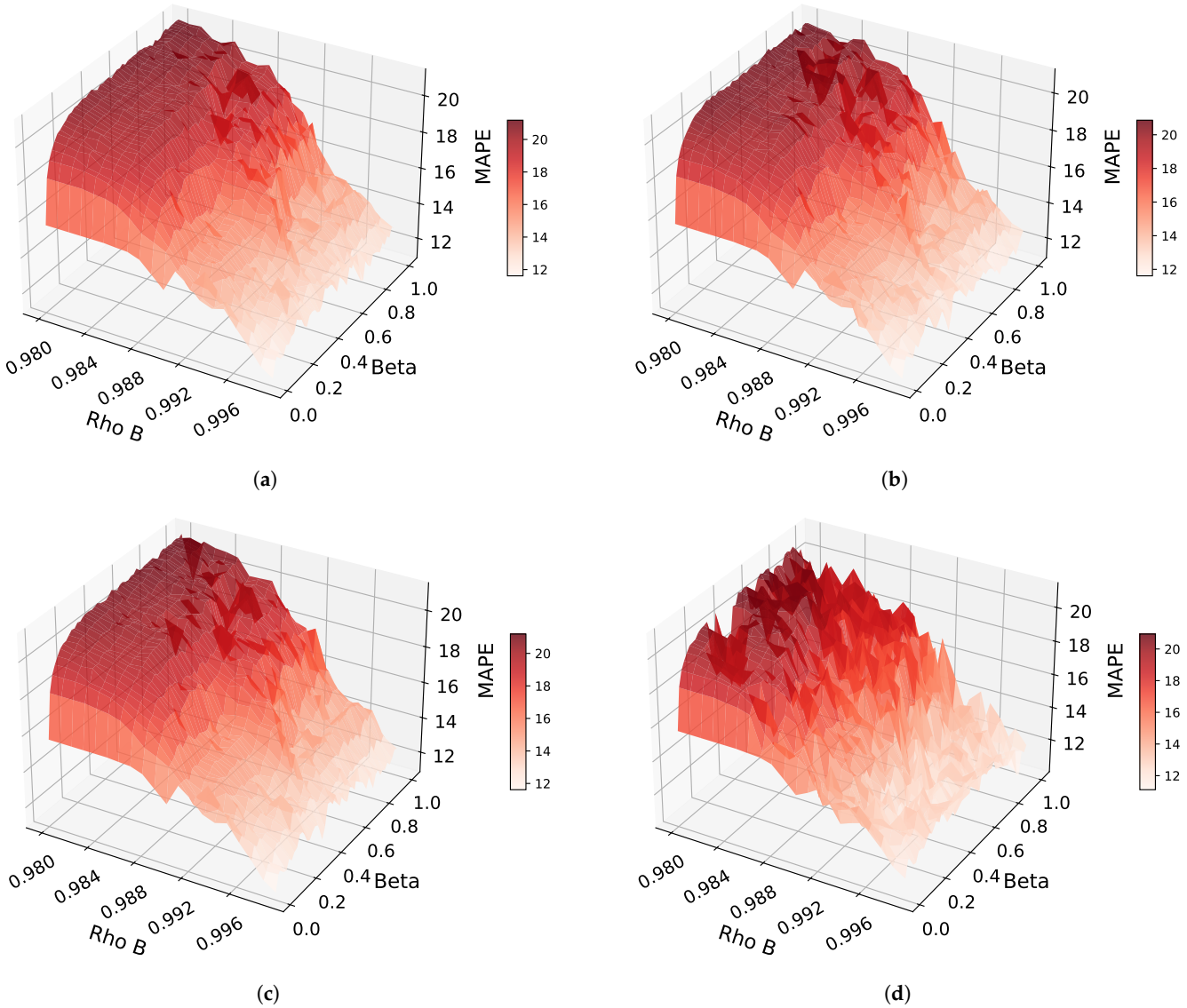


Figure 17. Three-dimensional visualization of the relation between parameters β, ρ_b, and MAPE: (**a**) maximum, (**b**) minimum, (**c**) mean, and (**d**) standard deviation.

4.4. Global Parameter Optimization Analysis

A global optimization was performed to evaluate whether a single set of parameters could provide satisfactory performance across all nine substations. The results are shown in Table 4. Using the standard deviation as the windowing feature yielded the lowest global MAPE of 9.98%.

The single best global parameter set optimized across all substations resulted in an MAPE of 14.85%. This is considerably higher than the substation-specific results, where

some values reached 2.72% (Table 2). This outcome confirms that the load characteristics varied significantly between substations.

Table 4. Global parameters for all substations.

Windowing Type	ρ_a	β	ρ_a	MAPE
Mean	0.99	0.94	0.999	13.44%
Maximum	0.95	0.54	0.998	13.43%
Minimum	0.99	0.84	0.997	12.41%
Standard Deviation	0.91	0.42	0.996	9.98%
Best Overall Global Set	0.91	0.02	0.99	14.85%

Although the global approach is less computationally demanding and may be suitable for large-scale grids, local optimization for each substation provides higher forecasting accuracy.

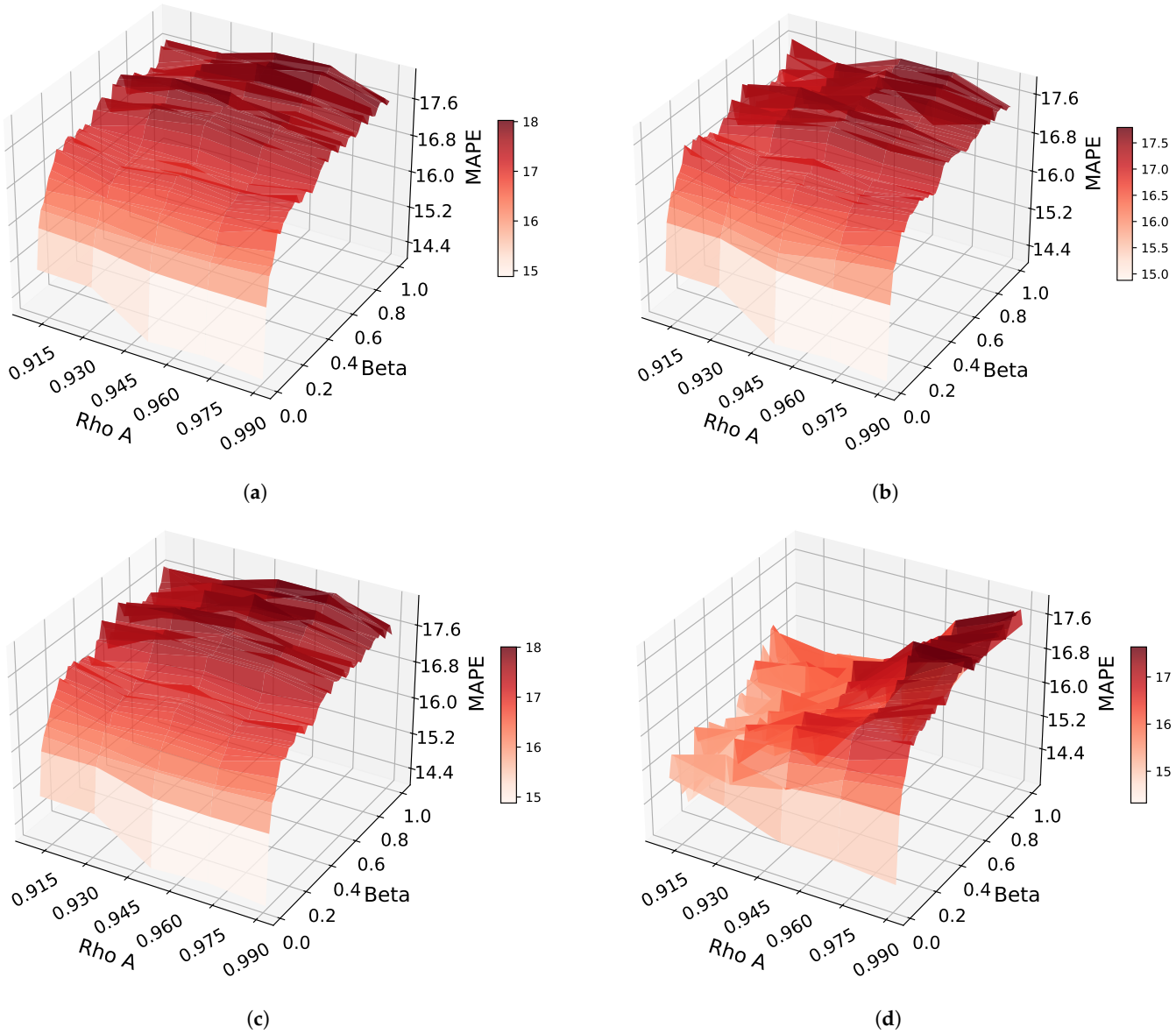
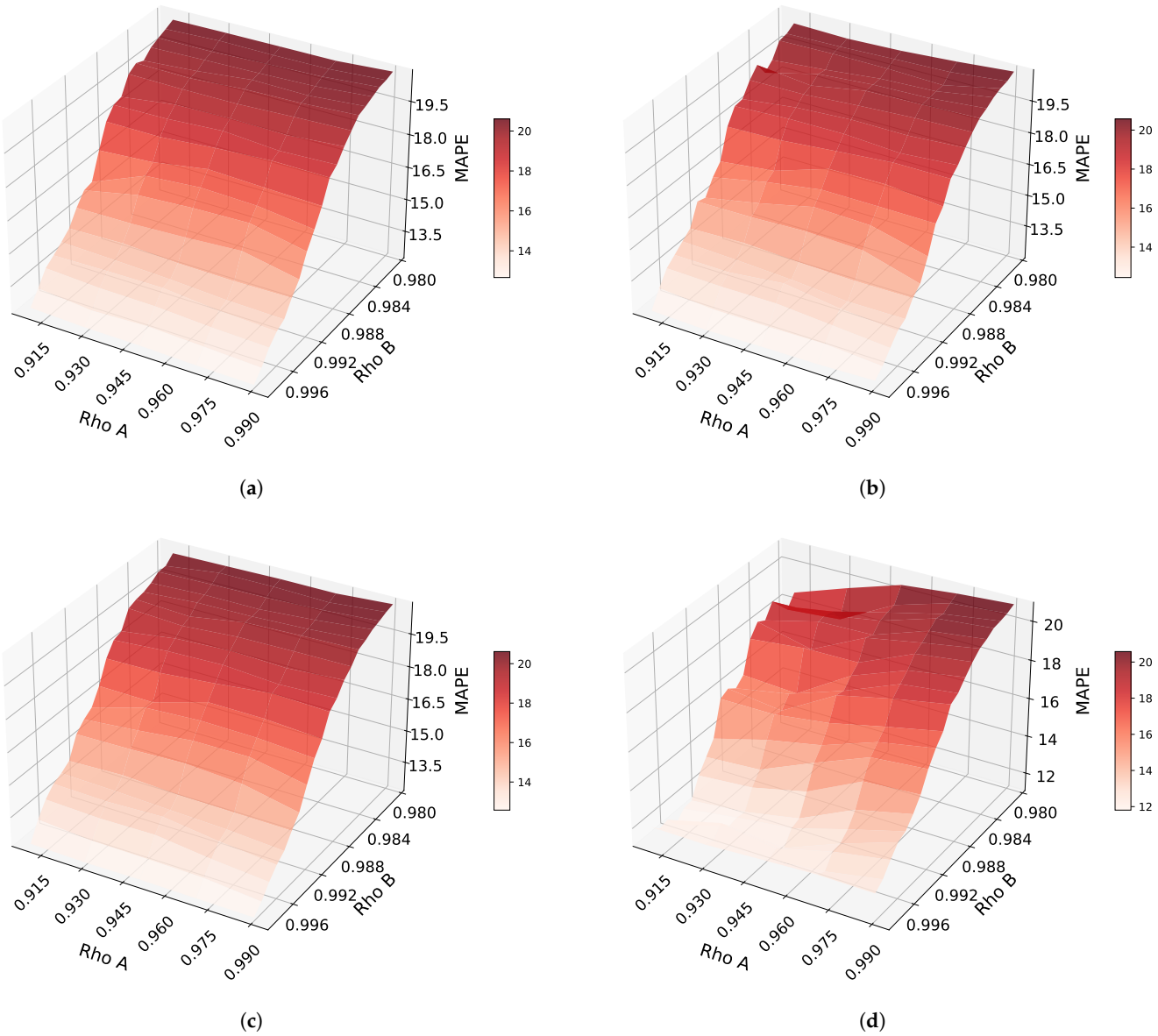


Figure 18. Three-dimensional visualization of the relation between the parameters β, ρ_a, and MAPE: (**a**) maximum, (**b**) minimum, (**c**) mean, and (**d**) standard deviation.



(**a**) (**b**)

(**c**) (**d**)

Figure 19. Three-dimensional visualization of the relation between the parameters ρ_a, ρ_b, and MAPE: (**a**) maximum, (**b**) minimum, (**c**) mean, and (**d**) standard deviation.

4.5. Comparison with Other Models

The Fuzzy ARTMAP model was evaluated against three established machine learning techniques: Multi-Layer Perceptron (MLP), Support Vector Regression (SVR), and Random Forest (RF). All models were applied to the same dataset using identical statistical windowing techniques, with performance assessed by MAPE as reported in Table 5.

Fuzzy ARTMAP achieved the lowest error in several substations. For example, at Waihou and Hamilton 11, the mean windowing method yielded MAPEs of 3.68%, outperforming all other models. Cambridge and Te Awamutu achieved the best results with the minimum windowing method, reaching 2.72% and 3.07%, respectively. For Kopu and Waikino, the standard deviation windowing method provided the lowest errors.

These characteristics make Fuzzy ARTMAP a reliable alternative for multinodal electrical load forecasting, consistently providing accurate predictions across substations with diverse load profiles. Its ability to adjust to different statistical features allows it to maintain high precision while remaining computationally efficient, confirming its suitability for real-time forecasting in multinodal power systems.

Table 5. MAPE (%) statistics by substation for MLP, SVR, and RF.

Substation	MLP				SVR				Random Forest			
	Mean	Max	Min	Std	Mean	Max	Min	Std	Mean	Max	Min	Std
Kopu	11.13	9.72	14.75	9.05	9.04	9.01	12.02	6.71	9.54	11.28	9.85	6.69
Waikino	9.04	9.04	9.04	9.04	8.70	9.09	9.17	9.01	7.09	7.03	6.19	6.69
Waihou	5.43	7.74	7.89	6.78	4.60	4.70	4.51	4.44	4.66	4.65	4.76	4.59
Hamilton 11	7.49	7.49	7.49	7.49	6.78	6.70	6.71	6.76	7.10	7.07	7.11	7.15
Hamilton 33	5.76	5.79	5.86	7.91	5.08	7.08	7.08	7.09	5.35	5.12	5.78	5.35
Cambridge	5.63	5.63	5.63	5.33	3.10	3.39	3.39	3.10	8.48	8.48	8.48	6.99
Te Awamutu	7.24	7.24	7.23	7.24	7.24	6.69	7.24	6.73	6.79	6.79	6.79	6.79
Hinuera	15.57	16.84	15.45	15.79	16.41	16.22	16.54	13.04	4.99	4.99	4.99	4.99
Kinleith	6.30	6.30	6.30	6.30	6.12	6.11	6.12	3.99	6.44	6.44	6.42	6.43

4.6. Advantages and Limitations of the Study

The methodology presented in this paper combines data enrichment with the Fuzzy ARTMAP network. Using statistical features such as the mean, standard deviation, maximum, and minimum to expand the input data allows the model to adjust to different load profiles, improving the precision of predictions. This feature enables more stable loads to benefit from the mean, while more volatile loads benefit from the standard deviation. These results demonstrate the method's adaptability to various conditions. Comparisons with other techniques such as MLP, SVR, and Random Forest show that Fuzzy ARTMAP achieves smaller errors in several substations.

Another clear advantage is its computational efficiency. The model makes predictions in microseconds without requiring specialized hardware or long training periods, which ensures its viability in real-time applications and scenarios with limited computational resources.

The flexibility gained from using statistics to enrich the data suggests that the proposed approach can be applied to other types of time series beyond electrical load data. There is potential for use in predicting vibrations in mechanical systems, as part of an equipment's useful life analysis, or in other data series such as network traffic or financial indicators. The ability to recognize patterns and group information in an adaptive way allows the proposed method to be explored in different domains with internal variability and dynamic behavior.

Despite its advantages, this study has limitations that open the door for new work. The dataset we used is from 2010, a period before the widespread adoption of renewable energy and smart meters, which have altered consumption patterns. This means that application of the proposed method in modern grids still needs validation. In addition, the work was performed on a limited number of substations, which raises the challenge of scalability. For larger networks, optimization by exhaustive search becomes unfeasible. An exploration of distributed computing or other more efficient adjustment methods to process data in parallel could be a solution.

Another point is that the model's performance on abnormal days such as holidays or under extreme weather conditions was not evaluated. These events generate load patterns that deviate from regular behavior. An analysis in these scenarios would be important to verifying the model's reliability. It is also relevant to note that while using multiple statistics enriches the data, it also increases the complexity of the input set. Future work could investigate which combinations of statistics are the most relevant for different load types, thereby reducing redundancies and maintaining efficiency.

Finally, integration with probabilistic methods which can provide confidence intervals for predictions would be a valuable enhancement. Expanding the study to larger networks or other time series domains would help to evaluate the generality of the proposed approach as well as the adjustments needed for different contexts.

5. Conclusions

Precise forecasting of electricity demand is an essential tool for the efficient and sustainable management of today's energy systems. This study applied the Fuzzy ARTMAP network to multinodal short-term electricity load forecasting at nine substations in New Zealand, presenting an approach that includes exhaustive parameter analysis and search along with the incorporation of statistics such as maximum, minimum, mean, and standard deviation.

The results showed that standard deviation was the most effective among the windowing techniques we explored in improving the precision of the forecasts, resulting in a reduction in Mean Absolute Percentage Error (MAPE) for most of the substations. In addition, our analysis of the Fuzzy ARTMAP parameters highlighted that the combination of the ρ_a and β parameters has a significant impact on the creation of categories, and consequently on the precision of forecasts.

The shaded area around the load curve, which represents the variability of the data and the uncertainty of the forecasts, was particularly impacted by the configuration of the parameters. Substations with higher values of ρ_a, β, and ρ_b, such as Waihou, Hamilton 33, Cambridge, and Te Awamutu, showed a significant reduction in the shaded area of the load curve, indicating greater stability in the demand forecast. On the other hand, stations with lower parameters, such as Waikino, Hamilton 11, and Kinleith, showed greater dispersion, reflecting a larger shaded area and less accurate forecasting.

This information indicates that fine adjustments to parameter settings, especially those such as ρ_b that are responsible for checking the similarity of data within clusters, have a direct effect on the model's ability. When ρ_a and ρ_b are lower, this allows for greater variation in the data, resulting in greater imprecision and a larger shaded area.

To assess its scalability and robustness of this methodology in different contexts, future work could explore its application to other data types, such as demand from industrial consumers or individual households. Furthermore, a comparative analysis with other advanced machine learning models such as transformers or hybrid models could be conducted to more comprehensively evaluate the performance of Fuzzy ARTMAP.

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Conflicts of Interest: The authors declare no conflicts of interest.

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Integrating autoencoders to improve fault classification with PV system insertion

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ABSTRACT

The extensive integration of distributed generation (DG) units leads to significant changes in the operation of power distribution systems (PDS). Integrating DG units contributes to meeting the growing energy demand, diversifying the energy matrix, and reducing power grid losses. In contrast, they can affect conventional protection systems in power grids by altering current flow, which affects the characteristics, direction, and amplitude of short-circuit currents. Consequently, improper operation of protection equipment can cause false positives, negatively affecting the detection, classification, and reliability of the power grid. This study addresses fault classification in PDS, considering the extensive integration of DG units, specifically PV systems. PDS is evaluated at various levels of PV insertion using different fault scenarios modeled in the IEEE 34-bus test system. This includes five scenarios with variations in the PV system insertion. Autoencoders are applied during the pre-processing phase, while eleven different algorithms are used in the classification stage to identify fault types. They can improve the performance of the classification system by reducing the size of input signals and extracting the most relevant features. The results reveal that the K-nearest neighbor (KNN) and random forest (RF) algorithms demonstrate the best performance, maintaining a minimum accuracy of 95.42% in all scenarios.

1. Introduction

Power distribution systems (PDS) have undergone significant transformations due to the incorporation of distributed energy resources, including distributed generation (DG) units such as photovoltaic (PV) systems, to meet decarbonization goals [1,2]. Concerns about reducing environmental impacts and meeting the growing demand for energy play an essential role in this scenario, driving investment to expand renewable energy sources. In Brazil, PV systems account for approximately 84% of DG installations [3].

Integrating DG units into PDS offers several advantages, including (i) reduction in greenhouse gas emissions, mainly from fossil fuel-based thermoelectric plants, (ii) diversification of the energy matrix, and (iii) reduction in power grid losses [4]. In contrast, this integration presents new challenges for PDS operation. The widespread and unregulated addition of DG units can lead to significant changes in traditional electrical system configurations, such as the loss of radial topology as multiple energy sources supply the system [1,5].

The rapid and often uncoordinated insertion of DG units as PV systems presents significant challenges to the PDS infrastructure [4]. The stochastic nature of solar generation makes it difficult to accurately predict the available energy at any given time, introducing operational uncertainties that can lead to more frequent system disturbances due to sudden changes in energy generation [6].

These disturbances can directly affect the performance of fault protection and classification algorithms, where the characteristics of the short circuit-current are modified, including its direction, amplitude, and distribution [7,8]. As PV penetration increases, short-circuit current profiles change compared to feeders without PV systems.

Under these conditions, the short-circuit current comprises contributions from the PV system and the conventional power grid. These contributions can sometimes maintain current levels relatively constant or even below the thresholds configured in overcurrent relays, compromising their operation [9].

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Given the complexity introduced by PV systems, this study is relevant as it evaluates the impacts of the widespread integration of PV into PDS. In this context, five scenarios with varying levels of PV penetration are considered. The analysis considers variations in the relevant parameters of the electrical system, such as fault resistance, insertion angle, short-circuit type, fault location bus, and the number of PV modules installed. A pre-processing technique based on autoencoders is applied to efficiently extract relevant features from the current signals, capturing the changes caused by PV systems.

Lastly, a comparative analysis is performed using eleven fault classification algorithms, including support vector machine (SVM), decision trees (DT), random forest (RF), extremely random trees (ERT), k-nearest neighbors (KNN), AdaBoost (AB), gradient boosting (GB), logistic regression (LR) and the artificial neural networks (ANN) multilayer perceptron (MLP), fuzzy ART (FA), and Euclidean ART (EA). These machine learning algorithms are applied to assess the classification efficiency of eleven types of single–, two–, and three-phase faults or their absence under different operating conditions.

1.1. Literature review

The increasing integration of DG units into PDS has been extensively discussed in the recent literature due to its impact on the operation and coordination of protection devices. Several studies have been conducted on fault detection and classification considering DG unit insertion [10]. These studies have developed robust methods to address the characteristics of DG units using soft computing algorithms, such as convolutional neural networks (CNN), discrete wavelet transform (DWT), deep belief networks (DBN), and fuzzy logic.

In [11], a CNN fault classification approach was presented, eliminating the need for a pre-processing phase. In this method, datasets comprising three-phase voltage and current signals were directly inputted into the classifier. The methodology's performance was assessed using confusion matrix metrics, including accuracy, sensitivity, specificity, precision, and F1-score.

Patcharoen and Ngaopitakkul [12] proposed a fault classification technique using DWT. Three-phase current signals extracted under nominal and fault conditions were processed via DWT to obtain the wavelet coefficients, which constitute the input set for the DT classifier.

A methodology based on DBN was presented in [13], which provides detailed voltage and current characteristics. It involves processing voltage and current signal data using min–max normalization and waveform splicing techniques to train the DBN. Furthermore, the authors implemented an SVM classifier for comparison purposes.

A methodology for detecting and classifying faults in PDS with PV system integration was presented in [2]. The authors introduced a fault index that combines two components: the Wigner distribution index, which is based on the energy density of the current signal, and the alienation index, which is evaluated using alienation coefficients derived from signal samples. This combined index is used to detect faults in the power grid. In addition, a new index was introduced called the ground fault index. This index is based on the Wigner distribution of the zero-sequence component of the current signal and aims to distinguish between double-phase and double-phase-to-ground faults. The fault classification was performed by counting the faulty phases identified by the fault index.

Cao et al. [6] proposed a fault detection and classification methodology using generative adversarial networks (GAN) combined with the social spider optimization (SSO) method. The PDS was represented through a digital twin with PV systems to collect data from various operational scenarios to train and evaluate the proposed technique. The authors compared the results of their approach with those of other techniques, demonstrating that their method outperforms the existing ones in fault detection and classification.

An unsupervised and automatic fault detection and classification methodology was presented in [14]. This technique employed a hybrid structure consisting of three subsystems: CNN, SVM and predictive residual analysis. To identify different configurations, CNN was trained using data windows containing current and voltage information. CNN output provided probability estimates for each configuration. Finally, SVM was applied for fault classification, using only fault-related data.

In [15], a method for fault detection and classification that combines wavelet singular entropy (WSE) and fuzzy logic was proposed. This technique analyzes the maximum WSE values, which indicate values that exceed the allowed limits for faulty phases. Consequently, the WSE of the three-phase currents was the input set for the classifier, whose purpose is to detect and classify faults in PDS.

A proposal based on deep learning and deep autoencoders was presented in [16]. The methodology involves two main steps: the former distinguishes between disturbances and faults, while the latter classifies the fault type. The fault diagnosis structure includes three neural networks trained using supervised and unsupervised learning. A deep autoencoder was used to eliminate misleading fault characteristics by selecting the most relevant information, thus enhancing the efficiency of the fault classification process.

In [17], a methodology was proposed for detecting and classifying short-circuit faults, considering the influence of DG units under various operating conditions. The Fortescue approach extracted the fault characteristics, considering the components of positive, negative, and zero-sequence currents. The authors introduced softmax regression to address the negative impact of transient data samples on fault classification.

In [18], a fault classification technique that combines fast dynamic time warping (Fast-DTW) and short-time Fourier transform (STFT) was implemented. The authors applied Fast-DTW to reduce the size of voltage and current data during the pre-processing phase. The signal features were enhanced through STFT to create a time-frequency energy map of the measured voltage and current signals. Different faults were classified using a CNN, with the input consisting of these energy maps.

In summary, Table 1 shows relevant characteristics of the evaluated references, including signal analysis method, DG units integration, sensitivity analysis in DG units, and classification method. To the authors' knowledge, our study is the most comprehensive among the evaluated references.

1.2. Contributions

This study introduces an efficient methodology for fault classification in the context of massive insertion of PV systems. The main contributions of the proposed approach are described below:

- The application of autoencoders in the pre-processing phase reduces data size and extracts relevant features from current signals. They offer a compact and efficient representation of signals while preserving the most essential information.
- This study considers the uncertainties associated with the generation of energy from renewable energy sources. It includes a sensitivity analysis of PV modules connected to PDS. This approach allows for the study of five scenarios with different levels of PV insertion.
- Each of the eleven classification algorithms is used to classify all short-circuit types (single–, two–, and three-phases) and identify faults or nominal operating conditions (NOC). In contrast, many studies applied different classifiers for each fault type. In this sense, our unified approach offers a more realistic and versatile solution to the management of PDS.
- This study distinguishes itself by using neural networks based on adaptive resonance theory (ART), specifically Fuzzy ART and Euclidean ART, for fault classification. These networks have unsupervised training, learning from the inherent characteristics of the data. One significant advantage of these architectures is that

Table 1
Overview of evaluated references.

Ref.	Signal analysis method	DG units integration	Sensitivity analysis in DG units	Classification method
[2]	Wigner distribution and alienation index	Yes	No	Accounting for defective phases identified by the fault index
[6]	No reported	Yes	No	GAN and SSO
[11]	Raw data	Yes	No	CNN
[12]	DWT	Yes	Yes	DT
[13]	Waveform splicing and min–max	Yes	No	DBN
[14]	CNN	Yes	No reported	SVM
[15]	WSE	Yes	No	Fuzzy logic
[16]	Deep autoencoder	No reported	No	CNN
[17]	Fortescue approach	Yes	No	Softmax regression
[18]	Fast-DTW and STFT	No	No	CNN
This work	Autoencoders	Yes	Yes	RF, KNN, SVM, GB, MLP, DT, ERT, LR, AB, FA, and EA

they avoid limitations common to other algorithms, such as slow learning and sensitivity to noisy data. Furthermore, they do not experience catastrophic forgetting, the sudden loss of previously acquired knowledge [19].

- Finally, our study analyzes and compares eleven classifiers applied in the classification stage. It also examines the impacts and limitations on the PDS resulting from the insertion of PV systems.

1.3. Paper structure

This study follows in Section 2, which addresses the fundamental theory of autoencoders. Section 3 details the proposed methodology, including extraction (Section 3.1) and pre-processing (Section 3.2) of current signals. The fundamental theory of eleven fault classification algorithms, including DT, RF, ERT, SVM, KNN, AB, GB, MLP, LR, FA, and EA, are addressed in Section 3.3. The cross-validation process is discussed in Section 3.4, followed by an overview of the confusion matrix metrics in Section 3.5. Section 4 presents the results, starting with the construction of fault scenarios (Section 4.1) and the classification of short-circuit faults (Section 4.2). This includes defining the parameters for the autoencoder (Section 4.2.1) and applying the eleven classifiers to identify faults or NOC in PDS (Section 4.2.2). The results of the previous section are analyzed in more depth in the discussion section – Section 5. Finally, Section 6 presents the conclusions of our study.

2. Autoencoders

Although working directly with data or signals in a specific domain is possible, representation learning is often more efficient. The performance of automatic classification models can depend significantly on the quality of the underlying features in the data [20]. Therefore, it is recommended that the data be preprocessed and transformed into a more abstract and meaningful representation.

Bengio and Lecun [21] discussed several challenges machine learning models face, including the curse of dimensionality. These challenges often make it advantageous to work with representations rather than raw data, as representations can more easily capture and express the general characteristics of machine learning tasks [22,23]. Bengio et al. [20] highlighted several benefits of using representation learning models. These benefits include extracting and exploring temporal and spatial coherence from data, mainly when the data consists of observations such as signals or time series. Representation learning also facilitates the association of relevant categorical concepts. Moreover, representation learning models often concentrate probability mass in regions of much lower dimensionality than the original data space, a characteristic explicitly applied by manifold-inspired algorithms like autoencoders.

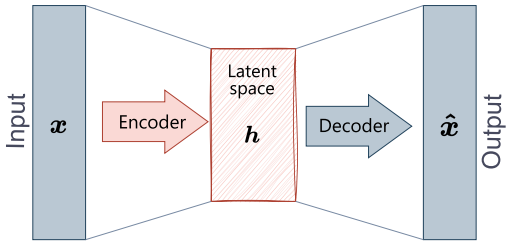


Fig. 1. Autoencoder architecture.

Autoencoders are a type of neural network architecture, specifically an MLP, applied to learning data representations. These models can extract and organize information through nonlinear transformations, enabling them to map input data to a more meaningful representation [23]. This representation is captured in a vector known as the latent space, which contains the essential information needed for subsequent classification tasks [22,23]. In addition to learning to encode the input data, autoencoders also feature a decoding function that reconstructs the input space from the latent representation, generating a new version of the original data.

The main feature of this architecture is that the output label corresponds directly to the input data. The layer arrangement is symmetric around the latent space, as shown in Fig. 1. The hidden layers preceding the latent space are known as the encoder, while the hidden layers following the latent space are called the decoder. The encoder compresses the input data into the latent space, and the decoder reconstructs the data from this latent representation. The information bottleneck concept, introduced by Tishby et al. [24], is fundamental to the architecture of autoencoders, particularly in the reduction of dimensionality and the elimination of noise. The central idea involves compressing the information into an intermediate layer with fewer neurons. This bottleneck compels the autoencoder to learn a more compact and efficient data representation [25]. By restricting the amount of information transmitted through the bottleneck, the autoencoder focuses on the most significant aspects of the data, eliminating redundancies and less relevant details [26]. Consequently, this process results in a latent representation that retains the essential characteristics of the original data, allowing for efficient reconstruction while reducing dimensionality and filtering out noise.

The autoencoders were introduced by [27], where a feature extraction function was defined, denoted as f_θ. It encodes the characteristic vector $\boldsymbol{h} \in \mathcal{F}$ from the input $\boldsymbol{x} \in \mathcal{X}$. Furthermore, the feature extraction function $f_\theta : \mathcal{X} \rightarrow \mathcal{F}$ maps the input space $\mathcal{X}$ to the feature space $\mathcal{F}$.

Consequently, the representation vector is defined by (1).

$$\boldsymbol{h}_i = f_\theta(\boldsymbol{x}_i), \quad (i = 1, 2, \ldots, N), \tag{1}$$

where $\boldsymbol{h}$ represents the feature vector also known as the representation, encoding, or latent space of the input $\boldsymbol{x}$, and N is the number of samples in the dataset. The decoding function $g_\phi : \mathcal{F} \to \mathcal{X}$ maps the latent space back to the input space, in order to produce a reconstructed version of the original input, as described in (2).

$$\hat{\boldsymbol{x}}_i = g_\phi(\boldsymbol{h}_i) = g_\phi(f_\theta(\boldsymbol{x}_i)) = (g_\phi \circ f_\theta)(\boldsymbol{x}_i), \quad (i = 1, 2, \ldots, N). \tag{2}$$

In this process, the autoencoder is trained and parameterized through its encoder and decoder components, where the parameters $\{\theta, \phi\}$ of both are learned simultaneously while reconstructing the original input. Specifically:

- $\theta = \{\boldsymbol{W}, \boldsymbol{b}\}$: parameters of the encoder, consisting of the weight matrix $\boldsymbol{W}$ and the bias vector $\boldsymbol{b}$;
- $\phi = \{\boldsymbol{W}', \boldsymbol{d}\}$: parameters of the decoder, consisting of the weight matrix $\boldsymbol{W}'$ and the bias vector $\boldsymbol{d}$.

The objective is to minimize the reconstruction error $L(\boldsymbol{x}, \hat{\boldsymbol{x}})$, which measures the discrepancy between the input data $\boldsymbol{x}$ and its corresponding reconstruction $\hat{\boldsymbol{x}}$. Thus, training the autoencoder involves adjusting the parameters $\{\theta, \phi\}$ to minimize the reconstruction error, as described in (3).

$$\underset{\theta,\phi}{\operatorname{argmin}} \sum_{i}^{N} L(\boldsymbol{x}_i, g_\phi(f_\theta(\boldsymbol{x}_i))). \tag{3}$$

The representation mapping from the inputs in the encoder and the input prediction from the latent space in the decoder is determined by f_θ and g_ϕ, according to (4):

$$f_\theta(\boldsymbol{x}) = \sigma_f(\boldsymbol{b} + \boldsymbol{W}\boldsymbol{x}), \qquad g_\phi(\boldsymbol{h}) = \sigma_g(\boldsymbol{d} + \boldsymbol{W}'\boldsymbol{h}), \tag{4}$$

where σ_f and σ_g are activation functions. The set of parameters $\boldsymbol{b}$ and $\boldsymbol{d}$ are the encoder and decoder bias vectors, and $\boldsymbol{W}$ and $\boldsymbol{W}'$ are the encoder and decoder weight matrices, respectively.

Autoencoders can be configured as deep architectures by adding hidden layers, which enables the extraction of increasingly abstract features. This configuration is achieved by the exponential growth in the number of paths the data can follow, allowing the model to capture more abstract concepts [22,23]. As a result, autoencoders become more robust to local changes in input, benefitting from the nonlinear functions incorporated into the model [28].

3. Methodology for fault classification

Feeder faults are events inherent to PDS operation and characterized by deviations from the system's standard operating conditions [29]. When a disturbance such as a short-circuit occurs, the affected phase current signals change in amplitude and frequency. During these events, the current in the affected phases increases significantly compared to the other phases that remain in NOC. In this sense, our study involves three main steps: (i) extraction of current signals, (ii) processing and analysis of these signals using autoencoders, and (iii) classification of short-circuit types.

3.1. Extraction of current signals

The technique relies on continuously analyzing current signals obtained from the substation output. A fixed window is defined as capturing a cycle of samples from the three-phase post-fault currents to extract various characteristics of different short-circuit types. The samples are collected at a sampling rate of 10.8 kHz. In addition, a data window that contains samples from the system under NOC establishes the baseline for the standard operating state of the system.

Table 2
Coding of the system operating state.

Label	System operating state
[1 0 0 0 0 0 0 0]	NOC
[0 1 0 0 0 0 0 0]	A_g
[0 0 1 0 0 0 0 0]	B_g
[0 0 0 1 0 0 0 0]	C_g
[0 0 0 0 1 0 0 0]	AC or AC_g
[0 0 0 0 0 1 0 0]	BC or BC_g
[0 0 0 0 0 0 1 0]	AB or AB_g
[0 0 0 0 0 0 0 1]	ABC or ABC_g

3.2. Signal pre-processing and feature extraction

After extracting the three-phase current signals under various fault conditions, windows containing one cycle of 180 samples from each phase are selected and preprocessed in three main steps, as shown in Fig. 2. In the first step, the windows from the three phases are concatenated to form a vector with 540 samples, representing a complete cycle of the three-phase signal. In the second step, the data are normalized using the Min-Max method, as described in (5). Finally, in the last step, the vector is applied as the autoencoder's input, which reduces the size of the signal window and extracts the most relevant features from the analyzed short-circuits.

$$x_{\text{scaled}} = \frac{x - \min(\boldsymbol{x})}{\max(\boldsymbol{x}) - \min(\boldsymbol{x})}. \tag{5}$$

It should be mentioned that normalization parameters (min and max values) are obtained only from training data and then applied consistently to training, validation, and test data to avoid data leakage.

Data labels are adjusted to identify faults in the phases, while grounding is ignored. Original labels included categories such as A_g, B_g, C_g, AB_g, AC_g, BC_g, AB, AC, BC, ABC_g, ABC, and NOC. They are refined to A_g, B_g, C_g, AB, AC, BC, ABC, and NOC, as short-circuit coding with and without grounding is identical and this differentiation is not considered in our study. Furthermore, since all labels are categorical variables, they are converted using the one-hot encoding technique, also known as dummy coding. In this process, each category is represented by a separate column, where the column corresponding to the active class is marked with one, while all other columns are marked with zero. Table 2 shows how the operating states of the system are encoded using one-hot encoding.

3.3. Faults classification via machine learning

This section addresses eleven algorithms to identify short-circuit types. The autoencoder provides the data window that contains the input set with essential information to differentiate between faults and the NOC. In addition, these algorithms determine the fault types and the phases involved. The machine learning algorithms applied in our study and their main characteristics are presented below.

3.3.1. Decision Tree (DT)

DT algorithms are supervised models that operate sequentially [30]. Their structure resembles a flowchart, allowing a series of tests in which a numeric attribute is compared with a threshold value according to logical rules. The data set is divided into homogeneous subsets (branches) by repeatedly selecting attributes until a final classification level (leaves) is reached [31]. According to [31], this model can simplify complex decision problems, making logical rules more interpretable. DT are helpful for clustering, feature selection, and classification tasks. However, they have limitations, including their complexity increasing with data size, leading to longer processing times. Furthermore, DT can be unstable, sensitive to minor variations in the data, and may exhibit bias toward dominant features [31].

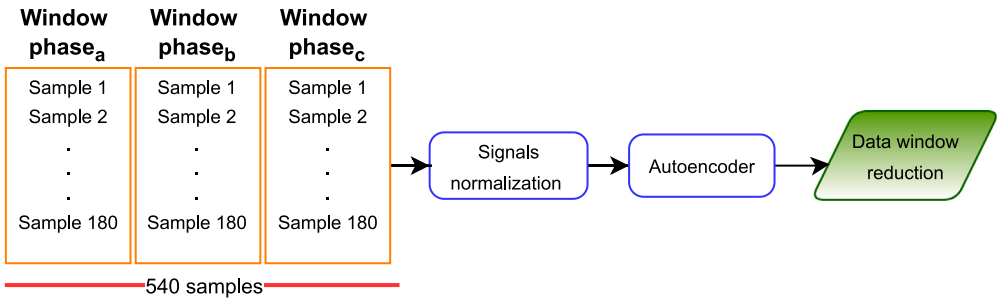


Fig. 2. Three-phase current data window.

3.3.2. Random Forest (RF)

RF algorithm was initially introduced by Breiman [32]. It combines multiple DT trained on different data subsets and features determined by independently sampled random vectors with the same distribution. RF's performance improves as the number of trees increases as the generalization error approaches a limit. This error is influenced by the strength of the individual trees and their correlations. RF can achieve error rates comparable to individual trees [31]. Notable features of this model include its ability to handle outliers and noise in the data and its reduced risk of overfitting as more trees are added [33]. However, its computational intensity and high memory consumption are significant limitations, especially when dealing with many trees and extensive datasets [31].

3.3.3. Extremely Randomized Trees (ERT)

ERT is a variation of the RF algorithm proposed by Geurts et al. [34]. Unlike traditional RF, which searches for the optimal split at each node, ERT introduces greater randomness in selecting features and cut-off points. This increased randomness helps reduce variance and overfitting and allows faster training because it avoids the exhaustive search for the best split at each node [34,35]. However, ERT may be less accurate than RF, and its performance can vary significantly depending on the nature of the data and the parameters used [36].

3.3.4. Support Vector Machines (SVM)

SVM is primarily a binary linear classifier, but can be applied to nonlinear data [37]. It aims to find a hyperplane that separates classes in a feature space while maximizing the margin between the closest points of each class, known as support vectors. By optimizing this margin, SVM aims to enhance model generalization and reduce overfitting, where its performance depends on carefully adjusting parameters, such as the limited penalty cost and kernel function. Additionally, SVM can be limited to large datasets due to its computational complexity and high memory requirements [38].

3.3.5. K-nearest neighbor (KNN)

The K-nearest neighbor (KNN) is a supervised classification technique that is easy to understand and implement [39]. It is used effectively in various applications, including text classification, pattern recognition, and image classification [40]. KNN relies on three fundamental elements: (i) a set of labeled points, (ii) a similarity metric to measure the distance between points, and (iii) the parameter k, which specifies the number of nearest neighbors to consider [39,40]. To classify an unlabeled point, KNN calculates the distance between it and all labeled points, identifies its k nearest neighbors, and assigns the class label based on the majority class among these neighbors. In this sense, KNN has a drawback: its performance is sensitive to the choice of k. If k is too small, the classification results may be influenced by noise in the data. In contrast, if k is too large, the algorithm can include points from different classes in the neighborhood, which can compromise the accuracy of the classification [39].

3.3.6. AdaBoost (AB)

AB, introduced by Freund and Schapire [41], was the first adaptive boosting algorithm. It adjusts its parameters based on the performance of each iteration and focuses on misclassified samples [42,43]. The algorithm sequentially trains weak classifiers (or base learners, typically DT), emphasizing patterns previously misclassified. At the end of the process, AB combines these weighted weak classifiers to create a robust classifier with improved generalization. However, it is sensitive to noise in the data and may be prone to overfitting [42].

3.3.7. Gradient boosting (GB)

GB is a supervised learning technique designed to improve the precision of a weak classification model by combining multiple models, typically DT, into a more accurate ensemble [44]. During each iteration of the learning process, a new model is trained to reduce the overall error of the ensemble. This new model focuses on the negative gradient of the loss function associated with the current predictions. As a result, the model progressively improves, providing more accurate estimates [45]. GB has high flexibility and ease of implementation, which makes it adaptable to various tasks. However, its sequential training process can be slow and resource intensive and require substantial memory. Furthermore, there is a risk of overfitting [43].

3.3.8. Multilayer perceptron (MLP)

MLP is an essential ANN class that includes three different types of layers: input, intermediate (or hidden) and output layers. MLP has a feedforward structure, in which the input signal propagates through the network layer by layer. The backpropagation algorithm, introduced by Rumelhart et al. [27], is commonly used for the supervised training of MLP. This algorithm is based on the error correction learning rule and involves two main steps: forward propagation and backward propagation. During forward propagation, the input data is propagated through the network to produce predictions. In backward propagation, the algorithm adjusts the network weights based on the error between the predicted and actual outputs, refining the model's accuracy. MLP is a versatile model for handling nonlinear relationships in data due to its ability to learn complex patterns in hidden layers, making it widely used in classification tasks. However, its training can be computationally expensive, especially with large datasets or complex architectures. Additionally, it is prone to overfitting if it is not properly regularized. Hence, its performance relies on the correct selection of hyperparameters, such as the number of neurons and layers. Proper tuning and regularization techniques are essential to prevent overfitting and to improve the generalization of the model.

3.3.9. Logistic regression (LR)

LR is an essential statistical technique commonly used in data mining and binary classification tasks. It falls into the broader category of generalized linear models (GLMs) [46]. GLMs have three components: a link function, an exponential probability density function, and a linear predictor [47,48]. LR uses the logit link function, which linearly combines the explanatory variables to estimate the probability

that an outcome belongs to a particular class. For LR, the binomial probability distribution function is the most suitable [47]. LR has been extensively used in various fields, such as biomedicine, credit management, genetics, and engineering. The ability to provide probabilistic estimates straightforwardly is its main advantage. Furthermore, it is flexible enough to handle multiclass classification problems [49,50]. This model is easy to interpret and computationally efficient. However, its performance can be limited when dealing with complex nonlinear relationships, high multicollinearity between variables, or when the dataset is very large and complex.

3.3.10. Fuzzy ART (FA)

FA ANN is part of the ART class, a cognitive theory inspired by brain function and introduced by Grossberg [51]. FA integrates fuzzy set theory, employing a fuzzy AND operator ($\wedge$) to generalize learning for analog and binary input patterns. It features an unsupervised, rapid, and stable learning process that facilitates category recognition in response to arbitrary input vectors. Its training algorithm includes a pre-processing phase in which the input data is normalized to prevent category proliferation. In addition, a complementary vector is constructed to preserve data amplitude. The training process is stable because the adaptive weights gradually reduce over time [52]. The integration of fuzzy set theory allows it to handle binary and analog patterns effectively. However, it is sensitive to the choice of appropriate parameters and high variability in input patterns, which can affect its performance.

3.3.11. Euclidean ART (EA)

EA is an unsupervised clustering ANN that uses the Euclidean distance to assess the similarity between the data points [53,54]. FA employs fuzzy operations to evaluate how well a pattern belongs to a cluster, while EA applies the Euclidean distance. This results in the formation of hyperspheres, where each hypersphere's center represents a category or class, as opposed to the hyper-rectangles formed in the FA. Additionally, the EA method does not have the pre-processing phase required in the FA algorithm, which includes normalization and encoding of input vectors. This omission reduces the dimensionality of input data by half, significantly improving computational efficiency [53,54]. EA can dynamically adapt to data, continuously learning, and adjusting its structure without needing prior labels. Despite being a relatively simple model, it is robust when dealing with high-dimensional data. However, as with other ART networks, its performance is sensitive to the choice of parameters, and due to its reliance on Euclidean distance, it may face difficulties when handling complex nonlinear patterns.

3.4. Cross validation

K-fold cross-validation is a well-known method for estimating and validating the performance of machine learning models [55]. It addresses overfitting and model variability, more accurately assessing a model's generalization ability to unseen data [56]. The technique divides the available data into k mutually exclusive subsets, or folds, where k represents the number of subsets used for cross-validation [56,57].

In each k iteration, one fold is used as the validation set, while the remaining $k-1$ folds are combined to form the training set. This process is repeated k times, ensuring that each subset is used as the validation set exactly once [57]. At the end of the process, the model's performance is evaluated by averaging the results from all k iterations [56], providing a more reliable and stable estimate of the model's performance compared to a single training and testing step.

Several studies discuss the optimal value of k, where many suggest that $k=5$ or $k=10$ offers a good balance between bias and variance [57]. These values are practical because they provide a favorable trade-off between computational time and estimation accuracy.

Table 3
Multiclass confusion matrix.

Confusion matrix		Predicted state				
		Class 1	Class 2	Class 3	…	Class k
Real state	Class 1	C_{11}	C_{12}	C_{13}	…	C_{1k}
	Class 2	C_{21}	C_{22}	C_{23}	…	C_{2k}
	Class 3	C_{31}	C_{32}	C_{33}	…	C_{3k}
	⋮	⋮	⋮	⋮	⋱	⋮
	Class k	C_{k1}	C_{k2}	C_{k3}	…	C_{kk}

In this sense, our study uses $k=5$ for k-fold cross-validation based on its widespread acceptance in the literature and its balance between computational complexity and estimates robustness [55]. The original data set is divided into 70% for training and 30% for testing. The model parameters are adjusted and optimized within the training set using k-fold cross-validation, ensuring consistent hyperparameter tuning. All reported results are based on the test set; therefore, the performance metrics reflect the model's ability to generalize to new data. Parameter optimization for each algorithm is performed using the grid search method, which tests all possible parameter configurations provided.

3.5. Metrics for evaluation

The performance of the eleven classifiers implemented is evaluated using AUC-ROC and confusion matrix metrics, as shown in Table 3. Since this study addresses a multiclass problem that involves classification into eight classes, macro and micro averages are obtained from the confusion matrix metrics, including accuracy, recall (sensitivity), specificity, precision, and F1-score, as detailed in (6)–(10), respectively. Although AUC-ROC is used more frequently in binary classification problems, it can be extended to multiclass problems by computing macro and micro AUCs using the one-vs-rest (OvR) strategy [58], where each class is compared with all other classes. The macro average consists of the arithmetic mean of the metrics across all classes, giving equal weight to each class regardless of its size. In contrast, the micro average aggregates the total counts of true positives, false positives, true negatives, and false negatives across all classes [59].

The accuracy in (6) provides the overall performance of the classifiers. It represents the ratio between the number of correctly classified cases and the total number of cases predicted.

$$\text{Accuracy} = \frac{\sum_{i=1}^{k} C_{ii}}{\sum_{i=1}^{k}\sum_{j=1}^{k} C_{ij}}. \tag{6}$$

Recall in (7) measures the classifier's ability to correctly identify positive cases within each class, known as true positives. Therefore, a high recall indicates that the classifier effectively detects all positive samples, whether in a short-circuit or under NOC.

$$\begin{aligned} \text{Recall}_{\text{ Macro}} &= \frac{1}{k}\sum_{i=1}^{k}\frac{C_{ii}}{C_{ii}+\sum_{j=1,j\neq i}^{k} C_{ij}}, \\ \text{Recall}_{\text{ Micro}} &= \frac{\sum_{i=1}^{k} C_{ii}}{\sum_{i=1}^{k}\left(C_{ii}+\sum_{j=1,j\neq i}^{k} C_{ij}\right)}. \end{aligned} \tag{7}$$

The specificity in (8) measures the classifier's ability to correctly identify all samples that do not belong to a specific class; that is, the proportion of negative cases correctly classified as negative by the model for each class – true negatives. It is complementary to recall and assesses the number of "false alarms" produced by the model.

$$\begin{aligned} \text{Specificity}_{\text{ Macro}} &= \frac{1}{k}\sum_{i=1}^{k}\frac{\sum_{j=1,j\neq i}^{k}\sum_{l=1,l\neq i}^{k} C_{jl}}{\sum_{j=1,j\neq i}^{k}\sum_{l=1,l\neq i}^{k} C_{jl}+\sum_{j=1,j\neq i}^{k} C_{ji}}, \\ \text{Specificity}_{\text{ Micro}} &= \frac{\sum_{i=1}^{k}\sum_{j=1,j\neq i}^{k}\sum_{l=1,l\neq i}^{k} C_{jl}}{\sum_{i=1}^{k}\left(\sum_{j=1,j\neq i}^{k}\sum_{l=1,l\neq i}^{k} C_{jl}+\sum_{j=1,j\neq i}^{k} C_{ji}\right)}. \end{aligned} \tag{8}$$

The precision in (9) refers to the ratio of true positives, cases correctly identified as positive for a specific class, to the total number of cases where the classifier predicted as positive for that class.

$$\text{Precision}_{\text{Macro}} = \frac{1}{k}\sum_{i=1}^{k}\frac{C_{ii}}{C_{ii}+\sum_{j=1,j\neq i}^{k}C_{ji}}, \qquad \text{Precision}_{\text{Micro}} = \frac{\sum_{i=1}^{k}C_{ii}}{\sum_{i=1}^{k}C_{ii}+\sum_{j=1,j\neq i}^{k}C_{ji}}. \tag{9}$$

The F1-score in (10) is the harmonic mean of precision and recall for each class. It balances the model's ability to minimize false positives (high precision) with its capacity to identify all positive samples (high recall).

$$\text{F1}_{\text{Macro}} = \frac{2}{k}\sum_{i=1}^{k}\frac{C_{ii}}{C_{ii}+\frac{1}{2}\left(\sum_{j=1,j\neq i}^{k}C_{ij}+\sum_{j=1,j\neq i}^{k}C_{ji}\right)}, \qquad \text{F1}_{\text{Micro}} = \frac{2\cdot(Precision_{\text{Micro}}\cdot Recall_{\text{Micro}})}{(Precision_{\text{Micro}}+Recall_{\text{Micro}})}. \tag{10}$$

The receiver operating characteristic (ROC) curve is a graphical tool commonly used to evaluate the performance of binary classifiers. It illustrates the ability of the model to differentiate between two classes by depicting the relationship between the true positive rate (TPR), also known as recall, and the false positive rate (FPR) [60–62]. These rates are obtained through (11).

$$\text{TPR}(\tau) = \frac{\frac{1}{n}\sum_{i=1}^{n}\mathbf{I}\{x_i\geq\tau, y_i=1\}}{\frac{1}{n}\sum_{i=1}^{n}\mathbf{I}\{y_i=1\}}, \qquad \text{FPR}(\tau) = \frac{\frac{1}{n}\sum_{i=1}^{n}\mathbf{I}\{x_i\geq\tau, y_i=0\}}{\frac{I}{n}\sum_{i=1}^{n}\mathbf{I}\{y_i=0\}}, \tag{11}$$

$\mathbf{I}\{\cdot\}$ is an indicator function that returns one if the condition in (11) is true and zero otherwise. The cut-off point $\tau \in \mathbb{R}$ represents the threshold above which observations are classified as belonging to the positive class, indicating the presence of faults. The variables x_i represent the probabilities associated with the observation ith, while y_i denotes the real classes of these observations: if $y_i = 1$, it indicates a positive class (presence of a fault); otherwise, if $y_i = 0$, it indicates a negative class (NOC). The summation is performed for all n observations in the dataset.

The ROC curve is performed using various cut-off points or thresholds at which the classifier assigns scores. TPR and FPR values are calculated for each cut-off point. The ROC curve is plotted on a graph where the x-axis represents FPR, and the y-axis represents TPR [63]. In this way, the set of ordered pairs $(\text{FPR}(\tau), \text{TPR}(\tau))$ for different values of τ forms the ROC curve [64].

An ideal ROC curve for a classifier with correct predictions passes through the point $(0, 1)$, indicating an FPR = 0 and a TPR = 1 [63]. In contrast, the straight line connecting the points $(0, 0)$ and $(1, 1)$ represents a random classifier, where the hit rate is equal to the error rate [63,64].

The area under the ROC curve (AUC) measures the probability that a positive instance receives a higher score than a negative instance [62]. It is widely used to evaluate the performance of classifiers and is an important metric because it is insensitive to label distributions and class costs [63]. The AUC is defined in (12) for binary classification problems.

$$\text{AUC}(0,1) = \frac{1}{n_0 n_1}\sum_{i=0}^{n_0}\sum_{j=0}^{n_1}\mathbf{I}\{x_{0i}<x_{1j}\}+\frac{1}{2}\mathbf{I}\{x_{0i}=x_{1j}\}, \tag{12}$$

Where $n_0 = \sum_{i=1}^{n}\mathbf{I}\{y_i=0\}$ and $n_1 = \sum_{i=1}^{n}\mathbf{I}\{y_i=1\}$ represent the number of instances in classes 0 and 1, respectively. On the other hand, the terms x_{0i} e x_{1j} denote the scores the classifier assigns to instances in classes 0 and 1.

Table 4
IEEE 34-bus test system parameters.

Parameters	Values
Fault resistance	1 to 40 Ω
Insertion angle	0°, 45°, 90°, and 120°
Location fault (Bus)	806, 814, 824, 828, 830, 840, 854, and 860
Short-Circuit type	A_g, B_g, C_g, AB, AC, BC, AB_g, BC_g, AC_g, ABC, ABC_g

The AUC can be evaluated for multiclass tasks using the OvR approach, as proposed by Provost and Domingos [58]. In this method, the average AUC is obtained for each class plus the set of remaining classes, according to (13) [55,65].

$$\text{AUC}^{\text{OvR}} = \frac{1}{k}\sum_{i=0}^{k-1}\text{AUC}(i, i^c), \tag{13}$$

Where k is the number of classes and i^c represents the set of classes different from i.

Fig. 3 shows a flowchart that describes the methodology, providing an overview of the processes performed. The methodology consists of four steps: (i) simulation of fault conditions in the IEEE 34-bus test system, (ii) extraction of three-phase current signal windows, (iii) pre-processing of signals using an autoencoder, and (iv) classification of faults.

The first step consists of several simulations considering different fault scenarios in the IEEE 34-bus test system. In the second step, signal windows corresponding to the three-phase currents are extracted to capture the fault characteristics. In the third step, the signals are preprocessed using an autoencoder, which reduces the window size and extracts relevant features from the data. Finally, in the last step, eleven algorithms are implemented for fault classification.

4. Results

This section evaluates the proposed methodology through several simulations performed on the IEEE 34-bus test system, modeled using ATP/EMTP and its graphical interface, ATPDraw [66]. The methodology employs autoencoders in the pre-processing phase to reduce the size of the signal windows. The electrical system is assessed under various conditions, including NOC (without anomalies) and during short-circuits.

The IEEE 34-bus test system operates at a nominal voltage of 24.9 kV, where its main feeder features branches with single- and three-phase loads. The main components include (i) two voltage regulators, (ii) a step-down transformer that reduces the voltage from 24.9 to 4.16 kV, (iii) unbalanced distributed loads and balanced concentrated loads, and (iv) shunt capacitors. The voltage regulators are excluded from the analysis as they do not affect the results of this study. The loads along the feeder are modeled using a constant impedance model.

4.1. Construction of fault scenarios

Different fault conditions are simulated at a sampling rate of 10.8 kHz and frequency of 60 Hz, resulting in 2160 signals over a simulation time of 200 ms. This case corresponds to 12 cycles, each cycle containing 180 signal samples. Table 4 shows several parameters whose values are modified for each simulated scenario, including short-circuit type, fault resistance, fault location (bus), and insertion angle.

Variation of these parameters is essential to construct a dataset that accurately represents the operating conditions of the PDS. Table 5 shows five simulated scenarios, including installed power and the connection bus for PV units. The objective is to evaluate how these changes affect the fault classification stage within the IEEE 34-bus test system. Modifications are made through adjustments in system parameters and the number of connected PVs, including bus locations and installed

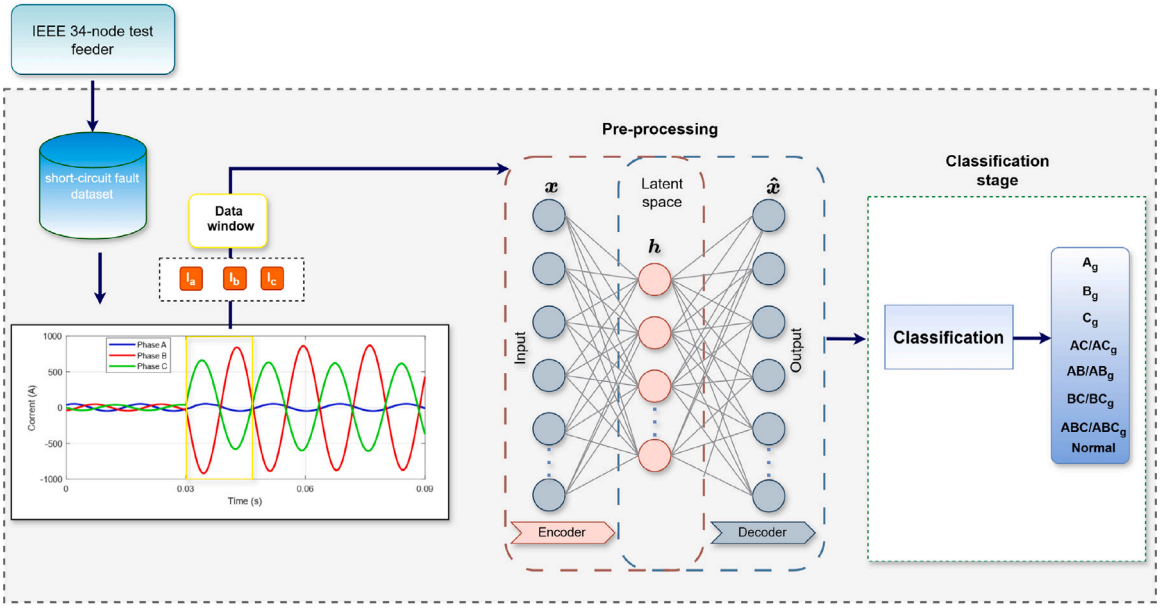


Fig. 3. Flowchart of the proposed methodology.

Table 5
Fault scenarios with PV system insertion.

Scenarios	Number of PV systems	Buses	Power installed
1	0	–	0 MW
2	2	840 and 848	2 MW
3	3	822, 840, and 848	3 MW
4	5	808, 822, 828, 840, and 848	5 MW
5	10	808, 812, 820, 822, 824, 828, 840, 844, 846, and 848	10 MW

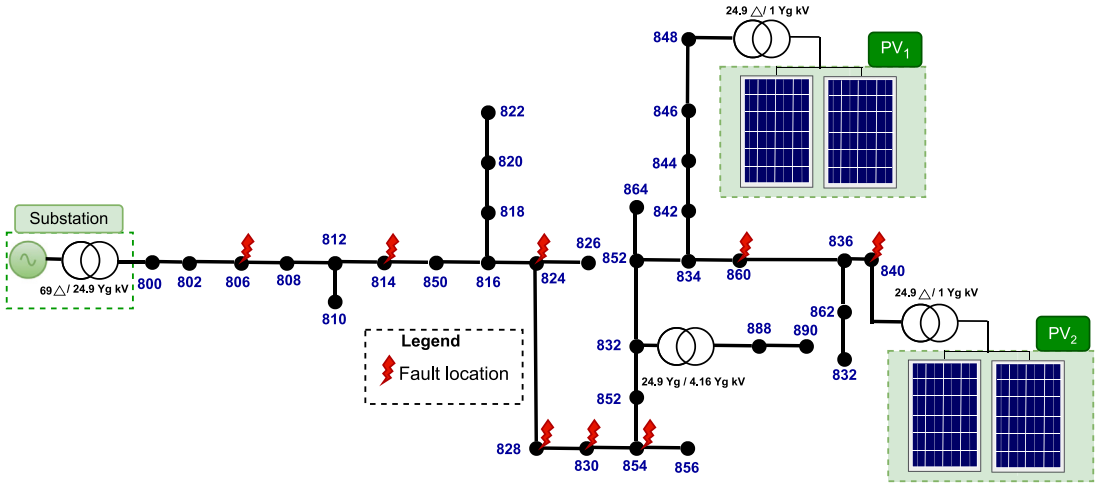


Fig. 4. IEEE 34-bus test system for the scenario 2.

power. In this sense, Fig. 4 shows the test system for scenario 2 with two PV units as shown in Table 5. It is worth mentioning that all PV units are modeled according to [67].

On the other hand, the allocation of PV systems is selected according to [68] for scenario 2. Additional buses for the other scenarios shown in Table 5 are strategically chosen in various buses within the IEEE-34 bus test system, including near the substation, along the feeder branches, and its ends to ensure the distribution of the PV system throughout the electrical system to assess its effects.

Fault location buses are also strategically selected at different points in the electrical system: near the substation, along the sides of the feeder, and at its ends. This approach ensures the analysis of a range of scenarios, providing a comprehensive representation of faults' effects on various parts of the electrical system. All fault scenarios shown in Table 5 use the same buses as fault occurrence locations. In each bus, eleven short-circuit types listed in Table 4 are considered, where each fault is individually evaluated.

The data generated from these simulations are essential for validating the algorithms used for fault classification. A fixed window containing a cycle of post-fault signal samples is applied to extract various characteristics of the short-circuits considered in this study. Similarly, data from the system during NOC are analyzed using a data window with three-phase current signals before a fault occurs. In total, 23760 simulations are performed.

Table 6
Classifiers performance in Scenario 1.

Model	RF	KNN	SVM	GB	MLP	DT	ERT	LR	AB	FA	EA
Accuracy	1.0000	0.9992	0.9790	0.9983	0.9891	0.9933	0.9916	0.9790	0.8544	0.9790	1.0000
Macro recall	1.0000	0.9997	0.9760	0.9962	0.9865	0.9930	0.9856	0.9760	0.7875	0.9760	1.0000
Micro recall	1.0000	0.9992	0.9790	0.9983	0.9891	0.9933	0.9916	0.9790	0.8544	0.9790	1.0000
Macro specificity	1.0000	0.9999	0.9956	0.9998	0.9978	0.9987	0.9988	0.9956	0.9779	0.9956	1.0000
Micro specificity	1.0000	0.9999	0.9970	0.9998	0.9984	0.9990	0.9988	0.9970	0.9792	0.9970	1.0000
Macro precision	1.0000	0.9990	0.9938	0.9981	0.9960	0.9972	0.9892	0.9938	0.8154	0.9938	1.0000
Micro precision	1.0000	0.9992	0.9790	0.9983	0.9891	0.9933	0.9916	0.9790	0.8544	0.9790	1.0000
Macro F1 score	1.0000	0.9994	0.9835	0.9971	0.9909	0.9950	0.9873	0.9835	0.7957	0.9835	1.0000
Micro F1 score	1.0000	0.9992	0.9790	0.9983	0.9891	0.9933	0.9916	0.9790	0.8544	0.9790	1.0000
Macro AUC	1.0000	1.0000	0.9972	1.0000	0.9998	0.9992	0.9921	0.9977	0.9745	0.9975	1.0000
Micro AUC	1.0000	1.0000	0.9994	1.0000	0.9999	0.9990	0.9952	0.9996	0.9805	0.9995	1.0000

Table 7
Classifiers performance in scenario 2.

Metrics	RF	KNN	SVM	GB	MLP	DT	ERT	LR	AB	FA	EA
Accuracy	0.9966	1.0000	0.9714	0.9983	1.0000	0.9806	0.9840	0.9790	0.6852	1.0000	0.9992
Macro recall	0.9923	1.0000	0.9394	0.9962	1.0000	0.9705	0.9735	0.9603	0.6736	1.0000	0.9981
Micro recall	0.9966	1.0000	0.9714	0.9983	1.0000	0.9806	0.9840	0.9790	0.6852	1.0000	0.9992
Macro specificity	0.9994	1.0000	0.9942	0.9996	1.0000	0.9971	0.9977	0.9967	0.9558	1.0000	0.9999
Micro specificity	0.9995	1.0000	0.9959	0.9998	1.0000	0.9972	0.9977	0.9970	0.9550	1.0000	0.9999
Macro precision	0.9983	1.0000	0.9894	0.9995	1.0000	0.9761	0.9764	0.9743	0.6536	1.0000	0.9990
Micro precision	0.9966	1.0000	0.9714	0.9983	1.0000	0.9806	0.9840	0.9790	0.6852	1.0000	0.9992
Macro F1 score	0.9952	1.0000	0.9619	0.9978	1.0000	0.9730	0.9744	0.9667	0.6391	1.0000	0.9985
Micro F1 score	0.9966	1.0000	0.9714	0.9983	1.0000	0.9806	0.9840	0.9790	0.6852	1.0000	0.9992
Macro AUC	1.0000	1.0000	1.0000	1.0000	1.0000	0.9877	0.9856	0.9982	0.9494	1.0000	0.9990
Micro AUC	1.0000	1.0000	1.0000	1.0000	1.0000	0.9908	0.9909	0.9992	0.9507	1.0000	0.9995

It is worth pointing out that all simulations are conducted on a personal computer with an Intel Core i7-13700 processor, 16 GB of DDR4 RAM running at 3200 MHz.

4.2. Short-circuit fault classification

Eleven algorithms are applied to identify short-circuit types. In this sense, the classification stage is initiated after the detection algorithm performed in [10] detects a fault in PDS.

An autoencoder processes a data window containing 180 points (one cycle) of current signals for each phase. The autoencoder's objective is to reduce the size of the current signals and extract their main features. Subsequently, a comparative analysis is performed among the eleven classifiers evaluated in the study.

Data are partitioned using the hold-out method, where 70% and 30% are allocated to training and testing of classifiers. The training set is divided using the 5-fold cross-validation method ($k = 5$) to optimize the selection of algorithm parameters, ensuring a robust cross-validation.

4.2.1. Autoencoder parameters

This study applies an autoencoder to obtain a compact and efficient data representation. This model consists of two main components: an encoder, which reduces the data's dimensionality by compressing them into a smaller latent space, and a decoder, which reconstructs them from a compact representation.

The encoder consists of four layers with 512, 256, 128, and 64 neurons, all using the ReLU activation function. After compression, the data are represented in a latent space with 32 dimensions according to experimental results. Tests with latent space dimensions 2^n are conducted, and the dimension of 32 consistently yielded the best performance for our dataset. The metric used to determine the optimal latent space dimension is the mean squared error (MSE), applied to the comparison between the real input data and the reconstructed output data. A dimension of 32 minimize the MSE, ensuring a balance between data compression and reconstruction accuracy. The decoder reconstructs the data from this latent space using four additional layers with 64, 128, 256, and 512 neurons, which also apply the ReLU activation function. The model is trained using the Adam optimizer with the MSE loss function for 100 epochs and a batch size of 16 samples.

4.2.2. Fault classification

In the classification stage, eleven methods are implemented and compared to identify fault types in the IEEE 34-bus test system. The input set consists of reduced data windows that contain three-phase current signals generated by the autoencoder. Several confusion matrix metrics, such as accuracy, recall, specificity, precision, F1-score, and AUC, are used to evaluate the classifiers' performance in each scenario, as shown in Table 5.

Table 6 shows the results obtained for the base scenario without PV insertion. Most classifiers accurately identify all short-circuit types, where RF and EA achieve a performance of 100%, followed by KNN and GB, which have accuracies close to 100%. In contrast, the AB algorithm achieves an accuracy of 85.44%, significantly lower than other models. Although it records a high specificity above 97%, AB presents a macro recall of 78.75%, indicating reduced efficiency in identifying all faults and classes in NOC.

Table 7 shows the results for scenario two, where RF, KNN, GB, MLP, FA, and EA algorithms achieve accuracy exceeding 99%, emphasizing KNN, MLP, and FA. In contrast, the SVM algorithm achieves an accuracy of 97.14%, which is lower than other models. However, it has an AUC of 1.0, indicating its ability to distinguish between short-circuit signals and those in NOC. This suggests that while the SVM may struggle in accurately classifying all fault cases, it is highly effective in class separation.

The DT, ERT, and LR algorithms also yield good results, although they are lower than previously mentioned. In contrast, the AB algorithm performs significantly worse, achieving an accuracy of 68.52%. This indicates a considerable rate of incorrect classifications compared to the other models. Its macro recall of 67.36% further highlights its poor ability to identify positive samples in all classes accurately, particularly in fault cases.

Table 8 presents the results for scenario three, where most of the algorithms implemented achieve satisfactory results. The KNN algorithm presents perfect performance, while the FA, EA, RF, MLP, and GB algorithms also show excellent results, with accuracies exceeding 99%. In particular, the KNN, MLP, and GB algorithms achieve an AUC of 1.0,

Table 8
Classifiers Performance in Scenario 3.

Metrics	RF	KNN	SVM	GB	MLP	DT	ERT	LR	AB	FA	EA
Accuracy	0.9975	1.0000	0.9630	0.9941	0.9975	0.9798	0.9823	0.9840	0.7955	0.9992	0.9992
Macro recall	0.9942	1.0000	0.9170	0.9882	0.9959	0.9679	0.9734	0.9775	0.7420	0.9990	0.9981
Micro recall	0.9975	1.0000	0.9630	0.9941	0.9975	0.9798	0.9823	0.9840	0.7955	0.9992	0.9992
Macro specificity	0.9995	1.0000	0.9925	0.9989	0.9995	0.9968	0.9973	0.9975	0.9675	0.9999	0.9999
Micro specificity	0.9996	1.0000	0.9947	0.9992	0.9996	0.9971	0.9975	0.9977	0.9708	0.9999	0.9999
Macro precision	0.9992	1.0000	0.9861	0.9958	0.9975	0.9726	0.9785	0.9825	0.7452	0.9981	0.9990
Micro precision	0.9975	1.0000	0.9630	0.9941	0.9975	0.9798	0.9823	0.9840	0.7955	0.9992	0.9992
Macro F1 score	0.9967	1.0000	0.9448	0.9919	0.9967	0.9702	0.9757	0.9798	0.7395	0.9986	0.9985
Micro F1 score	0.9975	1.0000	0.9630	0.9941	0.9975	0.9798	0.9823	0.9840	0.7955	0.9992	0.9992
Macro AUC	1.0000	1.0000	0.9998	1.0000	1.0000	0.9826	0.9853	0.9999	0.9540	0.9995	0.9990
Micro AUC	1.0000	1.0000	0.9999	1.0000	1.0000	0.9894	0.9899	0.9999	0.9591	0.9995	0.9995

Table 9
Classifiers Performance in Scenario 4.

Metrics	RF	KNN	SVM	GB	MLP	DT	ERT	LR	AB	FA	EA
Accuracy	0.9937	0.9968	0.8538	0.9926	0.9937	0.9359	0.9674	0.9159	0.9682	0.9874	0.9958
Macro recall	0.9919	0.9973	0.8376	0.9905	0.9919	0.9290	0.9598	0.9155	0.9771	0.9851	0.9951
Micro recall	0.9937	0.9968	0.8538	0.9926	0.9937	0.9359	0.9674	0.9159	0.6982	0.9874	0.9958
Macro specificity	0.9991	0.9996	0.9789	0.9989	0.9991	0.9909	0.9954	0.9877	0.9561	0.9982	0.9994
Micro specificity	0.9991	0.9950	0.9791	0.9989	0.9991	0.9908	0.9953	0.9880	0.9569	0.9982	0.9994
Macro precision	0.9943	0.9960	0.9099	0.9934	0.9940	0.9318	0.9634	0.9280	0.7214	0.9864	0.9951
Micro precision	0.9937	0.9968	0.8538	0.9926	0.9937	0.9359	0.9674	0.9159	0.6982	0.9872	0.9958
Macro F1 score	0.9930	0.9966	0.8586	0.9919	0.9929	0.9296	0.9612	0.9211	0.6846	0.9856	0.9951
Micro F1 score	0.9937	0.9968	0.8538	0.9926	0.9937	0.9359	0.9674	0.9159	0.6982	0.9874	0.9958
Macro AUC	1.0000	0.9990	0.9949	0.9999	0.9998	0.9865	0.9780	0.9934	0.9441	0.9977	0.9972
Micro AUC	1.0000	0.9988	0.9953	0.9999	0.9999	0.9905	0.9819	0.9936	0.9414	0.9981	0.9976

Table 10
Classifiers performance in scenario 5.

Metrics	RF	KNN	SVM	GB	MLP	DT	ERT	LR	AB	FA	EA
Accuracy	0.9642	0.9542	0.6712	0.9479	0.9048	0.9032	0.8496	0.7091	0.5986	0.8627	0.9153
Macro recall	0.9415	0.9261	0.3869	0.9127	0.8742	0.8526	0.7632	0.5057	0.3900	0.7689	0.8612
Micro recall	0.9642	0.9542	0.6712	0.9479	0.9048	0.9032	0.8496	0.7091	0.5986	0.8627	0.9153
Macro specificity	0.9928	0.9913	0.9213	0.9895	0.9838	0.9817	0.9720	0.9398	0.9259	0.9714	0.9824
Micro specificity	0.9949	0.9935	0.9530	0.9926	0.9864	0.9862	0.9785	0.9584	0.9427	0.9804	0.9879
Macro precision	0.9627	0.9524	0.8548	0.9491	0.8783	0.8818	0.8136	0.6717	0.4766	0.8696	0.9231
Micro precision	0.9642	0.9542	0.6712	0.9479	0.9048	0.9032	0.8496	0.7091	0.5986	0.8627	0.9153
Macro F1 score	0.9518	0.9374	0.4643	0.9300	0.8708	0.8663	0.7838	0.5605	0.4090	0.8128	0.8891
Micro F1 score	0.9642	0.9542	0.6712	0.9479	0.9048	0.9032	0.8496	0.7091	0.5986	0.8627	0.9153
Macro AUC	0.9979	0.9948	0.9283	0.9969	0.9929	0.9416	0.9319	0.8713	0.8874	0.9798	0.9773
Micro AUC	0.9987	0.9967	0.9580	0.9977	0.9942	0.9557	0.9536	0.9243	0.9128	0.9873	0.9842

indicating their efficiency in differentiating between classes. The SVM, DT, ERT, and LR algorithms also perform well, with accuracies above 96%. Specifically, the SVM shows an almost perfect AUC of 0.9999 despite an accuracy of 96.30%.

In contrast, the AB algorithm performs less than the other models. It has a significant rate of incorrect classifications with an accuracy of 79.55%. Its macro recall of 74.20% and macro precision of 74.52% further emphasize this assessment, indicating a reduced ability to accurately identify positive samples in all classes.

The results for scenario four are presented in Table 9, which includes the insertion of five PV units into the PDS. It shows that the KNN, EA, RF, MLP, and GB classifiers perform excellently, achieving accuracies above 99%. In particular, KNN excelled with an accuracy of 99.68% and an AUC of 0.999. RF also demonstrates outstanding performance, with an accuracy of 99.37% and an AUC of 1.0, indicating perfect class distinction and precise separation between short-circuits and those in NOC. MLP and GB follow closely, with an AUC of 0.9999 and an accuracy of 99. 37% and 99.26%, respectively.

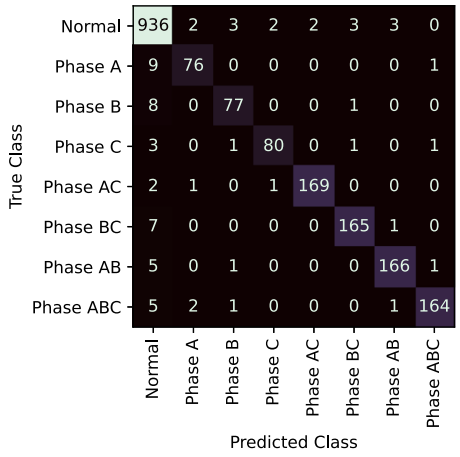
The AB algorithm shows better performance, achieving an accuracy of 96.82%. Although its AUC of 0.9441 is lower compared to other models, it indicates a reasonable ability to distinguish between classes. AB identifies true positives in all classes because it has a macro recall of 97.71% and a macro precision of 72.14%. This improvement suggests that the parameter optimization may not have been fully effective, indicating that the model might not have found the optimal parameters in the previous scenarios.

The SVM shows a significantly lower accuracy of 85.38% and a macro recall of 83.76%, indicating difficulty in correctly classifying the various classes, particularly short-circuits. However, the SVM still achieves an AUC greater than 0.994, demonstrating an excellent ability to distinguish between classes. The DT, ERT, and LR algorithms also yield reasonable results, with accuracy of 93.59%, 96.74%, and 91.59%, respectively. Although these values are lower than the other algorithms, they reflect an acceptable capacity to classify short-circuits and classes in NOC, where the ERT stands out as the most effective among the three.

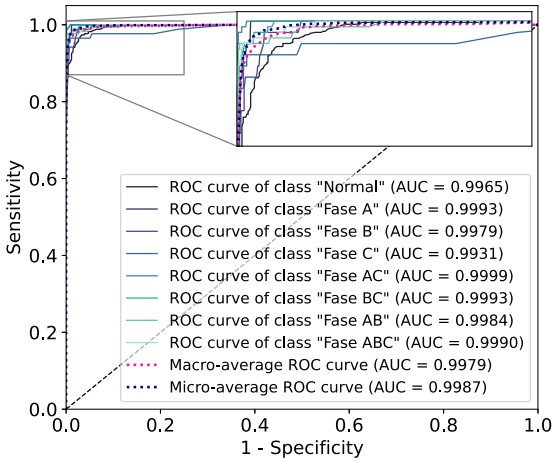
Finally, Table 10 shows the results for scenario five. The analysis uses the confusion matrix metrics and is supported by the ROC curve for the best (RF and KNN) and worst (SVM and AB) models. Table 10 also reveals that the classifier performance changes significantly, with the RF and KNN algorithms exhibiting the highest accuracies, achieving 96.42% and 95.42%, along with an excellent AUC of 0.9979 and 0.9948, respectively.

RF demonstrates the best performance among the eleven evaluated classifiers, as shown by the confusion matrix and the ROC curve in Fig. 5(a) and Fig. 5(b), respectively. RF accurately classifies most samples, achieving an AUC close to one.

Figs. 6(a) and 6(b) show the confusion matrix and the ROC curve for the KNN classifier. It performs similarly to RF; however, KNN incurs more errors than RF.

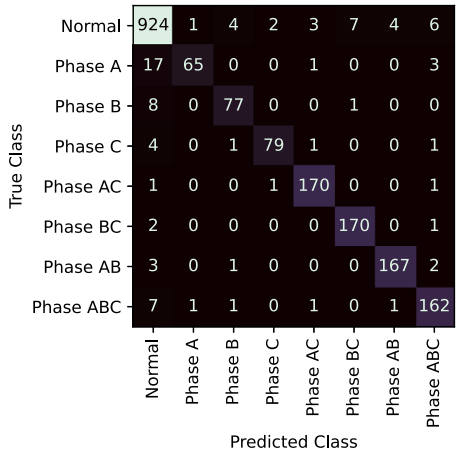


(a) Confusion matrix for RF.

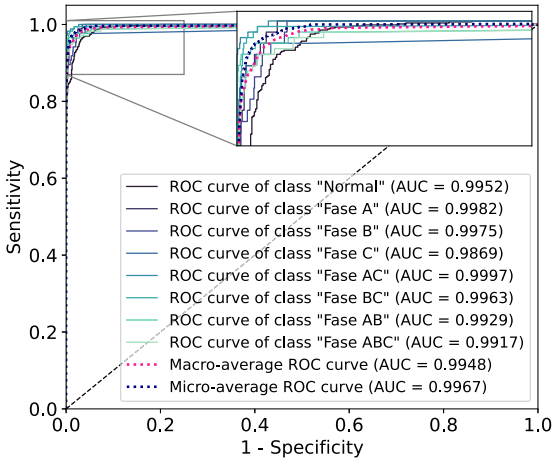


(b) ROC curve for RF.

Fig. 5. ROC curve and confusion matrix for RF classifier in scenario 5.



(a) Confusion matrix for KNN.



(b) ROC curve for KNN.

Fig. 6. ROC curve and confusion matrix for KNN classifier in scenario 5.

The MLP and GB models have 90.48% and 94.79% accuracy and AUC scores of 0.9929 and 0.9969, respectively. GB has a solid ability to differentiate among classes, with an AUC close to 1.0. In contrast, the DT, ERT, and LR algorithms perform moderately. Specifically, DT and ERT achieve an accuracy of 90.32% and 84.96%, respectively, while LR has a lower accuracy of 70.91%, indicating a limited detection capacity. ERT outperforms DT and LR due to an AUC of 0.9319.

The FA and EA models achieve accuracy of 86.27% and 91.53%, respectively, along with AUC scores of 0.9798 and 0.9773. Both models show relatively high AUC values, indicating their solid ability to distinguish among classes.

The SVM and AB algorithms performed poorly in scenario 5. The SVM has an accuracy of 67.12% and a macro recall of 38.69%, indicating significant challenges in accurately classifying short-circuit cases. These cases are often misclassified as cases in NOC, as shown in the confusion matrix in Fig. 7(a). The ROC curve for each class can be analyzed in Fig. 7(b).

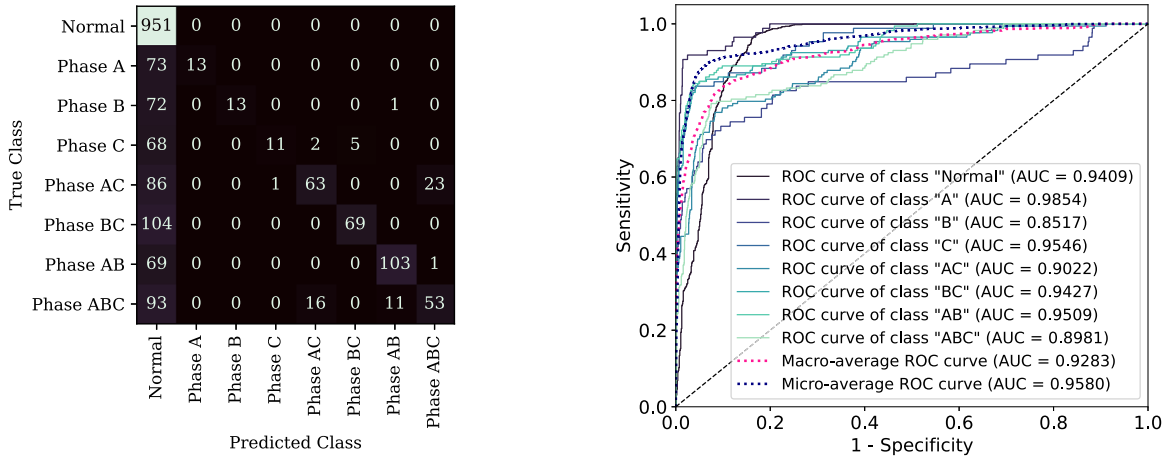
The AB algorithm performs even worse, with an accuracy of 59.86%. Although it has a moderate AUC of 0.8874, reflecting some ability to distinguish between classes, its accuracy remains low. The macro recall of 39.00% and the precision of 47.66% indicate that AB struggles to correctly identify short-circuits and samples in NOC, resulting in a high rate of misclassifications, as shown in the confusion matrix in Fig. 8(a). The lower AUC and ROC curves for each class are shown in Fig. 8(b).

In summary, the SVM and AB algorithms are less effective than the other algorithms analyzed. This may be due to the increased complexity and variability introduced by the large number of PV systems in PDS.

It should be noted that all the analyzes in this section are based on confusion matrix metrics, including macro precision, recall, f1-score, and AUC. These metrics provide a comprehensive view of the eleven classifiers' performance evaluated for recognizing short-circuits and classes in NOC. Although the classes are not overly imbalanced, there is a more significant number of samples in the NOC.

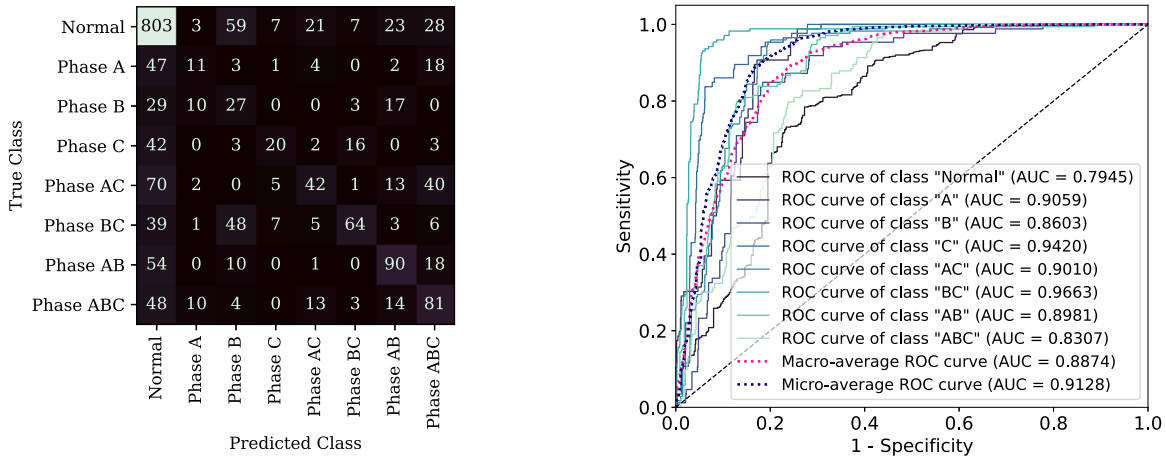
5. Discussion

The decrease in classifier performance in some scenarios is directly related to the changes in current levels caused by the increase in PV insertion. To better understand this effect on the data, we employed t-distributed stochastic neighbor embedding (t-SNE) to reduce the dimensionality of the autoencoder's latent space from 32 dimensions to 3. This reduction enables visualization through three-dimensional graphs. The t-SNE is a dimensionality reduction technique that preserves the proximity relationships between data points in a lower dimensional space, allowing a more precise visualization of the cluster and class separations [69].



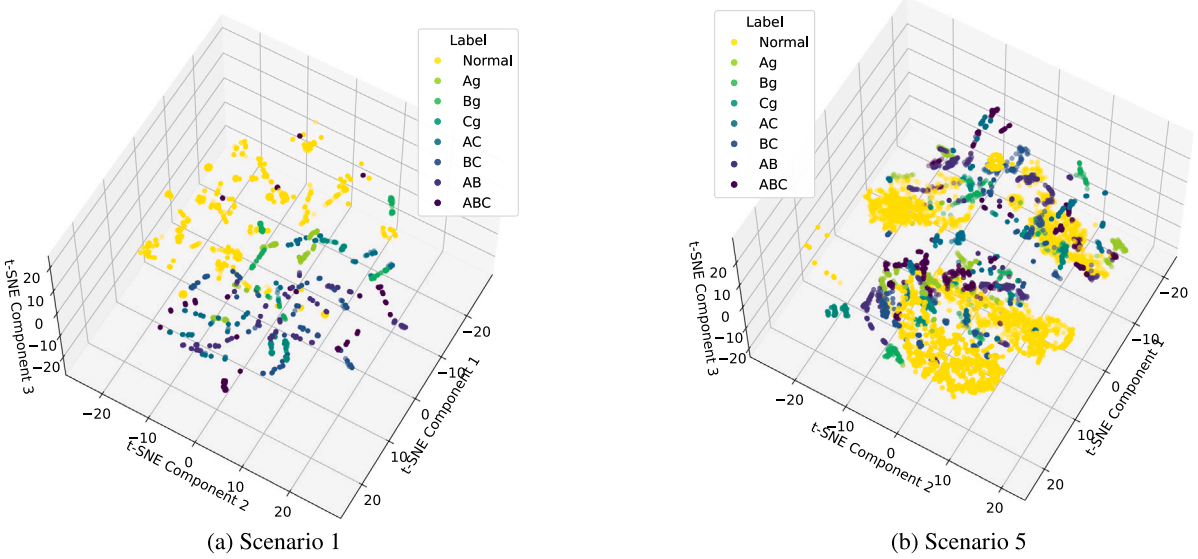
(a) Confusion matrix for SVM. (b) ROC curve for SVM.

Fig. 7. ROC curve and confusion matrix for SVM classifier in scenario 5.



(a) Confusion matrix for AB. (b) ROC curve for AB.

Fig. 8. ROC curve and confusion matrix for AB classifier in scenario 5.



(a) Scenario 1 (b) Scenario 5

Fig. 9. Three-dimensional visualization of the autoencoder's latent space for scenarios 1 and 5 via t-SNE.

Figs. 9(a) and 9(b) show the latent space of the autoencoder for scenarios 1 and 5. In scenario 1, without PV systems, the short-circuit classes are clearly distinguished from the ones in the NOC class. This well-defined separation in the latent space leads to more predictable behavior and reduced variability in current levels of PDS. As a result, short-circuit current and NOC signals have distinctive characteristics that facilitate their accurate identification.

After inserting ten PV modules into the system (scenario 5), some changes in the latent space are observed, as shown in Fig. 9(b). In this scenario, the NOC class overlaps other classes in some graph regions. This overlap accounts for the reduced performance of some classifiers in the last scenario. The current signal characteristics under fault and in NOC become very similar, leading to confusion among some samples in different classes.

Most of the classifiers that show significant reductions in confusion matrix metrics, such as macro recall in scenario 5, misclassified all short-circuit fault classes as NOC. This misclassification occurs due to the extensive integration of PV units, which reduces the contribution of the fault current to the power grid, increasing thus the risk of failure of the protection system [8]. As a result, the algorithm may not detect faults in the PDS, even when they are present.

Despite confusion between classes after the insertion of ten PV modules into the power grid, some classifiers demonstrate high accuracy, including GB (94.79%), KNN (95.42%), and RF (96.42%). These outstanding results can be attributed to some distinct characteristics of each algorithm that make them more effective. For example, GB works sequentially, correcting the errors of previous models as it builds new ones, resulting in a more robust model over time [45]. KNN, a instance-based method, uses the local proximity of samples to achieve more accurate classification, even in the presence of overlap, using the local structure to preserve essential information. RF is a set of DT, where each tree is trained in different data subsets, enhancing the robustness of the model by reducing the risk of overfitting [32]. Finally, RF evaluates the importance of the characteristic, prioritizing the most discriminatory ones [70].

On the other hand, MLP, DT, ERT, LR, FA, and EA achieve moderate results. ANNs, such as MLP, can learn complex data representations but are susceptible to overfitting and require careful hyperparameter tuning [71]. DT can capture nonlinear relationships in data, but their tendency to overfit can limit their performance [31]. ERT introduces additional randomness, which offers some resistance to overfitting, although their performance can be heavily influenced by the nature of the data and the parameters used [36]. Finally, LR is constrained by its linearity, which can limit its effectiveness in more complex scenarios.

ANN based on ART, such as FA and EA, adapt their categories to accommodate new patterns [52,54]. Although these algorithms can handle changes in data patterns, they face challenges when there is extreme class overlap. In the FA and EA classifiers, the label of each cluster is determined by most samples within it. Therefore, even if some noisy samples are incorporated into a cluster, they do not significantly affect the algorithm performance if they remain a minority compared to all samples. Moreover, these algorithms are designed to be robust against noise and data variations, focusing on the most relevant features while suppressing irrelevant features and noise [19].

Finally, SVM and AB showed poorer performance with massive insertion of PV systems. SVM relies on a clear margin of separation among classes, which becomes problematic due to its dependence on and sensitivity to kernel functions. These kernels may fail to capture complex relationships in the data, leading to reduced performance [72]. In contrast, AB has difficulty dealing with noisy and unbalanced data [73]. It tends to assign more weight to noisy samples, and this effect intensifies with increasing iterations, further degrading its performance [74].

It is worth pointing out that computational time is essential for applying fault classification algorithms in real-world scenarios. In this context, the autoencoder has an average training time of 77 s per scenario. However, real-time applications are not affected as training is done offline. The transformation stage for new samples, or prediction, is highly efficient, with a computational time of 75.33 μs per sample. RF stands out for balancing computational time and performance among the classifiers. It has an average training time of 0.81 s and a prediction time of 16.3 μs per sample. KNN has a minimal training time, but a moderate prediction time of 153 μs per sample. Finally, GB, while requiring a longer training time of 21.9 s, achieves a prediction time of 19.7 μs. It is essential to note that the methods analyzed are promising for real-time monitoring, mainly RF, which combines high accuracy with low latency.

6. Conclusion

This study analyzed the impact of photovoltaic (PV) systems integration on the short-circuit fault classification process in power distribution systems (PDS). The approach was based on the continuous analysis of current signals measured at the substation output. It involved two steps, including (i) feature extraction from the three-phase currents using an autoencoder and (ii) application of various algorithms for classifying short-circuit types. The effectiveness of the methodology was tested through extensive simulations in eleven classification algorithms, considering fault scenarios and nominal operating conditions (NOC).

In the classification process, a single classifier for each algorithm was implemented to identify all short-circuit types, constructing the problem as a multiclass task with eight classes. The classification results for different short-circuit types in five scenarios with varying levels of PV penetration highlighted the performance of the random forest (RF), gradient boosting (GB), and K-nearest neighbors (KNN) algorithms, consistently achieved accuracies above 94%. In contrast, algorithms such as the support vector machine (SVM) and AdaBoost (AB) performed worse, particularly in more challenging scenarios with higher PV insertion.

These results demonstrate the significant impact of PV integration in PDS, especially in scenarios with higher penetration levels, where the characteristics of the current signals were substantially altered. This change led to a decline in the performance of algorithms such as SVM and AB. At the same time, KNN, GB, and RF showed resilience, maintaining accuracy above 94%, even in the scenario with the highest level of PV penetration (scenario 5). Although most algorithms showed performance degradation, the KNN, GB, and RF classifiers remained effective, consistently distinguishing between phases in NOC and short-circuited, regardless of increased PV insertion.

Furthermore, artificial neural networks (ANNs) based on adaptive resonance theory (ART), such as fuzzy ART (FA) and Euclidean ART (EA), also obtained promising results. However, in the last scenario, they are less effective than RF, KNN, and GB. Across all scenarios, these ANNs stood out for their consistency, demonstrating significant potential for short-circuit detection. Future work could explore the extension of these ANNs for online training, enabling real-time short-circuit detection. This capability would allow continuous learning of new system characteristics, making it possible to identify faults regardless of PV penetration levels without requiring retraining for each newly detected characteristic.

Additionally, future studies could assess the impact of DG unit allocation using sensitivity analysis, considering additional DG types such as synchronous and wind generations together with PV systems. Comprehensive analyzes could evaluate individual impacts of each DG type and combined effects when all are integrated simultaneously into PDS.

CRediT authorship contribution statement

Andréia Silva Santos: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Reginaldo José da Silva:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Paula Andrea Montenegro:** Writing – original

draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Lucas Teles Faria:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Formal analysis. **Mara Lúcia Martins Lopes:** Writing – review & editing, Supervision, Formal analysis. **Carlos Roberto Minussi:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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Article

Machine Learning for Affordable Diagnosis of SARS-COV-2 from Hematological and Biochemical Tests

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Abstract: COVID-19, caused by the SARS-CoV-2 virus, has spread around the world and killed around 6.9 million people. Rapid and accurate diagnosis is essential for preventing and controlling the disease, reducing transmission and consequently saving lives. RT-PCR is the gold standard test used to detect the disease. However, the test is expensive and the result is time-consuming, which makes mass testing difficult, especially in countries with limited resources. In addition, the test has high analytical specificity and low diagnostic sensitivity, which leads to false-negative results. Several studies in the literature report the presence of hematological and biochemical alterations in infected patients and use these alterations with machine learning algorithms to help diagnose the disease. Therefore, this article presents the results obtained by different neural network architectures based on Adaptive Resonance Theory (ART) for the diagnosis of COVID-19. The study was conducted in two distinct stages: the first consisted of selecting the best ART network among several, using three open-access datasets and comparing the results with the literature. In the second stage, the chosen model was tested on a dataset containing patients from various hospitals in four countries. In addition, the model was subjected to external validation, including data from a country not present during the training and adjustment of the model, in order to validate the robustness and generalization capacity of the model. The results obtained by the ART networks in this study are promising, outperforming not only classical models, but also the deep learning models often used in the literature. Validation on data from different countries strengthens the model's reliability and effectiveness.

Keywords: adaptive resonance theory; Artificial Neural Networks; blood tests; coronavirus; COVID-19

1. Introduction

On Dec. 31 2019, the severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) causing coronavirus disease (COVID-19) was discovered in Wuhan in China. The virus has spread rapidly around the world and as of Apr. 21 2020 there were more than 2.3 million confirmed cases and a total of 162.000 deaths [1]. Currently, the virus has infected more than 775 million people and caused the deaths of 7 million worldwide [2]. Despite the high number of recorded deaths, the World Health Organization estimates the total number of deaths directly or indirectly from COVID-19 to be 14.9 million people [3], three times higher than recorded.

The symptoms of COVID-19 can be divided into three categories being common symptoms such as fever, cough, tiredness and loss of taste or smell, less common symptoms such as sore throat, headache, myalgia and diarrhea, and the critical symptoms which are dyspnea, loss of speech, mobility or confusion and chest pain [4]. Despite having several symptoms, a large proportion of infected people do not experience them. According to Wang et al. [5] the proportion of asymptomatic people may represent about a third or a fifth of the total number of cases.

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Although infected individuals develop mild or moderate disease and recover without hospitalization, a proportion of those infected may have severe symptoms, requiring hospitalization or intensive care unit (ICU) treatment.

To ensure the prevention of COVID-19, it is crucial to implement effective measures of detection, isolation and immediate medical care for infected people, in order to contain the spread of the virus [6–9]. This approach has proved effective in combating the spread of COVID-19. It is worth noting that, in addition to overloading the health system, the spread of the virus can have considerable economic impacts [10]. The costs associated with treating infected patients put significant pressure on health systems, especially in regions with limited resources [10]. Infected patients may require a diverse range of medical resources, from hospitalization in wards and ITUs to specialized interventions [11,12].

Among the various tests available to detect the disease, the reverse transcription polymerase chain reaction (RT-PCR) test, which detects the genetic material of the virus, is considered the gold standard for detecting COVID-19 and is carried out by collecting nasopharyngeal and/or oropharyngeal secretions [13]. Although RT-PCR has high analytical specificity, its diagnostic sensitivity is limited, especially in the early stages of infection, when the viral concentration may not be high enough to produce accurate results, leading to possible false negative results [14–17]. In addition, SARS-CoV-2 is an RNA virus prone to mutations, and since the beginning of the spread of the virus we have seen the emergence of several variants with different replication capacities and clinical manifestations [18]. Furthermore, the test has some limitations for testing patients on a large scale, requiring certified laboratories, expensive equipment and trained personnel, as well as the delay in obtaining results [19,20]. These limitations become even more serious in countries with limited resources, such as underdeveloped countries [21].

In addition to the RT-PCR mentioned above, computed tomography (CT) and X-ray have also been proposed to aid in the detection of COVID-19, playing a powerful role in detecting and judging the progression of the disease [22–27]. However, these tests are not efficient for the proper management of patients and the control of the spread of the virus, as their sensitivity and specificity are higher only in advanced stages of COVID-19 or other diseases [28–30], as these tests detect characteristics associated with COVID-19, such as lesions and lung inflammation, when the disease is already at an advanced stage of lung involvement. This is because the virus needs to replicate and cause significant damage to the lungs before changes are visible on the images. In addition, these exams subject patients to significant doses of ionizing radiation, which can pose major health risks [31,32].

Currently, machine learning has been an important tool to help doctors diagnose various diseases [33,34], including COVID-19 [35–37]. Studies have made use of this technology to detect COVID-19, either through images such as x-rays or CT scans, or through laboratory tests [38–40], since many clinical studies have shown hematological changes in infected patients [41–44]. While others focus on analyzing how these parameters vary throughout the course of the disease and their relationship with more serious clinical outcomes [45–47]. Given the need for efficient and more accurate diagnoses in the early stages of COVID-19, the relevant attributes of laboratory tests can be used to train machine learning algorithms, making it possible to predict the risk and evolution of the disease, as well as the early classification of infected patients. It is worth noting that laboratory tests have the advantage of being relatively more accessible in terms of cost compared to RT-PCR and imaging tests such as x-rays and CT scans, as well as avoiding exposure to radiation. This approach, based on hematological evaluations, could enable its use on a large scale, with specialist algorithms, which is especially relevant for developing countries with limited resources to carry out RT-PCR tests on the entire population, thus avoiding adverse social and economic impacts.

Several studies in the literature have used machine learning to detect COVID-19 from laboratory tests. One of the first articles, by Wu et al. [35], used the Random Forest (RF) algorithm with data collected from hospitals in China, achieving an AUC-ROC of 0.9926 in predicting patients with COVID-19. However, the model underperformed when tested with external samples. The authors Batista et al. [48] used data collected in a hospital in São Paulo, Brazil, and trained five algorithms for the detection

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of COVID-19 patients, Support Vector Machine (SVM) and RF were the ones that showed the best results, with an AUC-ROC of 0.847.

Sun et al. [36] employed the SVM algorithm to predict severe or critical cases of COVID-19 with patient data from a clinical center in China, obtaining a sensitivity of 77.5%, specificity of 78.4% and AUC-ROC of 0.9757. Tordjman et al. [49] developed a strategy using binary logistic regression (LR) to estimate the probability of a patient being diagnosed with COVID-19 before RT-PCR, achieving an AUC-ROC of 0.918, with sensitivity and specificity of 79% and 90%, respectively. Yang et al. [50] created a model using routine laboratory tests of patients at a medical center in the United States, and among the models tested, the Gradient Boosting Decision Tree (GBDT) obtained the best performance with an AUC-ROC of 0.854 in the internal validation and 0.838 in the external validation.

Joshi et al. [51] developed a decision support tool using LR to predict COVID-19 negative patients. The model achieved AUC-ROC of 0.78 for Stanford Health Care data, with AUC-ROC ranging from 0.75 to 0.81 at other centers. Goodman-Meza et al. [52] combined the probabilities of several algorithms to detect COVID-19 patients, achieving an AUC-ROC of 0.91 after adding certain tests such as ferritin, CRP and LDH. AlJame et al. [53] proposed ERLX, a model that stacks algorithms such as extremely randomized trees (ET), RF, LR and Extreme Gradient Boosting (XGB) to diagnose COVID-19 patients from blood tests, obtaining an AUC-ROC of 0.9938. Brinati et al. [21] proposed a model capable of predicting which patients were or were not infected with COVID-19, using data from patients admitted to the San Raffaele Hospital in Italy. Among the various algorithms tested, RF obtained the best performance, achieving an AUC-ROC of 0.84, sensitivity of 92% and specificity of 65%.

Alakus and Turkoglu [54] used a dataset containing hematological and biochemical attributes of patients from the Hospital Israelita Albert Einstein in São Paulo, Brazil, to train six deep learning architectures, with the aim of detecting patients with COVID-19. The best performance was obtained with Long Short-Term Memory (LSTM), achieving AUC-ROC of 0.625 in cross-validation and AUC-ROC of 0.90 in hold-out validation. Kukar et al. [55] created Smart Blood Analytics (SBA), a model for diagnosing COVID-19 from blood tests using algorithms such as RF, Deep Neural Network (DNN) and XGB. The best result with an AUC-ROC of 0.97 was obtained by XGB.

Using patient data from the Kepler University Hospital in Linz, Austria, Tschoellitsch et al. [56] applied the RF algorithm to predict the RT-PCR result using blood tests, achieving an AUC-ROC of 0.74. Soltan et al. [57] trained three models to identify COVID-19 patients in emergency departments and verify the need for hospitalization from patient data from Oxford University Hospitals in the UK. The best performance was obtained with XGB, achieving AUC-ROC of 0.939 in detecting patients and 0.94 in predicting hospitalization.

Cabitza et al. [58] evaluated several algorithms, including RF, Naive Bayes (NB), LR, SVM and KNN, to predict COVID-19 patients using data from patients admitted to the San Raphael Hospital in Italy. The models achieved AUC-ROC between 0.74 and 0.90, and during internal-external validation, they obtained AUC-ROC between 0.75 and 0.78. Alves et al. [59] developed a decision tree-based model to aid in the diagnosis of COVID-19 and provide explanations to healthcare teams, achieving AUC-ROC of 0.87, sensitivity of 67% and specificity of 91%. Rahman et al. [60] created the QCoVSML model, which uses stacked machine learning and a nomogram scoring system to detect COVID-19 based on biomarkers. Seven different datasets were used, and the combined model achieved an AUC-ROC of 0.96 and accuracy of 91.45%. There are many other studies that make use of laboratory tests and demonstrate the ability of machine learning algorithms to assist in medical decision-making, speeding up the diagnosis of the disease.

Although the algorithms used have achieved promising results, it is essential to recognize that they have some limitations, especially those based on backpropagation. According to Grossberg [61], these limitations include offline training, catastrophic forgetting, slow learning and sensitivity to outliers. To overcome these problems, Grossberg [62] developed Adaptive Resonance Theory (ART), which is described in detail in section 2.

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Therefore, this article presents the use of neural networks based on Adaptive Resonance Theory (ART) for the classification of COVID-19, using laboratory tests (blood tests) in order to improve the detection of infected patients. Comparisons were made between seven different ART architectures, including online training algorithms. The most successful architecture was subjected to a dataset containing laboratory tests from patients from different countries. This aspect is of paramount importance, as it allows us to evaluate the algorithm's ability to detect relevant patterns and characteristics in data from different sources, thus increasing its usefulness in a global detection context.

2. Adaptive Resonance Theory

Developed Adaptive resonance theory in order to solve the stability-plasticity dilemma [63], such a dilemma can be understood as the ability of the network to remain able to adapt or group input patterns, and continue learning even with the inclusion of new patterns, without catastrophically forgetting its past knowledge Carpenter and Grossberg [64], this being a fundamental characteristic of ART family networks.

The first architecture based on this theory was developed by Grossberg [62] and named Grossberg Network (GN), in 1987, the ART1 network was created with the purpose of binary grouping patterns and was based on the GN network of Grossberg [64]. In the same year, Carpenter and Grossberg [65] improved ART1 creating ART2. This new network had the purpose of processing patterns with binary or continuous values. Considering the need to solve problems with analog input patterns, in 1991 Carpenter et al. [66], based on ART1 [64], developed the ART Fuzzy neural network (Figure 1), which employs the fuzzy min operator ($\wedge$) in place of the intersection operator ($\cap$) present in ART1 network. The first ART network with supervised training was developed by Carpenter et al. [67] and has two interconnected ART modules, the ART_a module receives and processes the input pattern. In contrast, the ART_b module receives the desired response for the presented pattern, and the Inter-ART module is in charge of performing the "matching" between the two ART modules. The following year Carpenter et al. [68] presented the Fuzzy ARTMAP that also has supervised training. However, its calculations are based on Fuzzy Logic [69].

Different ART network architectures have been tested in order to find the one that best classifies the data. These architectures have different characteristics, whether in their topology (cluster geometry), training method (supervised or unsupervised) or online learning capacity (Continuous Training), these characteristics appear in isolation or combined in the models to optimize data classification. The networks used in this study were:

- Self-expanding Fuzzy ART Neural Network [66]
- Self-Expanding Euclidean ART neural network [70]
- Self-Expanding Fuzzy ARTMAP neural network [68]
- Self-expanding Fuzzy ART neural network with Continuous Training [71]
- Self-expanding Euclidean ART neural network with Continuous Training [71,72]
- Self-expanding Fuzzy ARTMAP neural network with Continuous Training [71]
- Self-expanding Euclidean ARTMAP neural network with Continuous Training [73]

Although the self-expanding mechanism is not present in the original structure of the first three models, it was used because it adds advantages, as will be discussed in subsection 2.5. The other characteristics will be detailed in subsection 2.1, 2.2, 2.3, 2.4 and 2.6.

In these architectures, some parameters play a fundamental role in controlling the functioning of the model, namely α, β and ρ. The α parameter, also known as the learning rate ($\alpha > 0$), is crucial, indicating the extent to which new patterns will influence the category when the weights are updated in the next training cycle. The β parameter, or training rate ($0 < \beta \leq 1$), regulates the speed of training, with lower values indicating a slower process and values close to 1 implying faster training. Finally, the ρ parameter, or vigilance parameter ($0 < \rho \leq 1$), plays a crucial role in determining the degree of similarity required to create a new category. These parameters are essential for adjusting the network's dynamics and adapting it effectively to the input data.

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2.1. ART - Unsupervised

Unsupervised ART models are effective in environments where the training data does not need to be labeled. The model's independence from labels allows it to autonomously explore the complexity of the data and identify relevant patterns. The basic architecture of an ART network can be seen in Figure 1, and consists of three levels, the F_1 comparison level, the F_2 recognition level and the F_0 level responsible for pre-processing the data, in which the input data is normalized and encoded (Fuzzy Topology). In addition, there is a reset mechanism responsible for controlling the degree of similarity of the patterns associated with the same category.

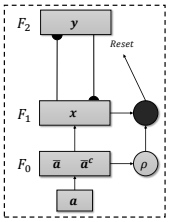

Figure 1. Fuzzy ART Architecture.

It is worth noting that, although they adopt an unsupervised approach, labels are assigned to each cluster after the model has been trained. This process is explained in detail in subsection 3.4.

Algorithm 1 Algorithm ART.

Input: Input data, parameters [β, ρ, α (Fuzzy)]

Output: Cluster

```
procedure
    if Topology for Fuzzy then
        Normalize and encode the input data by Equation 1 and 2
    end if
    while Data for training exists do
        Calculate the distance or similarity between the input data and all existing clusters
        Equation 3 or Equation 7
        while not finding a cluster to allocate the data do
            Select the cluster according to the equation Equation 4 or Equation 8
            if the Equation 5 or Equation 9 is satisfied then
                Update the weights according to Equation 6 or Equation 10
            else
                Remove the category from the search process
            end if
            The data will be allocated to the current cluster
        end while
    end while
end procedure
```

2.2. ARTMAP - Supervised

The ARTMAP model with supervised learning is made up of two ART modules, ART_a and ART_b, as shown in Figure 2. Unlike their unsupervised counterparts, these modules do not operate in the dark. They use both the input data and the corresponding labels, providing guidance during training. Furthermore, ARTMAP doesn't just passively accept information. It actively searches for meaningful connections. The inter-ART associative memory module is responsible for checking whether there is a match between the input patterns and the output. The vigilant match-tracking mechanism minimizes prediction errors and maximizes the model's ability to generalize. The weight matrices associated with the ART_a ($\boldsymbol{W}^a$), ART_b ($\boldsymbol{W}^b$) and Inter-ART ($\boldsymbol{W}^{ab}$) modules are initialized with values equal to 1, indicating the absence of active categories. However, as training takes place, there is resonance

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between the input patterns and the output, causing specific categories to be activated. This dynamic activation process allows ARTMAP to learn and adapt in a controlled manner.

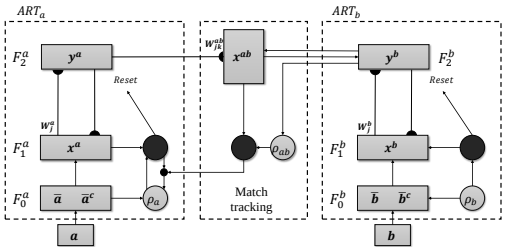

Figure 2. Architecture ARTMAP.

Although it is a powerful architecture, it is a model sensitive to noise in the labels due to supervised learning, accuracy can be compromised if the labels are inaccurate or noisy.

Algorithm 2 ARTMAP algorithm.

Input: Input data, parameters [β, ρ, α (Fuzzy)]

Output: Cluster

procedure
  if Topology for Fuzzy **then**
    Normalize and encode the input data by Equation 1 and 2
  end if
  while Data for training exists **do**
    Module B
    Calculate the distance or similarity between the input data and all existing clusters Equation 3 or Equation 7
    while not finding a cluster to allocate the data **do**
      Select cluster according to Equation 4 or Equation 8
      if the Equation 5 or Equation 9 is met **then**
        Update the weights of module B according to Equation 6 or Equation 10
      else
        Remove the category from the search process
      end if
    end while
    Module A
    Calculate the distance or similarity between the input data and all existing clusters Equation 3 or Equation 7
    while not finding a cluster to allocate the data **do**
      Select cluster according to Equation 4 or Equation 8
      if the Equation 5 or Equation 9 is met **then**
        Inter-ART module
        if The category found by module A matches the category of module B **then**
          Update the weights of module A (Equation 6 or Equation 10) and Inter-ART
        else
          Remove the category from the search process and increase the vigilance parameter of module A
        end if
      else
        Remove the category from the search process
      end if
    end while
    The data will be allocated to the current cluster
  end while
end procedure

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2.3. Fuzzy Topology in ART Networks

In the fuzzy topology, ART networks use the fuzzy logic connectives developed by [69]. These connectives are used both to assess the similarity between input patterns and in the vigilance test. Furthermore, in this topology the data needs to be normalized and coded by complement as specified by Equation 1 and 2, in order to avoid the proliferation of categories [68].

$$\bar{\boldsymbol{a}} = \frac{\boldsymbol{a}}{\left|\sum_{i}^{M} a_i\right|}, \tag{1}$$

$$\boldsymbol{I} = [\ \bar{\boldsymbol{a}} \quad \bar{\boldsymbol{a}}^c\ ], \tag{2}$$

The Equation 3 in turn is responsible for checking the similarity between the input pattern and the existing categories, the winning category will be the one with the greatest similarity to the input data as seen in the Equation 4. The j index refers to the cluster in which the calculations are being performed, while the J index identifies the winning category in this process.

$$T_j = \frac{|\boldsymbol{I} \wedge \boldsymbol{W}_j|}{\alpha + |\boldsymbol{W}_j|}, \tag{3}$$

$$J = arg\{\max\{T_j : j = 1, ..., \text{number of clusters}\}, \tag{4}$$

Resonance occurs if the vigilance criterion given by Equation 5 is met. This process acts as a filter preventing the unbridled creation of categories. It ensures that only significantly different patterns trigger the formation of new categories, preventing the network from becoming full of redundant categories. The ρ parameter sets the threshold for this filter, determining the level of similarity required between an input pattern and the existing categories for the network to accept it or not within the cluster.

$$\frac{|\boldsymbol{I} \wedge \boldsymbol{W}_J|}{|\boldsymbol{I}|} \geq \rho, \tag{5}$$

When resonance is achieved, the network starts a weight adjustment process, as detailed in Equation 6. In this way, the weights associated with the winning category are modified to accommodate the new pattern.

$$\boldsymbol{W}_J^{new} = \beta\left(\boldsymbol{I} \wedge \boldsymbol{W}_J^{old}\right) + \left(1 - \beta\right)\boldsymbol{W}_J^{old}, \tag{6}$$

By using fuzzy logic in the network calculations, together with normalization and complement coding, the clusters take the form of hyperrectangles, giving flexibility to the complexity of the input patterns.

2.4. Euclidean Topology in ART Networks

By adopting Euclidean topology, ART networks use Euclidean distance as a metric to measure the proximity between input patterns and existing categories. As a result, the network generates hyperspheres centered on the cluster representing the class, instead of hyperectangles. This approach eliminates the need to normalize and encode the input vector, thus reducing computational costs [72]. The Equation 7 indicates how the distance between the input pattern and the existing clusters is calculated. The category chosen is the one with the smallest distance according to Equation 8, in contrast to the fuzzy approach, where the choice is based on maximum activation.

$$T_j = \sqrt{\sum_{i=1}^{M} (a_i - W_{ji})^2}, \tag{7}$$

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$$J = arg\{\min\{T_j : j = 1, ..., numberofclusters\}\}, \tag{8}$$

The vigilance criterion will be given by Equation 9 and follows a similar pattern to that used in fuzzy topology, as detailed above in subsection 2.3.

$$\frac{T_J}{\max\left\{\sum_{i=1}^{M}(a_i)^2, \sum_{i=1}^{M}(W_{Ji})^2\right\}} \leq \rho, \tag{9}$$

Another change is in the Equation 10 responsible for the network's learning process. Here, the weight matrix ($\boldsymbol{W}_J$) is updated according to a weighted combination of the input pattern ($\boldsymbol{a}$) and the previous weight matrix ($\boldsymbol{W}_J^{old}$), controlled by the β parameter.

$$\boldsymbol{W}_J^{new} = \beta(\boldsymbol{a}) + (1 - \beta)\boldsymbol{W}_J^{old}, \tag{10}$$

2.5. Self-Expanding Mechanism

The self-expanding mechanism proposed by Moreno [71,72] starts training the network by adopting the first pattern as the center of the first cluster and expands according to the need to create new clusters, i.e. as the new input patterns do not fit into the pre-existing categories, a new cluster is created to allocate that pattern. This modification reduces the processing time and, consequently, the computational expense [71,72], as it only checks the similarity of active clusters, unlike traditional architectures, in which similarity is checked with all clusters, including inactive ones. Furthermore, when inputting data that is incompatible with any of the existing clusters, the network starts a new cluster with the new pattern that is not in resonance with the input data and the empty cluster. These changes do not alter the stability of the network or compromise its ability to recognize new information in the training data.

Algorithm 3 Self-expansion mechanism algorithm.

Require: Initialization of the weight matrix with the first pattern
1: **procedure**
2: **if** The surveillance criterion is not met (Equation 5 or Equation 9) **then**
3: **if** There are categories to be checked **then**
4: Remove the current category from the search process and search again
5: **else**
6: Create a new cluster to allocate the current data
7: The weight for this cluster will be the input pattern itself
8: **end if**
9: **end if**
10: **end procedure**

2.6. Continuous Training in ART Neural Networks

The continuous training strategy enables networks to learn online and continuously so that prediction and training are carried out constantly, making them more efficient. This approach eliminates the need to restart training when incorporating new data, resulting in a network that constantly adjusts to changes in the environment. The inclusion of this module in ART networks is only possible due to their characteristics of stability and plasticity [72]. With online learning, there is no need to restart training when new data becomes available; this module allows new patterns to be added to the network's memory permanently [72].

In order to incorporate this module, the network needs to have temporary and definitive categories, a necessary process so that data that appears rarely is not incorporated into the network's memory. In addition, two new parameters are added: the first, called the permanence parameter ($NMIN$), defines the number of patterns needed for a temporary category to become definitive. The

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second, called the novelty index (η), checks for the need to update the weights of the definitive category, discarding information that is not very relevant to the network, avoiding overloading the network's memory [72]. Whenever a pattern belongs to a temporary cluster, resonance will occur; on the other hand, if the category to which the pattern belongs is definitive, resonance will only occur if the distance/similarity between the entry and the center of the category is greater than the parameter (η) [72].

Thus, Equation 3 and Equation 7 are calculated for their respective topologies, for both temporary and permanent categories. The selection process remains the same. If the chosen category is definitive and passes the surveillance criterion, resonance will only occur if the condition imposed by Equation 11 (Fuzzy) or Equation 12 (Euclidean) is met, i.e. resonance will only occur if the pattern is significantly different.

$$T_J > \eta, \tag{11}$$

$$T_J < \eta, \tag{12}$$

3. Application of ART Networks in the diagnosis of COVID-19

The simulations were carried out using ten open-access datasets. These datasets contain information on clinical biomarkers collected from patients in hospitals located in Brazil, Spain, Italy and Ethiopia. The number of instances and features contained in each of the datasets can be seen in Table 1.

Table 1. Amount of data and features.

Available in:	Name	Country	Features	Positive	Negative
[21]	$Data_1$	Italy	14	177	102
[58]	$Data_2$	Italy	35	786	838
[54]	$Data_3$	Brazil	19	80	520
[74]	$Data_4$	Italy	22	163	174
[74]	$Data_5$	Italy	22	104	145
[74]	$Data_6$	Italy	22	118	106
[74]	$Data_7$	Brazil	23	334	11
[74]	$Data_8$	Brazil	23	352	949
[74]	$Data_9$	Brazil	23	375	1960
[74]	$Data_{10}$	Ethiopia	24	200	0
[74]	$Data_{11}$	Spain	24	78	42

The methodology used in this article consisted of two stages. In the first stage, the main objective was to select the best ART model from the seven architectures tested. In the second stage, the model selected in the previous stage was used to verify its generalization capacity in relation to data from different countries. Specific details of each stage are described in subsection 3.1 and subsection 3.2, respectively.

3.1. First Stage

In this stage, three different data sets were selected: $Data_1$, $Data_2$ and $Data_3$. This choice was made considering the diversity of attributes and the fact that they have been subjected to simulations with various algorithms, including deep learning models. This is to make it easier to compare the performance of the ART models with those in the literature. The Table 2 presents in detail the attributes present in each dataset used in this stage, providing an overview of the attributes used during the experiments conducted.

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Table 2. Attributes present in the data.

Data	Attributes
$Data_1$	Alanine Aminotransferase, Aspartate Aminotransferase, Basophils, Class, Eosinophils, Gamma GT, Gender, Age, Lactate Dehydrogenase, Leukocytes, Lymphocytes, Monocytes, Neutrophils, C-Reactive Protein, Platelets
$Data_2$	Alanine Aminotransferase, Aspartate Aminotransferase, Basophils ($10^9\ell^{-1}$), Basophils (%), Calcium, Average Corpuscular Hemoglobin Concentration, Creatine Kinase, Class, Creatinine, Eosinophils ($10^9\ell^{-1}$), Eosinophils (%), Alkaline Phosphatase, Gamma GT, Gender, Glucose, Average Corpuscular Hemoglobin, RBC, Hematocrit, Hemoglobin, Age, Lactate Dehydrogenase, Leukocytes, Lymphocytes ($10^9\ell^{-1}$), Lymphocytes (%), Monocytes ($10^9\ell^{-1}$), Monocytes (%), Neutrophils, Neutrophils (%), C-reactive protein, Platelets, Potassium, Suspected COVID-19, Urea, Mean Corpuscular Volume, Mean Platelet Volume
$Data_3$	Alanine Aminotransferase, Aspartate Aminotransferase, Basophils, Class, Creatinine, Eosinophils, Serum glucose, RBC, Hematocrit, Hemoglobin, Age, Leukocytes, Lymphocytes, Monocytes, Neutrophils, C-reactive protein, Platelets, Potassium, Sodium, urea

The simulations were conducted using the same attributes as those previously used by other researchers. More specifically, the $Data_1$ set underwent the same pre-processing carried out by Brinati et al. [21]. This set contains data from patients who were admitted to the IRCCS San Raffaele Hospital, located in Segrate, in the province of Milan, Italy, between February and March 2020.

The $Data_2$ set was used in the same way as in Cabitza et al.[58]. This set includes the results of routine blood tests carried out on 1925 patients admitted to the emergency room of San Raffaele Hospital between mid-February and May 2020.

For its part, the $Data_3$ set was made available pre-processed by Alakus and Turkoglu [54]. This set was obtained from data made available by the Hospital Israelita Albert Einstein[1]. The data was collected from 5644 patients and contains information from 111 laboratory tests carried out on patients treated at the hospital located in São Paulo, Brazil. It is worth noting that all datasets were used without any modification.

3.2. Second Stage

In this stage, $Data_2$, $Data_4$, $Data_5$, $Data_6$, $Data_7$, $Data_8$, $Data_9$ and $Data_{10}$ were merged to create a new dataset containing hematological information from different countries. This combination was carried out in order to provide more comprehensive information to verify the model's ability to predict COVID-19, since hematological parameters are influenced by individual factors such as age, gender and lifestyle, as well as population and ecological factors such as ethnic origin, climate, exposure to pathogens and altitude [75–77].

The Table 3 shows the averages and standard deviations of the common attributes between all the data sets. The Variance Inflation Factor (VIF) and the Mann-Whitney (MW) test [78] were used to carefully select the attributes for the simulations.

[1] The dataset is available at https://www.kaggle.com/einsteindata4u/covid19

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Table 3. Average and standard deviation of attributes.

Attributes	$Data_2$	$Data_4$	$Data_5$	$Data_6$	$Data_7$	$Data_8$	$Data_9$	$Data_{10}$
Age	60,55 ± 19,6	54,38 ± 25,0	66,35 ± 18,5	60,53 ± 20,2	54,40 ± 15,5	47,01 ± 17,4	42,87 ± 17,0	59,73 ± 13,1
BA (%)	0,34 ± 0,3	0,46 ± 0,4	0,18 ± 0,4	0,32 ± 0,3	0,30 ± 0,3	0,52 ± 0,3	0,48 ± 0,3	0,29 ± 0,4
BAT	0,02 ± 0,04	0,03 ± 0,04	0,02 ± 0,05	0,02 ± 0,04	0,02 ± 0,03	0,03 ± 0,02	0,03 ± 0,02	0,04 ± 0,1
EO (%)	0,83 ± 1,6	1,23 ± 2,1	0,74 ± 1,6	0,60 ± 1,2	1,05 ± 1,6	2,30 ± 2,4	2,27 ± 2,5	0,62 ± 1,1
EOT	0,06 ± 0,1	0,09 ± 0,2	0,06 ± 0,1	0,05 ± 0,1	0,06 ± 0,1	0,15 ± 0,2	0,17 ± 0,2	0,08 ± 0,3
HCT	39,41 ± 5,6	37,77 ± 7,4	38,20 ± 6,3	39,67 ± 6,1	40,75 ± 5,0	41,47 ± 4,0	41,18 ± 4,1	39,96 ± 7,9
HGB	13,21 ± 2,0	12,86 ± 2,7	13,21 ± 2,3	13,11 ± 2,2	13,74 ± 1,9	13,82 ± 1,5	14,11 ± 1,5	13,36 ± 3,2
LY (%)	18,54 ± 11,1	21,90 ± 14,5	16,56 ± 11,6	18,30 ± 11,9	23,14 ± 11,4	31,10 ± 11,2	27,57 ± 11,0	7,58 ± 7,1
LYT	1,37 ± 1,0	1,84 ± 4,9	1,63 ± 6,3	1,82 ± 6,1	1,31 ± 1,0	2,01 ± 1,9	2,00 ± 0,9	0,77 ± 0,6
MCH	29,20 ± 2,7	30,41 ± 2,9	29,62 ± 3,1	29,51 ± 2,4	29,60 ± 1,9	29,37 ± 1,9	29,59 ± 1,8	29,58 ± 2,8
MCHC	33,47 ± 1,4	33,98 ± 1,4	34,49 ± 1,5	33,00 ± 1,3	33,68 ± 1,1	33,31 ± 1,1	34,25 ± 1,0	33,09 ± 2,0
MCV	87,22 ± 7,1	89,44 ± 7,4	85,72 ± 8,0	89,41 ± 6,5	87,90 ± 5,1	88,19 ± 5,1	86,39 ± 4,7	88,69 ± 8,3
MO (%)	7,76 ± 3,9	8,86 ± 4,7	7,17 ± 3,9	8,13 ± 5,0	9,64 ± 4,4	9,29 ± 3,7	8,64 ± 3,2	5,74 ± 8,2
MOT	0,62 ± 0,5	0,73 ± 0,9	0,64 ± 0,4	0,64 ± 0,3	0,56 ± 0,3	0,59 ± 0,3	0,63 ± 0,3	0,63 ± 0,8
NE (%)	72,50 ± 13,3	67,54 ± 17,2	75,03 ± 14,2	72,48 ± 14,5	65,78 ± 14,1	56,79 ± 12,4	61,02 ± 12,4	85,77 ± 12,7
NET	6,47 ± 4,5	5,62 ± 4,3	7,47 ± 4,9	6,76 ± 4,3	4,35 ± 3,1	3,92 ± 2,1	4,82 ± 2,4	11,60 ± 8,5
PLT	234,49 ± 94,9	204,00 ± 113,7	220,23 ± 90,0	218,00 ± 84,8	199,81 ± 73,4	246,96 ± 70,3	239,92 ± 64,6	268,69 ± 146,8
RBC	4,55 ± 0,7	4,25 ± 0,9	4,49 ± 0,8	4,46 ± 0,8	4,65 ± 0,7	4,72 ± 0,5	4,78 ± 0,5	4,46 ± 0,9
WBC	8,69 ± 4,7	8,31 ± 7,1	9,81 ± 8,0	9,53 ± 7,5	6,30 ± 3,5	6,70 ± 3,0	7,66 ± 2,7	14,18 ± 17,8

BA: Basophils; BAT: Basophil Count; EO: Eosinophils; EOT: Eosinophil Count; HCT: Hematocrit; HGB: Hemoglobin; LY: Lymphocytes; LYT: Lymphocyte Count; MCH: *Mean Corpuscolar Hemoglobin*; MCHC: *Mean Corpuscular Hemoglobin Concentration*; MCV: *Mean Corpuscular Volume*; MO: Monocytes; MOT: Monocyte Count; NE: Neutrophils; NET: Neutrophil Count; PLT: Platelets; RBC: Red Blood Cells

The MW test is a non-parametric statistical technique used to compare the medians of two independent samples [78,79]. The test was used to check for significant differences between the classes in relation to each attribute analyzed. The determination of statistical significance was established at a confidence level of 0.05, which indicates that values of p below this threshold are considered statistically significant, indicating relevant differences between the medians of the compared groups.

On the other hand, VIF is a statistical measure used to assess multicollinearity between variables. It is obtained from the elements on the main diagonal of the inverse correlation matrix, as expressed in Equation 14 [80].

$$\boldsymbol{R}_{j\times j} = \left(\frac{1}{i-1}\cdot \boldsymbol{W}'_{j\times i}\cdot \boldsymbol{W}_{i\times j}\right), \tag{13}$$

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$$V_{j\times j} = \left(R_{j\times j}\right)^{-1}, \tag{14}$$

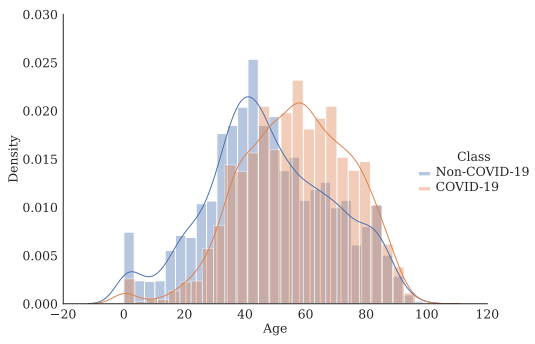
Where R represents the correlation matrix, and W is the matrix of centered and reduced data. It is important to note that the R matrix will be invertible if, and only if, the rank(W) is complete, i.e. the columns/rows of W are linearly independent [81]. The literature points out that VIF values above 5.0 or 10.0 indicate the presence of multicollinearity [82]. On this basis, it was decided not to include attributes with a VIF above 8.0 in the simulations.

The results obtained after the tests are shown in Table 4. The attributes highlighted were selected for the simulations because they were relevant in both tests. It is worth noting that attributes expressed as percentages were not included in the selection, even though they were indicated as relevant by the tests. The decision to exclude these attributes was based on the consideration that absolute values have clinical significance and provide more complete information than relative values (%) [83].

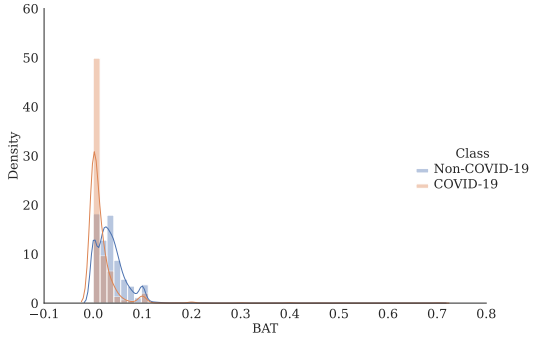
Table 4. Selection of attributes by different methods.

Attributes	*p*-value MW	VIF	Attributes	*p*-value MW	VIF
Age	0.0000	1.4213	MCHC	0.0000	10.5313
BA (%)	0.0000	2.8222	MCV	0.3976	81.7446
BAT	0.0000	2.6595	MO (%)	0.7285	12.1883
EO (%)	0.0000	10.5956	MOT	0.0000	2.8672
EOT	0.0000	7.2687	NE (%)	0.0000	140.5236
HCT	0.0003	400.5673	NET	0.0000	5.4626
HGB	0.0137	351.4449	PLT	0.0627	1.2310
LY (%)	0.0000	100.8982	RBC	0.0007	59.8770
LYT	0.0000	2.1397	WBC	0.0000	3.8370
MCH	0.0136	105.2796			

The selected attributes are part of the white blood series and play a fundamental role in the immune system, helping to fight infections, viruses and bacteria. Analyzing the attributes, it was possible to see some relevant information in relation to patients affected by COVID-19, such as the reduction in cell counts, such as basophils (Figure 3b), eosinophils (Figure 3c) and lymphocytes (Figure 3d). These factors are supported in the literature by authors such as Kermali et al. [84], Samprathi and Jayashree [85] and Keykavousi et al. [86].



(a) Age



(b) Basophiles

Figure 3. *Cont.*

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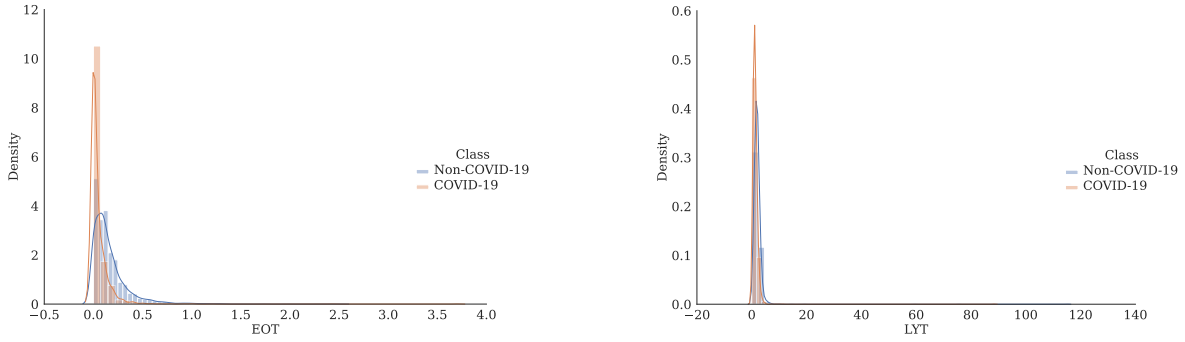

(c) Eosinophils **(d)** Lymphocytes

Figure 3. Histogram and estimate of the probability density function for each of the attributes separated by class.

After using the combined data set to train and test the model, it was subjected to an external validation process using the Data_1 and Data_{11} sets to assess its ability to generalize on data never before seen by the model. It is worth noting that the same attributes used in training and testing were also used in the external validation phase. Data1 was chosen because it comes from Italy, a country already present in the training data, but with data collected in a different period, while Data11 was selected because it comes from Spain, a different country to those included in the training.

3.3. Evaluation Metrics

To evaluate the performance obtained by each network some metrics frequently present in the literature were used such as accuracy (ACC), sensitivity (Se), specificity (Sp), F_1-Score (F_1), Matthews correlation coefficient (MCC) given respectively by (15), (16), (17), (18), and (19) [87–89].

$$\text{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \tag{15}$$

$$\text{Se} = \frac{TP}{TP + FN} \tag{16}$$

$$\text{Sp} = \frac{TN}{TN + FP} \tag{17}$$

$$\text{F}_1 = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \tag{18}$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP) \cdot (TP + FN) \cdot (TN + FP) \cdot (TN + FN)}} \tag{19}$$

Along with the metrics mentioned above, we use the ROC (Receiver Operating Characteristics) curve to evaluate the quality of the model classification. This curve is generated by plotting the Sensitivity and False Positive Rate ($1-$ Specificity) for different thresholds of the probability estimated by the model. Based on [90] work, the AUC (Area Under the Curve) can be understood as a summary of the ROC curve, the AUC value and the quality of the model can be seen in Table 5.

Table 5. Interpretation of AUC Values.

Range	Result	Range	Result
0.5 – 0.6	Bad	0.7 – 0.8	Acceptable
0.8 – 0.9	Good	> 0.9	Excellent

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3.4. Classifier

In the unsupervised learning models, it was necessary to use a classifier to convert the clusters to the respective classes of the problem, "COVID-19" and "non-COVID-19". The probability required to generate the ROC curves was also extracted in this step, in which the number of patterns contained in each cluster and the class to which it belonged were analyzed. For example, suppose 10 patterns are allocated to a certain cluster during training, of which 9 are from patients with COVID-19. In this case, this cluster will be labeled as COVID-19 by majority vote, the probability of being COVID-19 will be 0.9, and the complementary probability, that is, of the patient being non-COVID-19 will be 0.1, any pattern allocated to this cluster in the test stage will receive this probability. It is worth noting that for the ARTMAP models that have supervised learning it was not possible to extract the probability and consequently generate the ROC curve, due to the model having discrete output.

3.5. Validation

To partition the data, the hold-out cross-validation method was used, randomly dividing the data into training and test sets based on a pre-determined rate. In addition, the best model found during hold-out validation on each data set was subjected to the repeated hold-out method, also known as Monte Carlo cross-validation [91,92], was employed as a way to obtain a more consistent estimate of model performance. The data was partitioned using the 70–30 ratio, that is, 70% of the samples for training and 30% for testing. For Monte Carlo cross-validation, the process was repeated 50 times, with the data split randomly in each repetition.

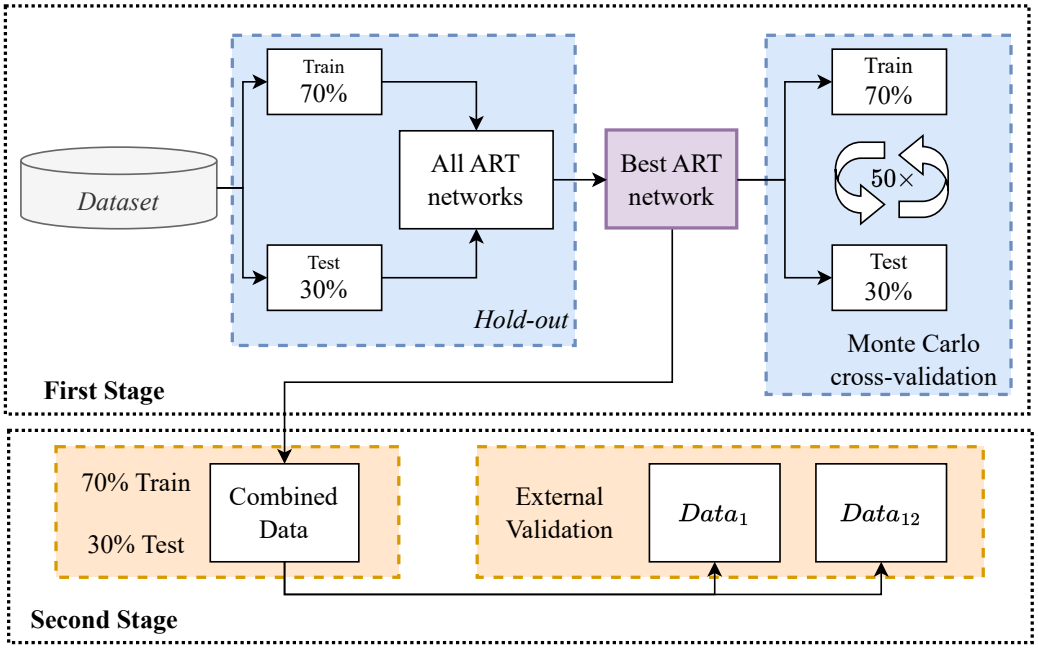

Figure 4. Validation Overview.

4. Computational Results

This section will present the results obtained for COVID-19 diagnostics using the ART networks on each dataset.

4.1. Primeita Etapa

4.1.1. $Data_1$

The performance of various ART networks was evaluated, and the results are presented in Table 6. The SE Fuzzy ART CT network showed the highest accuracy, reaching 0.871, and also demonstrated the greatest ability in correctly detecting patients with COVID-19, as shown by the sensitivity of 0.926.

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Notably, fuzzy logic-based networks were found to perform better than Euclidean distance-based networks for this particular dataset.

Table 6. Results obtained by each model for $Data_1$.

Model	ACC	Se	Sp	MCC
SE Fuzzy ART	0.800	0.870	0.670	0.561
SE Euclidean ART	0.706	0.759	0.613	0.370
SE Fuzzy ARTMAP	0.835	0.870	0.774	0.645
SE Fuzzy ART CT	**0.871**	**0.926**	**0.774**	**0.717**
SE Euclidean ART CT	0.741	0.778	0.677	0.450
SE Fuzzy ARTMAP CT	0.800	0.759	0.871	0.608
SE Euclidean ARTMAP CT	0.635	0.778	0.387	0.176

Considering that the network with the best performance in the simple hold-out validation was the SE Fuzzy ART CT, it was used in the Monte Carlo validation to evaluate the network's generalization ability. Table 7 shows the mean metrics as well as the 95% confidence interval (CI) for the Monte Carlo validation.

Table 7. Average result after Monte Carlo cross-validation for $Data_1$.

Metrics	Average result (CI 95%)
ACC	0.8061 (0.7932 – 0.8191)
Se	0.8211 (0.8049 – 0.8373)
Sp	0.7800 (0.7495 – 0.8105)
MCC	0.6555 (0.6264 – 0.6846)
F_1	0.8679 (0.8569 – 0.8788)

To illustrate the results obtained by the best network in each iteration of the Monte Carlo validation, Figure 5 shows the ACC, Se, and Sp for each iteration. It can be seen that the network's performance was not constant in all iterations, reaching a maximum accuracy of 0.9294 and a minimum of 0.7176. On the other hand, Figure 6 presents the average ROC curve and the average AUC obtained by the network in this validation. According to [90], an AUC score above 0.8 can be considered a good performance achieved by the network.

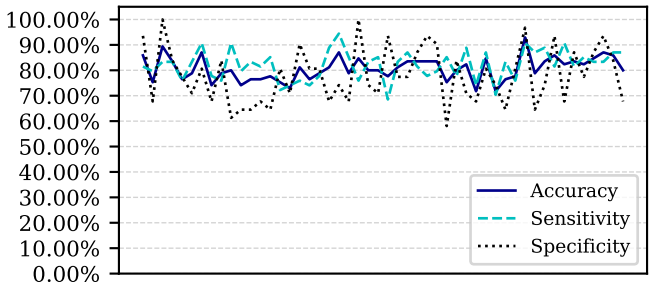

Figure 5. Performance for each of the 50 iterations of the Monte Carlo cross-validation for the $Data_1$.

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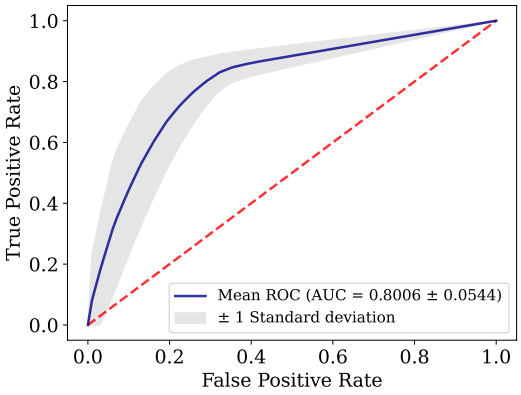

Figure 6. ROC curve for $Data_1$.

4.1.2. $Data_2$

Table 8 presents the results obtained by the networks for $Data_2$. The tested model's achieved approximate performance with MCC ≈ 0.5, and the SE Fuzzy ART CT network achieved the best result, almost perfectly detecting positive and negative patients for COVID-19. Although the SE Euclidean ART CT network achieved the best result in specificity, it was not the best model since it correctly detected only 54.7% of the COVID-19 patients.

Table 8. Results obtained by each model for $Data_2$.

Model	ACC	Se	Sp	MCC
SE Fuzzy ART	0.750	0.746	0.754	0.500
SE Euclidean ART	0.727	0.691	0.762	0.454
SE Fuzzy ARTMAP	0.783	0.792	0.774	0.566
SE Fuzzy ART CT	**0.984**	**0.970**	0.996	**0.967**
SE Euclidean ART CT	0.781	0.547	**1.000**	0.619
SE Fuzzy ARTMAP CT	0.738	0.712	0.762	0.474
SE Euclidean ARTMAP CT	0.697	0.712	0.683	0.394

The SE Fuzzy ART CT Network, which demonstrated the best performance, was subjected to Monte Carlo cross-validation to ensure the reliability of the results, reducing the risk of overfitting. The results are presented in Table 9, where the average results and 95% confidence intervals for the different metrics are presented. The network's performance was remarkable, with an average accuracy of 0.9348, indicating that it successfully classified the data. In addition, the network obtained an MCC of 0.8721, a vital performance measure that considers the balance between sensitivity and specificity.

Table 9. Average result after Monte Carlo cross-validation for $Data_2$.

Metrics	Average result (CI 95%)
ACC	0.9348 (0.9282 – 0.9413)
Se	0.8992 (0.8885 – 0.9098)
Sp	0.9682 (0.9594 – 0.9770)
MCC	0.8721 (0.8592 – 0.8850)
F_1	0.9301 (0.9231 – 0.9371)

Figure 7 illustrates the network's performance in ACC, Se, and Sp for each iteration of the Monte Carlo validation. The results showed that the network was highly stable performing well in almost all iterations. The network achieved a maximum accuracy of 0.9775 and a minimum of 0.875, and of the 50 iterations, only five were below 0.91 in accuracy. These results suggest that the network

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is highly robust and can be used effectively for the classification task. It should also be noted that the stability in the network's performance over the different iterations of the Monte Carlo validation strongly indicates its generalizability and ability to handle different types of input data.

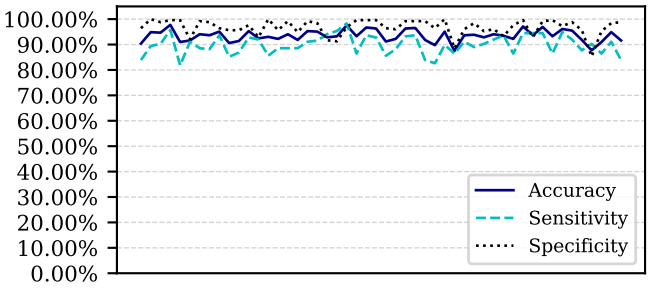

Figure 7. Performance for each of the 50 iterations of the Monte Carlo cross-validation for the $Data_2$.

In Figure 8, the ROC curve and the average AUC obtained after Monte Carlo validation are presented. The AUC value obtained was 0.9395, which suggests that the network has excellent discrimination ability, according to [90].

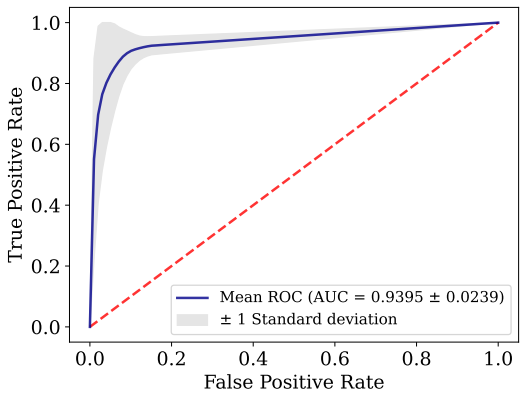

Figure 8. ROC Curve for $Data_2$.

4.1.3. $Data_3$

This dataset needs to more balanced, containing 13.33% data from patients with COVID-19 and 86.67% non-COVID-19 patients. As expected, this unbalance impaired the classification ability of the implemented models, resulting in a low predictive performance as can be seen in Table 10.

Table 10. Results obtained by each model for $Data_3$.

Model	ACC	Se	Sp	MCC
SE Fuzzy ART	0.889	0.208	0.994	0.382
SE Euclidean ART	0.878	0.500	0.936	0.452
SE Fuzzy ARTMAP	0.867	0.375	0.942	0.359
SE Fuzzy ART CT	0.944	0.583	1.000	0.740
SE Euclidean ART CT	**0.972**	**0.792**	**1.000**	**0.876**
SE Fuzzy ARTMAP CT	0.889	0.458	0.955	0.468
SE Euclidean ARTMAP CT	0.872	0.458	0.936	0.417

When working with unbalanced classes, evaluating the performance of a model can be challenging, as traditionally used metrics such as accuracy, sensitivity, and specificity can be misleading and may not provide an accurate representation of model performance [87,88]. Instead, it is recommended to

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use metrics suitable for unbalanced data, such as the Matthews Correlation Coefficient (MCC) and Area Under the ROC Curve (AUC) [88,89].

In this context, the best performing network for the dataset in question was the SE Euclidean ART CT, achieving an MCC of 0.876. This indicates that the model had a strong correlation between the predicted and actual classes, taking into account both true and false positives and negatives. Table 11 presents the average metrics as well as the 95% confidence interval obtained by the SE Euclidean ART CT network after Monte Carlo validation.

Table 11. Average result after Monte Carlo cross-validation for the $Data_3$.

Metrics	Average result (CI 95%)
ACC	0.9265 (0.9180 – 0.9350)
Se	0.4714 (0.4045 – 0.5382)
Sp	0.9965 (0.9951 – 0.9979)
MCC	0.6195 (0.5656 – 0.6734)
F_1	0.5924 (0.5268 – 0.6580)

By analyzing the results presented in Table 11, it can be seen that the specificity obtained by the network was very high, with a value of 99.65%. This result was expected since most of the dataset samples were from patients who were not infected with COVID-19. However, the average sensitivity obtained by the network was only 0.4714, which indicates that the network did not perform well in detecting COVID-19 positive cases.

Looking at the MCC, a suitable metric for unbalanced data sets, it is noted that the SE Euclidean ART CT network obtained an average MCC of 0.6195 after Monte Carlo validation. While this is not a high value, it is still considered acceptable. Comparing the results obtained in Table 10 and Table 11, it is evident that there was a reduction in the network's performance after Monte Carlo validation. The network could not correctly detect positive COVID-19 cases in some iterations of the validation, with the sensitivity falling below 30% in some iterations, as illustrated in Figure 9.

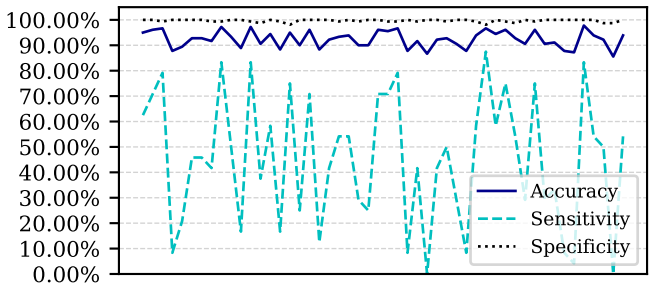

Figure 9. Performance for each of the 50 iterations of the Monte Carlo cross-validation for the $Data_3$.

The ROC curve and the AUC can be seen in Figure 10. By analyzing them, we can see that while specificity remained relatively stable at all cut-off points, there was considerable variation in sensitivity, i.e., the ability to detect positive patients correctly. The mean AUC was 0.7458, with a standard deviation of 0.1265, achieving acceptable performance according to [90].

Table 12 Comparison the results obtained by the neural networks used in this paper and the results found in the literature for the same data sets. The table includes metrics such as ACC, Se, Sp, MCC, F_1, and AUC (ROC).

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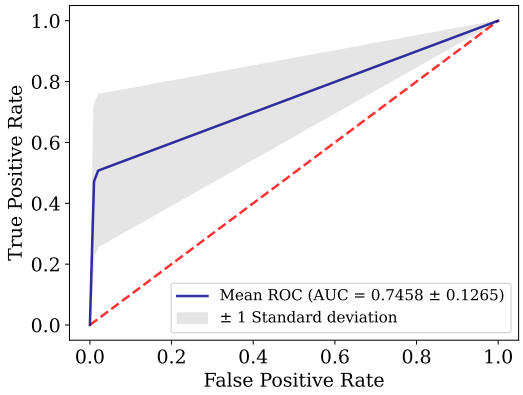

Figure 10. ROC curve for $Data_3$.

Table 12. Comparison of the results for each data set.

Author	Model	Data	Partition	ACC	Se	Sp	MCC	F_1-Score	AUC
This study	SE Fuzzy ART	$Data_1$	70–30	0.8000	0.8704	0.6774	0.5610	0.8468	0.7739
	SE Euclidean ART	$Data_1$	70–30	0.7059	0.7593	0.6129	0.3697	0.7664	0.6861
	SE Fuzzy ARTMAP	$Data_1$	70–30	0.8353	0.8704	0.7742	0.6446	0.8704	-
	SE Fuzzy ART CT	$Data_1$	70–30	0.8706	0.9259	0.7742	0.7170	0.9009	0.8501
	SE Euclidean ART CT	$Data_1$	70–30	0.7412	0.7778	0.6774	0.4496	0.7925	0.7276
	SE Fuzzy ARTMAP CT	$Data_1$	70–30	0.8000	0.7593	0.8710	0.6078	0.8283	-
	SE Euclidean ARTMAP CT	$Data_1$	70–30	0.6353	0.7778	0.3871	0.1763	0.7304	-
	SE Fuzzy ART CT	$Data_1$	Monte Carlo	0.8061	0.8211	0.7800	0.6555	0.8679	0.8006
[93]	ET	$Data_1$	4-Fold	0.8000	0.7500	**0.9444**	0.6133	0.8478	0.7710
	DNN	$Data_1$	4-Fold	**0.9211**	**0.9614**	0.8456	**0.8250**	**0.9388**	**0.9220**
	TWRF	$Data_1$	70-30	0.8600	0.9500	0.7500	-	-	-
[21]	RF	$Data_1$	80-20	0.8200	0.9200	0.6500	-	-	0.8400
This study	SE Fuzzy ART	$Data_2$	70–30	0.7500	0.7459	0.7540	0.4996	0.7426	0.7499
	SE Euclidean ART	$Data_2$	70–30	0.7275	0.6907	0.7619	0.4540	0.7102	0.7263
	SE Fuzzy ARTMAP	$Data_2$	70–30	0.7828	0.7924	0.7738	0.5659	0.7792	-
	SE Fuzzy ART CT	$Data_2$	70–30	**0.9836**	**0.9703**	0.9960	**0.9674**	**0.9828**	**0.9481**
	SE Euclidean ART CT	$Data_2$	70–30	0.7807	0.5466	**1.000**	0.6194	0.7068	0.7733
	SE Fuzzy ARTMAP CT	$Data_2$	70–30	0.7377	0.7119	0.7619	0.4745	0.7241	-
	SE Euclidean ARTMAP CT	$Data_2$	70–30	0.6967	0.7119	0.6828	0.3942	0.6942	-
	SE Fuzzy ART CT	$Data_2$	Monte Carlo	0.9348	0.8992	0.9682	0.8721	0.9301	0.9395
[93]	ET	$Data_2$	4-Fold	0.8498	0.8788	0.8221	0.7012	0.8509	0.8510
	DNN	$Data_2$	4-Fold	0.9316	0.9327	0.9302	0.8633	0.9263	0.9320
[58]	KNN	$Data_2$	*hold-out*	0.8600	0.8000	0.9200	-	-	0.8700
This study	SE Fuzzy ART	$Data_3$	70–30	0.8889	0.2083	0.9936	0.3824	0.3333	0.6010
	SE Euclidean ART	$Data_3$	70–30	0.8778	0.5000	0.9359	0.4524	0.5217	0.7179
	SE Fuzzy ARTMAP	$Data_3$	70–30	0.8667	0.3750	0.9423	0.3595	0.4286	-
	SE Fuzzy ART CT	$Data_3$	70–30	0.9444	0.5833	1.0000	0.7404	0.7368	0.7917
	SE Euclidean ART CT	$Data_3$	70–30	**0.9722**	0.7917	**1.0000**	**0.8758**	0.8837	0.8962
	SE Fuzzy ARTMAP CT	$Data_3$	70–30	0.8889	0.4583	0.9551	0.4685	0.5238	-
	SE Euclidean ARTMAP CT	$Data_3$	70–30	0.8722	0.4583	0.9359	0.4175	0.4889	-
	SE Euclidean ART CT	$Data_3$	Monte Carlo	0.9265	0.4714	0.9965	0.6195	0.5924	0.7458
[93]	ET	$Data_3$	4-Fold	0.9200	0.7143	0.9301	0.4530	0.4545	0.6590
	DNN	$Data_3$	4-Fold	0.9333	0.7705	0.9547	0.6904	0.7249	0.8597
[54]	LSTM	$Data_3$	10-Fold	0.8666	**0.9942**	-	-	0.9189	0.6250
	CNN-LSTM	$Data_3$	80–20	0.9230	0.9368	-	-	**0.9300**	**0.9000**

SE: Self-Expanding; **CT**: Continuous Training; **ET**: Extremely Randomized Trees; **DNN**: Deep Neural Network; **RF**: Random Forest; **TWRF**: Three-way Random Forest; **KNN**: K Nearest neighbors; **LSTM**: Long short-term memory; **CNN**: Convolutional Neural Networks.

The results obtained for $Data_1$ were quite promising, as the SE Fuzzy ART CT network achieved AUC of 0.8501 and 0.8006 in hold-out and Monte Carlo validation, respectively. These results were better than those obtained by commonly used models such as RF and ET, often used for classification tasks. Although the performance of the SE Fuzzy ART CT network was good, the DNN deep learning model used by [93] obtained even better results, achieving an AUC value of 0.9220.

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When examining the results for $Data_2$, it can be seen that the SE Fuzzy ART CT network achieved outstanding performance. It obtained an AUC of 0.9481 in the hold-out validation, indicating high accuracy in distinguishing between COVID-19 positive and negative patients. It outperformed machine learning models traditionally used in classification tasks such as ET, KNN, and even the deep learning algorithm DNN, which achieved an AUC of 0.9320. The sensitivity of the ART network was also significantly higher at 97.03%, indicating it is effective in correctly identifying positive patients. It is worth noting that even when the SE Fuzzy ART CT network was subjected to Monte Carlo cross-validation, it maintained its good performance with an AUC of 0.9395, outperforming DNN once again.

For $Data_3$, the SE Euclidean ART CT network achieved AUC of 0.8962, a result similar to that obtained with the CNN-LSTM model used by [54], which achieved AUC of 0.9000. However, after performing Monte Carlo cross-validation, a decrease in the performance of the SE Euclidean ART CT network was observed due to a significant imbalance between classes.

The Comparison of the computational time (in seconds) between the two best models implemented in this paper (SE Fuzzy ART CT and SE Euclidean ART CT), with the best model obtained by [93] the Deep Neural Network (DNN), can be seen in Table 13. It is worth noting that the computational time given in the table considers only the training and testing time. Also, the best algorithms for each dataset are highlighted in bold.

Table 13. Computational time (in seconds) for each dataset.

Model	$Data_1$	$Data_2$	$Data_3$
SE Fuzzy ART CT	1.25	**6.34**	5.14
SE Euclidean ART CT	1.22	0.62	**1.67**
DNN	**20.2**	40.32	21.01

When analyzing the computational time in Table 13, it is clear that the ART networks showed significantly faster convergence than the DNN. This is due to the nature of the network and accentuated by the self-expansion mechanism. In addition, it can be seen that the Euclidean distance-based neural network obtained a slightly shorter time than the fuzzy neural network. This difference can be attributed to the fact that the Euclidean network not requiring normalization and encoding of the input data. Consequently, the size of the input data entered into the Euclidean network is halved, leading to faster processing. Another relevant feature of ART networks is that they do not suffer from catastrophic forgetting, unlike those based on the back-propagation algorithm and deep learning models, as reported by [61].

4.2. Second Stage

In this second stage, we sought to compare the performance of the winning model from the previous stage, SE Fuzzy ART CT, with traditional models frequently used in the literature. To do this, the combination of data described in subsection 3.2 was used. Algorithms such as K-Nearest Neighbors (KNN), Random Forest (RF), Multi-layer Perceptron (MLP), Logistic Regression (LR), Extra Trees (ET), XGBoost (XGB) and Naive Bayes (NB) were implemented in Python with the Scikit-Learn library [94]. The parameters of each model were optimized using the GridSearchCV method, exploring various combinations of hyperparameters in order to identify the one that resulted in the best performance. The results obtained are presented in the Table 14, highlighting the superiority of the ART network over traditional models, achieving an MCC of 0.9628.

In addition, the Figure 11 shows the ROC curve and AUC for each model. The SE Fuzzy ART CT network obtained the best AUC, with a value of 0.9770, indicating that it has a high ability to distinguish between classes. Among the traditional models tested, the RF algorithm obtained the best performance, with AUC and MCC of 0.8504 and 0.5510, respectively. However, these results are

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significantly lower than those of the ART network. The RF algorithm's ability to distinguish between classes is around 12% lower than that of the ART network.

Table 14. Results obtained by each model for the combined data set.

Model	SE Fuzzy ART CT	KNN	RF	MLP	LR	ET	XGB	NB
ACC	**0.9828**	0.7277	0.7931	0.7825	0.7617	0.7748	0.7333	0.7429
Se	**0.9667**	0.5986	0.7014	0.6806	0.6389	0.5806	0.8722	0.6806
Sp	**0.9920**	0.8019	0.8458	0.8411	0.8323	0.8866	0.6534	0.7788
F_1-Score	**0.9762**	0.6162	0.7123	0.6955	0.6619	0.6531	0.7048	0.6590
MCC	**0.9628**	0.4059	0.5510	0.5267	0.4790	0.4988	0.5076	0.4536

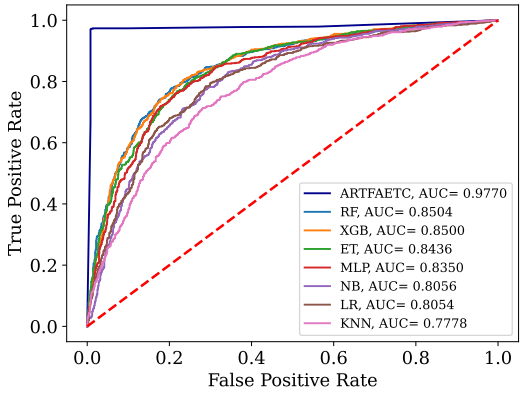

Figure 11. ROC curve and AUC for the combined data set.

The Table 15 and Figure 12 show, respectively, the metrics and the ROC curve along with the AUC of the models trained with the combined data and validated with $Dados_{12}$, i.e. the data collected in Spain.

Table 15. Results obtained by each model trained with the combined data and validated with $Dados_{12}$.

Model	SE Fuzzy ART CT	KNN	RF	MLP	LR	ET	XGB	NB
ACC	**0.9770**	0.7169	0.7834	0.7724	0.7547	0.7633	0.7308	0.7331
Se	**0.9661**	0.5834	0.6813	0.6625	0.6286	0.5609	0.8595	0.6637
Sp	**0.9838**	0.7991	0.8462	0.8400	0.8323	0.8879	0.6515	0.7759
F_1-Score	**0.9698**	0.6110	0.7057	0.6893	0.6614	0.6436	0.7087	0.6547
MCC	**0,9513**	0,3903	0,5355	0,5112	0,4714	0,4838	0,5076	0,4374

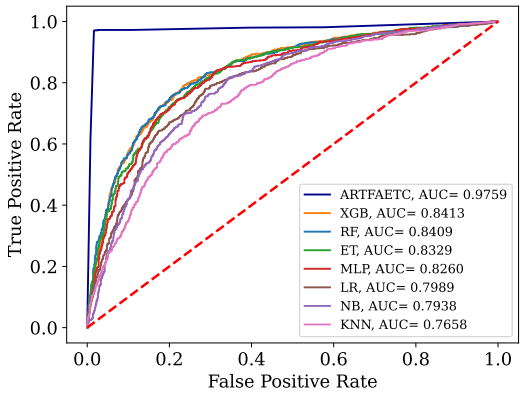

Figure 12. ROC curve and AUC obtained by each model trained with the combined data and validated by $Dados_{12}$.

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Although it showed a slight decline in performance when compared to the results presented in Table 15 and Figure 11, the SE Fuzzy ART CT network still outperformed the traditional models, which also showed a reduction in performance when tested with the new data. The model obtained an AUC of 0.9759, a significantly better performance than the traditional models, which obtained AUCs of 0.8413 for XGB and 0.8409 for RF.

When the SE Fuzzy ART CT network trained with the combined data was submitted to the $Data_1$ validation set, there was a reduction in its performance compared to the results obtained with the combined data. This reduction was only 1.71% in terms of AUC, from 0.9770 to 0.9606. The traditional models, on the other hand, showed an increase of less than 0.01% in relation to the AUC obtained with the combined data. However, even with this reduction in performance, the ART network still remained the most effective model in distinguishing between patients with and without COVID-19.

Table 16. Results obtained by each model trained with the combined data and validated by $Dados_1$.

Modelo	SE Fuzzy ART CT	KNN	RF	MLP	LR	ET	XGB	NB
ACC	**0,9645**	0,7326	0,7863	0,7699	0,7539	0,7654	0,7326	0,7370
Se	**0,9710**	0,6388	0,7168	0,6901	0,6589	0,6042	0,8796	0,7101
Sp	**0,9601**	0,7947	0,8323	0,8227	0,8168	0,8722	0,6352	0,7548
F_1-Score	**0,9561**	0,6556	0,7278	0,7050	0,6809	0,6725	0,7239	0,6827
MCC	**0,9266**	0,4377	0,5522	0,5168	0,4817	0,5011	0,5090	0,4597

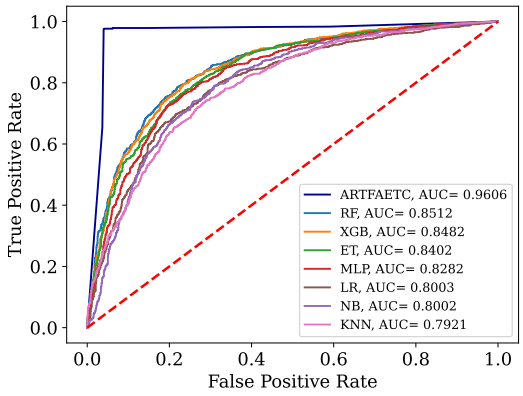

Figure 13. ROC curve and AUC obtained by each model trained with the combined data and validated by $Data_1$.

4.3. Discussão

The combined analysis of the two stages highlights not only the effectiveness of SE Fuzzy ART CT in detecting COVID-19, but also its ability to handle new patient data. In the first part, it did better than other models mentioned in the literature. In the second stage, when we compared conventional models with SE Fuzzy ART CT, we confirmed its superiority. The network's results remained consistent even with data from different countries. The online training capacity, combined with the unsupervised training, were the attributes responsible for the excellent performance and places it as an adaptable and promising tool for accurate, real-time classifications.

In addition to its positive results, the study also addresses some limitations listed by other articles. A study by [95], for example, cites the lack of patients from other geographical locations as a limitation. [95] states that it is necessary to consider patients from different demographic groups in order to establish reliable results. This study addresses this limitation by training and testing the models on data from several hospitals in four countries. This allows the results to be more representative of the population.

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In addition, the study tests models with unsupervised learning, a limitation pointed out in the work by [96]. Since unsupervised learning allows models to learn to identify patterns in the data without the need for labels.

5. Conclusion

This article presents the results obtained by neural networks based on Adaptive Resonance Theory (ART) for the classification of COVID-19, using laboratory tests (blood tests) in order to correctly classify infected and healthy patients. A comparison was made between seven different ART architectures, including online training algorithms. The most successful architecture was subjected to a dataset made up of laboratory tests from patients in different countries. When comparing the results of the first stage, it can be seen that the self-expanding fuzzy ART network with continuous training (SE Fuzzy ART CT) and the self-expanding Euclidean ART network with continuous training (SE Euclidean ART CT) outperformed the other ART networks implemented. In addition, in some cases, the networks outperformed the DNN and CNN deep learning models frequently used in the literature. The SE Fuzzy ART CT obtained an AUC of 0.8501 for $Data_1$, 0.9481 for $Data_2$ and 0.8962 for $Data_3$. When considering computational time, the self-expanding ART neural networks with continuous training were more efficient in the three data sets tested.

In the second stage, the self-expanding SE Fuzzy ART CT that won out in the first stage was subjected to a data set from various hospitals in four countries. This network was compared with traditional algorithms in the literature, maintaining an excellent performance that significantly outperformed all traditional models. Even during external validation with data from a different country to that included in the training, validating its generalization capacity.

In addition to the superior results compared to most of the models analyzed, the networks used do not face the main problems associated with traditional algorithms, such as slow learning, catastrophic forgetting and off-line training. Thus, these results highlight the potential of networks based on adaptive resonance theory, such as SE Fuzzy ART CT and SE Euclidean ART CT, as highly effective and reliable tools to aid in the detection of COVID-19, reducing the false negative rate of RT-PCR or even as a viable solution for detection in places with limited resources or underdeveloped countries.

Despite the promising results obtained by the ART networks, in future studies we intend to reduce the size of the data sets, selecting the most significant attributes to detect healthy and COVID-19 patients, and we can also add more information to aid classification, such as the use of imaging tests or symptoms. In addition, it is possible to make some adjustments to the models to improve the quality of the classification, such as modifying the normalization used, extending the research to the β parameter, using the k-Fold cross-validation method as a way of validating and evaluating the model's generalization capacity.

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RESEARCH ARTICLE

Estimation of Vulnerable Areas to Faults Caused by Tree Vegetation in Power Distribution Systems

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ABSTRACT Faults in power distribution feeders cause damage to power utilities due to the deterioration of reliability and power quality indexes and the displacement of field maintenance teams to replace or repair power grid equipment. Additionally, consumer units have energy supply interruptions for an undetermined time. Studies in specialized literature usually detect, classify, and locate faults after they occur. In contrast, preventing faults by estimating areas vulnerable to them is crucial to mitigate all inconveniences and additional costs after they occur. Tree vegetation is an essential factor contributing to faults. In this sense, an enhanced method for tree vegetation mapping by areas is developed using multilayer perceptron neural networks trained on high-resolution images from Google Earth. A geographic space is incorporated to estimate the regions vulnerable to failures due to tree vegetation. Geographically weighted spatial analysis is applied from local variables aggregated by areas. Spatial data analysis is used to real faults and tree vegetation data from a medium-sized Brazilian city via QGIS and R programming environments. As a result, thematic maps are produced with the areas whose feeders are vulnerable to faults, where there is a moderate positive correlation by regions between the faults in distribution transformers and tree vegetation in the northeast and southwest areas of the city under study.

INDEX TERMS Exploratory spatial data analysis, power distribution systems, spatial data analysis, steady-state fault.

NOTATION

The notation used throughout this paper is reproduced below for quick reference.

Acronyms:

ANN	Artificial Neural Network;
CTs	Census Tracts;
CUs	Consumer Units;
ESDA	Exploratory Spatial Data Analysis;
ESI	Energy Supply Interruptions;
ETT	Enhanced Tree Trimming
GIS	Geographic Information Systems;
GW	Geographically Weighted;
GWEA	Geographically Weighted Exploratory Analysis;
GWMs	Geographically Weighted Models;
MLP	Multilayer Perceptron;
PDS	Power Distribution Systems;
SDA	Spatial Data Analysis;
SWM	Spatial Weighting Matrix.

Indices:

i,j	Index for census tracts.

Parameters:

$\bar{X}$	Arithmetic mean;
σ	Standard deviation;
b	Bandwidth parameter for the Gaussian kernel function;

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d	Difference between the ordinal position of two variables in the same census tract;
d_{ij}	Distance between the centroids of census tracts i and j;
n	Number of census tracts;
$\boldsymbol{W}_{(n\times n)}$	Spatial weighting matrix;
w_{ij}	Weighting between the census tracts i and j;
y_i, z_i	Attributes associated with a census tract at i.
Set:	
Ω_k	Set of k classes used in the legend of thematic maps.
Statistical Metrics:	
$\rho(z_i, y_i)$	Geographically weighted Pearson's local correlation coefficient;
$c(z_i, y_i)$	Geographically weighted covariance;
$m(z_i)$	Geographically weighted mean;
r_s	Spearman's correlation coefficient;
$s\,(z_i)$	Geographically weighted standard deviation.
Variables:	
NFT	Number of faults in distribution transformers by census tracts;
PFT	Percentage of faults in distribution transformers by census tracts;
PTV	Percentage of tree vegetation by census tracts.

I. INTRODUCTION

Power distribution systems (PDS) are continually subject to events that can result in disturbances and, consequently, failures [1]. These failures cause numerous interruptions in the energy supply and negatively impact PDS's reliability and power quality, seriously threatening its operation [2], [3].

ESIs are associated with multiple factors such as (i) tree vegetation close to the overhead utility grid, (ii) adverse weather conditions (wind gusts, snow, storms, electrical discharges), (iii) damaged equipment (missing maintenance, obsolescence, or manufacturing defect), (iv) animals as birds and insects, (v) vehicle collision, (vi) overload on utility grid; (vii) clandestine connection or vandalism, (viii) human failures, (ix) protective equipment acting, and (x) kites [4].

Tree vegetation represents a relevant cause associated with faults among all the factors above, and its impact is a recurring problem for power utilities that results in significant economic losses [5], [6]. This impact is related to the uncontrolled growth of tree vegetation close to distribution lines, plus adverse weather conditions such as wind gusts and storms [7], [8]. Additionally, the contact of tree branches with energized conductors belonging to the utility grid often causes short circuits, failures, damage to system infrastructure, and risks to human health [9], [10].

Tree vegetation close to overhead distribution lines also can interfere with the operation of the utility grid in each area, making it more vulnerable to failures. The overlapping of tree branches over the cables can cause interference, ESIs, and compromise the safety of consumer units (CUs).

Faults resulting from tree vegetation can be classified into two categories: the first refers to interruptions caused by tree branches that meet utility grid cables as they grow close to the feeder. The second category is directly related to adverse weather conditions, which can result in trees falling onto distribution lines or forced contact of branches with the feeder.

Adopting preventive measures is crucial to mitigate the faults, inconveniences, and additional costs after they occur in PDS. To the best of the authors' knowledge, specialized literature studies aim to detect, classify, and locate faults after they occur – post-fault studies [11], [12]. On the other hand, estimating regions whose distribution feeders are vulnerable to faults can be a valuable tool to power utilities in planning and executing preventive actions to minimize the occurrence of faults and all maintenance costs related to them.

As mentioned above, several factors can make some regions or cities more vulnerable to ESIs; however, some may be more relevant than others. For example, wind gusts may be appropriate in coastal towns; conversely, tree vegetation can be applicable in heavily wooded cities. Thus, the variables associated with faults depend on the region or city under study. In this context, spatial data analysis (SDA) is a tool for preventing faults by estimating the city's regions whose feeders are vulnerable to them. Furthermore, the leading causes of faults can be identified in the city under study.

This work aims to incorporate the geographic space study to estimate areas whose utility grid is vulnerable to failures due to tree vegetation. The variables evaluated in this study are shutdowns in distribution transformers that caused steady-state faults and the percentage of area occupied by tree vegetation per census tracts (CTs), which are directly associated with faults in utility grid feeders. It is observed that all transformers and tree vegetation are georeferenced in a time window; that is, their geographic coordinates are known, enabling SDA application.

A. LITERATURE REVIEW

Faults in PDS related to tree vegetation are addressed in recent works. Cerrai et al. [5] assessed the impact of enhanced tree trimming (ETT) on ESI, in which two different methodologies were implemented. First, the authors applied a statistical analysis to verify the trend of reduction in interruptions as the number of ETTs increases. An interruption prediction model was addressed in the second stage, where the aim is to evaluate changes in the number of interruptions before and after ETT.

An approach for predicting vegetation-related interruptions was presented in [7]. They were categorized into two groups: those caused by tree growth and those caused by vegetation due to adverse weather conditions.

The predictions applied two models: a statistical approach based on time series algorithms and an approach based on non-linear machine learning.

In [11], the interruptions' frequency, extent, and duration were considered. The analysis of tree pruning operations was evaluated based on a set of actual data, and two autoregressive models were introduced: the spatial autoregressive model and the spatial Durbin model. The results demonstrated that tree pruning operations reduced interruptions and the number of affected CUs.

Chen and Kezunovic [13] implemented a predictive method for managing interruptions in transmission and distribution systems, where the impact due to the combination of winds and tree vegetation was considered. A geographic information systems (GIS) framework was introduced to correlate energy system data with multiple climate data layers, including wind and vegetation information. As a result, maps were presented with the areas vulnerable to interruptions.

Local variables associated with faults due to trees during storms were analyzed in [14]. Five fault risk models due to tree vegetation were developed and compared using a random forest algorithm, where the physical structure of trees plus local environmental factors such as utility infrastructure, faults history, tree pruning data, soil, and land cover information were considered.

Taylor et al. [15] evaluated the impact of vegetation management to reduce ESIs during storms. A machine learning model was applied, where data on vegetation management, soil cover, climate data, and electrical infrastructure were inserted. The study quantified the damage reduction in the power grid due to ETT during adverse weather events. It was found that annual reductions in hotspots varied between 25.7% and 42.5% due to ETT.

The authors empirically demonstrate the impact of ETT on reducing faults during storms [16]. ESI reduced significantly in areas where ETT treatment was applied, while more failures occurred in untreated lines.

The aforementioned studies evaluate the impact of ETT on reducing ESI. On the other hand, the relevance of our research consists of assessing the relationship between tree vegetation and ESI, typically caused by factors such as wind, snow/ice accumulation, and falling branches, considering the city's geographic space. Given that power utilities implement tree vegetation management programs such as ETT, this study becomes pertinent when there is a need to enhance the understanding or optimize strategies within these programs. For example, this study can help in the decision-making process to select critical areas that can be the targets of ETT programs.

In [17], a spatial fuzzy influence diagram was presented to identify vulnerable points in the power grid with a high risk of interruptions caused by trees during extreme weather conditions. Data such as forest resources, soil type, power grid, and history of interruptions were considered. As a result, the authors presented a map showing the areas whose power grid is vulnerable to interruptions.

Finally, Onaolapo et al. [18] presented an outage prediction model based on an artificial neural network (ANN). Two models were built, one with ANN and the other with multiple linear regression, where climate variables influencing the power grid were introduced. The ANN-based model improved performance by incorporating ten variables: (1) outage history, (2) number of clouds, (3) minimum and maximum temperatures, and (4) number of frost days. The study highlighted the complex relationship between weather events and failures in the power grid that result in ESI. On the other hand, the city's areas most vulnerable to climate events were not highlighted. For example, strong winds can cause trees and branches to fall onto the power grid. Therefore, the city's heavily wooded regions will be more vulnerable to faults.

B. CONTRIBUTIONS

This study performs exploratory spatial data analysis (ESDA) from tree vegetation data associated with faults and shutdowns in distribution transformers. Its main contributions are described below.

- Geographical space study is incorporated to produce thematic maps showing areas vulnerable to faults due to tree vegetation. These maps are an easy-to-interpret visual tool for power utilities to plan and execute actions to prevent ESIs. Therefore, all inconveniences and additional costs are minimized after the fault occurs in PDS.
- Geographically weighted exploratory analysis (GWEA) by regions uses variables associated with each CT: distribution transformer faults and tree vegetation. Local geographically weighted (GW) measurements, such as standard deviation and local correlations, are applied. GWEA considers the variable values for each CT and the influence of closest or neighboring areas.

C. PAPER STRUCTURE

This paper follows in Section II with SDA, spatial analysis with data aggregated by areas (Subsection II-A), Spearman's correlation coefficient (Subsection II-A.1), spatial weighting matrix (Subsection II-A.2), and GWEA – Subsection II-A.3. Subsection II-B presents tree vegetation mapping procedure. Section III presents the implementation of GWEA in a Brazilian city. Subsections III-A and III-B present tree vegetation mapping and database description, respectively. An ESDA is performed in Subsection III-C, from the percentage of tree vegetation (Subsection III-C.1) and faults in distribution transformers – Subsection III-C.2. GW statistical summary is performed in Subsection III-C.3 considering the weighting among areas. Finally, Section IV shows the conclusions of our work.

II. SPATIAL DATA ANALYSIS

SDA aims to measure the properties and relationships of events considering their spatial location – geographical coordinates. In this way, geographic space is incorporated into the

analysis; therefore, there is a visual perception of the spatial distribution of the event under study [19].

This study uses SDA metrics to perform an ESDA on data aggregated by areas named CTs. In this way, each location is associated with a value for each variable under study: faults percentage in distribution transformers and percentage of area covered by tree vegetation.

ESDA is the crucial step of a study that applies SDA. It consists of metrics to identify patterns based on the spatial distribution of the event under study. Thus, hypotheses can be formulated about the variables influencing its spatial distribution [20], [21].

ESDA application shows essential effects: spatial autocorrelation and spatial heterogeneity [22]. Spatial autocorrelation refers to the analysis of variation in spatial dependence based on comparing observations from a given geographic area and its neighbors. This concept was defined by Tobler, who stated the first Law of Geography: all things are similar; however, closer things look more than farther away [23]. Autocorrelation has negative or positive values, where positive values indicate that close observations tend to exhibit similar attributes; therefore, there is aggregation, similarity, or grouping of the event under analysis. On the other hand, spatial heterogeneity shows the variation of observations in geographic space [19], [22].

A. SPATIAL ANALYSIS WITH DATA AGGREGATED BY AREAS

This study is linked to an SDA branch with several methods for analyzing data aggregated by areas delimited by polygons. They are applied to studying events aggregated by areas where their location is unavailable. There is only one value for each variable associated with each region. Thus, it is possible to analyze patterns, trends, and relationships in data sets with specific information about geographic areas such as CTs, cities, or countries [19], [24].

The presentation of data aggregated by areas is prepared using colored maps (thematic maps) that display the spatial pattern of the event under analysis [19].

A branch of SDA with data aggregated by area allows the use of public information from the demographic census. They are grouped into small areas named CTs, whose area is a function of their population density; CTs with greater-density populations have smaller areas, and vice versa [4].

1) CORRELATION COEFFICIENT OF SPEARMAN

The Spearman's coefficient correlation is a metric that identifies whether two variables are associated, where its value belongs to the continuous interval $[-1, 1]$.

Negative correlation values indicate that the variables are inversely proportional. Positive values indicate that the variables are directly proportional. Correlation strength is evaluated as this metric approaches the two extremes $\{-1, 1\}$. On the other hand, positive or negative correlations close to zero indicate no significant correlation. Consider two hypothetical variables, A and B: if A is associated with B, it does not necessarily mean that A causes B and vice versa.

The r_s coefficient is obtained by ranking the values of both variables in ascending (or descending) order, where original values are replaced by a positive integer value representing the variable's ordinal position in the data series. The r_s coefficient is calculated by:

$$r_s = 1 - \frac{6 \sum d^2}{n(n^2 - 1)} \tag{1}$$

where n is the total number of CTs and d is the difference between the ordinal position of two variables in the same CT.

Spearman's coefficient is not sensitive to asymmetries or outliers in the data series. Furthermore, the pair of variables do not need to be linearly associated [25].

Finally, Spearman's coefficient is a global metric that does not consider the variables' space distribution. In contrast, the following two subsections present a spatial weighting matrix (SWM) representing the influence of closest or neighboring areas for obtaining GW metrics values.

2) SPATIAL WEIGHTING MATRIX

The neighborhood structure among areas establishes their influence relationship, and it is crucial in studies that apply SDA techniques with data aggregated by areas.

SDA techniques are widely applied in epidemiology, botany, soil studies, mineral resources prospecting, and criminology. A neighborhood structure among areas from Euclidean distance among centroid areas is usually applied in these knowledge areas, where closer areas have more significant influence than distant areas [23]. This same neighborhood structure is used in this study because an abrupt transition of tree vegetation or faults among neighboring areas is not expected.

In this sense, SWM is applied to estimate the spatial variability in data aggregated by areas. SWM $\boldsymbol{W}_{(n\times n)}$ is constructed from a discrete set of n areas $\{A_1, ..,A_n\}$, where each element w_{ij} represents the spatial dependence among the variables observed in A_i and A_j areas.

Fig. 1 shows an example of obtaining SWM. There are six CTs (A, B, C, D, E, and F) of the city in Fig. 1 (a); on the other hand, in Fig. 1 (b), there is the corresponding SWM $\boldsymbol{W}_{(6\times 6)}$, since $n = 6$ CTs and whose obtaining rule is according to (2).

For example, CT B has borders in common with two CTs: A and E. Thus, according to (2), SWM elements w_{21} (weighting between B and A) and w_{25} (weighting between B and E) are equal to $1/2$. The other areas do not have common borders with B; therefore, they have zero weighting – $w_{23} = 0$, $w_{24} = 0$, and $w_{26} = 0$.

$$w_{ij} = \begin{cases} \dfrac{1}{No.\ of\ Borders\ CTs}, & if\ i\ and\ j\ are\ borders\ CTs \\ 0, & otherwise \end{cases} \tag{2}$$

SWM elements can also be obtained via kernel functions. They are decreasing and monotone functions of the distance

(a)

	A	B	C	D	E	F
A	1	0.2	0.2	0.2	0.2	0.2
B	0.5	1	0	0	0.5	0
C	0.5	0	1	0.5	0	0
D	0.33	0	0.33	1	0	0.33
E	0.33	0.33	0	0	1	0.33
F	0.33	0	0	0.33	0.33	1

(b)

FIGURE 1. An illustrative example of obtaining a SWM with six CTs of the city (a) and its corresponding SWM (b).

between A_i and A_j centroids. Therefore, it follows Tobler's first law, where the weighting among nearby CTs is more significant than for distant CTs. Gaussian kernel is applied in our study to obtain SWM elements according to (3):

$$w_{ij} = \exp\left(-\frac{1}{2}\left(\frac{d_{ij}}{b}\right)^2\right) \tag{3}$$

where d_{ij} represents the distance between the centroids of CTs i, and j and b is a bandwidth parameter for the function's decay rate [26].

3) GEOGRAPHICALLY WEIGHTED EXPLORATORY

Geographically weighted models (GWMs) are tools belonging to a particular branch of non-stationary spatial statistics, which intuitively incorporate local spatial relationships into their structure. These techniques are helpful in contexts where a global model does not adequately describe spatial data, as they enable the estimation of local parameters in geographic space [26], [27].

The GW model outputs are mapped to provide a helpful tool that typically precedes more sophisticated statistical analysis.

The kernel function is crucial in GW modeling, which aims to quantify the spatial dependence relationship among the variables observed in CTs.

GW local summary statistic is obtained from a set of spatial data and a SWM. Consider the attributes z_i e y_i associated with a CT at i. The following metrics can be obtained via (4)–(7), respectively: GW mean, GW standard deviation, GW Pearson's local correlation coefficient, and GW covariance [26].

$$m(z_i) = \frac{\sum_{j=1}^{n} w_{ij} z_j}{\sum_{j=1}^{n} w_{ij}} \tag{4}$$

$$s(z_i) = \sqrt{\frac{\sum_{j=1}^{n} w_{ij}(z_j - m(z_i))^2}{\sum_{j=1}^{n} w_{ij}}} \tag{5}$$

$$\rho(z_i, y_i) = \frac{c(z_i, y_i)}{s(z_i)\, s(y_i)} \tag{6}$$

$$c(z_i, y_i) = \frac{\sum_{j=1}^{n} w_{ij} \left[(z_j - m(z_i))(y_j - m(y_i))\right]}{\sum_{j=1}^{n} w_{ij}} \tag{7}$$

where $\boldsymbol{w_{ij}}$ are SWM elements.

B. METHODOLOGY FOR TREE VEGETATION MAPPING

Tree vegetation mapping is performed via image classification of each CT by pixel identification. This process involves identifying the RGB pattern of each pixel, allowing precise identification of the area percentage occupied by tree vegetation.

Some remote sensing techniques are limited and cannot adequately identify tree vegetation due to satellite images' pixel length of around 30 m [28]. Therefore, an ANN multilayer perceptron (MLP) is applied to classify tree vegetation from the pixels' RGB pattern [29].

Fig. 2 shows an illustrative example of a CT with highlighted tree vegetation and the image pixels in a zoom version. In this sense, Fig. 3 shows a flowchart with the steps to determine the area percentage of each CT occupied by tree vegetation: (i) obtaining high-resolution images from Google Earth of each CT; (ii) extraction of the RGB pattern of each pixel; (3) Application of MLP to check the compatibility of the pixel's RGB pattern with tree vegetation; (4) Determination of CT area occupied by tree vegetation.

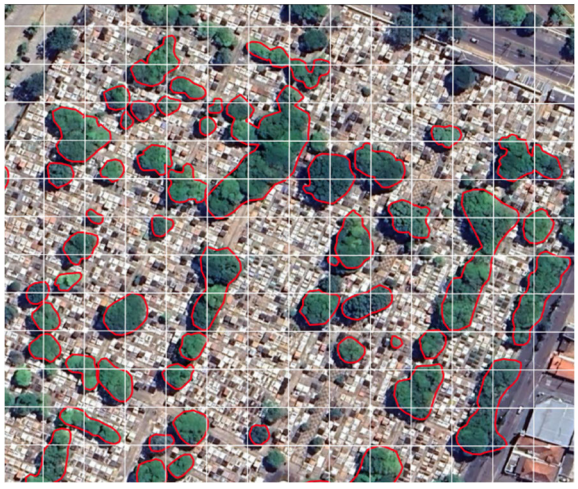

FIGURE 2. Illustrative example of tree vegetation mapping.

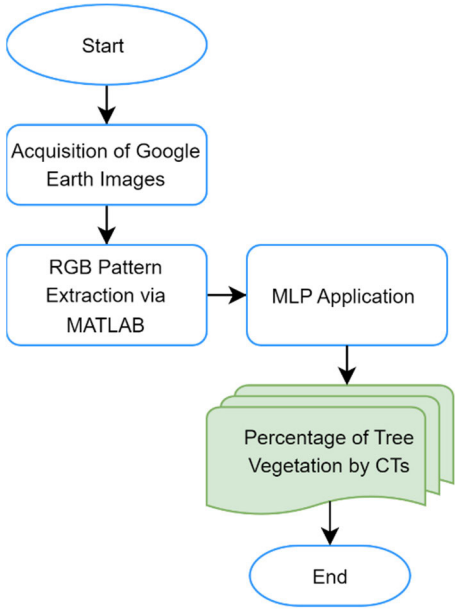

FIGURE 3. Flowchart of tree vegetation mapping.

III. EXPLORATORY SPATIAL DATA ANALYSIS IN A BRAZILIAN CITY

This work applies an ESDA to evaluate the influence of tree vegetation on making some city CTs vulnerable to steady-state faults.

In this sense, Fig. 4 shows some relevant factors that caused 9,266 ESI with more than a 1-minute duration over four years – 2009 to 2012. Tree vegetation occupies the second position in the ranking as the cause of 19% of ESI in the city under study. These data highlight the importance of developing effective strategies for adequately managing tree vegetation to mitigate their impact on interruptions, improving the PDS reliability and the service quality provided to CUs.

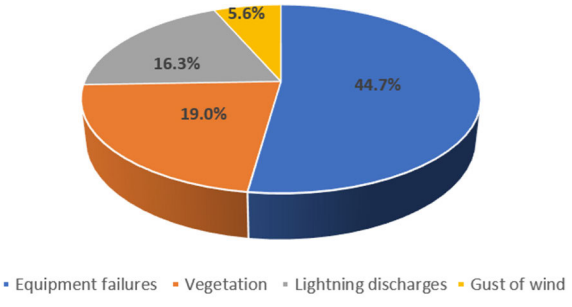

FIGURE 4. Relevant factors that caused steady faults in the city under analysis.

The power grid analyzed has feeders located in a Brazilian city. All simulations are performed via QGIS and R software version 3.30 and 4.1.2, respectively.

QGIS is a GIS software that enables visualization, editing, and SDA [30]. On the other hand, R is a free software for statistical and graphical computing, with several packages, including the GWmodel package applied in this study. Although not a GIS, R can perform similar functions [26], [31].

All simulations in our work are performed on a computer with a Windows operating system, an AMD Ryzen 7 3700X processor, 3.6 GHz, 64-bit, and 16 GB of RAM.

A. TREE VEGETATION MAPPING

An MLP is applied to identify the pixel's RGB pattern with tree vegetation from high-resolution images obtained from Google Earth. Image processing via MATLAB software is used to achieve this objective [32].

MLP has a topology with three inputs (RGB pattern), 20 neurons (intermediate layer), and one neuron in the binary output layer, where the RGB pattern of some pixels is compatible with tree vegetation. MLP training uses 815 samples with different shades, some compatible with tree vegetation and others non-compatible. Therefore, there is a clear distinction between tree vegetation and undergrowth.

Table 1 shows the MLP parameters applied to tree vegetation mapping for all city CTs [29].

TABLE 1. Multilayer perceptron parameters for tree vegetation mapping.

Parameter	Value
Training	Supervised
Architecture	Feedforward
Training Algorithm	Backpropagation
Topology	3–20–1
Constant Training	10^{-1}
Training Accuracy	10^{-6}
Training Set	815 samples (~70%)
Validation Set	349 samples (~30%)
Learning Type	Off-line

B. DATABASE DESCRIPTION

This section evaluates the percentage of tree vegetation (PTV) by CTs, which is responsible for permanent faults, and if it can make some city areas vulnerable to faults.

In this sense, the terminology used in spatial regression is applied where a set of independent variables explains a dependent variable's distribution in geographic space. In this work, the dependent or study variable evaluated is shutdowns in distribution transformers by CTs that caused steady-state faults. The faults are associated with distribution transformers because they are georeferenced, enabling the SDA application. An independent or explanatory variable evaluated, directly related to faults in utility grid feeders, is the percentage of area occupied by tree vegetation by CTs.

Consider a CT_i with $i = 1, \ldots, n$ where n is the number of city's CTs. The independent variable PTV_i is shown in (8). It is obtained from the ratio between the area occupied by tree

vegetation in CT_i and its area:

$$PTV_i = \frac{Tree\ Vegetation\ Area\ in\ CT_i}{CT_i\ Area}\ with\ i = 1, \ldots, n \tag{8}$$

The dependent variable percentage of faults in distribution transformers at CT_i named PFT_i is shown in (9). It is obtained from the ratio between the number of permanent faults at CT_i NFT_i that caused shutdowns in distribution transformers and the number of distribution transformers at CT_i named T_i.

$$PFT_i = \frac{NFT_i}{T_i} \text{with } i = 1, \ldots, n \tag{9}$$

PFT_i is a fault probability estimating in distribution transformers by CTs. CTs with more transformers T_i tend to have a more significant number of faults in transformers NFT_i. Therefore, PFT_i is a more effective variable than just the number of faults NFT_i because it represents a vulnerability measure of CTs to faults.

All variables applied in ESDA are described in Table 2.

TABLE 2. Variables description.

Description	Acronym	Type Variable
Number of Faults in Distribution Transformers by CTs	*NFT*	Dependent Variables
Percentage of Faults in Distribution Transformers by CTs	*PFT*	
Percentage of Tree Vegetation by CTs	*PTV*	Independent Variable

C. EXPLORATORY SPATIAL DATA ANALYSIS

An ESDA is performed in *NFT*, *PTV*, and *PFT* variables in the following two subsections, where the focus is on visualizing the variable distribution in the geographic space of the city via thematic maps.

Thematic map legends are performed via standard deviation intervals, which consider the statistical distribution of the variable to be mapped. This technique is more appropriate for use in data series that follow normal distribution. However, it can show the locational distortions on maps of some CTs concerning the global average [25], [33].

In this sense, the mean $\bar{X}$ and standard deviation σ of the distribution are calculated; then, the data series is segmented into class intervals whose limits are proportions of the variable's standard deviation. The set Ω_k of k classes can be generalized in (10). Alternatively, 0.5σ or 0.25σ can be used instead of σ for data series with small sparsity.

$$\Omega_k = \{\ldots, (\bar{X} - 2\sigma), (\bar{X} - \sigma), \bar{X}, (\bar{X} + \sigma), (\bar{X} + 2\sigma), \ldots\} \tag{10}$$

1) PERCENTAGE OF TREE VEGETATION

An ESDA is performed for the independent variable *PTV* in this section. Figs. 5 (a)–(c) show the *PTV* distribution by CTs from 2010 to 2012. The maps are very similar because tree vegetation usually changes very slowly. The CTs with the highest *PTV* are found on the city outskirts, mainly in the east, northeast, and southeast, and some CTs in the central and extreme west. As expected, the city's central region (red circle) has a low *PTV*.

Table 3 shows descriptive statistics metrics for *PTV* by CTs from 2010 to 2012. The metrics values are very similar for all years evaluated.

TABLE 3. Statistical summary for PTV by census tracts.

Parameter	Year Evaluated		
	2010	2011	2012
Max.	64.43	70.00	71.00
Min.	0.00	0.00	0.00
Mean	8.02	8.28	8.45
Standard Deviation	6.05	6.23	6.55

The numbers in parentheses in the Figs. 5 legend indicates the number of CTs whose *PTV* falls within the range. Figs. 5 (a)–(c) most CTs have $PTV \in [5\%, 8\%[$ for all years.

2) FAULTS IN DISTRIBUTION TRANSFORMERS

This section performs an ESDA for the dependent variables *NFT* and *PFT*. This study analyzed 2,848 interruptions caused by the failure of distribution transformers.

Figs. 6 (a)–(c) show the *NFT* distribution by CTs from 2010 to 2012. Fig. 6 (a) refers to *NFT* in 2010; 135 CTs have $NFT \in]0, 3]$. Most CTs have *NFT* in this range for all years.

The CTs with the highest *NFT* were found mainly in the south region in 2010 and the south plus northeast regions in 2011. *NFT* fell drastically in 2012. Power utilities periodically organize task forces to replace damaged equipment (such as transformers) in critical CTs. It cannot be conclusively stated that *NFT* fell in 2012 due to this, but it is a relevant hypothesis to consider. The city's central region has low *NFT* for all years.

Table 4 shows some descriptive statistics metrics for *NFT* by CTs from 2010 to 2012. On average, each CT has between two and four annual faults. However, in 2011, a single CT was the target of 27 shutdowns in distribution transformers.

TABLE 4. Statistical summary for NFT by census tracts.

Parameter	Year Evaluated		
	2010	2011	2012
Max.	21	27	13
Min.	0	0	0
Mean	3.55	3.83	2.08
Standard Deviation	3.63	4.28	2.35
Total	1069	1153	626

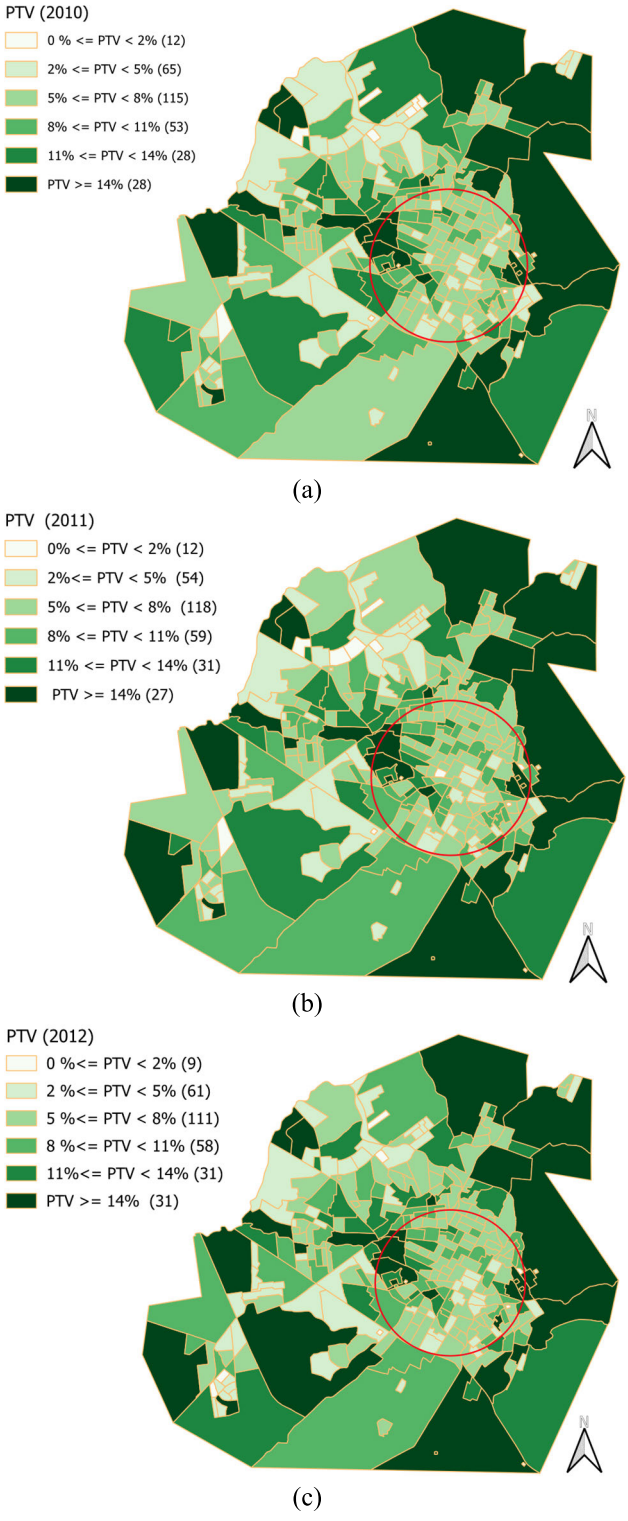

FIGURE 5. PTV by CTs for the years 2010 (a), 2011 (b), and 2012 (c).

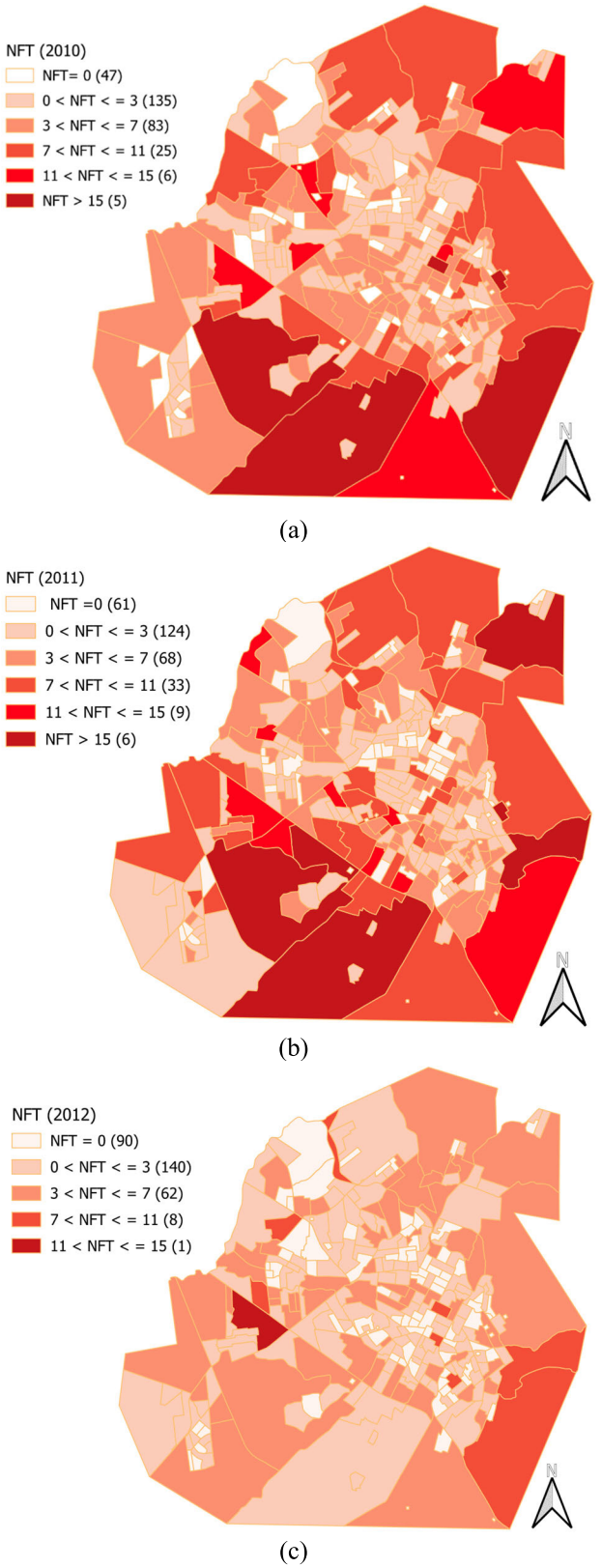

FIGURE 6. NFT by CTs for the years 2010 (a), 2011 (b), and 2012 (c).

On the other hand, thematic maps and statistical summaries are shown in Fig. 7 and Table 5, respectively, for the *PFT* variable.

Figs. 7 (a)–(c) shows *PFT* distribution by CTs from 2010 to 2012. CTs with the highest *PFT* were found in the southeast, west, and northwest regions in 2010 and in the southeast, west, northwest, and northeast areas in 2011.

PFT fell drastically in 2012, too, due to the *NFT* drop in 2012 – Fig. 6 (c).

Table 5 shows some descriptive statistics metrics for *PFT* by CTs from 2010 to 2012. It is worth pointing out that *PFT* has maximum values exceeding 100% for all years. This is possible because several shutdowns can occur in the same transformer over the year in a CT_i, that is, if $NFT_i > T_i$, then $PFT_i > 100\%$ according to (9).

TABLE 5. Statistical summary for PFT by census tracts.

Parameters	Year Evaluated		
	2010	2011	2012
Max.	400.00	366.67	300.00
Min.	0.00	0.00	0.00
Mean	48.39	49.30	30.83
Standard Deviation	50.11	53.22	45.34

Finally, there is a distinction between *NFT* and *PFT*. *NFT* only represents the number of transformer faults per CT. On the other hand, *PFT* is a measure of CTs' vulnerability to faults, as it considers the percentage of damaged transformers over the year for each CT.

3) GEOGRAPHICALLY WEIGHTED SUMMARY STATISTICS

Spearman's correlation global coefficient is shown in Table 6 between $PTV - PFT$ and $PTV - NFT$ variables for the years 2010 to 2012, according to (1). *PTV* and *NFT* have a moderate positive global correlation for all years. Thus, a cause-effect relationship can exist between them; therefore, an increase in *PTV* causes an increase in *NFT*. Conversely, there is no significant global correlation between *PTV* and *PFT* variables. Additional studies must be performed to confirm or refute this hypothesis.

TABLE 6. Spearman's global correlation coefficient.

Pair of Variables	Year Evaluated		
	2010	2011	2012
PTV and *PFT*	0.1261	0.1573	0.2405
PTV and *NFT*	0.3535	0.3390	0.4482

Noteworthy, global metrics do not consider neighborhood structure represented by SWM. Additionally, correlation at the global level represents all cities' CTs by a single numerical value. Therefore, global and local correlations can present different results, and no correlation at a global level may be confirmed at the local level or vice versa [33].

For a more detailed study, GWEA is performed considering neighborhood structure via SWM, obtained from the Gaussian kernel in (3). Thematic maps from GW metrics (4)–(7) are done in this section.

In this sense, Figs. 8 (a)–(c) show the standard deviation GW for *PTV* by CTs from 2010 to 2012. For all years, there has been a high local variability in peripheral regions in the extreme south, southeast, and northeast. Noteworthy, these regions are close to the city's rural area under study. High local variability indicates nearby CTs with very different *PTV* values.

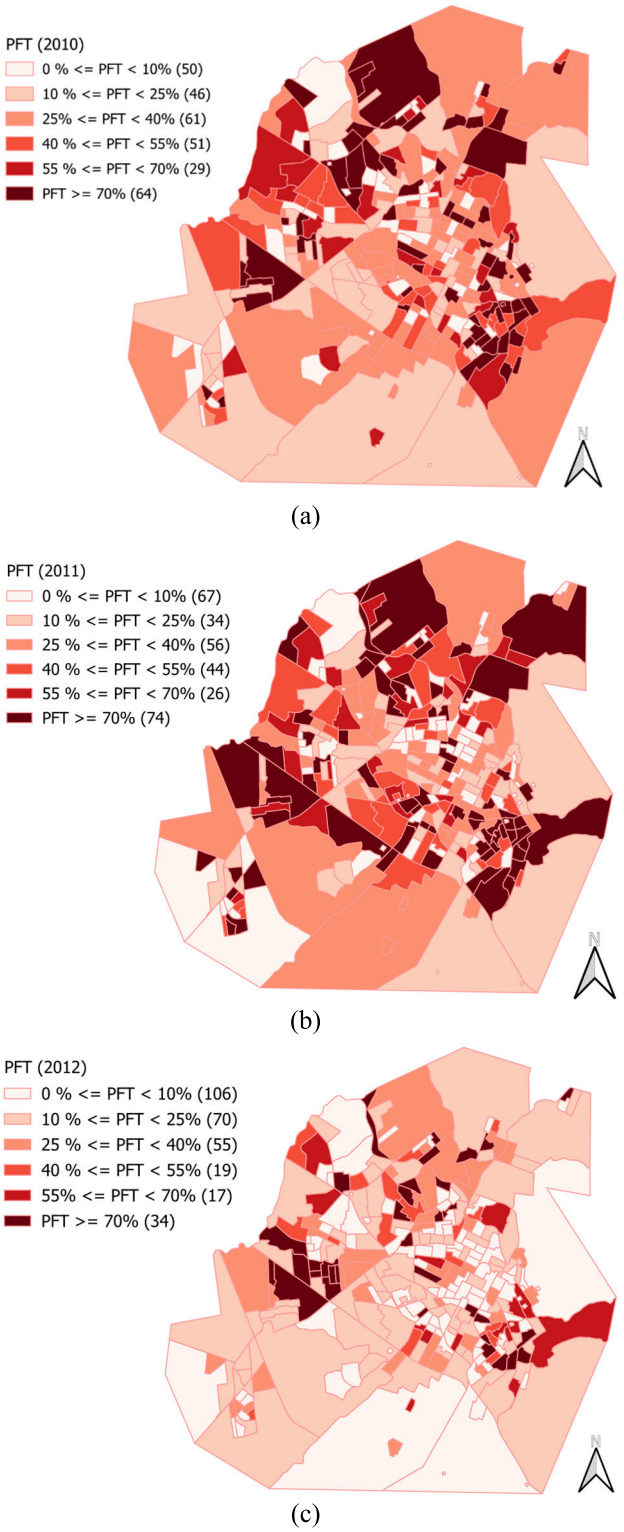

FIGURE 7. PFT by CTs for the years 2010 (a), 2011 (b), and 2012 (c).

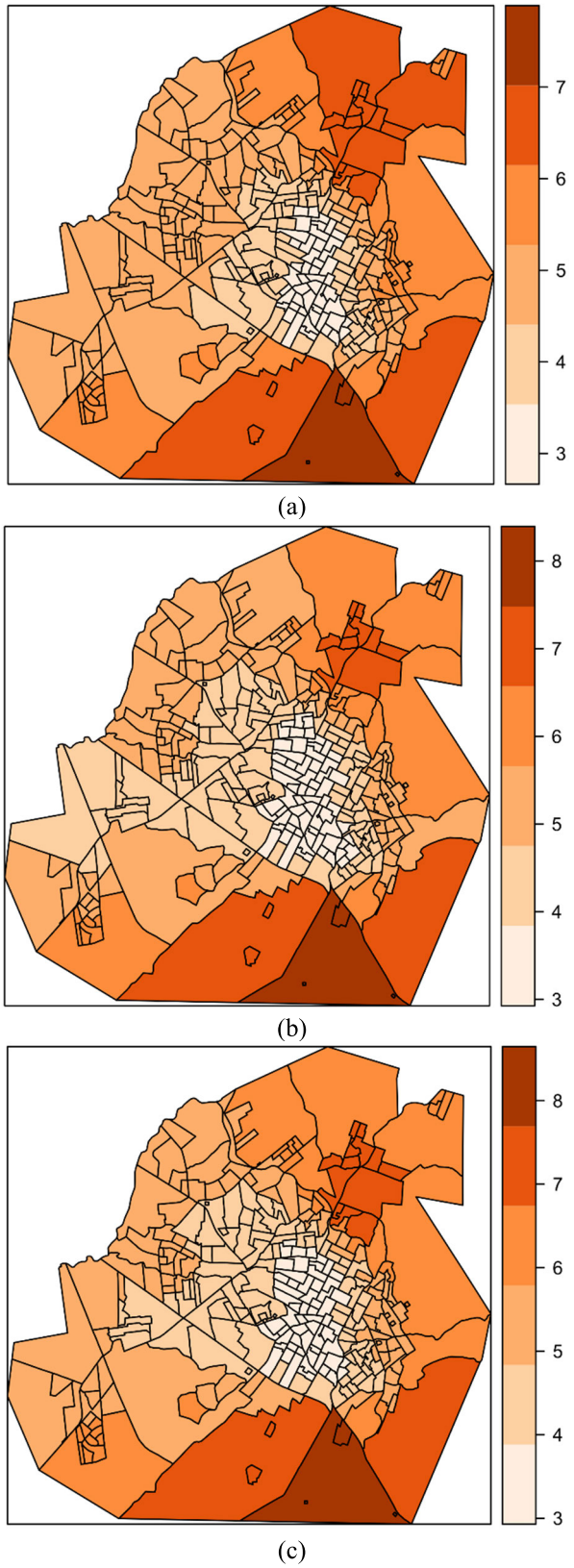

FIGURE 8. GW standard deviation for PTV by CTs for the years 2010 (a), 2011 (b), and 2012 (c).

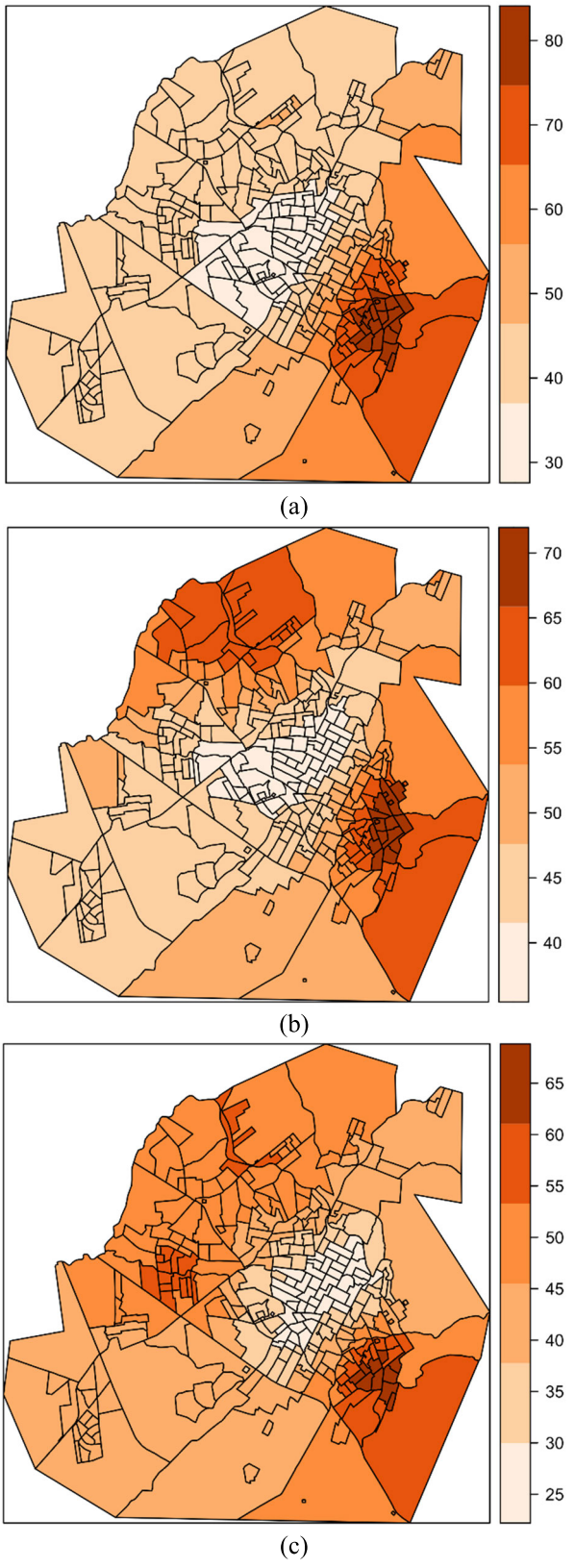

FIGURE 9. GW standard deviation for PFT by CTs for the years 2010 (a), 2011 (b), and 2012 (c).

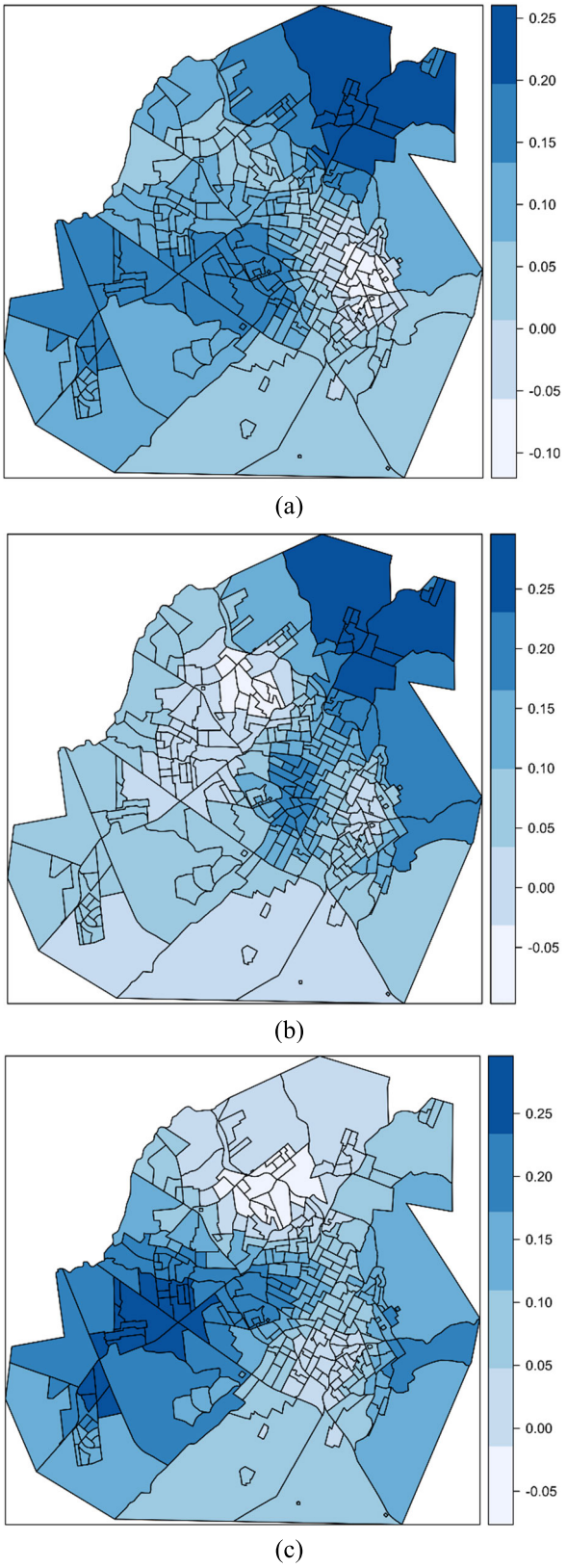

FIGURE 10. GW local Pearson's correlation between NFT and PTV for the years 2010 (a), 2011 (b), and 2012 (c).

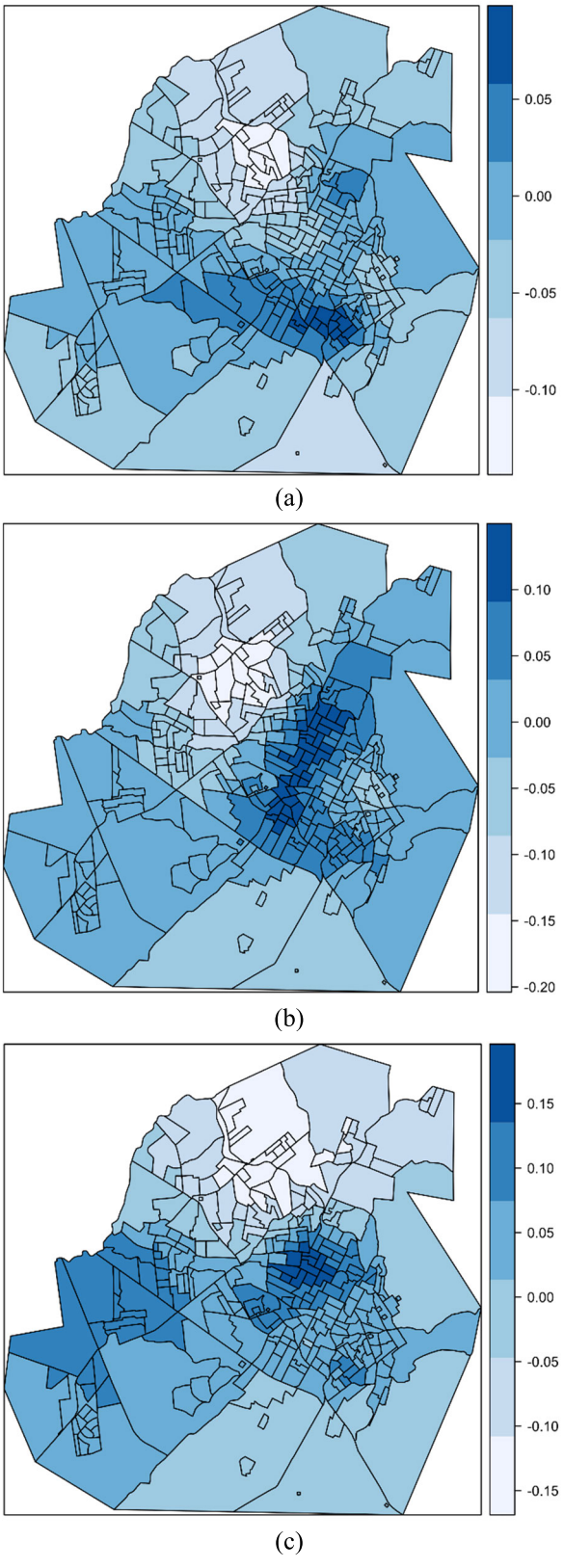

FIGURE 11. GW local Pearson's correlation between PFT and PTV for the years 2010 (a), 2011 (b), and 2012 (c).

Figs. 9 (a)–(c) show the standard deviation GW for *PFT* by CTs from 2010 to 2012. There is a high local variability in peripheral regions in the extreme southeast in 2010 and northwest and southeast regions in 2011 and 2012. Noteworthy is the attenuation of regional variability in 2012.

GW correlation is shown in Figs. 10 (a)–(c) between *NFT* and *PTV* from 2010 to 2012. Both variables have a non-stationary relationship with a moderate positive GW correlation in the northeast (in 2010 and 2011) and southwest (in 2012) regions. A positive correlation means that, in these regions, a rise in *NFT* is accompanied by an increase in *PTV* and vice versa.

Conversely, Figs. 11 (a)–(c) show the GW correlation between *PFT* and *PTV* from 2010 to 2012. Both variables have a non-stationary relationship; however, they have no significant correlation in all CTs and years evaluated.

Therefore, from Fig. 10, there is numerical evidence that tree vegetation (*PTV*) is a relevant factor associated with interruptions in the utility grid (*NFT*) in the northeast (in 2010 and 2011) and southwest (in 2012) regions.

On the other hand, the GW correlation between *PTV* and *PFT* is not significant from Fig. 11. Therefore, only the tree vegetation variable does not make regions vulnerable to faults in the city under study. According to Fig. 4, several other variables are associated with faults. Combining several variables could make some city regions vulnerable to faults.

ESDA applied in this study supports the execution of more sophisticated SDA models to add other variables that, together with *PTV*, can make some city areas vulnerable to faults.

Noteworthy, the non-stationary relationship between *PTV* and *NFT* and between *PTV* and *PFT* indicates that a global spatial regression model would not be appropriate for modeling faults; conversely, a local regression model would be more relevant to represent the non-stationarity at the local level [33].

Finally, an adaptive bandwidth with the influence of 30 closest CTs is considered in GWEA. The closest CT value corresponds to 10% of CTs in the city under study where there are $n = 301$ CTs [26].

IV. CONCLUSION

In this study, a fundamental step of spatial data analysis (SDA) called exploratory spatial data analysis (ESDA) was applied to produce thematic maps that showed the city's regions or census tracts (CTs) whose feeders are vulnerable to faults due to tree vegetation.

Actual data georeferenced by CTs were vital for this study: the dependent or study variables number of faults in transformers (*NFT*), percentage of faults in transformers (*PFT*), and independent or explanatory variable percentage of tree vegetation (*PTV*).

PTV variable was obtained from an enhanced method for tree vegetation mapping by areas using multilayer perceptron (MLP) artificial neural network (ANN) trained on high-resolution images from Google Earth.

ESDA was performed by applying descriptive statistics metrics and visualizing the spatial distribution of the variables in the city's geographic space. Geographically weighted (GW) summary statistics showed local spatial variability for *PFT* and *PTV* variables.

GW Pearson's local correlation showed a moderate positive correlation between *NFT* and *PTV* in the northeast (in 2010 and 2011) and southwest (in 2012) regions. Conversely, the GW correlation between *PTV* and *PFT* is insignificant for all evaluated years.

Therefore, tree vegetation (*PTV*) is a relevant factor associated with interruptions (*NFT*) in the utility grid for some regions. Conversely, only the tree vegetation variable does not become regions vulnerable to faults in the city. There are other variables traditionally associated with faults: electrical discharges, fires, equipment failures, and adverse weather conditions. The set of several variables can make some city regions vulnerable to faults.

ESDA performed in this study will support the implementation of more advanced models for estimating areas whose feeders are vulnerable to faults. Furthermore, incorporating other variables, such as those mentioned in the previous paragraph, will provide more robustness in future works.

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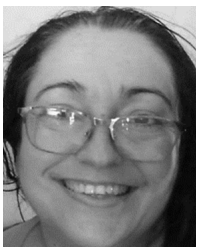

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Power Distribution Systems' Vulnerability by Regions Caused by Electrical Discharges

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Abstract: Energy supply interruptions or blackouts caused by faults in power distribution feeders entail several damages to power utilities and consumer units: financial losses, damage to power distribution reliability, power quality deterioration, etc. Most studies in the specialized literature concerning faults in power distribution systems present methodologies for detecting, classifying, and locating faults after their occurrence. In contrast, the main aim of this study is to prevent faults by estimating the city regions whose power grid is most vulnerable to them. In this sense, this work incorporates a geographical-space study via a spatial data analysis using the local variable electrical discharge density that can increase fault risks. A geographically weighted spatial analysis is applied to data aggregated by regions to produce thematic maps with the city regions whose feeders are more vulnerable to failures. The spatial data analysis is implemented in QGIS and R programming environments. It is applied to the real data of faults in distribution power grid transformers and electrical discharges in a medium-sized city with approximately 200,000 inhabitants. In this study, we highlight a moderate positive correlation between electrical discharge density and the percentage of faults in transformers by regions in the central and western areas of the city under study.

Keywords: power distribution systems (PDS); spatial data analysis (SDA); steady-state fault

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1. Introduction

Energy supply interruptions (ESIs) or faults are events associated with the operational process of power distribution systems (PDSs) [1]. They are classified into two categories: temporary and permanent. Transient faults occur in a very short time (in the order of milliseconds). After this, a PDS is automatically restored. On the other hand, there is no electrical system restoration in the occurrence of a permanent fault. In this case, the PDS returns to its nominal operating conditions after the fault location and removal [2].

An ESI causes substantial financial losses for power utilities and consumer units (CU) that share the same distribution power grid. In this sense, electrical discharges represent a relevant cause of an ESI; furthermore, they are associated with damage to electrical equipment [3]. With the energy market's development, an energy supply with good quality and reliability become a relevant factor for power grid operators [4]. Thus, an ESI depreciates the PDS indices of quality and reliability [5], where fault events represent 80% to 90% of the times without energy supply to CUs [6].

In this context, there are several factors associated with ESIs: (i) adverse weather conditions, (ii) fires, (iii) tree vegetation close to the overhead power distribution, (iv) human failures, and (v) equipment failures (due to manufacturing defect, lack of maintenance, or obsolescence) [2,7].

The National Electrical System Operator (ONS) presented a statistical report of the main factors responsible for faults in Brazilian power grids, as shown in Table 1. Adverse

weather conditions such as electrical discharges, heavy rains, storms, and wind gusts are the main ESI causes, with approximately 30% participation [7].

Table 1. Main causes of disturbances in Brazilian power grids.

Fault Reasons	Number of Annual Faults					
	2020	**%**	**2021**	**%**	**2022**	**%**
Adverse Weather Conditions	708	29.75%	704	29.82%	743	35.01%
Fires	587	24.66%	633	26.81%	250	11.80%
Equipment Failures	144	6.05%	167	7.10%	127	6.00%
Tree Vegetation	135	5.67%	87	3.70%	97	4.60%
Human Failures	109	4.58%	141	6.00%	136	6.41%

Additionally, Table 1 shows the reasons associated with ESI, representing an average among Brazilian cities. In this sense, consider a town on the northeast coast of Brazil, where wind gusts are the main reason for faults in distribution feeders. Meanwhile, they are not associated with ESIs for another city in the central region with different weather conditions such as those of coastal towns. Therefore, various factors cause ESIs in other areas. Thus, incorporating spatial data analysis (SDA) helps understand which factors are relevant to a city region becoming more vulnerable to ESIs.

To the best of the authors' knowledge, most studies in the specialized literature prioritize fault detection and location after their occurrence. For example, in [8–12], fault detection and classification were performed considering various fault scenarios. On the other hand, estimating regions in which the power grid distribution is more vulnerable to failures provides crucial information for planning measures to avoid inconveniences and additional costs arising after steady-state faults occur.

Reducing the time of ESIs is essential for improving the power quality in a PDS [13]. In this context, estimating regions vulnerable to failures assists in decision-making and identifying critical areas that must be prioritized for replacing damaged equipment. Thus, power supply with more excellent quality and reliability is ensured.

In this sense, this study aims to incorporate the analysis of geographic space for estimating city regions whose utility grid is more vulnerable to failures. It is assumed that some local factors or variables interfere with the utility grid, making it more vulnerable to failures. These local factors or variables are shown in Table 1: adverse weather conditions such as electrical discharges, tree vegetation, and fires.

In this context, geographic information systems (GISs) are essential to an SDA for estimating regions vulnerable to faults. They are used in managing, planning, and controlling power grids. They comprise geoprocessing tools for collecting, storing, analyzing, and displaying geographic data. A GIS application enables power grid visualization, which is affected by local factors that belong to the region where it is located, as well as other power utility assets, such as substations, transformers, and protection equipment [14,15].

According to [16], the first step in an SDA is to perform an exploratory analysis of spatial or georeferenced data. An exploratory analysis consists of the graphical representation of data available on thematic maps or heat maps and the incorporation of descriptive statistics metrics considering the geographic space.

Thus, this study performs a geographically weighted exploratory analysis (GWEA) on geographic data from a Brazilian city. Crucial for this study are the georeferenced data and the weighting matrix. In the former, georeferenced data are those whose geographical coordinates of the phenomenon under investigation are known. In the latter, the weighting matrix represents the urban zone's neighborhood structure because it allows for characterizing the influence among small regions or census tracts (CTs).

Therefore, the first step of an SDA with aggregated data in small regions or CTs is performed in this study: GWEA. It is performed based on a local factor or variable frequently associated with faults in a PDS: electrical discharges. It is determined whether these local variables become the utility grid in some city regions more vulnerable to failures.

Therefore, GWEA is applied to data aggregated by small areas or CTs to produce thematic maps with the city regions whose feeders are more susceptible to failures.

1.1. Literature Review

A fault incidence matrix method was introduced in [4] to evaluate the reliability of a medium voltage distribution grid. The authors presented a reliability calculation unit called M-Segment-N-tie-switch (MSNT), which included several distribution grid structures; thus, the method became applicable for different systems' operating conditions. Each MSNT was associated with a matrix of a power supply path obtained by inversing the node branch incidence matrix. Subsequently, it was used to obtain the fault incidence matrix. The reliability indices for each system load were calculated, where the fault incidence matrix and the vectors of fault events' characteristics were applied. The technique improved the reliability efficiency, as it avoided some procedures such as the fault events enumeration and the repetitive search for the fault influence range. Furthermore, the technique is intuitive and enables a sensitivity analysis and system vulnerability identification.

In [17], a GIS-based tool for spatial analysis in a PDS was presented. The authors applied spatial analysis techniques such as inverse distance weighting, slope, and contour maps. The load flow results were the input data for the ArcGIS, where several maps were produced to analyze the dynamic behavior of the PDS. The results showed that any disturbances occurring in the PDS were detected and controlled in real-time through the GIS application.

Chen and Kezunovic [18] proposed a fuzzy logic-based tool for the predictive analysis of ESIs. The authors highlighted the extreme climate impacts and their risk to the electrical system's operation. Fuzzy logic was introduced to process weather information. Wind speeds and gusts were input parameters, whereas the fuzzy inference system provides an alert level as the output. A risk map was produced from the vulnerability results and processed using GIS. The risk results were shown as power-cut probability maps. Weather forecasts and operational data belonging to the system improved the decision-making process.

In [5], a data mining-based tool was proposed to determine fault probabilities. The technique was based on the Naive Bayes model, incorporating weather information, assets, and outage history. Feeder sections were classified according to their fault probability using GIS.

Zhou et al. [1] presented a methodology for assessing and managing fault risks in PDSs with distributed generation units. An island partitioning strategy was proposed to improve accuracy and speed in the evaluation process. The best management strategy was calculated by applying an improved genetic algorithm.

Monte Carlo simulation (MCS) was introduced by Goerdin, Smit, and Mehairjan [19] for fault risk analysis in PDSs. A decision-making framework based on fault risk was built from the MCS results to guide asset management processes. The study performed fault predictions with a focus on medium-voltage cables and failures in cable joints.

Muñoz and García [20] presented a GIS-based tool for analyzing the impacts of floods on a PDS. The technique applies a flood risk analysis and the calculation of reliability indices for PDSs.

In [21], a strategy for analyzing the vulnerability of geographically correlated faults was proposed. The authors introduced a DC power flow and a cascade fault model. The detection of areas most vulnerable to faults was performed through the application of a network survival analysis using data obtained via GIS.

A study of impacts on energy supply was presented in [22]. The authors introduced an analysis of extreme weather events that directly affect Indonesia's electrical system. In this sense, electrical discharges represent an essential cause of ESIs in high-voltage transmission lines.

A method for forecasting vulnerability to faults in PDSs was proposed in [23]. Piping and cables on the verge of failure were identified. The authors applied geospatial data,

georeferenced faults clustering, and machine learning. The system components were classified according to the number of failures attributed to them. In this way, a ranking list was created where equipment with a high probability of failure occupied the first places.

An automatic system for locating faults in a PDS was presented by Albasri et al. [24]. The technique was based on a smart meter implementation. A GIS was applied to display the fault location on maps. The voltage sag method was selected to determine the fault location, where a correlation was found between the fault location and the voltage sag at the measurement points. This provided a reduction in the search area for a segment of the feeder with the anomaly.

In [25], an intelligent system for electrical grid monitoring was proposed based on the Internet of Things (IoT) and GIS. The fault state was determined via sensors whose objective was to collect current changes in the PDS. The technique combines information about faults, which is transmitted to the information center using ZigBee technology. A GIS was applied to obtain the disturbance's geographic location.

Finally, Chen et al. [26] presented a predictive method for managing interruptions in a PDS caused by wind gusts and tree vegetation. A GIS-based tool was introduced to correlate power system data with different climate data layers. Maps with areas vulnerable to faults caused by wind gusts and tree vegetation were presented.

1.2. Contributions

This study presents a method for estimating regions vulnerable to faults. The highlights of the proposed approach are described below.

(1) Incorporation of a geographic space to estimate regions vulnerable to failures in a PDS. Previous estimations of areas vulnerable to faults are essential information to aid decision-making and guide preventive actions by the power utilities. Such actions can avoid all inconveniences and additional costs after faults occur in a PDS.
(2) GWEA by regions is accomplished from local variables associated with faults and electrical discharges. GWEA allows for separate exploratory analyses of electrical discharge and faults and the search for local associations between these variables in each region of the city.

1.3. Paper Structure

This paper continues in Section 2 with the relationship between an ESI and electrical discharges. Section 3 presents an overview of an exploratory spatial data analysis (ESDA) (Section 3.1), Spearman's correlation coefficient (Section 3.1.1), spatial analysis with data aggregated by regions (Section 3.2), neighborhood structure with data aggregated by areas (Section 3.2.1), and GWEA (Section 3.2.2). Section 4 presents a case study of the implementation of GWEA in a Brazilian city. Sections 4.1 and 4.2 present a case study description. In Section 4.3, an ESDA is performed, where the variables electrical discharge density (Section 4.3.1) and the percentage of the faults in distribution transformers by regions (Section 4.3.2) are presented. Section 4.3.3 offers the geographically weighted (GW) statistical summary considering the neighborhood structure and weighting among areas. Finally, Section 5 shows the conclusions of this study.

2. Energy Supply Interruptions: Electrical Discharges

Overhead distribution grids are susceptible to ESIs due to adverse weather conditions such as electrical discharges, rains, storms, and wind gusts. Numerous electrical discharge protection measures, such as lightning rod installations, have been implemented, improving insulation levels. However, the ESI caused by electrical discharges is significant according to operational experience [27]. They contribute significantly to both temporary and permanent interruptions [28].

Electrical discharges occur directly and indirectly in transmission and distribution lines. Direct events, which are the most dangerous, happen when an electrical discharge directly strikes the line. On the other hand, indirect events, whose frequency is higher, are

classified as occurring in regions where the power grid is close to the ground. These events are critical for distribution systems since they are characterized by a low critical flashover voltage [29].

According to Teru and Okabe [30], the faults that occur in Japanese distribution lines are mainly caused by electrical discharges. They are the leading causes of ESIs in China, Japan, and Malaysia [28]. In this sense, Figure 1 shows the monthly average number of disturbances caused by electrical discharges over ten years in Brazil. It is observed that there are more discharges in the first and fourth quarters of the year because this period is more humid and rainy [7].

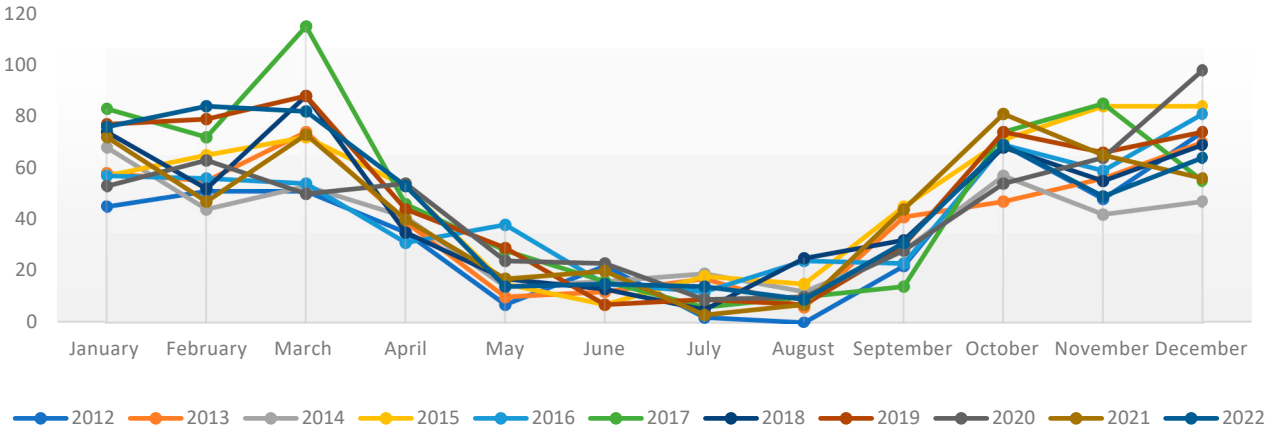


Figure 1. Disturbances caused by electrical discharges between 2012 and 2022 in Brazil [7].

3. Spatial Data Analysis

An SDA aims to measure properties and relationships considering the phenomenon of spatial location. In this way, the geographic space is incorporated into the study; thus, there is a visual perception via the spatial distribution of the problem under analysis [31].

3.1. Exploratory Spatial Data Analysis

An ESDA is a graphical presentation of georeferenced data on thematic maps or heat maps. An ESDA is a set of techniques to describe and explore spatial or georeferenced data [32]. In this context, a series of metrics derived from descriptive statistics are incorporated; thus, one can identify spatial patterns and formulate hypotheses related to data distribution in geographical space [31,33]. An ESDA is the first step toward a study in SDAs.

3.1.1. Spearman's Correlation Coefficient

Spearman's global correlation coefficient in (1) estimates whether two variables are associated; additionally, they do not need to be linearly associated [31].

The coefficient r_s is obtained by arranging the values of two variables in ascending (or descending) order, replacing each original value with a positive integer that represents the ordinal position of the original value in the data series. This correlation coefficient is calculated via (1):

$$r_s = 1 - \frac{6\sum d^2}{n(n+1)(n-1)} \quad (1)$$

where n is the number of CTs; d^2 is the square of the difference between the ordinal position of two variables in the same CT.

This coefficient varies in the range $[-1, 1]$, where a positive correlation coefficient implies a directly proportional relationship between the variables, while a negative coefficient implies an inversely proportional relationship.

Additionally, Spearman's coefficient is not sensitive to asymmetries in the data distribution or outliers. It indicates the degree of dependence between two variables and is an important tool for formulating hypotheses about spatial dependence and cause-and-effect relationships.

Finally, it is worth mentioning that Spearman's coefficient is a global metric whose influence on the neighborhood is not considered. In contrast, the next section presents local metrics, where the influence of neighboring CTs is represented by a spatial weighting matrix.

3.2. Spatial Analysis with Data Aggregated by Regions

This study is associated with an SDA using data aggregated by regions. It consists of methods that allow for the analysis of georeferenced data whose location is associated with areas delimited by polygons. It is applied to address events aggregated by municipalities, neighborhoods, or CTs, where the exact location of events is unavailable; however, an aggregated value by area is available. The visualization of data aggregated by regions is usually performed through thematic maps with the spatial pattern phenomenon under study [16,34,35].

With data aggregated by areas, an SDA allows for the use of public information by the Brazilian Institute of Geography and Statistics (IBGE). The IBGE does not disclose the private data of the individuals interviewed for confidentiality reasons. These data are grouped into small areas or polygons called CTs, whose area is a function of population density: CTs with a higher population density have a smaller area and vice versa.

3.2.1. Weighting Matrix

Studies in the specialized literature that apply SDA techniques commonly belong to epidemiology, botany, criminology, and mineral resources' prospecting. In these fields of study, there is a typical neighborhood structure based on spatial proximity or Euclidean distance among the centroids of areas; closer areas have more significant influence than more distant areas [36]. This neighborhood structure based on spatial proximity among areas is also applied in our study. Numerous other neighborhood structures are based on spatial proximity among areas evaluated in [37].

In this context, a spatial weighting matrix (SWM) $\boldsymbol{W}_{(n\times n)}$ is formed using weights w_{ij}, where the degree of spatial relationship or spatial dependence between the variables observed in the areas i and j is estimated. An SWM is an essential tool for GW modeling. Three key elements must be considered for SWM building: distance type, kernel function, and bandwidth [38,39].

The SWM $\boldsymbol{W}_{(n\times n)}$ is a symmetric matrix built for a set of n areas $\{A_1, \ldots, A_n\}$, where, from each element w_{ij}, the spatial relationship or spatial dependence between the variables observed in A_i and A_j areas are estimated.

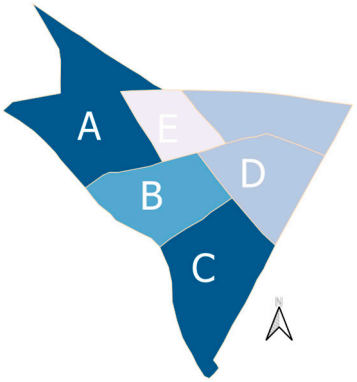
Figure 2 shows an illustrative example of obtaining a weighting matrix $\boldsymbol{W}_{(n\times n)}$. In Figure 2a, there is a city with five CTs (A, B, C, D, and E); on the other hand, in Figure 2b, there is a weighting matrix $\boldsymbol{W}_{(5\times 5)}$ for a city with $n = 5$ CTs and whose obtaining rule is according to (2). The CT A has borders in common with two other CTs: B and E. Therefore, the total number of borders of CTs for A is equal to two. Thus, the weighting matrix elements, w_{12} (weighting between A and B) and w_{15} (weighting between A and E), are equal to $1/2$ according to (2). Other CTs do not have borders in common with A (C and D for example); therefore, they have zero weighting: $w_{13} = 0$ and $w_{14} = 0$.

$$w_{ij} = \begin{cases} \frac{1}{Number\ of\ Borders\ CTs}, & if\ i\ and\ j\ are\ borders\ CTs \\ 0, & otherwise \end{cases} \quad (2)$$

3.2.2. Geographically Weighted Statistics Metrics

Geographically weighted models (GWMs) are techniques belonging to non-stationary spatial statistics, which incorporate local spatial relationships into their structure intuitively and explicitly [38,40]. Their application is suitable for situations where the spatial data need to be better described using a local model since they enable the estimation and mapping

of each location in the geographical space under study. GWMs' outputs are commonly mapped to provide a helpful tool that usually precedes a more sophisticated statistical analysis [38].



(**a**)

	A	B	C	D	E
A	1	0.5	0	0	0.5
B	0.25	1	0.25	0.25	0.25
C	0	0.5	1	0.5	0
D	0	0.33	0.33	1	0.33
E	0.33	0.33	0	0.33	1

(**b**)

Figure 2. An illustrative example of constructing a weighting matrix: (**a**) five census tracts of the city under study; (**b**) corresponding weighting matrix.

Spatial weighting functions are crucial elements in GW modeling, whose objective is quantifying the spatial relationship or spatial dependence between the observed variables via SWM [38].

In this context, in our study, SWM elements w_{ij} are obtained via a Gaussian kernel application in (3) and shown in Figure 3. It is a monotonic decreasing function of the distance among centroids of areas A_i and A_j. These functions have a b parameter for the bandwidth, which controls their decay rate [34,38].

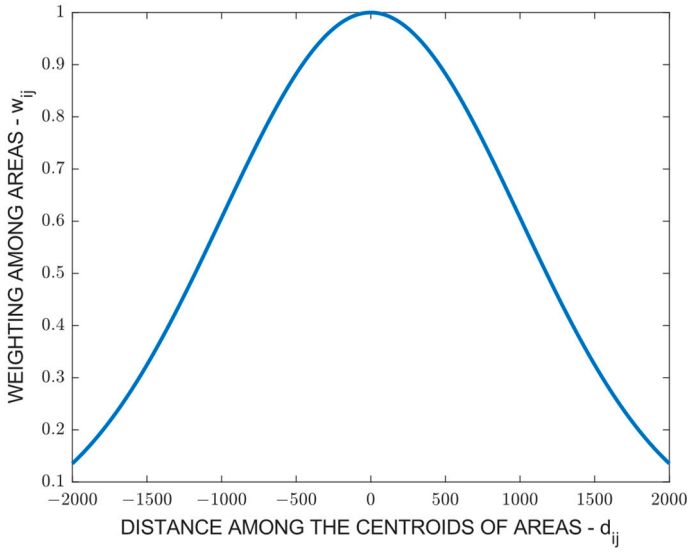


Figure 3. Gaussian kernel function with bandwidth b = 1000.

For example, Figure 3 shows the most conventional kernel function or weighting function from Gollini et al. [38]: the Gaussian kernel. It is obtained using (3) with an arbitrary bandwidth $b = 1000$.

$$w_{ij} = exp\left(-\frac{1}{2}\left(\frac{d_{ij}}{b}\right)^2\right) \quad (3)$$

where d_{ij} represents the distance between the centroids of areas A_i and A_j, and b is the bandwidth parameter.

It is worth pointing out that kernel the function selection is performed empirically. The Gaussian kernel is chosen because it follows Tobler's first law, where the weighting between nearby CTs is greater than for distant CTs [36]. Furthermore, as shown in Figure 3, the Gaussian kernel is a continuous function with a smooth decay. Additionally, the decay rate is regulated through a bandwidth parameter b. Other kernel functions with characteristics analogous to the Gaussian kernel could be applied.

A GW local summary statistic can be obtained from a spatial data set. Therefore, from the attributes z_i and y_i at any point i, the following metrics can be obtained through (4)–(7): mean GW, standard deviation GW, Pearson's correlation coefficient GW, and covarianceGW, respectively [38,41].

$$m(z_i) = \frac{\sum_{j=1}^{n} w_{ij} z_j}{\sum_{j=1}^{n} w_{ij}} \tag{4}$$

$$s(z_i) = \sqrt{\frac{\sum_{j=1}^{n} w_{ij} \left(z_j - m(z_i)\right)^2}{\sum_{j=1}^{n} w_{ij}}} \tag{5}$$

$$\rho(z_i, y_i) = \frac{c\ (z_i,\ y_i)}{s(z_i) s(y_i)} \tag{6}$$

$$c(z_i, y_i) = \frac{\sum_{j=1}^{n} w_{ij}\ \left(z_j - m(z_i)\right)\ \left(y_j - m(y_i)\right)}{\sum_{j=1}^{n} w_{ij}} \tag{7}$$

where w_{ij} is an element of SWM $\boldsymbol{W}_{(n \times n)}$.

4. Results and Discussion

In this study, the spatial regression terminology is applied where there is a dependent or study variable whose distribution in geographic space is partially explained through a set of independent or explanatory variables. In our study, the dependent variable consists of a percentage of damaged distribution transformers that resulted in steady-state failures.

According to [42], the power utility of Paraná State, Brazil (COPEL), presented an annual failure rate of 1.2% in distribution transformers, where electric discharges caused 30% of these occurrences.

In this sense, the independent or explanatory variable analyzed is the surface density of electric discharges per km^2. They are associated with faults in distribution feeders. It is worth pointing out that all transformers and all-electric discharges in the city under study were georeferenced; their geographic coordinates were known, making it possible to apply an SDA.

4.1. Case Study in A Brazilian City

This work performs an exploratory spatial data analysis to evaluate how electric discharges influence some city regions more vulnerable to steady-state faults. In this sense, Figure 4 shows an overview of the main steps of our study: the (1) acquisition of georeferenced data (damaged distribution transformers and electric discharges), (2) exploratory analysis of spatial data, and (3) display of the results in thematic maps.

The power utility applied in the simulations had a real feeder located in São Paulo State, Brazil. The simulations used QGIS version 3.30 and R software version 4.1.2 [37]. QGIS is a GIS that allows for the visualization, editing, and analysis of spatial data. QGIS allows for the checking, processing, previewing, updating, and presentation of spatial data. Different types of data can be displayed through maps, where it is possible to understand patterns, trends, and relationships [15].

On the other hand, R is a programming language and free statistical and graphical computing software. It contains many libraries or packages for numerical analysis, such as

the GWmodel version 2.3.1 applied in this study. Unlike QGIS, R was not initially created as a GIS; however, it can perform functions like a GIS.

All simulations in this study were performed on a computer with a Windows operating system; an Intel Core i7 processor, 1.8–2.3 GHz, 64-bit; and 16 GB of RAM.

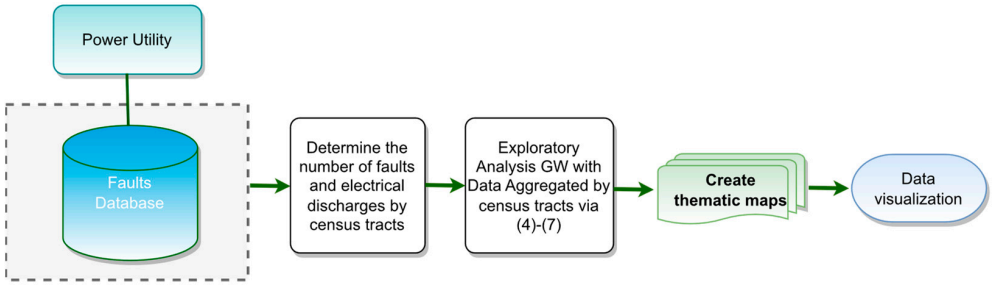


Figure 4. Overview of an exploratory analysis GW of spatial data.

4.2. Database Description

In this section, the variables electrical discharge density (EDD) and the percentage of permanent faults in distribution transformers by regions (NFT) are evaluated because electrical discharges can make some city regions more vulnerable to faults. NFT and DED are variables obtained in small areas called CTs. CTs show the public demographic census data produced by IBGE because individual data are confidential.

Consider a CT_i of the city under analysis with $i = 1, \cdots, NCT$, where NCT corresponds to the number of CTs. The dependent or study variable NFT_i is shown in (8). It is obtained from the ratio between the number of permanent faults per CT_i that caused faults in power distribution transformers F_i and the total number of transformers in CT_i represented by T_i. NFT represents the probability of faults in the transformer per CT. Thus, the number of failures in transformers of CT_i is approximately proportional to its total number of transformers. It is noteworthy that the faults in our study are associated with transformers because they are georeferenced:

$$NFT_i = \frac{F_i}{T_i} \; with \; i = 1, \cdots, NCT \tag{8}$$

The surface density of electrical discharges (EDD) is an independent or explanatory variable, and it is expressed in (9). It is the ratio between the number of electrical discharges L_i and the CT_i of area S_i per km^2. The EDD variable obtained is more effective than the total number of electrical discharges by CTs because the number of electrical discharges that reach a CT is approximately proportional to its area.

$$EDD_i = \frac{L_i}{S_i} \; with \; i = 1, \cdots, NCT \tag{9}$$

Figure 5 shows the main reasons that caused 9266 ESIs in more than one minute in a Brazilian city over four years. According to Figures 1 and 5, weather conditions are the leading causes of interruptions in the utility grid. Figure 5 shows that meteorological conditions such as wind gusts and electrical discharges accounted for 21.9% of ESIs.

However, it is worth noting that Figure 5 shows numerical values from the power utility database without a refined treatment. Thus, technicians in the field identified the percentages associated with the factors responsible for ESIs without further investigation. For example, the leading causes of ESIs were "equipment failures", with an occurrence of 44.7%. However, this is a "black box" because equipment failures can be caused by numerous factors such as equipment obsolescence, adverse weather conditions (rains, wind gusts, electrical discharges), overload, clandestine connections, or human failures. In this sense, the overview of the numerous factors responsible for ESIs is more important than the numerical values shown in Figure 5.

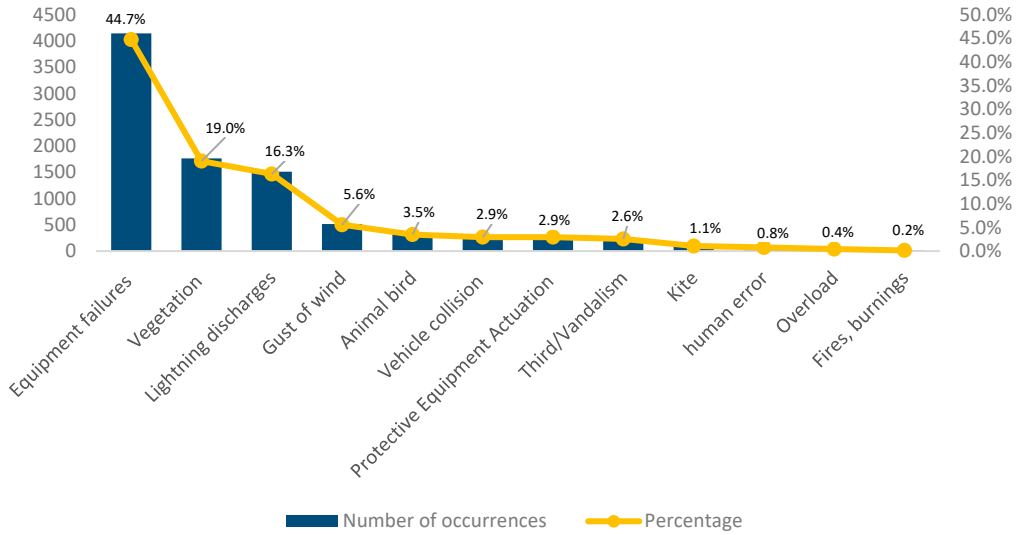


Figure 5. Main factors that caused permanent faults in a Brazilian city.

4.3. Exploratory Spatial Data Analysis

4.3.1. Electrical Discharges

Electrical discharges occasionally reach the utility grid; consequently, they cause disturbances and ESIs. Figure 6 shows a monthly distribution of 2036 electrical discharges that occurred in the city under study over four years. Table 2 shows some descriptive statistics metrics by CTs. From Table 2, each CT experiences between one and two electrical discharges by year on average. However, a single CT was the target of 40 electrical discharges in 2011.

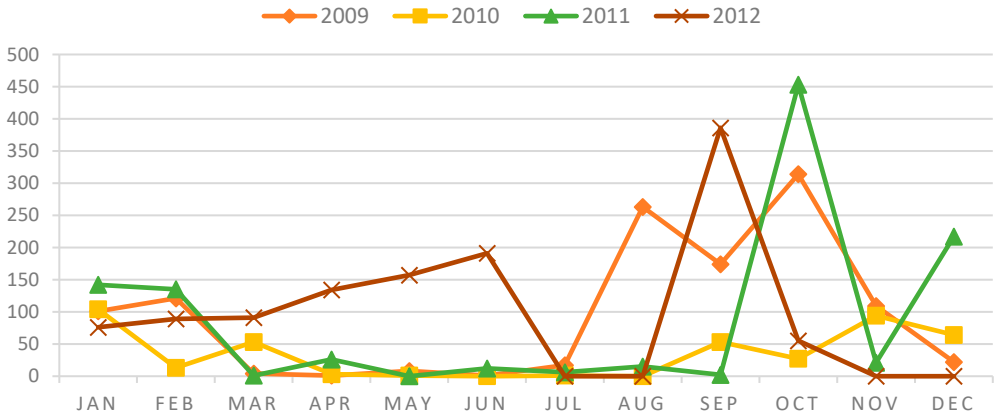


Figure 6. Monthly distribution of electrical discharges.

Table 2. Statistical summary of electric discharges by census tracts.

Parameters	Evaluated Years			
	2009	**2010**	**2011**	**2012**
Maximum	39	26	40	39
Minimum	0	0	0	0
Average	2.35	0.79	1.82	1.80
Standard deviation	5.78	2.52	4.55	5.03
Total number	707	239	548	542

Figure 7a–d show the heat maps obtained using the Gaussian kernel density considering the electrical discharge distribution for the years 2009, 2010, 2011, and 2012, respectively. The Gaussian kernel is based on electrical discharge clustering based on a predefined distance called bandwidth. Therefore, CTs with high concentrations of electrical discharges (indicated by red regions) have high Gaussian kernel values. The maximum Gaussian kernel value in 2010 corresponds to approximately half of the value observed in other years, for example. Therefore, this variation indicates a heterogeneous distribution of electrical discharges over the years.

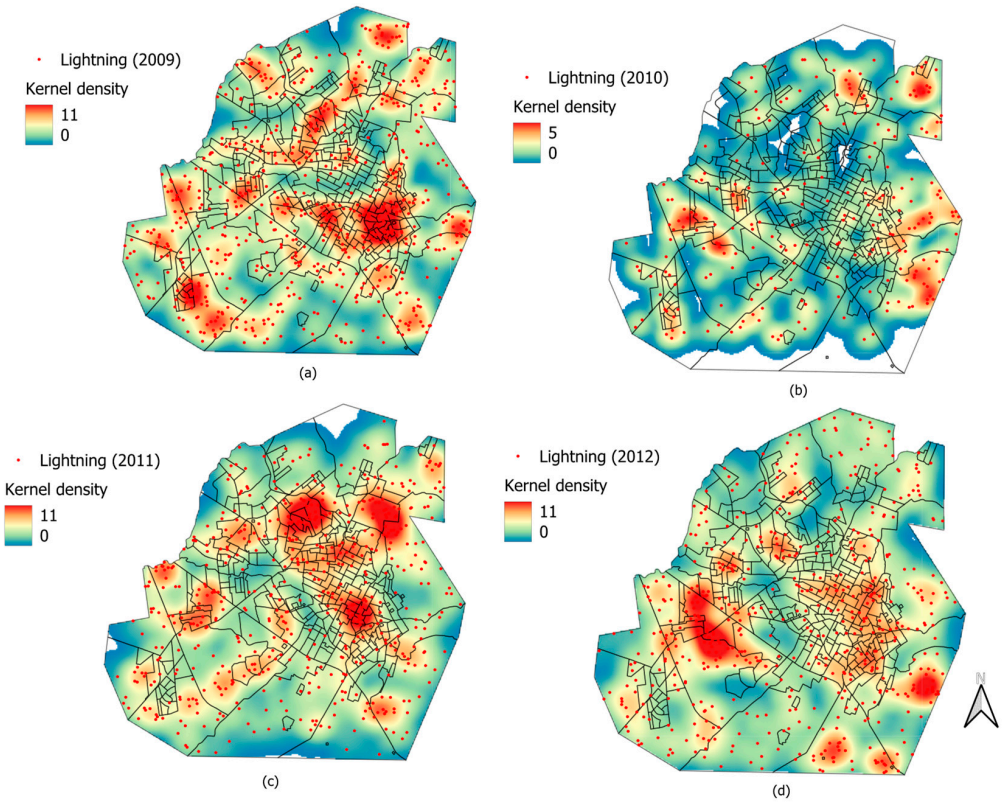


Figure 7. Heatmap considering electric discharge distribution by census tracts for the years: (**a**) 2009, (**b**) 2010, (**c**) 2011, and (**d**) 2012.

The *EDD* variable considers the total area of each CT in km^2. In many cases, the number of electrical discharges in a CT is proportional to its area. Figure 8a–d show thematic maps for *EDD* over the years 2009, 2010, 2011, and 2012, respectively. The numbers in parentheses indicate the CTs whose *EDD* is in the designated range. For example, in Figure 8a, which corresponds to the *EDD* in 2009, it is observed that 41 CTs have an *EDD* with a value between 2 and 6 electrical discharges per km^2. Most of these CTs belong to the city's periphery and are represented in light green.

According to Figure 8, the *EDD* has a similar pattern for all years analyzed. CTs with a high *EDD* belong to the central regions and move toward the center-southeast and center-northwest of the city. There is an exception in Figure 8b. According to Table 2, 2010 is an atypical year (outlier) with few rains and electrical discharges. The regions with a high *EDD* (central, northwest, and southeast) are more densely urbanized and have many elevated structures such as buildings, antennas, and tree vegetation that attract electrical discharges. Overhead lines also attract electrical discharges; thus, they have protective equipment to deal with them. On the other hand, CTs located on the city's periphery and, therefore, close to rural areas show a reduced *EDD*. These CTs contain extensive flat areas with open fields.

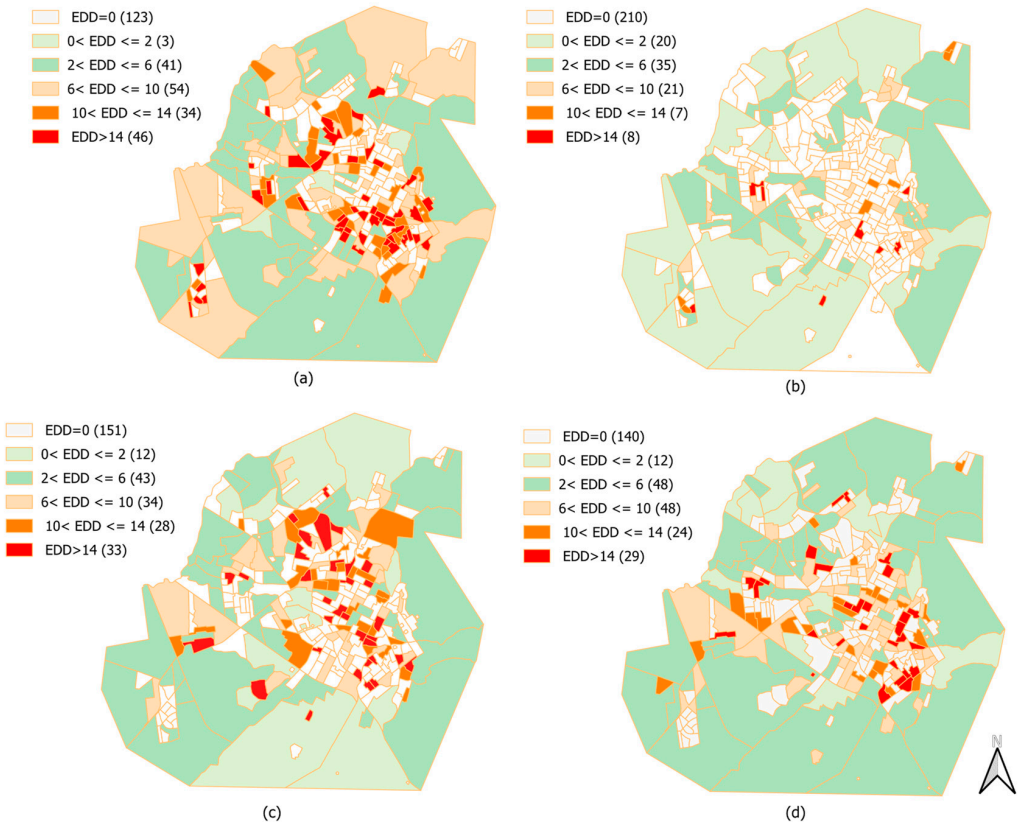


Figure 8. Electrical discharges density (*EDD*) by census tracts for the years 2009 (**a**), 2010 (**b**), 2011 (**c**), and 2012 (**d**).

4.3.2. Number of Faults in Transformers by Regions

The number of faults in transformers is the dependent or study variable. This work covers 3794 interruptions caused by distribution-transformer failures. This approach is adopted because transformers are georeferenced, and this is a condition for using SDA techniques.

Table 3 shows metrics of descriptive statistics applied to the annual number of interruptions by CTs caused by faults in power distribution transformers from 2009 to 2012. There were three yearly interruptions by CT on average. However, there were 42 interruptions in a single CT caused by transformer failures in 2009.

Table 3. Statistical summary of the number of interruptions in transformers by census tracts.

Parameters	Evaluated Years			
	2009	**2010**	**2011**	**2012**
Maximum	42	21	27	13
Minimum	0	0	0	0
Average	3.14	3.55	3.83	2.08
Standard deviation	4.08	3.63	4.28	2.35
Total number	946	1069	1153	626

Figure 9a–d show the heat maps where the interruptions in distribution transformers that occurred in 2009, 2010, 2011, and 2012 are represented. The main objective is to visualize the high concentration of interruptions in transformers in some city areas. The heat maps for all years have a similar pattern with a high fault concentration in the city's central region. This region has a high population density and, therefore, has many transformers. Figure 9c

shows the heat map for the year 2011, with warm areas in the center-southeast direction. Finally, Figure 9d shows the heat map for the year 2012, where there is a significant cluster of distribution transformer faults in the western, center, and southeast regions.

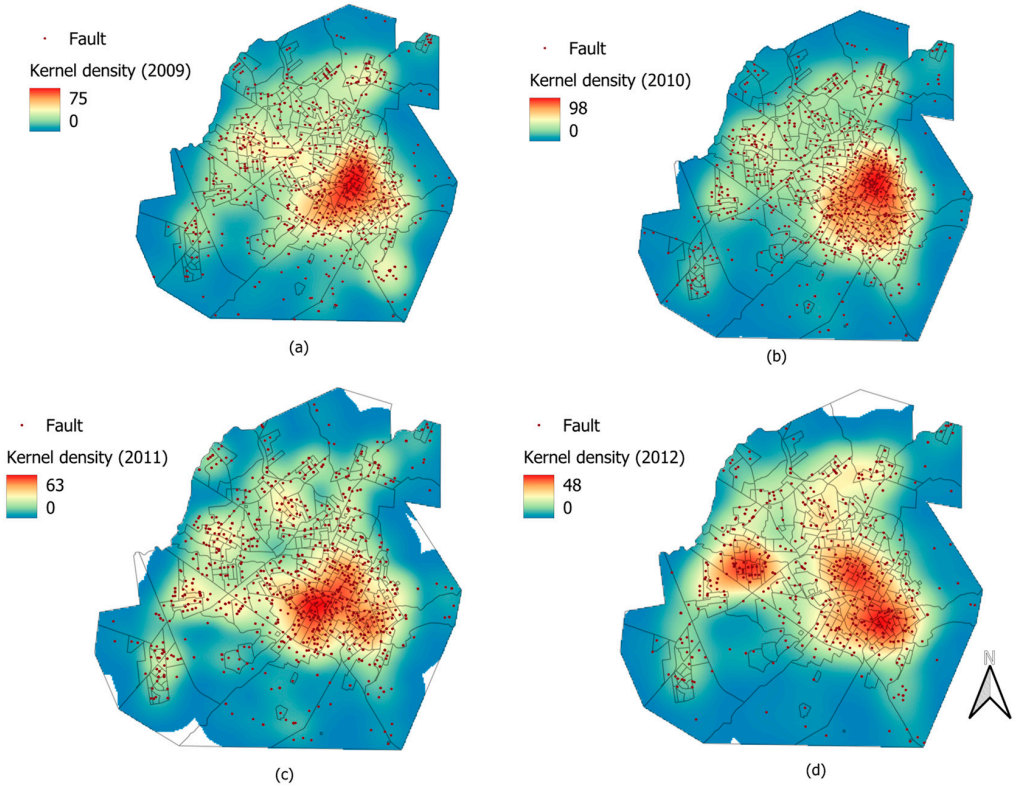


Figure 9. Heat map considering faults in transformers by census tracts in urban areas for the years 2009 (**a**), 2010 (**b**), 2011 (**c**), and 2012 (**d**).

In this sense, Figure 10a–d show thematic maps for the *NFT* dependent variable for the years 2009, 2010, 2011, and 2012, respectively. The maps visually represent interruptions, considering the total number of transformers installed in each CT. There is a contrast between Figures 9 and 10. Figure 9 shows a fault distribution pattern with a significant concentration in the central region. On the other hand, Figure 10 shows a variation in trends over the years studied.

It is noteworthy that Figure 10 shows information in parentheses with the total number of CTs whose *NFT* is within the indicated range. For example, Figure 10a shows 86 CTs with $NFT \in]20,\ 40]$, where most of the CTs have *NFT* in that range for all years evaluated.

In Figure 10a, there are 64 CTs with a high $NFT > 60$. Most are in the southeast, west, and north regions. In Figure 10b, there are 80 CTs with a high NFT. The southeast and west regions contain some CTs with a high *NFT*, and other CTs appear in the northwest region. Figure 10c has the largest number of CTs with a high *NFT*, being 93 CTs. The southeast, west, and northwest regions contain many CTs with a high NFT, and other CTs appear in the northeast and central regions. Lastly, Figure 10d has the smallest number of CTs with a high *NFT*: there are 41 CTs, and the majority are in the east, west, and northwest regions. Certainly, in 2012, the distributor performed the maintenance or replacement of many damaged transformers located in CTs with many faults.

According to Figure 10, there are some CTs with a high *NFT* located in peripheral city regions. On the other hand, these regions have a low *EDD* in Figure 8. Therefore, a preliminary assessment indicates that there are possibly other local variables that influence the high *NFT* in these regions.

In this context, peripheral regions close to rural regions are fire targets. However, we cannot conclude that the *NFT* is high in peripheral regions of the city due to fires. Further studies should confirm or refute that hypothesis.

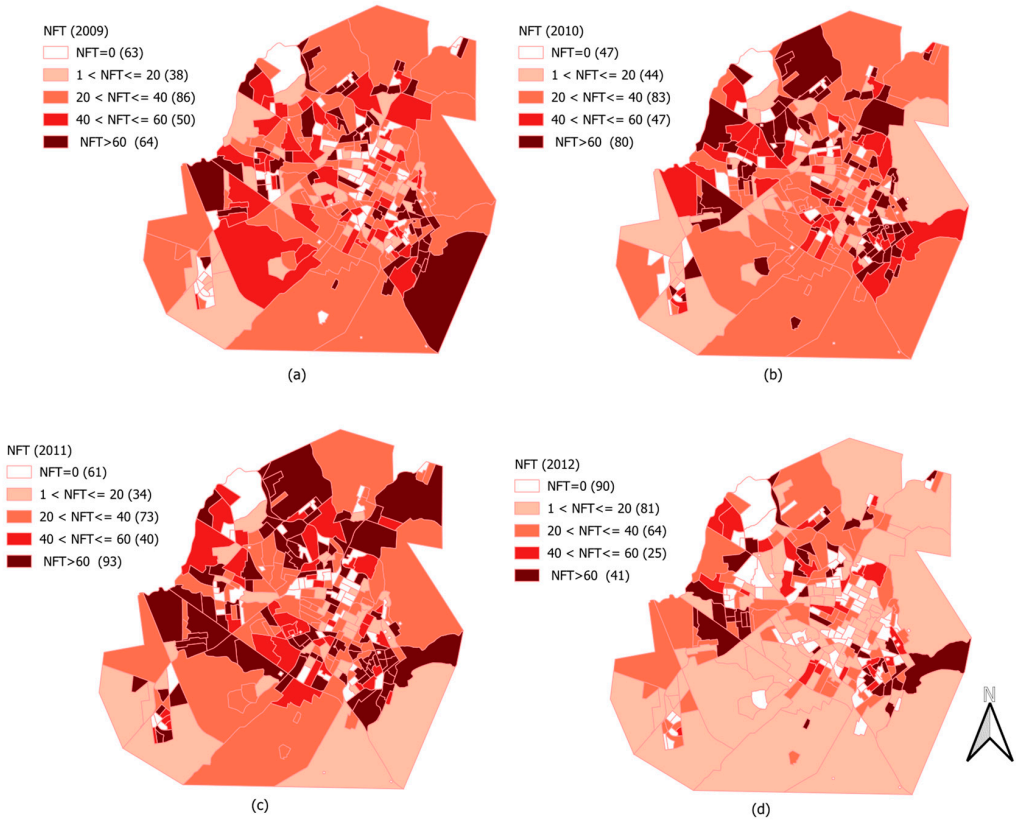


Figure 10. Percentage of faults in transformers (*NFT*) by census tracts for the years 2009 (**a**), 2010 (**b**), 2011 (**c**) and 2012 (**d**).

4.3.3. Geographically Weighted Summary Statistics

Spearman's correlation global coefficient between the *EDD* and *NFT* is shown in Table 4 for the years 2009 to 2012 according to (1). The *NFT* and *EDD* have a moderate positive global correlation for all years analyzed. Thus, a cause-and-effect relationship may exist between these variables; therefore, an increase in the *EDD* causes an increase in the *NFT*. Additional investigations must be performed to confirm or refute this hypothesis.

Table 4. Spearman's correlation coefficient between *EDD* and *NFT*.

Evaluated Years			
2009	**2010**	**2011**	**2012**
0.4407	0.6553	0.5073	0.5432

In contrast to local metrics, global metrics do not consider the influence of neighboring CTs, whose influence is represented by SWM. Thus, the correlation absence at the global level does not imply the correlation absence at the local level [31,37]. Therefore, global and local correlations can present different results because the correlation coefficient at a global level represents all CTs with a single numerical value.

Therefore, for a more detailed analysis, a GW local exploratory analysis is performed considering the influence of neighboring CTs. In this sense, this section presents thematic maps as the result of applying the GW metrics in (4)–(7). The neighborhood structure among the nth CTs influences these GW metrics. The neighborhood structure is represented

by the weighting matrix $\boldsymbol{W}_{(n\times n)}$ which is constructed from the Gaussian kernel in (3) and represented in Figure 3.

The neighborhood structure in this study follows Tobler's notion of spatial dependence, which states the first law of geography where all things are similar; however, things closer look more than distant things [36].

In this sense, the weighting assigned by the Gaussian kernel is inversely proportional to the Euclidean distance between the CT centroids. Thus, as the distance between CTs i and j is reduced, the weighting w_{ij} between them increases. The elements w_{ij} belong to the weighting matrix $\boldsymbol{W}_{n\times n}$, and it represents the neighborhood structure in an urban area of a Brazilian city with $n = 301$ CTs.

Therefore, if a CT has a high number of electrical discharges, it is likely that nearby CTs will show this same characteristic. On the other hand, if a CT has feeders with many ESIs, nearby CTs are expected to have this same problem because they share the same power grid. Therefore, the neighborhood structure based on Euclidean distance among CTs is suitable for estimating areas vulnerable to faults.

Figure 11a–d show the GW standard deviation for the independent variable *EDD* for the years 2009, 2010, 2011, and 2012, respectively. Figure 11a,c show high local variability for the central and southeastern regions. Figure 11b shows high local variability for the central-east region. Finally, Figure 11d shows high local variability for the central-eastern region. A high local variability indicates that there are nearby CTs that present very different *EDD* values. As shown in Figure 11, it is worth highlighting the high local variability in regions around the central area. It contains buildings and tree vegetation that attract electrical discharges.

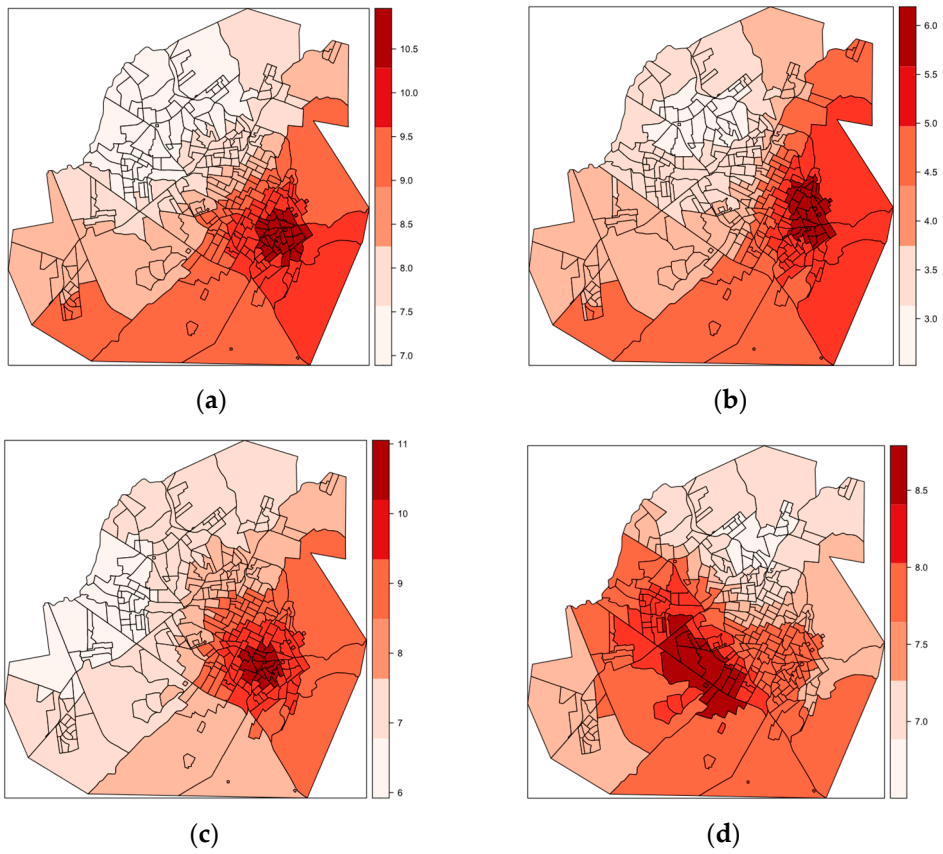


Figure 11. GW standard deviation for electrical discharges density (*EDD*) for the years 2009 (**a**), 2010 (**b**), 2011 (**c**) and 2012 (**d**).

On the other hand, Figure 12a–d show the GW standard deviation for the dependent variable *NFT* for the years 2009, 2010, 2011, and 2012, respectively. Figure 12a shows high

local variability for the northern regions. Figure 12b shows high local variability for the southeast region, and finally, Figure 12c,d show high local variability for the southeast and northwest regions. It is worth highlighting that there is a change in regions with greater local variability that is more pronounced for the *NFT* than the *EDD* over the years. Field teams work continuously to maintain and replace damaged transformers; therefore, there is greater variation in the *NFT* than the *EDD* by CTs over the years.

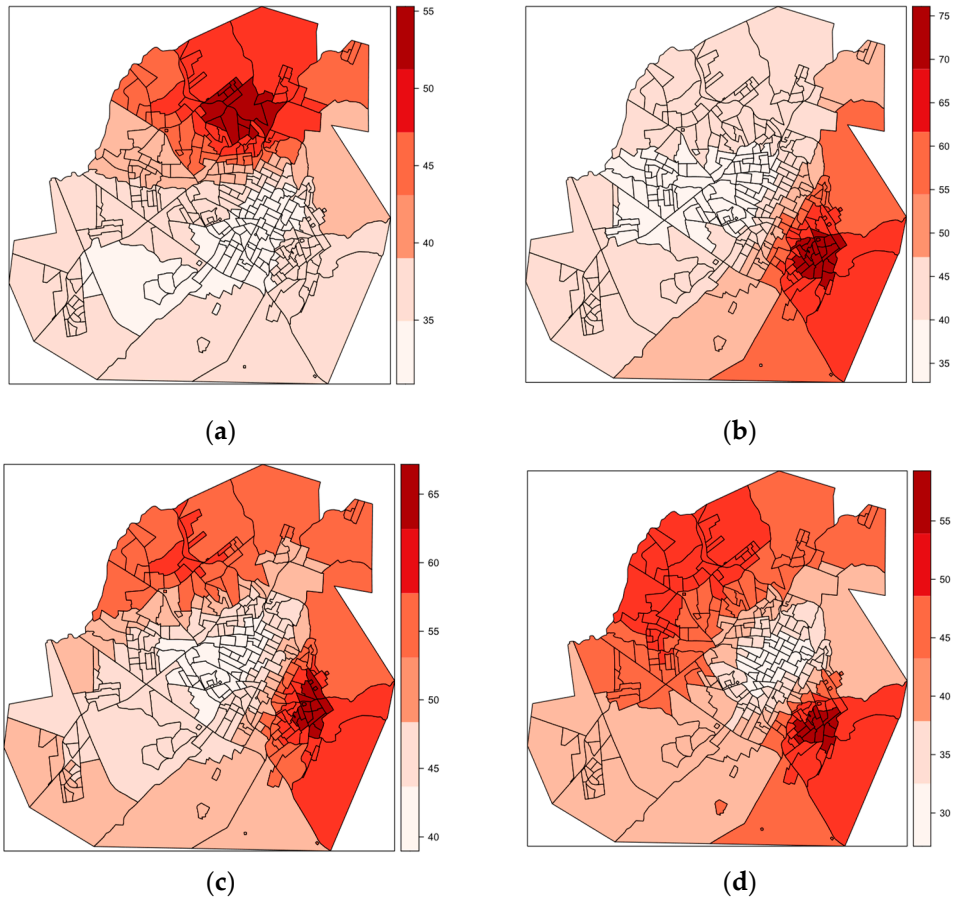


Figure 12. GW standard deviation for the percentage of faults in transformers (*NFT*) by CTs for the years 2009 (**a**), 2010 (**b**), 2011 (**c**), and 2012 (**d**).

Lastly, Figure 13a–d show a local correlation GW between the dependent variable *NFT* and the independent variable *EDD* for 2009 to 2012, respectively. There is a non-stationary relationship between the *NFT* and *EDD* variables with a moderate GW local correlation in the central (Figure 13a,d) and west (Figure 13b,c) regions.

The positive correlation means that, in these regions, the *NFT* and *EDD* are directly proportional variables. Therefore, there is first a numerical indication that electrical discharges are associated with ESIs in power grid transformers in these regions.

The non-stationary relationship between the *NFT* and *EDD* indicates that a global spatial regression model would not be suitable for estimating regions vulnerable to faults; on the other hand, a local regression model would be better suited to represent non-stationarity at the local level.

It is worth highlighting that GWEA is performed with an adaptive bandwidth where the influence of the 45 closest CTs is considered for the analysis of each CT. This closest CT value corresponds to 15% of the total CTs in the city under study [38].

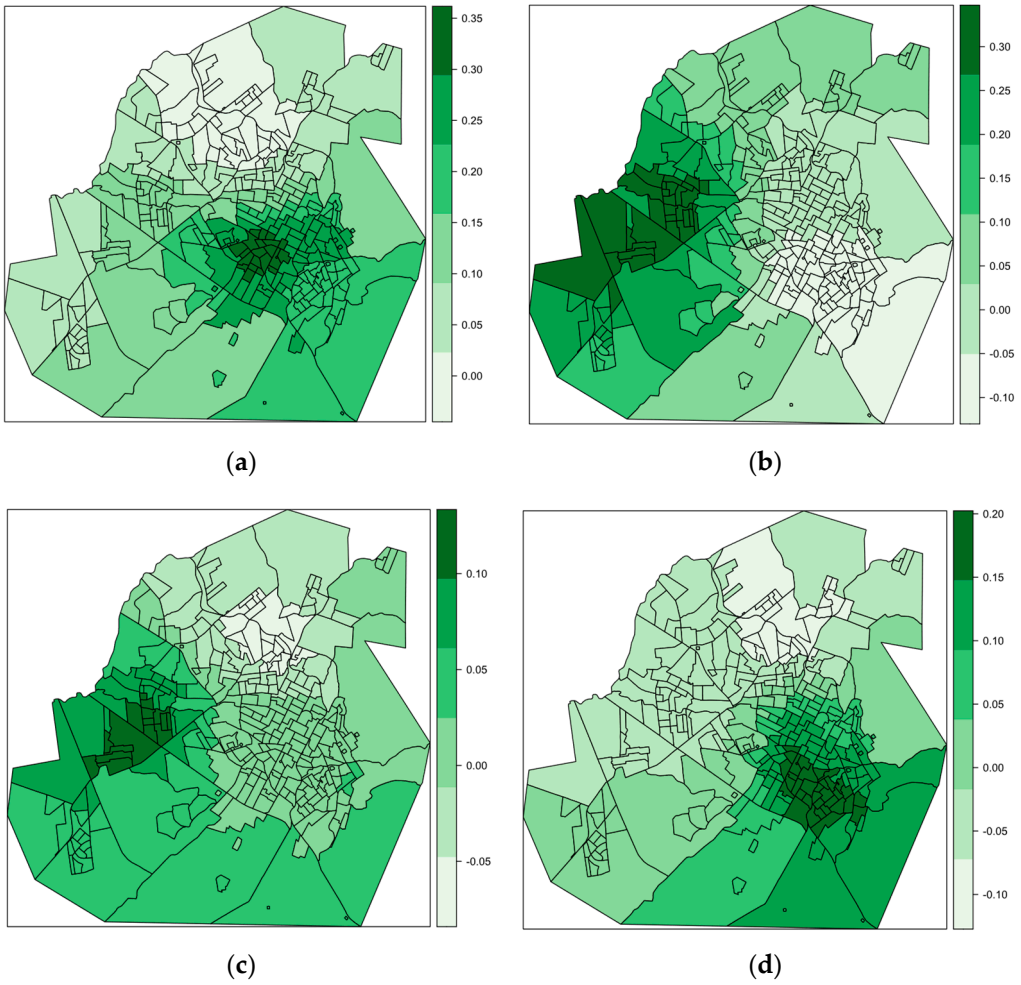


Figure 13. GW local correlation between *NFT* and *EDD* for the years 2009 (**a**), 2010 (**b**), 2011 (**c**), and 2012 (**d**).

5. Conclusions

In this study, a crucial step was performed that preceded the estimating of census tracts (CTs) vulnerable to faults: the geographically weighted (GW) exploratory spatial data analysis (ESDA). Essential for this analysis was the availability of georeferenced real data: the dependent variable, number of faults in distribution transformers (NFT) by census tracts (CTs) and the independent or explanatory variable, electrical discharges density (EDD).

An ESDA was performed using metrics of GW statistics with a visual presentation of variables in the city's geographical space. The GW statistics summarily demonstrated the spatial variability between the NFT and EDD variables.

The GW correlation showed a moderate positive correlation between the EDD and NFT in the central (in 2009 and 2012) and in the west (in 2010 and 2011) regions. Therefore, electrical discharges are associated with the power grid faults in these regions.

It is worth mentioning a limitation of this study: only faults in power distribution transformers were considered, which was because the geographical coordinates of the equipment were known.

The ESDA performed in this study is vital for implementing more sophisticated mathematical models in future studies to estimate regions vulnerable to faults. Furthermore, incorporating other variables, such as tree vegetation, will provide greater robustness to future analysis.

Author Contributions: Conceptualization, A.S.S. and L.T.F.; methodology, A.S.S. and L.T.F.; software, A.S.S. and L.T.F.; validation, A.S.S., L.T.F., M.L.M.L. and C.R.M.; formal analysis, A.S.S., L.T.F., M.L.M.L. and C.R.M.; investigation, A.S.S.; resources, C.R.M.; data curation, A.S.S.; writing—original draft preparation, A.S.S. and L.T.F.; writing—review and editing, A.S.S., L.T.F., M.L.M.L. and C.R.M.; visualization, A.S.S.; supervision, C.R.M.; project administration, C.R.M.; funding acquisition, C.R.M. All authors have read and agreed to the published version of the manuscript.

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Data Availability Statement: The data presented in this study are available on request from the corresponding author. The data are not publicly available because the authors are not allowed to publish them publicly.

Conflicts of Interest: The authors declare no conflict of interest.

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Using Aggregated Electrical Loads for the Multinodal Load Forecasting

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Abstract
Forecasting electrical loads is essential from a practical and economic point of view. With this forecast, it is possible to plan the supply of energy safely and continuously, and without interruption. In the literature, most of the works that perform electric load forecasting consider the global demand, that is, the sum of the total energy consumption. This work proposes to carry out the load forecasting along with the buses of a distribution system (multinodal forecasting) based on the use of the load aggregation concept. The proposed method uses a Fuzzy-ARTMAP neural network to forecast electrical loads in substations (multinodal forecasting) 24 h ahead, with the main objective of studying and identifying possible aggregations of multinodal loads, aiming at improving the multinodal load forecasting. The database used was from an electricity distribution subsystem, consisting of nine substations.

Keywords Artificial neural network · Adaptive resonance theory · Electrical system distribution · Multinodal load forecasting · Aggregate electric loads

Abbreviations

AL	Aggregated load
ALF	Aggregated load predicted
ALPF	Aggregated load participation factor
ALPFF	Aggregated load participation factor predicted
ANN	Artificial neural networks
ARIMA	Autoregressive integrated moving average
ART	Adaptive resonance theory
ARTMAP	Supervised ART ANN
FAM	Fuzzy-ARTMAP ANN
FL	Fuzzy logic
GL	Global load
GLF	Global load forecasting
GLFS	Global load forecasting system
GLFSS	Global load forecasting by the sum of substations
GPF	Global participation factor
MAPE	Mean absolute percentual error
ML	Multinodal load
MLF	Multinodal load forecasting
MLFS	Multinodal load forecasting system
MLFWAL	Multinodal load forecasting with aggregated loads

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1 Introduction

Electrical load forecasting is very important for energy providers, once this information determines the quantity of energy that is demanded each bus, besides anticipating decisions for an adequate providing of energy. This is a very studied matter since adequate providing can generate maximum profits and minimum errors (Moreira-Júnior, 2020).

The load forecasting problems have been studied a long time ago as for example by statistical and stochastic techniques (O'Donovan, 1983), by ARIMA of Box-Jenkins (Box & Jenkins, 1976), and linear or multiple regression among others.

Lately, techniques based on machine learning (Haykin, 2008; Zadeh, 1965), have been emphasized to solve load fore-

casting problems. Considering Artificial Intelligence, these methods provide knowledge to computers by dataset. Artificial neural networks (ANN), and fuzzy logic (FL) are some examples (Haykin, 2008).

The literature emphasizes global load forecasting using ANN (Taylor & Buizza, 2002), Fuzzy-ART&ARTMAP ANN (Lopes et al., 2005), ANN with Levenberg–Marquardt training (Saini & Soni, 2002), ANN with descent gradient training (Malki et al., 2004), FL (Kocak, 2017; Pandian et al., 2006), Genetic Algorithms (Liao & Tsao, 2006); Classical procedures and ARIMA (Box & Jenkins, 1976; O'Donovan, 1983; Moghram & Rahman, 1989; Gross & Galiana, 1987; El Desouky & El Kateb, 2000).

However, there are studies for the electrical system operation, as for example, load flow, angle stability, etc. that need the multinodal load study (buses, substations, feeders, transformers, etc.). Few works are found in the literature dealing with this approach when compared to global forecasting. As follows, there are some works for substations load forecasting.

Chen et al. (1996) develop a prediction method for three substations attending consumers for different needs (residential, commercial, and industrial). Altran (2010) used a multilayer perceptron trained by backpropagation (Werbos, 1974) modifying the activation function by a radial-based one for multinodal load forecasting (MLF). Nose-Filho (2011) proposed a model that can be used for multinodal load forecasting using a general regression neural network. Kong et al. (2017) sets a methodology to predict electric current in several transmission lines based on statistics. Abreu et al. (2018) used a Fuzzy-ARTMAP (FAM) (Carpenter et al., 1992) artificial neural network to predict the energy consumption of nine substations in a New Zealand system.

This work presents a methodology that effectuates short-term load forecasting 24 h ahead analyzing several points of the system such as substations, transformer, and feeders. MLF is realized by a FAM-ANN (Carpenter et al., 1992). This ANN is based on adaptive resonance theory (ART) (Grossberg, 2013), that contains important characteristic such as the dilemma stability-plasticity.

The proposed method is realized in four steps: (i) global load forecasting; (ii) participating factors prediction for each substation and in the sequence the local load forecasting (substations); (iii) identification of possible aggregations for the multinodal loads to improve the local and global forecasting; (iv) by aggregating the substations realize the multinodal load forecasting referred to substation block aggregated to predict the individual substation loads.

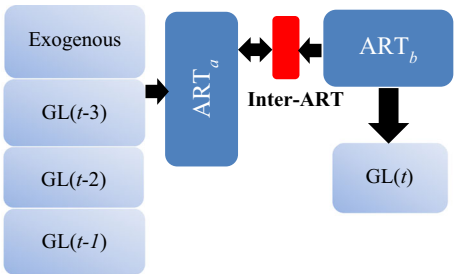

Fig. 1 Global load forecasting structure

2 Proposed Methodology

This paper proposes a new approach for multinodal load forecasting considering several points of the electric energy system. Load forecasting in substations is realized using intelligent techniques based on load forecasting in blocks (substation groups).

The steps for the proposed methodology are presented as follows:

1.Realize the global load forecasting (GLF);
2.Effectuate the load forecasting in several buses of the system, as for example the substations multinodal load forecasting;
3.Realize a study of aggregating some buses of the system (substations) and verify how the aggregated loads behavior influences the system;
4.Based on the study of step 3, predict the nodal loads (substations) from the aggregated loads.

These load predictions (global, aggregated, and nodal) are realized using a FAM-ANN.

2.1 Global Load Forecasting

Global load forecasting is realized by a system based on a FAM-ANN named the global load forecasting system (GLFS). The input dataset is composed of 48 load data (half-hour measured). The input dataset of ARTa contains exogenous data (day of the week, hour of day, holiday, and daylight saving time) and global load (GL), for example, in three previous times t-1, t-2, and t-3. This representation of GL includes the windowing (Altran, 2010) and aims to identify the behavior of the load profile. The output of FAM-ANN is represented by the GL in the subsequent instant 't".

Figure 1 presents the strategy used for global load forecasting with FAM-ANN (Moreira-Júnior et al., 2020).

The exogenous data are presented in binary coding.

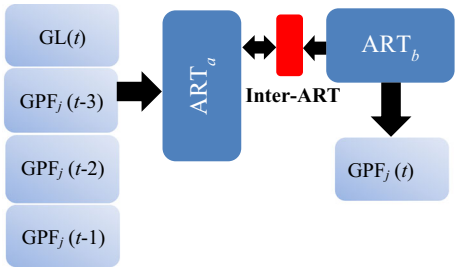

Fig. 2 Multinodal load forecasting system

2.2 Multinodal Load Forecasting

The formulation of the model is based on the use of as well FAM-ANN.

This predictor system is composed of predictor modules that act independently, such as each module has a FAM-ANN effectuating prediction in specific points of interest as substations, transformers, feeders among others. To realize the substation load forecasting the MLFS (multinodal load forecasting substations) uses the global participation factor (GPF) referred to as the load act percentual of each substation on the system defined by Eq. (1) (Altran, 2010; Nose-Filho et al., 2011):

$$\mathrm{GPF}_j(t) = \frac{\mathrm{ML}_j(t)}{\mathrm{GL}(t)} \tag{1}$$

where

$\mathrm{ML}_j(t)$: Multinodal load referred to the j-th substations in time t;

$\mathrm{GL}(t)$: Global load, which corresponds to sum of loads of each substation in time t.

Therefore, for each substation of the approached system is realized a load forecasting composing the multinodal prediction. The FAM-ANN input dataset for each substation j is composed of the global load in time t, $\mathrm{GL}(t)$, and the participation factors of substation j in times t-1, t-2, and t-3. The neural network output for MLFS is the participation factor in time t.

Figure 2 shows the MLFS (Moreira-Júnior et al., 2020).

Once predicted the $\mathrm{GPF}_j(t)$ in time t, it is necessary to identify which load value of substation j, is associated with this percentual. This way, the multinodal load is calculated associating the participation factor predicted for substation j, in time t, $\mathrm{GPF}_j(t)$, and the global load predicted in time t, $\mathrm{GLF_j}(t)$. Therefore according to (Abreu et al., 2019; Altran, 2010):

$$\mathrm{MLF}_j(t) = \mathrm{GPF}_j(t) \times \mathrm{GLF}_j(t) \tag{2}$$

2.3 Substations Aggregation Analysis in Forecasting

Considering the multinodal forecasting where the load is forecasted for each bus, follows (Altran, 2010):

$$\sum_{j=1}^{n} GPF_j(t) = 1 \tag{3}$$

where

$\mathrm{GPF}_j(t)$: Global participation factor of substation j in time t;

n: The total quantity of substations.

When the system contains many substations, it is thought that the global participation factor of some substations is lower, and therefore interferes with the performance of the FAM-ANN (Abreu et al., 2019). This performance is related to the diagnosis (prediction) of the ANN and the processing time. Therefore, this strategy can represent an improvement in processing time (Abreu et al., 2018).

After realizing the multinodal load forecasting, the objective is to detect on the system, the buses that can be grouped to minimize the discrepancies of the participation factor, improving the multinodal load forecasting and consequently the global load forecasting.

Groups were generated to preserve the distribution of the system to assure concise load forecasting and becoming more precise.

The aggregated load (AL) is represented by a set, ALi, which contains k buses (substations) aggregated in this set. The index i shows that there can be more than one aggregated set to realize the system analysis. Considering n the number of substations of the system and for $2 \leq ns < n$, then:

$$\mathrm{AL}_i = \sum_{k=1}^{ns} \mathrm{ML}_k(t) \tag{4}$$

where

$\mathrm{ML}_k(t)$: Multinodal loads (substations) aggregated on the set, AL_i in time t.

ns: Quantity of aggregated substations.

The other multinodal loads are approached individually, considering the global load of the system is constituted by individual multinodal loads and the sets of aggregated loads.

2.4 Multinodal Load Forecasting Realized by Aggregated Load Forecasting

Multinodal load forecasting is realized by the participation of each bus according to Fig. 2. However, as the multinodal loads are disaggregated loads, the predictions are less precise. To improve the predictions, the grouping strategy of the more sensible substation is employed. Nevertheless, as

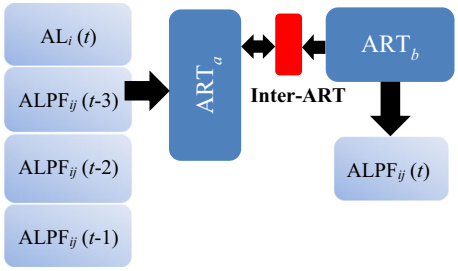

Fig. 3 Multinodal forecasting system used for multinodal load forecasting of aggregated substations

the substation set is grouped, the information referred to the substations aggregated is lost.

This way, the multinodal load forecasting with aggregated loads (MLFWAL) is used, which predicts only the loads of the substations used on the aggregated set given in Eq. (4).

The MLFWAL has the same structure as the MLFS with one difference: the global load is referred to as aggregated load of the set, AL_i. The aggregated participation factor ($ALPF_{ij}$) is calculated according to Eq. (5):

$$\mathrm{ALPF}_{ij}(t) = \frac{\mathrm{ML}_{ij}(t)}{\mathrm{AL}_i(t)} \tag{5}$$

where

$\mathrm{ML}_{ij}(t)$: multinodal load referred to the j-th substation contained on the i-th set of the aggregated substation in time t;

$\mathrm{AL}_i(t)$: i-th set of the aggregated substations (sum of the ns aggregated substations) in time t.

The MLFWAL aims to effectuate the prediction of the aggregated loads of set $\mathrm{AL}_i(t)$. The input set used on FAM-ANN for each substation j forming the set $\mathrm{AL}_i(t)$.is composed by the value of grouped load in time t, $\mathrm{AL}_i(t)$, and the participation factors of the j-th substation of $\mathrm{AL}_i(t)$ in times t-1, t-2 and t-3. The output of the FAM-ANN for the MLFWAL is given by the value of the aggregated load participation factor in time t.

Figure 3 shows the MLFWAL by FAM-ANN.

Once the participation factors of the aggregated loads are obtained for each substation j, the multinodal load forecasting with aggregated loads (MLFWAL) is calculated associating the aggregated participation factor predicted for j-th substation contained on the i-th set of the aggregated substation, in time t, ALPFF_{ij} (t), and the i-th set of the aggregated load predicted in time t, ALF_i (t), according to Eq. (6)

$$\mathrm{MLFWAL}_{ij}(t) = \mathrm{ALPFF}_{ij}(t) \times \mathrm{ALF}_i(t) \tag{6}$$

This process provides the forecast of the local load through the load forecast carried out for a set of aggregated substations. This strategy is called load forecast with return.

In the forecasts of local loads, the errors obtained are greater than the predictions of loads carried out globally. In this way, by aggregating the substations, a smaller forecast error is obtained compared to local forecasts. However, once aggregated, it is not possible to obtain the local forecast of the aggregated block.

Therefore, through this technique it is possible to return the local forecast, that is, obtain the load forecast of the substations that make up each aggregated block.

It is emphasized that the methodology is applied to a distribution system of New Zealand.

3 Results and Discussions

The dataset used on the Fuzzy-ARTMAP neural network training is from a New Zealand electric company (Nose-Filho et al., 2011), which contains nine substations such as Kopu, Waikino, Waihou, Hamilton 11, Hamilton 33, Cambridge, Te Awamutu, Hinuera, and Kinleith.

The data used refer to historical data of local loads composed of active power values generated by each substation, measured every half hour, taken from the Centralized Dataset provided by the Electricity Commission of New Zealand (disponível: https://www.ea.govt.nz/).

The training phase of the ANN is composed of data referred to the period between December, 8th 2007 and January, 7th 2008, with 1488 data.

The output of the diagnosis phase corresponds to the load prediction of January, 8th 2008 with 48 load values (half-hour) representing the load forecasting one day ahead (24 h).

Several tests were realized to find the parameter values in each phase, and the most adequate results were chosen by trial and error.

The increase, or decrease in the size of the database used has not significantly influenced the quality of the results (Abreu et al., 2018).

3.1 Comparison Criterion of the Results

Mean absolute percentual error (MAPE) (Park et al., 1991) is used to evaluate the performance of the methodology according to Eq. (7):

$$\mathrm{MAPE} = \frac{1}{n_i}\sum_{n=1}^{n_i}\left\{\frac{\left|L(h) - \underline{L}(h)\right|}{L(h)}\right\} \times 100 \tag{7}$$

where

n_i: Quantity of information of the time series;

$L(h)$: Real load in time h;

$\underline{L}(h)$: Predicted load in time h;

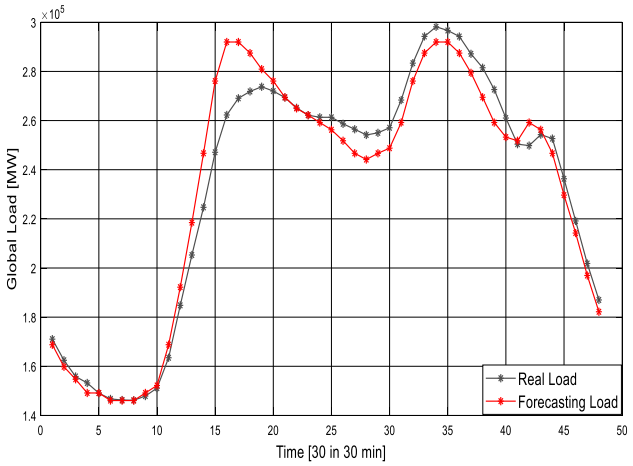

Fig. 4 Real global load and predicted load curves by FAM-ANN

3.2 Global Forecasting Results

The parameter values used in the FAM-ANN for global forecasting are: $\beta = 1, \alpha = 0{,}7, \rho a_{\text{baseline}} = 0.99, \rho_b = 0.99, \rho_{ab} = 1$, and $\varepsilon = 0.005$.Where

$\alpha > 0$: Parameter of chosen;

$\beta \in [0,1]$: Rate of training;

$\rho \in [0,1]$: Parameter of vigilance;

ε: Incremental parameter of the initial value of ρa;

ρa_{baseline}:Initial value of ρa.

The indices "a" (input), "b" (output), and "ab" (inter-ART) indicate that they are associated with the ARTa, ARTb and Inter-ART modules, respectively. These three modules are interconnected forming the FAM-ANN, algorithm details of which can be found in reference Carpenter et al. (1992).

The vigilance parameters of the ART_a, ART_b and inter-ART modules are set close to the maximum value allowed so that in both modules it would be able to adequately discriminate the categories. The choice parameter was obtained through numerous tests.

Figure 4 shows de predicted and real global load curve for January 8th, 2008.

3.3 Results of Substation Forecasting

Multinodal forecasting uses the same parameters used for global forecasting.

Table 1 presents the MAPE for the multinodal forecasting and also for the global load forecasting by the sum of substations (GLFSS) predicted by FAM- ANN.

3.4 Results for the Aggregated Load Forecasting

Analyzing Table 1, some substations have MAPE greater than 6% (Waikino, Hamilton 11, Hinuera, and Kinleith); to improve the results for the prediction of these substations, the grouping of multinodal loads was executed aiming to realize a less disaggregated prediction and consequently more efficient.

Table 1 MAPEs of the substations by FAM-ANN

Substations	MAPE (%)
Kopu	5.9102
Waikino	12.2909
Waihou	3.5178
Hamilton 11	6.0190
Hamilton 33	4.2064
Cambridge	4.7037
Te Awamutu	4.9943
Hinuera	6.4905
Kinleith	10.8801
Global Load	2.9139
GLFS	1.3336

Table 2 MAPE of the individual and aggregated substations by FAM-ANN

Substations	Individual MAPE (%)	*MAPE* (%) with Aggregation
Kopu	5.9102	5.9102
Waihou	3.5178	3.5178
Hamilton 11	6.0190	4.3982
Cambridge	4.7037	
Kinleith	10.8801	
Hamilton 33	4.2064	4.2064
Waikino	12.2909	3.2321
Hinuera	6.4905	
Te Awamutu	4.9943	
GLFSS	1.3336	1.1298

Several tests were executed using two, three and even four aggregated substations (Moreira-Junior, 2020). An analysis of the several aggregations is presented (Moreira-Junior, 2020).

Aggregation Analysis:

- AL_1 = {Hamilton 11, Cambridge and Kinleith}
- AL_2 = {Waikino, Hinuera and Te Awamutu}

Table 2 presents the MAPE for the individual and aggregated substations. Two aggregation blocks were elaborated in the analysis, each one containing three substations. The aim to reduce the prediction errors is attained, once the literature suggests an error less than 5% for a good prediction (Ranaweera et al., 1997).

Observing Table 2, for each aggregated block, there is a substation of a great error, and realizing the aggregation, the results show a considerable improvement.

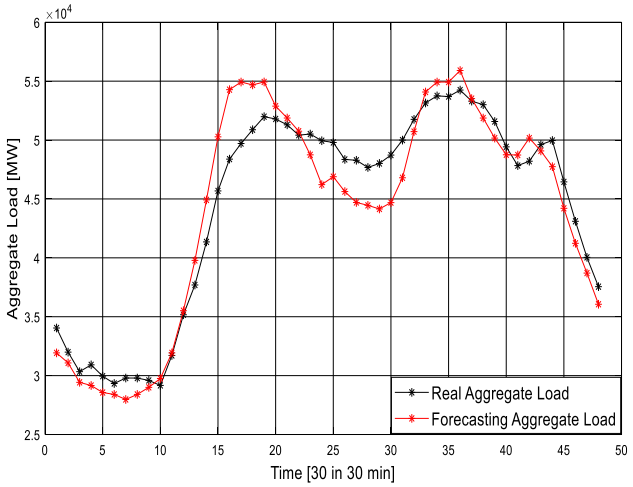

Fig. 5 Real aggregated load and aggregated load predicted FAM-ANN for block 1

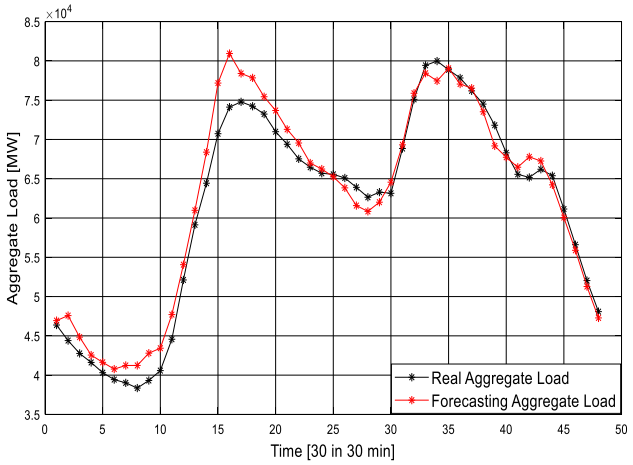

Fig. 6 Real aggregated load and aggregated load predicted FAM-ANN for block 2

Figures 5 and 6 show that the aggregation for block 1 as well as for block 2 set a good aggregation with the substations. Block 1 contains the following substations Hamilton 11, Cambridge, and Kinleith, and block 2 by aggregating Waikino, Hinuer, and Te Awamutu.

Figure 5 shows the real load (load grouped by substations of block 1) and the predicted load by FAM-ANN. The behavior of the predicted aggregated load is similar to the real load. This is a signal that the substations chosen to be aggregated in the same group are adequate.

Figure 6 shows the real load (aggregated load for block 2) and aggregated load predicted by Fuzzy-ARTMAP ANN. The same behavior with the real and predicted load is observed for block 2, showing that the substations grouped for block 2 are adequate.

Table 3 Parameters of the FAM-ANN for forecasting substations aggregated loads—analysis—AL_1

Parameter	Substations		
	Hamilton 11	Cambridge	Kinleith
β	1.0000	1.0000	1.0000
α	0.9000	0.9900	0.9900
ρa_{baseline}	1.0000	1.0000	1.0000
ρ_b	0.9992	0.9999	0.9995
ρ_{ab}	1.0000	1.0000	1.0000
ε	0.5000	0.9000	0.0500

Table 4 Parameters of the fuzzy-ARTMAP ANN for forecasting substations aggregated—analysis—AL_2

Parameter	Substations		
	Waikino	Hinuera	Te Awamutu
β	1.0000	1.0000	1.0000
α	0.7000	0.7000	0.0500
ρa_{baseline}	0.9000	0.9000	0.9000
ρ_b	0.9990	0.9950	0.999049
ρ_{ab}	1.0000	1.0000	1.0000
ε	0.0005	0.0005	0.5000

3.5 Results of the Substation Forecasting from Aggregated Load Forecasting

When the multinodal load is aggregated, the individual substation analysis is lost, i.e. it is not possible to predict the load for each substation of the aggregated block. Thus, the strategy used is to consider as global load the aggregated load of the blocks aiming to forecast the individual loads that were aggregated on the set $AL_i(t)$.

As two aggregations were done, AL_1 and AL_2, there are two MLFWAL, one for AL_1 and the other for AL_2. The MLFWAL_1 will forecast the loads for substations Hamilton 11, Cambridge, and Kinleith, and the MLFWAL_2 for substations Waikino, Hinuera, and Te Awamutu.

The parameters used for MLFWAL_1 and MLFWAL_2 are presented in Tables 3 and 4, respectively.

Table 5 shows the MAPE for the substations that were contained in the analysis.

Figures 7, 8, 9, 10, 11, 12 show the curves related to the return of the loads for each one of the substations (analysis AL_1 and AL_2), using the proposal compared with the real local loads and initially predicted local loads (without aggregation and return).

Figure 7 shows a load of substation Hamilton 11, where the predicted load by the proposal (load forecasting by aggregated load), i.e., predicted load with the return is closer to the real load. Therefore, the proposal provides an improve-

Table 5 MAPE for invidual and aggregated substations and substations obtained by AL_1 and AL_2

	MAPE (%)		
Substations	Indivi-duals	Aggregated	Proposal
Kopu	5.9102	5.9102	5.9102
Waihou	3.5178	3.5178	3.5178
Hamilton 11	6.0190	4.3982	4.6677
Cambridge	4.7037		3.9570
Kinleith	10.8801		6.5646
Hamilton 33	4.2064	4.2064	4.2064
Waikino	12.2909	3.2321	4.0280
Hinuera	6.4905		7.8663
Te Awamutu	4.9943		4.6354
GLFSS	1.3336	1.1298	1.2469

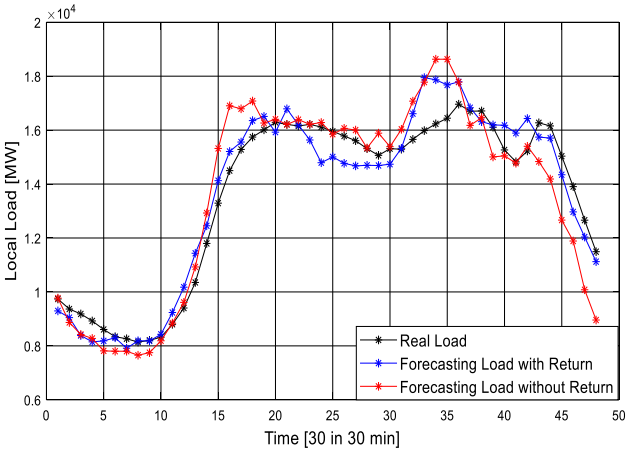

Fig. 7 Real local load, predicted with return and predicted without return for substation Hamilton 11

ment in multinodal forecasting which is verified with the results presented in Table 5. For individual multinodal forecasting, the MAPE is 6.0190% and the error of the proposal is 4.6677% confirming an improvement in predicting the load by the aggregation.

Figure 8 presents the loads for the substation Cam-bridge. Both prediction present the same behavior of the real load and by Table 5, there is an improvement as well, once the MAPE decreases from 4.7037 to 3.9570%.

Figure 9 shows the loads for substation Kinleith where it is observed the proposal without return presents a marked variation on the predicted curve. This variation is smoother for the proposal. The improvement for these substations is verified in Table 5, where the MAPE is reduced from 10.8801 to 6.5646%.

Substation Waikino is in Fig. 10, where a considerable improvement is observed, once the real load and the predicted one are very close. The MAPE is reduced from 12.2909 to 4.0280% using the proposal.

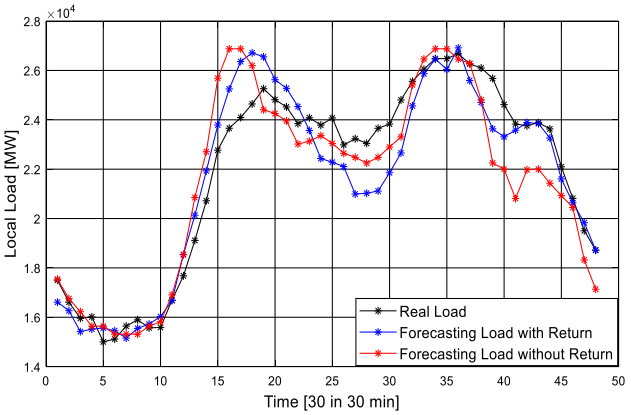

Fig. 8 Real local load, predicted with return and predicted without return for substation Cambridge

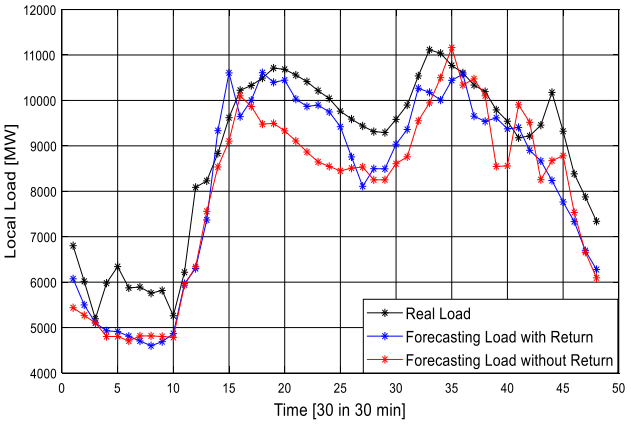

Fig. 9 Real local load, predicted with return and predicted without return for substation Kinleith

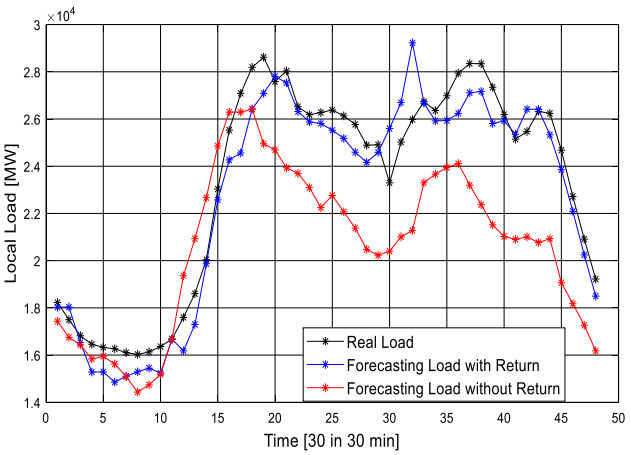

Fig. 10 Real local load, predicted with return and predicted without return for substation Waikino

Figure 11 shows the loads for substation Hinuera. It is observed that the curves obtained with and with-out return in determining time are very far from the real load, and consequently, the MAPE is increased. It is emphasized that this

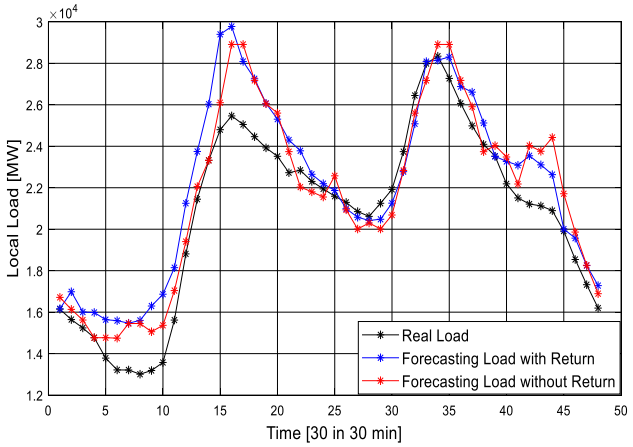

Fig. 11 Real local load, predicted with return and predicted without return for substation Hinuera

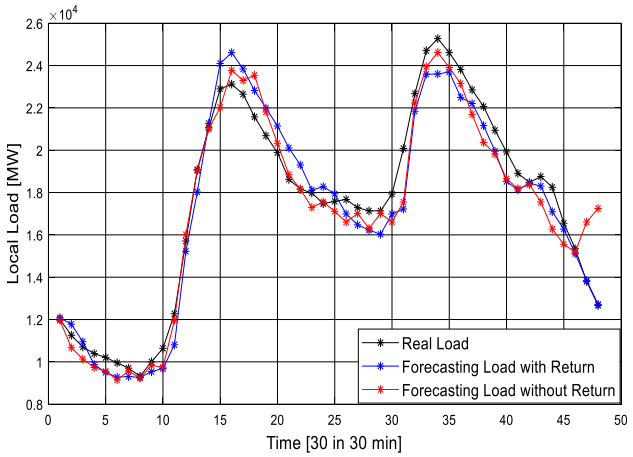

Fig. 12 Real local load, predicted with return and predicted without return for substation Te Awamutu

substation was the only one that there is no improvement when compared with the proposal without return. However, this difference with the two methodologies is not significant and therefore does not disqualify the proposal.

Substation Te Awamutu is represented in Fig. 12. The curves are very similar and by Table 5 the MAPE considering the two methodologies are very close, and the last one is 4.6354% for the proposal.

Analyzing the results, the individual multinodal load forecasting presents good behavior, except for the Hinuera substation. However, there is an expressive improvement for substations Waikino and Kinleith, once they were the most difficult to verify among the nine selected.

4 Conclusion

Th his work presents the use of Fuzzy-ARTMAP ANN to implement global and load prediction 24 h ahead (with half-hour measures). The advantage to using this ANN is based on the principal characteristics such as plasticity and stability, assuring convergence. Literature has been approaching the multinodal load forecasting (with the participation factor of global load) and recently some works deal with aggregated forecasting to improve the computational time and improve global forecasting. However, once using the aggregated substations it is not possible to analyze the substations individually. Therefore, the innovation to predict the substation load is based on multinodal forecasting. Nevertheless, considering as global load the aggregated load by substations and the use of the participation factor of these substations on the aggregated system, there is a new dataset (one for each aggregated substation on the system) to be used to predict the individual loads of the substations that were aggregated. This methodology provides the prediction for every substation of the system reducing the MAPE. Several analyzes were effectuated, however, only one was totally presented. The load aggregations were done in different forms, such as observing the electrical system or the region where they are located or the proximity with them. Every case presents an improvement in multinodal load forecasting comparing to the initial multinodal predictor as the MAPE were lower and assuring the proposal as a good strategy in effectuating local load predictions.

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Efficient Methodology for Detection and Classification of Short-Circuit Faults in Distribution Systems with Distributed Generation

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Abstract: Fault detection and classification are crucial procedures for electric power distribution systems because they can minimize the occurrence of faults. The methods for fault detection and classification have become more problematic because of the significant expansion of distributed energy resources in distribution systems and the change in their currents due to the action of short-circuiting. In this context, to fill this gap, this study presents a robust methodology for short-circuit fault detection and classification with the insertion of distributed generation units. The proposal methodology progresses in two stages: in the former stage, the detection is based on the continuous analysis of three-phase currents, whose characteristics are extracted through maximal overlap discrete wavelet transform. In the latter stage, the classification is based on three fuzzy inference systems to identify the phases with disturbance. The short-circuit type is identified by counting the shorted phases. The algorithm for short-circuit fault detection and classification is developed in MATLAB programming environment. The methodology is implemented in a modified IEEE 34-bus test system and modeled in ATPDraw with three scenarios with and without distributed generation units and considering the following parameters: fault type (single-phase, two-phase, and three-phase), angle of incidence, fault resistance (high impedance fault and low impedance fault), fault location bus, and distributed generation units (synchronous generators and photovoltaic panels). The accuracy is greater than 94.9% for the detection and classification of short-circuit faults for more than 20,000 simulated cases.

Keywords: distribution systems; distributed generation; fuzzy logic inference; wavelet transform; multi-resolution analysis; short-circuit fault detection; short-circuit fault classification

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1. Introduction

Electric power distribution systems (EPDS) have the continuous supply of electric energy to consumer units as their main objective. However, these systems are exposed to several factors that can cause disturbances, steady-state faults, or blackouts. The faults are classified as temporary or permanent. Temporary faults cause a transient disturbance in EPDS, which automatically returns to its normal operating state (NOS). On the other hand, permanent faults cause an interruption in the energy supply; thus, EPDS will return to its NOS after the short-circuit repair. The faults are characterized by impermissible deviation from the standard operating conditions of EPDS [1].

Distribution systems have changed with the massive insertion of distributed energy resources (DERs) based on renewable energy [2]. DERs expansion is essential to meet the growing energy demand. In addition, environmental concerns have been necessary to encourage renewable energy expansion. The increase in renewable distributed generation (DG) units reduce greenhouse gases [3].

On the other hand, the arbitrary and massive insertion of DG units into EPDS causes significant changes in its configurations. The electrical system loses its radial characteristic as multiple power sources power it. The main impact of these changes is related to the conventional protection system, since features such as direction, amplitude, and short-circuit currents are changed [3,4].

1.1. Literature Review

Several studies in the specialized literature developed tools for fault analysis in EPDS by inserting DG units. They addressed the impact on the conventional protection system due to the significant growth of DG units into EPDS. The following studies discussed different methodologies for fault detection and classification in EPDS.

Rai et al. [5] proposed a convolutional neural network technique for fault classification in EPDS with DG units. The proposed method does not have the pre-processing phase; thus, three-phase currents and voltage signals were applied directly as inputs.

A new method for short-circuit faults detection and classification in balanced and unbalanced distribution systems was presented by Zhang et al. [6]. The current's positive, negative, and zero sequence components characterized the faults. The operating modes of DG units were considered using the Fortescue approach. A softmax regression model was introduced to minimize the impact of transient signals on the fault classification module and applied only two resistors with maximum value during the simulations.

The discrete wavelet transform (DWT) was used in [7] on three-phase current signals, whose values were applied to formulate a decision tree to perform the fault classification. Decision tree input was formed by a comparison parameter obtained using the maximum value of three-phase current signals and zero-sequence. Those parameters were applied as a reference to detect the fault phases. The fault resistance variation was not considered.

In [8], a strategy based on fuzzy logic was used for fault detection and classification. Chaitanya et al. implemented two fuzzy inference systems (FIS). They were built to detect and classify low-impedance faults (LIF) and high-impedance faults (HIF). A Teager Energy Operator (TEO) was applied to extract three-phase current signal characteristics, which composed the FIS input set.

In [9], fault detection was performed by applying the power spectral density calculated from the wavelet covariance matrix. The signal information was extracted through a wavelet transform (WT), where the signals were decomposed into three levels using the *db4* mother wavelet.

Elnozahy et al. [10] proposed a new method for detecting and classifying single-phase faults using DWT and artificial neural networks (ANN). The transient signals were analyzed based on the *db4* mother wavelet to extract their characteristics.

A method based on DWT and ANN for HIF detection was presented by Silva et al. [11]. The authors focused on incremental learning procedures to find new fault patterns. DG units were not considered.

Decanini et al. [12] presented a method for automatic fault diagnosis. The detection process was based on statistical and direct analysis of three-phase currents and voltage signals in the wavelet domain. DWT and multi-resolution analysis (MRA) were introduced to extract the signal characteristics. The short-phase classification was performed by a set of Fuzzy-ARTMAP ANN. DG units were not considered in the system modeling.

A combination of maximum overlap discrete wavelet packet transform (MODWPT) and empirical mode decomposition (EMD) were applied for HIF detection in [13]. That methodology was based on the estimation of fault current signals inter-harmonic through MODWPT.

In [14], a method based on an adaptive neural fuzzy inference system was proposed for fault classification. Three-phase currents and voltage signals measured at the substation output were evaluated. The signal transient components were extracted via WT.

Different fault types were detected and classified in [15] using a combination of wavelet singular entropy theory (WSE) and fuzzy logic. The algorithm was based on the singular

wavelet values of each phase, since a phase with an anomaly presents values outside of the allowed limit.

A proposal based on deep learning and deep belief networks (DBN) was implemented by Hong et al. [16] for fault classification in EPDS with and without DG units. After the DBN training with current and voltage signals, it was possible to obtain the signal characteristics and classify the fault types that occurred in EPDS.

Ola et al. [17] presented an algorithm for fault classification, based on the fault index. It was calculated from the three-phase current signals, in which the Wigner distribution function and the decomposition based on the coefficient of alienation were applied. The anomaly phases were identified from the fault index. The fault classification was carried out from the count of phases with anomaly.

A methodology for HIF detecting was presented in [18]. The technique was based on EMD with adaptive noise applied to decompose the zero-sequence current signals to obtain the intrinsic mode function (IMF). TEO was applied to identify changes in characteristics of IMF waveforms. HIF identification was performed by calculating the signal WSE. WSE for faulty signals are higher than that one for normal signals.

Sekar et al. [19] proposed an intelligent method based on WT and data mining for HIF detection. The main characteristics of three-phase current signals were extracted via WT.

In [20], the authors presented a technique for identifying HIF. A method based in Adaline ANN was applied to extract the third harmonic angle (THA) from current signals. Adversarial generative conditional was applied to produce input data from THA. Finally, a convolutional neural network was applied to separate HIF from others transient events.

Wontroba et al. [21] proposed a methodology based on the analysis of harmonic and symmetrical components of currents to detect HIF due to cable break. HIF detection process has two steps: (i) cable break detector and (ii) HIF detector. In the former step, the identification of a cable break was carried out by applying the phase current phasors. In the latter step, HIF detection was carried out via application of neutral current and analysis of its harmonic components.

Yuan et al. [22] presented a method for detecting faulty feeders after a single line-to-ground fault (SLGF). It was based on waveform recognition, where zero-sequence voltage (ZSV) on bus and local zero-sequence current (ZSC) were applied. An image for each feeder was constructed via ZSV and ZSC superposition. Finally, a convolutional neural network was applied to perform image recognition, where all fault features were extracted adaptively.

1.2. Contributions

This study presents a robust methodology for fault detection and classification in EPDS with DG units. The highlights of the proposed approach are described below.

- Development of a simple and effective methodology for short-circuit faults detection and classification of distribution systems that included the DG units.
- Sensitivity analysis to assess the impacts on fault detection and classification in each proposed scenario. There are synchronous generators and photovoltaic (PV) panels with different levels of DG units' insertions into EPDS.
- Maximum overlap discrete wavelet transforms (MODWT) and FIS for faults detection and classification, respectively.

1.3. Paper Structure

The rest of this paper is as follows: Section 2 explains the techniques used in short-circuit fault detection: problem description (Section 2.1), HIF (Section 2.2), MODWT (Section 2.3), the steps for fault detection (Section 2.4), and the confusion matrix metrics (Section 2.5). Section 3 addresses the methodology used for short-circuit fault classification: a FIS for short-circuit faults classification (Section 3.1), FIS rules set (Section 3.2), and the steps for short-circuit faults classification (Section 3.3). Section 4 presents the results and discussions. All simulated cases are applied in the modified IEEE-34 bus test system ad-

dressed in Section 4.1. The results for detection and classification of short-circuit faults and the validation for both are explained and discussed in Sections 4.2–4.4, respectively. Finally, Section 5 presents the conclusions.

2. Short-Circuit Fault Detection

2.1. Problem Description

Energy supply interruptions occur due to several factors such as: equipment failures, atmospheric discharges, human errors, contact of trees with the overhead distribution network, and fires in the vicinity of the electrical network.

EPDS are susceptible to four fault types: line-to-ground fault (SLGF), double line-to-ground fault (DLGF), line-to-line fault (LLF), and three-phase-ground fault (LLLGF). SLGF is the most frequent fault type, with 70% of classified cases [1]. Figure 1 shows an example to illustrate the fault types, where there are 12 cycles (200 ms) of three-phase currents with a frequency of 60 Hz. Figure 1a shows an SLGF fault, where a short-circuit is applied at $t = 66$ ms (4th cycle) in phase C with ground: C_g. After the disturbance, the fault phase current has its value increased. Thus, it is easy to note the difference in current magnitude between the normal phases A and B with respect to phase C where there is an anomaly. Figure 1b shows a DLGF fault where the phases A and B are short-circuited: AB_g. We observe that the magnitude of phase C current in NOS is lower than that ones in phases A and B, because they have anomalies. Finally, Figure 1c shows an LLLGF fault. At $t = 66$ ms (4th cycle) a three-phase short-circuit is applied, which causes an increase in the current magnitude after the disturbance.

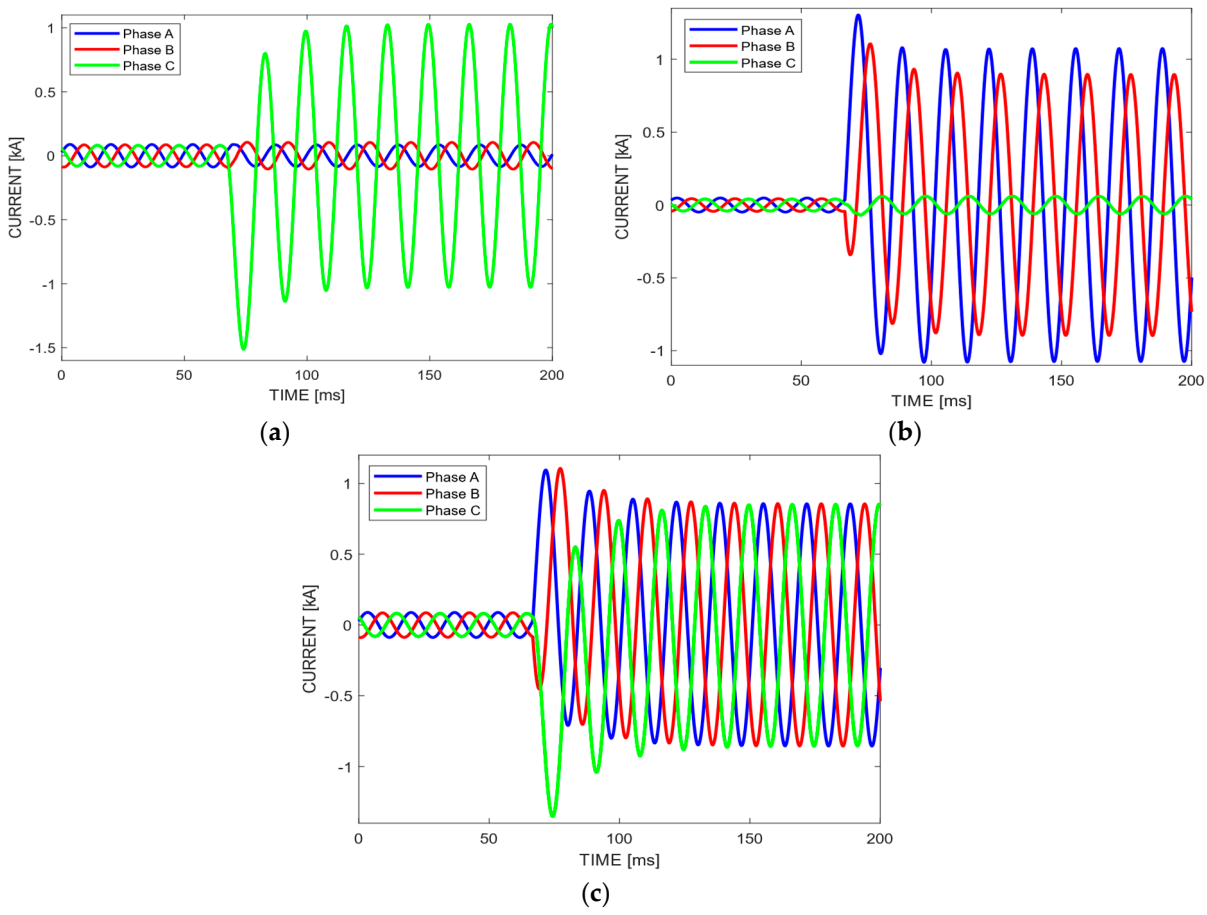


Figure 1. Fault types: (**a**) line-to-ground fault (SLGF); (**b**) double line-to-ground fault (DLGF); and (**c**) three-phase-ground fault (LLLGF).

2.2. High-Impedance Faults

HIFs are caused by direct contact of electrical wires from the distribution network with high-impedance objects or surfaces such as concrete, sand, asphalt roads, and tree branches. They are characterized by low magnitudes of fault currents, which may be insufficient to excite an overcurrent relay. Thus, fault detection becomes very difficult, since the conventional protection devices are based on overcurrent. During HIF, the short-circuit current presents values equivalent to the rated current of the electrical system at the fault location [11,23]. On the other hand, LIFs can be detected easily by established methodologies in specialized literature.

In this study, we apply a HIF model with two diodes in parallel [23]. As shown in Figure 2, the model is composed of two DC sources, V_p, and V_n, and two resistors, R_p and R_n, that represent the fault resistances.

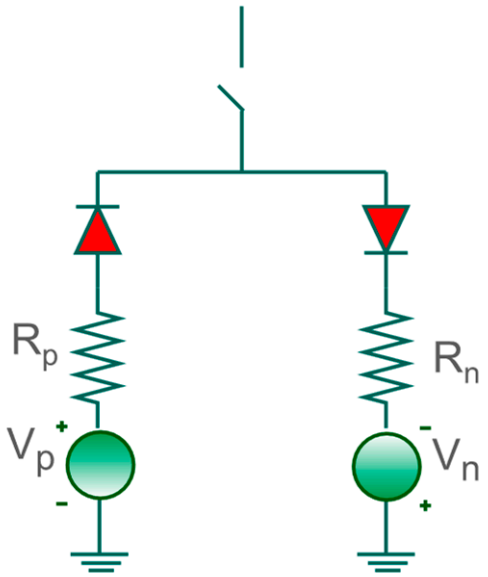


Figure 2. Model adopted for HIF [23].

2.3. Maximum Overlap Discrete Wavelet Transform

A MODWT performs a linear filtering operation, where a series is transformed into coefficients associated with variations over a scale set. It was developed to act similarly to DWT [24]. Unlike DWT, MODWT can be applied to all sample sizes and it has flexibility in the initial point selection, where the coefficients do not change as a result of changes in the initial point [25].

The filtering process consists of applying a high-pass wavelet filter at a given decomposition level j to produce a set of wavelet coefficients and a low-pass scaling filter to produce a set of scaling coefficients [26]. After the filtering process, the data size becomes twice the size of original signal.

On the other hand, DWT is applied in a simplified filtering process. This process is performed through the down-sampling operator whose objective is to reduce the number of samples at the filter output. MODWT does not apply the down-sampling operator; thus, at each level j of decomposition, the wavelet coefficients will have the same length as the original signal as shown in Figure 3 [26,27].

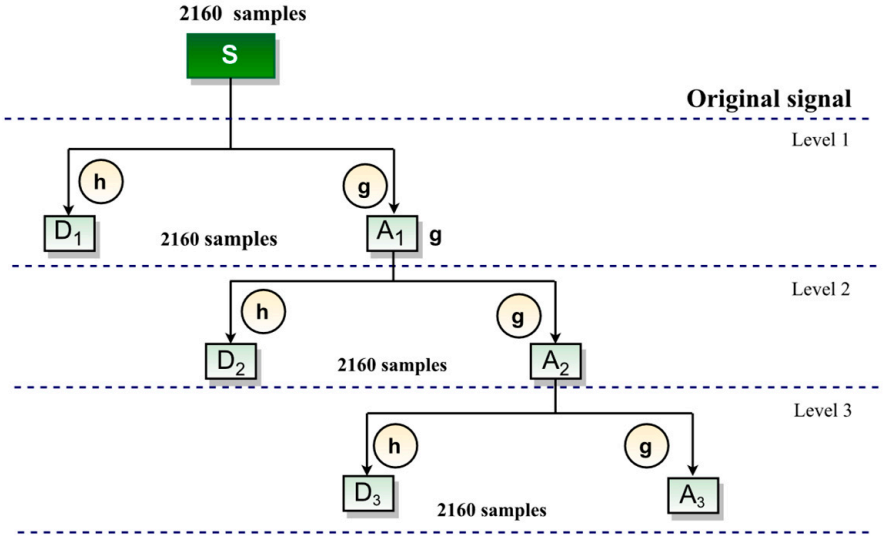


Figure 3. Illustrative example of MODWT filtering process [28].

The MODWT scale filter $\widetilde{h}_{j,\ell}$ and the wavelet filters $\widetilde{g}_{j,\ell}$ are defined in (1) and (2) based on the DWT filters $g_{j,\ell}$ (low-pass) and $h_{j,\ell}$ (high-pass) [24–27]:

$$\widetilde{h}_{j,\ell} = \frac{h_{j,\ell}}{\sqrt{2}} \tag{1}$$

$$\widetilde{g}_{j,\ell} = \frac{g_{j,\ell}}{\sqrt{2}} \tag{2}$$

where $\ell = 0, \ldots, L-1$ and L represents the filter length. The subscript j is the level to which the signal will be decomposed.

For a given level j, the MODWT coefficients can be defined as the convolution of the time series X_n according to (3) and (4) [25]:

$$\widetilde{W}_{j,n} = \sum_{\ell=0}^{L-1} \widetilde{h}_{j,\ell} X_{n-\ell} \ \ mod\ N \tag{3}$$

$$\widetilde{V}_{j,n} = \sum_{\ell=0}^{L-1} \widetilde{g}_{j,\ell} X_{n-\ell} \ \ mod\ N \tag{4}$$

By means of (5) and (6), we obtain for the *nth* signal sample at the *j*th decomposition level the approximation and detail components $\widetilde{A}_j$ and $\widetilde{D}_j$ respectively [25]. Where $\widetilde{g}^0_\ell$ and $\widetilde{h}^0_\ell$ are periodized of $\widetilde{g}$ and $\widetilde{h}$ to length N, respectively. N corresponds to the signal sample size:

$$\widetilde{A}_{j,n} = \sum_{\ell=0}^{L-1} \widetilde{g}^0_{j,\ell} \widetilde{V}_{j,\ n+\ell} \ \ mod\ N \tag{5}$$

$$\widetilde{D}_{j,n} = \sum_{\ell=0}^{L-1} \widetilde{h}^0_{j,\ell} \widetilde{W}_{j,\ n+\ell} \ \ mod\ N \tag{6}$$

Figure 3 shows a filtering process via MODWT, in which the original signal is decomposed into three levels of resolution. At each level j, the approximation and detail coefficients A_j and D_j are the same size as the original signal, because there is a retention of reduced values at each level of decomposition which are normally discarded by the down-sampling operator applied in DWT. Therefore, these values are preserved in MODWT [22].

2.4. Fault Detection Methodology

The fault detection is based on the real-time monitoring of three-phase current signals measured at the substation output. The signal sample characteristics are extracted via MODWT, where the detail and approximation coefficients are obtained from signal decomposition. Signals are decomposed into three levels of resolution by applying a fourth-order filter from Daubechies family: db4. There are high and low-frequency signals in detail and approximation coefficients.

NOS of electrical system is defined via normalized energy of detail coefficients at the third level and by means of current signals. The difference between NOS and a fault condition is determined via a comparative analysis of energies and a normalized mean with a previously established threshold of variation for each phase. If the normalized energy or normalized mean of at least one phase exceeds the threshold, then an anomaly or fault was identified in the electrical system.

The magnitudes of faulted phase currents are significantly higher than the ones in NOS as shown in Figure 1. Thus, the current means that belong to defective phases will be greater than the ones that belong to normal phases.

The algorithm for faulted phase detection is divided into five steps: (i) signal decomposition; (ii) energy signal calculation; (iii) three-phase currents mean; (iv) normalization; and (v) faulted phase detection.

2.4.1. Step 1: Signal Decomposition

The data window consists of one post-fault cycle, and it is decomposed as follows:

1. Data window is decomposed into three resolution levels j via MODWT using *db4* mother wavelet of Daubechies family.
2. Applying the fourth order filter of Daubechies family, we obtain the detail coefficients with three resolution levels.

2.4.2. Step 2: Signal Energy

The energy of current signals for each phase is obtained via (7) according to the Parseval's theorem [29]:

$$E_{i,j} = \sum_{n=1}^{N} \left| d_{i,j}^{n} \right|^2 \; with \; j = 3 \tag{7}$$

where $E_{i,j}$ is the energy for the detail coefficients to phase i current signal and $d_{i,j}^{n}$ represents the nth sample of detail coefficients for the phase i at the third level of signal decomposition—$j = 3$.

Figure 4 shows an illustrative example of normalized energies obtained for all fault types considered in this study. The energies values of phases with a disturbance present are greater than phases without anomaly.

The energy vector is shown in (8). It is composed by energies of I_a, I_b, and I_c three-phase currents:

$$\boldsymbol{E} = \begin{bmatrix} E_{a,j} & E_{b,j} & E_{c,j} \end{bmatrix} \text{ with } j = 3 \tag{8}$$

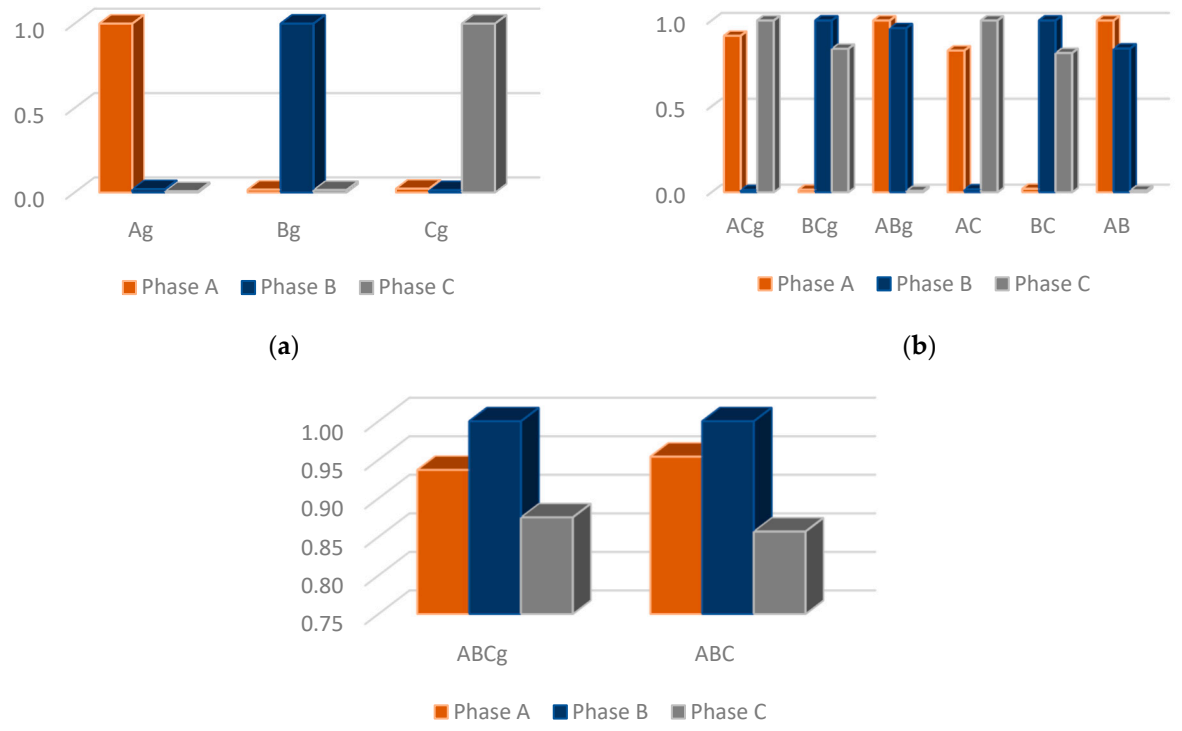


Figure 4. Illustrative example with the normalized energies. Short-Circuits: (**a**) single-phase; (**b**) two-phase; and (**c**) three-phase.

2.4.3. Step 3: Three-Phase Currents Mean

The currents mean of each phase are obtained via (9) from a data window composed of three cycles with 180 samples (one cycle) extracted from post-fault current signals. The vector of three-phase currents mean is shown in (10). Figure 5 shows an illustrative example, where the defective phases present a higher mean of normalized current signal than that ones for normal phases.

$$M_i = \frac{\sum M_i}{180}, \text{ with } i \in \{a, b, c\} \tag{9}$$

$$\boldsymbol{M} = \begin{bmatrix} M_a & M_b & M_c \end{bmatrix} \tag{10}$$

2.4.4. Step 4: Normalization

The normalization $\overline{M}_i$ and the mean of energy signals $\overline{E}_i$ are performed via (11) and (12), respectively.

$$\overline{M}_i = \frac{M_i}{\max(\boldsymbol{M})}, \text{ with } i \in \{a, b, c\} \tag{11}$$

$$\overline{E}_i = \frac{E_{i,j}}{\max(\boldsymbol{E})} \text{ with } j = 3 \tag{12}$$

2.4.5. Step 5: Detection

The system operation state is defined via (13):

$$\text{Electrical System} = \begin{cases} \text{With Anomaly,} & \text{if } Max\left[\overline{\boldsymbol{E}}\right] > LD_i \text{ and } Max\left[\overline{\boldsymbol{M}}\right] > LD_i \\ \text{Without Anomaly,} & \text{if } Max\left[\overline{\boldsymbol{E}}\right] \leq LD_i \text{ or } Max\left[\overline{\boldsymbol{M}}\right] \leq LD_i \end{cases} \tag{13}$$

where the $Max[\cdot]$ function returns the maximum value in an array. The limit for detection (LD_i) is that one allowed for the phase i current.

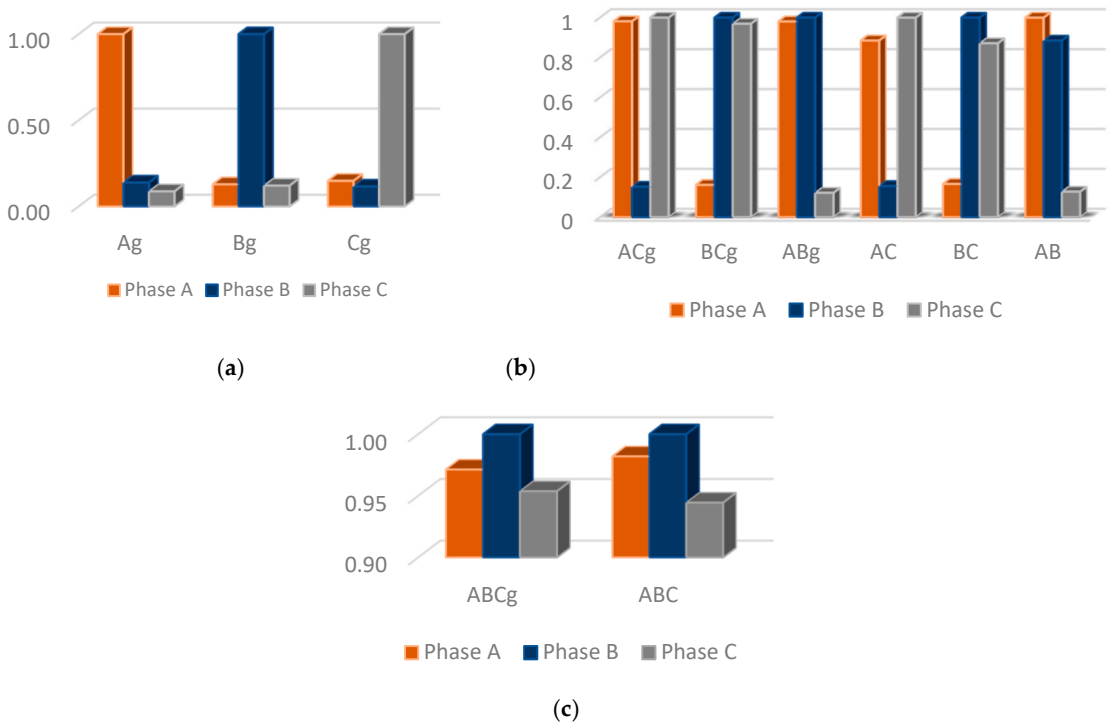


Figure 5. Illustrative example with the means of normalized current signals. Short-Circuits: (**a**) single-phase; (**b**) two-phase; and (**c**) three-phase.

2.5. Detection Validation

In this section, the well-known confusion matrix and the metrics obtained from its elements are presented to evaluate the efficiency of the short-circuit fault detection stage.

2.5.1. Confusion Matrix

The confusion matrix metrics are applied to evaluate the detection stage performance. It is worth pointing out that the confusion matrix application is feasible because the output of the detection stage is binary (absence or presence of short-circuit faults).

Table 1 shows the structure of confusion matrix adapted for this study.

Table 1. Confusion matrix structure.

Confusion Matrix		**Predicted State**	
		N (Normal)	**F (Fault)**
Real State	N (Normal)	Q_{NN}	Q_{NF}
	F (Fault)	Q_{FN}	Q_{FF}

Follow below a description of confusion matrix elements:

- Q_{NN}: number of normal cases (without fault) and whose detection was correct (true-positive cases);
- Q_{NF}: number of normal cases (without fault); however, they were incorrectly detected as fault (false-negative cases);
- Q_{FN}: number of fault cases; however, they were incorrectly detected as normal (without fault)—false-positive cases;
- Q_{FF}: number of fault cases and whose detection was correct (true-negative cases).

2.5.2. Confusion Matrix Metrics

In this section, the accuracy, precision, recall, and f1-score metrics obtained from the confusion matrix are presented to evaluate the short-circuit detection stage.

Accuracy in (14) is the ratio between the number of hits (true-positive and true-negative cases) and the number of all simulated cases. This metric alone is insufficient for evaluating problems where there is an imbalance between the simulated cases (normal and fault cases). Therefore, other metrics such as precision, recall and f1-score are evaluated together with the accuracy to support the evaluation of the short circuit detection stage proposed in this study.

Precision in (15) is the ratio between the number of real cases without fault (true-positive) and the number of erroneously predicted cases by detection algorithm as without fault (false-positive). That metric evaluates the algorithm's coverage, i.e., its ability to identify faults.

Recall or sensitivity in (16) is the measure of normal cases (without fault) over the number of real cases without faults. This metric allows evaluating the algorithm's ability to detect real cases of faults. It is related to the 'false alarm', i.e., the fault assignment to normal cases.

Finally, f1-score in (17) is a metric that incorporates both precision and recall. It is a harmonic mean between these two metrics.

$$\text{Accuracy} = \frac{Q_{NN} + Q_{FF}}{Q_{NN} + Q_{NF} + Q_{FN} + Q_{FF}} \tag{14}$$

$$\text{Precision} = \frac{Q_{NN}}{Q_{NN} + Q_{FN}} \tag{15}$$

$$\text{Recall} = \frac{Q_{NN}}{Q_{NN} + Q_{NF}} \tag{16}$$

$$\text{F1} - \text{Score} = \frac{2(\text{Precision} \times \text{Recall})}{(\text{Precision} + \text{Recall})} \tag{17}$$

3. Short-Circuit Faults Classification

The fault classification algorithm is activated after the fault identification by the detection algorithm. The characteristics of different fault types are applied in a classification process. They are obtained from the signal samples of three-phase post-fault currents [30]. NOS and the disturbance state of each phase are determined from the properties of fault types. Thus, the fault phases are identified, and the short-circuit types are calculated via count of fault phases.

The relationships with the currents are calculated according to (18)–(20), where the maximum absolute value of currents for each phase is applied [30]:

$$r_1 = \frac{\max\{abs(I_a)\}}{\max\{abs(I_b)\}} \tag{18}$$

$$r_2 = \frac{\max\{abs(I_b)\}}{\max\{abs(I_c)\}} \tag{19}$$

$$r_3 = \frac{\max\{abs(I_c)\}}{\max\{abs(I_a)\}} \tag{20}$$

The normalization of the relationships with the currents is performed in (21) [30]:

$$r_{in} = \frac{r_i}{\max\{r_1,\ r_2, r_3\}} \text{ with } i \in \{1, 2, 3\} \tag{21}$$

The behavior indices with the properties of three-phase currents are obtained via (22)–(24) [30]. The indices C_1, C_2, and C_3 contain the characteristics referring to the currents of phases A, B, and C, respectively [30]:

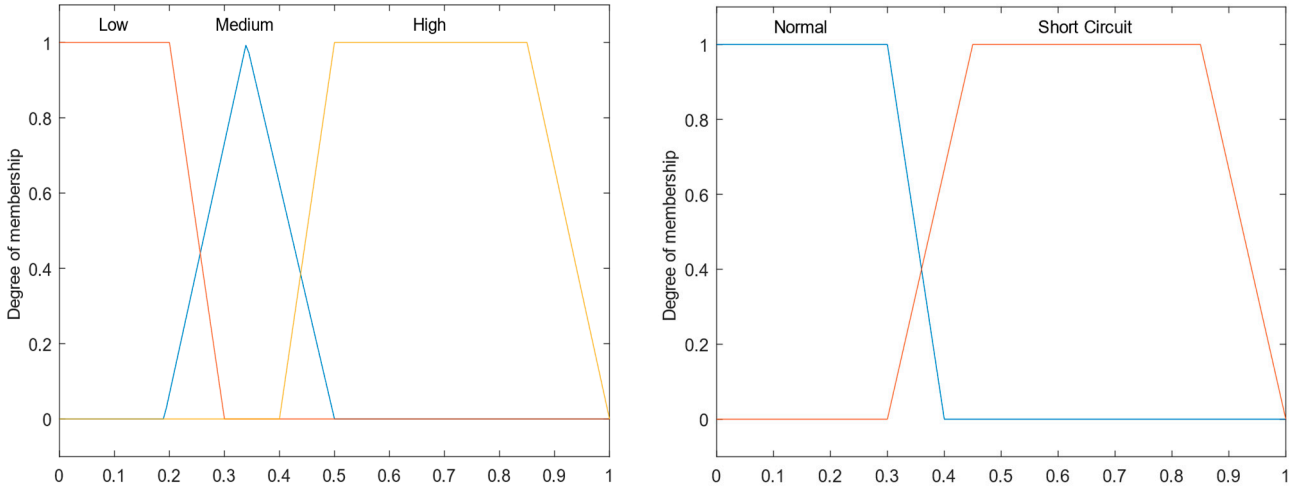
$$C_1 = r_{1n} - r_{2n} \tag{22}$$

$$C_2 = r_{2n} - r_{3n} \tag{23}$$

$$C_3 = r_{3n} - r_{1n} \tag{24}$$

3.1. Fuzzy Inference System for Short-Circuit Faults Classification

The FIS allows us to intuitively incorporate the empirical knowledge of fault experts via fuzzy rules set. In this work, the short-circuit faults classification is performed via three FIS; therefore, an FIS is implemented for each fault type. FIS has three input variables: $phase_a$, $phase_b$, and $phase_c$. Each input variable is associated with three membership functions of triangular and trapezoidal types with intervals [0, 1] as shown in Figure 6a. Each input variable is represented by three linguistic variables: *low*, *medium*, and *high*. The absolute values of short-circuit behavior indices obtained in (18)–(20) are the FIS inputs.



(**a**) (**b**)

Figure 6. Linguistic variables and membership functions. Linguistic variables: (**a**) Input variables; and (**b**) Output variables.

The operating state of each phase is determined through the FIS's outputs. The FIS contains three linguistic output variables: $Phase_a$ *State*, $Phase_b$ *State*, and $Phase_c$ *State*. The FIS' outputs are represented by two trapezoidal membership functions: Normal and Short-Circuit as shown in Figure 6b.

3.2. Fuzzy Inference System Rule Set

A rule set is defined for each FIS, where linguistic variables are applied to identify the fault phases. The rules are built from the absolute values of C_1, C_2, and C_3 behavior indices of currents in (18)–(20). They are calculated from of three-phase current signals measured at the substation output.

Table 2 shows a rule set to classify a single-phase short-circuit. When there is a fault in phase A, the C_1 index associated with that one phase has medium or high values. The C_2 index has low values because the phases B and C have no disturbance. Finally, the C_3 index has a negative sign, since it is obtained by subtracting between r_{3n} and r_{1n}, and r_{1n} has a high value, because phase A is shorted.

The behavior indices associated with the phases with disturbances normally present medium and high values in two-phase short-circuits, while the phases without anomaly have low values. Thus, a rule set is elaborated considering these characteristics as shown in Table 3. Finally, Table 4 shows the rule set for three-phase short-circuits, where the behavior indices have high or medium values.

Table 2. FIS rules set to classify single-phase short-circuit.

Rules	Short-Circuit	Behavior Indices of Three-Phase Currents		
		C_1	C_2	C_3
1st	A_g	High	Low	High
2nd	A_g	Medium	Low	High
3rd	A_g	High	Low	Medium
4th	B_g	High	High	Low
5th	B_g	High	Medium	Low
6th	B_g	Medium	High	Low
7th	C_g	Low	High	High
8th	C_g	Low	Medium	High
9th	C_g	Low	High	Medium

Table 3. FIS rules set to classify two-phase short-circuit.

Rules	Short-Circuit	Behavior Indices of Three-Phase Currents		
		C_1	C_2	C_3
1st	AB_g	High	High	Low
2nd	AB_g	High	Medium	Low
3rd	AB_g	Medium	High	Low
4th	AC_g	High	Low	High
5th	AC_g	Medium	Low	High
6th	AC_g	High	Low	Medium
7th	BC_g	Low	High	High
8th	BC_g	Low	Medium	High
9th	BC_g	Low	High	Medium
10th	AB	High	High	Low
11th	AC	High	Low	High
12th	BC	Low	High	High

Table 4. FIS rules set to classify three-phase short-circuit.

Rules	Short-Circuit	Behavior Indices of Three-Phase Currents		
		C_1	C_2	C_3
1st	ABC	Medium	Medium	Medium
2nd	ABC_g	High	Medium	Medium
3rd	ABC_g	High	Medium	High

3.3. Short-Circuit Classification

The short-circuit classification is performed after the identification of the phases with anomaly. The fault type is determined from the count of defective phases. The FIS outputs belong to the range [0, 1]. Thus, there is a post-processing step, where FIS outputs become binary via application of (25). Bit 0 indicates the normal state phase, while bit 1 indicates the shorted phase.

$$S_i = \begin{cases} 0, & Phase_i\ State \leq LC_i \text{ where } i \in \{a, b, c\} \\ 1, & \text{otherwise} \end{cases} \quad (25)$$

where: subscript i represents the abc phases. S_i are the FIS outputs associated with the $Phase_i$ and LC_i are the limits for abc phases classification. The short-circuit types are identified using (26):

$$\text{Short Circuit Type} = \begin{cases} \text{single}-\text{phase}, & \text{if } \sum_{i=1}^{3} S_i = 1 \\ \text{two}-\text{phase}, & \text{if } \sum_{i=1}^{3} S_i = 2 \\ \text{three}-\text{phase}, & if \sum_{i=1}^{3} S_i = 3 \\ \text{Absent}, & \text{if } \sum_{i=1}^{3} S_i = 0 \end{cases} \tag{26}$$

Figure 7 shows a flowchart with an overview of the proposed methodology, addressing the steps for detection and classification of short-circuit faults.

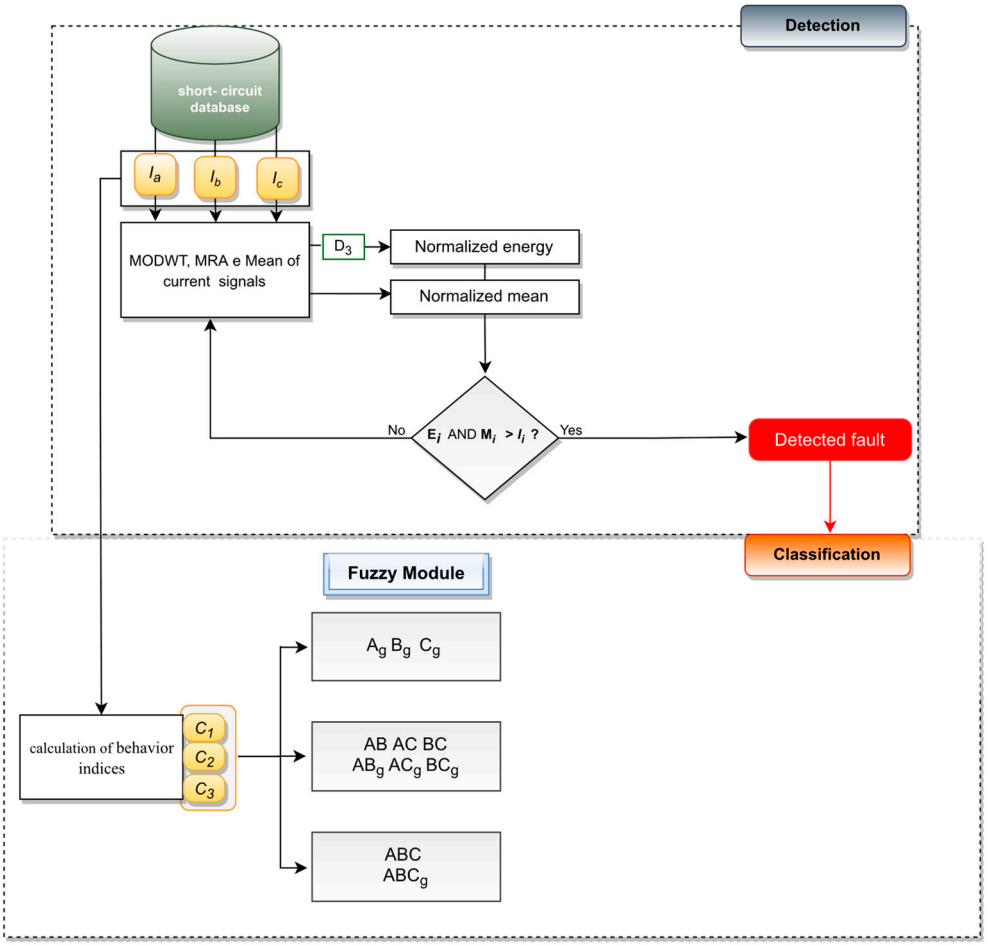


Figure 7. Flowchart with an overview of proposed methodology.

4. Results and Discussions

4.1. Test System Modelling IEEE-34 Bus

Different fault configurations are simulated in IEEE-34 bus test system. It consists of an extensive, lightly loaded North American real feeder with 24.9 kV nominal voltage. Its main feeder has three-phase branches, single-phase side branches, and single-phase and three-phase loads. The IEEE-34 bus main characteristics are listed as [31]: (i) two voltage regulators, (ii) an in-line transformer for voltage reduction from 24.9 kV to 4.16 kV, (iii) unbalanced distributed loads and balanced concentrated loads, and (iv) shunt capacitors.

The modified IEEE-34 bus is modeled in ATP/EMTP and its ATPDraw graphical interface. All fault combinations are simulated with 10.8 kHz sampling rate. The loads are modeled with a constant impedance model [32]. Short-circuit detection and classification algorithms are developed in MATLAB® environment. All simulations are performed on a computer with an Intel Core i7 processor; 1.8–1.99 GHz and 8 GB of RAM.

Three different scenarios are modeled in the IEEE-34 bus system to evaluate the proposed methodology. Different short-circuit conditions are performed considering the variation of the parameters: fault resistance, insertion angle, short-circuit type, fault location bus, and DG units.

In the first scenario, IEEE-34 bus test system is modeled without DG units as shown in Figure 8a. In the second scenario, according to Figure 8b, two synchronous generators described in [33] are inserted at buses 840 and 848 with 220 V and 480 V of nominal voltage, respectively. Finally, Figure 8c shows the third scenario, where a PV generator is introduced at bus 848. The PV model applied in this study is based on [34] with 1 MVA of installed power.

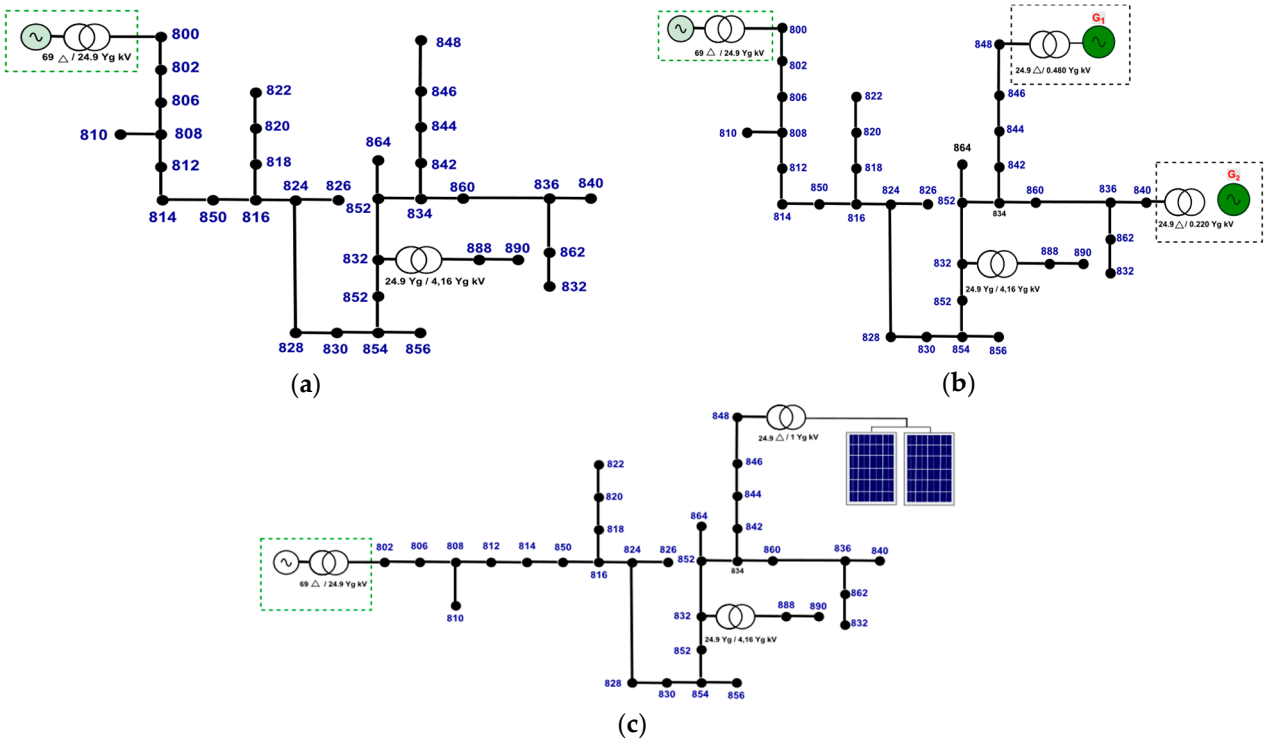


Figure 8. IEEE-34 bus test system. Fault Simulation: (**a**) Without DG unit (Scenario 1); (**b**) With two synchronous generators (Scenario 2); and (**c**) With PV panels (Scenario 3).

Eleven types of simulated short-circuits are considered in several buses along the feeder. The system parameters applied for different fault scenarios are presented in Table 5. Short-circuits with resistance values greater than 40Ω are considered as HIF, where a specific model for it is introduced as shown in Figure 2.

Table 5. IEEE-34 bus test system parameters for fault simulations.

Parameters	Configurations
Fault types	A_g, B_g, C_g, AB, AC, BC, AB_g, AC_g, BC_g, ABC, ABC_g
Fault Location bus	806, 810, 814, 824, 828, 830, 840, 850, 854, 860
DG Units	Two synchronous generators and PV panels
Fault resistance	1 Ω to 40 Ω (LIF) and 40 Ω to 300 Ω (HIF)
Fault insertion angle	0°, 30°, 45°, 60°, 90°, 120°, and 150°

4.2. Short-Circuit Fault Detection

The fault state can be determined by comparative analysis of energies and normalized mean with a variation limit for each phase. They are empirically determined from a test set, in which a discrepancy between the phase values under NOS and with disturbance is observed. The accuracy and the number of simulated cases are shown in Figure 9 for each scenario.

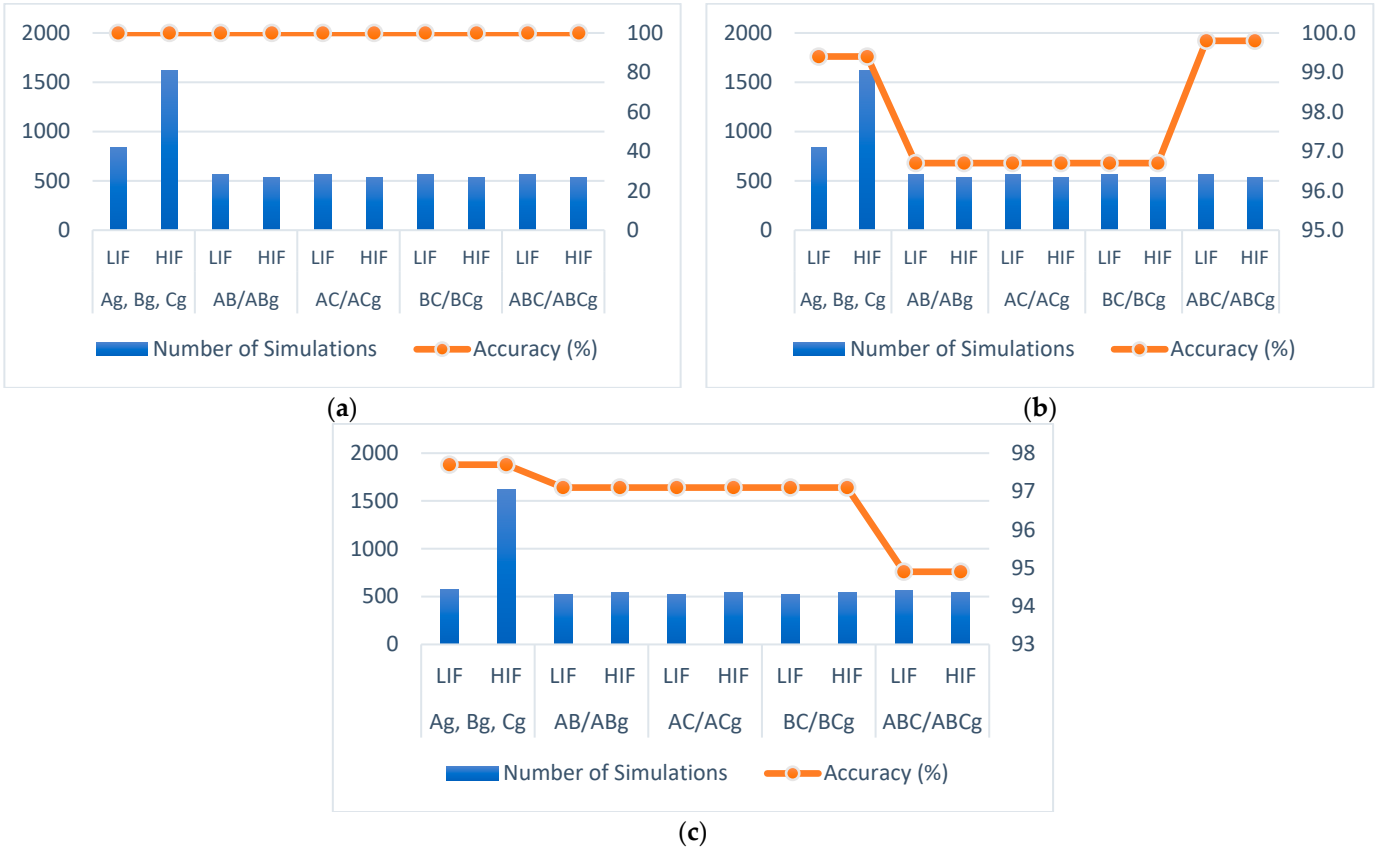


Figure 9. Short-circuit detection for three scenarios: (**a**) without DG units (scenario 1), (**b**) with two synchronous generators (scenario 2), and (**c**) with PV panels (scenario 3).

The input signal is evaluated at a sampling rate of 10.8 kHz which is equivalent to 180 samples per cycle. The simulation time for each short-circuit type is 0.2 s; therefore, there are 12 cycles and 2160 signals. Lower and upper limits are established for each data window to ensure the reliability of the acquired samples; thus, approximately 5% of initial and final data are removed.

The number of simulations and the accuracy for short-circuit fault detection are shown in Figure 9. Single-phase, two-phase, and three-phase short-circuits are detected for the scenarios 1, 2, and 3. Noteworthy that after the insertion of DG units, there is a reduction in accuracy.

The values obtained for normalized energy or for normalized mean of three-phase currents in some phases with NOS are high in some parameter combinations; therefore, the pre-established variation limit LD_i is exceeded in some simulated cases. Consequently, the algorithm wrongly identifies some perturbations.

4.3. Short-Circuit Fault Classification

The classification process is performed after the identification of a fault by the detection algorithm. The classification consists of identifying the phases involved in short-circuit. The operational status of each phase is obtained from FIS outputs. The fault classification is obtained from the number of defective phases.

As shown in Figure 6, the input variable $Phase_i$ Input assumes one among three linguistic values represented by membership functions, where two that are trapezoidal are named "low" and "high", and the other one is triangular, named "medium". The parameters of these membership functions applied for short-circuit classification are shown in Table 6.

Table 6. Membership functions parameters for FIS input variables.

Fault Types	Membership Functions Parameters		
	Low	Medium	High
Single-Phase	[0.00, 0.00, 0.1, 0.4]	[0.20, 0.34, 0.50]	[0.35, 0.50, 0.85, 1.00]
Two-Phase	[0.01, 0.10, 0.20, 0.35]	[0.3, 0.40, 0.50]	[0.40, 0.50, 0.80, 1.00]
Three-Phase	[0.00, 0.00, 0.10]	[0.00, 0.10, 0.50]	[0.20, 0.45, 0.55, 1.00]

The phase operational state is obtained from the FIS outputs. It is represented by the variable $Phase_i$ State that assumes two linguistic values associated with two trapezoidal membership functions: "Normal" and "Short-Circuit". The parameters of these membership functions are shown in Table 7.

Table 7. Membership functions parameters for FIS output variables.

Fault Type	Membership Functions Parameters	
	Normal	Short-Circuits
Single-phase	[0.00, 0.00, 0.10, 0.40]	[0.200, 0.400, 0.850, 1.000]
Two-phase	[0.00, 0.00, 0.30, 0.40]	[0.300, 0.400, 0.700, 1.000]
Three-phase	[0.00, 0.00, 0.10]	[0.046, 0.094, 0.600, 0.950]

Table 8 shows the number of simulations performed for each scenario. For each fault type, thousands of simulations are performed along the feeder for both LIF and HIF.

Table 8. Number of simulations per scenario and fault type.

Fault Type	Class	Number of Simulations		
		Scenario 1	Scenario 2	Scenario 3
$A_g/B_g/C_g$	LIF	840	840	580
	HIF	1620	1585	1620
AB/AB_g	LIF	560	560	520
	HIF	540	540	540
AC/AC_g	LIF	560	560	520
	HIF	540	540	540
BC/BC_g	LIF	560	560	520
	HIF	540	540	540
ABC/ABC_g	LIF	560	560	560
	HIF	540	540	540
Total		6860	6825	6480

Figure 10a shows the results for the short-circuit classification for scenario 1. In total, 6860 simulations were carried out, with 2460, 3300, and 1100 FIS simulations for classification of single-phase, two-phase, and three-phase short-circuits, respectively. The proposed methodology presents satisfactory performance for LIF and HIF.

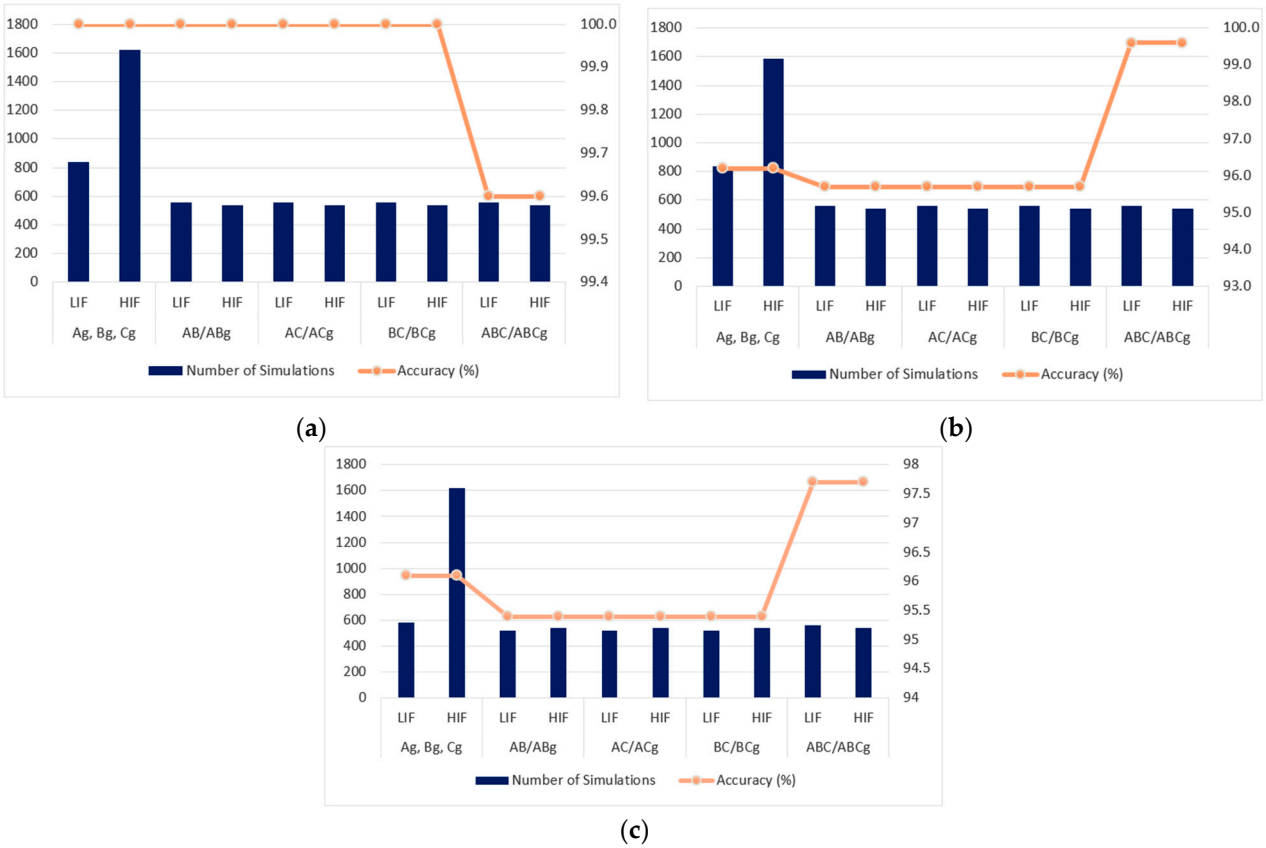


Figure 10. Short-circuit classification for three scenarios: (**a**) without DG units (scenario 1), (**b**) with two synchronous generators (scenario 2), and (**c**) with PV panels (scenario 3).

The results obtained for the short-circuit classification for scenario 2 with two synchronous generators DG units are shown in Figure 10b, where 6825 simulations are performed with 2425, 3300, and 1100 SIFs' simulations for classification of single-phase, two-phase, and three-phase short-circuits, respectively.

Finally, Figure 10c shows the results for scenario 3, where DG units with PV panels are considered and 6480 simulations are performed with 2200, 3180, and 1100 SIFs' simulations for classification of single-phase, two-phase, and three-phase short-circuits, respectively.

According to Figure 10, there is a drop in accuracy for the faults at scenarios 2 and 3, where DGs units are introduced: two synchronous generators and PV panels.

Additionally, in some parameter combinations, especially that ones with high resistances for single-phase and two-phase short-circuits, the behavior indices of normal phases present high values; therefore, they exceeded the limit established LC_i for each phase. Consequently, the classifier erroneously identifies a normal phase as defective phase in some cases.

On the other hand, the classifications with error occur in three-phase short-circuits due to behavior index with low values for the phases with disturbance; therefore, the classifier identifies them as normal, since the tolerance limits LC_i are not violated. It is worth noting that the same FIS set is used for all simulated scenarios (with and without DG units); therefore, there are no changes in membership functions parameters.

4.4. Methodology Validation for Detection and Classification of Short-Circuits Faults

This section presents metrics for evaluating the proposed methodology for detecting and classifying short-circuit faults for scenarios 2 and 3. It is noteworthy that scenario 1 has an accuracy of 100% for short-circuit fault detection and classification; therefore, it is not mentioned in this section.

4.4.1. Detection Stage Validation

The short-circuit detection stage performance is evaluated in this section for scenarios 2 and 3 for single-phase, two-phase, and three-phase short-circuit faults. In this sense, cases with short-circuit absent (NOS) and short-circuit fault are simulated to obtain the confusion matrix and its metrics to evaluate the detection stage performance.

Table 9 shows the confusion matrix elements and its metrics to evaluate the detection stage performance. It is noteworthy that the accuracy and precision have high values for all cases simulated. Therefore, the detection stage has a good coverage, i.e., it identifies most short-circuit faults.

Table 9. Confusion matrix metrics to evaluate the detection stage for scenarios 2 and 3.

Detected Fault	Scenarios	Confusion Matrix Elements				Accuracy	Precision	Recall	F1-Score
		Q_{NN}	Q_{NF}	Q_{FN}	Q_{FF}				
Single-Phase	2	290	17	0	2591	99.4%	100.0%	94.5%	97.2%
	3	241	57	0	2200	97.7%	100.0%	80.9%	89.4%
Two-Phase	2	461	122	6	3284	96.7%	98.7%	79.1%	87.8%
	3	616	56	57	3143	97.1%	91.5%	91.7%	91.6%
Three-Phase	2	196	2	0	1100	99.8%	100.0%	99.0%	99.5%
	3	210	70	0	1100	94.9%	100.0%	75.0%	85.7%

On the other hand, the recall has a lower value than the accuracy and precision metrics. Thus, in some cases, there is the fault assignment to normal cases or "false alarms", especially for the three-phase short-circuit at scenario 3. F1-score metric incorporates the outcomes of precision and recall metrics.

The insertion of synchronous generators (scenario 2) and PV panels (scenario 3) DG units changes the short-circuit currents of IEEE 34-bus test system and, in some cases, the detection stage assigns fault in normal cases, resulting in "false alarms".

4.4.2. Classification Stage Validation

The short-circuit classification stage performance is evaluated in this section for scenarios 2 and 3 for single-phase, two-phase, and three-phase short-circuit faults. Classification stage is applied after the fault detection stage.

There is an FIS classifier for each single-phase, two-phase, and three-phase fault type. Table 10 shows the FIS classifier performance for scenarios 2 and 3. For example, the FIS classifier for single-phase faults has 2365 hits (faults classified correctly) and 94 errors (faults classified incorrectly) for scenario 2. Thus, its accuracy is 96.2%. There are 52 faults classified incorrectly as single-phase. In this sense, suppose that there is a fault at phase A; however, according to FIS classifier, the fault is in B or C phases. Therefore, that is an illustrative example of single-phase fault classified incorrectly.

Table 10. FIS classifier performance for scenarios 2 and 3.

Scenario	FIS Classifier	Hits	Errors: Faults Classified Incorrectly			Accuracy
			Single-Phase	Two-Phase	Three-Phase	
2	Single-Phase	2365	52	0	42	96.2%
	Two-Phase	3154	0	138	5	95.7%
	Three-Phase	1096	0	3	1	99.6%
3	Single-Phase	2112	52	14	20	96.1%
	Two-Phase	3050	0	122	26	95.4%
	Three-Phase	1075	1	24	0	97.7%

It is worth pointing out that the accuracy metric has high value, and it is greater than 95.4% for all simulated cases.

5. Conclusions

A methodology for detection and classification of short-circuit faults in electric power distribution systems (EPDS), considering the insertion of distributed generation (DG) units, was presented in this work. The fault detection algorithm was based on continuous analysis of current signals measured at the substation output, and whose characteristics were extracted through the application of maximum overlap discrete wavelet transform (MODWT) and the mean of current signals. The system operating state was defined via comparative analysis of the normalized energy and mean values of currents, with variation limits previously established for each phase. Three fuzzy inference systems (FIS) were implemented to classify the faulty phases. The behavior indices of three-phase currents were the inputs of each FIS.

The proposed methodology was evaluated in the modified IEEE 34-bus test system and represented in ATPDraw. The proposed methodology was evaluated via 20,000 simulated cases considering three scenarios with and without DG units and considering variations of the following parameters: type of fault (single-phase, two-phase, three-phase), angle of incidence, fault resistance (high impedance fault (HIF) and low impedance fault (LIF)), fault location bus, and types of DG units (synchronous generators and photovoltaic (PV) panels).

In the fault detection stage, greater than 94.9% of short circuits were correctly detected in all scenarios with and without DG units. On the other hand, in the fault classification stage, the classifier achieved a high accuracy percentage, where 100% of short-circuits were classified correctly for scenario 1 (without GD units). In scenarios 2 and 3 (with GD units), the accuracy was greater than 95.4% for all simulated cases. The insertion of DG units increases the short-circuit current; therefore, in some cases, the classification system may erroneously indicate faults on a feeder in normal operating state (NOS).

The proposed methodology is efficient. It presented accuracy greater than 94.9% for detection and classification of all short-circuit faults, even after the insertion of DG units.

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A new approach to online training for the Fuzzy ARTMAP artificial neural network

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ABSTRACT

The evolution of internet resources has led to an increase in the flow of data and, consequently, the need for classification or forecasting models that support online learning. The Fuzzy ARTMAP neural network has been used in the most areas of knowledge; however, few have explored real-time applications that require continuous training. In this work, a Fuzzy ARTMAP neural network with continuous training is proposed. This new network can acquire knowledge via classification or prediction. Modifications made to the architecture and learning algorithm enable online learning from the first sample of data and perform the classification or forecasting at any time during training. To validate the proposed model, three experiments were performed, one for forecasting and two for classification. Each experiment used benchmark databases and compared its final results with the results of the original Fuzzy ARTMAP neural network. The results demonstrate the ability of the proposed model to acquire knowledge from the first data samples in a stable and efficient way. Thus, this study contributes to the evolution of the Fuzzy ARTMAP neural network and introduces the continuous training method as an effective alternative to real-time applications.

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1. Introduction

The artificial neural network Fuzzy ARTMAP (FAM ANN) has been frequently used to solve problems of the most areas since its first proposed version in [1]. This is due to important advantages over other ANN models such as fast training, ease of parameterization, debugging and implementation [1,2]. These capabilities are derived from Adaptive Resonance Theory, which is based on Adaptive Resonance Theory (ART) ANNs and allows them to learn in an incremental and stable way [3]. The FAM ANN has a dynamic structure, developed to overcome the dilemma of stability–plasticity; e.g., as ANN learns from new samples, it preserves the knowledge acquired with previous data [3].

Thus, FAM ANN was used in several different applications, such as structural damage monitoring [4], failure diagnosis of bearings [5,6], prediction of electric load [7,8], handwriting recognition [9], classification of hand movements [10], characterization of wave regime [11], classification of images [12] and recognition of audio-visual emotions [13].

In some of the works cited above, modifications and/or additions of other methods, such as hybrid methodology, were used to increase the efficiency of the model. In others, deeper modifications were performed with the objective of treating possible situations of FAM ANN deficiency such as proliferation of categories, sensitivity to noisy data and learning depending on the order of presentation of the samples.

Moreover, with the rise of industrial and scientific data, problems involving large volumes of data, often in real time, have increased exponentially. Consequently, new versions of the FAM ANN were proposed to deal, not only with these new challenges, but also with others not previously explored with ANN, thus contributing to its evolution.

The most recent proposed modifications to FAM ANN often addresses these new challenges via methodologies focused on the problem of category proliferation, order-dependent learning and reduction of training time and classification. In Benites (2017) [2], a new layer is added to the FAM ANN structure, with the goal of creating a hierarchical architecture that allows only a portion of the categories to compete. In the work of Pourpanah (2019) [14], the FAM is improved using a combination of Q-learning and Bayesian formalism to reduce the number of categories. The categories with low Q-value are removed and consequently the complexity of the FAM ANN is decreased. In the work of Anitha (2015) [12], a new version of FAM ANN is used as an image recovery system. The proposed version, called NPFAM, aims to avoid the proliferation of categories by storing additional probabilistic information in the Inter-ART module and

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one-to-many relationships with the ART_a and ART_b modules. In the work of Anitha (2019) [15], after training FAM using different data set samples, changing the order of presenting the samples, the differential evolution algorithm (DE) is adopted to build FAM architecture optimized, generalized and with the reduced category proliferation problem. In the work of Zhang (2014) [16], the process of competition between categories for activation was modified to reduce the category proliferation problem. The mechanism combines the Q-max dynamic rules and later probability rules to perfect the category choice and avoid overlapping areas. In the work of Jin (2013) [17], a group of FAM ANNs receive the same training data set, yet the ordering is unique for each of them. Then, the Bayesian belief method is used to select the classification with better precision among the FAM ANNs. In this way, the influence of order in the samples is reduced. Mokhtar (2013) [18] replaces the Min–Max learning rule in Fuzzy ARTMAP with a Gaussian classifier, thus establishing Gaussian ARTMAP. This modification resulted in a reduction of the category proliferation problem. In the work of Vigdor (2006) [19], the performance of FAM ANN is investigated by using average order strategies and voting for the presentation of training data and modes by season, including validation and conclusion. In the work of Parrado-Hernández (2003) [20], distributed learning was used to attenuate the problem of category proliferation. Here, the choice function activates a group of neurons for distributed learning. In the work of Gómez-Sánches (2002) [21], to address the category proliferation problem, the math-tracking mechanism was removed and the associations between the unsupervised modules maintained. Off-line learning is restarted with a new value of the monitoring parameter ρ until a suitable predefined error value is reached. In the work of Dagher (1999) [22], an algorithm was presented to optimize the ordering of the input patterns based on the Min–Max grouping method.

Thus, proposals with the aim of contributing to the evolution of online learning for real-time problems are not often discussed in the literature. In the work of Wong (2015) [23], the author introduces an online learning methodology that combines FAM ANN and OELM (Online Extreme Learning Machine). This model allows online learning to begin from the first data sample and permits classification at any time during training. In the work of Wang (2014) [24], a modified FAM ANN was used in the classification of tool conditions. One of the modifications of ANN occurred in its architecture; the module ART_b, and the categories directly related by the mapping, were removed. Additionally, the author presents incremental learning online, so that it is possible to predict categories and maintain incremental learning simultaneously. Lim (2003) [25] conducted an experimental study of the Hybrid ANN PFAM (probabilistic Fuzzy ARTMAP), which originated from the combination of FAM and probabilistic ANNs (PNN), to classify online standards in non-stationary environments. The forecast is performed for each presented sample sequentially in the training. Thus, the same sample is used for training and testing.

The need for models that support online learning is essential. The evolution of internet resources and the growing number of connected devices, for the most diverse purposes, contributes to the increase in flow of information in various areas of knowledge. In this way, it has driven research into systems capable of learning and adjusting its structure in real time. In this sense, the proposal of an online learning algorithm applied to an FAM ANN architecture is indeed important, not only for its improvement, but to develop better performance among the new real-time problem challenges.

In this way, this work presents a new online learning methodology for FAM ANN. A new layer F_2 is introduced in the architecture to represent new created neurons. These neurons, called temporary neurons, can or not pertain to the definitive knowledge of the ANN due to the defined criteria on the learning algorithm. Besides, a decrement mechanism of Rho parameter is implemented to assure the classification or prediction, in cases the ANN finds no corresponding neuron. These innovations, with adaptations in the learning algorithm, can provide the online learning from the first data sample in a stable way during the sample processing.

Three experiments with benchmark problems were performed to validate the proposed model, one for forecasting and two for classification.

2. Fuzzy ARTMAP artificial neural network

The Fuzzy ARTMAP neural network consists of two Fuzzy ART modules called ART_a and ART_b in addition to an Inter-ART module. Fig. 1 shows the Fuzzy ARTMAP architecture, where the input is in the ART_a module and the output is in the ART_b module. The inter-ART module determines the input–output pattern mapping [1].

2.1. Fuzzy ART module

Each Fuzzy ART module is composed of three layers: the input layer F_0, the comparison layer F_1, and the recognition layer F_2.

(1) Input Layer F_0*:* the input vector a, with dimension M, is normalized and codified by its complement, resulting in the activity vector I [1], according to the following equation:

$$I = \frac{a}{|a|} \tag{1}$$

where:

a = input vector

$I = [I_1 I_2 \ldots I_M]$

The complement codification is given by Eq. (2).

$$I = \left(a, a^c\right) = (a_1, \ldots, a_M, a_1^c, \ldots, a_M^c) \tag{2}$$

where:

$$a_1^c = 1 - a_1 \tag{3}$$

(2) Comparison Layer F_1: It verifies the compatibility with the input pattern ($I = (a, a^c)$) and an already existing category, which is represented by the weight vector, w_j. The vector x, results from the AND operation with the vectors I and w_j, characterized by Eq. (4) [1]:

$$x = \begin{cases} I \\ I \wedge w_{ji} \end{cases} \tag{4}$$

(3) Recognition Layer F_2: The chosen process of an existing category is initiated when a new pattern is presented to the input layer. The chosen function T_j is calculated for every jth category of layer F_2 according to Eq. (5):

$$T_j(I) = \frac{|I \wedge w_j|}{\alpha + |w_j|} \tag{5}$$

where:

α = Chosen parameter ($\alpha > 0$).

The chosen category is indexed by J in Eq. (6) and the weight vector w_j is used in the comparison layer F_1.

$$T_J = \max\{T_j : j = 1 \ldots N\} \tag{6}$$

where:

N = the total quantity of categories in the recognition layer F_2 [4].

The similarity grade with the input pattern, represented by the input vector I of layer F_0, and the chosen category J, determines

if there is resonance (weight updating) associated with the layers and consequently, the learning. The vigilance parameter ρ is used to indicate the similarity grade necessary to induce resonance. The vigilance test, with the input pattern I and the weights w_j of the chosen category, is given by Eq. (7) [1]:

$$\frac{|I \wedge w_J|}{|I|} \geq \rho \tag{7}$$

If the vigilance test is not satisfied, the chosen function rejects the reproved category and continues to search for another J category that satisfies the test (4).

2.2. Inter-ART module

After the resonance process with ART_a and ART_b modules, the Inter-ART module verifies the correspondence with the active category ART_a and ART_b modules, resulting in the Inter-ART module vector x^{ab} of Eq. (8):

$$x^{ab} = \begin{cases} y^b \wedge w_J^{ab} & F_2^a \text{ active and } F_2^b \text{ active} \\ w_J^{ab} & F_2^a \text{ active and } F_2^b \text{ inactive} \\ y^b & F_2^a \text{ inactive and } F_2^b \text{ active} \\ 0 & F_2^a \text{ inactive and } F_2^b \text{ inactive} \end{cases} \tag{8}$$

The binary Inter-ART module weight matrix w_J^{ab} maps the categories of the ART_a module with the respective category in the ART_b module, while the Inter-ART module confirms the correspondence with the matrix w_J^{ab} categories. Otherwise, the vector x^{ab} assumes the value 0, and a new search process in ART_a is executed until the correspondence between input and output is validated. This mechanism is called Match Tracking [1].

(1) Match Tacking: Its objective is to change the category in the ART_a module by force, thus answering the corresponding error of the active categories in ART_a and ART_b. The active category of ART_a has satisfied the vigilance test, i.e., the similarity grade with the input pattern of ART_a, vector I^a, and the active category of ART_a, J^a, which is greater than the exigency of the vigilance parameter ρ_a. Thus, to force the neural network to choose another category, the parameter ρ_a is increased by a predefined incremental parameter ε until the current active class no longer satisfies the vigilance test [1].

At the beginning of each input pattern presented to the ART_a module, the vigilance parameter ρ_a is equal to the base vigilance value $\overline{\rho}_a$. If equation (9) is satisfied, then ρ_a is incremented by ε until equation (10) is satisfied [1].

$$|x^{ab}| < \rho_{ab}|y^b| \tag{9}$$

where: ρ_{ab} = the vigilance parameter of Inter-ART module

$$|I \wedge w_J^a| < \rho_a|I| \tag{10}$$

When this occurs, the ART_a module searches for another category to activate and the Inter-ART module performs a new corresponding verification with the input and output patterns.

After realizing the resonance process for the ART_a and ART_b modules and the *Match Tracking* in the Inter-ART module, the neural network is ready to acquire new information by adapting the weights w_J^a and w_{Jk}^{ab}, referred to as the ART_a and Inter-ART modules, according to Eq. (11), [1]:

$$w_J^{new} = \beta\left(I \wedge w_J^a\right) + (1-\beta)w_J^a \tag{11}$$

$$w_{JK}^{new} = 1 \tag{12}$$

where:

β = training rate ($\beta \in [01]$).

The adaptation of the set of ART_b weights is independent of the Inter-ART module after the active class in ART_b remains stable until the next input. Thus, the adaptation of the weights is realized after the ART_b resonance, according to equation:

$$w_k^{new} = \beta\left(I \wedge w_K^b\right) + (1-\beta)w_K^b \tag{13}$$

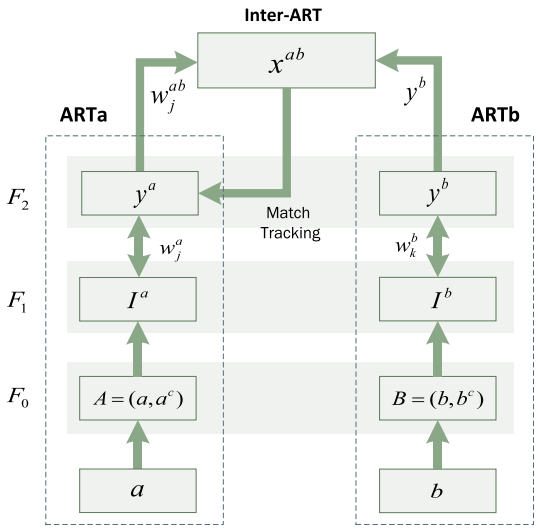

Fig. 1. Fuzzy ARTMAP architecture flow chart.

3. Continuous training

Continuous training is a mechanism that provides FAM ANN with the ability to perform its learning online from the first data samples presented to the network. In this mechanism, for each entry pattern, the ANN performs forecasting or classification, and in sequence, the training. Therefore, the same input pattern is used for both diagnosis and training.

According to Fig. 2, in the off-line training, the diagnosis is initiated only after ending the phase training. In online mode, the training and diagnosis are executed simultaneously.

Online learning allows the continuous extraction of knowledge relative to the problem addressed, favoring the adaptation of ANN to possible changes in behavior of the input data during the diagnosis. However, it requires that the model has a flexible architecture and some mechanism that avoids the proliferation of categories.

The FAM ANN, due to its architecture originated from ART, has the essential characteristics of a model with online learning, although it presents the problem of proliferation of categories, a result of the overlapping representation areas of each category [20,21].

In this proposal, a second layer F_2 is introduced in ART_a module of ANN. Besides the changing in the ANN architecture, the learning algorithm is adapted to allow the continuous flow of samples without occurring the proliferation of categories, and consequently an impracticable computational cost. Among the principal adaptations in the learning algorithm, it is emphasized the automatic adjustment of the ρ_a parameter in the diagnosis phase, and the monitoring of the availability of input and output data executed by Inter-ART module. The absence of data in ART_a or ART_b module, identified in Inter-ART module determines the execution of diagnosis or training. Therefore, modifications were performed in the FAM ANN architecture and in its learning algorithm to select the data according to their relevance and ensure good quality in the ANN training process.

The concept of continuous training was originally proposed by Marchiori (2011) [26] for the transient stability analysis of

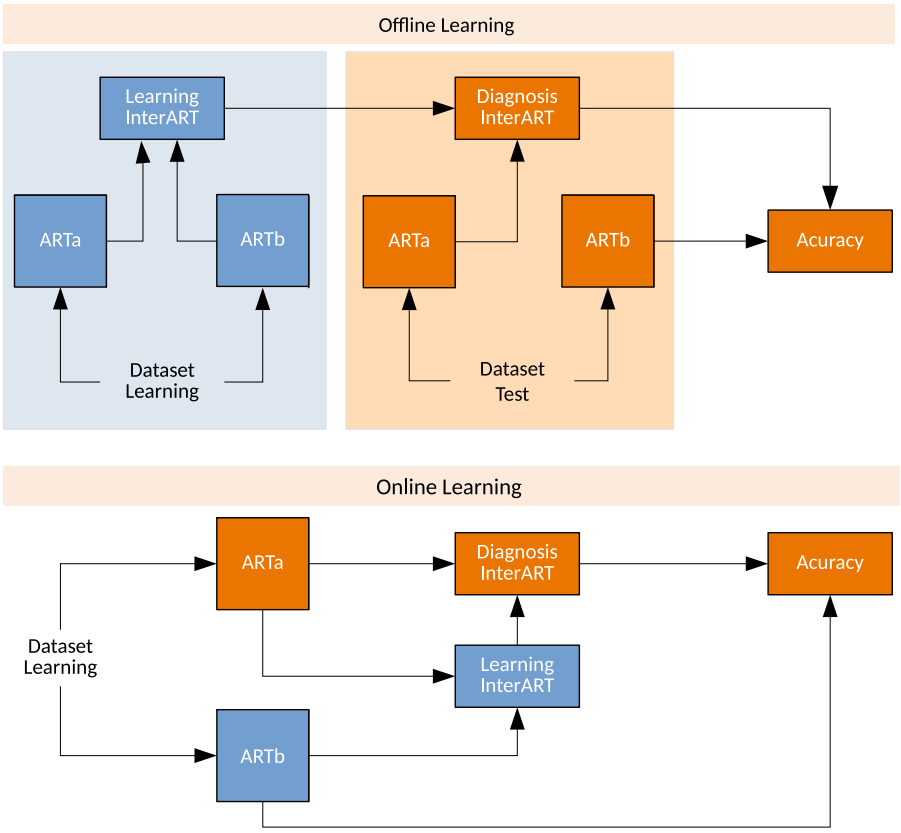

Fig. 2. Block diagram of online and offline learning. In online learning, for each input pattern, the ANN performs prediction or classification, and in the sequence, the training.

electric power systems. However, important changes were made in the process. In the work of Marchiori (2011) [26], continuous training was implemented under the Euclidian ARTMAP ANN architecture, which also belongs to the ART family. During the training process, part of the set of samples is intended for offline learning; categories are created in the training phase of the modules. Later, in the diagnosis phase, ANN allows the weights of the categories belonging to the module to be changed, once new relevant information is identified.

Other works have contributed to the evolution of the Fuzzy ARTMAP ANN online learning, emphasizing the works of Lim (2003) [25], Wang (2014) [24] and Wong (2015) [23]. In Lim (2003) [25], it is necessary the off-line training phase to build the knowledge base before initializing the online learning. In the present proposal and in the works of Wang (2014) [24] and Wong (2015) [23], the online learning can be initiated from the first data samples. Among other similarities, there is the capacity of realizing diagnosis at any moment of the training phase.

The novelties of this proposal, besides the cited characteristics, are the flexible architecture and the adaptation of the learning algorithm, which provide the following advantages to the ANN:

- Maintenance of the neurons that represent the present data flow and removing the neurons with less relevance or that become outdated face to new samples. This characteristic reduces the problem of proliferation of categories and supports the application in forecasting problems, which is few explored in literature, considering the ART family ANN.
- Monitoring the availability of data in modules ART_a and ART_b to determine the supervised and non-supervised training.
- Automatic adjustment of parameter ρ_a in diagnosis phase to assure the correspondence with new samples and an inexistent class in ANN.

3.1. FAM ANN architecture: Temporary F_2 layer

The change of the ANN architecture occurs in the layer F_2 of module ART_a, which in addition to the existing F_2 layer, is added the temporary layer F_2, responsible for temporary categories.

Since ANN creates a new neuron in ART_a, because it does not find a category compatible with the input pattern, this new category is called temporary and is represented by a vector of temporary weights, w_{temp}, belonging to the temporary F_2 layer. The new category can be incorporated into the ANN after being selected to represent new patterns introduced to ANN a minimum number of times, as predefined through the parameter N_{min}, or permanently excluded if the number of standards introduced to the network exceeds the value stipulated by parameter N_{max}. This series of steps occurs without using the temporary category. Therefore, temporary categories that meet the criteria established by the parameters, N_{min} and N_{max}, are incorporated into the definitive F_2 layer and become part of the permanent knowledge of ANN. These actions prevent the proliferation of categories, and consequently a possible memory overhead, in addition to reducing computational cost and improving ANN performance [3]. The new architecture is shown in Fig. 3.

Temporary categories represent recent information presented to ANN. The influence of the temporary category on the performance of the model is not known, since increasing the frequency of inputs with similar characteristics presupposes the tendency towards a certain behavior of the problem. Therefore, it is characterized as good-quality information to associate with the permanent knowledge of ANN. Single or low-frequency input patterns with similar characteristics presupposes that they are noisy and therefore have low quality to positively influence the good ANN performance. The N_{min} and N_{max} parameters determine

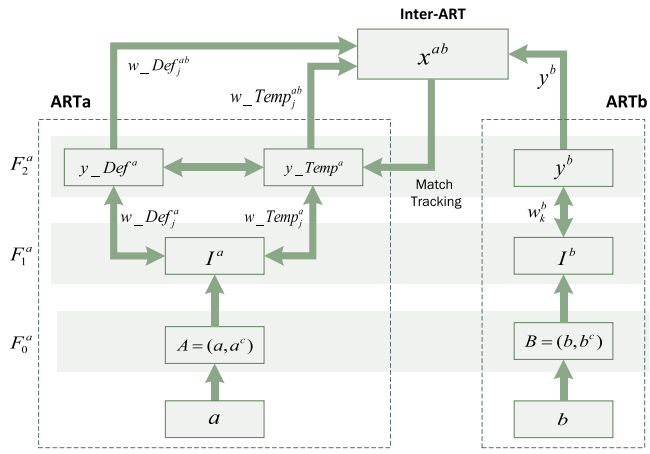

Fig. 3. Block diagram of FAM CT ANN architecture. The ART_a module receives a new F_2 layer responsible for the temporary categories.

the minimum and maximum frequencies for the acquisition of new information by ANN.

- N_{min}: Minimum number that a category must be selected as the winner to become a definitive category [26].
- N_{max}: Number of entries that a category remains as temporary without having reached the value of N_{min}. After that period, the category is rejected [26].

3.2. Monitoring of data through the Inter-ART module

The Inter-ART module specifies the combinations with the categories of ART_a and ART_b modules through their weights obtained in the training phase. In addition, in the FAM ANN with continuous training, the Inter-ART module will monitor the availability of input patterns, in the modules ART_a and ART_b, to activate the following procedures:

- Diagnosis and supervised training;
- Diagnosis and unsupervised training.

FAM ANN, in its original form, only has supervised training. For this to happen, input patterns must always be available to ART_a and ART_b modules. If the input pattern is the only pattern available to the ART_a module, the FAM ANN will start the diagnosis phase. Therefore, the availability of input patterns for the ART_b module determines whether ANN is in the supervised training or diagnosis phase. In the FAM ANN with continuous training, the same input pattern is used for both diagnosis and training, unifying the procedures. In this way, the Inter-ART module monitors the modules to activate the sequence of procedures correctly via Eq. (14):

$$x^{ab} = \begin{cases} y^b \wedge w_j^{ab} & F_2^a \text{ active and } F_2^b \text{ active} \\ w_j^{ab} & F_2^a \text{ active and } F_2^b \text{ inactive} \end{cases} \quad (14)$$

Eq. (14) shows the output given by the Inter-ART module according to the data availability in ART_a and ART_b modules. Thus, if there are current input patterns in both modules, both will be enabled, and ANN will perform the following steps:

Step 1. Execution of the diagnosis procedure using the existing ANN categories.

Step 2. Execution of the ANN supervised training procedure using the current data available in the ART_a and ART_b modules.

Once the module ART_a is activated, that is, there is no available input pattern in the ART_b module, ANN will perform the following steps:

Step 1. Execution of the diagnosis procedure using the existing ANN categories.

Step 2. Execution of the training procedure of the ART_a module (update of the temporary or definitive weights).

The monitoring process of the Inter-ART module also contributes by adding improvements to the implementation of online learning. This provides FAM ANN with a greater flexibility with the data flow, since it does not necessarily require the continuous flow of data in the modules ART_a and ART_b. The execution of supervised training, unsupervised training, and diagnosis procedures may alternate in response to the input patterns available for both modules.

3.3. Learning algorithm and diagnosis of continuing training

In online learning, the training and diagnosis procedure are performed together, so that, for the same input pattern, the two procedures are executed in sequence. In the diagnosis, ANN activates the category that best represents the input pattern through the choice of function and verifies whether the degree of similarity with the input pattern and the active category meets the value defined by the parameter ρ_a. In this process, entry patterns with characteristics very different from the existing categories are expected to arise so that none of the categories may be similar enough for approval in the vigilance test.

Therefore, a new mechanism for the automatic reduction of parameter ρ_a was implemented. Thus, if none of the existing categories have sufficient compatibility with the new standard required by parameter ρ_a, the value is gradually decreased until the next category becomes compatible.

The execution of FAM CT ANN (Fuzzy ARTMAP Neural Network with Continuous Training) can be described by the following steps:

Step 1. Presentation of the input vector for FAM CT ANN.

Step 2. If no category is compatible with the current input pattern, then the ρ_a parameter is iteratively decreased by ε until one of the existing categories is reached in the vigilance test. The diagnosis is made sequentially.

Step 3. If, while choosing the winning category in step 2, the ρ_a parameter has decreased, then a new temporary category is created. Otherwise, the winning category weights are updated. If there is input data in module ART_b, the training will be supervised.

Step 4. If the winning category is defined as temporary, check the number of times it won. If this value is equivalent to the value of the parameter N_{min}, the weights become definitive.

Step 5. Temporary categories are eliminated if they do not meet the criteria established by N_{max} and N_{min} parameters.

The continuous training flowchart is shown in Fig. 4.

4. Computational results

The performance of FAM CT ANN is evaluated by three benchmark experiments, the Spambase data set and Satellite Image data set of the UCI Machine Learning repository [27] and the Mackey–Glass time series data set [28]. To select the databases, the focus are both in classification and forecasting to better evaluate the model.

To make the comparison with the FAM ANN and FAM CT, each data set will be divided into two parts, training and testing.

The ANNs were implemented in the Java 8 programming language and processed on a Microcomputer with Intel Core i7-4500U 1.8 GHz processor and 8.00 GB of RAM.

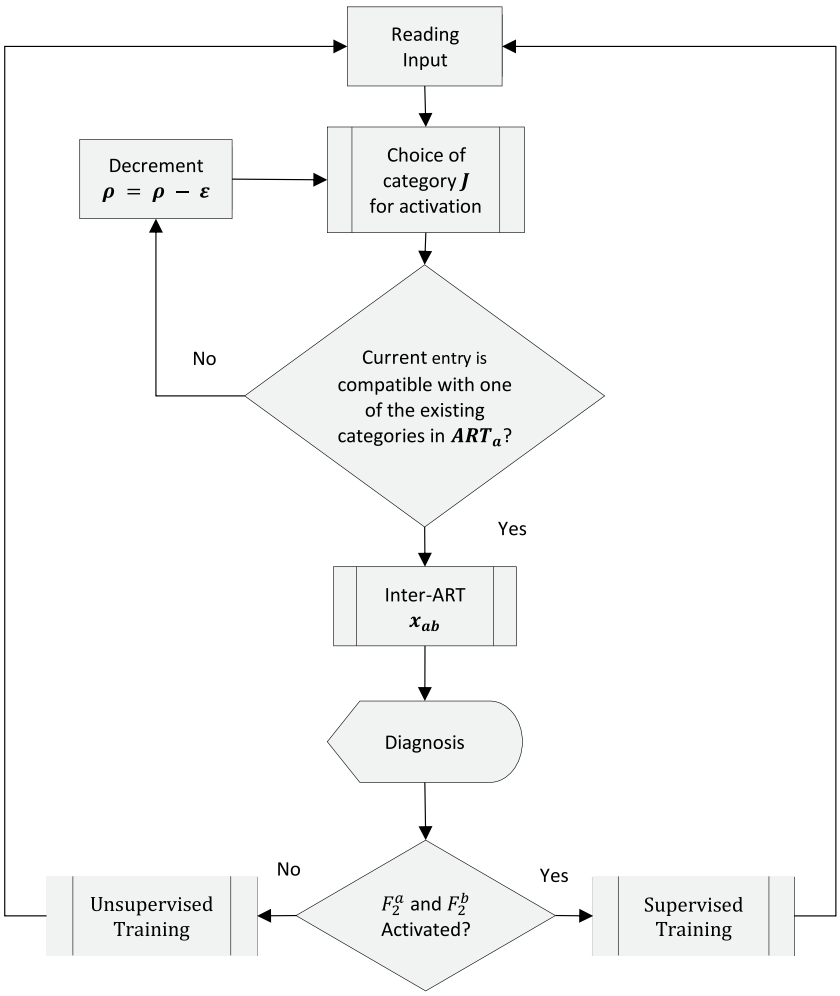

Fig. 4. Continuous training flowchart.

4.1. Classification of spams

The Spambase data set [27] was generated in the Hewlett-Packard lab and has 4601 samples, of which 1813 are spams. The samples are composed of 57 attributes that correspond to words, characters, and consecutive sequence lengths of capital letters. The value of each attribute is set according to its frequency in the email.

The good performance of the classification task has a strong dependence on the quality of the data used in the training phase. The relationship with the characteristics found in the samples that were used in the training, and the samples used in the test phase, defines the quality of the data and its complexity. Similar characteristics with the training and test bases facilitate the classifier task. However, in regard to real problems, it may not always be possible to forecast the next samples to be classified, which makes classification complex.

To better evaluate FAM CT ANN and to compare it to FAM in the classification task, three tests were performed using different samples in the FAM ANN training data set. In test 1, the training set consisted of 30% of the spam set and 30% of the legitimate e-mail set. In test 2, 10% of the spam set and 50% of the legitimate e-mail set were used. In test 3, 50% of the spam set and 10% of the legitimate e-mail set were used.

The variation in the number of spam and legitimate emails in the training set tends to influence the rate of correctness during the sample classification, which is classified according to the most frequent characteristics in the training set.

Table 1
Training data set.

	Test 1	Test 2	Test 3
Spams	544 (30%)	181 (10%)	906 (50%)
Legitimate	836 (30%)	1394 (50%)	279(10%)
TOTAL	1380	1575	1185

Table 2
Training and test data set.

	Test 1	Test 2	Test 3
Training Data Set	1380 (30%)	1575 (34%)	1185 (26%)
Test Data Set	3221 (70%)	3026 (66%)	3416 (74%)

The sequence of samples in the training set was created by alternating between legitimate e-mails and spam. For the test set, the first samples are formed by legitimate e-mails and then by spam.

Table 1 presents the data used in the FAM ANN training data set in the three tests performed.

This way, the quantity of data used in training and test are described in Table 2.

The parameters used in FAM CT ANN and FAM are presented in Table 3:

Fig. 5 compares the evolution of accuracy of each sample classified by FAM CT ANN and FAM in the three tests performed. The FAM CT starts the classification from the first sample and

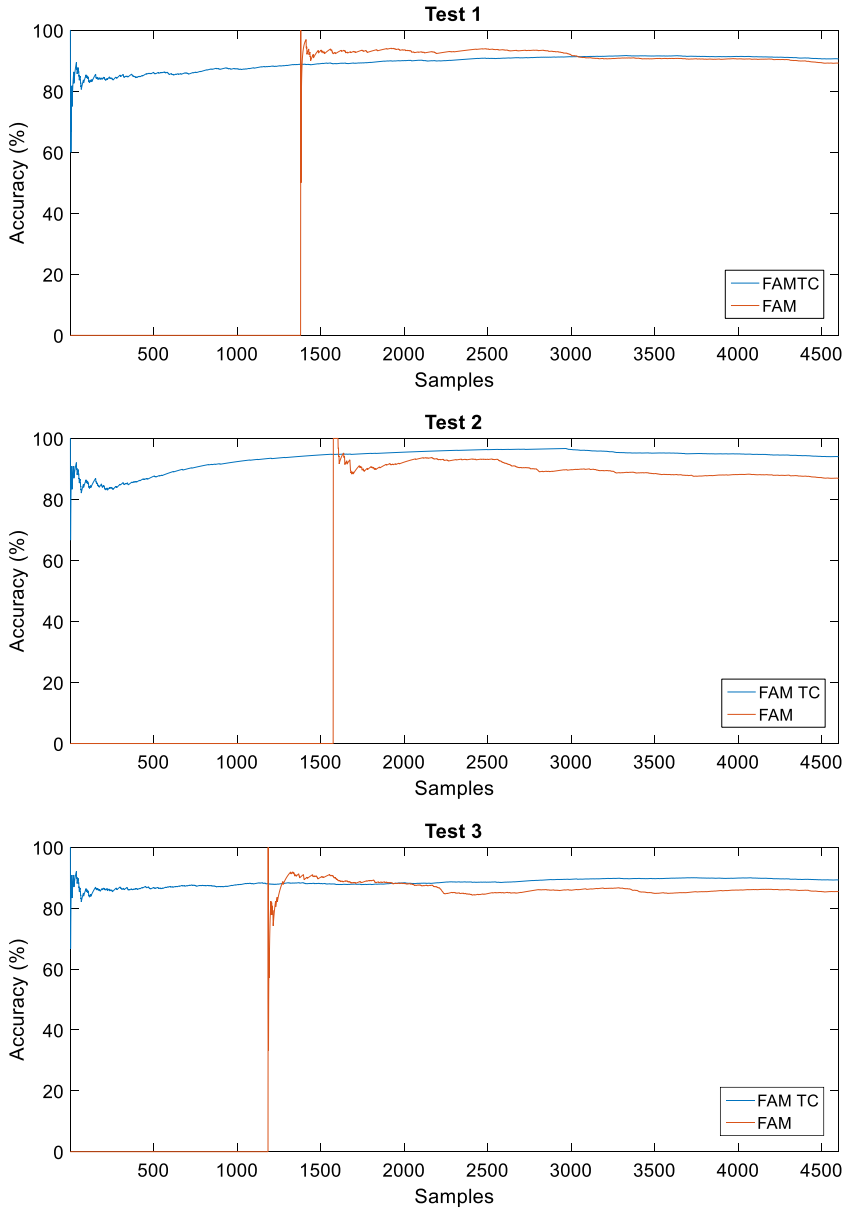

Fig. 5. Evolution of accuracy for each sample of the FAM CT (blue) and FAM (red) ANNs.

Table 3
Parameters of FAM CT and FAM.

	Test 1		Test 2		Test 3	
	FAM CT	FAM	FAM CT	FAM	FAM CT	FAM
$\overline{\rho}_a$	0.82	0.88	0.80	0.91	0.76	0.72
Nmin	3	–	6	–	3	–
Nmax	600	–	200	–	1100	–

FAM begins after the training phase is finished. In test 1, the FAM CT maintains its stable performance throughout the test and shows an improved rate of correctness with each increasing sample, since its learning remains active. The FAM ANN starts the classification at sample 1381, and although it exceeds FAM CT at the beginning of the classification, its performance tends to decrease during the test, since the knowledge acquired in the training phase becomes outdated. In this test, FAM CT ANN obtained accuracy of 91.4, while the FAM ANN reached 89.26.

In test 2, only 10% of the spam was integrated into the training set; the remainder consisted of 50% of the legitimate e-mails. This set revealed FAMs limitations, from its inability to correctly classify spam to its struggle with maintaining performance in the face of new information. FAM classification started from the 1576th sample, and despite having a training set with more samples than in Test 1, it could not improve performance. FAM CT ANN, on the other hand, showed little influence in the classification of spam due to the low number of spam samples at the beginning of the classification. Its hit rate kept evolving and showed a subtle drop from the 3000th sample. It is emphasized that the training base for test 2, 10% of spam set and 50% of legitimate set, is the closest of the real behavior of the problem. In this test, the FAM CT ANN obtained accuracy of 93.65, while the FAM ANN reached 86.95.

In test 3, only 10% of the legitimate e-mails were part of the training set, the rest being 50% spam. Again, the low number of samples from one of the classes in the training set makes it difficult to classify by FAM ANN. The classification started from sample 1185, and in the course of classification, there was an increase in errors committed by ANN; although it counts a training set with more samples than in test 1, it cannot improve the performance. FAM CT ANN, on the other hand, presents stability

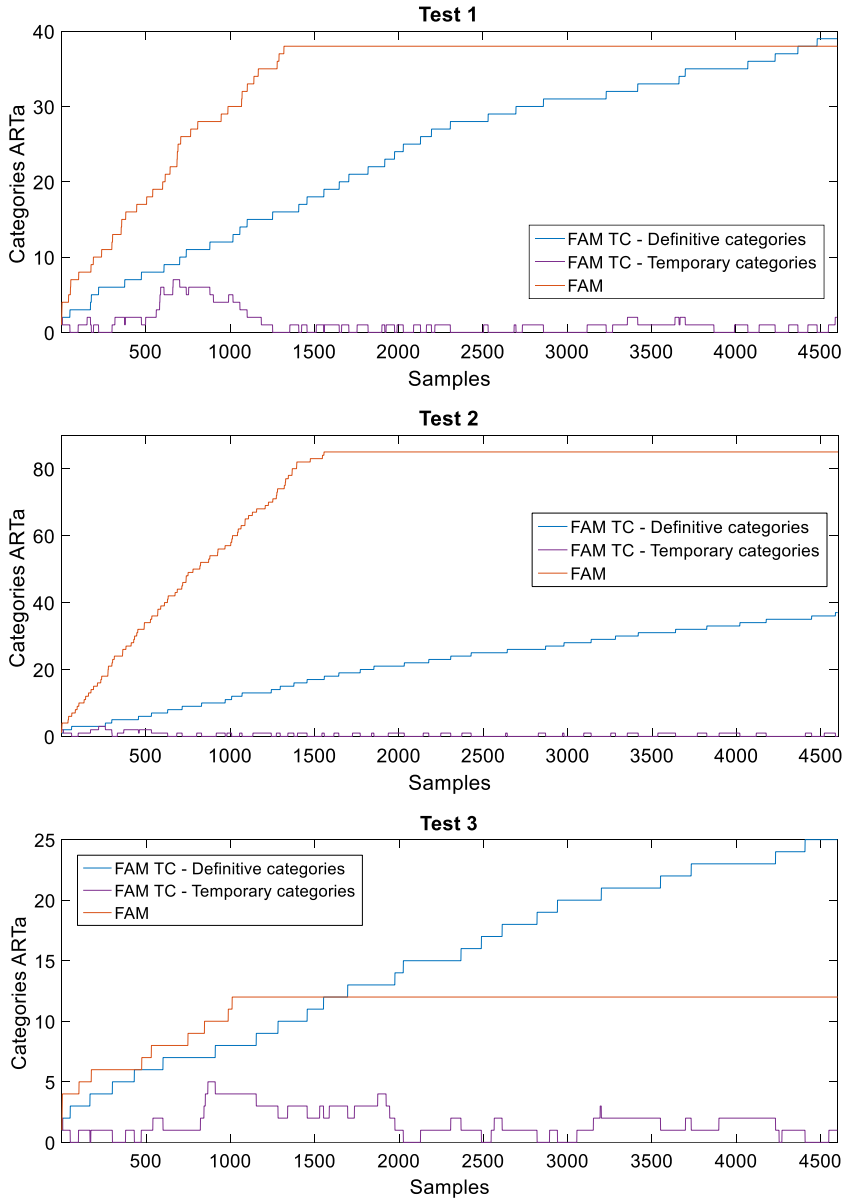

Fig. 6. Numbers of neurons created in the ART_a module of the FAM CT (blue) and FAM (red) ANNs.

and improves its accuracy rate for each sample presented. In this test, the FAM CT ANN achieved an accuracy of 89.7, while the FAM ANN achieved an accuracy of 85.48.

To ensure continuous training, the ANN architecture must evolve according to the knowledge acquired from the samples. Fig. 6 shows the categories created by the ANNs during the three tests. Note that the creation of FAM categories occurs only in the training phase. The performance of the FAM categories is limited by the quality and quantity of the training set data, whereas the FAM CT keeps their learning active, selecting the relevant information from the N_{min} and N_{max} variables.

Table 4 presents a summary of the results obtained by the FAM CT and FAM ANNs in the spam classification.

4.2. Satellite image

The Satellite Image database [27] was generated from the scanner multi-spectral Landsat. It consists of images of neighborhoods represented by sub-areas composed of 82 × 100 pixels, each sample of the set corresponding to a region of 3 × 3 pixels. The objective is to recognize the central pixel of one of the six existing regions. These include red soil, cotton crop, gray soil, moist gray soil, stubble vegetation (stubble) and very humid gray soil, according to the multispectral value of each region. The training and test sets contain 4435 and 2000 samples, respectively.

They are: red soil, cotton crop, gray soil, moist gray soil, stubble and very wet gray soil, according to the multispectral value of each region.

The parameters used in FAM CT ANN were as follows:

- Initial monitoring parameter of the module ART_a $(\overline{\rho}_a) = 0.992$;
- Monitoring parameter of the module ART_b $(\rho_b) = 1$;
- Monitoring parameter of the module Inter-ART $(\rho_{ab}) = 1$;
- Training rate $\beta = 1$;
- Parameter of choice $\alpha = 0.1$;
- Parameter of increment/decrement $\varepsilon = 0.1$;
- Parameter $Nmin = 2$;
- Parameter $Nmax = 100$;

The parameters used in FAM ANN were as follows:

Table 4
Spam classification.

	Model	Categories ARTa		Accuracy		Time (ms)/Sample
		Definitive	Temporary	Training	Test	
Test 1	FAM CT	39	2	88.84	91.40	0.028
	FAM	36	–	–	89.26	0.012
Test 2	FAM CT	37	0	94.73	93.65	0.028
	FAM	85	–	–	86.95	0.012
Test 3	FAM CT	25	1	88.10	89.70	0.028
	FAM	12	–	–	85.48	0.012

- Initial monitoring parameter of the module ART_a $(\overline{\rho}_a) = 0.96$;
- Monitoring parameter of the module ART_b $(\rho_b) = 1$;
- Monitoring parameter of the module Inter-ART $(\rho_{ab}) = 1$;
- Parameter of increment $\varepsilon = 0.1$;
- Training rate $\beta = 1$;
- Parameter of choice $\alpha = 0.001$;

Fig. 7 shows the evolution of the accuracy to each sample classified by the FAM CT ANN and FAM. FAM CT starts the classification from the first samples, and the FAM from sample 4436, after the training phase ended. The FAM CT ANN maintains its stable performance throughout the test, obtaining the average accuracy of 92.50, while the FAM ANN reached 90.80.

In Fig. 8, note the ability of the model to select the relevant information through the variables Nmin and Nmax while maintaining its active learning during the processing of all data base. As shown in Table 5, it were necessary 1777 categories for the representation of the training set by the FAM ANN. While the FAM CT ANN only 3 definitive categories and 100 temporary categories. In this way, it significantly reduces the need for computational resources.

4.3. Forecasting the time series Mackey–Glass

The Mackey–Glass benchmark time series [28] is used in this work to evaluate the performance of the model's ability to forecast a long-term time series. The values of the series are obtained via the following differential equation in time (16) [29]:

$$\frac{dx(t)}{dt} = \frac{0.2x(t-\tau)}{1+x^{10}(t-\tau)} - 0.1x(t) \tag{15}$$

To find the numerical solution of the equation, the fourth-order Runge–Kutta method was used. Assuming the step size h = 0.1, $x(0) = 1.2$, $\tau = 17$ and $x(t) = 0$ for $t < 0$. The objective is to predict the value of x_{t+85} for any value of t, considering the inputs x_{t-18}, x_{t-12}, x_{t-6}, and x_t [20].

The data set consists of 3500 samples. Of those samples, 3000 samples were obtained in the interval [201, 3200], and 500 samples were obtained in [5001, 5500] [29].

The parameters used in FAM CT ANN were as follows:

- Initial monitoring parameter of the module ART_a $(\overline{\rho}_a) = 0.97$;
- Monitoring parameter of the module ART_b $(\rho_b) = 0.99$;
- Monitoring parameter of the module Inter-ART $(\rho_{ab}) = 1$;
- Training rate $\beta = 1$;
- Parameter of choice $\alpha = 0.001$;
- Parameter $Nmin = 2$;
- Parameter $Nmax = 1500$.

The results obtained were compared with the FAM ANN. To accomplish this, the samples in the interval from$t = 1$ to$t = 875$ were reserved to train the specific ANN.

Fig. 9 compares the actual value and predicted value by the FAM CT and FAM ANN of the first 1000 prediction points. Fig. 9(a) shows the forecast performed by FAM CT, and Fig. 9(b) shows the forecast performed by FAM ANN.

In Fig. 10, the error in the prediction of each point is presented. Fig. 10(a) shows the ability of FAM CT ANN to maintain good performance during the forecasting, since the error tends to decrease as knowledge is maintained through online learning. However, Fig. 10(b) shows a gradual increase in error from the beginning of the FAM ANN prediction. Such behavior can be justified by the lack of knowledge acquired in the training phase, since the predictions of the distant points of the samples used for learning tend to exhibit different characteristics.

Online learning requires a flexible ANN architecture. Figs. 11 and 12 shows the evolution of the model structure for each input. It is important to note that continuous training enables category creation, whenever necessary, as well as the exclusion of temporary neurons. In the ART_b module, FAM CT ANN stabilized approximately 85 categories, while FAM stabilized 161 categories. The ART_a module of the FAM CT ANN stabilized at approximately 190 definitive categories and 1000 temporary categories, while FAM ANN created 789 categories.

Table 6 shows the results obtained by FAM and FAM ANNs in the time series forecast.

4.3.1. Reduction of category proliferation in forecasting the time series

The proliferation of categories is one of the problems of FAM ANN frequently addressed in literature and consists of overlapping areas of representation of categories [12,14,16,20,21]. The main consequence of the proliferation of categories is the memory overhead that contributes to the increase in the computational cost and in the reduction of the performance of the FAM ANN in general.

The Mackey–Glass reference series [28] is used to evaluate the performance of the model's ability to reduce the category proliferation problem. The objective is to predict the value of x_{t+85} for any value of t, considering the inputs x_{t-18} and x_t. In these tests, only two entries were used in order to make graphic representation viable.

The data set consists of 3500 samples. Of those samples, 3000 samples were obtained in the interval [201, 3200], and 500 samples were obtained in [5001, 5500] [2].

The parameters used in FAM CT ANN were as follows:

- Initial monitoring parameter of the module ART_a $(\overline{\rho}_a) = 0.97$;
- Monitoring parameter of the module ART_b $(\rho_b) = 0.99$;
- Monitoring parameter of the module Inter-ART $(\rho_{ab}) = 1$;
- Training rate $\beta = 1$;
- Parameter of choice $\alpha = 0.001$.

Fig. 13 shows the representation areas of the categories created in the ART_a module by the FAM ANN. It is observed the overlapping of the categories representation areas, causing several categories to represent the same point on the chart. This phenomenon limits the performance of FAM ANN not only in the computational cost but also in the accuracy of the results.

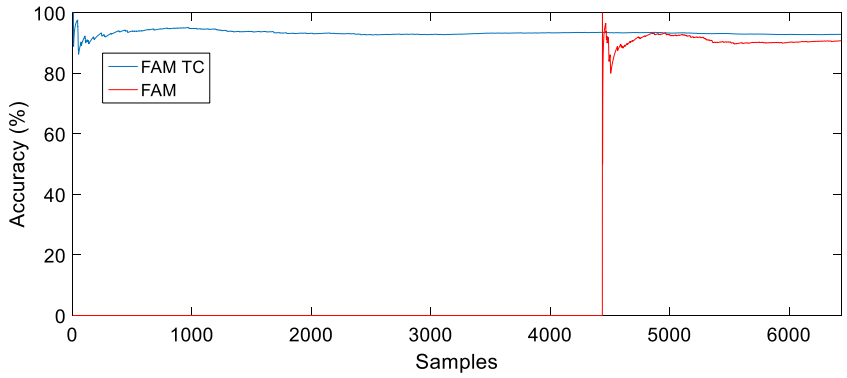

Fig. 7. Evolution of accuracy for each sample of the FAM CT (blue) and FAM (red) ANNs.

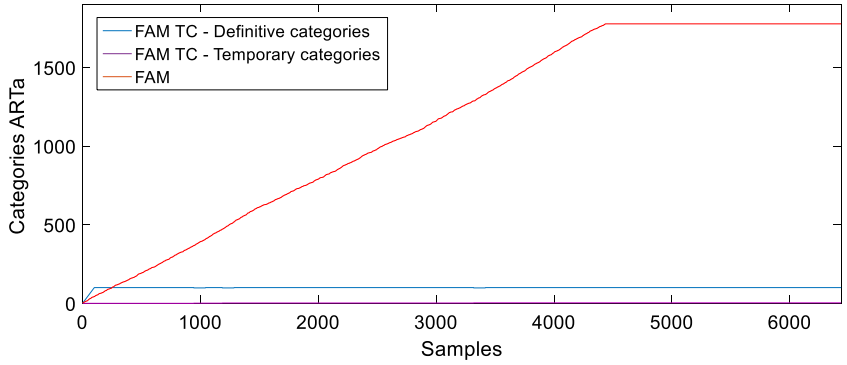

Fig. 8. Numbers of neurons created in the ART_a module of the FAM CT (blue) and FAM (red) ANNs.

Table 5
Satellite image.

Model	Categories ARTa		Accuracy		Time (ms)/Sample
	Definitive	Temporary	Training	Test	
FAM CT	3	100	93.51	91.50	0.457
FAM	1777	–	–	90.80	0.242

Table 6
Forecasting the time series Mackey–Glass.

Model	Category ARTa		Category ARTb	RMSE		Time (ms)/Sample
	Definitive	Temporary		Training	Test	
FAM CT	171	1099	89	0.0437	0.0279	0.436
FAM	789	–	161	–	0.1826	0.254

Fig. 14 shows the areas of representation in the categories created in the ART_a module by the FAM CT ANN. It is possible to notice the reduction in the overlap of the areas in the representation of the categories. The *Nmin* and *Nmax* parameters are responsible for regulating the creation of the definitive categories in the FAM CT controlling the insertion of new categories, disregarding obsolete or low frequency data in the data flow. Thus, it contributes to the problem of proliferation of categories in FAM ANN.

5. Conclusion

In this work, a new online learning approach for FAM ANN was presented. The model introduced the concept of continuous training, which enables ANN to acquire knowledge throughout the diagnosis from the first samples. To accomplish this, modifications were made in the architecture and in the FAM ANN algorithm. They are: the inclusion of a new layer in the ART_a module, a new strategy of adjusting the parameter ρ_a according to the input pattern, selection of the relevant categories through the parameters N_{min} and N_{max}, and automatic activation of supervised training due to the availability of data in the ART_b module.

The model was evaluated using the Mackey Glass time series forecast, Spambase classification data set and classification data set Satellite Image. In both applications, the performance of FAM CT ANN was compared to that of FAM ANN to validate the contributions of the proposed work. The results confirm the ability of FAM CT ANN to acquire knowledge along the samples presented in a stable and efficient manner, guaranteeing the quality of the diagnosis through the selection of relevant information for its long-term performance.

Given this and the increasing need for models that support online learning, FAM CT ANN proves to be a good alternative to use for applications and studies that require the capacity to maintain knowledge efficiently. In addition, it contributes to the evolution of the Fuzzy ARTMAP ANN.

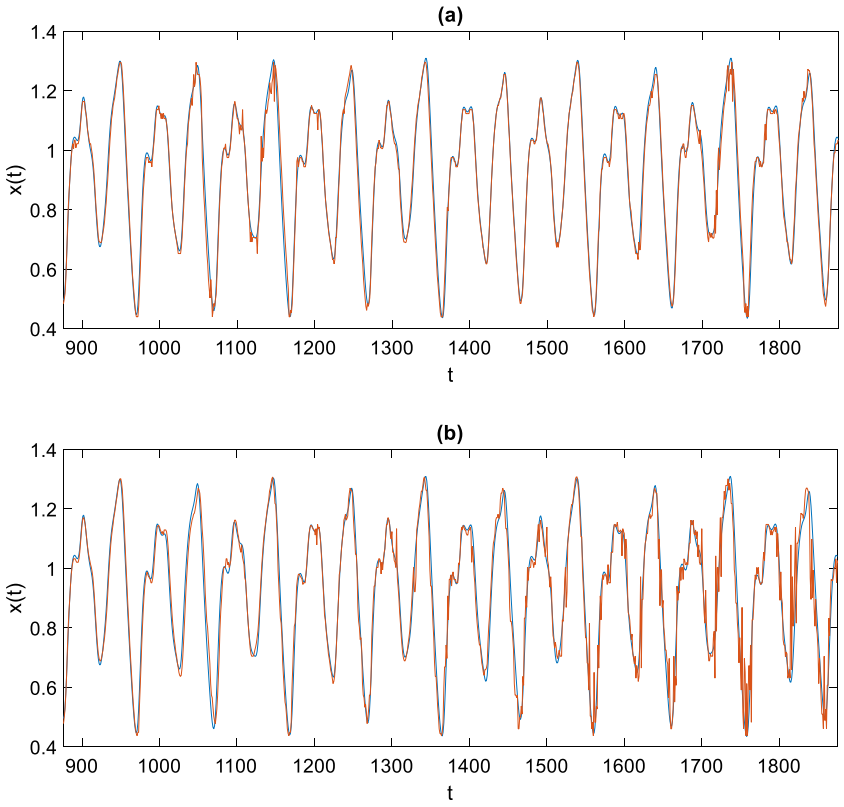

Fig. 9. (a) Forecasting for the time series Mackey Glass by FAM CT ANN. (b) Forecasting for the time series Mackey Glass by FAM ANN. The graph compares the actual output of the data set (blue) with the output predicted by the model (red) for $t = 876$ to $t = 1875$.

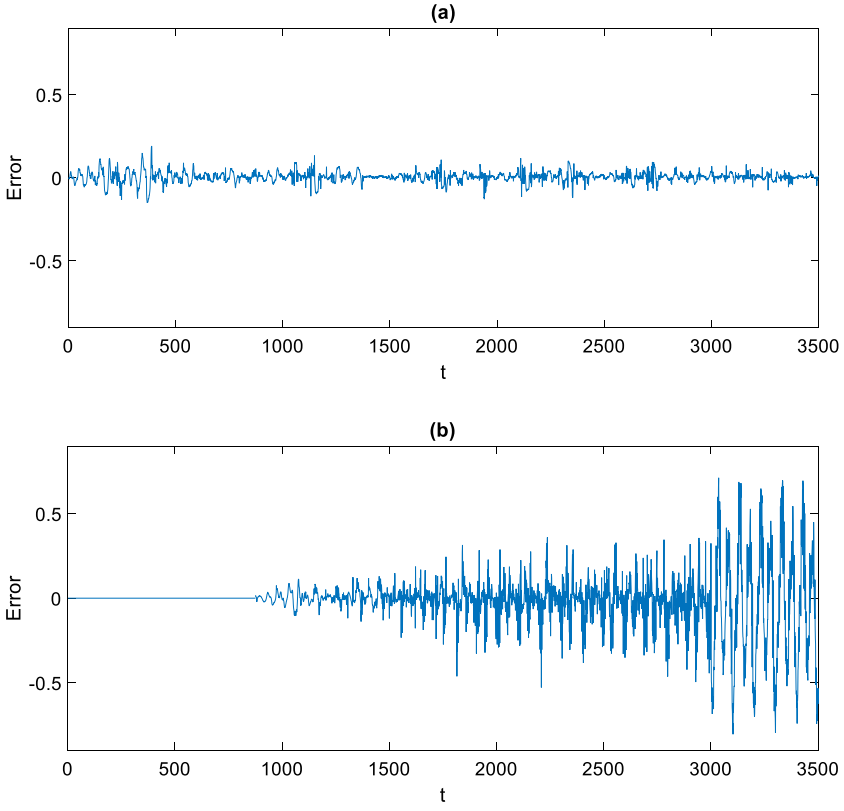

Fig. 10. Error in predicting the Mackey Glass time series (a) FAM CT ANN and (b) FAM ANN.

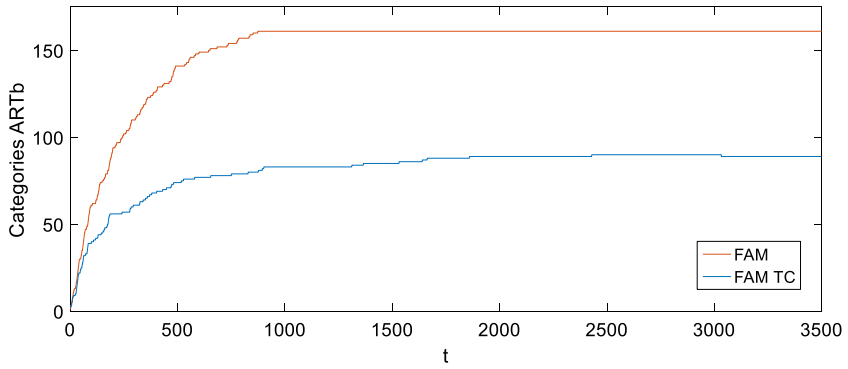

Fig. 11. Numbers of neurons created in the ART_b module of the FAM CT (blue) and FAM (red) ANNs.

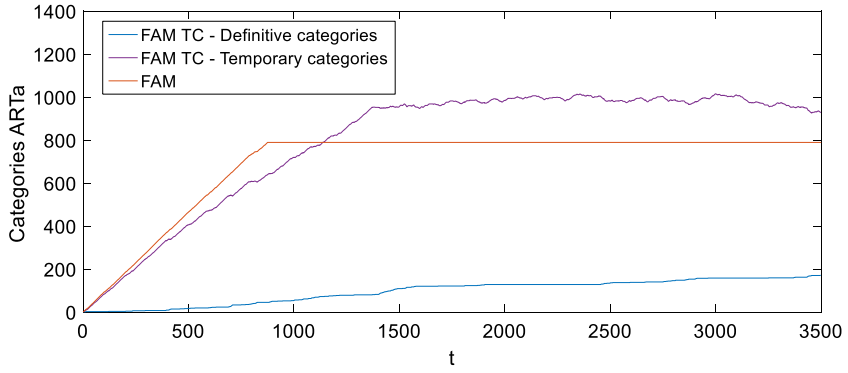

Fig. 12. Numbers of neurons created in the ART_a module of the FAM CT (blue) and FAM (red) ANNs.

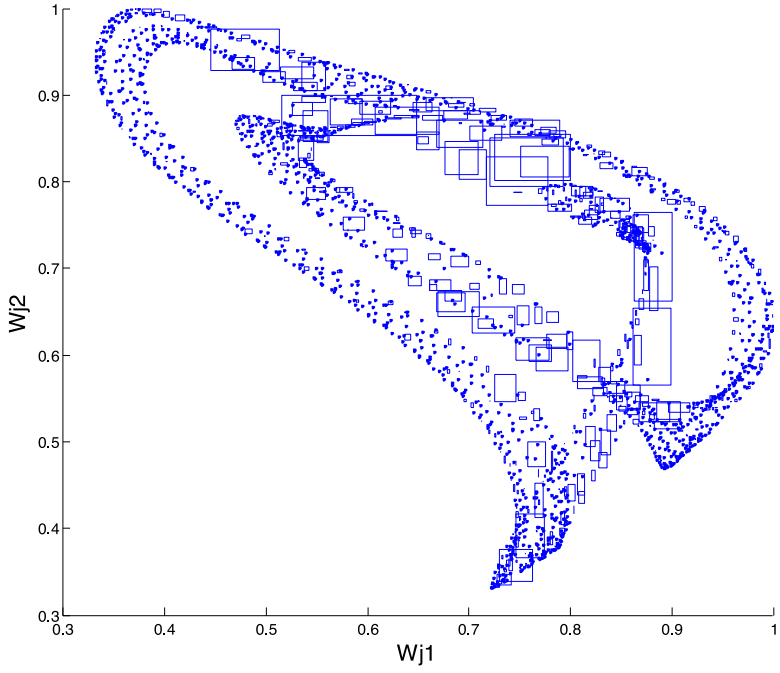

Fig. 13. Areas of representation in the categories created in the ART_a module of the FAM ANN.

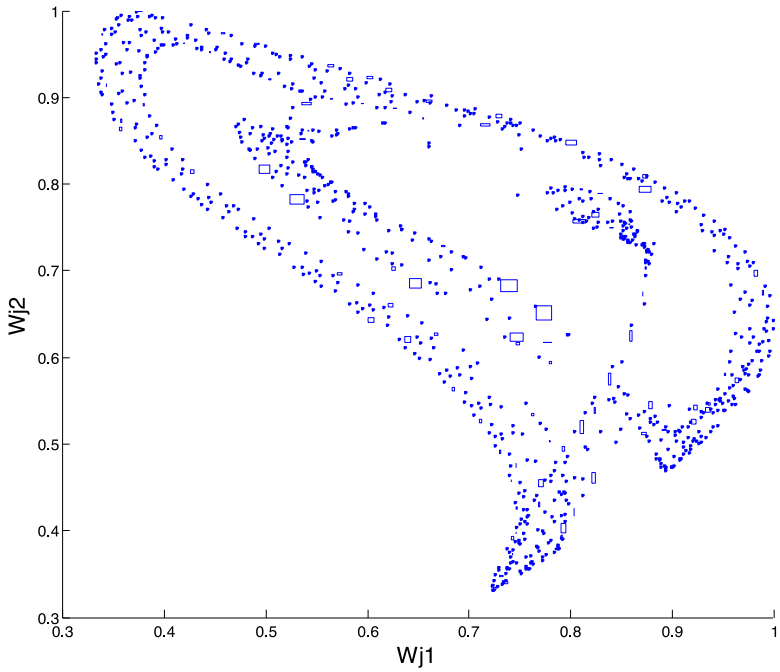

Fig. 14. Areas of representation of the categories created in the ART_a module in FAM CT ANN.

CRediT authorship contribution statement

Carlos R. Santos-Junior: Conceptualization, Formal analysis, Writing – original draft, Writing – review & editing. **Thays Abreu:** Conceptualization, Writing – original draft, Writing – review & editing. **Mara L.M. Lopes:** Conceptualization, Writing – original draft, Writing – review & editing. **Anna D.P. Lotufo:** Conceptualization, Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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MONITORAMENTO DE FALHAS EM SISTEMAS DINÂMICOS UTILIZANDO UM MÉTODO DE ANÁLISE DE DADOS BASEADO NO ALGORITMO DE SELEÇÃO NEGATIVA

Monitoring Failures in Dynamic Systems Using a Data Analysis Method Based on the Negative Selection Algorithm

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RESUMO – Este trabalho apresenta a aplicação de um método de monitoramento e diagnóstico de falhas em estruturas mecânicas que se baseia na teoria de sinais de vibrações e nos Sistemas Imunológicos Artificiais para auxiliar no processamento de dados. Utiliza-se do Algoritmo de Seleção Negativa como ferramenta para identificação de amostras de falhas extraídas dos sinais simulados em laboratório de um rotor dinâmico. Esta metodologia pode auxiliar profissionais de manutenção de estruturas mecânicas, facilitando a tomada de decisões. O conjunto de dados empregado no processamento do sistema inteligente foi gerado por meio de experimentos. Para as condições normais (*base-line*) utilizaram- se os sinais do rotor em funcionamento livre, ou seja, sem acréscimo de massa desbalanceadora e para as condições de falha foram adicionadas massas desbalanceadoras ao sistema. Os resultados são satisfatórios, apresentando precisão e robustez.
Palavras-chave: Algoritmo de Seleção Negativa, Rotor de Jeffcott, Monitoramento de Integridade Estrutural, Estruturas Mecânicas, Maquinas Rotativas.

ABSTRACT – This work presents the application of a method for monitoring and diagnosing failures in mechanical structures based on the theory of vibration signals and on Artificial Immune Systems to assist in data processing. It uses the Negative Selection Algorithm as a tool to identify fault samples extracted from the laboratory simulated signals of a dynamic rotor. This methodology can help mechanical structure maintenance professionals, facilitating decision-making. The data set used in the processing of the intelligent system was generated through experiments. For normal (base-line) conditions, the signals of the rotor in free operation were used, that is, without the addition of unbalance mass, and for the fault conditions, unbalance masses were added to the system. The results are satisfactory, showing precision and robustness.
Keywords: Negative Selection Algorithm, Jeffcott Rotor, Structural Integrity Monitoring, Mechanical Structures, Rotating Machines.

1. INTRODUÇÃO

Maquinas rotativas (Rotores) seguem em evolução desde a invenção da roda, assim como a humanidade, sejam em aplicações como nos automóveis, ventiladores, maquinas de lavar, ou na indústria em complexas bombas centrifugas, turbinas eólicas, compressores, eixos de transmissão, etc.

Equipamentos de grandes dimensões como redutores, turbinas, geradores, estão sujeitos a vibrações excessivas que podem ser transmitidas ao piso de fabrica, trabalhadores e máquinas vizinhas e ainda podem gerar ruídos acentuados, o que pode afetar a saúde e conforto das pessoas, além de antecipar o desgaste de componentes. A longo prazo pode ocasionar falhas como desbalanceamento e desalinhamento de eixos, folga nos mancais e implicar no mal funcionamento do maquinário (SOTELO JUNIOR, 2006).

O cenário da indústria e serviços atuais exige máxima eficiência de processos, produtos e estruturas, isto inclui menor custo de produção, maior agilidade do processo, maior qualidade, confiabilidade e segurança, tanto para a produção quanto para o produto final. Acompanhando este processo a indústria como um todo tem investido no aperfeiçoamento de tecnologias que proporcionem boa parte destes predicados, de forma rápida e eficiente. O monitoramento de integridade de estruturas (*Structural Health Monitoring - SHM*) é uma tendência ao que se refere a técnicas direcionadas a manutenção preditiva e detecção de falhas (BALAGEAS; FRITZEN; GUEMES, 2006).

A expansão da chamada indústria 4.0 e da sociedade tem expressado a necessidade de inovações tecnológicas para monitoramento de estruturas em tempo real, técnicas que possuam a capacidade de antecipar falhas, que apresente respostas rápidas de modo a possibilitar tomadas de decisões eficazes. O emprego de soluções embasadas em processos naturais como Redes Neurais Artificias, Lógica fuzzy, Sistemas Imunológicos Artificiais e etc., tem aperfeiçoado processamento de dados o que otimiza os sistemas de monitoramento de estruturas (SANTOS; LOPES, 2020).

Embora as técnicas que utilizam os Sistemas Imunológicos Artificiais sejam relativamente novas, o emprego destes algoritmos em análise de dados, reconhecimento de padrões, segurança computacional, etc., tem sido pesquisado em diversas áreas, conforme as características do problema (DE CASTRO; TIMMIS, 2002).

Neste artigo propõe-se o uso do Algoritmo de Seleção Negativa num Sistema de Monitoramento de Integridade Estrutural, a partir de dados de um rotor acoplado a um motor elétrico, a fim de verificar o desempenho do Algoritmo proposto por Forrest *et al*. (1994), como ferramenta aplicada à diferenciação de sinais produzidos em laboratório a partir do acréscimo de massa desbalanceadora no disco fixado ao rotor. O uso deste algoritmo é justificado por sua capacidade de reconhecimento de padrões e por apresentar eficiência em outros trabalhos que exigiram estas mesmas características (LIMA *et al*., 2013).

As demais seções deste trabalho estão organizadas da seguinte maneira: Na seção 2 é apresentada uma revisão sobre sistemas dinâmicos, balanceamento e desbalanceamento de rotores; Na seção 3 é introduzido o Monitoramento da Integridade de Estruturas *(Structural Health Monitoring — SHM);* Na seção 4 é apresentada uma revisão do Algoritmo de seleção Negativa proposto por Forrest *et al.* (1994); Na seção 5 é apresentada a metodologia proposta no trabalho e a descrição do experimento realizado para coleta de dados; Na seção 6 são apresentados os resultados obtidos; Por fim na seção 7 são apresentadas algumas conclusões e também algumas sugestões para trabalhos futuros.

2. SISTEMAS DINÂMICOS

Da perspectiva da física um sistema pode ser compreendido como uma coleção de elementos interligados que apresentem relações de causa e efeito, de modo a atender determinado objetivo. Dividido em dois estados, este sistema pode ser estático onde suas propriedades descritivas não variam com o tempo ou em sistemas dinâmicos quando estas propriedades variam com o tempo. O espaço pode variar em ambos os casos (OUTA, 2017).

2.1. Vibrações Mecânicas

Segundo Rao (2009), vibração ou oscilação é definida por qualquer movimento repetido após um intervalo de tempo. O estudo das vibrações aborda movimentos oscilatórios e as forças associadas a corpos.

Um sistema vibratório é considerado um sistema dinâmico, pois suas variáveis como as excitações e respostas são inerentes a variação

do tempo. De modo geral, um sistema vibratório engloba a transferência e armazenamento de energia, seja ela de energia potencial (mola ou elasticidade) para energia cinética (massa ou inércia) e vice-versa (RAO, 2009).

2.2. Desbalanceamento

O desbalanceamento é um evento identificado pela presença de desequilíbrio de massa relacionadas ao eixo de rotação. Segundo Garcia (2005), os rotores apresentam desbalanceamento residual, decorrente das tolerâncias das máquinas de usinagem, processos de fabricação ou até devido a montagem. Um rotor desbalanceado além de causar vibrações, pode causar o encurtamento do ciclo de operação do equipamento e causar danos.

O balanceamento pode ser caracterizado, a partir do acréscimo ou remoção de massa, de maneira que anule o efeito de desbalanceamento. Desta forma é possível garantir o funcionamento adequado do equipamento (GARCIA, 2005).

2.3. Rotores Dinâmicos

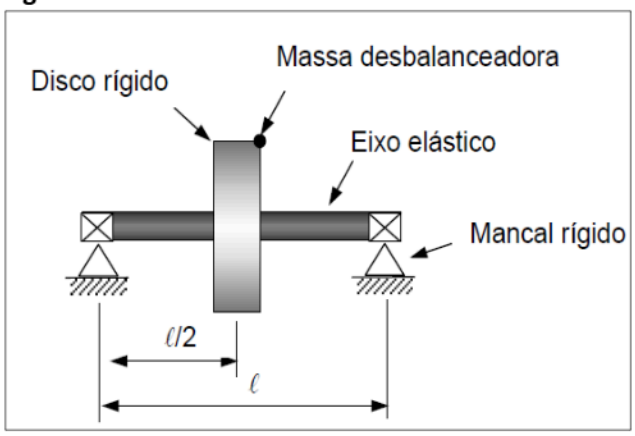
De acordo com Peres (2016), os rotores são quase que exclusivamente constituídos por mancais, eixo e disco, os sistemas rotativos desempenham dois movimentos rotativos superpostos:

- Rotação em torno de si próprio.
- Rotação do eixo defletido em torno de sua configuração não defletida (precessão).

Os movimentos de precessão podem ser sincronizados, quando a trajetória que o centro geométrico desempenha acompanha o sentido de rotação própria do rotor, ou não sincronizado, neste caso a trajetória que o centro geométrico do rotor realiza está no sentido oposto a rotação própria do rotor; no geral precessões síncronas são causadas por desbalanceamento no rotor (PERES, 2016).

Segundo Outa (2017), a teoria de vibração para sistemas de rotores dinâmicos é conhecida como *Jeffcott rotors.* O rotor de Jeffcott ou rotor de Laval é caracterizado por um eixo flexível e um disco central apoiado por mancais análogos como exemplificado na Figura 1. Refere-se a um modelo desenvolvido para evidenciar o aparecimento de velocidades críticas em rotores, compreender como a amplitude se torna máxima na velocidade crítica e como se movimenta a massa desbalanceadora internamente a órbita do rotor (OUTA, 2017).

Figura 1. Modelo rotor de Jeffcott.



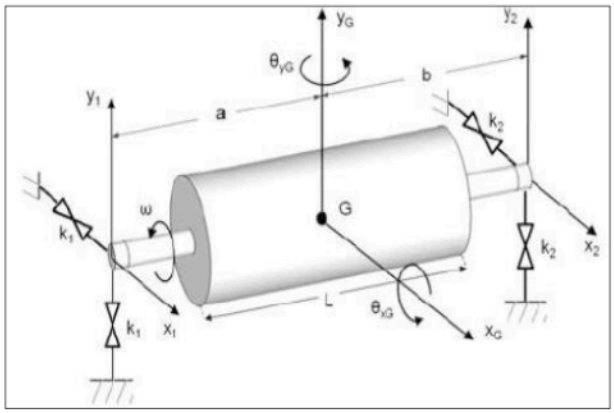
Fonte: (Peres, 2016).

No rotor de Jeffcott a vibração é dependente do movimento e a força excitadora é interna, o que em algumas situações podem tornar o eixo instável e isso pode aumentar de maneira contínua a amplitude do deslocamento em relação ao tempo (OUTA, 2017).

2.4. Efeito Giroscópio

Segundo Outa *et al.* (2015), o desalinhamento do eixo, relativo a seus componentes de apoio ocorre por causa do efeito giroscópio. As consequências do efeito giroscópio se dão pela rotação do eixo em torno do eixo x e y e no centro da massa do disco como mostrado na Figura 2, onde as constantes de rigidez são representadas por $K1$,$K2$, e, a e b, e caracterizam a distancia entre o centro eixo de massa G e o fim do eixo onde L representa a distancia absoluta entre os rolamentos (OUTA, 2017).

Figura 2. Eixo sob efeito giroscópico.



Fonte:(Outa, 2017).

São formados os ângulos θyG e θxG em decorrência das forças giroscópicas no disco com centro de massa G. O eixo tem influência de duas forças giroscópicas, a primeira é a inércia giroscópica, que atua diretamente na rigidez do rotor (disco) a partir do princípio da inércia, o que resulta na modificação do eixo de rotação relativo a sua referência. A segunda é a precessão, que é desencadeada quando a velocidade angular em torno do próprio eixo altera sua orientação o forçando a movimentar-se sempre em ângulo reto (OUTA, 2017).

3. SHM

Inserido nos contextos atuais de manutenção preditiva e projeto de maquinas, o Monitoramento da Integridade de Estruturas *(Structural Health Monitoring — SHM)* tem como finalidade identificar falhas em fases iniciais, baseado na variação das propriedades de determinada estrutura (BALAGEAS; FRITZEN; GUEMES, 2006).

Segundo Worden e Dulieu-Barton (2004), o SHM pode ser caracterizado como um método para resolução de problemas de reconhecimento de padrões. E pode ser elaborado em algumas etapas:

1. *Investigação Operacional*- Nesta etapa são questionadas as justificativas econômicas, ou seja, os custos e condições relevantes para a operação.
2. *Aquisição da Informação*- aqui são definidos os tipos de sensores necessários para obtenção dos sinais, o tipo de processamento e o armazenamento que serão utilizados;
3. *Triagem*-, as características apresentadas pela estrutura que demonstrem propriedades de falhas e característica que não apresentem falhas, são extraídas;
4. *Processamento e análise*- Realiza-se o processamento dos dados, a partir de algoritmos que permitam apontar a real situação da estrutura.

De acordo com worden e Dulieu-Barton (2004), um sistema de monitoramento eficiente deve acumular informações sobre o dano de maneira a permitir que, ações corretivas sejam tomadas para restaurar a estrutura ou sistema de operação. E que garanta confiabilidade e segurança ao sistema, além de assegurar que o monitor receba apenas as informações necessárias para tomada de decisão.

Segundo Rytter (1993), o acúmulo de danos leva a falhas, que não permitem a estrutura atuar de maneira satisfatória, deste modo estabelece - se quatro níveis para avaliação de danos (FARRAR; WORDEN, 2007):

- Detecção: qualifica que tipo de dano é apresentado na estrutura;
- Localização: Informações sobre a possível posição do dano;
- Avaliação: estima-se a extensão do dano;
- Predição: estimativa da vida útil da estrutura, informações sobre a segurança da estrutura.

O bom desempenho de um SHM garante segurança e confiabilidade a estrutura, além disto, assegura maior vida útil aos componentes, menor custo com manutenção. O uso do SHM permite a execução ideal da estrutura, possibilita substituir a manutenção programada pela baseada no desempenho, o que reduz o tempo de parada e erros humanos (BALAGEAS; FRITZEN; GUEMES, 2006).

Na literatura é possível encontrar diversas pesquisas que apresentam bons resultados e precisão, utilizando sistemas inteligentes para no monitoramento de falhas.

Chandrashekhar e Ganguli (2009) apresentam um sistema fuzzy para detecção de falhas estruturais usando curvaturas das formas modais de vibração.

Xiang-Jun, Zhan-Feng e Qiang (2010) propõe um modelo para avaliar a integridade de estruturas de pontes através dos sinais de vibração utilizando transformada Wavelet.

Oliveira, Chavarette e Outa (2020) apresentam uma metodologia para diagnóstico de configurações de danos apresentados a um rotor, que utiliza os algoritmos de Seleção Negativa e Seleção Clonal.

4. ALGORITMO DE SELEÇÃO NEGATIVA

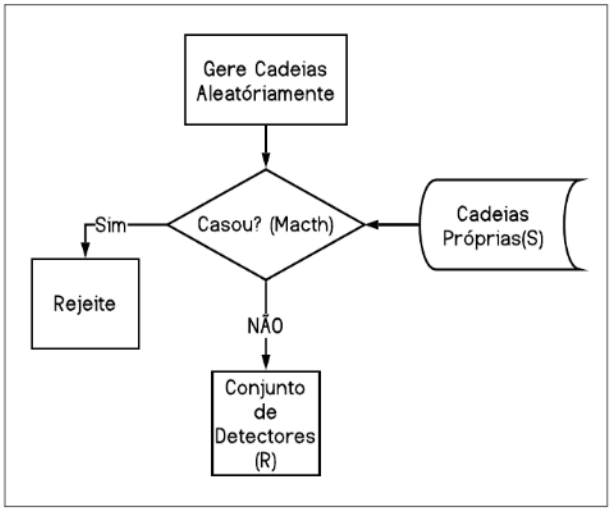
Proposto por Forrest et al. (1994), o Algoritmo de Seleção Negativa (*ASN*) foi desenvolvido inicialmente para a detecção de anomalias em sistemas computacionais. Inspirado no processo de seleção negativa dos linfócitos tipo T que ocorre no timo, o algoritmo possui a capacidade de reconhecimento de padrões (FORREST *et al*., 1994).

Análogo ao sistema de defesa biológico, um conjunto de cadeias próprias (S) é definido, assim é realizada a fase de censoriamento, depois cadeias aleatórias são geradas e avalia-se a afinidade (*match*) entre as cadeias próprias (S) e

as geradas aleatoriamente, para o caso da afinidade ser superior ao valor pré-definido a cadeia é rejeitada caso contrario é armazenada num conjunto (R) de detectores. Enfim dado o conjunto de cadeias que se deseja proteger, a afinidade entre cada uma delas e o conjunto de detectores é validada. Se a afinidade for superior ao valor pré-estabelecido, então um elemento não próprio é identificado.

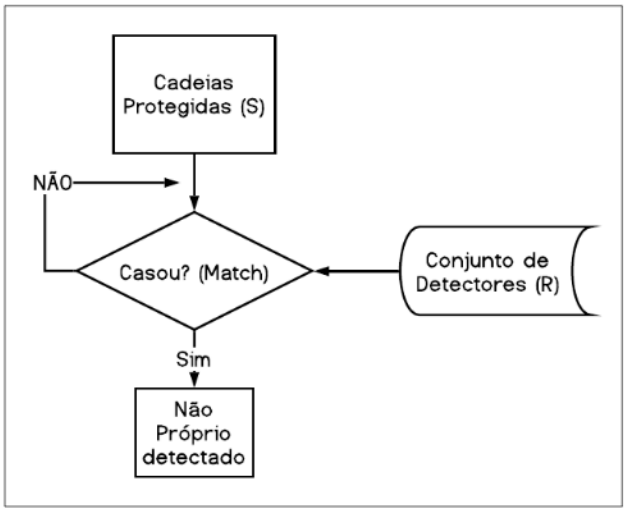
A seguir nas Figuras 3 e 4 estão as representações dos fluxogramas do censoriamento e monitoramento respectivamente (CASTRO, 2001; LIMA *et al.*, 2013).

Figura 3. Fase censoriamento.



Fonte :(Castro, 2001).

Figura 4. Fase Monitoramento.



Fonte :(Castro, 2001).

A afinidade entre as cadeias é definida a partir do critério conhecido como casamento ou combinação, de modo que o casamento pode ser classificado como perfeito ou parcial (LIMA *et al.*, 2015). Para este trabalho foi adotado o casamento parcial, de maneira que uma quantidade de pontos entres os padrões deve possuir o mesmo valor para que haja a confirmação do casamento. Tal quantidade é denominada taxa de afinidade, deve ser previamente definida. A taxa de afinidade define o grau de semelhança mínimo para que se determine o casamento entre as cadeias em análise (BRADLEY; TYRRELL, 2002).

A taxa de afinidade é calculada a partir da Equação 1 (BRADLEY; TYRRELL, 2002) :

$$TAf = \left(\frac{An}{At}\right) \times 100 \qquad (1)$$

Sendo:

TAf : taxa de afinidade;

An: número de cadeias normais no problema (cadeias próprias);

At: número total de cadeias no problema (cadeias próprias e não próprias).

Para quantificar a afinidade total entre os padrões, utiliza-se a Equação 2.

$$Af_T = \frac{\sum_{i=1}^{L} Vc}{L} \times 100 \qquad (2)$$

Sendo:

Af_T : relação percentual de afinidade entre os padrões analisados;

L: quantidade total de variáveis;

Vc: variáveis casadas;

$\sum_{i=1}^{L} Vc$: somatória (quantidade) de variáveis casadas.

Para que o casamento entre os dois padrões seja estabelecido Af_T deve ser maior ou igual a TAf, desta maneira consideram-se semelhantes.

5. METODOLOGIA

5.1. Experimentos

As especificações dos equipamentos utilizados são:

- 1 Motor elétrico KOHLBACH de indução trifásico tipo gaiola de 220V, 0,5cv frequência nominal de 60 Hz, com inversor de frequência Weg CFW-10 Clean podendo variar de 0 a 60 Hz;
- Eixo de aço 1045, diâmetro de 12 mm, distância entre eixos 128 mm;
- Dois mancais de rolamento P205 UC20;
- 01 acelerômetro MPU6050;
- 01 placa de microcontrolador Arduíno UNO;

- 01 computador Intel Core i5 8th Gen, 8GB de memória RAM, sistema operacional Windows 10, 64 bits, com os softwares GNU Octave® e Arduino IDE (*Integrated Development Environment*) instalados;
- 01 disco raiado com furos e três massas cilíndricas de 50g cada uma.

Foram coletados os sinais em estado normal, ou seja, quando o rotor funciona livremente sem o acréscimo das massas. Utilizaram-se três massas de aproximadamente 50 gramas cada uma, para simular o desbalanceamento do rotor, estas foram acopladas a um disco de fixação junto ao eixo uma a uma. Os sinais foram coletados a uma variação de frequência de 4 a 16Hz de duas em duas, para cada frequência foram coletados 10 sinais. As frequências foram definidas a partir de testes, nos quais se atentou à inércia rotacional e à trepidação (devido à vibrações excessivas) da máquina. Durante o experimento notou-se que acima de 16Hz o sistema entra em ressonância, dessa maneira os sinais foram coletados até a frequência de 16Hz para cada configuração de falha. Desta maneira foi definida a faixa de frequência de 4Hz a 16Hz que possibilitou a maquina operar sem ocorrências de trepidação.

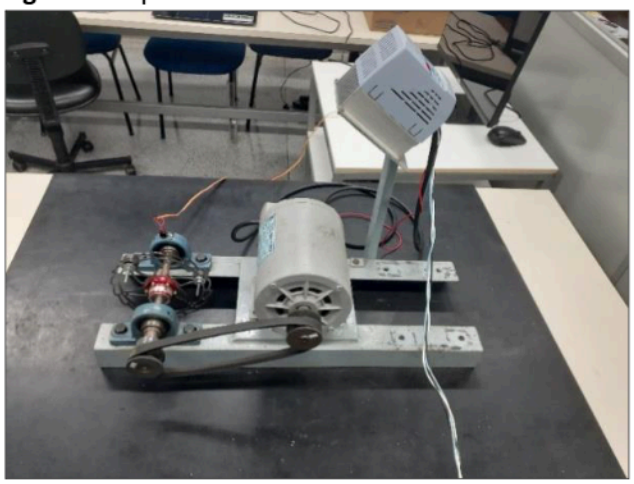
Dessa forma coletou-se 70 sinais para cada massa adicionada e 70 sinais para configuração de normalidade, quando não havia nenhuma massa acoplada ao disco. Na Figura 5 estão dispostas as três massas utilizadas para simular o desbalanceamento do rotor neste experimento.

Figura 5. Massas desbalanceadoras.

Fonte: Autoria própria

A Figura 6 mostra como foi montado o modelo experimental, elaborado a partir de um motor elétrico trifásico do tipo gaiola, ligado por meio de duas polias e uma correia a um eixo com disco centralizado e apoiado a dois mancais (rolamentos), junto ao sistema tem-se o inversor de frequências caracterizando o rotor.

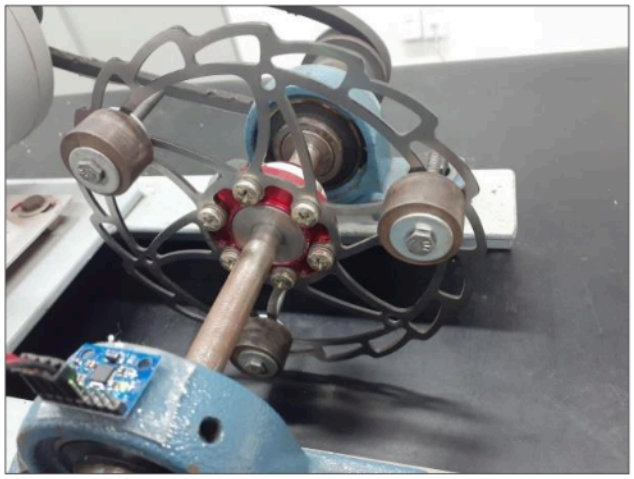
Figura 6. Experimento montado.



Fonte: Autoria própria.

O acelerômetro foi fixado num dos mancais de apoio com auxilio de fitas dupla face como apresentado na Figura 7. Através de cabos o acelerômetro é conectado ao microcontrolador Arduíno, que por sua vez, está conectado ao computador, via porta USB para estabelecer a comunicação serial e a energia a placa. As frequências foram ajustadas através do inversor de frequências acoplado ao motor a cada medição.

Figura 7 Configuração do rotor.



Fonte: Autoria própria.

5.1.1 Transmissão e Processamento dos Sinais

Os softwares GNU Octave® e Arduino IDE (*Integrated Development Environment*) são executados para permitir a comunicação serial entre o sensor com o microcontrolador e do

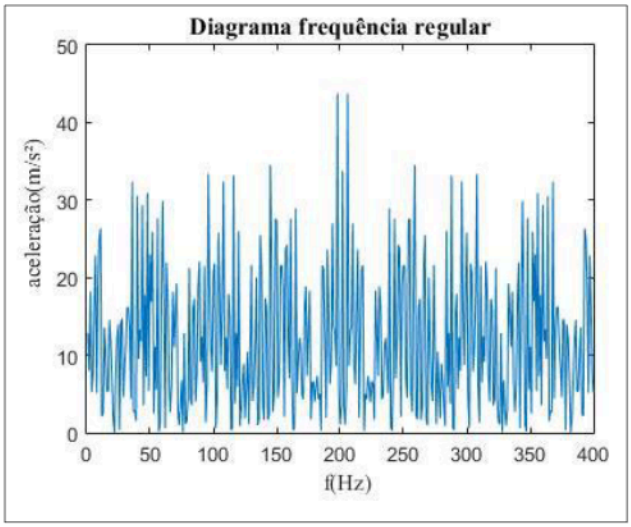
microcontrolador com computador, possibilitando a captação dos sinais.

O sensor capta os sinais e imprime os valores de aceleração relativos aos eixos coordenados X,Y,Z; em formato de texto. O software GNU Octave® executa o programa que realiza a leitura dos dados, os converte para o formato *float* e armazena-os em vetores conforme à referência. Assim, o resultado da aquisição são três vetores: um preenchidos com os valores da aceleração em X, outro da aceleração em Y e o último com as acelerações em Z.

Logo após, os dados captados são processados, assim é criado um único vetor para a norma dos valores de aceleração, em seguida a FFT (Transformada Rápida de Fourier) é aplicada para extrair o valor absoluto, onde se obtém sua magnitude. Os vetores criados são salvos em arquivos distintos separados por suas configurações. Após coletados todos os sinais das 4 configurações são organizados em duas matrizes de dados, formando o banco de dados SN com 70 sinais de características normais (*baseline*) e SF com 210 sinais característicos de falhas sinais da máquina, sinais estes que compõe o banco de dados aplicados ao Algoritmo de Seleção Negativa para a análise dos resultados desta proposta de monitoramento de falhas.

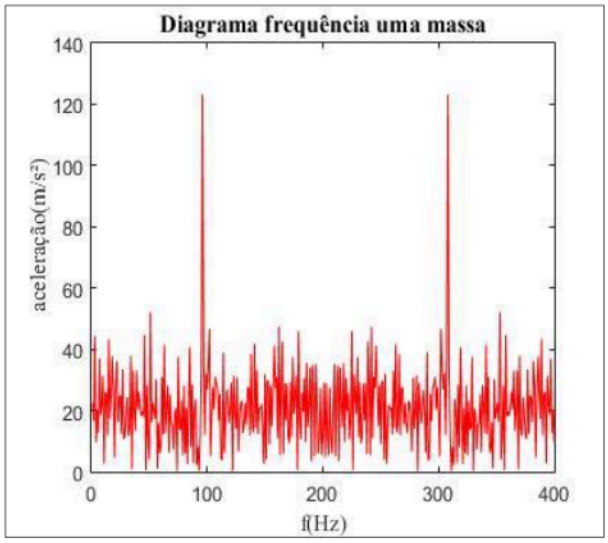
Nas Figuras 8 e 9 estão representadas as características dos sinais normais e dos sinais que apresentam falhas respectivamente, no qual é possível observar a variação no comportamento dos sinais, relativos a aceleração e frequência.

Figura 8. Gráfico representativo de um sinal emitido pela estrutura em seu estado normal.



Fonte: Autoria própria.

Figura 9. Gráfico representativo do sinal em condições de falha.



Fonte: Autoria própria.

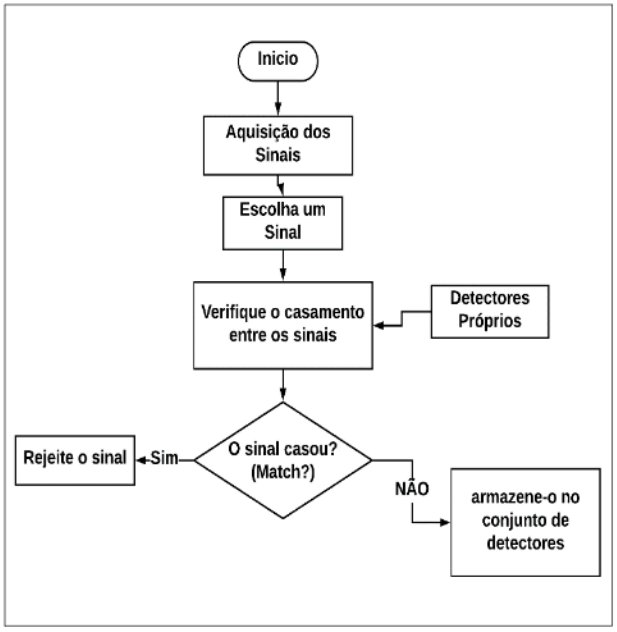
5.2 Metodologia proposta

O Presente trabalho propõe evidenciar a eficiência do ASN, aplicado ao monitoramento e detecção de falhas em uma estrutura mecânica rotativa (Rotor), a fim de identificar falhas a partir dos sinais obtidos em laboratório como descrito anteriormente. O método divide-se nas fases de censoriamento e de monitoramento:

5.2.1 Censoriamento

Esta fase é a responsável pela extração das características normais da estrutura e por armazená-las numa memória para a realização das detecções. Geram-se os detectores próprios (*base-line*) (Sinais que se deve proteger), sinais de vibração que apresentam a normalidade da estrutura, ou seja, não apresentam características de falhas. As cadeias (sinais) estão armazenados em formato de vetores, onde os valores discretizados são ordenados na posição de cada vetor. Os sinais são comparados ponto a ponto. Se houver casamento entre a cadeia gerada e um detector próprio este será rejeitado, caso contrário será aceito como um detector e armazenado num conjunto de detectores de falhas. Ocorre a repetição desta sequencia até que se atinja a quantidade de detectores (SANTOS *et al.*, 2021). O fluxograma do módulo censoriamento é apresentado na Figura 10 (LIMA *et al.*, 2013):

Figura 10. Módulo Censoriamento do ASN.



Fonte: (Adaptado de Lima *et al.*, 2015).

5.2.2 Monitoramento

Na fase de monitoramento calcula-se a taxa de afinidade, que neste trabalho utiliza-se do critério proposto por Bradley and Tyrrel (2002). Afim de dinamizar e otimizar o processo, é sugerido a implementação de um desvio vinculado ao padrão detector (anticorpo) de maneira a adicionar uma tolerância ao critério de combinação entre os padrões na avaliação da afinidade (LIMA et al., 2015). A partir da Equação 3 o desvio é empregado a cada posição *i* do vetor, desse modo o casamento é avaliado ponto a ponto (SANTOS et al., 2021).

$$\underline{Ab_i} \leq Ag_i \leq \overline{Ab_i} \qquad (3)$$

Sendo:

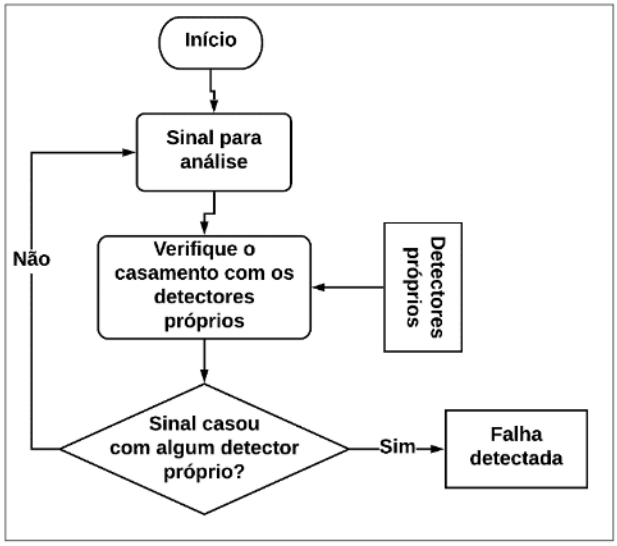
$\underline{Ab_i}$: Valor nominal da posição *i* menos o desvio adotado no anticorpo (padrão detector);

Ag_i: Valor nominal da posição *i* do antígeno (padrão em análise);

$\overline{Ab_i}$: Valor nominal da posição *i* mais o desvio adotado no anticorpo (padrão detector).

Para este trabalho emprega-se o desvio de 3% para os detectores. Geram-se os vetores para análise e verifica-se a afinidade entre estes e o conjunto de detectores próprios, desta forma caso ocorra o casamento entre eles uma falha foi detectada. A Figura 11 ilustra o fluxograma do módulo de monitoramento (SANTOS et al., 2021).

Figura 11. Módulo Monitoramento do ASN.



Fonte:(Adaptado de Lima *et al.*, 2015).

6. RESULTADOS

Todas as simulações foram realizadas em linguagem MATLAB, utilizando um PC Intel Core i5 8th Gen., 8GB de memória RAM, sistema operacional Windows 10, 64 bits. O algoritmo proposto foi aplicado aos dados experimentais realizados em parceria com Laboratório de Sistemas Complexos-SISPLEXOS.

Com objetivo validar o *ASN*, foram feitos testes de processamento, onde se alterou a quantidade de sinais em cada teste, afim de avaliar o comportamento do sistema com relação a quantidade de dados disponibilizados, tanto nas características de referência, quando o rotor trabalha em condições de normalidade, quanto nas informações de falhas. A finalidade é verificar a eficiência, precisão, a taxa de afinidade entre os detectores e o tempo de processamento em diferentes situações, recordando que são considerados a taxa de afinidade mínima de 70% e desvio de 3% para este trabalho. Os resultados e quantidade de sinais manipulados em cada teste estão apresentados na Tabela 1 a diante.

Tabela 1. Resultados obtidos.

Sinais Normais	Sinais Falha	Taxa de Afinidade	Acerto em (%)	Tempo de processamento (s)
70	21	76,9	100,00	4,44
70	63	70,00	100,00	7,61
70	210	70,00	100,00	3,6
65	21	75,00	100,00	2,99
35	105	70,00	100,00	3,01
35	210	70,00	100,00	9,49
07	21	70,00	100,00	3,8

Fonte: Autoria própria.

A fim de certificar a precisão dos resultados, os testes foram executados 25 vezes para cada configuração. Foi possível observar que o algoritmo de Seleção Negativa apresenta satisfatório índice de acerto, igual a 100%.

7. CONCLUSÕES E TRABALHOS FUTUROS

O presente trabalho foi desenvolvido com base na análise de dados extraídos a partir do comportamento de um sistema vibratório auto-exitado, caracterizado pela operação de um rotor dinâmico em desbalanceamento induzido. A proposta foi gerar um banco de dados, por meio de simulação em laboratório e aplica-lo a um método de monitoramento de integridade de estruturas, baseado nos sistemas imunológicos artificiais que utiliza como instrumento de processamento o algoritmo de seleção negativa. O critério proposto apresentou bons resultados, obtendo um índice de acerto de 100% em todos os testes, apresentando um bom tempo de processamento, sendo executado rapidamente (menos de 100 milésimos de segundo).

Sendo assim, é possível concluir, que proposta do trabalho foi atingida, o ASN foi capaz de detectar 100% dos sinais não-próprios dentre os analisados, isto demonstra a eficiência do algoritmo no reconhecimento de padrões, destaca aplicabilidade do meio empregado na reprodução do banco de dados, de maneira a caracterizar a estrutura monitorada.

Como sugestão para trabalhos incluem-se:

- O aperfeiçoamento do método, de maneira que o algoritmo seja capaz de classificar as falhas detectadas.
- A possibilidade que o método, apresente a localização da falha e exponha o nível do comprometimento da estrutura.

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Electrical Customer Profile Using Fuzzy Logic Theory

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***Abstract*—Considering the increasing electrical energy demand in residences, it is necessary to know the detailed pattern of electricity use, to change the behavior of the final consumer and to reduce the global consumption. Knowing the load curve profile in advance is very important for detecting the peaks and valleys to change habits of energy consumption, principally during periods when the prices are lower, i.e., following the electrical power industry opportunities. This work studies how the consumer behaves individually, as well as identifies similarities with other consumers. Thus, we herein propose a procedure for obtaining the residential load profile using fuzzy-logic theory. As the residential electrical energy consumption is highly correlated to the active occupation, the quantity of dwellers and the different periods of the day for typical consumers are considered. However, the ideas presented in this work can be applied to other regions in different countries. To verify the efficiency of the proposed system, the obtained results are compared with real load curves.**

***Index Terms*—Fuzzy logic theory, Peak time, Residential consumer, Residential load curve, Residential load profile.**

I. Introducão

A previsão de cargas elétricas tem grande utilidade no gerenciamento da demanda de energia elétrica. É indispensável em consequência do crescimento do número de usuários, principalmente no setor residencial [1 - 3]. O setor elétrico residencial representa 30% do total de consumidores brasileiros [4] e tem impactos significativos nos períodos de pico de demanda de energia elétrica ao longo do dia. Esses picos de carga afetam consideravelmente a eficiência do sistema de energia, resultando no congestionamento da rede elétrica [5, 6]. Assim, faz-se necessário ter-se um melhor conhecimento da curva de carga elétrica no setor residencial. Essa informação permite, aos usuários, detectar o uso indevido de energia elétrica, visando reduzir o seu consumo e viabilizando o uso da energia nas horas do dia em que a tarifa de energia elétrica possui um preço menor [7].

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Na literatura são encontradas diversas publicações que buscam a modelagem da carga elétrica residencial. Esses modelos são classificados em dois grupos: (*a*) de cima para baixo (*top-down*) e (b) de baixo para cima (*bottom-up*) [1].

Os modelos *top-down* não implicam numa grande complexidade, mas para realizar sua implementação, faz-se necessário utilizar uma extensa base de dados empíricos. Os dados utilizados são informações gerais, como: produto interno bruto (PIB), taxa de desemprego, apresentação estatística sobre a população-alvo, evolução predita, taxas de saturação de aparelhos etc. Este conceito também pode ser interpretado como de previsão agregada [8].

Os modelos *bottom-up* começam a partir das menores unidades possíveis de um sistema (residências) e, sucessivamente, vão agregando estas unidades para alcançar níveis mais elevados do sistema elétrico, obtendo o consumo de energia no setor residencial, de acordo com as suas características. Os dados de entrada são caracterizados por aparelhos domésticos, informações meteorológicas, conta de energia elétrica das famílias, comportamento humano etc. Neste caso é o que se chama de desagregado.

Usualmente os previsores são aplicados em dados de demanda agregados, em uma abordagem *top-down*, porém nos últimos anos, graças à popularização dos *smart meters*, dados de consumidores individuais passaram a estar cada vez mais disponíveis, permitindo que a previsão seja realizada para cada consumidor individualmente. Então, a demanda agregada obtida, a partir da soma das previsões de demanda individuais. Esta abordagem é denominada *bottom-up.*

Os modelos *top-down* apresentam maior vantagem por causa do seu nível de detalhes [1, 7].

Neste artigo, é utilizada uma estratégia *bottom-up*, sendo desenvolvido um modelo para obter o perfil da curva de carga elétrica residencial baseada na lógica *fuzzy* (LF) [9]. Esta abordagem é adotada tendo em vista a existência limitada (ou insuficiente) de dados no ambiente residencial que descrevem residências brasileiras, e a complexidade de modelar o comportamento humano. A partir do uso da lógica *fuzzy*, será estimado o perfil da carga baseado na observação comportamental dos usuários residenciais.

Obter o perfil de consumo de carga elétrica, em residências, não é uma tarefa fácil, pois mesmo que os clientes tenham as mesmas características gerais, (por exemplo, duas residências distintas com o mesmo número de habitantes, o mesmo tipo de habitação, número iguais de cômodos, aparelhos eletrônicos iguais etc.) é improvável que as duas curvas de demanda de energia serem iguais, ou muito similares. Essa é a razão pela qual a modelagem da demanda de energia em uma área específica é uma tarefa árdua.

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Considerando que o perfil da demanda de eletricidade no setor residencial está altamente correlacionado com o tempo de ocupação das residências, no modelo proposto consideram-se a quantidade de habitantes em uma residência em diferentes períodos do dia, assim como as atividades diárias que caracterizam como as pessoas usam o seu tempo. Neste modelo, considera-se a ocupação como ativa os períodos em que os usuários estão na residência acordados ou dormindo. Nas principais publicações disponibilizadas na literatura [7, 10] são consideradas a ocupação como ativa apenas o tempo em que os usuários estão na residência e acordados.

O perfil do consumo de energia utilizando um sistema inteligente pode simular as atividades diárias dos ocupantes em suas residências e, também, contribuir no gerenciamento de demanda de energia [11-13], na resposta da demanda [11], [14-16] e, futuramente, no projeto da rede inteligente (*smart grid*) [17-19]. A rede inteligente constitui-se num novo paradigma de relacionamento entre sistema, ambiente e os consumidores. A produção de energia é caracterizada por várias fontes de geração, como, por exemplo, hidráulica, térmica, eólica, solar etc., na qual, o consumidor terá livre escolha do fornecimento de energia, podendo consumir e/ou produzir energia elétrica.

Portanto, nesta pesquisa, baseado num sistema *fuzzy* (SF), as curvas de carga residencial são "construídas" considerando as características de um grupo de domicílios brasileiros, considerando-se os períodos do dia e o número de ocupantes na casa.

Este texto encontra-se organizado do seguinte modo. O sistema *fuzzy* utilizado, para a determinação do perfil de carga residencial, é apresentado na Seção II. Na Seção III são relacionados os resultados e as discussões. Finalmente, as conclusões são apresentadas na Seção IV.

II. Sistema *Fuzzy* Utilizado para a Determinação do Perfil de Carga Residencial

A. Construção da Curva de Carga Residencial

Usualmente, na literatura, o problema de previsão de carga é abordado considerando a carga global do sistema, ou seja, todos os consumidores (residenciais, comerciais e industriais). Nessa condição, os dados históricos disponíveis são usados. Portanto, é estatisticamente provável que o perfil de carga seja repetido. O problema não se limita a valores repetidos. Outras informações (exógenas) são igualmente importantes e, juntamente com os dados históricos, são utilizados como entrada do sistema de previsão. Este passo geralmente é desenvolvido usando uma alternativa clássica, por exemplo, a abordagem de Box-Jenkins [20]. Recentemente, os pesquisadores preferem usar técnicas inteligentes, por exemplo, redes neurais artificiais [21] e lógica *fuzzy* [9]. A carga global prevista está dividida entre várias barras do sistema para investigar o fluxo de energia, análise de estabilidade etc., que são estudos necessários para o planejamento e a operação do sistema de energia elétrica. No entanto, em relação às barras (dividindo a carga global), a previsão é complexa, pois o comportamento da carga não segue estatísticas regulares, isto é, no caso de residências, essa complexidade aumenta.

O caso do consumo interno é intrigante e desafiador. Neste estudo pretende-se contribuir para a previsão do consumo de energia residencial. O objetivo é propor um padrão de comportamento dos usuários residenciais e, em particular, fornecer uma ferramenta para racionalizar o uso da energia no país. Portanto, foi desenvolvido um procedimento baseado em lógica *fuzzy* [9] que precisa de pouca informação para executá-lo. É robusto e confiável, ou seja, algo simples e útil para o usuário e capaz de receber inovações.

Na metodologia proposta, emprega-se a teoria da lógica *fuzzy* para modelar o comportamento residencial com objetivo de obter o perfil da curva de carga residencial. A grande vantagem da lógica *fuzzy* é que não necessita muitas informações do hábito dos consumidores. É importante enfatizar, que este trabalho tem como objetivo fornecer a curva de perfil de consumo em residências, e não realizar a previsão de carga elétrica e calcular erros percentuais.

Para construir o perfil de consumo de energia, usando LF, foram realizadas pesquisas sobre moradores que trabalham 8 horas por dia, que é a rotina da maioria dos brasileiros.

Este estudo foi elaborado de modo empírico, observando o comportamento das famílias brasileiras em duas cidades: uma de pequeno porte e outra de grande porte. O levantamento dos dados de consumo das duas residências foi realizado de hora em hora para gerar a curva diária e comparar com a obtida pelo sistema *fuzzy*.

Isso envolveu uma pesquisa com perguntas sobre, por exemplo, os hábitos diários e quais aparelhos elétricos estão disponíveis na residência. Assim, uma entrada do FS é o parâmetro "Períodos do dia", e o outro parâmetro é definido como sendo a "Ocupação". O resultado (saída) é "Consumo de energia".

Se for desejado obter o perfil de carga de um grupo diferente de consumidores, é necessário conhecer a rotina dos novos consumidores e alterar as regras difusas usando essas informações. À medida que mais informações são obtidas dos moradores em estudo, tão próximo ao perfil de energia real é o perfil de carga por lógica *fuzzy*.

O modelo proposto é implementado no ambiente MATLAB.

B. Entradas do SF

Para formular os conjuntos de entrada do sistema *fuzzy*, foram consideradas residências brasileiras específicas, com no máximo cinco habitantes, que realizam uma jornada de trabalho de 8 horas por dia. Foram criados dois conjuntos de entradas "Períodos do Dia" e a "Ocupação", que serão utilizadas como variáveis de entrada do sistema *fuzzy*.

A função de pertinência utilizada para representar os conjuntos de entrada foi à função sino (gaussiana), sua escolha se deve ao fato desta função interpretar de melhor forma a representação dos períodos e das ocupações na residência.

1) Períodos do dia

Para definir os diferentes períodos do dia a serem usados como conjunto de entrada SF, foi realizada uma pesquisa domiciliar (PD) para obter informações sobre a rotina diária das pessoas que trabalham 8 horas por dia, de segunda a sexta-feira. Esta é a rotina básica da maioria dos brasileiros. Considerando

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esta rotina diária, oito períodos diferentes entre 0 e 24 h são definidos.

O primeiro período considerado corresponde ao primeiro período de sono. Neste período, normalmente, são utilizados os aparelhos de ar-condicionado e/ou ventiladores quando a temperatura está alta.

O segundo período, refere-se ao café da manhã. Neste período, os habitantes acordam e começam a ligar os diversos aparelhos elétricos, como: iluminação, chuveiro elétrico, micro-ondas, entre outros.

O próximo período considerado é o primeiro período de trabalho, sendo este o terceiro período do dia. Nestes períodos, os habitantes deixam suas casas para irem para o trabalho desligando todos aparelhos elétricos, excetos os aparelhos de consumo contínuo, como geladeira e freezer.

O quarto período é o do almoço. Trata-se de um período em que as pessoas dispõem de um intervalo de 2 horas (em média), para almoçarem. Neste período, consideram-se dois casos distintos, ou seja, o caso das pessoas que voltam para suas respectivas casas e, assim, consomem energia elétrica em suas residências, caso as pessoas que não retornam para suas respectivas casas.

O quinto período considerado é o segundo período de trabalho, que é o momento em que os habitantes retornam para seus respectivos trabalhos.

Ao acabar o expediente de trabalho, os moradores voltam para suas respectivas casas, ativando diversos aparelhos elétricos, como chuveiro elétrico, iluminação, TV e já começam a preparar o jantar, ou algo para se alimentarem. Esse é o sexto período considerado.

O sétimo período considerado é o descanso, sendo este o momento em que os moradores descansam após um longo dia de trabalho, ativando algum aparelho para refrigeração de ar quando a temperatura esta alta, o chuveiro elétrico quando a temperatura está amena ou fria, aparelho de TV, entre outros.

O último período utilizado é o do segundo período do sono. Neste período os habitantes estão se preparando para dormir, desativando praticamente todos os aparelhos elétricos, exceto os de consumo contínuo e os equipamentos de climatização caso a temperatura esteja alta (ar condicionado e/ou ventiladores).

Assim, oito funções de pertinência no intervalo de [0, 24] horas são definidas, o que corresponde a 24 horas do dia, para identificar os diferentes períodos do dia, ou seja, os períodos em que os habitantes estão em suas respectivas casas utilizando aparelhos elétricos ou dormindo e também os períodos que estão fora da casa.

Logo, a variável linguística "Períodos de Dia" possui oito valores linguísticos, onde cada termo é caracterizado por uma função de pertinência. de acordo com a Fig. 1.
no qual:

- Primeiro período do sono (PS1) : [0h, 6h];
- Café da manhã (CM) : [6h, 8h];
- Primeiro período de trabalho (T1) : [8h, 12];
- Almoço (AL) : [12h, 14h];
- Segundo período de trabalho (T2) : [14h, 18h];
- Jantar (JA) : [18h, 21h];
- Descanso (DE) : [21h, 23h];
- Segundo período do sono (PS2) : [23h, 24h].

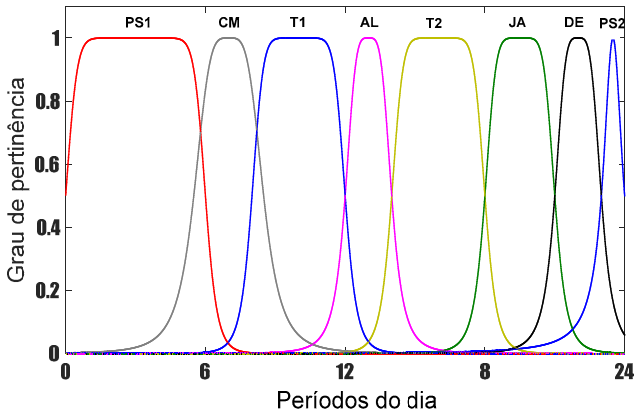

Fig. 1 . Funções de pertinência “períodos do dia”.

2) *Ocupação*

Como consequência da alta correlação que existe entre o perfil da demanda de eletricidade no setor residencial, com o tempo de em que os habitantes estão nas residências, é criada a variável de entrada ocupação.

A variável linguística "Ocupação" é definida por 7 valores linguísticos que determinam se a ocupação é alta, média ou baixa, dependendo da quantidade de usuários disponíveis para utilizar aparelhos elétricos. Este trabalho propõe uma residência com um máximo de 5 habitantes. A ocupação é considerada alta quando 4 ou 5 habitantes estão na residência, médio, quando há 2 ou 3, e baixa quando há apenas um ou nenhum habitante. Cada valor linguístico é caracterizado por uma função de pertinência ilustrada na Fig. 2.

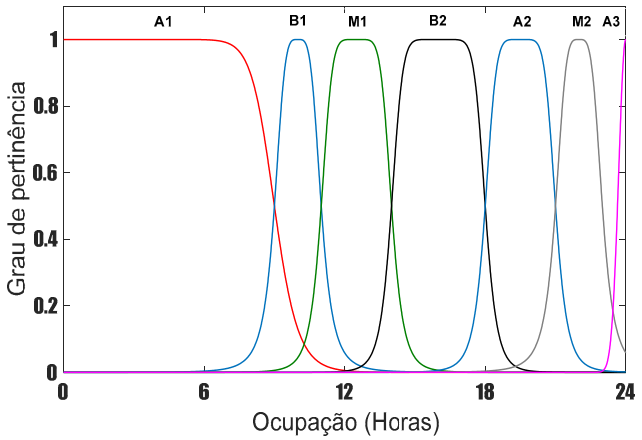

Fig. 2 . Funções de pertinência “ocupação”.

sendo:

- Alta 1 (A1) : ocupação alta no período [0h, 8h];
- Baixa 1 (B1) : ocupação baixa no período [8h, 12h];
- Media 1 (M1) : ocupação média no período [12h, 14h];
- Baixa 2 (B2) : ocupação baixa no período [14h, 18h];
- Alta 2 (A2) : ocupação alta no período [18h, 21h];
- Media 2 (M2) : ocupação média no período [21h, 23h];
- Alta 3 (A3) : ocupação alta no período [23h, 24h].

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C. *Saída do SF*

A saída FS é representada pelo consumo de energia elétrica em um período de 24 h. O objetivo é ilustrar os perfis de consumo dos usuários durante todo o dia, indicando os tempos com mais e menos uso dos aparelhos elétricos.

1) *Consumo de energia elétrica*

Para representar a saída do SF "Consumo de energia elétrica", cinco funções de pertinência são definidas dentro do intervalo [0, 1]. Na Fig. 3, são ilustradas as funções de pertinência, que são determinadas pelos valores linguísticos ("Muito Alto", "Alto", "Médio", "Baixo" e Muito Baixo"). Estas informações são convertidas em valores reais entre 0 e 1. Um valor próximo de 1 significa que várias pessoas na casa para ativar os aparelhos eletrônicos e a iluminação. Um valor próximo de zero significa que há poucas pessoas e que o potencial de consumo de energia é pequeno.

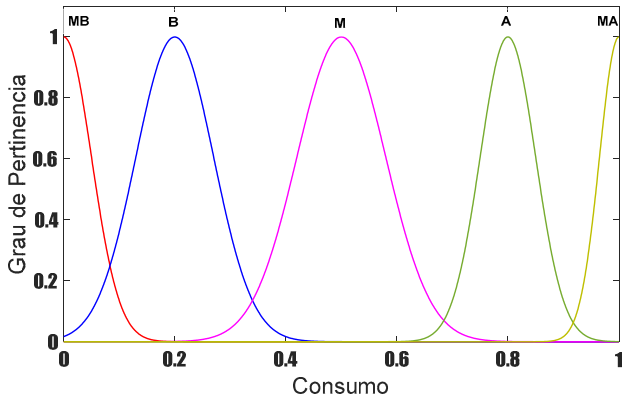

Fig. 3 . Funções de pertinência "consumo de energia elétrica".

sendo:

- "Muito Baixo" (MB) : muito baixo consumo;
- "Baixo" (B) : baixo consumo;
- "Médio" (M) : médio consumo;
- "Alto" (A): : alto consumo;
- "Muito Alto" (MA) : muito alto consumo.

A função de pertinência gaussiana é escolhida porque possui a melhor representação dos valores linguísticos.

D. *Base de Regras do SF*

Depois de definidas as entradas e as saídas pelas variáveis linguísticas, é possível criar um conjunto de regras para a obtenção da curva de consumo de energia elétrica. A regra base é um dos principais componentes do SF [22]. As regras são definidas de acordo com o conhecimento de um especialista e são eficazes para obter bons resultados.

As regras utilizadas neste trabalho foram definidas de acordo com a seguinte relação:

$$se\ \ X_1 \text{ é } A \ e\ X_2 \text{ é } B, então, Y \text{ é } C$$

sendo: X_1 (período do dia) e X_2 (ocupação) são as entradas do sistema; Y, o consumo de energia, é a saída; e A, B e C são os valores assumidos para a variáveis linguísticas. Esta relação pode ser expressa da seguinte forma: "*SE* antecedente, *ENTÃO*, consequente".

O procedimento utilizado nas entradas do SF é convertido pelas funções de pertinência para as saídas. Ao aplicar as regras, a saída é obtida, que é um número *fuzzy*, que deve ser convertido em um valor real (*crisp*). Este processo é conhecido como defuzificação. No presente estudo, emprega-se o método centroide, que é definido do seguinte modo [22]:

$$CA = \frac{\sum_{i=1}^{n} \mu_i v_i}{\sum_{i=1}^{n} \mu_i} \tag{1}$$

na qual:

n : número de regras *fuzzy*;
μ_i : grau de pertinência da regra i;
v_i : atividade de controle recomendada referente a regra i.

Na Fig. 4, é representado o processo de inferência *fuzzy* em relação ao horário 17h30min, sendo consideradas as entradas como sendo as informações "Períodos do Dia" e "Ocupação". A saída corresponde ao "Consumo de Energia Elétrica". As regras (i) e (ii) são ativas neste horário. Considerando a entrada "Períodos do Dia", observa-se que às 17h30min, a variável linguística T2 possui um valor de pertinência aproximadamente igual a 0.9. A variável linguística JA possui um valor de pertinência igual a 0.15. A entrada "Ocupação", neste mesmo horário, possui a variável linguística B2 com pertinência igual a 0.9 e a variável linguística A2 com pertinência 0.15.

Utilizando a regra (i), tem-se que a pertinência, em relação à saída, é igual variável linguística B ("Baixo"):

$$\mu_i = \min\{\mu_{T2}\,,\ \mu_{B2}\} = \min\{0.9\,,\ 0.9\} = 0.9$$

Ao utilizar a regra (ii), é obtida a pertinência, em relação à saída, igual à variável linguística A ("Alto"):

$$\mu_{ii} = min\{\mu_{JA}\,,\ \mu_{A2}\} = min\{0.15\,,\ 0.15\} = 0.15$$

O valor real (*crisp*) *P*, obtido no processo de defuzificação, em relação à hora 17h30min, considerando as regras (i) e (ii):

$$P = \frac{\mu_i * v_B + \mu_{ii} * v_A}{\mu_i + \mu_{ii}} = \frac{0.9*0.2 + 0.15*0.8}{0.9+0.15} = 0.3$$

E. *Base de Regras*

A variável climática "Temperatura" é considerada visando definir as regras básicas que influenciam a carga residencial [23]. A "Estação do Ano" pode ser considerada como uma variável; No entanto, considerando que o Brasil é um país tropical, as quatro estações não ocorrem em períodos regulares. Em várias regiões, a temperatura no inverno é tão alta quanto a do verão. Assim, a estação é considerada uma variável climática e não fornece boas informações. As regras *fuzzy* usam a "Temperatura" como variável, o que influencia a ativação de aparelhos elétricos de alto consumo.

Diferentes regras de base são criadas em relação à variação de temperatura. Quando a temperatura é inferior a 15°C, esta é classificada como "Muito Fria" / "Fria". Se for entre 15 e 25°C, é classificada como "Muito Agradável"/ "Agradável". Se a temperatura estiver acima de 25°C, esta é classificada como "Quente" / "Muito Quente".

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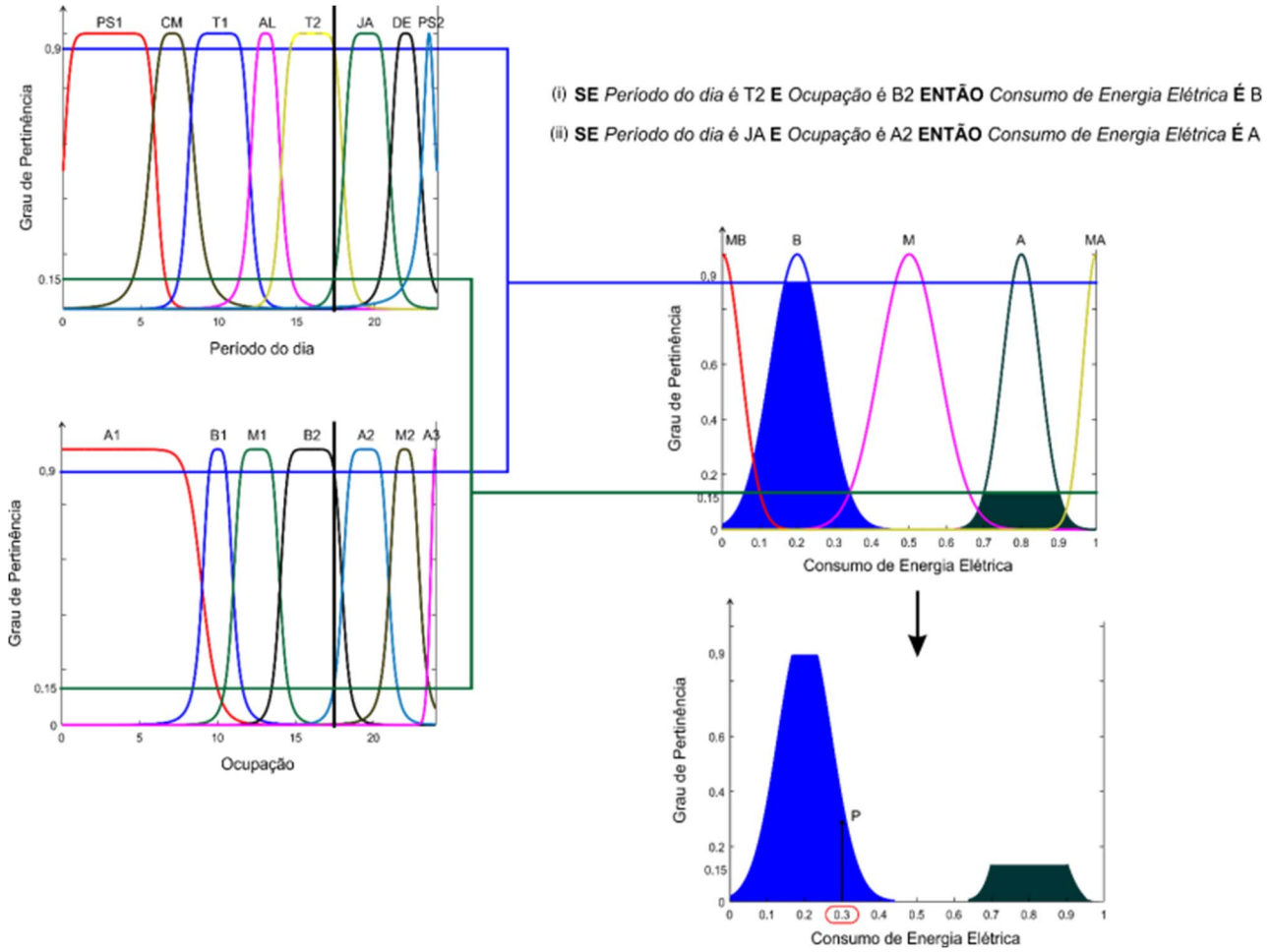

Fig. 4. Processo de inferência *fuzzy*.

O maior desafio é encontrar regras básicas que melhor representem o consumo de carga residencial considerando as diferentes temperaturas. Portanto, um PD é realizado em diferentes residências que correspondem ao perfil do consumidor estudado nesta pesquisa. Perguntas são feitas sobre os hábitos diários dos habitantes, quais aparelhos estão disponíveis nas residências.

O PD revela que os hábitos das pessoas que vivem em grandes cidades diferem daquelas pessoas que vivem em pequenas cidades, especialmente quando for o horário do almoço. Assim, são criadas duas base de regras diferentes, de acordo com as rotinas dos moradores em grandes cidades e em cidades pequenas.

Os aparelhos disponíveis na maioria das residências (de acordo com o PD), para a geração das regras são: refrigeradores, aparelhos de ar-condicionado, televisores, máquinas de lavar roupa, chuveiros elétricos e lâmpadas.

Nesta pesquisa serão mostrados os perfis residenciais obtidos, via execução do SF, considerando-se a temperatura acima de 25°C, ou seja, "Quente"/ "Muito Quente". Este caso foi escolhido, pelo fato do Brasil ser um país tropical. Assim, a maioria de dos dias durante o ano apresenta temperaturas elevadas.

Serão apresentados dois conjuntos de regras, considerando-se que são distinguidos por habitantes que moram em grandes cidades, e normalmente não retornam para suas respectivas residências no período do almoço, e o outro é definido por habitantes que moram em pequenas cidades, e assim retornam para suas residências para almoçar, ou descansar. Na Tabela I, apresenta-se o "Conjunto de Regras (A)", que foi criado baseado em informações e observações do hábito de habitantes que vivem em pequenas cidades, enquanto na Tabela II apresenta-se o "Conjunto de Regras (B)", sendo estas regras desenvolvidas considerando-se um habitante de grandes centros, que apresentam hábitos um pouco diferentes das pessoas que moram em pequenas cidades.

F. Geração do Banco de Dados

As regras básicas são constituídas por informações disponíveis para usuários interessados em receber benefícios fornecidos pelas empresas de energia [24]. O usuário fornece pouca informação, uma vez que os dados necessários para implementar o programa estão disponíveis no banco de dados das empresas elétricas (informações obtidas desde que a pessoa se tornou um usuário). Ao implementar este sistema, a previsão do consumo residencial com precisão aceitável pode ser realizada ajustando a curva de carga inicialmente obtida de acordo com a predição disponível em pontos próximos aos usuários, por exemplo, no transformador de alimentação onde a carga está conectada.

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TABELA I
BASE DE REGRAS (A)

Regra	Período do dia	Ocupação	Consumo
01	PS1	A1	A
02	PS1	A1	M
03	CM	A1	B
04	CM	B1	B
05	CM	M1	M
06	T1	A1	B
07	T1	B1	B
08	AL	B1	M
09	AL	M1	M
10	AL	B2	M
11	T2	B2	B
12	T2	B2	B
13	JA	A2	M
14	JA	M2	A
15	DE	M2	M
16	DE	A3	M
17	PS2	A3	A
18	S2P	A3	M

TABELA II
BASE DE REGRAS (B)

Regra	Período do dia	Ocupação	Consumo
01	PS1	A1	A
02	PS1	A1	M
03	CM	A1	M
04	CM	B1	M
05	CM	M1	M
06	T1	A1	B
07	T1	B1	B
08	AL	B1	B
09	AL	M1	B
10	AL	B2	B
11	T2	B2	B
12	T2	B2	B
13	JA	A2	M
14	JA	M2	A
15	DE	M2	A
16	DE	A3	A
17	PS2	A3	A
18	S2P	A3	M

A carga no transformador pode ser prevista usando a proposta apresentada em [25] (previsão de carga multinodal). O ajuste pode ser realizado iterativamente, de modo que a carga total dos usuários conectados ao transformador seja igual à carga prevista do transformador (com uma tolerância predefinida). Uma vez que esta previsão está disponível, o custo computacional adicional, para ajustar a demanda de carga residencial, é basicamente insignificante. Este sistema é útil para ambientes de redes inteligentes, onde os medidores inteligentes fornecem informações abundantes [26].

Ressalta-se que uma forma de inferir a qualidade das curvas obtidas é empregar uma métrica de inferência. A mais popular forma empregar o conceito *mean absolute percentual error* MAPE) [27]:

$$\text{MAPE} = \frac{1}{K}\sum_{k=1}^{K}\left|\frac{y_{real}(k) - y_{calculado}(k)}{y_{real}(k)}\right| x\ 100 \qquad (2)$$

Sendo K o número de amostras no período considerado; k o tempo discreto; $y_{real}(k)$ o valor efetivamente observado/medido no tempo discreto k; e $y_{calculado}(k)$ o valor obtido via aplicação da proposta apresentada neste artigo no tempo discreto k.

III. RESULTADOS E DISCUSSÕES

A. Resultados

Os resultados foram obtidos de acordo com as definições dos conjuntos de entrada e de saída, bem como as regras baseadas no comportamento humano e conhecimento sobre as atividades diárias obtidas por meio do PD.

Para validar o perfil de carga residencial obtido pela execução do SF proposto, o consumo de energia elétrica foi medido considerando-se duas residências diferentes: o experimento foi conduzido levando-se em conta um conjunto de residências da cidade de Ilha Solteira (SP) (25.000 habitantes) e da cidade de São Paulo (SP) cerca de 15 milhões de habitantes). A rotina dos habitantes e a quantidade de habitantes para essas residências foram descritas nas entradas do SF proposto. O consumo de eletricidade foi medido hora a hora durante 24 h em um dia de trabalho (Quinta-Feira). A temperatura média nesse momento era de 28°C.

A qualidade e quantidade de informações, por certo, é parte essencial para o desenvolvimento desta pesquisa. Contudo, neste artigo, foi usado um conjunto basilar de informações, sendo que novas informações poderão ser incluídas visando melhorar o desempenho do sistema. Neste sentido, está-se trabalhando com vistas à realização de um módulo adicional para a extração do conhecimento permanentemente (de forma incremental) de modo semelhante ao modelo apresentado na referência [28] baseado na teoria de sistemas imunológicos artificiais (SIA).

As Fig. 5 e 6 ilustram o perfil de consumo obtido. Duas bases de regras diferentes foram criadas, considerando dois grupos de pessoas com rotinas diferentes.

Na Fig. 6 mostra-se o perfil de consumo durante 24 h, que foi obtido usando o SF e também os dados reais. Considera um dia de trabalho, para um grupo de residências em pequenas cidades, nos quais os moradores retornam para casa durante o período do almoço. Considerando que a clima em análise é quente, o maior consumo ocorre durante a noite (primeiro e segundo período de sono), por causa do uso de aparelhos de ar-condicionado, que possuem alto consumo de energia.

Para realizar a comparação, o consumo medido é normalizado considerando os aparelhos disponíveis em cada residência (potência total); isto é, a comparação é ajustada de acordo com a unidade da demanda elétrica.

Assim, a carga diária estimada total depende dos valores obtidos pela execução do SF multiplicado pela energia total consumida pela residência.

Na Fig. 5, o perfil de carga gerado pelo SF é otimista, acompanhando a curva de perfil real de consumo de energia praticamente nas 24 horas. Os picos locais ocorrem no horário de almoço entre às 12h e às 14h e outro ocorre a partir das 19h, ou seja, quando os moradores chegam em casa.

Além disso, o maior consumo ocorre entre as 23h e as 5h, tanto nos perfis reais como *fuzzy,* por causa do uso de aparelhos de ar-condicionado, formando um platô correspondente a uma

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carga quase constante.

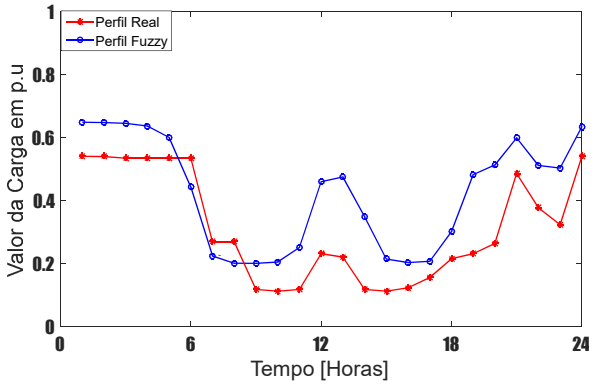

Fig. 5 . Perfil de consumo residencial (a).

Na Fig. 6 ilustra-se o perfil de consumo de energia elétrica obtido pelo SF e o perfil real (medido hora a hora) durante 24 h em uma residência na cidade de São Paulo (SP).

Na Fig. 6, nota-se que o perfil do SF é o mesmo perfil do consumo real basicamente em todos os horários. O pico mais alto da carga real ocorre às 21h e o pico mais alto gerado pelo SF está entre 21h e 22h, permanecendo alto e constante das 24h às 5h nas duas curvas de carga. Considerando que os consumidores não retornam para casa durante a hora do almoço, há um vale entre às 9h às 18h.

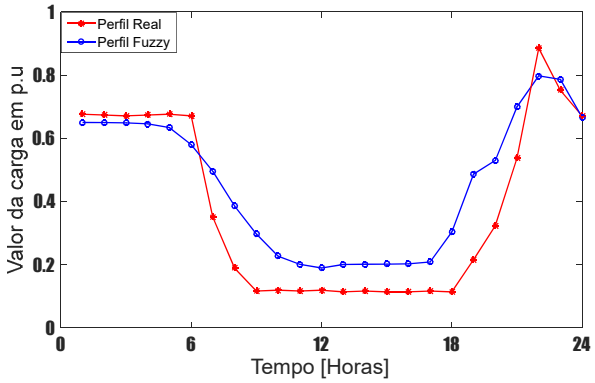

Fig. 6 . Perfil de consumo residencial (b).

Na Fig. 5, bem como na Fig. 6, os resultados obtidos tomando-se como base a metodologia proposta, foram comparados com resultados conhecidos, a partir da conta de energia elétrica do cliente-alvo. Este procedimento é usado com vistas a evidenciar a qualidade das soluções, ou seja, a precisão obtida. Esta métrica é similar ao emprego métrica MAPE, a qual se constitui numa métrica *benchmark* na área de previsão de carga em sistemas elétricos, como em (2).

B. Discussões

Nesta pesquisa propõe-se o uso de um SF para obter o perfil de consumo de energia elétrica residencial de acordo com a rotina de um grupo de consumidores.

Uma das principais contribuições desta pesquisa, é considerar como ocupação ativa o tempo que os moradores permanecem na residência, independentemente de estarem dormindo ou acordados. Os estudos relatados [8] consideraram apenas as pessoas em casa acordadas como ocupação ativa. No Brasil, um país tropical com um clima quente quase todo o ano é necessário considerar os momentos em que as pessoas estão em casa dormindo, quando é necessário usar aparelhos com alto consumo de energia, por exemplo, aparelhos de ar condicionado.

Para confirmar que o SF pode gerar perfis satisfatórios, o consumo de duas residências diferentes é medido de hora em hora. As residências estão localizadas nas cidades da Ilha Solteira (SP) e de São Paulo (SP) O dia de trabalho selecionado é uma quinta-feira. Estas duas cidades são escolhidas por suas características muito diferentes: a primeira é pequena, com poucos habitantes e a segunda é grande, com milhares de habitantes.

As Fig. 5 e 6 evidenciam que os perfis de carga obtidos pelo SF são satisfatórios, combinando as curvas reais em quase todos os casos.

Conforme supracitada, a estimativa de carga diária e é determinada considerando os valores obtidos pelo SF multiplicado pela energia total consumida pela residência. Assim, o FS proposto pode ser interpretado como sendo uma previsão de como os moradores de uma residência usam a energia elétrica. De acordo com essa previsão, podem ser obtidas outras informações, levando a benefícios econômicos e ao uso racional da energia.

Esta predição contém imprecisões inerentes à estocasticidade do problema. A proposta encontra-se no estágio inicial e deve ser melhorada.

Alternativamente, pode-se melhorar o desempenho deste sistema via implementação de sintonia dos pesos das regras, empregando, por exemplo, o conceito do controle *fuzzy* adaptativo [29].

Ressalta-se que na literatura especializada [30], o número de funções de pertinência recomendado está compreendido entre 3 e 7 regras. Após vários experimentos, constata-se ser mais adequado usar 5 funções de pertinência nesta pesquisa. Isto evita um aumento considerável do número de regras e mantém uma boa precisão dos resultados auferidos. Desta forma, há pouca chance de produzir uma explosão de regras. Se o número de regras for bastante elevado, pode-se recorrer a um processo de redução do número de regras, por exemplo, eliminando regras redundantes, ou proximamente redundantes. Este dispositivo tem sido sugerido em várias publicações, e.g., na referência [30].

Também, podem-se obter melhorias, de modo contínuo, na qualidade das soluções mediante a inclusão de um módulo semelhante ao que foi proposto, por exemplo, na referência [28], evidentemente com as devidas adaptações em razão de que na referência [28], na qual o método usado refere-se aos SIA. Neste artigo, emprega-se a teoria de sistemas *fuzzy*.

IV. Conclusão

O estudo abordando a previsão de carga elétrica residencial no Brasil é escasso, por causa, a princípio, da falta de dados reais. Na Espanha, estudos com modelos estocásticos estão bem avançados. Aqui, propomos o uso de um sistema inteligente que com poucos dados de entrada pode obter o perfil de carga

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residencial. Assim, um SF é proposto para obter perfis de carga residencial diária e incentivar os consumidores a mudar seu consumo nos horários de pico para momentos em que a capacidade de distribuição está aliviada.

Foram desenvolvidas duas bases de regras diferentes: uma para pessoas que vivem em grandes cidades que não voltam para casa durante o período do almoço e outra para pessoas que vivem em cidades pequenas que retornam para casa durante a hora do almoço para almoçar ou descansar. Se as pessoas retornam para casa afetam o consumo nesse período, o que é interessante para a gestão da demanda. As Fig. 5 e 6 comparam o perfil de consumo obtido usando o SF com o consumo real. Os perfis reais foram obtidos medindo a energia durante 24h nas duas cidades acima mencionadas, quando a temperatura média era de 28°C. Os perfis de consumo são semelhantes aos obtidos pelo SF e a carga real que foi medida, confirmando a confiabilidade do modelo proposto.

Com o perfil de consumo, é possível identificar os períodos de pico e evitar o congestionamento e interrupções da transmissão de energia elétrica. Além disso, essa informação pode ajudar os consumidores residenciais a economizar energia durante os horários de pico, horários em que a energia é mais cara. Por se tratar de uma versão "embrionária", a sua concepção contempla essencialmente a modalidade em *software*. Em versões subsequentes, pretende-se, a princípio, torná-la mais eficientes contemplando a integração *hardware* e *software*.

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Impact wave predictions by a Fuzzy ARTMAP neural network

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ABSTRACT

Impact waves caused by landslides, rock blocks, and avalanches in lakes and dam reservoirs have provoked several disasters since the dawn of time; further, they have caused considerable number of deaths and property loss. Understanding the behavior and characteristics of these waves, principally their height, propagation velocity, and energy, are key for determining engineering parameters, and therefore developing alert systems and evacuation plans. Several studies have investigated impact waves using experimental, mathematical, and numerical models; however, few have used artificial neural networks. Considering the learning and prediction characteristics of neural networks, this work explores their application in the prediction of impact waves. The main objective is to verify the capability of a fuzzy ARTMAP neural network in predicting the evolution of the maximum wave height, one of the main parameters of impact waves. The experimental data used in this paper are from the work developed by Huber (1980) when studying the impact waves generated by dropping deformable material (granular) in a channel. This study uses different forms of normalization, as well as different training parameters. The Fuzzy ARTMAP neural network predicts adequately the evolution of the maximum wave heights, becoming useful for this kind of application.

1. Introduction

Water waves generated by landslides or rockfalls in reservoirs of dams can lead to partial or complete destruction of their margins, leading to catastrophic consequences in valleys downstream. This is a problem affecting the society and is of particular concern to hydraulic engineers as well as electrical energy suppliers.

Risk factors in regions where there is a high probability of seismic or sliding activity, particularly in regions with unfavorable geological conditions, must be investigated during construction planning. The processes of impact wave generation and propagation that lead to the collapse of a dam can occur in a very short period, not giving time for an efficient evacuation procedure. Thus, a construction project must consider the most unfavorable hypothesis with respect to accidents (rockfalls into reservoir, dam failure) and evaluate the wave generation and propagation processes and consequent actions on the dam and adjacent structures, besides predicting the high-risk zones.

Several studies in the literature have investigated such hazards where a significant number of victims were reported including the landslides that occurred in Goldau in 1806 (Thuro and Hatem, 2010), in Davos lake in 1923 (Huber, 1980), Lituya bay in 1958 (Huber, 1980), Pontesi reservoir in 1959 (Ponziani and Gardoni, 2017), Vajont dam in 1963, (Slingerland and Voight, 1979), and Furnas reservoir (Brazil) in 1975 (Souza, 2007).

Recently, some efforts have been made to mitigate or prevent the impacts of hazards. These efforts were made based on physical models (scaled models of the dam and reservoir) or by considering non-deforming materials studied by (Souza, 2007), (Kamphuis and Bowering, 1970) and (de Freitas Maciel and do Nascimento, 2002), besides granular materials sliding on inclined planes into wave channels, studied by (Huber, 1980) and (Fritz et al., 2003).

Several analytical or semi-empirical solutions have been proposed (based on dimensional analyses), providing information on the height of waves and the energy dissipated by the impact including the classical works of (Kamphuis and Bowering, 1970); (Noda, 1974); (Slingerland and Voight, 1979), and (Huber, 1980). Fritz et al. (Fritz et al., 2003), examined these physical phenomena, with a focus on the impact zone, using an artificial sliding material composed of barium sulfate and

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poly-propylene and particle image velocimetry measurement techniques.

Other studies (such as (de Freitas Maciel and do Nascimento, 2002), (Raney and Butler, 1976), (Heinrich, 1992), (Zweifel et al., 2007), (Serrano-Pacheco et al., 2009), (Cremonesi et al., 2011), (Tinti et al., 2011), (Bosa and Petti, 2013), (Shi et al., 2016)) have described the drag effect caused by the solid material plunging into the reservoir using numerical and mathematical models. Moreover, authors have also investigated the evolution of numerical techniques used to solve the impulse wave problem, such as, the finite difference method, the finite element method, the finite volume with free-surface tracking technique (VOF), and more recently the Lagrangian method (Vasco et al., 2011).

Mathematical, numerical, and experimental models have been traditionally used to solve these particular problems. However, new approaches based on big data sets are available and are considered to be an important tool to predict impact waves, as for example the studies developed by (Santos et al., 2016) and (Panizzo et al., 2005).

Intelligent systems and artificial neural networks (ANN) can analyze, learn, memorize, and predict solutions to several problems. There are several studies in the literature concerning solitary waves ((Wei et al., 2010) using back-propagation), coastal waves ((Agrawal and Deo, 2004) using back-propagation (Santos et al., 2016), using Fuzzy ARTMAP), sea waves ((Sánchez et al., 2018) using back-propagation (Kumar et al., 2016), using sequential learning neural network (Prahlada and Deka, 2015), using wavelet decomposed neural network), tsunamis ((Romano et al., 2009) using multilayer perceptron and back-propagation) and impact waves ((Panizzo et al., 2005) using multilayer perceptron) among others.

Fuzzy ARTMAP neural networks are used in several research fields. The following examples show some of their applications. Mokhtar and Howe (2013) applied such a network to the management of energy demand in buildings; Pourpanah et al. (Pourpanah et al., 2016) used a hybrid model with fuzzy ARTMAP neural network and a genetic algorithm to classify data and extract rules; Marchiori et al. (Marchiori et al., 2011) used Euclidian ARTMAP for electrical energy systems; Barros et al. (Barros et al., 2015) used modified Euclidian ARTMAP with continuous training for detecting and classifying voltage disturbances in electrical energy systems.

The use of the adaptive resonance theory (ART) family of neural networks to predict impact waves is an unexplored field; however, there are several works using numerical methods validated with experimental tests, including one with multi-layer perceptron neural networks (Haykin, 1998).

Sections 2 to 2.1.2 present a Fuzzy ARTMAP neural network to predict impact waves, the Huber's data (Huber, 1980) processing, the nondimensional parameter calculus and the filtering processing of the data set. Sections 2.2 to 2.3 describe the training phase of Fuzzy ARTMAP, the input matrices used in each situation studied, the values of the training parameters and the diagnosis phase.

Sections 3 to 3.2 describe the evaluation criteria of the proposed methodology and the normalizations applied. Sections 3.3 to 3.3.4 present the results of the best and worst prediction obtained in each situation studied. Sections 3.4 to 3.4.5 present the studies on how the different normalizations can modify the results of the impact wave predictions.

At last, section four presents the conclusions and Appendix A presents the Fuzzy ARTMAP algorithm.

2. Fuzzy ARTMAP neural network applied to predict impact waves

This work proposes to apply a Fuzzy ARTMAP neural network to predict impact waves according to the following steps: Data preprocessing; Fuzzy ARTMAP neural network training; Diagnosis phase and Methodology evaluation, as shown in Fig. 1.

The application method is divided into the steps shown in Fig. 1 and the following sections describe them with details including information about the data set used.

There are two phases involved: the first one is the training phase and the second one is the diagnosis phase. The flowchart shown in Fig. 3 and the algorithm presented in Appendix A describe the two phases of this ANN. MATLAB® 2015a version (64-bit) is used as the simulation environment to implement the ANN.

2.1. Data set treatment from Huber, A. (1980)

This work uses the adaptive resonance theory (ART) family of neural networks (Carpenter et al., 1992a) to predict impact waves, based on the experimental study realized by Huber (1980), of Zurich University. His studies contain experiences with impact waves performed on a masonry horizontal channel (dimensions: 30.33 m, 0.50 m and 0.5 m), with a variable declivity ramp (28^0 to 60^0) upstream and a wave-damping beach (declivity of 1:12 to 1:0) downstream, allowing the dissipation of the generated waves (Fig. 2).

The sliding material used on the test was from granular nature (mass of 5, 15, 30 and 50 kg) that by impact have produced waves on the channel. The tests generated waves presenting oscillatory pattern until bores waves, measured by capacitive sensors along the channel (positions x'/h: 5, 10, 15 and 20). The depths tested on the channel were of 12, 18, 27 and 36 cm.

The total quantity of tests executed in (Huber, 1980) were 704 consisting of four sliding masses, six declivity ramps, four impact velocities, four damping shore declivities and four channel depths.

Impact waves are modeled by several independent parameters, such as the nondimensional slide volume (M); nondimensional impact velocity (V_e) of the sliding material or Froude impact number (F); plan inclination (sliding slope) (α); propagation relative distance (x'/h), where (h) is the still water depth in the channel; downstream inclination (β); wave propagation direction (γ); and relative density, i.e., ratio of the density of the sliding material to the density of water (ρ_s/ρ); and dependent parameters such as the maximum height of the relative wave (H/h); wave relative length (λ/h); wave steepness (Hi/λ); Ursell number (U); nondimensional wave velocity propagation (Fw); and the wave period ($Ti\sqrt{(g/h)}$), where (g) is the gravitational acceleration.

Huber, A. (Huber, 1980), presented correlation laws with the nondimensional parameters ϕi (M, F, α, x'/h, β) = 0, for a channel and classified waves according to an oscillatory pattern, such as, sinusoidal, cnoidal, solitary, and bore waves.

This work uses data referred to Huber (1980), considering the depth to be h = 0.12 m (with nondimensional H/h varying from 0.023 to 0.76) and considering 158 experiences. These selected data require pre-processing; this is divided into two steps: pre-calculation of nondimensional parameters (M and F) and data filtering.

2.1.1. Pre-calculation

The nondimensional parameters (M and F) are calculated according to equations (1) and (2) before inserting the data into the input matrix.

$$M = \text{mass} \,/\, \left(\rho_s bh^2\right) = \text{Volume} \,/\, \left(bh^2\right), \tag{1}$$

where

Volume: slide volume;
ρ_s: density of the sliding material;
b: width of the channel;

$$F = V_e \Big/ \sqrt{gh} \tag{2}$$

2.1.2. Data filtering

Certain data presented non-typical behavior without explanation even after an analysis was performed; therefore, they were excluded from the data set. This non-typical behavior is graphically verified in this paper and includes a general analysis of all parameters. The excluded

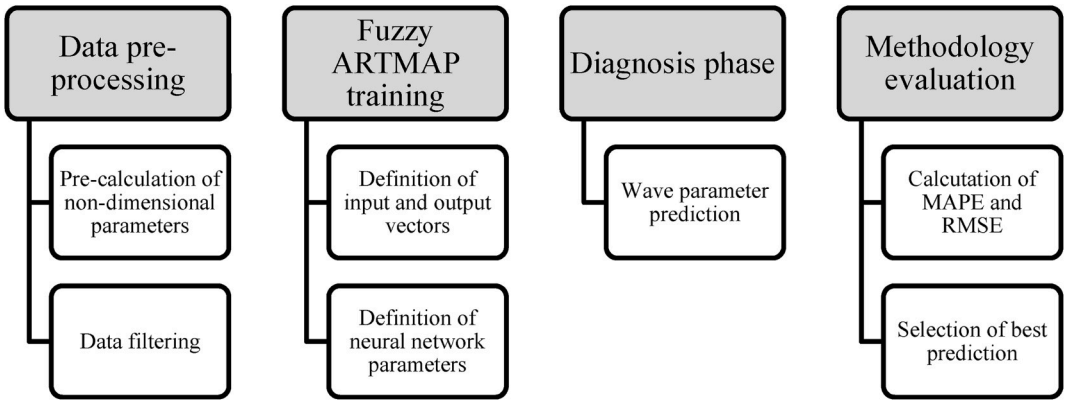

Fig. 1. Steps for applying a fuzzy ARTMAP neural network to predicting impact wave parameters.

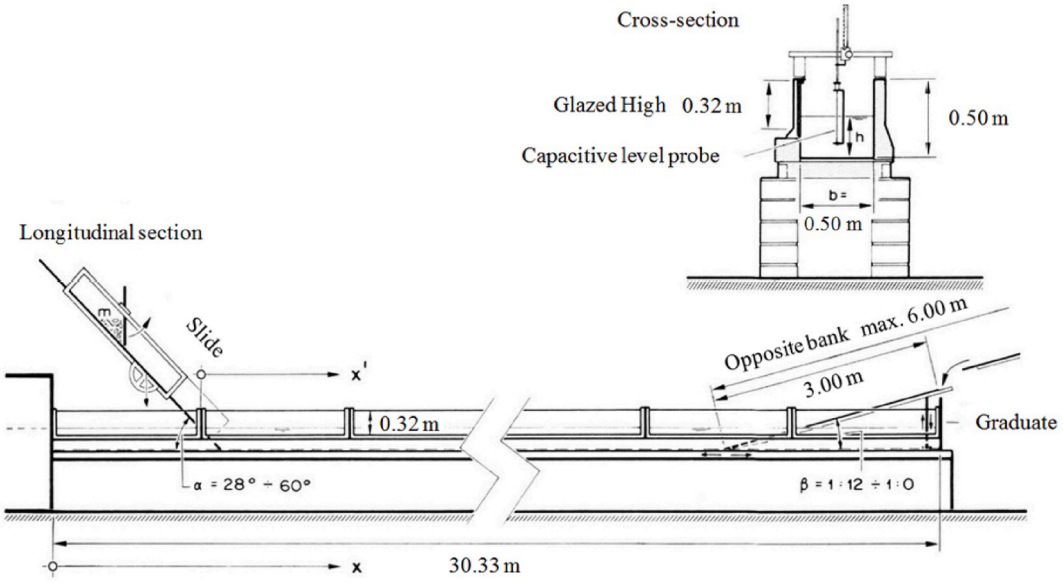

Fig. 2. Measuring channel, longitudinal and cross-section with lane and opposite bank.
Source: Modified from Huber (1980)

data contain no information about the relative height or the length period. With the filtering process, the data set with 158 experiences was reduced to 74 experiences: 50% of it (37 experiences) is used for training and the other 50% (37 experiences) for prediction. For tests considering the influence of normalization on the modified parameters (as given in Table 6), 80% of data (59 experiences) is used for training and 20% (15 experiences) for prediction.

2.2. Fuzzy ARTMAP neural network training phase

A Fuzzy ARTMAP neural network is used to predict the maximum wave height. The architecture of this network can be adapted to several problems and applications. Therefore, despite the high complexity of the present problem, this ANN can interpret the parameters (referred to as impact wave characteristics) and provide an adequate analysis.

The input data for training (37 experiences) are randomly selected from the 74 experiences. For this step, the lines of the matrix were numbered and the numerical order was randomly changed. Furthermore, this work intends to investigate the robustness of this neural network.

2.2.1. Fuzzy ARTMAP neural network

This network is supervised; it is composed of two ART modules called ARTa and ARTb connected by an inter-ART module based on fuzzy logic (Carpenter and Grossberg, 1992), and the principal characteristics of Fuzzy ARTMAP are plasticity, stability and malleability.

According to (Lopes, 2005) the plasticity is the capacity to continue learning when including new patterns, without losing the memory in relation to the previous patterns, while the stability is the capacity to learn by adjusting the weights. These two characteristics provide to the neural network a stable learning and a fast convergence. Furthermore, the malleability or flexibility is the capacity of the neural network to be applicable in every area of knowledge (Lopes, 2005).

2.2.2. Definition of input and output vectors

The input data of the neural network are in a matrix form, where each vector represents one experience; the same is valid for the output data.

$$X_i = [h,\ F,\ M,\ \alpha,\ \beta];\ i \in [1;N]; \tag{3}$$

$$Y_i = [H/h1,\ H/h2,\ H/h3,\ H/h4,\ H/h5];\ i \in [1;N]; \tag{4}$$

$$\text{Input matrix } Nxme = [X1;\ X2;\ X3;\ \ldots;\ XN]; \tag{5}$$

$$\text{Output matrix } Nxms = [Y1;\ Y2;\ Y3;\ \ldots;\ YN]; \tag{6}$$

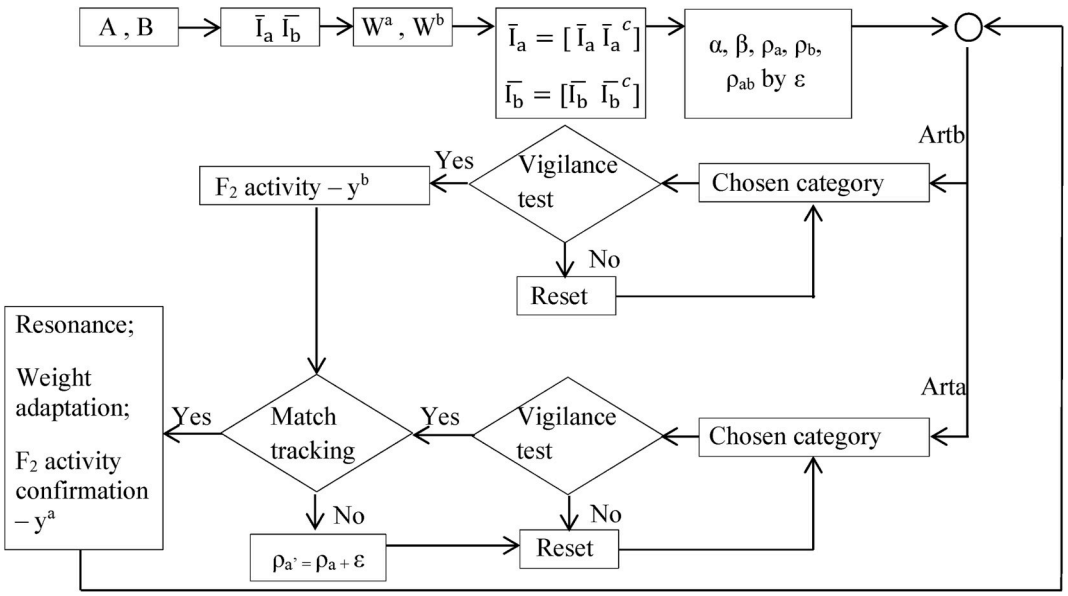

Fig. 3. Fuzzy ARTMAP ANN Training phase.
Source: Adapted from (Lopes, 2005)

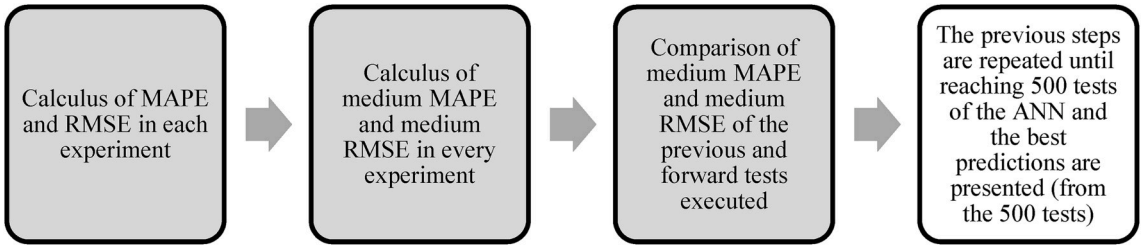

Fig. 4. Steps realized for selecting the best prediction.

where

X_i: input matrix; $X \in R^{Nxme}$;
N: number of experiences of impact waves;
Y_i: output matrix; $Y \in R^{Nxms}$;
me: number of considered parameters of the impact waves;
ms: number of measuring points along the channel (ms = 5);

2.2.3. *Fuzzy ARTMAP neural network parameters*

Appendix A describes the algorithm of the neural network according to (Carpenter et al., 1992a), (Carpenter and Grossberg, 1992), (Carpenter et al., 1992b), (Carpenter and Ross, 1995) and (Müller, 2018), and a flowchart is shown in Fig. 1. MATLAB was chosen as the simulation environment. Table 1 lists the training and prediction parameters, which were used for all experiences, except for the cases investigating the influence of normalization. Appendix A contains the variations on the training parameters.

Table 1
Parameters for training the ANN.

Parameter	Value
α	0.10
β	1.00
ρ_a	0.50
ρ_b	1.00
ρ_{ab}	1.00
ρ_d	0.50
ε	0.05

In Table 1, α is the chosen parameter of the neural network (α > 0); this value can be close to zero, providing little variation for the vigilance test according to (Lopes, 2005); β is the training rate ([0, 1]); ρ_a, ρ_b, ρ_{ab} and ρ_d are the vigilance parameters of modules ARTa, ARTb, InterART and Prediction, respectively, pertaining to the interval (0, 1]. These values, when close to 1, force the neural network to be more demanding, leading to lose some categories and a low generalization. Meanwhile, values close to zero group similar categories, that are not necessarily equal, according to (Lopes, 2005). ε is the increment of the vigilance parameter.

The parameters presented in Table 1 were obtained after several tests with the ANN such as the least values provide good results, allowing more malleability, instead of increasing the exigency grade according to what is shown in section 3.4.

2.3. *Diagnostic phase of the fuzzy ARTMAP neural network*

After the training phase, where the neural network groups the input patterns in categories, representing by updated weight matrix, the diagnosis phase begins. During this step, new input data are presented and evaluated by the neural network, classifying them according to the adapted weight matrix obtained on the training phase.

The diagnosis phase is conducted according to item IX of the algorithm presented in Appendix A. At this step, the ANN can predict the maximum relative height of the impact waves, and the results are evaluated in the following section.

Table 2
Experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (SF).

Experimental	Prediction	Experimental	Prediction
0.438	0.435173	0.251	0.44408
0.354	0.354586	0.279	0.367952
0.296	0.298174	0.309	0.304512
0.240	0.240151	0.354	0.250588
0.208	0.207916	0.393	0.218868
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.329729	0.001618	36.756	0.131323

Table 3
Experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (FI).

Experimental	Prediction	Experimental	Prediction
0.540	0.54351	0.310	0.240571
0.440	0.43927	0.225	0.206638
0.370	0.3673	0.170	0.181315
0.298	0.29781	0.125	0.159537
0.258	0.2581	0.110	0.15194
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.32944	0.00201	20.59386	0.04058

3. Results and discussion

Comparisons were made between the data of (Huber, 1980) and the predictions obtained by the Fuzzy ARTMAP ANN. The results are presented in Tables 2–13 and Figs. 5–13. The values presented refer to the minimum and maximum Mean Absolute Percentage Error (MAPE) of the best prediction selected according to a certain criterion from 500 tests. The best medium MAPE is presented on the analysis executed in sections 3.3 to 3.4.5.

The values of RMSE (Root Mean Square Error) presented in Tables 2–13 are referred to the maximum and minimum MAPEs. The RMSE measures the behavior of the data. Furthermore, a less value of RMSE is not at all associated to a less value of MAPE, it only means that the prediction curve is similar to the experimental data.

The tests realized do not measure overestimation or under estimation of the predictions, however, observing Figs. 5–13 is possible to visualize the overestimations and under estimations.

3.1. Criterion for evaluation of methodology

To evaluate the methodology, 500 random tests are conducted on the ANN. This is done to analyze the robustness of the ANN. The difference with each one of the 500 tests is due to the input position of each line of the matrix, once the data are randomized. The evaluation criterion is set according to the following procedure for each execution:

Additional steps

a) Calculation of MAPE and RMSE (root mean square error) in each experience:

$$MAPE_{nd} = \left[\left(\sum_{i=1}^{ms} |SR_i - SP_i| / SR_i \right) \Big/ /ms \right] * 100 \quad (7)$$

$$RMSE_{nd} = \left[\sum_{i=1}^{ms} (SR_i - SP_i)^2 / ms \right]^{1/2} \quad (8)$$

where

nd: number of inputs presented in the diagnosis phase, $nd \in [1; 15]$;
SR: value obtained in the experience;
SP: value obtained in the prediction;

Table 4
Experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (SH).

Experimental	Prediction	Experimental	Prediction
0.634	0.62784	0.480	0.399766
0.576	0.574331	0.370	0.345818
0.524	0.524389	0.290	0.298787
0.462	0.464935	0.205	0.247606
0.420	0.424505	0.160	0.213024
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.608709	0.003736135	16.04086	0.048427905

Table 5
Experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (AM).

Experimental	Prediction	Experimental	Prediction
0.404	0.403121	0.470	0.373328
0.320	0.319495	0.366	0.341771
0.253	0.253023	0.292	0.312595
0.185	0.185479	0.230	0.278061
0.146	0.146882	0.202	0.254244
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.249439	0.000637898	16.20025	0.055490495

Table 6
Original and changed values of parameters used for training the ANN.

Parameter	Original value	Changed value
α	0.1	0.1
β	1.0	1.0
ρ_a	0.5	0.97
ρ_b	1.0	1.0
ρ_{ab}	1.0	1.0
ρ_d	0.5	0.8
ε	0.05	0.05

Table 7
Values of MAPE (%) obtained on tests with normalization of input and output matrices (original parameters and 50% of data used for training).

	Normalization – original parameters				
MAPE/test	Me – SM	MCe – SMC	SCe – SSC	MLe – SML	SLe - SN
Maximum	95.10	45.72	127.61	16.86	20.59
Minimum	1.15	0	0	0.70	0.32
Medium	16.21	15.41	14.90	6.97	5.39

b) Calculation of MAPE and RMSE in the current analysis:

$$MAPE_{medium} = \left(\sum_{i=1}^{nd} MAPE_{nd} \right) \Big/ nd \quad (9)$$

$$RMSE_{medium} = \left(\sum_{i=1}^{nd} RMSE_{nd} \right) \Big/ nd \quad (10)$$

c) Criterion for selecting the best prediction:

The best prediction comes from the comparison with the previous test and the forward test using the value obtained according to equation (11) as comparison criterion. At the first test, as there is no previous one, the best result is considered, and from this one the comparison begins.

$$\text{Best prediction} = \min MAPE_{medium} \,\&\, \min RMSE_{medium} \quad (11)$$

The comparison criterion is chosen after executing tests with the ANN using other criteria of selecting the best prediction.

Table 8
Values of MAPE (%) obtained on tests with normalization of input and output matrices (changed parameters and 80% of data used for training).

	Normalization – changed parameters[a]				
MAPE/test	Me – SM	MCe – SMC	SCe – SSC	MLe – SML	SLe - SN
Maximum	33.35	29.44	16.81	11.09	8.95
Minimum	1.79	0	1.79	0.70	0.32
Medium	12.35	10.30	7.72	2.33	3.82

where,
SN: normalization of the output matrix according to (Mokhtar and Howe, 2013).
Me: normalization by the maximum of the input matrix.
MCe: normalization by the maximum of the columns of the input matrix.
MLe: normalization by the maximum of the rows of the input matrix.
SCe: normalization by the sum of columns input matrix.
SML: normalization by the maximum of the rows of the output matrix.
SM: normalization by the maximum of the output matrix.
SMC: normalization by the maximum of the columns of the output matrix.
SSC: normalization by the sum of the columns of the output matrix.
SLe: normalization of the input matrix according to (Mokhtar and Howe, 2013).
[a] according to Table 6.

Table 9
Changed conditions for experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (MES).

Experimental	Prediction	Experimental	Prediction
0.400	0.400	0.289	0.480
0.290	0.290	0.250	0.370
0.215	0.212	0.216	0.290
0.150	0.145	0.179	0.205
0.118	0.123	0.154	0.160
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
1.793194	0.003435	33.35	0.106835

Table 10
Changed conditions for experimental and prediction results with respect to the minimum (right) and maximum (left) MAPEs (MC).

Experimental	Prediction	Experimental	Prediction
0.352	0.352	0.280	0.380
0.290	0.290	0.200	0.265
0.240	0.240	0.146	0.190
0.190	0.190	0.102	0.130
0.162	0.162	0.084	0.102
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0	0	29.44616	0.058769

Table 11
Changed conditions for experimental and prediction results related to the minimum (left) and maximum (right) MAPEs (SC).

Experimental	Prediction	Experimental	Prediction
0.400	0.400	0.380	0.450
0.290	0.290	0.265	0.322
0.215	0.212	0.190	0.226
0.150	0.145	0.130	0.150
0.118	0.123	0.102	0.112
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
1.793194	0.003435	16.81328	0.044598

At first time, the minimum MAPE with each test is used as the criterion, however it was not a good criterion once it presents only a very low MAPE. At second time, the minimum medium MAPE is used, nevertheless, even with a low medium MAPE, the graphical behavior not always matches the experimental data.

At third time, the minimum RMSE is used, however it only indicates that the curves have a similar behavior, which not necessarily was associated to low MAPEs neither a good prediction. At fourth time, the medium RMSE is used, nevertheless the same situation presented when using the minimum RMSE was observed.

Table 12
Changed conditions experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (ML).

Experimental	Prediction	Experimental	Prediction
0.438	0.438	0.405	0.405
0.354	0.356889	0.318	0.294092
0.296	0.300111	0.250	0.220569
0.240	0.241711	0.192	0.160754
0.208	0.209267	0.168	0.134585
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.705379	0.00244	11.09092	0.026573

Table 13
Changed conditions experimental and prediction results with respect to the minimum (left) and maximum (right) MAPEs (CA).

Experimental	Prediction	Experimental	Prediction
0.540	0.543508	0.352	0.312207
0.440	0.439273	0.290	0.277474
0.370	0.367302	0.240	0.246254
0.298	0.297813	0.190	0.21074
0.258	0.258104	0.162	0.187324
MAPE (%)	**RMSE**	**MAPE (%)**	**RMSE**
0.329729	0.001618	8.955552	0.023879

At last, the option was to join the medium MAPE and the medium RMSE and verify if they were enough to indicate the best prediction within the 500 tests. This is confirmed with the chosen criterion presented according to equation (11).

Fig. 4 shows the previous steps, observing that the steps described in Fig. 4 occur after the conclusion of the diagnosis.

3.2. Normalization

The present work also analyzes the influence of the normalization process for the input and output matrices of the ANN. Besides, normalization according to (Mokhtar and Howe, 2013), the following normalizations were used: maximum value of each column, equation (12); maximum of the matrix, equation (13); summation of each column sum, equation (14); and maximum of each row (15).

a) Normalization by the maximum value of each column:

$$\bar{I} = \mathrm{I} \,/\, \max|\mathrm{I}_j| \tag{12}$$

b) Normalization by the maximum of the matrix:

$$\bar{I} = \mathrm{I} / \max|\mathrm{I}| \tag{13}$$

c) Normalization by summation of each column:

$$\bar{I} = \mathrm{I} \,/\, \Sigma \mathrm{j}|\mathrm{I}_j| \tag{14}$$

d) Normalization by the maximum of each row:

$$\bar{I} = \mathrm{I} \,/\, \max|\mathrm{I}_i| \tag{15}$$

where:
I: input or output matrix;
i: number of rows of matrix I;
j: number of columns of matrix I;
An increase of 20% above the maximum values for the standardized

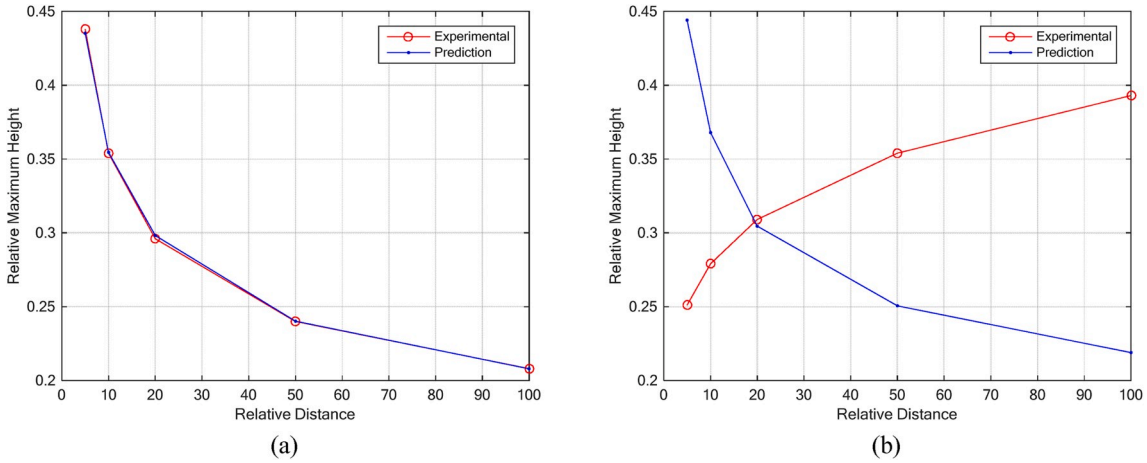

Fig. 5. Comparison of the experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (SF) and RMSE correlated.

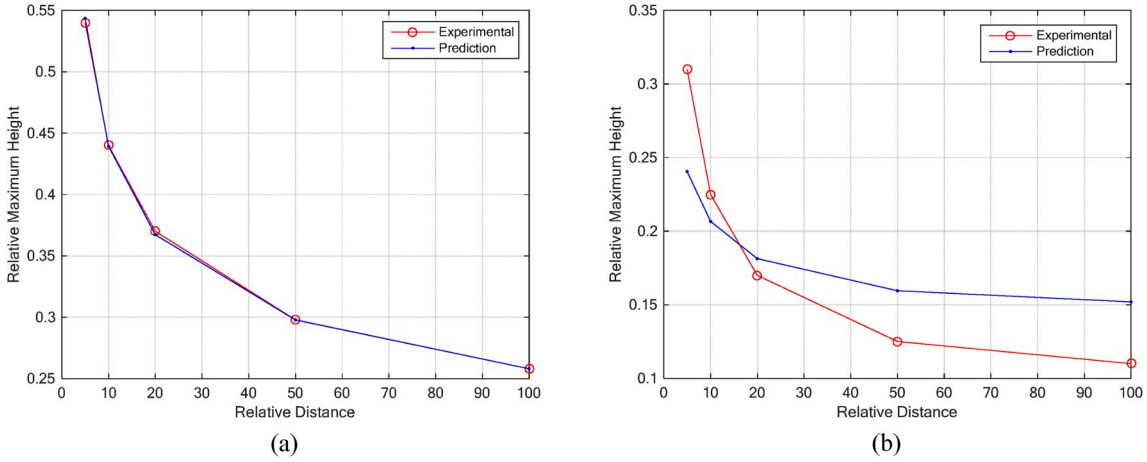

Fig. 6. Comparison of the experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (FI) and RMSE correlated.

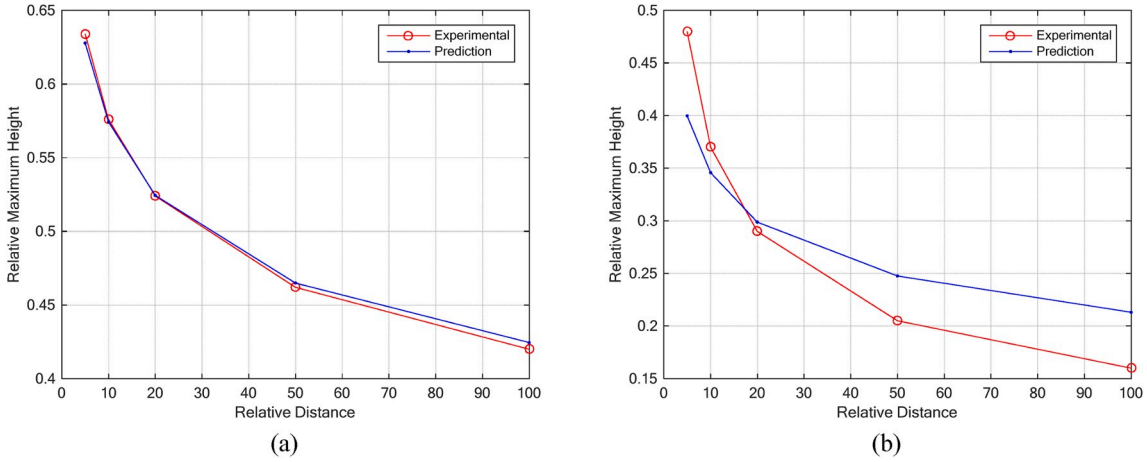

Fig. 7. Comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (SH) and RMSE correlated.

data is considered. This procedure allows the ANN to analyze situations with values higher than the maximum presented in the training data set.

3.3. Prediction of the maximum relative height for $h = 0.12$ m

3.3.1. Results for fuzzy ARTMAP ANN without data filtering (SF)

This test executes 158 experiences, instead of 74 experiences chosen after filtering, where 50% of data are for training and 50% are for prediction; the parameters of the ANN are according to Table 1 and the input matrix is given in equation (3). Table 2 presents the results of the minimum and maximum MAPEs of the best prediction result of the ANN among 500 tests.

Although the ANN could predict part of the data, there were some inconsistencies. This is due to the presence of non-typical data. It is

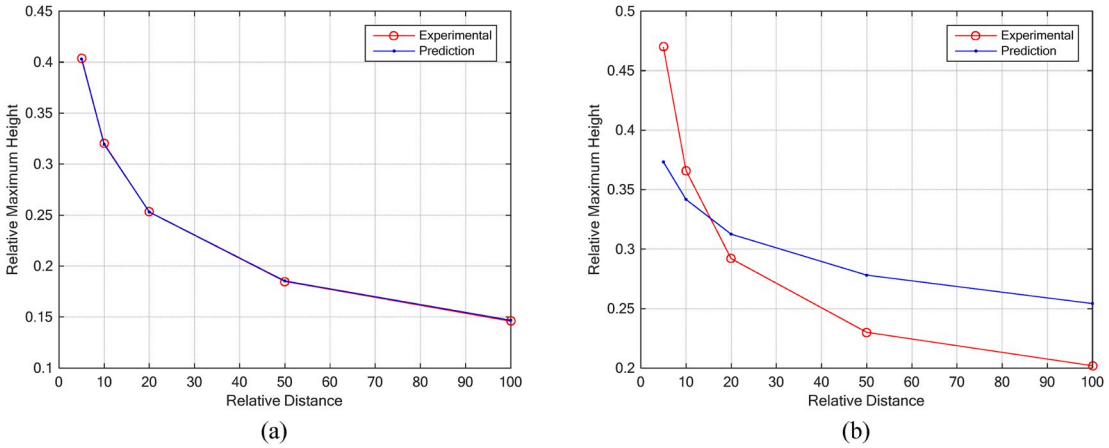

Fig. 8. Comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (AM) and RMSE correlated.

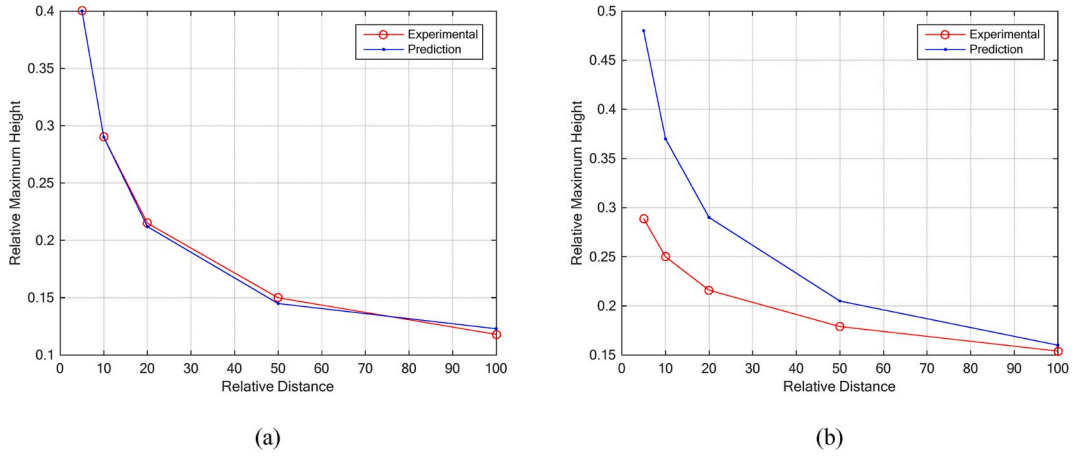

Fig. 9. Changed conditions for comparison of the experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (MES) and RMSE correlated.

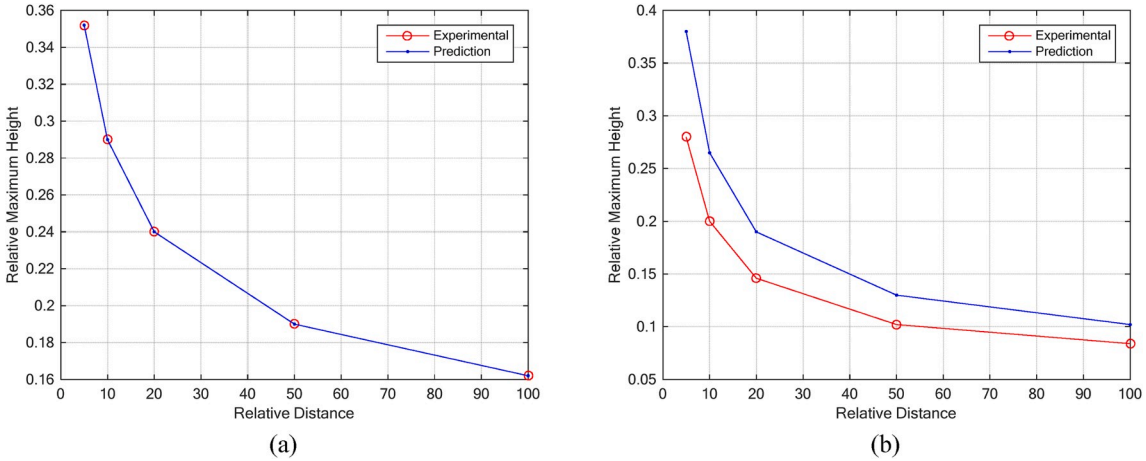

Fig. 10. Changed conditions for comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (MC) and RMSE correlated.

expected that the maximum heights exhibit a descending trend (Fig. 5 (a)), i.e., as the impact wave evolves and is attenuated on the channel, the waves tend to decrease, unlike what is shown in Fig. 5 (b) for the experimental results.

The maximum MAPE is 36.7%, the minimum is 0.329%, and the medium is 6.55%, indicating the need to filter the input data. An analysis of the RMSE shows that the best prediction has an RMSE value close to zero, indicating that the ANN could predict the trend of the data nicely; however, when its predictions become inconsistent, the RMSE value is close to 0.13 according to Table 2.

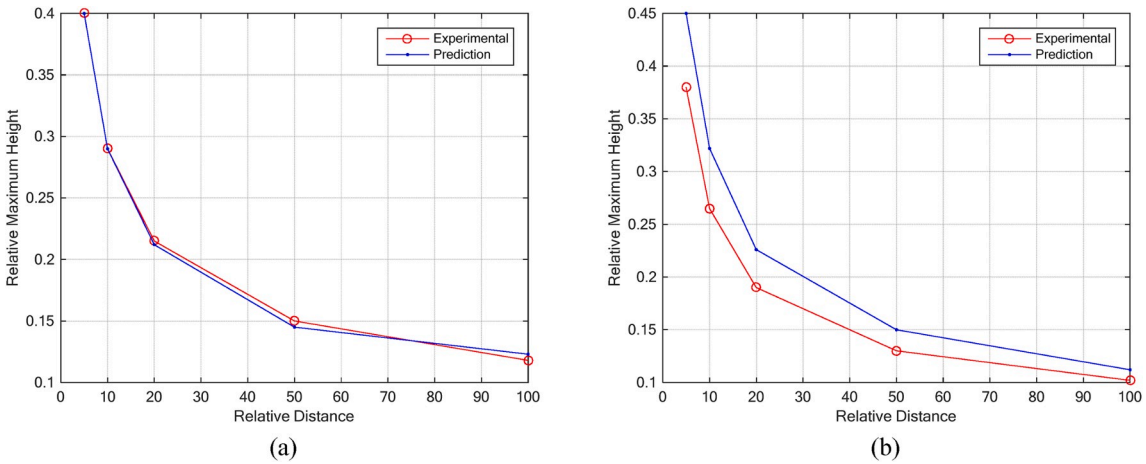

Fig. 11. Changed conditions for comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (SC) and RMSE correlated.

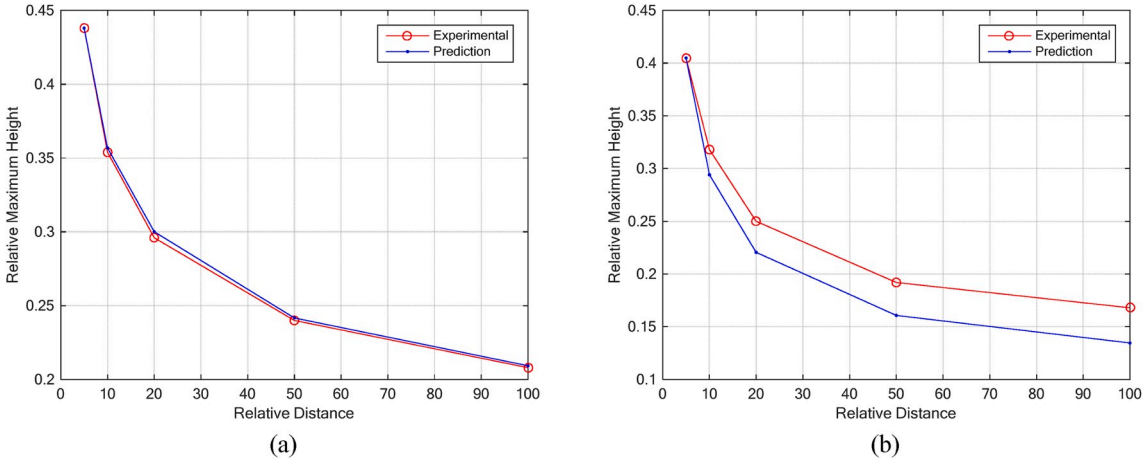

Fig. 12. Changed conditions for comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (ML) and RMSE correlated.

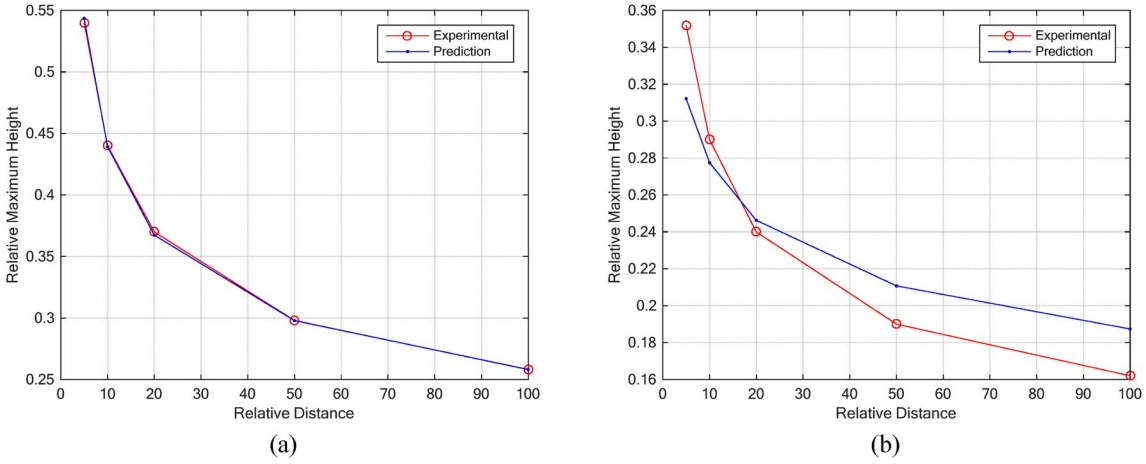

Fig. 13. Changed conditions comparison of experimental and prediction results in relation to the minimum (a) and maximum (b) MAPEs (CA) and RMSE correlated.

3.3.2. Fuzzy ARTMAP ANN with data filtering (FI)

Further tests were conducted after filtering the data with 74 experiences, 50% of which were used for training and 50% for prediction. The training parameters and the matrices were unchanged. Table 3 presents the results of the minimum and maximum MAPEs.

An analysis of the results of this test and those of the previous one shows a reduction in the maximum MAPE from 36.75% to 20.59%, the minimum MAPE is maintained in 0.32%, and a reduction in the medium

MAPE from 6.55% to 5.39%. This reduction in the medium MAPE indicates that more data are close to the expected values than in the previous case. An analysis of the RMSE shows that predictions are in close agreement to the trends in experimental data, as shown in Fig. 6 (a) and 6 (b).

3.3.3. Results of fuzzy ARTMAP ANN with filtering and without the input of water depth (h)(SH)

This test also uses 74 experiences, out of which 50% data are used for training and 50% are used for prediction; however, the input matrix given in equation (3) is modified by eliminating the column referred to as (h). Therefore, the simulation executes for (F, M, α, β) without the still water depth input (h). This parameter is removed to verify the parameters that have a greater influence on the predictions and to demonstrate the behavior of the ANN. Table 4 presents the results referring to the minimum and maximum MAPEs.

Little improvement is observed in the prediction results in comparison with the previous analysis, considering a maximum MAPE of 16.0%, a minimum MAPE of 0.60% and a medium MAPE of 5.29%. According to Fig. 6, the behavior improves, as the RMSE is closer to zero with respect to the minimum MAPE (Fig. 7(a)) and close to 0.05 with respect to the maximum MAPE (Fig. 7 (b)). Based on this information, the parameter h presents no significant influence on the prediction.

Other tests were also conducted by eliminating parameters related to the inclined slope and downstream area; these results also show no significant influence on the predictions and therefore were considered not important for analysis.

3.3.4. Results of the fuzzy ARTMAP ANN with filtering and only the nondimensional slide volume (M) (AM)

This test is realized only with the nondimensional slide volume (M), as a last search for the nondimensional parameter that is useful for prediction. The input matrix is modified to one with 74 rows and single column. The percentages of data used for training and prediction and the ANN parameters remain unchanged. The minimum and maximum MAPEs are presented in Table 5.

The ANN exhibits a maximum MAPE of 16.2%, minimum of 0.249%, and medium of 5.12%, showing little improvement when compared with the previous tests. Considering the behavior of the curves, the RMSEs are close to zero and 0.05, as seen in Fig. 8 (a) and 8 (b), respectively; these results are similar to those of the previous analysis.

The influence of Froude number (F) on the ANN behavior in predicting the maximum height was also verified. Although the ANN could predict the data, the maximum MAPE was 24.94%, the minimum MAPE was 0.06%, and the medium MAPE was 8.48%. These results demonstrate that this parameter, on its own, is not enough to obtain a satisfactory prediction, when compared to the results obtained using only the nondimensional slide volume (M), particularly in the case of impact waves generated by the fall of granular material.

Considering both the analyses, it is observed that information on the nondimensional slide volume (M) was enough to undertake the predictions; which confirms the conclusion drawn by (Huber, 1980), who stated that the maximum height of a wave could be predicted only using this parameter.

3.4. Influence of normalization on the determination of maximum heights

The influence of normalizing the input and output matrices of the ANN was investigated. The matrices were normalized according to the five forms described in section 3.2 by assuming that the input and output matrices must be normalized using the same form. The analyses were conducted with filtered data and the results of these tests are presented in sections 3.4.1 to 3.4.5.

Tests were conducted considering the normalization of the input and output matrices with different parameters, as given in Table 6.

The changed conditions correspond to the values that the ANN requires to obtain an accurate prediction in comparison with the original conditions. Note that, in these changed conditions, 80% of data are used for training and 20% for prediction. The values of the MAPEs are listed in Tables 7 and 8 along with a comparison of the five forms used to normalize the parameters (Table 6).

Based on the previous analysis, normalization defined by (Mokhtar and Howe, 2013) as well as MLe-SML normalization present good results when compared to the others.

3.4.1. Normalization by the maximum of input and output matrices (MES)

Normalization is conducted according to equation (13). At first, a test was realized with original parameters of the ANN using 50% of the data for training. The maximum MAPE is 95.10%, the minimum MAPE is 1.15%, and the medium MAPE is 16.21%; these are not satisfactory when compared with the normalization chosen for the approached problem (normalization SLe-SN).

To obtain a better prediction, the training parameters were changed according to Table 6, while the percentage of data used for training was increased to 80%. Table 9 presents the values of MAPE for the changed conditions and a comparison of experimental and prediction curves is presented in Fig. 9 (a) and (b).

In the changed situation, the maximum MAPE is 33.35%, the minimum MAPE is 1.79%, and the medium MAPE is 12.35%; these are higher than the expected values in comparison with the chosen normalization (SLe-SN). This behavior is not considered satisfactory; therefore, a second test using a different normalization form is executed.

3.4.2. Normalization realized by the maximum of each column (MC)

This test realizes normalization by the maximum of each one of the columns, either for the input or the output matrix according to equation (12). The maximum MAPE is 45.72%, the minimum MAPE is zero, and the medium MAPE is 15.41%. Even with minimum MAPE equal to zero, the results are far from expected, when analyzing the value of medium MAPE. In this way, a changed situation was also analyzed; the results are presented in Table 10 and Fig. 10 (a) and (b).

Considering the changed situation, the ANN presents a maximum MAPE of 29.44%, a minimum of zero, and a medium of 10.30%. Based on these results and data presented in Table 10, this normalization, when compared with the chosen one (SLe-SN), has higher MAPEs.

3.4.3. Normalization by the sum of the column (SC)

The third investigation is realized considering normalization by the sum of each column of the input and output matrices, according to equation (14). The maximum MAPE is 127.6%, the minimum is zero, and the medium is 14.90%, proving that this normalization is not adequate for the problem due to the high values of maximum and medium MAPEs. The change in the parameters of the ANN was also investigated and the results are presented in Table 11 and Fig. 11 (a) and (b).

This changed situation is better than the previous one, with a maximum MAPE of 16.81%, minimum of 1.79%, and medium of 7.72%. However, compared with the chosen normalization, the normalization by the sum of the columns is not adequate for the problem, even with the changed conditions.

Table 11 (left side) and Table 9 (left side) are identical; this is due to an increase in the training percentages and the modification of the training parameters of the ANN.

3.4.4. Normalization by the maximum of the row (ML)

The fourth investigation was the normalization of the maximum of each row, either on the input or output matrices according to equation (15). Considering the pattern, the maximum MAPE of the ANN is 16.86%, minimum is 0.70%, and medium is 6.97%. Compared with the other normalizations, this shows the best results; however, the medium MAPE shows that the predictions are still far from the expected. The results for the changed conditions were also analyzed and are presented

in Table 12 and Fig. 12 (a) and (b).

For the changed conditions, the ANN obtains a maximum MAPE of 11.09%, a minimum of 0.70%, and a medium of 2.33%. There is an improvement with respect to the original conditions when compared to the medium MAPE of the chosen normalization, but, the minimum and maximum MAPEs are still higher.

3.4.5. Normalization according to Carpenter (1992) (Mokhtar and Howe, 2013) with modified parameters (CA)

For comparison, the chosen normalization is analyzed for the changed conditions, and the results are presented in Table 13 and Fig. 13 (a) and 13 (b).

Comparing the two last normalizations (by the maximum of the rows and according to (Mokhtar and Howe, 2013)), the (Mokhtar and Howe, 2013) normalization is better when analyzing the maximum MAPE (11.09 and 8.99, respectively), minimum (0.70 and 0.33) and medium (2.33 and 3.82), both are adequate to the problem under analysis. It is emphasized that, when analyzing all the situations, normalization (Mokhtar and Howe, 2013) is the chosen one.

4. Conclusions

This study is based on the works developed by (Huber, 1980), (Santos et al., 2016) and (Panizzo et al., 2005) with a new approach using Fuzzy ARTMAP neural network to determine the evolution of the maximum heights of impact waves, instead of using multilayer perceptron, a common approach to sea wave problems.

The experimental data used for the input and output matrices come from (Huber, 1980), which study contains one of the most important experimental database for impact waves. After some preliminary analysis, it was observed the need for filtering, once there were some inconsistencies.

After executing the first tests, considering the criterion of the best prediction (the minimum MAPE), it is verified that this criterion is not adequate for the problem. New criteria were investigated until obtaining the adequate one (the medium MAPE, the minimum RMSE, medium MAPE and the medium RMSE). The criterion chosen for the best prediction is the medium MAPE and medium RMSE.

Another important procedure to improve the results is to randomly input the data to the ANN. Although they are not presented in this study, better and worse situations existed, depending on the randomness. However, even after several executions, the ANN is stable and exhibits consistent results.

Literature provides several works using the normalization according to (Mokhtar and Howe, 2013). However, this work opts to investigate other normalization forms, such that lead to best predictions. Five normalizations were tested, including the one of (Mokhtar and Howe, 2013). The objective was to verify, if changing the training parameters, they could lead to best answers.

Executing the tests with the original and modified parameters, considering the five normalizations applied, the conclusion is that normalizations Me-SM, MCe-SMC, SCe-SSC were not adequate for the approaching problem. Nevertheless, normalizations MLe-SML and SLe-SN present good results, either with the original or modified parameters. These two normalizations present the least maximum, medium and minimum MAPEs, therefore, they are adequate for the approaching problem.

It was observed that in several analyzed situations the medium MAPE presented value close to the minimum MAPE, indicating that the predictions obtained by ANN were very close to the experimental data, but the training parameters need to be further explored and the process to better answers needs to be further analyzed.

The RMSE values show that the chosen ANN is able to make the prediction curves consistent with the experimental ones, thus, predicting the wave behavior. These two conclusions emphasize the ability of the Fuzzy ARTMAP neural network to predict the behavior of the impact waves.

Comparing the results obtained from the experiences of (Huber, 1980) with the predictions by Fuzzy ARTMAP ANN, the latter was found to predict the data related to the maximum waves heights with good precision.

The results and predictions of maximum heights could be applied to other depths based on the data that were explored and presented in non-dimensional form, and in situations of generating waves in channel (1D) such as: H/h as function of M/bh2. The proposed paper considers the analysis of H/h (h fixed in 12 cm depth). The same way, it is emphasized that such results must be used for predicting maximum waves within the non-dimensional parameter range tested experimentally by Huber (1980), such as: $0.5 \leq Fr \leq 4$ and $0.023 \leq Hi/h \leq 0.76$.

Fuzzy ARTMAP neural network is a powerful tool to predict the characteristic parameter of impact waves: evolution of the maximum heights. Although, the ANN is able to obtain significant results, other situations still need to be analyzed and explored; thus, more significant results can be obtained. This study can, in the future, help to develop an alert system that predicts the occurrence of catastrophes caused by impact waves, therefore, becoming a system that helps to prevent disasters.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

P.B.A. Oliveira: Conceptualization, Formal analysis, Writing - original draft, Writing - review & editing. **A.D.P. Lotufo:** Conceptualization, Writing - original draft, Writing - review & editing. **M.L.M. Lopes:** Conceptualization, Writing - original draft, Writing - review & editing. **G.F. Maciel:** Conceptualization, Funding acquisition, Writing - original draft, Writing - review & editing.

Appendix A

The Fuzzy ARTMAP artificial neural network algorithm presented as follows is based on the works of (Carpenter et al., 1992a), (Carpenter and Grossberg, 1992), (Carpenter et al., 1992b), (Carpenter and Ross, 1995) and (Müller, 2018).

Steps of the algorithm:

I. Choose the parameters

Chosen parameters of the network ($\alpha > 0$): α

Training rate [0, 1]: β

Vigilance parameters of modules ARTa, ARTb, InterART and Prediction, respectively (0, 1]:

ρ_a, ρ_b, ρ_{ab} and ρ_d

Increment of the vigilance parameter: ε

II. Input data treatment

Read input and output vectors, respectively:

$Ia = [a_1, a_2, \ a_3, \ldots, \ a_n]$ and $Ib = [b_1, \ b_2, \ b_3, \ldots, \ b_n]$

Normalization of the input and output vectors, respectively:

$\overline{I_a} = Ia/|Ia|$ and $\bar{I}_b = Ib/|Ib|$

where: $|I_a| = \Sigma i|I_{ai}|$ and $|I_b| = \Sigma i|I_{bi}|$

III. Weight initialization

$w_j^a = 1$; $w_k^b = 1$ and $w_j^{ab} = 1$

Value equal to 1 shows that there is no active category.

IV. Complement of the input and output vectors, already normalized

$I_a = [\bar{a} \ \bar{a}c]$ and $I_b = [\bar{b} \ \bar{b}c]$

V. Module ARTb

Choose of category (K):

$T_k^b = \left| \ I_b \ \wedge \ w_k^b \right| / (\alpha + \left| \ w_k^b \right|)$

$T_K^b = \max\{T_k^b : k = 1, \ \ldots, \ N_b\}$

Vigilance Test:

$|x_b| = \left|I_b \wedge w_K^b\right| / |I_b| \geq \rho_b$

If the vigilance criterion is not attained, there is the Reset: $T_K^b = 0$

If attained, there is resonance and the weights are adapted and the activity vector in F2 is calculated:

$w_K^{new} = \beta(I_b \wedge w_K^{old}) \ + (1 - \beta) w_K^{old}$

$y_{Kk}^b = 1; \ \ y_{Kk}^b = 0 \ \ k \neq K$

Inter-ART module follows

VI. Module ARTa:

Choose of category (J):

$T_j^a = \left| \ I_a \ \wedge \ w_j^a \right| / (\alpha + \left| \ w_j^a \right|)$

$T_J^a = \max\{T_j^a : j = 1, \ \ldots, \ N_a\}$

Vigilance Test:

$|x_a| = \left|I_a \wedge w_J^a\right| / |I_a| \geq \rho_a$

If the vigilance criterion is not attained, there is the Reset: $T_J^a = 0$

If attained, there is resonance and calculate the activity vector in F2:

$y_{Jj}^a = 1; \ \ y_{Jj}^a = 0 \ \ j \neq J$

Inter-ART module follows

VII Inter – ART module

Vigilance Test– Match Tracking

$|x_{ab}| = \left|y^b \wedge w_J^{ab}\right| / \left|y^b\right| \geq \rho_{ab}$

If the vigilance criterion is not attained, there is the Reset and the increment of the vigilance parameter, respectively:

$T_J^a = 0$

$\rho_a = \left|I_a \wedge w_j^a\right| / |I_a| \ + \ \varepsilon$

If attained, there is resonance and the weights of module ARTa and Inter-ART are adapted, respectively:

$w_J^{new} = \beta(I_a \wedge w_J^{old}) + (1 - \beta) w_J^{old}$

$w_{JK}^{ab} = 1 \ w_{jk}^{ab} = 0$ if $j \neq J$ and $k \neq K$

VIII. End of training

If all training pairs are processed, the training ends and prediction follows, otherwise:

$\rho_a = \rho_a^{original}$

Back to modules ART and the processing continues.

IX. Prediction

Read input vectors

$Id = [d_1, \ d_2, \ d_3, \ldots, \ d_n]$

Normalization of input vectors

$\overline{I_d} = Id/|Id|$

where: $|Id| = \Sigma i|Idi|$

Choose the category (L):

$T_l^d = \left| \ I_d \ \wedge w_l^a \right| / (\alpha + \left| \ w_l^a \right|)$

$T_L^d = \max\{T_l^d : l = 1, \ldots, N_d\}$

Vigilance Test:

$|x_d| = |I_d \wedge w_L^a| / |I_d| \geq \rho_d$

If the vigilance criterion is not attained, there is the Reset: $T_L^d = 0$

If attained, there is resonance and the activity vector in F2 is calculated:

$y_{Ll}^d = 1; \; y_{Ll}^d = 0 \; l \neq L$

Calculus of the activation vector yb and determine the active category (K):

$y^b = y^d * w^{ab}$

$K = \max\{y_i^b : i = 1, \ldots, N_b\}$

Determination of the output pattern related to the input vector Id:

$Sd = w_K^b$ (without the complement)

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Water tariff forecasting models applied to municipal and private companies in the south and southeast regions of Brazil

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Abstract This paper has as a main goal to evaluate how models of the forecast will work with a group of variables that were selected based only on their correlation with the average tariff variation. Two forecast models are used, the first based on multiple linear regression techniques and the second based on the application of artificial neural networks (perceptron). We intend to use those models to reach the current water tariff based on the historic variation of the charge and the selected variables applied to municipal and private companies that operate water supply and wastewater systems in the South and Southeast regions of Brazil. The subsidiary data for the elaboration of the models were obtained through the National Sanitation Information System (SNIS). The obtained results indicated that the forecasting processes, in both models used, were able to forecast with high accuracy the fees, and guaranteed the maintenance of the surplus for the analyzed systems.

Keywords Water tariff · Artificial neural network · SNIS, forecast models

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Introduction

Basic sanitation, in Brazil (2007), represents the set of infrastructures, operating facilities, and services of:

1. Drinking water supply
2. Sanitary sewage and treatment
3. Urban cleaning and solid waste management
4. Stormwater drainage and management

The Brazilian Ministry of Cities, in 2010a, b, expected to reach universal sanitation in 2020, since the annual investment, since 2000, was around 0.45% of gross domestic product (GPD). However, investments in this period totaled approximately 0.2% of GDP. The same Ministry, in 2014, projected that the universalization will be achieved in 2033, provided that the investment reaches about R$304 billion in a 20-year horizon (de Araújo and Bertussi 2018).

The universalization of sanitation in Brazil can result in (i) a decrease in the number of hospitalizations, resulting in savings of R$27.3 million annually; (ii) 15.5% of reduction in mortality from gastrointestinal infections; (iii) reduction in the number of leaves of work, avoiding a loss of R$258 million per year; (iv)

gain in payroll, resulting in payroll growth of R$105.5 billion per year; (v) a long-term increase in the payroll of around R$31.6 billion per year, due to an improvement in productivity due to a decrease in education backlog; (vi) real estate appreciation around R$178.3 billion; and (vii) increase in the number of workers in the tourism sector, generating R$7.2 billion per year in salaries. Totaling a value of approximately R$323 billion (de Araújo and Bertussi 2018).

A review of the literature shows that the focus of researches in basic sanitation, in Brazil, have been guided to the operational efficiency of the systems (Fonseca and Gabriel 2015), (de Oliveira Pereira et al. 2016), (da Silva Carvalho et al. 2014), (de Lourdes Neto 2003).

Within this framework we asked, besides the demand for investments in the implementation of new systems, what is the financial outlook for systems operating in Brazil?

Federal Law n°11.445 established guidelines of great importance for the exercise of the regulatory function (de Araújo and Bertussi 2018). Articles 21 and 22 stand out:

I. Decision independence
II. Technicity and objectivity of decisions
III. Establishment of standards for the adequate provision of services
IV. Guarantee of compliance with the established conditions and goals
V. Prevention and repression of the abuse of economic power
VI. Tariff setting through mechanisms to induce efficiency and effectiveness of services

According to Chap. VI, of Law n°11.445, "sanitation services will be remunerated for their operation." In this way, the tariff is defined as the instrument whose objective is to ensure that sanitation service providers receive remuneration, generating revenue that, at last, covers your expenses. From the consumer's point of view, the tariff is the amount paid for the service.

It is worth mentioning that the tariff refers to the amount paid for the service, and not for the natural resource, in this case, water (Checco 2017).

According to the guidelines defined by the regulatory decree of the National Basic Sanitation Plan (PNSB) (Bazil 2010), there are advances in establishing that tariffs must guarantee:

I. Priority for the fulfillment of essential public health functions
II. Increased access of low-income citizens and localities to services
III. Generation of the necessary resources for the investments, aiming at the accomplishment of the goals and objectives of the plan
IV. Inhibition of superfluous consumption and waste of resources
V. Recovery of costs incurred in providing the service on an efficient basis
VI. Adequate remuneration of the capital invested by the contracted service providers
VII. Encouraging the use of modern and efficient technologies, compatible with the required levels of quality, continuity, and safety in the provision of services
VIII. Encourage the efficiency of service providers

Such tools indicate that the tariff is directly linked to the improvement of the national sanitation framework.

The public service regulation instruments have historically been guided by economic aspects to the detriment, in most cases, of the public and social interest (Brazil 2014). However, the experience of regulation in the basic sanitation sector has not reached the institutional level, the legal-administrative qualification and the necessary technical performances (Brazil 2014).

For public sanitation services, in Brazil, there is no specific methodology that defines the amount to be charged for the service. There is a set of guidelines and legal principles that are associated with specific rules and regulations established by the holder, define the service's financing tool. These guidelines can best be explained in (Brazil 2014).

According to data from SNIS, for the period 2013–2017, when direct operating revenues and total expenses with services are observed, the findings are not optimistic, as shown in Fig. 1. It is possible to verify that the companies that exclusively attend water supply or wastewater treatment, in the mentioned period, presented revenue values lower than the total system expenses. For systems that operate simultaneously, water supply and wastewater, the situation is reversed.

According to the Brazilian Institute of Geography and Statistics (IBGE), in 2010, the southern and southeastern regions of Brazil represent 60.7% of the country's total population. It represents 115.7 million inhabitants. The southern region consists of the states of Rio Grande do Sul (RS), Paraná (PR), and Santa Catarina (SC). By the way, the southeast regions are composed of the states of São Paulo (SP), Rio de Janeiro (RJ), Minas Gerais (MG), and Espírito Santo (ES)[1].

Predictions can be defined as "a methodological process for determining future data based on statistical, mathematical or econometric models or subjective models supported by a clear and previously defined working methodology." Forecasting methods can be divided into two groups: qualitative and quantitative. The first make predictions based on opinions and feelings, representing, in essence, a subjective approach. While the second analyzes the data through statistics so that they can make predictions and even allow different experts to obtain identical predictions. Quantitative methods are subdivided into the following: time series and causal models. The first subgroup examines the behavioral pattern of a phenomenon over time and uses the analysis to predict the future behavior of that phenomenon. The second subgroup sometimes called explanatory, calculates the predictions through the elaboration of econometric models (Devore 1972).

The main methodological tools used in the forecasting models are as follows (Devore 1972): arithmetic moving average, exponential smoothing means, trend and seasonality models, ARIMA models (Box-Jenkins), linear regression models (Nnamoko and Korkontzelos 2020), (Qi et al. 2006), models econometric and, more recently, artificial neural networks (Sun et al. 2017), (Wang et al. 2017).

Modeling through artificial neural networks has been consolidated as one of the most used data processing techniques, due to the performance of the computer that allows the processing of large databases (Devore 1972).

Therefore, the forecast should be an auxiliary tool for decision making, not an end in itself, because it assumes that the future forecast will be a projection of the available past data.

From this point of view, it is believed that an approach focused on the study of self-sufficient systems could be developed and used as the following direction for the next studies on charges for water supply and sewage treatment systems. Then, a simplified analysis of the variation of the indicators of the systems, face the average tariff practiced, and the effects of the application of these indicators in the forecast models found in correlated bibliographies are proposed.

[1] In order to facilitate comparison and reduction of the vertical scale of the graph, the values for water and wastewater providers simultaneously represent 10% of the actual value.

Material and methods

Data collect

It was defined that all the data used for the composition of the primary database would be extracted from the SNIS "Historical Series App" (Brazil 2019), from 1995 to 2017.

For a better representation and to avoid the occurrence of a false indication of outliers, as a result of distortions due to a high population amplitude among the verified cities, a population stratification criterion was adopted, closely as described in (Hamdan 2016).

- Class I: less than 10,000 inhabitants
- Class II: between 10,000 and 25,000 inhabitants
- Class III: between 25,000 and 50,000 inhabitants
- Class IV: between 50,000 and 100,000 inhabitants
- Class V: over 100,000 inhabitants

The interquartile range analysis technique was used to identify possible outliers and to minimize the spread of them, for each of the variables, based on the available data, for each system (Nnamoko and Korkontzelos 2020), (Qi et al. 2006), (Sun et al. 2017), (Wang et al. 2017).

It is important to explain that for the forecasting models, the database is divided into two groups: diagnostic and learning. The first one is composed of the current indicators available on the SNIS, for the most recent year. The second group is composed of the past years.

Finally, it was proposed to group the systems into three subgroups, according to the relationship between their direct and indirect revenues and total expenses, in the year of 2017:

I. Direct revenue ≥ total expense (scenario 01)—allowed data, on the training database, where direct revenue was lower than total expenses
II. Direct revenue ≥ total expense (scenario 02)—not allowed data, on the training database, where direct revenue was lower than total expenses

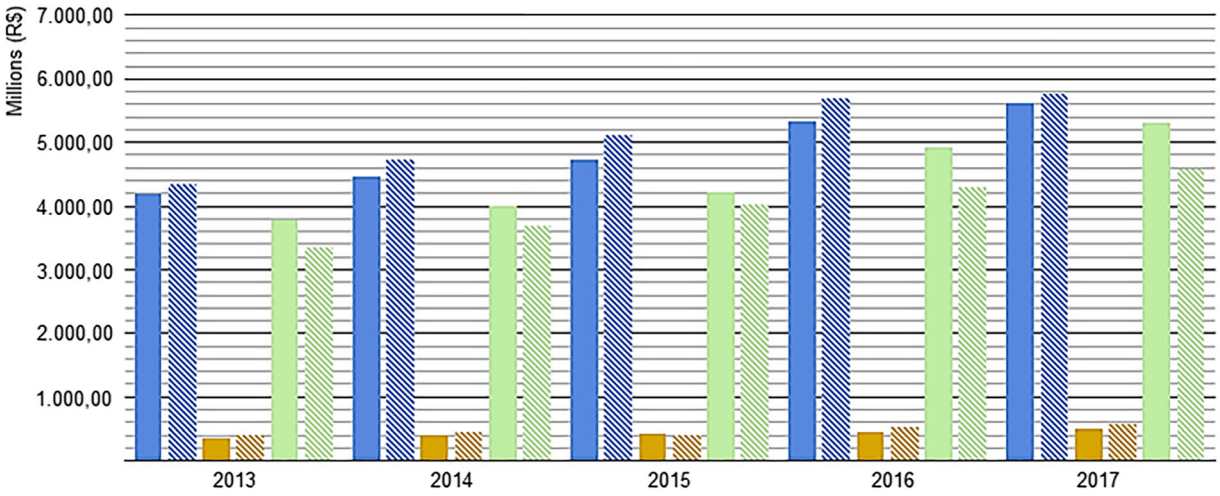

Fig. 1 Total revenue and expenses of systems operating between 2013 and 2017 in Brazil (source: Own Author, 2019)"The concept of association, represented by the correlation coefficient (r), is fundamental in the regression analysis, to describe the relationship between two variables." Also, when changes in one variable are associated with changes in the other variable, it is said that these two variables are correlated (Devore 1972)

III. Total revenue ≥ total expense (scenario 03)—disregarded annual records for a given system whose total revenue was less than the total expenses within the training database.

This subclassification was proposed to verify the effect of indirect revenue on the financial performance of the systems and its impact on the selection of variables.

Availability of information

Observations that present a certain distance from others, in some cases, can be excluded from the information base are called outliers (Hamdan 2016). Besides, to outliers, there is the occurrence of unfilled data by service providers in the city.

Table 1 Total distribution of historical records for information on water supply and wastewater systems, 1995–2017 (source: Own Author, 2019)

State	Local authority	Private company
PR	226	20
RS	138	7
SC	241	13
ES	283	20
MG	1072	25
RJ	88	197
SP	1259	169
Total	3307	451

It proposes to verify what was called analytical data index (IDA), for each city (Hamdan 2016). This index is given by Eq. 1:

$$IDA = 1 - \frac{Q.D.O. + Q.D.N.P.}{T.D.} \quad (1)$$

where *Q.D.O.* is the outlier data, *Q.D.N.P.* is the unfilled data, and *T.D.* is the total data.

The average value of the IDA was calculated, based on the IDA values obtained for each variable from every city, of all systems, for the same legal class of the provider. Thus, we defined the total of variables, whose data filling was the most representative and reliable, thus reducing the number of variables to be applied for the composition of the training database.

The variables that best explain a specific system may not be adequate to describe another. This statement reinforces the need to segment the systems according to the size of the population of the city served, as well as the legal class of the provider (Florêncio 2016).

The correlation analysis was applied between each of the variables and the "average tariff" (IN-004), according to the three scenarios described.

All selected variables had a "strong" correlation coefficient (*r*), according to the proposed classification (Kow et al. 2020), as long as they were common to all stratifications within the same legal class.

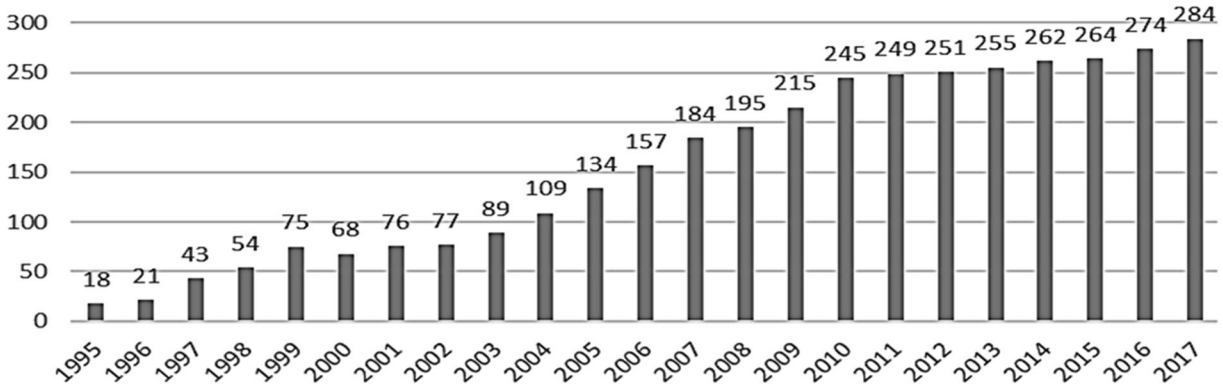

Fig. 2 Total city with information and indicators available from SNIS implementation (source: Own Author, 2019)

The ideal situation for a researcher would be to have several independent variables highly correlated with the dependent variable, but with little correlation between themselves (Hair Jr Joseph et al. 2005). Multicollinearity is the name given to the correlation between the independent variables. There are several proposed tools for solving the multicollinearity problem, for this study, it stands out "using the model with highly correlated independent variables only for the forecast, that is, not interpreting the regression coefficients" (9).

Forecasting models

After selecting the variables of interest, modeling procedures were initiated through multiple linear regression processes. For this process, the sets of variables strongly correlated with the average tariff practiced for each legal class were selected, and the regression was calculated for each municipality contained in the training database.

Linear regression analysis is defined as "a statistical technique that can be used to analyze the relationship between a single dependent variable (criterion) and multiple independent variables (predictors)" (Hair Jr Joseph et al. 2005).

The multiple linear regression analysis consists basically of performing a statistical analysis to verify the existence of a functional relationship between a dependent variable with one or more independent variables.

The decision making depends on accurate forecasts aimed at not allowing misuse of resources or elimination of opportunities (Porto and Philippi 2018). Also, neural networks have been widely used as forecasting models of the most varied demands (tourist, financial, and economic) and their forecasting performance can reach the highest levels of accuracy.

The artificial neural network used in this work was the Perceptron with backpropagation learning algorithm. To speed up the learning phase, the input data was normalized (Li and Liu 2011). For the

Table 2 Average annual IDA evolution for the selected systems between 2007 and 2011 (source: Own author, 2019)

Year		2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Local authority	Class I	83%	92%	92%	93%	94%	93%	95%	95%	94%	93%
	Class II	87%	92%	93%	94%	93%	95%	95%	95%	94%	94%
	Class III	76%	89%	90%	91%	91%	92%	92%	92%	92%	92%
	Class IV	84%	91%	90%	91%	91%	92%	92%	91%	91%	91%
	Class V	80%	83%	83%	82%	81%	81%	81%	79%	79%	78%
Private company	Class I	–	–	–	–	–	–	–	–	–	–
	Class II	83%	90%	93%	94%	96%	95%	96%	95%	94%	94%
	Class III	78%	78%	73%	78%	83%	83%	82%	92%	92%	91%
	Class IV	80%	87%	87%	90%	92%	93%	94%	93%	94%	93%
	Class V	85%	91%	91%	92%	92%	91%	92%	91%	91%	91%

Table 3 Number of systems elected to apply tariff forecasting models (source: Own Author, 2019)

	Scenario 01	Scenario 02	Scenario 03
Local authority	76	54	107
Private company	19	10	13

normalization, max normalization process was adopted (Singh and Singh 2019); this process was used to all indicators listed for each system individually, dividing all values of a given variable by the maximum value increased by 3%.

Proposal validation

When using forecast models, it is necessary to understand that we do not achieve a perfect forecast of the dependent variable; it is very useful to estimate the range of forecasted values, not relying on a single estimate (Hair Jr Joseph et al. 2005).

The MSE, according to Eq. 2, is a tool frequently used to verify the accuracy of numerical models (Kow et al. 2020) (Ciulla and D'amico 2019) and the absolute mean percentage error (MAPE) (Liu 2020) (Ciulla and D'amico 2019), presented by Eq. 3.

Table 4 Variables selected for the water supply and wastewater systems contained in scenario 01. (Source: Own Author, 2019)

	Selected indicator
Local authority	FN001—Total direct operating revenue
	FN002—Direct operating revenue from water
	FN003—Direct operating revenue from wastewater
	FN005—Total operating revenue (direct + indirect)
	IN003—Total expenditure on services per m^3 billed
	IN004—Average tariff
	IN005—Average water tariff
	IN006—Average wastewater tariff
	IN026—Operating expense per m^3 invoiced
Private company	AG001—Total population served with water supply
	AG021—Number of total water connections
	FN006—Total collection
	IN005—Average water tariff
	IN004—Average tariff

Table 5 Variables selected for the water supply and wastewater systems contained in scenario 02 (source: Own Author, 2019)

	Selected indicator
Local authority	AG002—Number of active water connections
	AG021—Number of total water connections
	ES002—Number of active wastewater connections
	ES009—Number of total wastewater connections
	FN002—Direct operating revenue from water
	FN003—Direct operating revenue from wastewater
	FN005—Total operating revenue (direct + indirect)
	FN006—Total collection
	FN010—Own personnel expense
	FN015—Operating expenses (DEX)
	FN017—Total expenditure on services (DTS)
	IN003—Total expenditure on services per m^3 billed
	IN005—Average water tariff
	IN008—Average annual expense per employee
	IN026—Operating expense per m^3 invoiced
	IN004—Average tariff
Private company	AG001—Total population served with water supply
	AG021—Number of total water connections
	FN006—Total collection
	IN005—Average water tariff
	IN004—Average tariff

$$MSE = \frac{1}{n}\sum_{1}^{n}\left(x_{real} - x_{calculated}\right)^2 \quad (2)$$

$$MAPE = \frac{1}{n}\sum_{1}^{n}\frac{\left(x_{real} - x_{calculated}\right)}{x_{real}} \times 100 \quad (3)$$

For this study, it was considered:

X_{real} = the real value of the tariff for the year 2017.

$X_{calculated}$ = the value forecasted by the proposed models.

An analysis was also used for each system, which consisted of analyzing the application of the predicted average tariff (TMP), determined by each of the methods used, on billed water volumes, exported gross water volume, treated water volume, exported, billed

Table 6 Variables selected for the water supply and wastewater systems contained in scenario 03 (source: Own Author, 2019)

	Selected indicator
Local authority	FN001—Total direct operating revenue
	FN002—Direct operating revenue from water
	FN003—Direct operating revenue from wastewater
	FN005—Total operating revenue (direct + indirect)
	FN006—Total collection
	FN015—Operating expenses (DEX)
	IN003—Total expenditure on services per m^3 billed
	IN005—Average water tariff
	IN006—Average wastewater tariff
	IN026—Operating expense per m^3 invoiced
	IN004—Average tariff
Private company	AG001—Total population served with water supply
	AG021—Number of total water connections
	FN006—Total collection
	IN005—Average water tariff
	IN004—Average tariff

sewage, and imported primary sewage volume, respectively, "AG011," "AG017," "AG019," "ES007," and "ES013" at SNIS database. To better understand the definition of the variables, all the information is available at (de Oliveira Pereira et al. 2016).

Results

The first filter selection adopted to deal with the definition of the region and the legal classes to be considered for the study. The database was reduced to the systems from the south and southeast regions and the two studied legal classes, according to Table 1.

Due to the number of systems listed, the use of the entire time interval would require substantial work, due to the high amount of data to be analyzed. Figure 2 shows the total number of cities whose accession to the system has been found for the transfer of data on information and indicators, for the region under study, since the implementation of the system with the beginning of the release of annual reports in 1995.

According to Table 2, it was evident that the choice of the period from 2011 to 2016 presented an IDA value of over 90% for most of the listed categories. The exceptions are 2011, 2012, and 2013 for private companies operating in a municipality with a population between 25,000 and 50,000 inhabitants and also for those operating in a city with a population of over 100,000 inhabitants, in all years observed.

Thus, the databases were finally defined as follows: (1) training bench: containing the values referring to the variables selected for each system, in each scenario studied, between 2011 and 2016; (2) diagnostic database: containing the values referring to the variables selected for each system, in each scenario studied, in 2017.

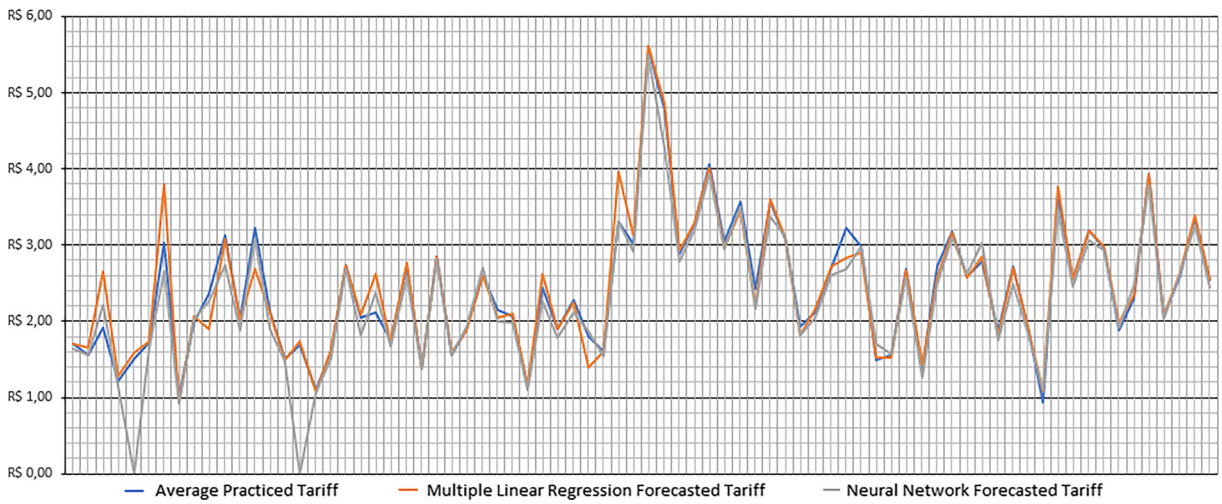

Fig. 3 Comparison of forecasted tariffs for municipal authorities under scenario 01 (source: Own Author, 2019)

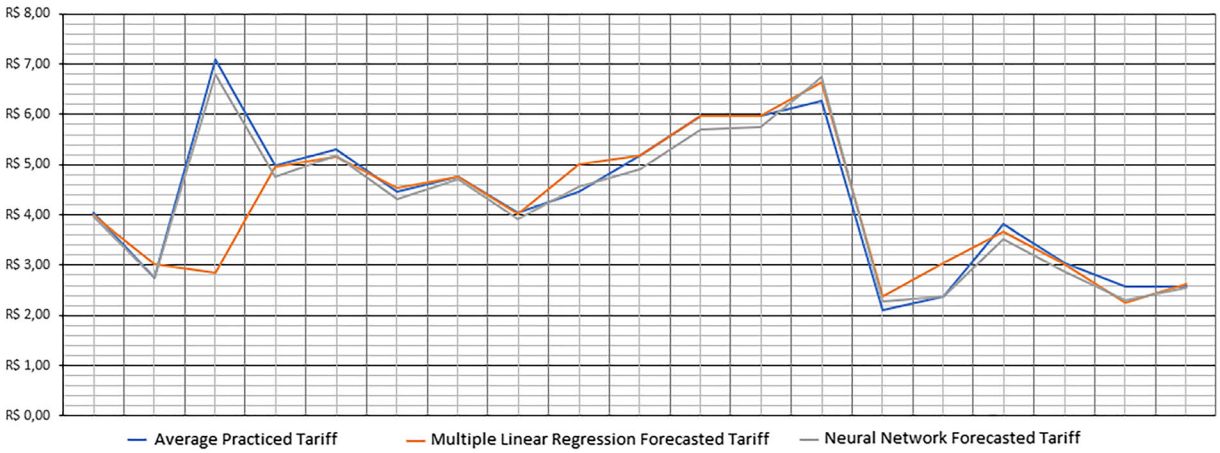

Fig. 4 Comparison of forecasted tariffs for private companies under scenario 01 (source: Own Author, 2019)

Finally, applying the proposed methodology for grouping and selecting systems, we obtained the set described in Table 3.

Applying the correlation analysis, as proposed, to all systems presented in Table 3, respecting the groupings, we obtained the following variables to apply the proposed forecasting models, as shown in Tables 4, 5, and 6.

The removal of historical records whose revenue/expenditure ratio showed a financial deficit (scenario 02), reflected in the local authority class, reducing the

Table 7 Direct revenue obtained by the application of the forecasted tariffs to private companies (source: Own Author, 2019)

	Direct revenue by multiple linear regression	Direct revenue by neural network	Indirect revenue—FN004 (year 2017)	Total expenses with the service—FN017 (DTS) (year 2017)
Baixo Guandu	R$6.105.694,67	R$5.865.367,49	R$140.844,92	R$6.110.693,58
Colatina	R$25.504.117,89	R$24.022.136,57	R$2.026.504,47	R$22.130.694,28
Guaçuí	R$4.213.558,87	R$3.505.384,82	R$	R$2.407.068,00
Jerônimo Monteiro	R$1.739.363,04	R$1.553.613,65	R$41.750,02	R$1.214.163,19
Linhares	R$28.587.598,68	R$27.118.763,54	R$1.242.465,50	R$19.591.420,50
Marataízes	R$11.959.668,93	R$8.332.138,60	R$451.096,65	R$7.911.080,29
Mimoso do Sul	R$2.215.173,34	R$2.180.545,69	R$35.881,82	R$2.153.574,22
Aimorés	R$5.476.794,30	R$5.393.172,52	R$176.827,00	R$4.600.504,00
Barbacena	R$24.901.185,79	R$22.106.106,84	R$930.332,79	R$21.191.677,73
Cambuí	R$5.489.866,04	R$5.136.326,55	R$412.765,51	R$3.766.681,92
Campo Belo	R$10.339.161,17	R$11.865.924,51	R$696.529,12	R$8.081.052,31
Campo do Meio	R$1.720.673,40	R$1.548.224,36	R$	R$1.512.888,46
Carangola	R$6.427.488,50	R$6.413.772,81	R$196.954,21	R$6.036.245,84
Elói Mendes	R$3.600.838,72	R$3.571.998,29	R$675.166,92	R$3.269.974,64
Formiga	R$12.478.570,34	R$11.593.416,52	R$2.724.572,50	R$9.283.654,14
Governador Valadares	R$77.765.494,16	R$76.342.936,88	R$6.136.374,83	R$76.715.505,20
Guanhães	R$5.263.220,10	R$4.628.672,63	R$541.022,36	R$4.951.581,05
Itabira	R$33.493.956,66	R$30.219.919,15	R$1.208.728,77	R$21.244.052,36
Lagoa da Prata	R$12.890.401,32	R$12.597.823,09	R$977.299,35	R$7.168.749,36

Table 8 Mean values of the mean square error (MSE) calculations for the systems that fit the proposed classifications (source: Own Author, 2019)

Multiple linear regression						
Class		I	II	III	IV	V
Local authority	PR	0.000051		0.023553	0.019156	
	RS					0.000209
	SC	0.000967	0.004627	0.000418	0.000086	0.000130
	ES		0.002459	0.038511		0.001653
	MG	0.017489	0.000868	0.000386	0.005328	0.007632
	RJ					0.001409
	SP		0.000002	0.003569	0.000146	0.001422
	Average	0.006169	0.001989	0.013287	0.006179	0.002076
Private company	PR					
	RS					
	SC				0.002720	
	ES					0.000148
	MG		0.009274			
	RJ			0.168353	0.000000	0.002991
	SP		0.075264		0.007627	0.006015
	Average		0.042269	0.168353	0.003449	0.003051

Table 9 Mean values of the mean square error (MSE) calculations for the systems that fit the proposed classifications (source: Own Author, 2019)

Artificial neural network						
Class		I	II	III	IV	V
Local authority	PR	0.001217		0.002407	0.002259	
	RS					0.005429
	SC	0.000318	0.006187	0.000999	0.002765	0.000875
	ES		0.002475	0.009577		0.001024
	MG	0.001240	0.005246	0.002347	0.002461	0.005502
	RJ					0.001146
	SP		0.001616	0.005597	0.003245	0.003744
	Average	0.000925	0.003881	0.004185	0.002683	0.002953
Private company	PR					
	RS					
	SC				0.004575	
	ES					0.000409
	MG		0.000024			
	RJ			0.001415	0.002299	0.000791
	SP		0.000000		0.006679	0.004157
	Average		0.000012	0.001415	0.004518	0.001786

Table 10 Mean values of mean percentage error (MAPE) calculations for systems that fall within the proposed classifications (source: Own Author, 2019)

Multiple linear regression						
Class		I	II	III	IV	V
Local authority	PR	0.7381%		11.4188%	9.6538%	
	RS					1.3274%
	SC	3.2022%	4.5276%	0.4514%	0.9545%	1.1739%
	ES		5.3440%	13.7874%		3.2043%
	MG	8.4375%	1.2503%	0.1047%	1.8242%	4.7472%
	RJ					3.8669%
	SP		0.1452%	4.6331%	0.7599%	1.2301%
	Average	4.1259%	2.8168%	6.0791%	3.2981%	2.5916%
Private company	PR					
	RS					
	SC				5.8444%	
	ES					1.2520%
	MG		9.9193%			
	RJ			28.9202%	0.0276%	1.7278%
	SP		28.2573%		6.4591%	4.1009%
	Average		19.0883%	28.9202%	4.1106%	2.3602%

Table 11 Mean values of mean percentage error (MAPE) calculations for systems that fall within the proposed classifications (source: Own Author, 2019)

Artificial neural network						
Class		I	II	III	IV	V
Local authority	PR	3.5932%		1.0120%	3.4962%	
	RS					6.1294%
	SC	1.8373%	6.7240%	3.3577%	5.4162%	3.0475%
	ES		5.5898%	2.8697%		2.4546%
	MG	3.6276%	6.0886%	3.4059%	3.8115%	1.2436%
	RJ					3.4865%
	SP		4.1409%	1.9162%	0.5819%	2.6190%
	Average	3.0194%	5.6358%	2.5123%	3.3265%	3.1634%
Private company	PR					
	RS					
	SC				7.5798%	
	ES					2.0834%
	MG		0.5071%			
	RJ			3.8522%	4.9266%	1.9305%
	SP		0.0379%		8.0291%	0.3164%
	Average		0.2725%	3.8522%	6.8452%	1.4434%

number of explanatory variables to be used in the model forecasts. In contrast, for private companies, on the other hand, they presented the same group of explanatory variables for all three observed scenarios.

Forecast results

From the application of the forecasting models, the best results were obtained for scenario 01, for both legal classes analyzed. For this reason, only these results will be displayed in Figs. 3 and 4. The results from all scenarios are available in de Oliveira Bezerra (2019). Table 7 shows the direct revenues obtained, for private companies, as a result of the application of the forecasted tariff from both processes. The results from all scenarios are available in de Oliveira Bezerra (2019).

Conclusions

The conclusions are derived from the analyses based only on the present study, meeting the criteria imposed by the methodology employed.

Organization and selection of data

When analyzing the annual IDA values, it is noted that the reliability of the data presented has grown considerably over the years, for both legal classes studied. These results can be explained by public policies that conditioned the transfer of government funds to compliance with the SNIS system. It also shows an increase in technicality operations in the sector.

The selection among SNIS's 225 variables, using the correlation criterion as a tool, presented a remarkable difference between the two legal classes studied, in all financial scenarios analyzed. It was expected that there would be a difference between the variables selected according to the legal class. However, when observed, in the case of the local authority, a differentiation is evident between the indicators selected according to the financial scenario adopted, with a difference in the management and monitoring models of the indicators of the systems operated by them. As warned by (de Farias et al. 2016), the use of highly correlated variables is not indicated for explanatory models. In this case, we suggest other ways of selecting indicators for this purpose and analyzing their application in the two forecasting models proposed.

Performance—multiple linear regression and artificial neural network

Based on the results obtained for the mean square and percentage errors, as shown in Tables 8, 9, 10, and 11, both models can give good result for the tariff forecast. As shown in Fig. 3, there are two predictions made that results in the null tariff. This occurrence represents one of the main problems related to the neural network forecasts model called "paralysis of the algorithm." It is important to notice that the same variables used in the regression approach did not cause an issue.

From the perspective of mean square error (MSE), the mean value for the artificial neural network is lower than the same indicator obtained through the regression model. When observing the mean percentage errors (MAPEs), it appears that the artificial neural network presented an average value of the referred error, lower than the value obtained through the multiple linear regression model, indicating less likely error due to their forecasts. It is noted that the errors found for the forecasting processes performed by linear regression were much higher than those found for the results obtained through the neural network. Due to the artificial neural network ability to generalize nonlinear data, as reported in (Silva and Schimidt 2016), the low error is expected.

As shown in Table 7, the forecasts made by the neural network resulted, generally, in a direct revenue lower than those resulting from the forecasts made by the multiple linear regression. However, these revenues were sufficient to cover all costs of services, reinforcing the concept that models should be seen as an auxiliary tool for the system operator.

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TECHNICAL PAPER

Damage diagnosis in an isotropic structure using an artificial immune system algorithm

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Abstract
This work proposes a recent methodology for developing structural health monitoring based on intelligent computing techniques, with the purpose of detecting structural damages in aircrafts using artificial immune systems with negative selection. To assess this methodology, an experimental setup was built with piezoelectric transducers attached to an aluminum plate (which represents a wing of an airplane), which work both as actuators and sensors, where signals were acquired in normal and damage situations. The results show robustness and accuracy for the new methodology proposed.

Keywords Structural health monitoring · Artificial immune systems · Negative selection algorithm

1 Introduction

Structural health monitoring (SHM) consists of detecting damages in their initial stages, intervening on their propagation and preventing the structure from stopping or getting damaged. The inherent challenge for applying SHM is to identify and analyze the types of changes in the structure, and the architecture of this system depends on the characteristics of the structure, which can be defined in several ways.

SHM must meet certain requirements, such as: perform data acquisition and processing, signal validation and analysis, damage identification and characterization; interpret harmful changes in the structure; and help in decision making [1]. It is a process under development, which currently seeks to level up from controlled tests in laboratories to applications based on factors as: results of predictive models, experimental and environmental conditions, among others.

Thus, the basic objective of any SHM system is to detect, in real time or not, structural damage. However, according to [2] a complete and advanced SHM system consists of five stages, which are as follows:

- Detect the existence of damage;
- Locate it in the structure;
- Identify your type;
- Evaluate its extent;
- Determine the remaining life time of the structure.

The phases for implementing an SHM system are: In the first phase, acquisition of data or signals emitted by the structure to be monitored is carried out; in the second phase, it is necessary to work on the raw data; and the third phase consists of transmitting the data to the proposed system, which will interpret them. During the process of damage diagnosis, the system analyzes the data and then indicates whether the structure presents damages or not [3].

The technique applied for monitoring a structure depends on the constructive configuration, the experimental environment, and the type of structure, such technique is based on nondestructive techniques tests, named NDE. Among them, we may cite: magnetic particle inspection, Eddy currents, deformation analysis, techniques using optical fibers,

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acoustic emissions, comparative vacuum, X-rays, liquid penetrant inspection, wave propagation (Lamb waves), and electromechanical impedance [4].

Lamb waves are plane deformation waves that occur on the surface (upper and bottom) of a plate and are used for structural monitoring due to the extensive coverage of the set of sensors and actuators in big structures. Lead zirconate titanate (PZT) works both as a sensor and actuator in experimental setups, which can be attached to the structure to produce or collect Lamb waves, thus determining chemical or physical changes in the structure through wave attenuation and dispersion. PZT presents a high performance in high-frequency applications, such as the case of Lamb waves and electromechanical impedance [5]. PZT is applied in order to convert the mechanical energy resulting from vibrations of the structure into electrical energy [6].

After the signals are acquired, it is possible to process them using artificial intelligent systems, which, based on the information generated by deviations of the signals, can determine whether damages are present or not, their location and severity [4].

To develop modern and efficient SHM systems, it is necessary to use intelligent sensors and/or intelligent computing techniques (artificial neural networks [5], fuzzy logic [7], artificial immune systems [8]), which make possible to extract information about complex processes, analyze signals, as well as promote automated decision making.

In [9], a civil structural health monitoring (SHM) was proposed using artificial immune systems (AIS) for a three-story model and many damage scenarios. The methodology yielded satisfactory results in terms of detection and location of damages in many cases.

In [10], a damage detection methodology based on artificial immune systems (AIS) was presented as a technique for pattern recognition and matching portions to discriminate different structural states. The results show that the methodology could detect damages accurately.

In [11], a civil structural health monitoring (SHM) was proposed using the NDE method, artificial immune systems (AIS) and the wavelet transform for a four-story model with 120 degrees of freedom. The results obtained through computational simulations were satisfactory.

In [12], a new hybrid methodology was proposed to diagnose failures in aeronautical structures using as a tool the perceptron multi-layer artificial neural networks and ART-MAP fuzzy and the wavelet transform.

In [13], a work with a dynamic model was proposed to represent a building structure using the ARTMAP Fuzzy RNA methodology to identify faults.

The SHM is based on traditional techniques, as seen in [4], and this work aims to present a methodology for developing SHM based on intelligent computing techniques, with the purpose of detecting structural damages in aircrafts using artificial immune systems, more precisely, the negative selection algorithm. The negative selection algorithm (NSA) is suitable due to its good learning and pattern recognition skills, as well as its good performance in other types of problems related to pattern recognition and diagnosis, as highlighted by the authors [14, 15]. This work applies NSA aiming to meet levels 1, 2 and 3 in the process of SHM, which means detecting, locating and classifying different types of damage efficiently and quickly, considering the inherent dynamism of the aeronautical industry.

2 Negative selection algorithm (nsa)

The negative selection algorithm (NSA) is the main method of artificial immune systems (AIS), inspired by the proper and nonproper discrimination observed in mammals immune systems (MIS). Also, it may be considered as a branch of bioinspired computing with applications in different fields, especially in intrusion detection [16].

NSA was proposed by [17] to detect changes in states of computer systems. This technique is inspired in the process of negative selection performed by the *T* lymphocytes, which occurs in the thymus; this process represents the discrimination the body performs with its cells, dividing them into proper and nonproper. The NSA algorithm is a technique based on this process of pattern recognition performed by the biological immune system, but developed as a computational model.

The NSA algorithm presents approaches that make it different from other intrusion approaches:

- The algorithm detects unknown intrusions without any previous knowledge;
- Based on the algorithm, it is possible to adjust the detection ratio, defining the number of detectors generated;
- Each detector can detect abnormalities independently, without communication with other detectors;
- The detection is local: Each detector can detect any change in the dataset;
- The detector defined in each local may be unique, and this increases the robustness of the algorithm;
- The auto-adjustment and the set of detectors are mutually protectors [18].

The algorithm is executed in two phases, described as follows [8, 19].

- Censoring: Define the set of proper chains (*S*) that one wishes to protect; generate random chains and evaluate the match between each one of them and the proper chains. If the match is higher than a threshold previously

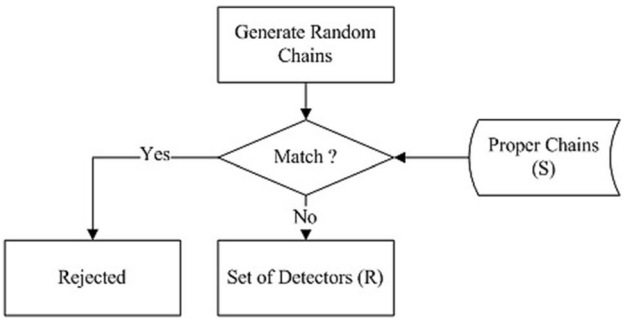

Fig. 1 Flowchart of the censoring phase of the NSA

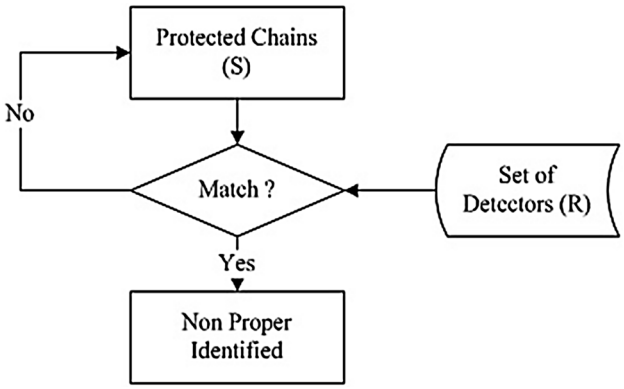

Fig. 2 Flowchart of the monitoring phase of the NSA

set, reject the chain. Otherwise, store it in a set of detectors (R);

- Monitoring: Given the set of chains to be protected, evaluate the match between each of them and the set of detectors. If the match is higher than a threshold previously set, a nonproper element is identified.

The censoring and monitoring phases of the negative selection algorithm are seen in Figs. 1 and 2, defined by the authors [8, 19].

In the censoring phase, the proper detectors are defined, which represent the normal condition of the body, known as proper chains (S), with the aim of generating a set of detector patterns (R), which can identify the nonproper pattern in the monitoring phase. From reading the data, random chains are chosen, and the match is verified by a comparison between these chains and the set of proper chains (S). If the match is higher than a threshold previously set (match ratio), this chain is rejected. Otherwise, this chain is accepted to the set of detectors (R) and will be used in the process of classification during the data monitoring. The detectors can detect any nonproper element, a modification or error in the data that will be monitored [19].

In the monitoring phase, the data are monitored aiming to identify changes in the behavior of the dataset and classify these changes using the set of detectors created in the censoring phase. The proper chains (S) are compared to the set of detectors (R) and the match if evaluated. If the match between the chains is higher than a threshold previously set, the nonproper element is identified and classified [8].

The chains in the NSA algorithm represent the patterns by means of data structures (vectors).The chains in the NSA algorithm can be classified in two types: antigens (Ag) and lymphocytes (Ab). Ag is the signal to be analyzed or unknown by NSA, and is represented by Eq. 1. The detectors, or lymphocytes (Ab), can be expressed according to Eq. 2, defined by [8, 19].

$$\mathrm{Ag} = \mathrm{Ag}_1, \mathrm{Ag}_2, \mathrm{Ag}_3, \mathrm{Ag}_4, \ldots, \mathrm{Ag}_\mathrm{L} \tag{1}$$

$$\mathrm{Ab} = \mathrm{Ab}_1, \mathrm{Ab}_2, \mathrm{Ab}_3, \mathrm{Ab}_4, \ldots, \mathrm{Ab}_\mathrm{L} \tag{2}$$

where L is the dimension of the space of lymphocytes (Ab) and antigens (Ag). It must be stressed that the censoring and monitoring phases are carried out offline and online, respectively [8].

2.1 Matching criterion and match ratio

To evaluate and verify match between the chains (proper and nonproper), as well as state whether they are similar, a criterion known as matching criterion is applied. This matching can be perfect or partial; it is perfect when two chains analyzed are the same. In this work, the concept of partial matching was chosen, that is, only a certain number of positions between the patterns must have the same value in order to confirm the matching. This number is defined as match ratio [20].

The match ratio (TAf) represents the degree of similarity necessary for match to occur between two chains under analysis. It is defined according to the following Eq. 3:

$$\mathrm{TAf} = \left(\frac{\mathrm{An}}{\mathrm{At}}\right) * 100 \tag{3}$$

where TAf: match ratio; An: number of normal chains in the problem (proper chains); At: total number of chains in the problem (proper and nonproper chains).

In the model proposed by [19], there is a deviation associated with the detector pattern, with a tolerance (plus or minus), from which it is possible to accept the combination of patterns. This deviation acts individually in each position i of the vector, thus making possible the assessment of matching at each point, as shown by Eq. 4:

$$\underline{\mathrm{Ab}_i} \leq \mathrm{Ag}_i \leq \overline{\mathrm{Ab}_i} \tag{4}$$

where Ag_i: nominal value of the position i of the antigen (pattern under analysis); $\underline{\mathrm{Ab}_i}$: nominal value of the position i minus the deviation adopted in the antibody (detector

pattern); $\overline{\text{Ab}}_i$: nominal value of the position i plus the deviation adopted in the antibody (detector pattern).

If the value of the position i of the vector is within the range of the equation, matching can be considered for such position, and it is possible to quantify the match between the patterns by analyzing each position. The measure of total match used to evaluate the patterns under analysis is expressed by the following Eq. 5, as proposed by [20].

$$\text{Af}_T = \frac{\sum_{i=1}^{L} \text{Pc}}{L} * 100 \tag{5}$$

where Af_T: of match between the patterns analyzed; L: total number of positions; Pc: position where match occurred; $\sum_{i=1}^{L}$ Pc: sum (number) of positions where match occurred.

When Af_T is higher or equal to TAf, matching occurs, and they are considered similar, that is, the signals under analysis are considered the same; therefore, it is classified as the normal condition of the structure, because it presents characteristics of the proper detectors. If Af_T is lower than TAf, the detector does not recognize the pattern, and thus matching does not occur between the signals, the signal is then classified as an abnormality, that is, a structural damage. This module returns a diagnosis of the signals analyzed based on proper (normal condition of the structure) and non-proper (structure with characteristics of damages).

In this work, during the censoring phase, it was possible to generate the set of proper detectors and damage detectors that were used for monitoring the data. The set of proper detectors was composed of signals that present characteristics of a normal (or baseline) condition of the structure, that is, without damages. As for the damage detectors, they were composed of damage signals used as detectors during the monitoring phase to characterize and detect the five different damage situations. The procedure was divided into three steps: signal reading, censoring module, and verification of matching in relation to the set of proper detectors. If the matching criterion was met, the signals were rejected, because they presented proper characteristics. Otherwise, they were stored in the set of damage detectors, which were used for monitoring the system.

The monitoring phase was divided into three steps: input module, detection module, and module of signal classification according to the five different damage situations.

3 Materials and methods

In this work, the experimental measurements obtained by [21] were used. Figure 3 shows the equipment for the experimental tests. This equipment consisted of: a sample: an aluminum plate; 9 PZT transducers, model T110-H4e-62, made by Piezo Systems, Inc.; aeronautical epoxy of the type Henkel Hysol EA 9320NA, to attach the transducers; adhesive mass of the type Hexcel, to simulate the delamination; acquisition board of the type Acellent ScanGenie II, and the software ACCESS 2.1, for signal acquisition; Protoboard MSB-400 for channels input; MATLAB®, version R2013a, as the processing software, and, at last, a climatic chamber Thermotron SM-8, as the experimental environment. In this work, the temperature variability was not taken into account.

In the experiment, the following were used: an aluminum plate whose dimensions were $500 \times 500 \times 2$ mm, 9 PZT transducers, which cover the surface of the structure, as shown in Fig. 6. The measurements used 1 PZT as the actuator and the other 8 as sensors; it was possible to cover the whole surface of the structure, as illustrated in Fig. 4.

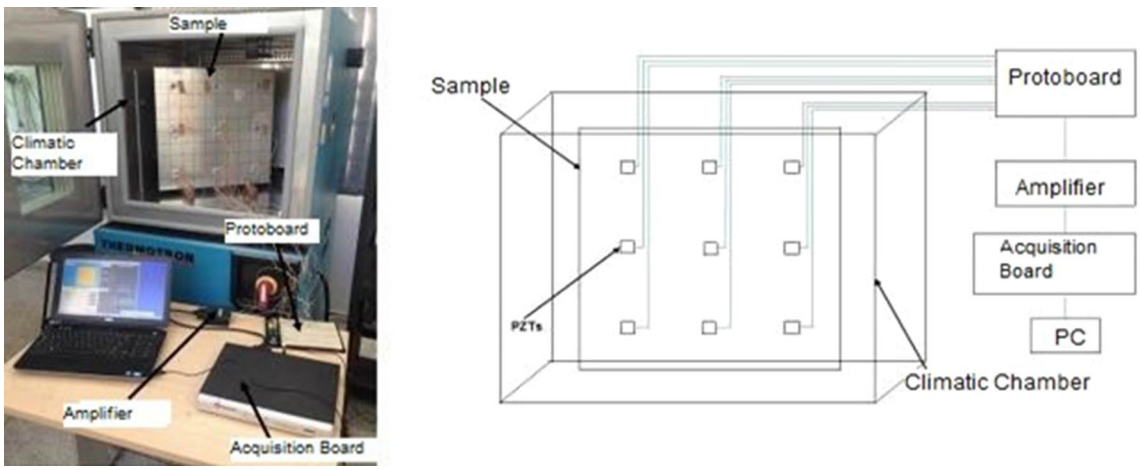

Fig. 3 Materials and equipment used in the experiment with an isotropic material, and illustration of the materials and equipment used in the experiment

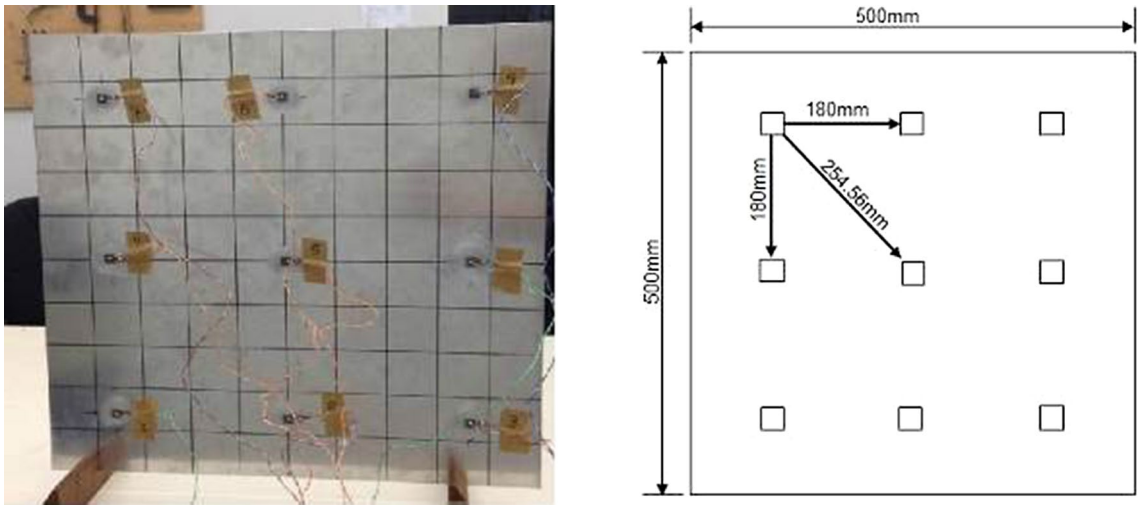

Fig. 4 Picture of the plate used in the experiment and an illustration of it showing the positioning of transducers

For the damage location analysis, five points were defined where the damages were inserted in five different situations. The damages were simulated by adding an adhesive mass of 15 g. The placement of the damages is depicted in Fig. 5. Figure 5a shows the position of damage 1, Fig. 5b, damage 2, Fig. 5c, damage 3, Fig. 5d, damage 4, and Fig. 5e, damage 5.

To identify the location of the damages in the structure, each signal generated by the Lamb waves was analyzed for each arrangement. For signal generation and acquisition, the plate Accelent, model ScanGenie II Battery Edition, was used in conjunction with the software ACCESS 2.1.

The signals were obtained at the frequencies 200, 250, 300 kHz, for five different damage situations, but the damages used for the tests consisted of the signals obtained at the frequency of 250 kHz. The excitation signal was Burst 5, that is, a sinusoidal signal windowed with 5 cycles, with a sampling frequency of 3 MHz and 1000 acquisition points for all tests. For each measurement, a mean of 10 signals was considered.

To manipulate the data, the first packet of the signals was defined, that is, only the time in which the signal reached the sensor, without reflections, as well as a time equal to the excitation time, was considered. The signals were obtained in the baseline situation (without damages), and, afterward, in the five different damage situations.

The first packet, that is, the first 81 points of the sinusoidal signal windowed with 5 cycles, was defined due to reflections that it may undergo in the structure; the rest of the signal, besides reflections, presents other wave modes that reach the sensor, thus making the analysis difficult. Afterward, graphics were generated, which clearly show when there are differences between the baseline and the actual signal (or signal to be analyzed), captured by the sensor.

Figure 6 presents the first packet of the signal P_{59}, that is, a signal emitted by the actuator 5 and received by sensor 9. This signal was chosen to show the influence of damage 4 on this signal. The red signal represents the baseline signal, and the blue one, the signal with damage.

Figure 7 shows the first packet of the signal P_{12}, that is, signal emitted by the actuator 1 and received by sensor 2. This signal was chosen to show the low influence of damage 4 on this signal, because this damage is located far from the signal chosen. The red signal represents the baseline, and the blue one, the signal with damage.

Figure 8, in turn, presents the first packet of the signal P_{68}, that is, a signal emitted by the actuator 6 and received by sensor 8. This signal was chosen to show that, although the damage is near the signal, the variation between the two signals is not expressive. The red signal represents the baseline, and the blue one, the signal with damage.

After defining the first packet for all signals, an artificial immune system was developed, more precisely, the negative selection algorithm. The SHM analysis was divided into 3 modules: data acquisition, censoring, and monitoring phases of the negative selection algorithm.

- Data acquisition: The data used were obtained by [22]. Then, they were used in the censoring and monitoring phases of the NSA algorithm;
- Censoring phase: The set of proper detectors was generated, consisting of 81 signals that present the baseline condition of the structure, that is, without damages and 405 signals that have a fault characteristic, that is, the five different damage situations as shown in Fig. 5.
- Monitoring phase: This phase was divided into three steps: The first step corresponds to the input module, in which the acquisition of data was carried out randomly

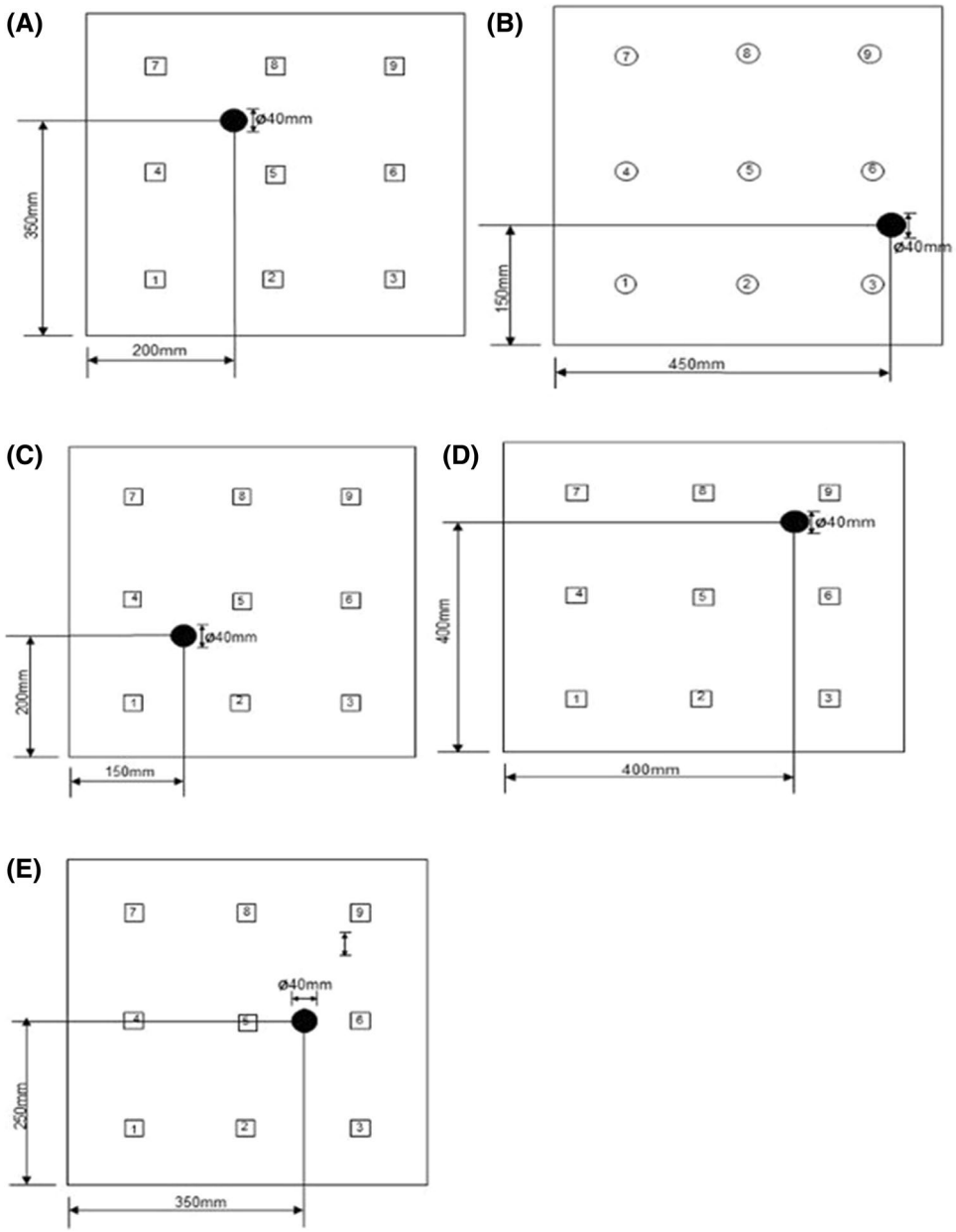

Fig. 5 Illustration of the positioning of damages: **a** damage 1, **b** damage 2, **c** damage 3, **d** damage 4 and **e** damage 5

from the dataset; the second step corresponds to the detection module, in which the signals under analysis were compared to the proper detectors (baseline and damage signals), with the purpose of identifying matching between the signals, a 1.5% deviation was used in this step to verify the signals; the third step, at last, corresponds to classifying the signals in normal and damage signals, identifying the damage situations.

Figure 9 depicts a flowchart of how the negative selection algorithm works in SHM. In this work, proper detectors and damage detectors with five different situations were defined, according to the model proposed by [15].

According to the methodology proposed by [15], it was possible to detect and locate the five different damage situations and calculate the scores of accuracy, and in the cases in which the percentage was not equal to 100%, show which signals the algorithm failed to locate the damages. With the purpose of enhancing the robustness of the methodology, it was possible to quantify the signals according to the degree of influence of the damages (low, medium, and high) for the five situations. The degree of influence is determined by Eqs. 7 and 8, which represent the influence of the damage in the signals, that is, the higher the difference in amplitude of the signals, the higher the degree of influence is. This quantification evidences how many signals were classified with low, medium, and high

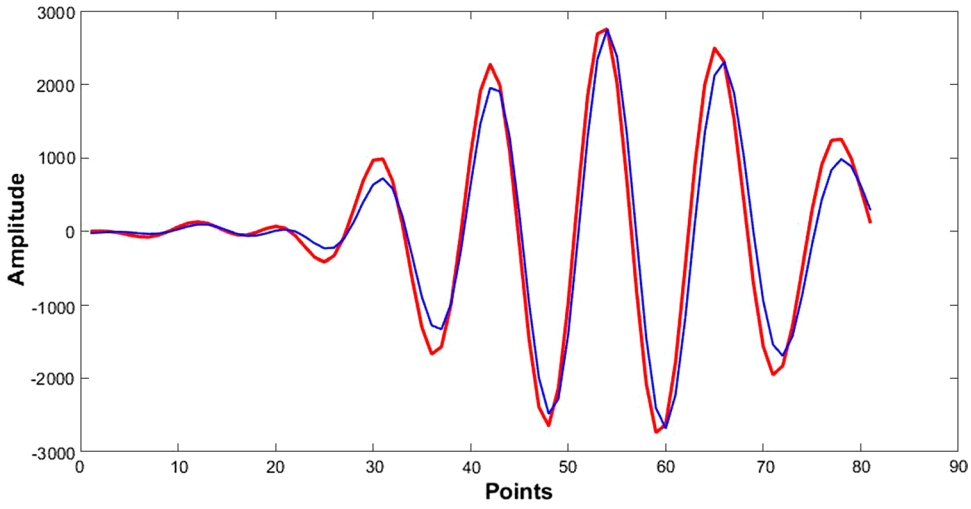

Fig. 6 First packet of the signal emitted by the actuator PZT 5 for 250 kHz and signal acquired by the sensor PZT 9 for a baseline situation and for a situation with damage 4

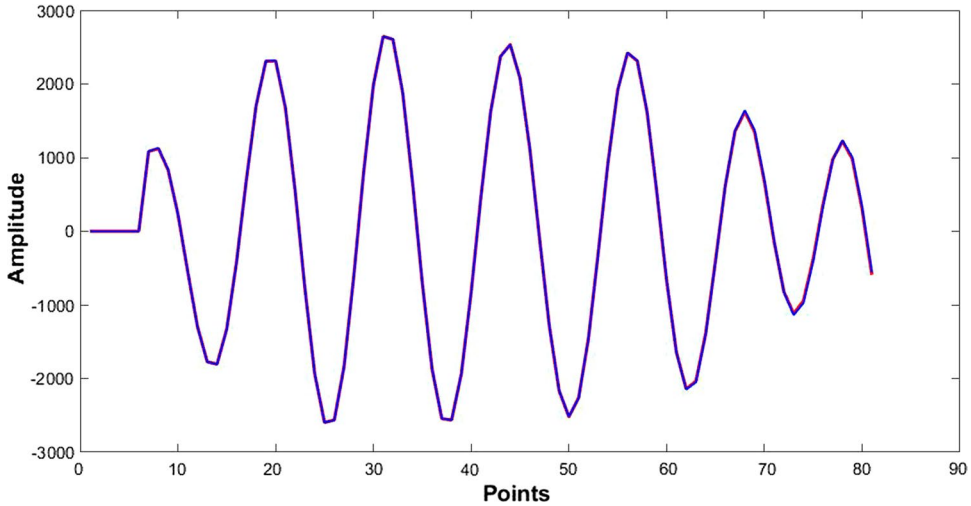

Fig. 7 First packet of the signal emitted the actuator PZT 1 for 250 kHz, and the signal acquired by the sensor PZT 2 for the baseline situation and for a situation with damage 4

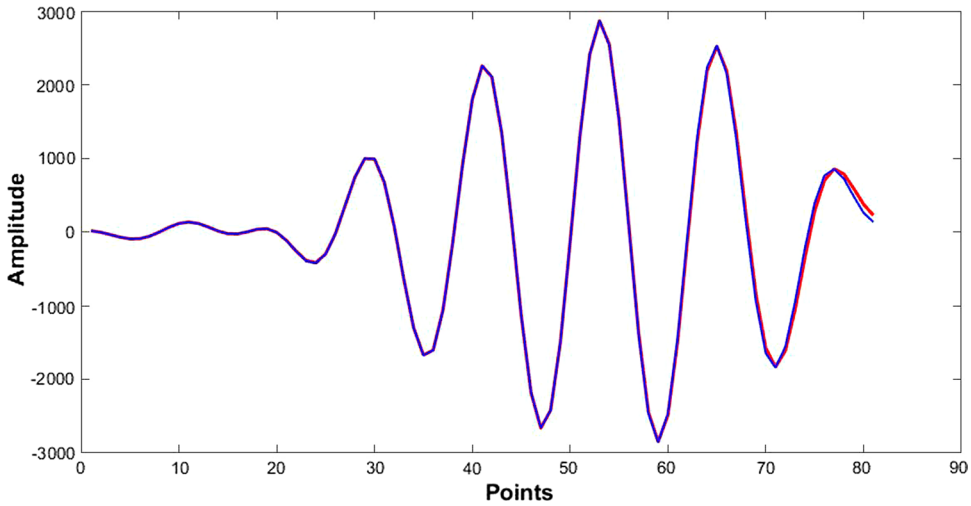

Fig. 8 First packet of the signal emitted by the actuator PZT 6 for 250 kHz and signal acquired by the sensor PZT 8 for a baseline situation and for a situation with damage 4

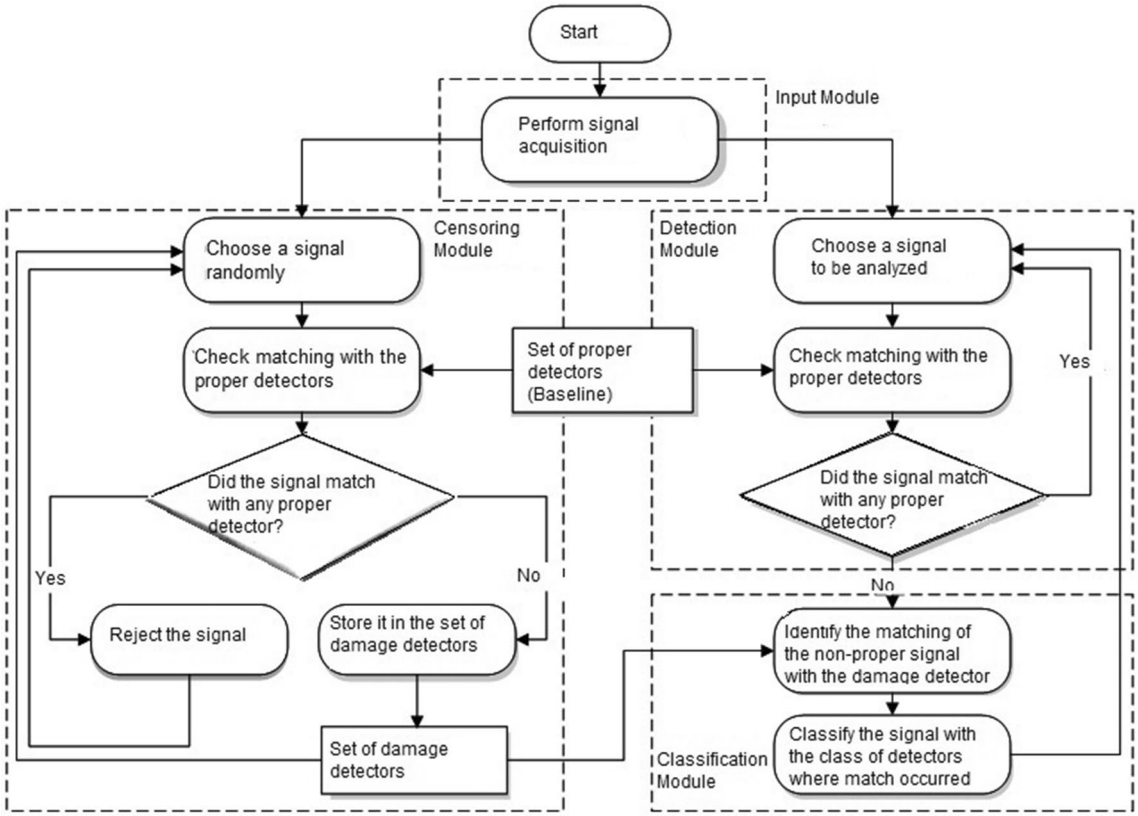

Fig. 9 Flowchart of the negative selection algorithm for SHM [15]

degrees of influence for the five damage situations, and does not account for the severity of the damage.

In this phase, it was possible to identify the score of accuracy, and in the cases in which this score was not 100%, show which signals were misclassified during the classification. Also, it was possible to classify all signals according to the degree of influence of the damages (low, medium, and high) for all situations.

For such classification, a metric named RMSD was defined for the five different damage situations. This metric provides good results for signal manipulation, as described by [22] to the following Eq. 6. The index i represents the lines, and the index j represents the columns of the normalization matrix.

$$\text{Soma} = \text{Abs}\left(\text{SD}(j,i)^2 - D(i,j)^2\right) \tag{6}$$

where Soma: sum; Abs: absolute value; SD: signal without damage; D: signal with damage.

After calculating the quadratic value of the signals, their square root was calculated, and the normalization was defined by the RMSD metric, according to the following Eq. 7:

$$Métrica = Raiz\ (Soma/81) \tag{7}$$

where Métrica: RMSD metric; Raiz: square root.

Afterward, the maximum and minimum amplitude values of the signals were defined for the five different damage situations, and then divided in three parts, as shown by Eq. 8:

$$Métrica = (Máximo - Mínimo)/3 \tag{8}$$

where: Métrica: metric; Máximo: maximum value of the RMSD metric; Mínimo: minimum value of the RMSD metric.

Therefore, Eq. 5 determines the normalization of the data by the RMSD metric, and the normalized values are used in Eq. 6 to determine the metric according to the degree of influence of damage.

4 Results and discussion

To carry out the tests in the censoring and monitoring phases, the experiment data obtained by [21] for the five different damage situations were used, which corresponds to 486 signals, 72 of which are proper (baseline signals) and 405 nonproper (damage signals).

For the negative selection algorithm to present a good accuracy in recognizing the signals, a 1.5% deviation was adopted due to the characteristics of the problem, that is, the signals are similar to each other; there are few variations.

To obtain the results, the system was subjected to a testing phase and parameter setting. The result was obtained through a cross-validation test, in which the system was executed 15 times to ensure accuracy to the results. It is possible to verify that the structural health analysis presents a good performance, with a score of accuracy equal to 99.25% in the diagnosis. With the purpose of obtaining a better analysis of the methodology proposed, three tests were carried out to verify the robustness of the negative selection algorithm in detecting damages in the system. The tests were carried out as shown in Table 1. It is possible to see that the system of structural health analysis performed in an isotropic material (aluminum plate) presents a good performance (score of accuracy equal to 99.38% in the best configuration), and that the number of detectors influences directly on the process of damage recognition. Damage detection for real applications can be obtained by collecting signals.

For each signal collected, the system created a detector in the censoring phase, that is, as it is created in the biological immune system, and afterward the location was performed in the monitoring phase, comparing the damage signals with all the detectors, as illustrated by Fig. 9, thus allowing for damage detection and location. In this phase, it was possible to detect and locate the five different damage situations, presenting the scores of accuracy, and in the cases in which the percentage was not equal to 100%, show for which signals the algorithm failed to locate the damages.

The detection of damages for application in the work of [22] was used the algorithm of propagation of affinity for the same problem where it was verified that the Learning of Unsupervised Machine needed adjustments that can be improved with algorithm of optimization. The present work had as proposal to use semi-supervised machine learning using the negative selection algorithm as supervised machine learning and the clonal selection algorithm as unsupervised machine learning, thus obtaining robust results. Actual results can be obtained with the collection of signals.

The novelty of this work consists of quantifying the degree of influence of the damages using the RMSD metric and the amplitudes of the signals, as demonstrated by Eq. 8, according to three levels: low, medium, and high. The results for the classifications of the signals concerning the degree of influence (low, medium, and high) yielded 100% of accuracy for the five different damage situations. For damage 1, 6 signals were classified with a high degree of influence, 22 with a medium degree of influence, and 44 with a low degree of influence, as shown in Table 2.

For damage 2, 3 signals were classified with a high degree of influence, 6 with a medium degree of influence, and 66 with a low degree of influence, as presented in Table 3.

For damage 3, 2 signals were classified with a high degree of influence, 1 signal with a medium degree of influence, and 69 with a low degree of influence, as shown in Table 4.

For damage 4, 4 signals were classified with a high degree of influence, 2 signals with a medium degree of influence, and 66 with a low degree of influence, as shown in Table 5.

For damage 5, 5 signals were classified with a high degree of influence, 6 signals with a medium degree of influence, and 61 with a low degree of influence, as presented in Table 6.

By using the negative selection algorithm, it was possible to meet the level 3 proposed by [3], which aims to detect, locate, and quantify damages. This methodology can be regarded as a new and robust methodology applied to SHM systems, in accordance with the results obtained in [22], but now including location and quantification of the degree of influence for the five different damage situations.

Table 2 Classification regarding the degree of influence for damage 1

Damages	Number of signals	Classification	Score of accuracy (%)
1	6	Grande	8.33
1	22	Médio	30.55
1	44	Pequeno	61.11

Table 3 Classification regarding the degree of influence for damage 2

Damages	Number of signals	Classification	Score of accuracy (%)
2	3	Grande	4.16
2	6	Médio	8.33
2	63	Pequeno	87.5

Table 1 Classification damage

Tested samples	Classification (proper)	Classification (nonproper)	Correct classification	Score of accuracy (%)	Processing time (ms)
399	69	330	396	99.25	17.29
421	71	350	418	99.29	16.86
486	81	405	483	99.38	16.66

Table 4 Classification regarding the degree of influence for damage 3

Damages	Number of signals	Classification	Score of accuracy (%)
3	2	Grande	2.77
3	1	Médio	1.38
3	69	Pequeno	95.83

Table 5 Classification regarding the degree of influence for damage 4

Damages	Number of signals	Classification	Score of accuracy (%)
4	4	Grande	5.55
4	2	Médio	2.77
4	66	Pequeno	91.66

Table 6 Classification regarding the degree of influence for damage 5

Damages	Number of signals	Classification	Score of accuracy (%)
5	5	Grande	6.94
5	6	Médio	8.33
5	61	Pequeno	84.72

5 Conclusion

In this work, a novel structural health monitoring system was developed, which is capable of detecting, locating and quantifying damages in an aeronautical structure through artificial immune systems based on the negative selection algorithm.

Based on the tests carried out, it was possible to verify, in the NSA, that the higher the knowledge obtained in the censoring phase is, the higher the efficiency in the process of recognition and classification of the algorithm in the monitoring phase will be.

The number of detectors influences directly on the process of damage diagnosis, and the cross-validation test provides reliability and speed. It was possible to obtain a score of accuracy equal to 100% concerning the location and classification of signals with damage.

Therefore, we conclude that the algorithm proposed in this work, which is based on intelligent computing techniques, is reliable, robust, and precise for structural health monitoring in isotropic structures, and that the work contributes to the field of study related to SHM, presenting a novel approach using artificial immune systems, more precisely, the negative selection algorithm.

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Forecasting the spatiotemporal variability of soil CO_2 emissions in sugarcane areas in southeastern Brazil using artificial neural networks

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Abstract Carbon dioxide (CO_2) is considered one of the main greenhouse effect gases and contributes significantly to global climate change. In Brazil, the agricultural areas offer an opportunity to mitigate this effect, especially with the sugarcane crop, since, depending on the management system, sugarcane stores large amounts of carbon, thereby removing it from the atmosphere. The CO_2 production in soil and its transport to the atmosphere are the results of biochemical processes such as the decomposition of organic matter and roots and the respiration of soil organisms, a phenomenon called soil CO_2 emissions (FCO_2). The objective of the study was to investigate the use of neural networks with backpropagation algorithm to predict the spatial patterns of soil CO_2 emission during short periods in sugarcane areas. FCO_2 values were collected in three commercial crop areas in the São Paulo state, southeastern Brazil, registered through the LI-8100 system during the years 2008 (Motuca), 2010 (Guariba city), and 2012 (Pradópolis), in the period after the mechanical harvesting (green cane). A neural network multilayer perceptron with a backpropagation algorithm was applied to estimate the FCO_2 in 2012, using data from 2008 and 2010 as training for the neural network. The neural network initially presented a mean absolute percentage error (MAPE) of 18.3852 and a coefficient of determination (R^2) of 0.9188. Data obtained from the observed and estimated values of FCO_2 present moder-

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ate spatial dependence, and it is observed from the maps of the spatial pattern of the CO_2 flow that the results from the neural network show considerable similarity to the observed data. The model results identify the higher and lower characteristics in sample points of CO_2 emissions and produce an overestimation of the range of spatial dependence (0.45 m) and an underestimation of the interpolated values in the field ($R^2 = 0.80$; MAPE = 12.0591), when compared to the actual soil CO_2 emission values. Therefore, the results indicate that the artificial neural network provides reliable estimates for the evaluation of FCO_2 from data of the soil's physical and chemical attributes and describes the spatial variability of FCO_2 in sugarcane fields, thereby contributing to the reduction of uncertainties associated with FCO_2 accountings in these areas.

Keywords Neural techniques · Soil respiration · Green harvest

Introduction

The human influence on the climate system is clear and recent levels anthropogenic greenhouse gas emissions are the highest ever (IPCC 2014). In addition, policies governing agricultural practices and the conservation and management of forests are most effective when they involve mitigation and adaptation, thus providing opportunities to reduce impacts and increase risk management. The increase of anthropogenic emissions of greenhouse gases (GHGs) is the dominant cause of global warming. Brazil, despite not having binding targets to reduce CO_2, made a voluntary commitment to reduce their emissions by the year 2020, being a signatory to several multilateral agreements to the global effort to mitigate GHGs. Unlike other countries, whose burning of fossil fuels is the main source of GHG emissions, Brazil's emissions are mainly from changes in land use and deforestation.

Globally, agriculture is responsible for significant amounts of anthropogenic emissions of the greenhouse gases, namely, CO_2, methane gas (CH_4), and nitrous oxide (N_2O), to the atmosphere, contributing 11%, 47%, and 58% of total greenhouse gas emissions, respectively (IPCC 2007). In Brazil, the agricultural share of CO_2 emissions represents 75% of the total anthropogenic emissions, due of the use of and changes in the land associated with such activity (Cerri et al. 2007). However, depending on the use and management of agricultural land, it can act as a source or a sink of atmospheric carbon (Ussiri and Lal 2009). According to Rayment and Jarvis (2000), the carbon balance of the ecosystem is represented by the quantity of carbon absorbed by photosynthesis and the amount of carbon lost in the soil respiration process. Therefore, small increases in respiration rates may be sufficient to change an ecosystem from a sink to a source of carbon to the atmosphere. Frequently, soil emissions associated with the loss of soil carbon via CO_2 emissions in agricultural areas are not considered because of the emission's large timeline variation and because it is a phenomenon resulting from a complex interaction of physical, chemical, and biological attributes and climatic variables (Epron et al. 2006; Figueiredo and La Scala Júnior 2011; Teixeira et al. 2012; Moitinho et al. 2013).

In the agricultural context, Brazil stands out as the world's largest producer of sugarcane (*Saccharum* spp.) with a production of 625.96 million tons for the 2018/2019 crop (CONAB 2018). The harvested area was estimated at 8.61 million hectares, distributed in all producing states, with the State of São Paulo being the largest producer, accounting for 54% of the processed sugarcane. Thus, estimates of soil CO_2 emission in sugarcane areas are important to the reduction of spatiotemporal uncertainties associated with the carbon balance calculation process for the country. In southeastern Brazil, changes in the management of sugarcane crop practices are important because large areas are being converted from a burnt cane management system to a green harvest system.

The addition of larger mass organic residues in the green harvest system (green cane) causes a larger interaction between organic and mineral fractions of soil as well as the protection of the ground surface against the impact of rain drops, thus reducing the erosion process. The return of crop residues to the soil surface favors the accumulation of organic matter and a reduction in gas emissions compared to the burned cane system (Razafimbelo et al. 2006; Cerri et al. 2007; Luca et al. 2008). The use of new methodologies to forecast the spatiotemporal patterns of FCO_2 is necessary to improve the quality of current estimates. Thus, the use of artificial neural networks is important in this task.

Neural networks

Artificial neural networks (ANNs) are an excellent alternative for modeling nonlinear systems that are too complex to be described by analytical methods, and ANNs are able to recognize patterns and perform diagnoses and predictions (Haykin 1999). ANNs are defined as adaptive communication networks linking a cost function to a desired output (Kartalopoulos 1996). ANNs can also be defined as a network that represents a dynamic process that can be modeled by a set of coupled nonlinear differential and/or algebraic equations (Kartalopoulos 1996). However, ANNs need to learn the task for which they were designed, and this is associated with the weights that connect the network. Therefore, knowledge is acquired by the network through a learning process and connection strengths between neurons (synaptic weights), which are then used to store the acquired knowledge (Haykin 1999).

Neural networks may have a single layer or multiple layers. The neural network used in this work is a multilayer neural network called the multilayer perceptron (MLP) trained by the algorithm backpropagation (BP). MLP is a type of feedforward neural network which is composed of many simple perceptron in a hierarchical structure forming a feedforward topology with one or more layers (hidden layers) between input and output layers (Kartalopoulos 1996; Haykin 1999).

The ability to learn through example and generalize the learned information make ANNs an interesting proposition for solutions in various areas of knowledge, such as engineering, medicine, meteorology, and agricultural science. One of the main neural network characteristics is the ability to learn and improve its performance. The network training introduces an input data set that produces an estimated or at least a consistent output, according to Wasserman (1989). Next, the calculated output is compared with a desired output, and thus an error is produced. This error is minimized in backward sense using the gradient descent algorithm until the minimum one is acquired (Haykin 1999). The weights are recalculated in each iteration, and this is the training process known as BP, introduced by Werbos (1974) and later improved by Widrow and Lehr (1990).

The Levenberg-Marquardt training algorithm is similar to backpropagation but the gradient descent minimization algorithm is replaced by a second-order Newton method (second derivative) (Liu 2010).

GRNN (general regression neural network) neural network (Spetch 1991) has a different concept when compared to a MLP, once it is based on a probability density function. It has four layers, one being the input, two hidden, and one output. The quantity of neurons in each layer depends on the quantity of inputs, outputs, and the training dataset. This neural network is trained in one presentation of the training patterns.

The MLP has been successfully used to solve problems in several areas involving high degrees of nonlinearity (Haykin 1999). Therefore, the use of an ANN is currently very significant. Thus, the works described below emphasize applications in agricultural.

Lotufo et al. (2007) introduced a procedure for the transient stability analysis and preventive control of electric power systems, and this procedure was formulated with a multilayer feedforward neural network. The neural network training was performed using the backpropagation algorithm with a fuzzy controller, and the most accurate results were obtained with this procedure compared to the traditional backpropagation algorithm. The work of Moretti et al. (2016) with the modulus of elasticity and compressive strength of concrete specimens used artificial neural networks with the backpropagation technique to predict the values of the elasticity modulus and resistance to concrete compression. Lentzsch et al. (2005) used a regression model and an artificial neural network with a set of 89 data points to evaluate the soil microbial biomass (SMB) of 30 sites in Germany, using a multilayer feedforward neural network built and tested with the Levenberg-Marquard algorithm for training. This model used a configuration with two or four hidden neurons and one output neuron. The results showed that both the linear full-factorial regression model and the neural network model are promising tools for predicting the SMB.

Merdun (2011) used a self-organizing neural network (SOM of Kohonen) to analyze the effects of the physical properties of soil on chemical/hydraulic processes and to diagnose the interrelationships of the multivariable soil data in the vadose zone. His conclusion shows that this neural network is effective for analyzing and diagnosing the dynamics of soil. Kim and Gilley (2008) used a multilayer perceptron neural network trained by backpropagation to estimate dissolved P (DP) and NH4–N concentrations in the soil erosion runoff from a land application site near Lincoln, NE, USA. The results showed that these estimations are

reliable. Song et al. (2014) analyzed the FCO_2 in Chinese forests using soil attributes and meteorological variables. They observed that a MLP neural network with three layers, trained with backpropagation algorithm and the gradient descent method, was an effective approach to predict FCO_2 in all Chinese forest ecosystems. This paper showed that the average FCO_2 of each type of forest in China was close to the average reported in similar forest ecosystems in other regions around the world. The results also showed that an increase in air temperature can greatly increase root respiration and microbial activity in these regions, leading to a high FCO_2, thus contributing to an increase in the greenhouse effect. Dahikar and Rode (2014) used a MLP neural network trained by backpropagation to predict agricultural crop yield. In this work, they also considered several soil parameters, such as the type of soil, the pH of the soil, and the nitrogen content. Deneshkumar et al. (2015) presented a multilayer feedforward neural network for agricultural forecasting and compared the results with the traditional autoregressive integrated moving average (ARIMA) of Box and Jenkins, concluding that the ANN provides an attractive alternative. Pandey and Mishra (2017) used two different neural networks (GRNN and radial basis neural network [RBF]) for evaluating the yield of potatoes. According to their conclusion, both ANNs perform accurately, but the GRNN provides better results than the RBF. Many other studies can be found in the literature; however, the application proposed in this work is a novelty.

A soil's CO_2 emission is the result of complex interactions between the physical and biological transport phenomena of gas production in the soil, which are influenced by various physical and chemical properties of soils. The objective of this study is to investigate the use of a MLP neural network with BP (backpropagation algorithm) to predict patterns of spatiotemporal variability for short periods of soil CO_2 emission in areas of sugarcane using a green harvest system in the southeastern section of the State of São Paulo, considering the physical and chemical attributes of the soil.

Data collect and experience

The present study compiles the previous findings regarding the emission of CO_2 from soil from experiments conducted in three commercial areas in São Paulo State, Brazil. The data originated from sugarcane areas in the green harvest, called green cane management, without burning practices, using mechanical harvesting that leaves large amounts of crop residues on the soil surface (mean of 12 tons ha^{-1} $year^{-1}$). These areas are as follows: Motuca (7 years of conversion from burned to green management), where the evaluation was carried out on October 25 and November 17 of 2008; Guariba (8 years of conversion from burned to green management), where the evaluation was carried out on July 14 and July 26 of 2010; and Pradópolis (15 years of conversion from burned to green management), where the evaluation was carried out on August 27 and September 14 of 2012 (Fig. 1). The soil CO_2 emission, soil temperature, and soil moisture were evaluated along short periods using sample grids, from which soil physical and chemical attributes were also evaluated (Panosso et al. 2012; Bicalho et al. 2014). The soils of all experimental areas were classified as Oxisol (Santos et al. 2013) and Eutrustox (USDA Soil Taxonomy). The climate, labeled Aw (according to Köppen), has a rainy tropical summer and a dry winter. The average rainfall is 1425 mm, concentrated between the months of October and March, with an average annual temperature of 22.2 °C recorded in the region for the past 30 years.

The CO_2 soil emission was registered by the LI-COR system (LI-8100), after the harvesting operation and during the early stages of crop growth. The soil temperature was monitored simultaneously with the CO_2 emission evaluation, using a temperature sensor that is part of the LI-8100 system. The soil moisture was determined with TDR equipment (time-domain reflectometry—Hydrosense TM, Campbell Scientific, Australia). Following the measurement period, soil samples were collected from 0 to 0.10 m in depth. The following routine tests were carried out: determining the content of organic matter (OM), the available P, K, Ca, Mg, and H + Al, and establishing the calculation of the sum of the bases and the exchange cation capacity (Van Raij et al. 2001). The bulk density was determined in undisturbed soil samples (EMBRAPA 1997). The total pore volume was distributed by size and macro- and micro-porosity and was determined using a fritted glass funnel under a pressure of a 60 cm height of water column (EMBRAPA 1997).

Data from 2012 (observed and estimated by the network) was used to analyze the experimental variogram (Webster and Oliver 1990), so that the spatial variability patterns of FCO_2 could be characterized, by

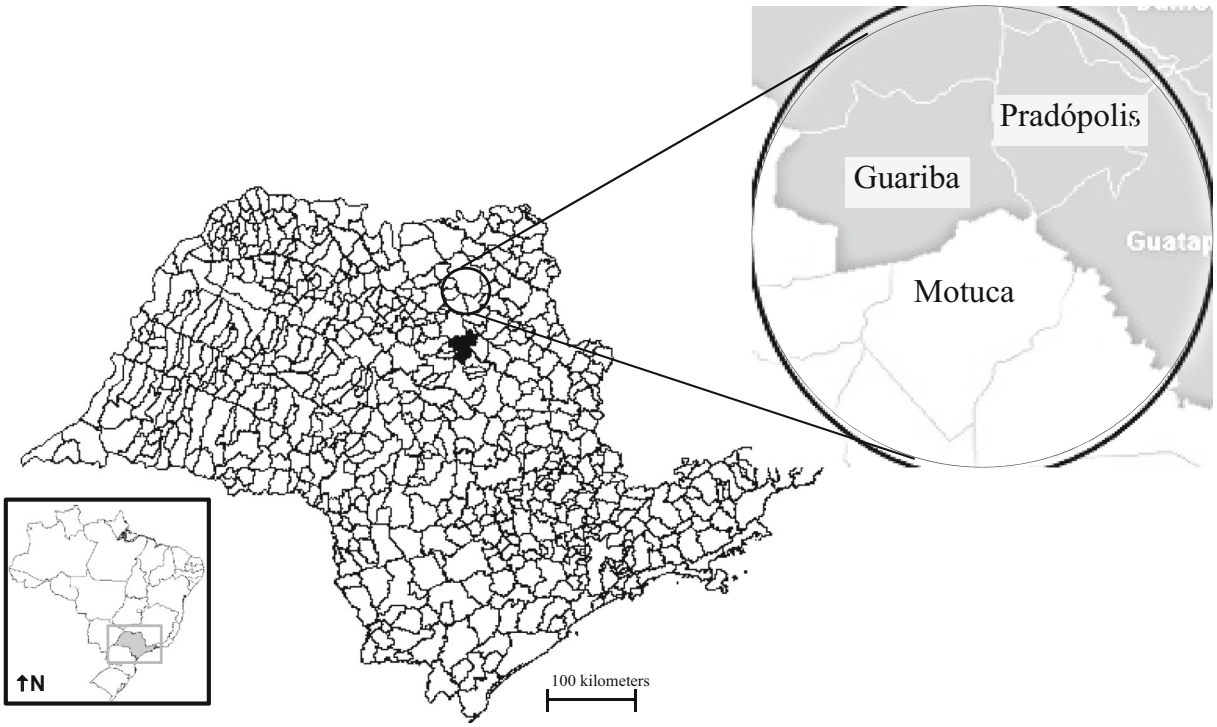

Fig. 1 Map showing the site locations of the studies, Motuca in 2008, Guariba in 2010, and Pradopolis in 2012

estimating the semi-variance in a given distance. This estimate was determined by Eq. (1):

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{i-1}^{N(h)} [Z(x_i) - Z(x_i + h)]^2 \quad (1)$$

where h is the distance between pairs of points; $N(h)$ is the number of pairs of points separated by the distance h; $Z(x_i)$ is the FCO_2 value at the point x_i; and $Z(x_i + h)$ is the FCO_2 value at the point $x_i + h$.

The choice of the model that best fits the experimental variograms (observed and estimated by the network) was made by cross-validation techniques, which involves the removal of each observation pertaining to the data set by estimating the value and interpolation method (data not shown). The chosen model best predicted the observed values, so that a linear regression equation between these values was produced, depending on the estimated values closer to the bisect—intercept equal to zero and angular coefficient equal to unity (Isaaks and Srivastava 1989). Theoretical models were used, as in Eqs. (2) to (4):

Exponential model:

$$\hat{\gamma}(h) = C_0 + C_1\{1 - \exp[-(3h/a)]\} \quad (2)$$

where $h > 0$.

Spherical model:

$$\hat{\gamma}(h) = C_0 + C_1\left[3/2(h/a) - 1/2(h/a)^3\right] \quad (3)$$

where $0 < h < d$, $\hat{\gamma}(h) = C_0 + C_1$, and $h > a$.

Gaussian model:

$$\hat{\gamma}(h) = C_0 + C_1\left\{1 - \exp\left[-3\left(h/a^2\right)\right]\right\} \quad (4)$$

where $0 < h < d$, and d is the maximum distance at which the variogram was defined.

There are three important parameters in the variogram: nugget (C_0); sill ($C_0 + C_1$); and range (a). The semi-variance value found at the intersection with the Y-axis, symbolized by C_0, is defined as the nugget and is the variability of FCO_2 in smaller distances than those sampled. Ideally, the amount of semi-variance would be zero when the separation distance h is zero. In practice, this usually does not occur due to experimental errors and the occurrence of spatial variability in distances smaller than those used in the samples' grid (Wang et al. 2002). The sill ($C_0 + C_1$) is the point at which the variogram model reaches a plateau and describes the variation of style. The range (a) is the distance at which the landing occurs and from which there is spatial independence and no spatial autocorrelation

between observations. It is expected that differences in $Z(x_i) - Z(x_i + h)$ progressively decrease so that h decreases, i.e., FCO_2 values (observed and estimated) located next to each other are more like each other than those separated by great distances. Thus, it is expected that $\hat{\gamma}(h)$ increases with the distance (h).

The parameters of the models fitted to the experimental variograms that were used to estimate FCO_2 in unsampled

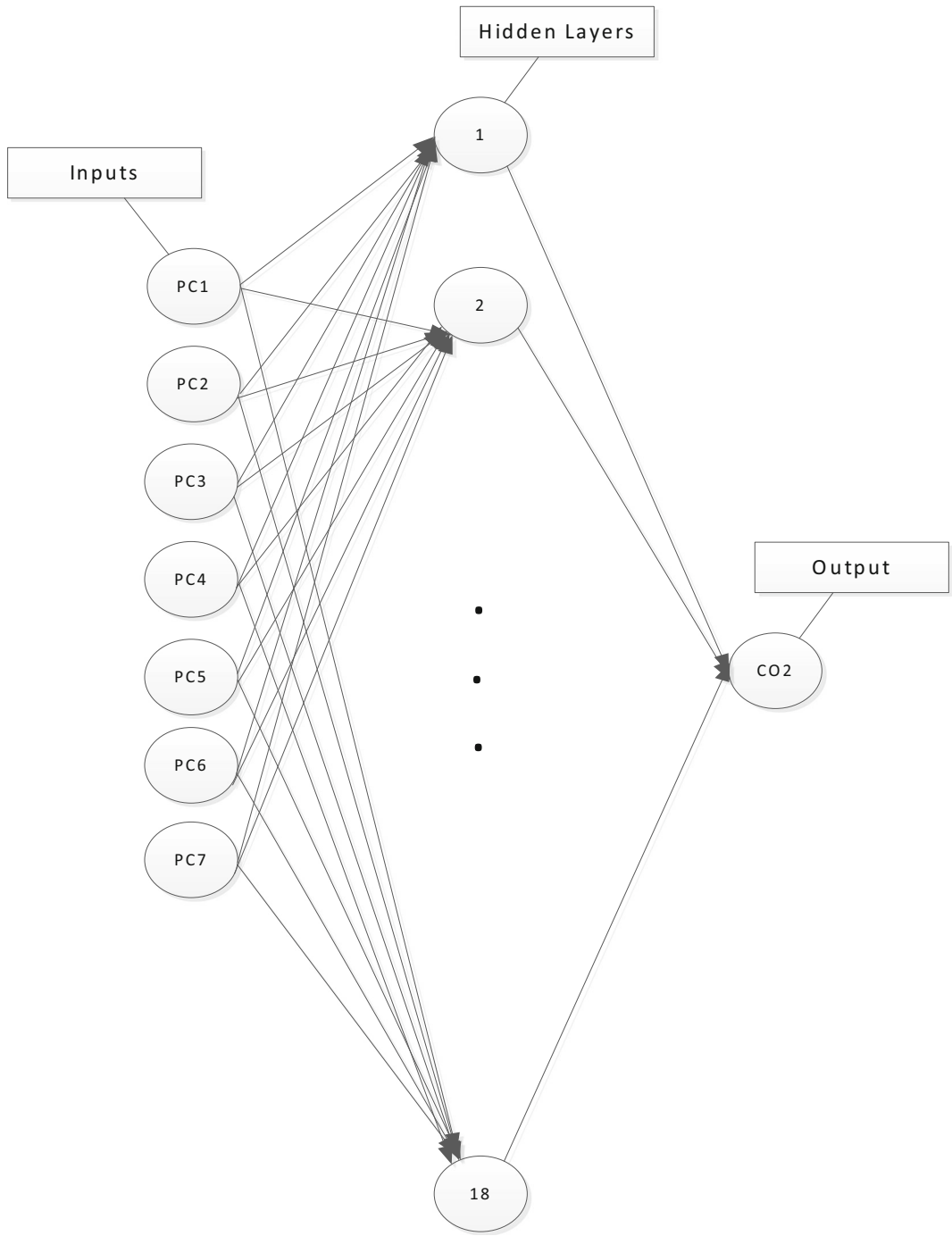

Fig. 2 Architecture of the MLP neural network

Table 1 Real outputs obtained through the MLP neural network representing the identification of CO_2 emissions

Am	SRM	SO	Am	SRM	SO
500	0.126	0.087	510	0.103	0.085
501	0.110	0.071	511	0.195	0.167
502	0.126	0.082	512	0.113	0.108
503	0.100	0.080	513	0.111	0.066
504	0.072	0.072	514	0.402	0.369
505	0.143	0.082	515	0.194	0.099
506	0.171	0.166	516	0.091	0.107
507	0.220	0.160	517	0.135	0.097
508	0.130	0.086	518	0.083	0.078
509	0.169	0.092	519	0.103	0.131

locations were used in the construction of spatial pattern maps with the ordinary kriging technique, Eq. (5):

$$Z^*(x_0) = \sum_{i=1}^{N} \lambda_i Z(x_i) \tag{5}$$

where z^* is the value to be estimated at the point not sampled x_0; N is the number of measured values; $Z(x_i)$ is involved in the estimate; and λ_i are the weights associated with each measured value $Z(x_i)$.

This study used 2377 recorded observations in 2008, 2010, and 2012, and were divided into two groups:

1) 1314 registered data points from 2008 and 2010 were chosen randomly and used in the neural network training phase.
2) 1063 registered data points from 2012 were used for the diagnostic phase of the neural network, or neural network classification, i.e., the estimation of FCO_2.

The neural network used had three layers. The input layer was composed of 7 neurons, the intermediate layer had 18 neurons, and the output layer had only 1 neuron corresponding to the neural network diagnosis, i.e., the soil CO_2 emission as shown in Fig. 2.

The neural network input variables were determined by a principal component analysis (PCA), whose main objective was to identify, in a data set, a variable that can explain a significant part of the variance observed in the system through linear correlations (Lawless and Heymann 2010). The PCA is a technique that transforms a large number of variables into a smaller set of non-correlated variables without a loss of information related to the initial variables, which are called the principal components (PCs). This smaller set of variables is generated by linear combinations of the original variables through the eigenvalues of the covariance matrix. These PCs work as axes, and each group applied to the component's equation yields the coordinates (scores), from which one can obtain its location in space. From a principal component, it is possible to obtain k variables, so:

$$\begin{aligned} \mathrm{PC}p &= a_{11}X_1 + a_{21}X_2 + \ldots + a_{p1}X_p \\ \mathrm{Unity}\,(1) &= a_{11}X_1 + a_{21}X_2 + \ldots a_{p1}X_p \\ \mathrm{Unity}\,(2) &= a_{11}X_1 + a_{21}X_2 + a_{p1}X_p \\ &\vdots \\ \mathrm{Unity}\,(k) &= a_{11}X_1 + a_{21}X_2 + \ldots + a_{p1}X_p \end{aligned} \tag{6}$$

Considering the samples originating from the sugarcane area: organic matter (OM) content, cation exchange capacity (CEC), carbon stock (CS), average

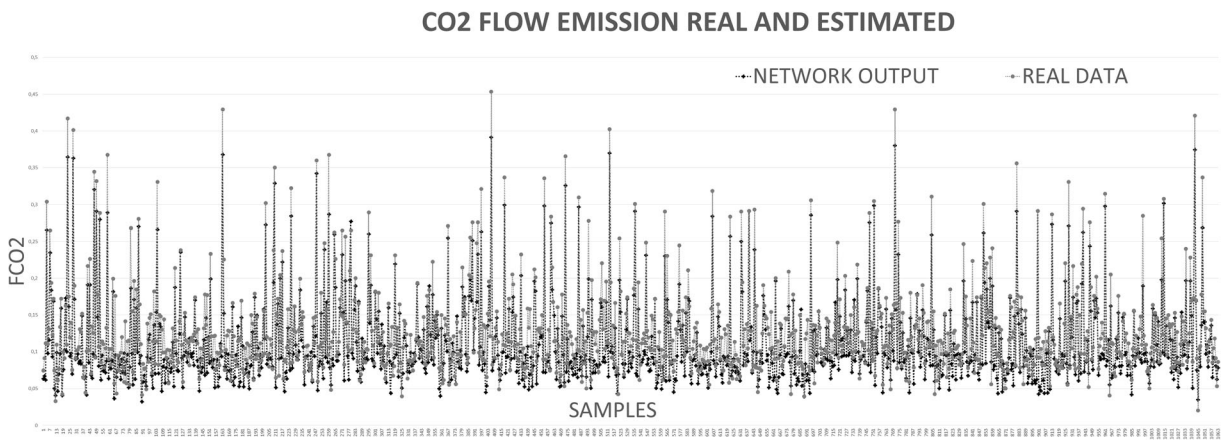

Fig. 3 Results from the 1063 data samples of soil CO_2 emission (FCO_2) presented by the neural network

temperature (avgT), average moisture (avgH), average free water pore space (avgPLA), hydrogenionic potential (pH), bulk density (BD), and total pore volume of the soil (TPV), were determined to be the principal components PC_1, PC_2, PC_3, PC_4, PC_5, PC_6, PC_7, respectively, and were the inputs of the neural network. The input data were normalized between [0, 1] in order to avoid saturation of the neural network.

For an accurate analysis of CO_2 emission, the mean absolute percentage error (MAPE) and the coefficient of determination R^2 were determined, respectively, by Eqs. (7) and (8):

$$\mathrm{MAPE} = \frac{1}{N}\left\{\sum_{i=1}^{N}\left(|ER - EO_{\mathrm{MLP}}|\right)/ER\right\} X\ 100 \qquad (7)$$

$$R^2 = \sum_{i=1}^{N}\left(EO_{\mathrm{MLP}} - ER_{\mathrm{avg}}\right)^2 / \sum_{i=1}^{N}\left(ER - ER_{\mathrm{avg}}\right)^2 \qquad (8)$$

where ER = real CO_2 emission registered; $EO_{\mathrm{MLP}} = CO_2$ emission from the MLP; ER_{avg} = real CO_2 emission average; and N is the number of samples.

Results and discussion

This work used the structure of a neural network (MLP feedforward neural network) composed of three layers (input, hidden, output) with 7, 18, and 1 neurons, respectively. The basic training algorithm chosen is backpropagation algorithm, which algorithm is presented in the Appendix. This simulation used the following parameters (see Appendix):

$\lambda = 0.3$ (inclination of sigmoid function)
$\gamma = 0.1$ (training rate)
and $\eta = 0$ (momentum rate)

Another training algorithm, such as the Levenberg-Marquardt (Liu 2010), was also used to comparison, as well as another neural network called general regression neural network (GRNN) (Specht 1991).

The MLP neural network output was represented by the average soil CO_2 emission. Table 1 shows a small sample of the1063 observations in the database for 2012 and the output obtained from the MLP neural network classification stage (diagnostic phase). The sample

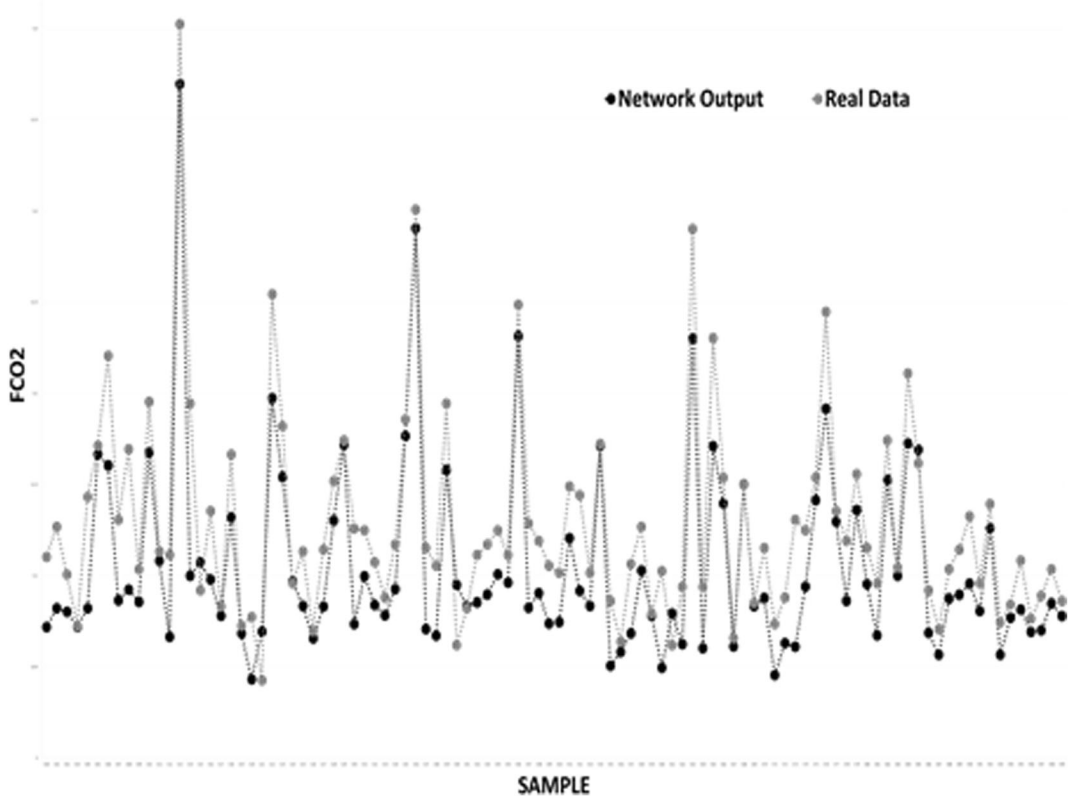

Fig. 4 Results from the 100 data samples of soil CO_2 emission (FCO_2) presented by the neural network

(Am) was used to compare the measured real output data (SRM) with the obtained outputs (SO). The real outputs are the information obtained through the instruments during the experimental phase. The obtained outputs are the information estimated by the network proposed in this paper. Figure 3 shows the 1063 actual results from the application and the results obtained by the neural network for a sample of data collected from the rating group, and Fig. 4 presents 100 data samples (samples numbering from 500 to 599) to facilitate the data reading and the analysis of the behavior of the neural network. The precision analysis of the determined CO_2 emissions represents a MAPE of 18.2852% and a coefficient of determination R^2 of 0.9188. When comparing to the Levenberg-Marquard training algorithm, MAPE is equal to 34.2984%, and in GRNN neural network, MAPE is equal to 43.0225%.

The Levenberg-Marquardt training is faster than backpropagation but depending on the size of the Hessian matrix, it has a disadvantage due to the need to invert this matrix, which demands time.

GRNN neural network, in general, is also very fast, but it was not adequate to solve the present problem when comparing the performance analyzing the MAPE.

That is the reason to choose backpropagation algorithm as the best one for the present problem (CO_2 emissions).

The emission of soil CO_2 varies in space and time, so the estimates of spatial dependence FCO_2 for the days studied in 2012 are presented in Table 2. The models fitted to the experimental variogram FCO_2 were mostly spherical, except for day 240, the first day of the evaluation, whose spatial variability structure was best described by the Gaussian model. The estimated values of soil CO_2 emission were similarly described in their spatial variability structure by spherical models, except for day 240 (exponential) and days 247 and 251 (Gaussian). The separation distances that comprise the autocorrelation data for the observed FCO_2 ranged from 8.34 m (day 255) to 21.66 m (day 249). Similar results were observed for the FCO_2 data estimated by the network, for which the parameter values varied from 6.34 m (day 255) to 23.2 m (day 249). Several models are used to describe the spatial variability of FCO_2, with spherical models (La Scala et al. 2000; Kosugi et al. 2007; Konda et al. 2008; La Scala et al. 2010; Herbst et al. 2012; Tavares et al. 2016) or alternations between spherical and exponential models (Tedeschi et al. 2006; Ohashi and Gyokusen 2007). The higher mean range value of the spatial variability structure indicates a more homogeneous distribution of FCO_2. La Scala et al. (2000), valuating the FCO_2 in a grid of 100 × 100 m in a bare soil, obtained range values from 20.2 to 58.4 m. Teixeira et al. (2011) observed values ranging from 22.39 to 54.31 m with smaller sample sizes (60 × 60 m) in areas with soil surface residues. Changes in the range of spatiotemporal variability models of FCO_2 have been observed between seasons (Ohashi and Gyokusen 2007), between months (Stoyan et al. 2000), after precipitation events (La Scala et al. 2000), and even according to the grid sample size (Konda et al. 2008; Herbst et al. 2009).

Analyzing the data obtained from the observed FCO_2 and the estimated FCO_2, it is possible to verify that the coefficient of the nugget effect has

Table 2 Models and estimates of parameters adjusted to the experimental variograms of soil CO_2 emission (observed and estimated by the network), on different days of assessment 2012

Day July	Model	C_0	$C_0 + C_1$	a (m)	r^2
FCO_2 observed					
240	Gau	0.119	1.197	14.96	0.72
242	Esf	0.066	0.148	11.6	0.75
244	Esf	0.058	0.159	19.98	0.85
247	Esf	0.063	0.150	19.16	0.79
249	Esf	0.113	0.321	21.66	0.86
251	Esf	0.111	0.239	17.27	0.72
255	Esf	0.050	0.128	8.34	0.68
258	Esf	0.052	0.140	17.12	0.80
Average	Esf	0.1179	0.272	18.87	0.67
FCO_2 estimated					
240	Exp	0.058	0.187	11.10	0.77
242	Esf	0.040	0.123	11.53	0.75
244	Esf	0.059	0.143	20.16	0.73
247	Gau	0.053	0.123	15.32	0.83
249	Esf	0.129	0.260	23.20	0.73
251	Gau	0.116	0.203	13.59	0.75
255	Esf	0.054	0.116	6.34	0.53
258	Esf	0.044	0.114	18.20	0.64
Average	Esf	0.047	0.211	19.32	0.77

Exp exponential, *Esf* spherical, *Gau* Gaussian, r^2 coefficient of determination of the model fit

moderate spatial dependence, i.e., the random component is important.

Figure 5 shows maps of the spatial pattern of the observed soil CO_2 flow and the values estimated by the neural network, and these data are of considerable similarity. By analyzing the grade, it is possible to see that the characteristics that point to higher emissions as well as to lower CO_2 emissions have been identified by the neural network with similar values.

The used network achieved satisfactory results in determining FCO_2 in sugarcane areas in a green harvest system, considering the diversity presented by the data and the fact that the results were determined by the main components according to the obtained MAPE and the correlation coefficient.

The neural network completed the learning process with data collected in 2008 and 2010 in different months and days and classified the data

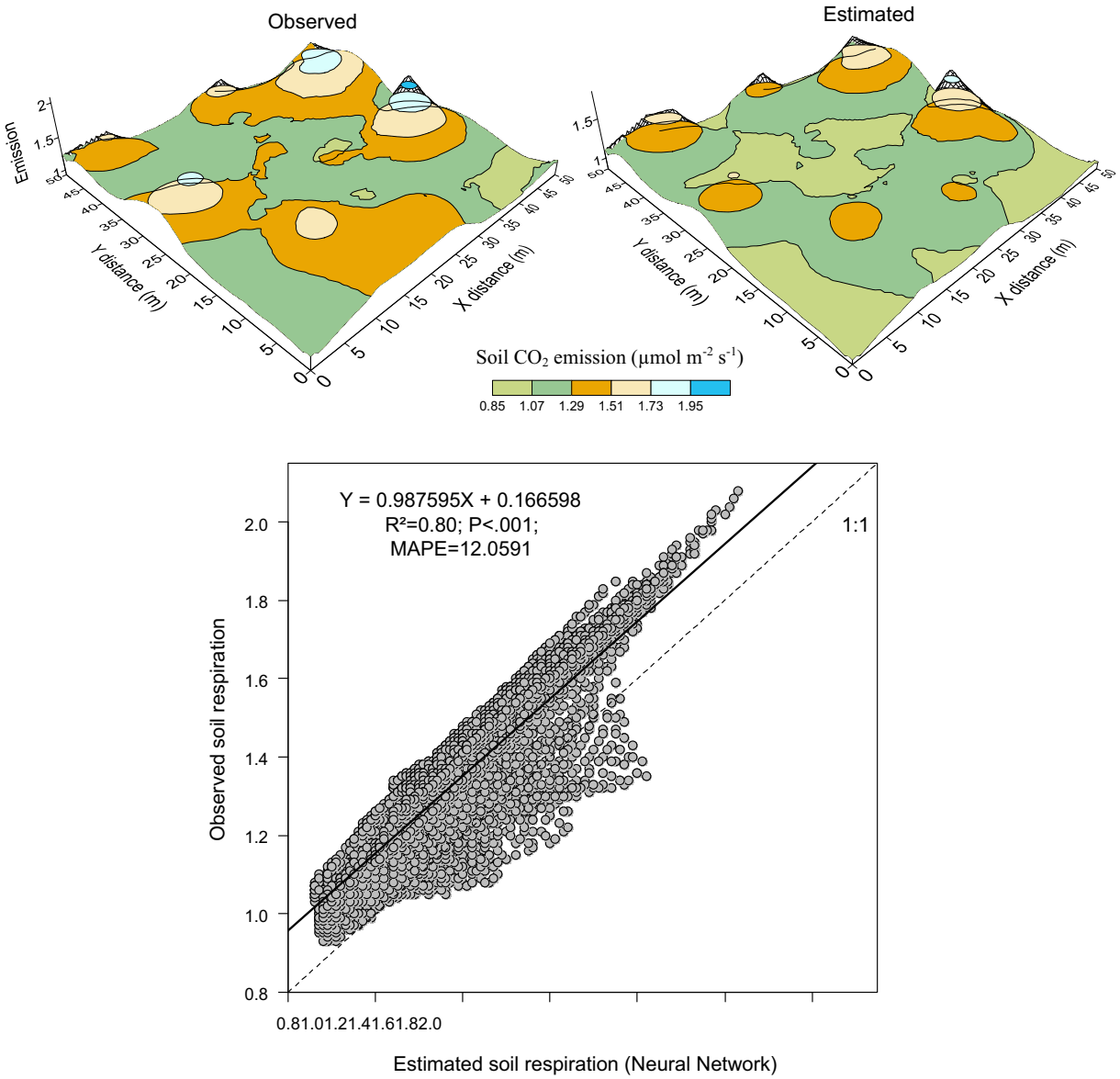

Fig. 5 Maps of the spatial pattern of soil CO_2 flux (FCO_2) observed and estimated together with a map comparison scatterplot between observed and estimated FCO_2 values in kriging maps

collected in 2012. Even with the interference from several natural factors in this period, the network obtained a good prediction of the actual values, which shows that the capacity of the neural network for learning and adaptation is ideal for forecasting the CO_2 emission.

Conclusions

Forecasting FCO_2 with artificial neural networks in sugarcane areas can contribute significantly to soil management adjustments to reduce the emission of CO_2 into the atmosphere and thus reduce the additional greenhouse effect. In addition, the use of this methodology can be a tool to reduce the uncertainties associated with estimates of this important component of the carbon cycle in the atmosphere.

Depending on the nature of the application and the consistency of the input data standards, it is possible for a network to train well and obtain excellent estimates. Thus, these neural networks can be applied to problems of dynamics and nonlinear relationships. They supply an analytical alternative to conventional techniques that are limited, often by strict assumptions of normality, linearity, and variable independence.

The FCO_2 forecasting results obtained by the MLP network and backpropagation algorithm are close to the experimentally measured values, proving an excellent efficiency of the use of artificial neural networks in agriculture.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Backpropagation algorithm

The initial weights are usually adopted as random numbers (Widrow and Lehr 1990). The backpropagation (BP) algorithm consists of adapting the weights such that the network quadratic error is minimized. The sum of the instantaneous quadratic error of each neuron of the last layer (network output) is given by (Widrow and Lehr 1990):

$$\varepsilon^2 = \sum_{i=1}^{no} \varepsilon_i{}^2 \tag{9}$$

where

ε_i $d_i - y_i$;
d_i desired output of the ith element of the last layer;
y output of the ith element of the last layer; and
no number of neurons of the last layer.

Considering the ith network neuron and using the descent gradient method, the weight adjustments can be formulated by (Widrow and Lehr 1990):

$$\boldsymbol{\Gamma}_i\,(r+1) = \boldsymbol{\Gamma}_i\,(r) + \theta_i\,(r) \tag{10}$$

where

$\theta_i\,(r)$ $-\gamma\,[\nabla_i\,(r)]$;
γ stability control parameter or training rate;
$\nabla_i\,(r)$ quadratic error gradient related to neuron i weights; and
$\boldsymbol{\Gamma}_i$ = vector containing neuron i weights $[w_{0i}, w_{1i}, w_{2i} \ldots w_{ni}]^T$.

The adopted direction in Eq. (1012) to minimize the objective function of the quadratic error corresponds to the gradient opposite direction. The γ parameter determines the vector length $\phi_i(r)$. The sigmoid function is defined by (Widrow and Lehr 1990):

$$y_i \Delta y_i(\lambda, \vartheta_i) = \left\{\left(1 + exp(-\lambda\vartheta_i)\right\} / \left\{\left(1 + exp(-\lambda\vartheta_i)\right\}\right. \tag{11}$$

or

$$y_i \Delta y_i(\lambda, \vartheta_i) = 1 / \left\{\left(1 + exp(-\lambda\vartheta_i)\right\}\right. \tag{12}$$

where

λ constant that determines the slope of the function y_i.

The variation of Eqs. (11) and (12) are (− 1,+ 1) and (0,+ 1), respectively.

Next, calculating the gradient as shown in Eq. (1012) and considering the sigmoid function defined by Eq. (11) or (12) and the momentum term, the following adaptation weight scheme is obtained (Lopes et al. 2003):

$$\Pi_{ij}(r+1) = \Pi_{ij}(r) + \Delta\Pi_{ij}(r) \tag{13}$$

where

$$\Delta\Pi_{ij}(r) = 2\gamma(1-\eta)\beta_j x_i + \eta\Delta\Pi_{ij}(r-1); \tag{14}$$

Π_{ij} weight corresponding to the connection with the *i*th and the *j*th neuron;

γtraining rate; and

ηmomentum constant ($0 \le \eta < 1$).

If the *j*th element is in the last layer, then:

$$\beta_j = \sigma_j \varepsilon_j \tag{15}$$

where

σ_j $\underline{\Delta}$ sigmoid function derivative, given by Eq. (11) or (12), respectively, related to ϑ_j

$$= \lambda/2\left(1-y_j^2\right). \tag{16}$$

$$= \lambda y_j\left(1-y_j^2\right). \tag{17}$$

If the *j*th element is in other layers, we have:

$$\beta_j = \sigma_j \sum_{k\in\Gamma(j)} w_{jk}\beta_k \tag{18}$$

$\Gamma(j)$ set of the element indices that are in the next layer to the *j*th element layer and are interconnected to the *j*th element.

The γ parameter that is used as a stability control for the iterative process is dependent on λ. The network weights are randomly initialized from the interval [0,1]. For convenience, the parameter γ (training rate) can be redefined by the following (Lopes et al. 2003):

$$\gamma = \gamma^*/\lambda \tag{19}$$

Replacing Eq. (19) in Eq. (12) will "cancel" the amplitude dependency of σ related to λ. The σ amplitude will be maintained constant to every λ. This alternative is important considering that λ will only actuate in the left and right tails of σ. Equation (14) can then be written as the following:

$$\Delta\Pi_{ij}(r) = \left\{2\gamma^*(1-\eta)\beta_j/\lambda\right\}x_i + \eta\Delta\Pi_{ij}(r-1). \tag{20}$$

The BP algorithm executes as follows (Widrow and Lehr 1990):

Step 1. Present a pattern *X* to the network, which provides an output *Y*.

Step 2. Calculate the error (difference with the desired value and the output) for each output.

Step 3. Determine the backpropagated error by the network associated with the partial derivative of the quadratic error.

Step 4. Adjust the weights of each element.

Step 5. Present a new pattern to the network and repeat the process until the convergence is attained (according to a predefined tolerance).

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REDE NEURAL ARTIFICIAL ARTMAP-FUZZY APLICADA NO RECONHECIMENTO DE FALHAS ESTRUTURAIS

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Resumo – A fim de desenvolver uma forma eficiente para a identificação e caracterização de falhas estruturais, este artigo teve por objetivo demonstrar uma aplicação de Redes Neurais Artificiais (RNAs) para tal problema. Portanto, através de um modelo dinâmico para a representação de uma estrutura predial, desenvolveu-se uma RNA ARTMAP-*Fuzzy* por ser uma rede flexível e estável em relação à sua habilidade em se adaptar às mudanças imprevistas do ambiente externo, para identificar tais falhas.

Palavras chave – Redes Neurais Artificiais, Reconhecimento de falhas, ARTMAP-*Fuzzy*.

1. INTRODUÇÃO

As falhas estruturais são danos na estrutura que implicam em mudanças nas suas propriedades físicas, e consequentemente nas características dinâmicas. Essas estruturas podem ser prédios, pontes, plataformas, etc., em que tais falhas podem influenciar no desempenho geral da estrutura, bem como causar consequências desastrosas, de acordo com [1]. Devido a isso, muitas empresas e indústrias aplicam a manutenção preventiva não apenas para evitar estas falhas, mas também para reduzir gastos e aumentar a confiabilidade e segurança destas estruturas, conforme [2].

A necessidade de desenvolver métodos eficazes para a detecção de falhas estruturais vem crescendo no decorrer dos anos. Segundo [3], estes métodos baseiam-se no fato de que os parâmetros modais (frequências, amortecimento e formas modais) representam funções das propriedades físicas da estrutura (massa, amortecimento e rigidez). Portanto, quando estas propriedades físicas sofrem alguma alteração, por exemplo, a redução de rigidez, as propriedades modais apresentarão alterações detectáveis.

Assim sendo, a aplicação de técnicas de monitoramento da integridade de estruturas em geral torna-se uma questão muito importante para a segurança da população, pois através destas técnicas, falhas podem ser detectadas em seu estágio inicial, auxiliando e promovendo a melhor forma de manutenção destas estruturas, garantindo que as mesmas continuem em pleno funcionamento.

Em [4] citam que o processo de implementação de estratégias na identificação destas falhas, seja na engenharia mecânica, civil ou no ambiente aeroespacial, é conhecido como Monitoramento da Saúde Estrutural (SHM). Em [5], são citadas algumas técnicas de SHM, como: emissão acústica, raios X, correntes de Eddy, métodos de campo magnético e outros. Os principais benefícios que os sistemas de SHM apresentam são: a diminuição dos custos com manutenção e a maior confiabilidade das estruturas projetadas, gerando uma maior segurança de vida.

De acordo com [2], apesar das técnicas tradicionais de SHM possuírem bons equipamentos e serem bem formuladas, elas não satisfazem as necessidades atuais das indústrias, principalmente quando se refe-

re à detecção de falhas em tempo real. Portanto, novas técnicas veem surgindo utilizando-se dos conceitos de redes neurais artificiais.

Em geral as Redes Neurais Artificiais (RNAs) são algoritmos computacionais utilizados na resolução de problemas complexos. Seu processamento baseia-se no funcionamento do sistema nervoso biológico. Segundo [6] as RNAs possuem uma habilidade natural em armazenar conhecimento experimental e disponibilizá-lo para uso posteriormente.

Uma das arquiteturas de redes neurais que vem se destacando na literatura são as redes neurais baseadas na teoria da ressonância adaptativa denominada redes ART (Adaptive Resonance Theory). Segundo [7,8] esta arquitetura preserva duas características fundamentais para o processamento da rede: a estabilidade e a plasticidade que proporcionam à rede uma aprendizagem estável e convergência rápida.

Este trabalho tem como objetivo utilizar a rede neural artificial ARTMAP *Fuzzy* como um sistema de detecção de falhas estruturais, sendo que os dados (com e sem falha) utilizados na fase de treinamento e diagnóstico da rede neural são gerados a partir de um modelo matemático de um edifício.

2. REDE NEURAL ARTIFICIAL ARTMAP *FUZZY*

Para [9] a rede neural ARTMAP *Fuzzy* é uma rede mais sofisticada, que emprega o treinamento supervisionado, ou seja, é baseado em estímulos de entrada-saída. É composta por dois módulos ART *fuzzy*: ARTa e ARTb interconectados através de módulo de memória associativa inter-ART, Fab, que possui um mecanismo auto-regulador denominado match tracking, que tem a incumbência de maximizar a generalização e minimizar o erro da rede.

A rede neural ARTMAP *Fuzzy* incorpora a teoria de conjuntos *fuzzy* em seus cálculos e, como tal, é capaz de aprender categorias estáveis em resposta a padrões de entradas analógicos ou binários.

A estrutura da rede neural ARTMAP *Fuzzy* é mostrada na Fig. 1.

O fluxograma do algoritmo da rede neural ARTMAP *Fuzzy* é ilustrado na Fig. 2. Através do Fluxograma é possível verificar os passos a serem executados para o funcionamento da rede.

3. METODOLOGIA

A metodologia proposta é baseada em três etapas:

1. Aquisição dos dados;

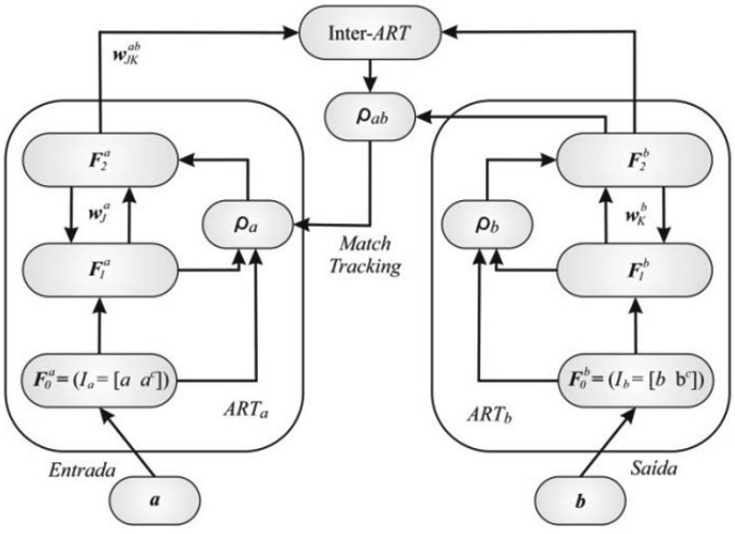

Fig. 1. Rede neural ARTMAP *Fuzzy* ([2]).

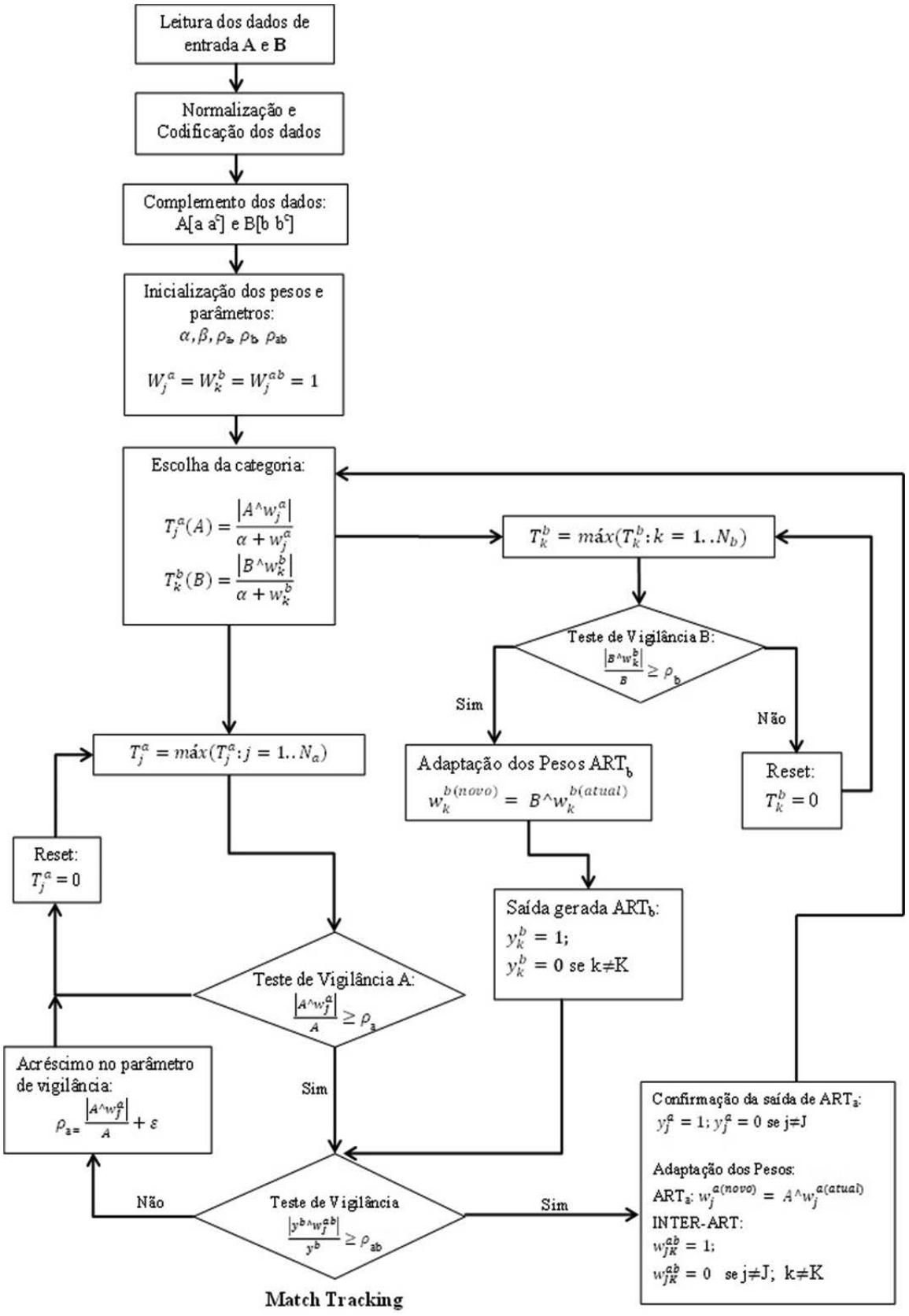

Fig. 2. Fluxograma da rede neural ARTMAP Fuzzy ([2]).

2. Treinamento da rede ARTMAP *Fuzzy*;
3. Diagnóstico da rede ARTMAP *Fuzzy*.

Para a execução do algoritmo da rede ARTMAP *Fuzzy*, tanto para o treinamento quanto para o diagnóstico, utilizou-se a plataforma Matlab®, por ser uma plataforma da fácil manuseio e por ser específica na resolução de problemas matemáticos.

3.1. Aquisição dos Dados

A aquisição dos dados foi feita por meio de um modelo matemático formado por um sistema de equações diferenciais modelando uma estrutura predial, sendo utilizado o integrador numérico *Runge Kutta* de quarta ordem. Para realizar as simulações das condições normais (*base-line*) e anormais (falha estrutural) no edifício, alteraram-se as massas (m_1 e m_2) dos andares na estrutura.

Conforme [10] a Fig. 3 ilustra o modelo que representa o comportamento dinâmico de um edifício, onde se derivam as equações do movimento para um pórtico plano simples, com consideração do encurtamento das barras devido à flexão das mesmas, sob excitação na direção vertical.

Os parâmetros utilizados neste modelo foram:

- m_1, m_2 - Massa da Estrutura
- k_1, k_2 - Coeficiente de Rigidez Elástica da Estrutura
- c_1, c_2 - Coeficiente de Amortecimento viscoso da Estrutura
- q_1, q_2 - Coordenadas Generalizadas
- $\dot{x_1}, \dot{x_3}$ - Velocidade
- $\ddot{x_2}, \ddot{x_4}$ - Aceleração
- S - Excitação Externa
- $\dot{S}$ - Velocidade da ação do sismo
- $\ddot{S}$ - Acelerograma da ação do sismo
- ω_1, ω_2 - Frequência Natural
- f - Frequência
- τ - Amplitude da Excitação Externa

em que:

$$x_1 = a + S + q_1$$

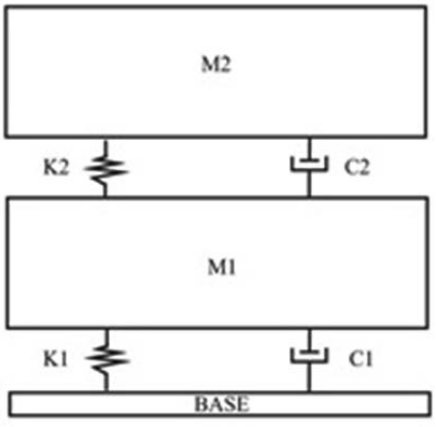

Fig. 3. Modelo Dinâmico ([10]).

$$x_2 = b + S + q_2$$

$$S = A\cos(\lambda t)$$

A energia cinética total (T) do sistema é:

$$T = \frac{1}{2}[m_1(\dot{S} + \dot{q}_1)^2 + m_2(\dot{S} + \dot{q}_2)^2] \tag{1}$$

A energia potencial total (V) do sistema é:

$$V = \frac{1}{2}[k_1 q_{1^2} + k_2(q_2 - q_1)^2] \tag{2}$$

O Lagrangeano (L=T-V) é:

$$L = \frac{1}{2}[m_1(\dot{S} + \dot{q}_1)^2 + m_2(\dot{S} + \dot{q}_2)^2 - k_1 q_{1^2} - k_2(q_2 - q_1)^2] \tag{3}$$

A equação de Lagrange para a coordenada generalizada q_1 é:

$$m_1(\ddot{q}_1 + \ddot{S}) + k_1 q_1 - k_2(q_2 - q_1) = -c_1\dot{q}_1 \tag{4}$$

A equação de Lagrange para a coordenada generalizada q_2 é:

$$m_2(\ddot{q}_2 + \ddot{S}) + k_2(q_2 - q_1) = -c_2(\dot{q}_2 - \dot{q}_1) \tag{5}$$

Assim, o sistema pode ser modelado pelo sistema de equações:

$$\begin{aligned} m_1\ddot{q}_1 + k_1 q_1 - k_2(q_2 - q_1) + c_1\dot{q}_1 &= -m_1\ddot{S} \\ m_2\ddot{q}_2 + k_2(q_2 - q_1) + c_2(\dot{q}_2 - \dot{q}_1) &= -m_2\ddot{S} \end{aligned} \tag{6}$$

ou

$$\begin{aligned} \ddot{q}_1 + \frac{k_1}{m_1}q_1 - \frac{k_2}{m_1}(q_2 - q_1) + \frac{c_1}{m_1}\dot{q}_1 &= -\ddot{S} \\ \ddot{q}_2 + \frac{k_2}{m_2}(q_2 - q_1) + \frac{c_2}{m_2}(\dot{q}_2 - \dot{q}_1) &= -\ddot{S} \end{aligned} \tag{7}$$

Tomando

$$\omega_1^2 = \frac{k_1 + k_2}{m_1}$$

$$\omega_2^2 = \frac{k_2}{m_2}$$

Tem-se

$$\begin{aligned} \ddot{q}_1 + \omega_1^2 q_1 - \frac{k_2}{m_1}q_2 + \frac{c_1}{m_1}\dot{q}_1 &= -S \\ \ddot{q}_2 + \omega_2^2(q_2 - q_1) + \frac{c_2}{m_2}(\dot{q}_2 - \dot{q}_1) &= -S \end{aligned} \tag{8}$$

E fazendo as seguintes considerações

$$x_1 = q_1$$

Tabela 1. Parâmetros do Modelo Matemático.

Parâmetros	Valores
m_1	7 kg
m_2	7 kg
c_1	0.6 N.s/m
c_2	0.7N.s/m
x_1	0.01 m
x_2	0.01 m
x_3	0.01 m
x_4	0.01 m
ω_1	1 rad/s
ω_2	2 rad/s
K_2	0.8 N/m
S	$Acos(\tau t)$
A	10^{-5}
τ	1

$$x_2 = \dot{q}_1$$

$$x_3 = q_2$$

$$x_4 = \dot{q}_2$$

Tem-se o sistema de equações descrito pela Equação (9).

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= -\omega_1^2 x_1 + \frac{k_2}{m_1} x_3 - \frac{c_1}{m_1} x_2 - S \\ \dot{x}_3 &= x_4 \\ \dot{x}_4 &= \omega_2^2 (x_1 - x_3) + \frac{c_2}{m_2} (x_2 - x_4) - S \end{aligned} \tag{9}$$

sendo S uma excitação periódica.

Os valores dos parâmetros iniciais utilizados no modelo matemático estão descritos na tabela 1, sendo aplicado o integrador Runge-Kutta de quarta ordem na equação (9) e para as condições iniciais x_1=0; x_2=0; x_3=0 e x_4=0 no tempo considerando amostras de 0 a 200, sendo A uma constante e τ é a frequência de excitação.

3.2. Treinamento da Rede ARTMAP *Fuzzy*

O treinamento de uma rede neural artificial é feito através do ajuste ou adaptação dos pesos sinápticos.

A arquitetura da ARTMAP *Fuzzy* é composta por um par de redes ART, também chamadas de módulo ARTa e ARTb. Durante o treinamento, o módulo ARTa recebe padrões de entrada ($a^{(p)}$) e ARTb, padrões de entrada ($b^{(p)}$), onde $b^{(p)}$ é a previsão correta dada aos padrões de entrada $a^{(p)}$.

Os módulos ARTa ARTb estão ligados por um terceiro módulo denominado inter-ART, que corresponde ao módulo de memória associativa.

Se as duas entradas no módulo inter-ART combinam, ou seja, se o prognóstico da rede é confirmado pela categoria selecionada, a rede aprenderá modificando os protótipos das categorias escolhidas em ARTa e ARTb , segundo um critério de treinamento.

Se as duas entradas não combinam, um sinal de *reset* é gerado e um processo denominado *match tracking* que ativará uma nova busca na qual escolherá uma nova categoria no módulo ARTa, cujo prognóstico será combinado novamente com a categoria atual do ARTb.

Todos os pesos relacionados ao ARTa, ARTb e ao inter-ART inicialmente possuem valor "1" indicando que as categorias não estão ativas, ou seja, inicialmente nenhuma categoria foi selecionada. Após a seleção das categorias elas tornam-se ativas.

Além do vetor de pesos sinápticos para cada módulo (wa, wb e wab), de acordo com [2] para o treinamento da rede ARTMAP *Fuzzy* os seguintes parâmetros são utilizados:

- Parâmetro de escolha;
- Taxa de Treinamento;
- Parâmetro de vigilância (P.V.) de ARTa;
- Parâmetro de vigilância (P.V.) de ARTb;
- Parâmetro de vigilância (P.V.) de ARTab.

Os padrões de entrada da rede neural são representados pelos deslocamentos, x_2, avaliado no tempo, t, e os padrões de saída da rede neural são representados pelo estado do sistema, ou seja, sistema normal ou sistema com falha. Matematicamente, tem-se:

$$X(h) = [x_2]$$

$$Y(h) = \begin{cases} 1 & \text{sistema normal} \\ 0 & \text{sistema com falha} \end{cases}$$

sendo:

X(h) : entrada da rede ARTMAP *Fuzzy*;
Y(h) : saída da rede ARTMAP *Fuzzy*;
h : quantidade de padrões.

3.3. Diagnóstico da Rede ARTMAP *Fuzzy*

Para realizar o diagnóstico da rede, foi apresentado um novo conjunto de dados, diferente do conjunto utilizado no treinamento, pois, uma vez treinada, a rede neural ARTMAP *Fuzzy* deve ser capaz de através do reconhecimento de uma categoria, classificar os dados não treinados, identificando assim, um sinal de condição normal ou anormal.

Durante o diagnóstico é realizado apenas o processo do módulo ARTa. A saída gerada por este módulo é então, comparada com o conhecimento adquirido durante o treinamento e é feito o reconhecimento e, consequentemente, encontrada a saída desejada.

4. RESULTADOS

Para a realização do treinamento e diagnóstico da rede ARTMAP *Fuzzy*, obteve-se 221 sinais, sendo que, 9 sinais representavam a condição normal e os outros 212 sinais representavam a condição de falha, sendo 170 sinais representando falha no 1º andar e 42 representando falha no 2º andar. A quantidade de sinais gerada foi obtida ao alterar a massa da estrutura m_1 de 8 até 9 com variação de 0,5 e a m_2 de 1 a 7 com variação de 0,5.

Portanto, o teste realizado consistiu em classificar os sinais em sinal normal, sinal com falha no primeiro andar e sinal com falha no segundo andar. E para realizar a identificação desta falha, ou seja, se a falha se encontra no primeiro ou segundo andar, adotou-se a seguinte estratégia:

- Os sinais que representavam a condição normal do edifício (*base-line*) tiveram uma saída desejada com uma representação binária composta por dois "*bits"*, ou seja, 0 0;
- Para os sinais que representam a condição de falha no 1º andar, a saída desejada é representada pelos "*bits"*, 0 1;
- Para os sinais que representam a condição de falha no 2º andar, a saída desejada é representada pelos "*bits"*, 1 0;

Tabela 2. Parâmetros do Treinamento da rede ARTMAP *Fuzzy*.

Parâmetro de Escolha	Taxa de Treinamento	P. V.1 ARTa	P. V. ARTb	P. V. Inter-ART
0,1	1,0	0,7	1,0	1,0

Tabela 3. Parâmetros da ARTMAP *Fuzzy* para o melhor resultado.

Treinamento	
Parâmetro de Escolha	0,1
Taxa de Treinamento	1,0
P. V. ARTa	0,7
P. V. ARTb	1,0
P. V. ARTab	1,0
Diagnóstico	
P. V. ARTa	0,7
Acertos (%)	100

O conjunto de dados para o treinamento foi formado por 80% dos sinais normais e 80% dos sinais com falha selecionados aleatoriamente. Os valores dos parâmetros da rede ARTMAP *Fuzzy* utilizados na fase de treinamento são mostrados na tabela 2.

O conjunto de dados para o diagnóstico foi composto por 20% dos sinais normal e 20% dos sinais com falha que não foram utilizados na fase de treinamento. Os parâmetros utilizados na fase de treinamento e diagnóstico da rede ARTMAP *Fuzzy* que geraram o melhor resultado é ilustrado na tabela 3.

A partir do resultado mostrado na tabela 3, realizou-se um teste cruzado no qual o treinamento da rede neural ARTMAP *Fuzzy* foi realizado 100 vezes e, observou-se que a rede neural apresentou uma média de porcentagem de acerto de 93,89% como melhor resultado.

Observou-se que apesar da quantidade pequena de sinais (padrões) a rede neural mostrou um ótimo desempenho. O melhor resultado obtido no primeiro teste foi de 100% de acerto, ou seja, a rede neural ARTMAP *Fuzzy* conseguiu identificar um sistema normal e um sistema com falha, além de classificar onde a falha se encontrava.

5. DISCUSSÕES E CONCLUSÕES

Este trabalho expôs a importância da detecção de falhas em estruturas prediais e apresentou um método utilizando uma rede neural ARTMAP *Fuzzy* como uma forma de realizar o Monitoramento da Saúde Estrutural para um modelo matemático de um edifício.

Observou-se através dos testes realizados que a rede neural comportou-se de forma esperada, apresentando um resultado positivo em relação ao reconhecimento e classificação de sinais.

Em relação ao teste em que a rede deveria classificar os sinais em sinal normal, sinal de falha no 1º andar e sinal de falha no 2º andar, a rede neural ARTMAP *Fuzzy* demonstrou um ótimo desempenho ao obter 100% de acerto. Outro ponto positivo observado foi o resultado do teste cruzado, que apresentou uma porcentagem de acerto de 93,89% de acerto.

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ARTMAP-FUZZY ARTIFICIAL NEURAL NETWORK APPLIED IN RECOGNITION OF STRUCTURAL FAULTS

Abstract – In order to develop an efficient way to identify and characterize structural faults, this paper aimed to demonstrate an application of Artificial Neural Networks (ANNs) for this problem. Therefore, through a dynamic model for the representation of a building structure, an ARTMAP-Fuzzy RNA has been developed because it is a flexible and stable network in relation to its ability to adapt to unforeseen changes in the external environment to identify such failures.

Keywords – Artificial Neural Networks, Fault Recognition, ARTMAP-Fuzzy.

AVALIAÇÃO DO MODELO DE REGRESSÃO LINEAR MÚLTIPLA E REDES NEURAIS ARTIFICIAIS NA PREVISÃO DO GANHO DE MASSA EM ANIMAIS

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RESUMO

A massa de um animal torna-se uma variável importante a ser estudada, visto que a partir dela, pode-se avaliar o crescimento e o estado nutricional do animal, administrar adequadamente remédios e parasiticidas e estabelecer o valor de venda do animal no mercado. O artigo aborda o desenvolvimento de uma proposta para previsão do ganho de massa em animais através do método de regressão linear múltipla e de uma técnica baseada na inteligência artificial, mais especificamente, as redes neurais artificiais. O objetivo é desenvolver os dois métodos e aplicá-los na análise de ganho de massa, para isto, levou-se em consideração dados de massa adquiridos pelo animal e o valor de escore de condição corporal obtidos em um determinado período. A análise foi feita utilizando dados de 12 animais avaliados em três períodos distintos e considerando três tipos de alimentação: 1 - alimentação convencional (controle), 2 - caroço de algodão e 3 - farelo de algodão e milho-triturado. Através dos resultados obtidos das análises do erro percentual médio absoluto e do coeficiente de determinação pode-se observar que ambos os métodos demonstram eficiência na previsão de ganho de massa em animais.

Palavras-chave: Estimação do Ganho de Peso. Modelo Estatístico. Perceptron Multicamadas. Algoritmo Backpropagation

AVALIATON OF GAIN WEIGHT ON ANIMALS THROUGH MULTIPLE LINEAR REGRESSION AND ARTIFICIAL NEURAL NETWORKS

ABSTRACT

The weight of an animal becomes an important variable to be studied, from it, we evaluated the growth and nutritional status of the animal, administer drugs and parasiticides and establish the sales value of the animal market. In this paper we approach the development of a proposal for the prediction gain weight in animals by the method of multiple linear regression and a technique based on artificial intelligence, more specifically artificial neural networks. The goal is to develop both methods and apply them in the analysis of weight gain, for this, we took into consideration data acquired by the animal weights and the value of body condition score obtained in a given period. The analysis was performed using data from 12

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animals evaluated in three different periods and considering three types of feed: 1- conventional feed (control), 2 - cottonseed and 3 - cottonseed meal and corn-ground. By means of the results of the analysis of the absolute mean percentage error and the coefficient of determination, it can be observed that both methods demonstrate efficiency in the prediction of mass gain in animals

Keywords: Weight Gain Estimation. Statistical Model. Multilayer Perceptron. Backpropagation Algorithm.

INTRODUÇÃO

Segundo Festing (2010) os principais motivos de se determinar a massa de um animal podem ser reduzidos à necessidade da utilização dessa variável em alguns estudos. Dentre elas tem-se: avaliação da taxa de crescimento do animal; determinação do seu valor no mercado; seleção de geração de raças; cálculos da dosagem de medicamentos e obtenção de uma relação da saúde geral do animal.

Usualmente, na literatura existem técnicas que são utilizadas para prever ganho de massa de animais e dentre elas podem-se destacar os modelos estatísticos, como por exemplo, regressão linear, que pode ser simples ou múltipla dependendo da quantidade de variáveis envolvidas no problema (ASSIS, 2013; NGO et al., 2013; SANGUN et al., 2009), no qual, pode-se obter uma relação matemática que permita estimar o desempenho do animal com relação às variáveis independentes utilizadas em sua modelagem (ASSIS, 2013).

Na literatura vem se destacando o uso de sistemas inteligentes e dentre eles destacam-se as redes neurais artificiais (RNAs) (SILVA et al., 2010). As RNAs são concepções fundamentadas no mecanismo do cérebro, sendo, por conseguinte, capazes de aprender com a experiência (SILVA et al., 2010) e constituem-se em uma excelente alternativa para a obtenção de soluções de problemas de um grande número de áreas do conhecimento humano (engenharia, biologia, economia, medicina, etc.), tais como: reconhecimento de padrão, análise, previsão, identificação, controle, etc. A grande vantagem do uso desta tecnologia refere-se à modelagem de problemas complexos, a qual é determinada pelo mecanismo de treinamento (aprendizado) supervisionado (estímulo entrada/saída) ou não supervisionado (baseado somente em estímulos de entrada), ou seja, uma vez concluído o treinamento, a rede armazena, em sua estrutura, o modelo do problema abordado, a princípio, não importando quanto complexo ele é (HAYKIN, 2008).

A proposta deste trabalho foi efetuar o estudo do ganho de massa em animais utilizando Regressão Linear Múltipla (RLM) e Redes Neurais Artificiais (RNAs), mais especificamente, a rede neural artificial *Perceptron* Multicamadas (MLP) com algoritmo *Backpropagation* (WIDROW AND LEHR, 1990) e, posteriormente, comparar os resultados obtidos pelos dois métodos.

Para desenvolver os métodos propostos foram levadas em consideração como variáveis independentes na análise: a variação de massa que o animal ganhou em 1 mês (02/09/2009 a 02/10/2009) e a variação de massa que o animal ganhou em 1 mês e 28 dias (02/09/2009 a 30/10/2009). A variável dependente representa o ganho de massa do animal em 9 meses e 17 dias, ou seja, quanto de massa o animal ganhou entre os dias 02/09/2009 a 19/05/2010 (ao final do experimento).

Os resultados obtidos são comparados com os valores reais de ganho de massa de cada animal e, então, é analisado o erro percentual médio (MAPE) e o coeficiente de determinação que indica o quanto a curva de ganho de massa prevista está próxima da curva de ganho de massa real. Esta análise se torna

importante, ao passo que, através da análise do ganho de massa pode-se estimar o quanto de ração deve ser fornecida ao animal para que ele obtenha este ganho de massa em um determinado período de tempo.

Portanto, a análise comparativa fundamenta que as redes neurais artificiais podem realizar a previsão de ganho de massa sem a necessidade da análise dos dados (correlação, autocorrelação, etc.) e sem a determinação de um modelo matemático, o que normalmente, é realizado através dos métodos estatísticos. Desta forma, tem-se um procedimento alternativo que irá auxiliar na análise de forma simples e precisa.

MATERIAL E MÉTODO

Os dados coletados foram obtidos de experimento realizado na UNESP de Ilha Solteira (Zootecnia) em 2009. Os animais usados no experimento foram bovinos o qual foram divididos em grupos e cada grupo foi nutrido por uma ração diferente: ração convencional, caroço de algodão e farelo de algodão e milho triturado. Os animais foram pesados em quatro dias distintos, sendo que, para cada dia de pesagem foram obtidos do experimento: o peso (massa) dos animais, expressos em kg, e o índice de escore de condição corporal (ECC) (FERNANDES et al., 2012).

O escore de condição corporal (ECC) é uma medida subjetiva baseada na classificação dos animais em função da cobertura muscular e da massa de gordura, desta forma, ele estima o estado nutricional dos animais por meio de avaliação visual e/ou tátil (palpação) realizado por um profissional da área (MACHADO et al., 2008). Existem diferentes escalas de escores, as quais variam no conceito, na topologia dos pontos de observação e na espécie animal à qual são aplicados. Neste trabalho, a escala utilizada foi a escala modificada por Spitzer (SPITZER, 1986), com aplicação de valores de 1 a 9 como mostra a Tabela 1.

Tabela 1. Escore de condição corporal (SPITZER, 1986)

ECC	Condição	Características
1	Debilitada	A vaca está extremamente magra, sem nenhuma gordura detectável sobre os processos vertebrais espinhosos e os processos transversos, e sobre os ossos da bacia e as costelas. A inserção da cauda e as costelas estão bastante proeminentes.
2	Pobre	A vaca ainda está muito magra, mas a inserção da cauda e as costelas estão menos projetadas. Os processos espinhosos continuam proeminentes, mas nota-se alguma cobertura de tecido sobre a coluna vertebral.
3	Magra	As costelas ainda estão individualmente perceptíveis, mas não tão agudas ao toque. Existe gordura palpável sobre a espinha, sobre a inserção da cauda e alguma cobertura sobre os ossos da bacia.
4	Limite	A individualização das costelas é menos óbvia. Os processos espinhosos podem ser identificados com o toque, mas percebe-se que estão mais arredondados. Existe um pouco de gordura sobre as costelas, sobre os processos transversos e sobre os ossos da bacia.
5	Moderada	O animal possui boa aparência geral. A gordura sobre as costelas parece esponjosa à palpação e as áreas nos dois lados da inserção da cauda apresentam gordura palpável.
6	Moderada boa	É preciso aplicar pressão firme sobre a espinha para sentir os processos espinhosos. Há bastante gordura palpável sobre as costelas e ao redor da inserção da cauda.
7	Boa	A vaca tem aparência gorda e claramente carrega grande quantidade de gordura. Sobre as costelas sente-se uma cobertura esponjosa evidente e também ao redor da inserção da cauda. Começam a aparecer "cintos" e “bolas” de gordura. Notase alguma gordura ao redor da vulva e na virilha.
8	Gorda	A vaca está muito gorda. É quase impossível palpar os processos espinhosos. O animal possui grandes depósitos de gordura sobre as costelas, na inserção de cauda e abaixo da vulva. Os "cintos" e as "bolas" de gordura são evidentes.
9	Extremamente Gorda	A vaca está nitidamente obesa, com a aparência de um bloco. Os "cintos" e as "bolas" de gordura estão projetados. A estrutura óssea não está muito aparente e é difícil de senti-la. A mobilidade do animal está comprometida pelo excesso de gordura.

Os valores da escala são definidos de acordo com a quantidade de reservas teciduais, especialmente de gordura e de músculos, em determinadas regiões do corpo, frequentemente associadas a marcos anatômicos específicos, como por exemplo, costelas, processos espinhosos da coluna vertebral, processos transversos da coluna vertebral, etc..

A análise do ganho de massa foi realizada em 36 bovinos o qual foram subdivididos em três grupos. A escolha do animal foi realizada aleatoriamente e desta forma cada grupo era composto por 12 animais. Cada grupo foi tratado com uma ração diferente, ou seja,

- Grupo 1 - ração convencional - controle;

- Grupo 2 - caroço de algodão - CA;
- Grupo 3 - farelo de algodão e milho triturado - FA.

Foram feitas quatro pesagens dos bovinos, sendo a primeira realizada no início do experimento e as três demais pesagens como descrito:

1. **Início do experimento - Primeira pesagem:** 02/09/2009 - P1;
2. **Segunda pesagem:** 02/10/2009 - P2;
3. **Terceira pesagem:** 30/10/2009 - P3;
4. **Final do experimento - Quarta pesagem:** 19/05/2010 - P4.

Os dados utilizados para análise do ganho de peso obtidos do experimento estão ilustrados nas Tabelas 2, 3 e 4, respectivamente. A Tabela 2 mostra a massa e o índice de escore de condição corporal (ECC) obtido cada bovino nos respectivos períodos de pesagem, para os animais que pertencem ao grupo 1, ou seja, que foram tratados com a ração convencional (grupo controle).

Tabela 2. Massa do animal e escore de condição corporal obtidas do experimento - controle

Animal	Períodos							
	P1		P2		P3		P4	
	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC
1	429,0	6	430,0	5	423,5	5	485,0	7
2	378,0	4	380,0	4	380,0	4	502,0	8
3	431,0	7	437,0	7	410,0	7	462,0	8
4	366,0	6	375,0	5	360,0	5	431,0	6
5	382,0	4	377,0	3	369,0	3	500,0	8
6	461,0	5	410,0	4	383,0	3	501,0	7
7	406,0	5	418,0	5	412,5	5	502,0	7
8	492,0	6	473,0	5	478,0	5	590,0	9
9	412,0	5	411,0	5	397,0	4	502,0	8
10	398,0	5	403,0	5	391,5	4	468,0	7
11	406,0	6	410,0	6	399,5	5	393,0	5
12	250,0	3	267,0	3	255,0	2	486,0	8

ECC - Escore de Condição Corporal.

É possível observar na Tabela 3 o valor de massa e o índice de escore de condição corporal (ECC) para cada bovino nos diferentes períodos de pesagem. É importante ressaltar que os animais que participaram deste grupo são distintos dos animais que estão descritos na Tabela 2. Estes animais pertencem ao grupo 2 no qual tem a condição imposta de serem alimentados com caroço de algodão (CA).

Tabela 3. Massa do animal e escore de condição corporal obtidas do experimento - CA

Animal	Períodos							
	P1		P2		P3		P4	
	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC
13	397,0	5	442,0	6	461,0	7	493,0	8
14	352,0	5	370,0	5	380,0	5	387,0	5
15	410,0	7	485,0	7	504,0	8	540,0	9
16	367,0	4	408,0	5	430,0	6	438,0	6
17	389,0	5	440,0	6	470,0	7	521,0	9
18	378,0	6	407,0	5	437,0	6	497,0	7
19	440,0	7	481,0	8	487,0	8	484,0	8
20	423,0	6	465,0	7	488,0	7	502,0	8
21	407,0	6	453,0	6	427,0	6	484,0	7
22	348,0	5	397,0	5	324,0	4	388,0	5
23	269,0	3	380,0	6	500,0	8	492,0	8
24	328,0	5	299,0	2	410,0	6	415,0	6

ECC - Escore de Condição Corporal.

A Tabela 4 apresenta o valor de massa e o índice de escore de condição corporal (ECC) para cada bovino nos períodos P1, P2, P3 e P4, respectivamente. Os animais que participaram deste grupo são dessemelhante dos animais que estão descritos nas Tabelas 2 e 3, respectivamente. Estes animais pertencem ao grupo 3 no qual tem a condição imposta de serem alimentados com farelo de algodão e milho triturado (FA).

Tabela 4. Massa do animal e escore de condição corporal obtidas do experimento - FA

Animal	Períodos							
	P1		P2		P3		P4	
	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC	massa (kg)	ECC
25	431,0	6	461,0	7	481,5	7	442,0	7
26	399,0	5	416,0	5	434,0	6	446,0	7
27	468,0	7	478,0	7	501,5	8	488,0	7
28	463,0	6	465,0	6	501,0	7	522,0	8
29	433,0	5	462,0	6	527,5	7	546,0	9
30	428,0	6	449,0	6	478,5	6	529,0	8
31	367,0	4	400,0	4	436,0	6	474,0	6
32	370,0	5	353,0	4	458,0	4	439,0	6
33	366,0	4	360,0	4	374,5	4	462,0	7
34	330,0	3	335,0	3	326,0	3	448,0	6
35	360,0	5	379,0	5	383,5	6	409,0	7
36	419,0	6	403,0	6	406,0	6	467,0	8

ECC - Escore de Condição Corporal.

Os métodos utilizados na análise do ganho de massa em animais são: Redes Neurais Artificiais e Regressão Linear Múltipla. Cada método será devidamente

apresentado, bem como, as estratégias utilizadas pelos métodos para efetuar a previsão do ganho de massa.

Redes Neurais Artificiais

As redes neurais artificiais são técnicas computacionais que apresentam um modelo matemático inspirado na estrutura neural de organismos inteligentes (neurônios) e que adquirem conhecimento através da experiência. A saída de um neurônio artificial é, então, dada pela Equação (1) (SILVA et al., 2010)

$$s_i = \sum_{j}^{n} w_{ij}\ x_j \tag{1}$$

em que:

s_i - saída linear do neurônio *i*;

x_j - entradas do neurônio;

w_{ij} - pesos do neurônio.

A saída não linear é obtida através da função sigmóide dada pela Equação (2)

$$v_i = \frac{1-e^{-\lambda s_i}}{1+e^{-\lambda s_i}} \tag{2}$$

no qual:

y_i - saída não linear do neurônio *i*;

λ - constante que determina a inclinação da função sigmóide.

A rede neural artificial utilizada foi a *Perceptron* Multicamadas (MLP) com algoritmo de treinamento *Backpropagation* (WIDROW AND LEHR, 1990). Este algoritmo possui um procedimento sistemático de adaptação de massa baseado no método do gradiente descendente, de forma que, o erro quadrático da rede neural seja minimizado.

Considerando-se o neurônio de índice *i* da rede, e utilizando-se o método do gradiente descendente (WIDROW AND LEHR, 1990), o ajuste de massa é formulado pela Equação (3):

$$V_i\,(k+1) = V_i\,(k) + 2\,\gamma\,\beta_i\,X_i \tag{3}$$

em que:

γ - parâmetro de controle da estabilidade ou taxa de treinamento;

X_i - vetor contendo as entradas do neurônio *i*;

V_i - vetor contendo os pesos do neurônio *i;*

$= [w_{0i}\ \ w_{1i}\ \ w_{2i}\ \ \ldots\ \ w_{ni}]^{T}$.

A Equação (4) descreve β_i quando o elemento *i* encontra-se na última camada, então

$$\beta_i = sgm'(s_i)\,\varepsilon_i \tag{4}$$

no qual:

$sgm'(s_i)$- derivada da função sigmóide com relação a s_i;

ε_i = $d_i - y_i$;

d_i - saída desejada do neurônio *i*.

Se o elemento *i* encontrar-se nas demais camadas, β_i é determinado pela Equação (5):

$$\beta_i = sgm'(s_i) \sum_{k \in R(j)} w_{ik}\ \beta_k \quad (5)$$

no qual:

$sgm'(s_i)$ - derivada da função sigmóide com relação a s_i.

$R(j)$ - conjunto dos índices dos elementos que se encontram na fileira seguinte à fileira do elemento i e que estão interligados ao elemento i.

Previsão do ganho de massa utilizando rede neural artificial

A rede neural *Perceptron* Multicamadas com algoritmo *Backpropagation* foi usada para realizar a previsão do ganho de massa em bovinos submetidos a três tipos de alimentação. Para compor o conjunto de dados de entrada da rede neural foram utilizados os valores de massa dos animais e o escore de condição corporal (ECC) analisados em um determinado período de tempo.

O conjunto de dados de entrada da rede neural MLP via *Backpropagation* contêm dados que compreendem dois períodos (1º período e 2º período), e o terceiro período completa o banco de dados de entrada da rede neural para o diagnóstico (3º período), ou seja, previsão do ganho de massa compreendido do início ao fim do experimento. Os três períodos avaliados na análise de massa do animal são:

1º período: primeira pesagem (02/09/2009) até a segunda pesagem (02/10/2009);

2º período: primeira pesagem (02/09/2009) até a terceira pesagem (30/10/2009);

3º período: primeira pesagem (02/09/2009) até a quarta pesagem (19/05/2010).

Os vetores de entrada, $\boldsymbol{X}(h)$, e a saída, $\boldsymbol{Y}(h)$, da rede neural são definidos, respectivamente, pelas Equações (6) e (7), respectivamente.

$$\boldsymbol{X}(h) = [P_n(i)\ \ P_n(i+h)\ \ G_{ECC}]^T, \quad (6)$$

$$\boldsymbol{Y}(h) = [GP_n] \quad (7)$$

em que:

i - índice que corresponde primeira pesagem = 1;

h = período de tempo = 1 e 2;

P_n - massa normalizado do animal;

G_{ECC} - ganho de escore de condição corporal

$$= ECC(i + h) - ECC(i); \quad (8)$$

GP_n - ganho de massa do animal normalizado avaliado no período h

$$= P_n(i + h) - P_n(i). \quad (9)$$

Observa-se que o vetor $\boldsymbol{X}(1)$ refere-se às informações do 1º período e o vetor $\boldsymbol{X}(2)$ refere-se ao 2º período. Portanto, tem-se que o conjunto de dados de entrada da rede neural para a fase de treinamento contempla informações dos dois períodos distintos, $\boldsymbol{X}(1)$ e $\boldsymbol{X}(2)$, sendo que em cada período a análise foi efetuada levando em consideração a massa de 12 bovinos. Desta forma, a quantidade de dados que serão utilizados na fase de treinamento da rede neural contém um total de 24 informações (12 dados referentes ao primeiro período e 12 dados relacionados ao segundo período). O conjunto de dados de saída da rede neural para fase de treinamento é representado pelos vetores: $\boldsymbol{Y}(1)$ que é a saída correspondente ao vetor $\boldsymbol{X}(1)$ e $\boldsymbol{Y}(2)$ que é a saída referente ao vetor $\boldsymbol{X}(2)$. Ambos os vetores de saída, $\boldsymbol{Y}(1)$ e $\boldsymbol{Y}(2)$, representam o quanto o

animal terá de ganho de massa nos 1^{o} e 2^{o} períodos, respectivamente.

O vetor ***X***(3) pertence ao conjunto de dados de entrada da rede neural na fase de diagnóstico da rede neural e este é caracterizado pelo conjunto de dados que contêm a variação de peso dos animais do início até o final do experimento, o qual corresponde ao 3^{o} período. Consequentemente, o vetor ***Y***(3) será o conjunto de dados de saída a ser previsto, ou seja, o ganho de massa do animal.

Portanto, a partir das informações cedidas a rede neural pelos dados de entrada, ela terá a capacidade de generalizar o ganho de massa do animal para um determinado período de tempo (período este que não está presente no conjunto de dados de entrada da rede neural) utilizando períodos anteriores a este. A rede neural MLP com algoritmo *Backpropagation* foi programada em Fortran.

Regressão Linear Múltipla

O método de regressão linear, simples ou múltipla, obtém uma reta ou curva que se aproxima melhor dos dados que são fortemente correlacionados. Como as variáveis independentes associadas ao problema do ganho de massa são: massa do animal, no início e fim do período, e escore de condição corporal, a equação de regressão linear que melhor irá se ajustar a este conjunto de dados é a regressão linear múltipla (ASSIS, 2013).

Uma equação de regressão linear múltipla é dada pela Equação (10):

$$\hat{y} = m_1 x_1 + m_2 x_2 + \cdots + m_v x_v + b \tag{10}$$

no qual, x_1, x_2, …, x_v, são variáveis independentes, b é a interseção com a variável dependente y. Os coeficientes m_1, m_2, …, m_v são estimados através do método dos mínimos quadrados.

Previsão do ganho de massa utilizando regressão linear múltipla

Os dados utilizados no cálculo da equação de regressão linear múltipla foram idênticos ao utilizado pela rede neural artificial. Logo, para obter a curva de regressão foram adotados: a massa do animal no início do período, a massa do animal no fim do período e o ganho de escore de condição corporal, Equação (8), e estão denotados pelas Equações (11), (12) e (13), respectivamente, e a variável dependente que corresponde ao ganho de massa do animal, Equação (9), está representado na Equação (14):

- $x_1 = \mathrm{P}_n(i)$ (11)
- $x_2 = \mathrm{P}_n(i+h)$ (12)
- $x_3 = \mathrm{G}_{ECC}$ (13)
- $y = f(x_1, x_2, x_3) = \mathrm{GP}_n$ (14)

para h = 1, 2 e 3, e portanto, tem-se a equação de regressão dada pela Equação (15)

$$\hat{y} = m_1 x_1 + m_2 x_2 + m_3 x_3 + b \tag{15}$$

no qual os coeficientes m_1, m_2 e m_3 serão determinados através do método dos mínimos quadrados, para assim, obter a equação que melhor se ajusta ao conjunto

de dados (controle, caroço de algodão e farelo de algodão com milho triturado).

Critérios para análise da previsão do ganho de massa

Para uma análise da precisão do ganho de massa previsto, são definidos o erro percentual absoluto médio (*MAPE*) e o coeficiente de determinação R^2, respectivamente, pelas Equações (16) e (17).

$$MAPE = \frac{1}{N} \{ \sum_{i=1}^{N} (|GP_{real} - GP_{MLP}|) / GP_{real} \} \times 100 \quad (16)$$

$$R^2 = \sum_{i=1}^{N} (GP_{MLP} - GP_{médio})^2 / \sum_{i=1}^{N} (GP_{real} - GP_{médio})^2 \quad (17)$$

em que:

GP_{real} - ganho de massa real do animal avaliado no terceiro período;

$GP_{médio}$ - média do ganho de massa real do animal avaliado no terceiro período;

GP_{MLP} - ganho de massa do animal obtido pela MLP avaliado no terceiro período;

N - número de animais.

RESULTADOS E DISCUSSÃO

Este trabalho realizou a previsão do ganho de massa de um animal utilizando duas metodologias: redes neurais artificiais e regressão linear múltipla.

A primeira técnica utilizada na previsão do ganho de massa em animais (bovinos) foi à rede neural artificial *Perceptron* Multicamadas (MLP) com algoritmo *Backpropagation*. A Tabela 5 apresenta os parâmetros utilizados para o treinamento da rede neural MLP com algoritmo *Backpropagation* aplicado ao problema de previsão do ganho de massa.

Tabela 5. Parâmetros da Rede Neural MLP com Algoritmo *Backpropagation*

Parâmetros da Rede Neural Artificial	Alimentação		
	Controle	CA	FA
Número de dados	24	24	24
Número de camadas	3	3	3
Número de neurônios por camada	3-8-1	3-8-1	3-8-1
Tolerância	$2x10^{-3}$	$2x10^{-3}$	$2x10^{-3}$
Taxa de treinamento	0,5	1,5	1,5
Termo momento	0,9	0,9	0,9

CA - Caroço de Algodão; FA - Farelo de Algodão e Milho Triturado.

A fase de diagnóstico teve o propósito de estimar o ganho de massa do animal obtido durante todo o experimento, ou seja, do início ao final do experimento. Portanto, a quantidade de dados de entrada da rede neural MLP utilizados na fase de diagnóstico é doze (“12”) que corresponde ao total de bovinos usados no experimento (para cada tipo de alimentação). Para validar a metodologia fez-se uma análise levando em consideração os três tipos de alimentações (controle, CA, FA).

Nas Figuras 1 a 3 observa-se um diagrama de barras no qual ilustram os ganhos de massa, em kg, dos animais submetidos ao experimento levando em consideração, respectivamente, as diferentes alimentações: controle, caroço de algodão e farelo de algodão e milho

triturado. No gráfico a barra em azul ilustra o ganho de massa real e a barra vermelha o ganho de massa previsto pela RNA.

Na Figura 1 é possível observar que os valores previstos pela rede neural artificial estão coerentes com os valores de ganho de massa real referente a cada animal analisado, ou seja, 12 bovinos que foram alimentados com a ração convencional.

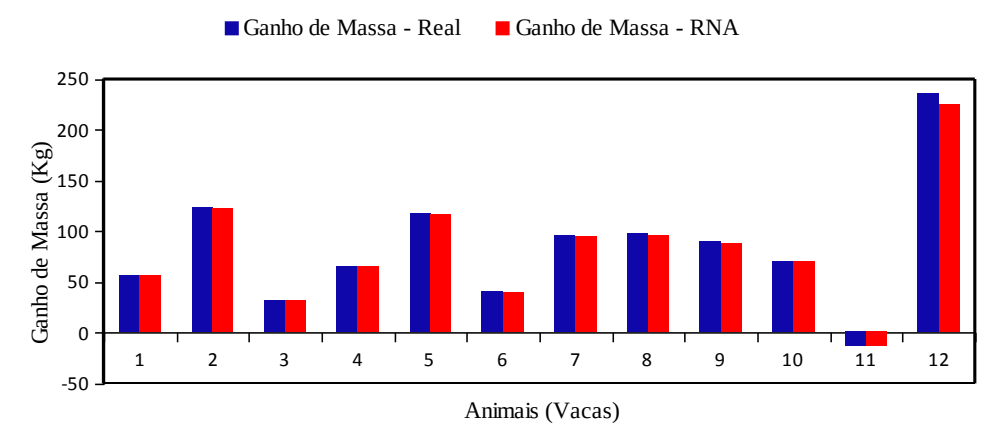

Figura 1. Visualização do ganho de massa real e obtida pela RNA – controle

Para os animais tratados com caroço de algodão, observam-se pela Figura 2 que os valores previstos pela rede neural artificial estão muito próximos dos valores de ganho de massa real.

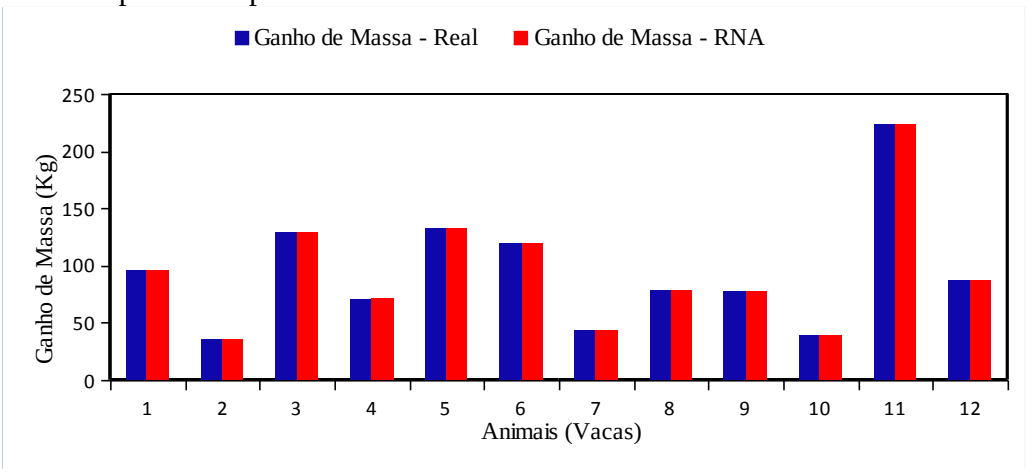

Figura 2. Visualização do ganho de massa real e obtida pela RNA – CA

Na Figura 3 pode-se notar o mesmo comportamento dos resultados obtidos na Figura 2, ou seja, os valores previstos pela rede neural artificial estão bem próximos dos valores de ganho de massa real. Os resultados obtidos foram para os animais tratados com farelo de algodão e milho triturado.

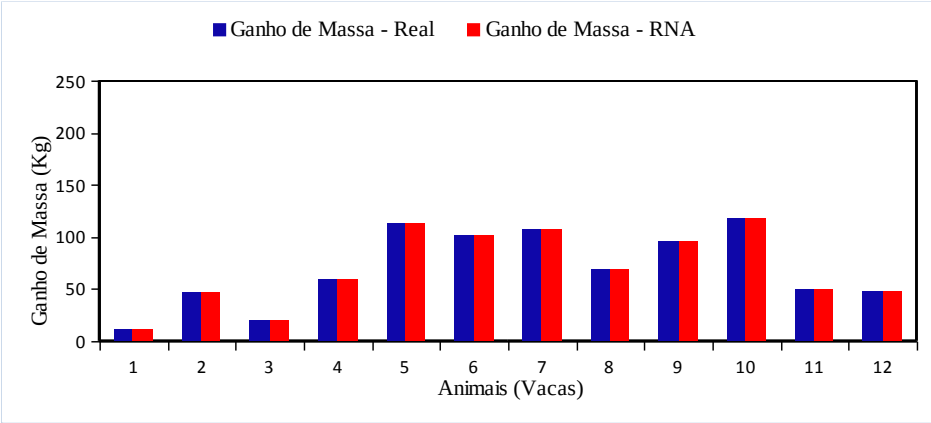

Figura 3. Visualização do ganho de massa real e obtida pela RNA - FA

A segunda técnica usada foi o modelo de regressão linear múltipla sendo que as equações de regressão foram obtidas utilizando os dados correspondentes a massa do animal no início do experimento e no período em análise e o índice de escore de condição corporal.

Levando em consideração os dados referentes à massa do animal no início do experimento e no período em análise (1º e 2º períodos) e o índice de escore de condição corporal têm-se a equação de regressão linear múltipla para o grupo 1 (ração convencional – controle) obtido pelo método dos mínimos quadrados, cujo modelo foi desenvolvido no LibreOffice versão 5.1.6.2, dada pela Equação (18).

$$\hat{y} = -1{,}000276x_1 + 0{,}999814x_2 - 0{,}000016x_3 + 0{,}000215 \quad (18)$$

Substituindo os dados das variáveis definidas pelas Equações (11), (12) e (13), referentes ao 3º período, na Equação (18) têm-se os valores de ganho de massa ilustrados na Figura 4. Nota-se que os valores de ganho de massa obtidos pela curva de regressão são muito próximos do valor de ganho de massa real.

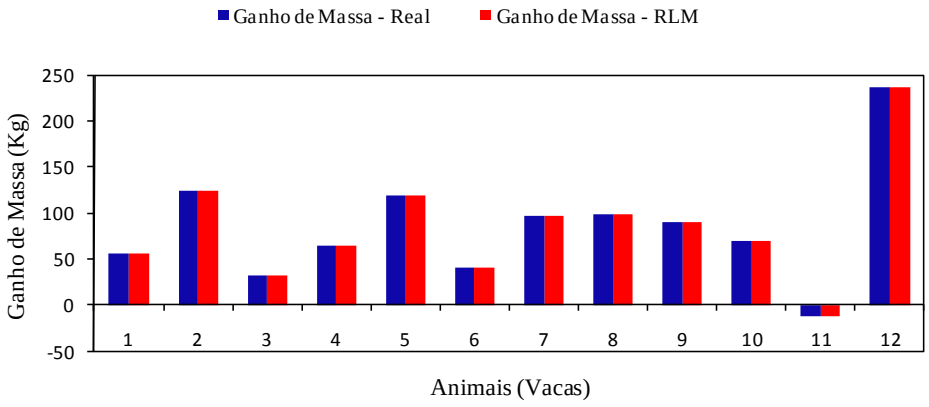

Figura 4. Visualização do ganho de massa real e obtida pela RLM – controle

Para os dados referentes à massa do animal no início do experimento e no período em análise (1º e 2º períodos) e o índice de escore de condição corporal têm-se para o grupo 2 (caroço de algodão) a equação de regressão linear múltipla definida pela Equação (19).

$$\hat{y} = -0{,}999668x_1 + 0{,}999928x_2 + 0{,}000011x_3 - 0{,}000121 \quad (19)$$

Empregando os dados das variáveis definidas pelas Equações (11), (12) e (13), referentes ao 3º período, na Equação (19) é obtido os valores de ganho de massa que pode ser visto na Figura 5 e através desta é possível verificar a proximidade dos valores reais.

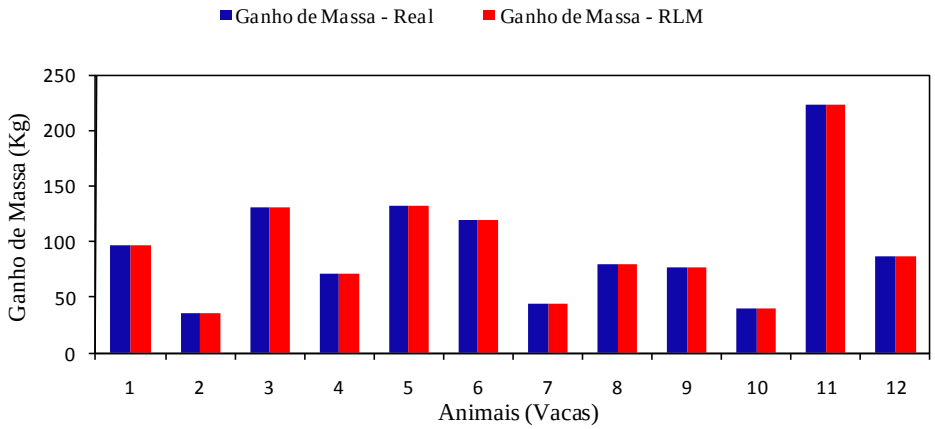

Figura 5. Visualização do ganho de massa real e obtida pela RLM – CA

Utilizando o mesmo procedimento mencionado para os grupos 1 e 2, tem-se a determinação da equação de regressão linear múltipla correspondente ao grupo 3 (farelo de algodão e milho triturado) dada pela Equação (20).

$$\hat{y} = -1{,}000111x_1 + 0{,}999972x_2 - 0{,}000015x_3 + 0{,}000060 \quad (20)$$

As variáveis definidas pelas Equações (11), (12) e (13), referentes ao 3º período, usadas na Equação (20) fornece os valores de ganho de massa para o grupo 3 que pode ser visto na Figura 6. Para este grupo constata-se que os valores obtidos da equação de regressão dada pela Equação (20) são muito semelhantes aos valores de ganho de massa real.

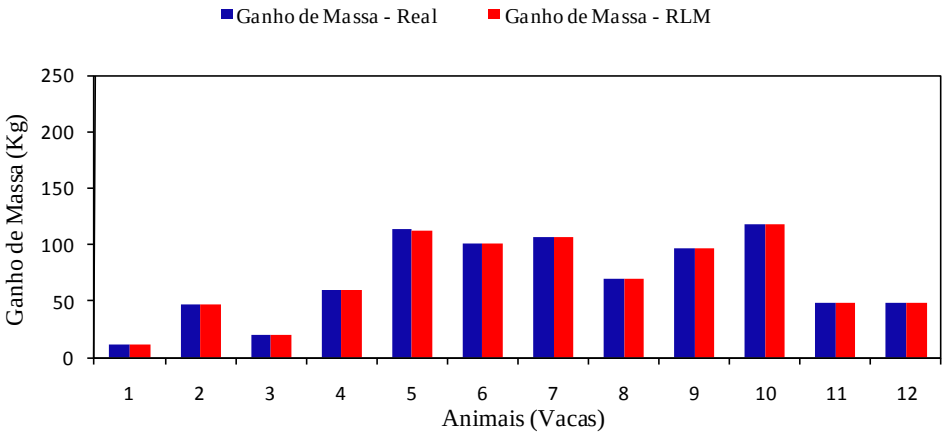

Figura 6. Visualização do ganho de massa real e obtida pela RLM - FA

Pode-se observar nas Figuras 1 e 4 que os ganhos de pesos reais variam entre a perda de massa de 13 kg a um ganho de massa de 236 kg. Já nas Figuras 2 e 5 as

variações limitam-se apenas aos ganhos de massa com um ganho mínimo de 35 kg a um ganho máximo de 223 kg. Através das Figuras 3 e 6, nota-se que os animais alimentados com o farelo de algodão e milho triturado, obtiveram os menores ganhos de massa que variam 11 kg e 118 kg.

A partir dos resultados mostrados pela Tabela 6, observa-se através do MAPE uma precisão na previsão do ganho de massa dos animais, ou seja, os resultados obtidos pela rede neural *Perceptron* Multicamadas com algoritmo *Backpropagation* são aproximadamente idênticos aos ganhos de massa real dos animais com um erro percentual médio, MAPE variando entre 1,4% a 0,10%. Também, pode-se observar o valor do MAPE obtido pela equação de regressão linear múltipla e estes demonstram uma boa precisão na previsão do ganho de massa dos animais, uma vez que, foi possível observar que estes valores de erros refletem nas Figuras 1 a 6, valores de ganhos previstos muito próximos de valores de ganhos reais. Nota-se que o erro percentual médio, MAPE variando entre 0,04% a 0,09%.

Tabela 6. Análise comparativa dos ganhos massa real e obtido pelos métodos: RNA e RLM

Método Utilizado	**Avaliação**	**Alimentação**		
		Controle	**CA**	**FA**
RNA	Erro Percentual Médio - *MAPE* (%)	1,43	0,18	0,10
	Coeficiente de Determinação - R^2	0,92	1,00	1,00
RLM	Erro Percentual Médio - *MAPE* (%)	0,07	0,04	0,09
	Coeficiente de Determinação - R^2	1,00	1,00	1,00

RNA - Rede Neural Artificial; RLM - Regressão Linear Múltipla, CA - Caroço de Algodão; FA - Farelo de Algodão e Milho Triturado.

A Figura 7 ilustra o ajuste dos valores de ganhos de massa previstos pela RLM e pela RNA.

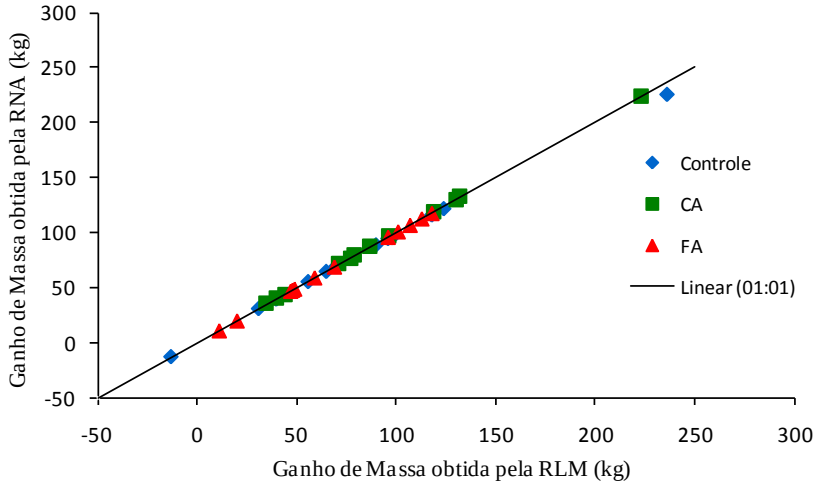

Figura 7. Diagrama de dispersão dos ganhos de massa obtidos pela RLM e pela RNA para as três alimentações (controle, caroço de algodão, farelo de algodão e milho triturado)

Observa-se na Figura 7 que as previsões realizadas por ambas combinam, ou seja, são muito parecidas. Isto pode ser evidenciado pelo coeficiente de determinação, R^2, mostrado na Tabela 7.

Tabela 7. Análise comparativa dos valores de ganhos de massa obtidos pela RNA e RLM

Avaliação	Alimentação		
	Controle	CA	FA
Erro Percentual Médio - *MAPE* (%)	1,35	0,18	0,11
Coeficiente de Determinação - R^2	0,92	1,00	1,00

CA - Caroço de Algodão; FA - Farelo de Algodão e Milho Triturado.

Na Tabela 7 nota-se que os resultados obtidos de ganho de massa com relação ao caroço de algodão e farelo de algodão com milho triturado para os dois métodos (RNA e RLM) foram muito semelhantes, ou seja, ambos os métodos fornecem resultados de previsão parecidos ou muitos próximos. O que não ocorre com relação ao grupo o qual foi alimentado pela ração controle. Esta diferença está associada a um possível erro experimental que pode ser observado na Tabela 8.

Tabela 8. Massa animal para o grupo 1 tratado com a ração convencional

Animal	Massa do animal (Kg)			
	Início do Experimento	1ª Pesagem	2ª Pesagem	3ª Pesagem
8ª	492	473	478	590
12ª	250	267	255	486

Nota-se que os animais selecionados (aleatoriamente) para compor o grupo 1, ou seja, o grupo que recebeu a ração convencional possuem uma grande variação de massa, sendo que, entre os 12 bovinos que fazem parte do grupo 1, nota-se através da Tabela 2 que o animal mais leve tem a massa inicial de 250 kg e o animal mais pesado a massa de 492 kg. Nos demais grupos esta diferença entre as massas dos animais (mais leve e mais pesado) é em torno de 28,75% a menos para o grupo alimentado com caroço de algodão (CA), e por volta de 42,5% a menos para a alimentação feita por farelo de algodão com milho triturado (FA). Constata-se também, que o animal mais pesado do grupo ao final do experimento tem um ganho de massa de 98 kg e o animal mais leve do grupo tem um ganho de 236 kg. Esta diferença existente entre os animais enfoca uma variação que não existe com os demais dados dentro do grupo. Portanto, esta diferença acaba interferindo na generalização realizada pela rede neural, uma vez que, esta não irá ocorrer novamente.

Observando os resultados obtidos pela rede neural artificial nota-se que a melhor previsão do ganho de massa foi para o grupo farelo de algodão com milho triturado e esta está associada ao fato do grupo de animais serem mais homogêneo, na qual, a variação da massa foi a menor o possível. Isto representa uma característica imprescindível para o funcionamento de uma rede neural artificial, uma vez que, ela tem a capacidade de generalizar o aprendizado para um determinado comportamento.

Os resultados obtidos demonstram que as duas metodologias propostas são eficientes e precisas, uma vez que, o erro percentual médio, MAPE, varia em torno de 0,10% a 1,4% para a previsão feita através da rede neural artificial e em torno

de 0,04% a 0,09% para a previsão realizada pela regressão linear múltipla. Através do coeficiente de determinação, R^2, observa-se a garantia de ajuste ou precisão da previsão comparada aos ganhos de massa reais para as duas técnicas utilizadas.

CONCLUSÕES

As metodologias propostas para estimar o ganho de massa de peso animal se basearam na regressão linear múltipla e redes neurais artificiais.

Na utilização dos métodos estatísticos (regressão linear múltipla) deve-se dispor de um modelo estatístico e matemático. Então, a qualidade das soluções depende de uma boa escolha do modelo, baseado, principalmente, na determinação das influências de cada variável adotada no modelo considerado. Esta atividade é bastante complexa e exige, portanto, um grande empreendimento de esforços, conhecimento e habilidade dos profissionais da área. Com os dados coletados do experimento notou-se que o modelo que mais se adequava era o modelo de regressão linear múltipla, e isto se torna evidente, uma vez que, constatam-se as variações na massa e no ECC do animal a cada pesagem (P1, P2, P3 e P4). Avaliou-se a correlação existente entre os dados reunidos do experimento, para assim, selecionar as variáveis a serem utilizadas como variáveis independentes no modelo de regressão linear múltipla.

Já as redes neurais artificiais têm se mostrado de forma eficiente no desenvolvimento de séries temporais, principalmente, porque não há necessidade de se conhecer o modelo matemático, ou seja, as correlações entre variáveis são estabelecidas de forma autônoma. A escolha da rede neural *Perceptron* via algoritmo *Backpropagation*, deve-se ao fato, desta arquitetura ser considerada na literatura um *benchmark* em termos de precisão (WIDROW AND LEHR, 1990).

Os resultados obtidos demonstram que as duas metodologias propostas são eficientes e precisas, uma vez que, o erro percentual médio, MAPE, varia em torno de 0,10% a 1,4% para a previsão feita através da rede neural artificial e em torno de 0,04% a 0,09% para a previsão realizada pela regressão linear múltipla. Através do coeficiente de determinação, R^2, observa-se a garantia de ajuste ou precisão da previsão comparada aos ganhos de massa reais para as duas técnicas utilizadas.

Com isso, pode-se concluir que ambas as técnicas utilizadas produziram resultados satisfatórios e muito semelhantes, mesmo tendo abordagens diferentes, sendo assim, ressalta-se que a escolha da estratégia para a obtenção destes resultados dependerá do perfil do profissional.

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An artificial immune system with continuous-learning for voltage disturbance diagnosis in electrical distribution systems

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ABSTRACT

This paper presents a new artificial immune algorithm with continuous-learning, which is inspired by the biological immune system, to realize the voltage diagnosis in electrical distribution systems. This conception allows one to compose a diagnosis system that can continuously learn without reinitialization when new disturbances occur due to the evolution of the electrical system. Two artificial immune algorithms, which are the negative selection algorithm and the clonal selection algorithm, are used for the pattern recognition process and the learning process, respectively. The principal application of this new method aids the operation during failures, supervises the protection system, and can evolve with the power systems to continuously acquire new knowledge. This new methodology has a direct impact in the area of diagnosis in electrical systems, as well as, in the pattern recognition problem, because the main contribution and novelty of this method is the continuous learning capability, which enables the system to learn unknown patterns without having to restart the knowledge. This is the major advantage of this methodology. To evaluate the efficiency and performance of this new method, failure simulations were performed in a real distribution system with 134 buses using the EMTP software. The results show robustness and efficiency.

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1. Introduction

Nowadays, a new concept of electrical power systems called Smart Grids requires several investments to transform the electrical systems that provide modern technologies to generate, transmit and principally distribute electrical energy systems (Dongli, Meng, & Song, 2011), (Gungor et al., 2011). The smart grid concept is based on an intense usage of automation, computation and telecommunication technologies to monitor and control electrical systems that allow the implementation of new control, communication, protection and optimization strategies to improve the efficiency of the currently available systems (Gungor et al., 2011).

Thus, several technologies are used, particularly digital and information technology (Alag et al., 2001), which allow the development of integrated systems that combine acquisition, analysis and data processing techniques, to provide the necessary assistance to automation, control and decision-making processes (Northcote-Green & Wilson, 2007).

Considering the failure diagnosis for the smart grid, those systems must be based on auto-restoration idea, i.e., they must be able to automatically detect, analyze, respond and restore failures (Dongli et al., 2011) without human intervention. Thus, using intelligent techniques (artificial neural networks, fuzzy logic, artificial immune systems, etc.) is an alternative to the diagnosis problem. Several techniques based on this concept have been used to aid the operators in executing routines in electrical systems, which provides security, velocity and efficiency in planning corrective actions.

However, most failure-diagnosis methods based on intelligence proposed to the actual electrical systems have a learning strategy to obtain the knowledge, on which all efficiency and robustness depend. Normally, these strategies (training or learning algorithms) must always reinitialize the learning process when the system is executed or a new disturbance occurs; therefore, the system is not significantly intelligent. The system does not constantly learn, and when a new pattern appears, every learning process is executed again.

A modern and intelligent failure-diagnosis system to the smart grids must continuously learn, follow innovations and modernizations of the electrical system, which include new types of prominent failures, learn with experience and constantly evolve.

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Developing a system with these characteristics is a complex task. In the literature, many studies present methods to solve the problem of the diagnosis of disturbances, as in (Uyar, Yildririm, & Gencoglu, 2008), (Oleskovicz et al., 2009), (Zhang, Li, & Hu, 2011), (Lima, Lotufo, & Minussi, 2014), (Lima, Minussi, Bessa, & Fidalgo, 2015), however these approaches do not consider the minimum requisites to the smart grids, i.e., the objective to provide continuous-learning to a pattern recognition system. This requisite based on auto-restoration idea, i.e., continuous learning provide the ability to perform the detection, analyze, respond and classify failures without human intervention.

In the literature, there are a few approaches that propose methodologies to develop a system with continuous training to realize diagnosis. For example, in Pham and Cham (2007) is proposed an online learning asymmetric boosted classifiers for object detection. This approach is one of the first to try the continuous learning in techniques of pattern recognition. In the reference (Marchiori, Silveira, Lotufo, Minussi, & Lopes, 2011) the authors presents a methodology to analyze the transient stability in electrical energy systems using, a Euclidean ARTMAP neural network with continuous training. In this paper the Euclidean ARTMAP neural network was modified to learning continuously and identify novelties in the analysis of the transient stability in electrical systems, i. e., the Euclidean ARTMAP neural network was able to identify unknown patterns, which were not learned yet in the training process. In Barros, Tonelli-Neto, Decanini, and Minussi (2015) the authors present a method to detection and classification of voltage disturbances in electrical power systems using a modified Euclidean ARTMAP neural network with Continuous Training. In this strategy the neural network is initially trained to obtain the knowledge. In the online process, if a pattern in analysis is unknown by the diagnostic system, is executed an online learning phase, and starting this, the diagnostic system is able to identify and classify the pattern that was unknown to the system. This method presents a precision of 90.32% in the voltage disturbance diagnosis.

In this sense, this work aims to propose a new approach to develop a system with continuous training to realize the voltage disturbances diagnosis in electrical distribution systems based on the Biological Immune System (BIS). Two immune algorithms were used to compose this system: the Negative Selection Algorithm (NSA) (Forrest, Perelson, Allen, & Cherukuri, 1994) and the clonal selection algorithm (CLONALG) (de Castro & Von Zuben, 2000).

The artificial immune algorithm, which is inspired by the BIS, is a promising technique in Intelligent Computing (IC). The BIS computationally reproduces the principal characteristics and the propriety functionalities and abilities. It has a stable, reliable, and adaptive architecture, allowing the continuous inclusion of the training module. This propriety allows the system to introduce new experiences and knowledge without the need to reinitialize the immune memory of the system. This is one advantage of using Artificial Immune Systems, which allow the possibility of continuous training unlike in other techniques such as using Artificial Neural Networks (ANNs). In some ANNs, to improve the training, it is necessary to reinitialize the process, consequently destroying the knowledge previously acquired, except for a few architectures that have the plasticity propriety as in the neural networks of the ART (Adaptive Resonance Theory) family (Carpenter, Grossberg, Markuzon, Reynold, & Rosen, 1992). The main contribution and novelty of the approach proposed in this paper is the paradigm shift, that is, enable continuous learning for the pattern recognition methods. This ability allows great evolution to intelligent systems, where the methods are able to learn in online mode, acquiring knowledge of the unknown patterns for the system.

To evaluate and validate the efficiency of this new approach, the results were compared with a conventional diagnosis system based only in the NSA, proposed and presented in Lima, Lotufo, and Minussi (2013).

This paper is organized as follows. Section 2 discusses the BIS, Section 3 focuses on the NSA, and Section 4 tackles the CLONALG. The modeling and simulations is presented in Section 5. The proposed methodology is presented in Section 6, and the applications and results are discussed in Section 7. The conclusions are presented in Section 8.

2. Biological immune system

The BIS is the principal defense of organisms against several infective agents that invade/infect the human body. In this case, the BIS must instantaneously act by effectively responding to the invading agents and identifying them to protect the human body (indicating the disease). There are two types of responses: the innate immune system and the adaptive immune system.

The innate immune system is the first defense line with a fast response characterized by dendritic cells (APC- Antigen Presenting Cell) and phagocyte (Granulocytes, Macrophages, etc.), which are responsible for ingesting strange particles to the organism and other types of defenses, such as physical barriers (skin) and chemicals (de Castro, 2001).

The adaptive immune system is the second level, which can recognize microorganisms, such as viruses, bacteria, fungi, protozoa, helminthes, and some types of worms. The adaptive immune system is responsible for realizing the learning process on the infective agents on the first contact with the antigens, i.e., the first exposure to the infective agent. In addition, it is responsible for creating memory immune cells from the first exposure to the infective agent to accelerate a response to this type of infective agent in future exposure (de Castro & Timmis, 2002).

2.1. Operation of the pattern recognition of the BIS

The biological organism, particularly the human body, is composed of several cells and molecules that work in harmony and respond to strange agents that enter the organism, which are called antigens.

The BIS has fundamental characteristics such as pattern recognition, learning and neutralization of infective agents. These steps represent an immune response, which is illustrated on Fig. 1.

Step (I) represents the BIS exposure to an antigen (infective agent). The BIS can react with two types of responses: the innate immune response, which begins at step (II), and the adaptive immune response, which begins at step (VII). If the process begins with the innate immune system (step (II)), the infective agent is considered to be an unknown agent by the BIS, and the infective cell in the organism must be identified and informed *a posteriori*; then, the adaptive immune system begins an adaptive response (learn and generate antibodies to neutralize the antigen). If the process begins with the adaptive immune system (step (VII)) the antigen is known by the BIS, and the detection process is realized by the memory lymphocytes in the organism.

In step (II), the innate immune process begins when some antigen (infective agent) is ingested by a dendritic cell (APC- Antigen Presenting Cell). At this phase, the antigens are ingested and disintegrated into antigen peptides. In step (III), the pieces of peptides are linked to the MHC (Major Histocompatibility Complex) molecules and presented to the surface of the dendritic cell.

In step (IV), the T lymphocytes, which have receptive molecules at the surface, can recognize/identify different antigen MCH/peptides that are processed by the dendritic cells, i.e., when there is a link (combination), the antigen is recognized,

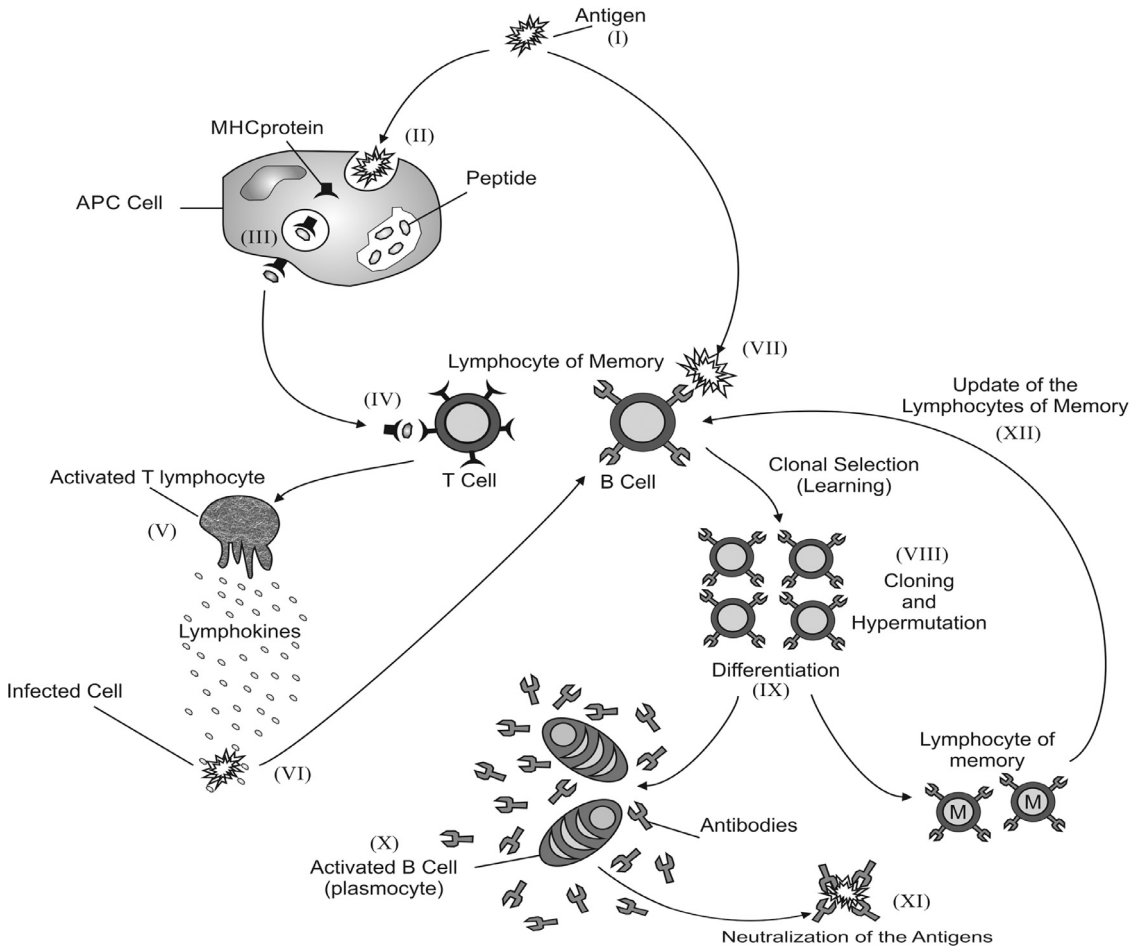

Fig. 1. Representation of the BIS.

and the lymphocyte state is activated. This step represents the proper/non-proper discrimination of the organism, which distinguishes proper cells from the infective agents (de Castro, 2001), (Dasgupta, 1998). Based on this pattern recognition principle, reference (Forrest et al., 1994) proposes the NSA. After the antigen detection, at step (V), the lymphocytes T are activated and secrete chemical signals (lymphokine) that suggest to other BIS components that a cell is infected.

After an infected cell is identified (step (VI)), B lymphocytes can recognize the infected cell and start the learning process, which is more commonly known as the principle of clonal selection (step (VII)). In this case, B lymphocytes can recognize free identified antigens without ingesting and digesting the presenting cells (APC).

When B cells recognize an antigen (by indication or memory), the clonal selection process begins, which is also known as learning process (step VIII). In this step, the lymphocytes pass under a clone and hyper-mutation process, which generates a set of B lymphocytes in the organism.

Then, step (IX) is a discerning process, where B lymphocytes with high affinity are separated to pertain to the memory set, and those with low affinity are activated and transformed in plasmocytes (step X) to secrete antibodies. In step (XI), the BIS-detected antigens are neutralized by the generated antibodies, which destroys the risk (disease).

In step (XII), the high-affinity lymphocytes in the distinguishing process are transformed and pertained to memory lymphocytes to substitute for those with low affinity. The new memory lymphocytes begin to circulate in the organism to ensure an efficient and fast response to a future exposure to the same antigen (infective agent).

It is emphasized that the entire process is realized with the cooperation of the cells that form the BIS, each of which is responsible for a relatively simple function, and the set that realizes an extremely complex work (de Castro, 2001), (Dasgupta, 1998).

This paper was inspired by the BIS operation, pattern recognition and learning processes.

3. Negative selection algorithm

The NSA is based on the pattern recognition process of the biological immune system and elaborated as a computational model. The NSA is proposed in Forrest et al. (1994). In this algorithm (negative selection) the T lymphocytes that occur at a specific time and discriminates the proper/non-proper cells in the organism. The algorithm is executed in two phases, such as presented in Figs. 2 and 3 (de Castro, 2001):

The censor phase of the NSA consists primarily of generating a detector set from the data that were randomly chosen and verifying which data can recognize a non-proper pattern. The detectors are similar to the mature T cells that can recognise pathogenic agents (Lima et al., 2013).

The monitoring phase consists of monitoring a system to identify a change in behaviour; thus, this monitoring phase classifies the change using the detector set that was created in the censor phase. The censor phase occurs in the off-line mode, and the monitoring phase occurs in real-time mode (de Castro & Timmis, 2002), (Dasgupta, 1998).

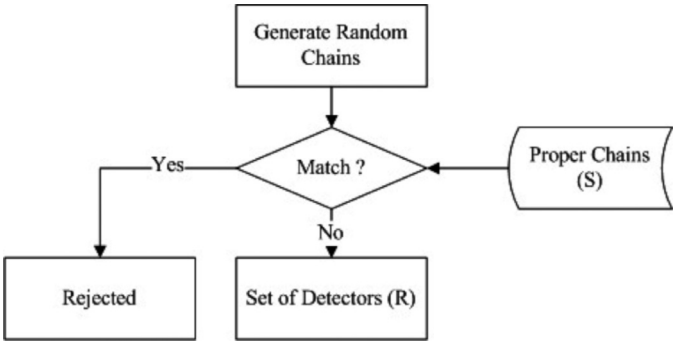

Fig. 2. Flowchart of the censor phase of the NSA.

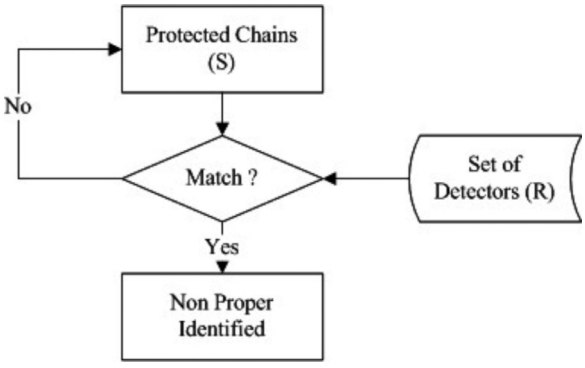

Fig. 3. Flowchart of the monitoring phase of the NSA.

The chains at the NSA represent the patterns by data structures (vectors). These chains can be classified in two types of NSA: the antigens (*Ag*) and the lymphocytes (*Ab*). The *Ag* is the signal to be analyzed, unknown by the NSA and represented by expression (1). The detectors, that is, the lymphocytes (*Ab*), are expressed according to expression (2) (Lima et al., 2013), (de Castro & Timmis, 2002):

$$Ag = Ag_1, Ag_2, Ag_3, \ldots, Ag_L \tag{1}$$

$$Ab = Ab_1, Ab_2, Ab_3, \ldots, Ab_L \tag{2}$$

where:

L is the dimension of the lymphocytes (*Ab*) and antigens (*Ag*).

It is emphasized that the censor and monitoring phases of the NSA are executed off-line and on-line, respectively (de Castro & Timmis, 2002).

3.1. Matching criterion and affinity rate

A criterion called matching or combination is used to evaluate the affinity with the chains (antigens and lymphocytes) and determine their similarity. The matching can be perfect or partial.

This paper uses the partial matching and affinity proposed in Bradley and Tyrrell (2002). The affinity rate represents the similarity grade with the two analyzed chains. According to Bradley and Tyrrell (2002), the affinity is defined as follows:

$$TAf = \left(\frac{An}{At}\right) 100 \tag{3}$$

where:

TAf	affinity rate;
An	number of normal chains in the problem (proper chains);
At	total number of chains in the problem (proper and non-proper chains).

The precise affinity rate can be calculated using Eq. (3), where a statistical relation with the samples is proposed.

A deviation, i.e., a tolerance rate, is proposed where the combination with the patterns is acceptable and can improve the diagnosis (Lima et al., 2013):

$$\underline{Ab}_i \le Ag_i \le \overline{Ab}_i \tag{4}$$

where:

Ag_i	nominal value of position *i* of the antigen (pattern under analysis);
$\underline{Ab}_i$	nominal value of position *i* minus the deviation adopted in the antibody (detector pattern);
$\overline{Ab}_i$	nominal value of position *i* plus the deviation adopted in the antibody (detector pattern).

Expression (5) represents the method to quantify the total affinity with the analyzed patterns (Bradley & Tyrrell, 2002), (Lima & Minussi, January-2014):

$$Af_T = \frac{\sum_{i=1}^{L} Pc}{L} 100 \tag{5}$$

where:

Af_T	percentage of affinity with the analyzed patterns;
L	total number of positions;
Pc	matching position;
$\sum_{i=1}^{L} Pc$	sum (number) of the matching positions.

Thus, if Af_T is greater than *TAf*, then the combination/matching with the patterns occurs, i.e., the patterns are considered equal/similar. Otherwise, there is no matching with the patterns.

4. Clonal selection algorithm

The clonal selection algorithm, CLONALG, was originally proposed in de Castro and Von Zuben (2000). There are two available versions of the algorithm in the literature, one of which solves machine learning problems and pattern recognition, and the other solves optimization problems (de Castro, 2001).

The CLONALG for recognition problems and machine learning has the following steps (de Castro, 2001), (de Castro & Timmis, 2002):

Step I: Initialization: generate a random population (Ab=$Ab_{\{M\}}$+$Ab_{\{R\}}$) with *N* lymphocytes for each antigen (*Agi*). *N* is given by *M*+*R*;

Step II: Affinity evaluation: each antigen (*Agi*) is presented to every *Ab* of the population (*Ab*) in a process of affinity evaluation. A vector *f* of affinity is determined;

Step III: Selection: the *n* *Ab*s with high affinity *f* to *Agi* are selected to compose a sub population ($Ab_{\{n\}}$);

Step IV: Cloning: the *n* selected *Ab* s proliferate (clone) proportionally to the affinities of the antigen (*Agi*) and generate a *C* population of clones in each selected *Ab*;

Step V: Hyper-mutation: the *C* population of clones is submitted to the maturation affinity process, which generates a new population, C^*, where each *Ab* will mutate with a rate inversely proportional to the affinity *f*;

Step VI: Affinity evaluation: determine the affinity f^* with the set C^* of mutated clones and the antigen *Agi*;

Step VII: Re-selection: from the mature population C^*, re-select the *n* best mature *Ab* s to make a sub-population ($Ab_{\{n\}}$). From this sub-population, choose the best *Ab*s to enter the memory set ($Ab_{\{n\}}$). A *Ab* enters the memory set if it has high affinity rate and can substitute a *Ab* of the memory;

Step VIII: Meta-dynamics: substitute d antibodies of ($Ab_{\{R\}}$) by new individuals ($Ab_{\{d\}}$) to make the repertoire diverse. The anti-bodies with the lowest affinity are substituted;

Step IX: Repeat steps II-VIII until the stop criterion is satisfied.

At the end of the iterative process, the memory set ($Ab_{\{M\}}$) has M *Abs* with high affinity rates to the antigen (*Agi*). This memory set can be used by the NSA to detect and classify learned antigens in the clonal selection process.

For learning problems, step VIII (meta-dynamics) is not executed; therefore, $d=0$.

The number N_c of clones generated in Step IV for each *Abi* is given by Eq. (6) (de Castro, 2001):

$$N_c^i = round\left(\frac{\beta N}{i}\right) \tag{6}$$

where:

β is a multiplicative factor between [0,1], N is the total number of *Abs* of population *Ab*, and *round*(.) is the approximation operator to the closest integer.

The mutation rate (α) of each clone is defined according to Eq. (7) (de Castro, 2001):

$$\alpha = \exp(-\rho f^n) \tag{7}$$

where:

ρ is a control damping parameter of the exponential function and f^n is the normalized value of affinity f, which is calculated using Eq. (8).

$$f^n = \frac{f}{f_{\max}} \tag{8}$$

Then, each clone mutates according to (9) (de França, Von Zuben, & de Castro, 2005):

$$m = round\{\alpha N(0,1)\} \tag{9}$$

where:

m is the number of mutations that each clone has, *round*(.) is the approximation operator to the closest integer, α is the mutation rate, and $N(0,1)$ is the Gaussian random variable with mean zero and $\sigma = 1$.

5. Modelling and disturbance simulation

Although there was no database of distribution systems under perturbations, the data to evaluate the methodology was provided simulating events in test systems.

Using the routine *models* (Dubé, 1996) from the EMTP software (EMTP-RV 2011), a real distribution system with 134 buses was modeled. A sample frequency of 15.36 kHz, which is equivalent to 256 samples per cycle, was used as a parameter. The simulation time was 200 ms.

For the voltage disturbances, the theoretical model proposed by Abdel-Galil, Kamel, Youssef, El-Saadany, and Salama (2004) was used for the simulations, where each disturbance was generated by varying the parameters in the equations that represented the waveform of the disturbances.

In total, 624 simulations were executed for the 134-bus system (LaPSEE 2011). The simulations considered the phase where the disturbance occurred, 50–120% to loading system, and the parameters of the theoretical model. Table 1 presents the quantity of executed simulations for each class of disturbance.

Table 1
Number of simulations executed.

Voltage disturbance	134 buses
Outage	48
Harmonic	144
Swell	72
Sag	72
Swell with harmonic	96
Sag with harmonic	96
Oscillatory transient	96
Total	624

6. Proposed methodology

This section describes the methodology proposed to realize the voltage disturbance diagnosis in electrical distribution systems using the artificial immune algorithm with continuous-learning.

The voltage disturbance diagnosis system with continuous-learning in this section is inspired by the pattern recognition and learning process of the BIS, which was described in Section 2.*A* of this work. To computationally formulate and reproduce this biological process, the NSA proposed by Forrest et al. (1994) and the CLONALG proposed by de Castro and Von Zuben (2000) were used. The NSA realizes the voltage diagnosis disturbance and the CLONALG as a strategy for learning.

The diagnosis system with continuous-learning has five principal steps: NSA (off-line) censoring, novelty detection, continuous-learning, NSA monitoring and knowledge updating. The block diagram is shown in Fig. 4. According to Fig. 4, in the off-line process called censor module, the proper and disturbance detectors for the NSA are defined, which form the set of NSA detectors.

Thus, from the data acquisition system (SCADA (Wylie & Pleydell, 2008)), voltage readings are executed at the electrical distribution substation to obtain the oscillographs. Then, the censor module defines the proper detector set. For electrical distribution systems, the proper signals have the normal operation characteristics. Then, 256 window samples of the signal that represents the normal system operation are filed as proper detectors.

Subsequently, the censor module is executed to generate the disturbance detectors, and for each type of disturbance, the censor module is executed once.

In this phase, the detectors are randomly generated from a set of signals that represent the disturbed system. Thus, 1 cycle (256 points) in the simulated data is windowed and these data are compared window by window with the proper detectors. If there is matching, the random vector is rejected. Otherwise, it is accepted and filed as a disturbance detector in the set of the disturbance detectors. The censor process is illustrated in Fig. 5.

The system operator defines the number of detectors (proper or disturbances). The detector set is used for the novelty detection and monitoring steps of the NSA.

The monitoring process (on-line) is executed after the off-line process. The test set (antigens), which is obtained from the acquisition system (SCADA (Wylie & Pleydell, 2008)), is formed using every available signal in the database. Then, a random signal is selected to be analyzed.

The novelty detection step is executed when a signal is selected to be analyzed, and the analyzed signal is compared with the detector set of the NSA. The matching of the signal with some detector is verified. If they match, the analyzed signal is known by the system, i.e., there is no novelty. Otherwise, the signal is considered a novelty or unknown by the system.

When the system identifies a novelty, the continuous-learning step is executed to learn about the new antigen (analyzed signal), and the learning process occurs, which is realized by the CLONALG.

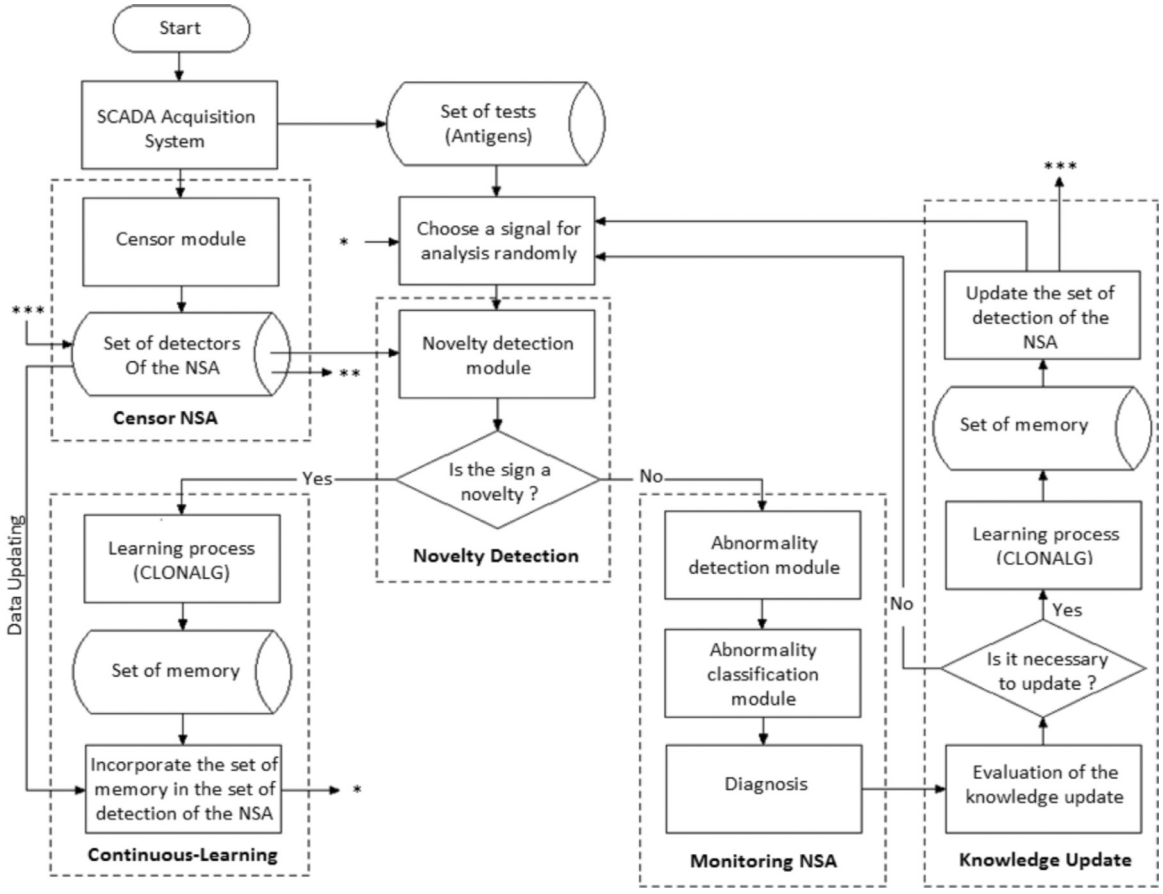

Fig. 4. Diagnosis system with continuous-learning.

The objective of the CLONALG is to generate a memory set (knowledge) from an unknown signal (antigen). This process follows the steps in Section 4 of this work. Initially, a population of lymphocytes is randomly generated. To randomly generate a lymphocyte, the maximum and minimum values of the voltage signal to be detected (antigen) are identified, and a vector with 256 positions is created and randomly filled with real values between the maximum and minimum voltages, which generates a random signal.

Eq. (5) is used to quantify the affinity with the lymphocytes of the population (*Ab*) and the antigen. Then, the *n* best lymphocytes with highest affinity values to the antigen are selected for the clonal process.

To quantify the affinity with the lymphocytes of the population (*Ab*) and the antigen, Eq. (5) is used. Then, the ***n*** best lymphocytes with highest affinity values to the antigen are selected for the cloning and hyper-mutation processes. The number of clones is calculated using Eq. (6), and the number of mutations to be realized on the clones depends on the calculus of Eqs. (7)–(9). Mutation is an evolutionary process to execute little modifications in the lymphocyte structure to increase the affinity to the antigen.

Due to the analyzed signals are expressed by vectors with real positive and negative numbers, this method uses the inductive mutation techniques (Wylie & Shakhnovich, 2012), which is an evolutionary-algorithm technique to solve this type of problems. Thus, the mutation process consists of realizing a mutation using Eq. (10) in a random position of the signal (lymphocyte).

$$Ab_i' = \begin{cases} Ab_i + \alpha^*(Ab_i - Ag_i), Ab_i > 0 \\ Ab_i + \alpha^*(Ag_i - Ab_i), Ab_i < 0 \end{cases} \quad (10)$$

where: Ab_i' is the mature lymphocyte, $Ab_i$is the position to be mutated, $Ag_i$is the goal (learning pattern), and α is a random number between [0, 1].

After the lymphocytes are matured, the *n* best matured clones are re-selected to be included in the population. The selected lymphocytes replace the worst lymphocytes of the population. Furthermore, the best lymphocytes are separated for the memory set. The process is repeated until the stop criterion is satisfied. In this work, the stop criterion is that every lymphocyte of the memory set reaches at least 92% affinity with the antigen (signal to be learned).

Fig. 6 illustrates the learning process of CLONALG to learn a harmonic signal. Fig. 6(a) illustrates the signal to be learned. Fig. 6(b) shows the memory antibody randomly generated in red in the iteration 1. Fig. 6(c), (d), (e), (f), (g), (h) and (i) shows the memory anti-body after 50, 100, 150, 200, 250, 300 and 350 iterations, respectively. Finally, Fig. 6(j) illustrates the memory anti-body after the stop criterion is satisfied, i.e., the affinity with the random signal and the pattern to be learned are greater than 92%.

At the end of the CLONALG process, there is a memory set (knowledge) related to the unknown antigen. This knowledge set is included to the detector set of the NSA to provide knowledge to the system. For example, in another analysis of the same type, the system can recognize and classify the disturbance. This characteristic is known as continuous-learning.

When the novelty detection step does not identify a novelty, i.e., when the antigen is known by the NSA, the process occurs according to Fig. 7, similarly to the conventional NSA. Thus, the signal under analysis must be compared with the detector set of the NSA and evaluate the matching and affinity to detect the abnormality and later classify the signal according to the class of disturbance where matching occurs.

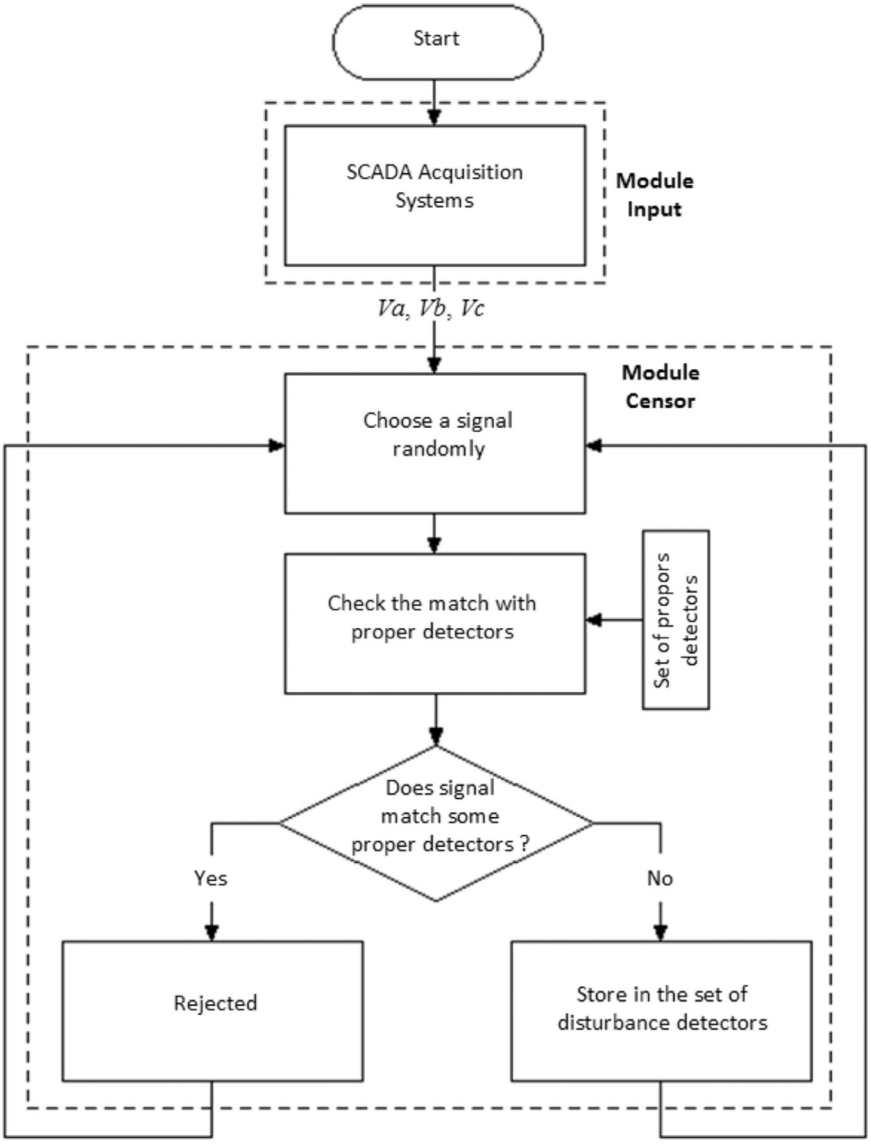

Fig. 5. Censor Module of NSA.

After the signal is analyzed, the monitoring process of the NSA finishes.

Furthermore, the knowledge module is updated, which is responsible for evaluating whether there is a requirement to update the detector set of the NSA (knowledge) to improve the disturbance detection and classification processes. This case uses the matching of the detectors with the antigen signal (signal under analysis) as the criterion. If 70% of the disturbance detectors match the analyzed signal, it is not necessary to update the detection set of the NSA for the signal classified. If less than 70% of the detectors do not match with the classified pattern, the knowledge-updating process occurs. When the NSA cannot recognize the signal with less than 70% of the detectors, it is necessary to update the detector set of the NSA to improve the recognition process and use the antigen to learn (obtain knowledge).

Therefore, it is necessary to update the knowledge of the CLONALG learning process, which is executed according to the previous description and identical to the process realized by the continuous-learning module.

After the learning process and a set of memory lymphocytes of the antigen signal (signal under learning) is obtained, the best lymphocyte of the memory set must be selected and updated to the detector set of the NSA. The updating occurs on the substitution of one detector that does not match the antigen pattern. This detector is substituted by the best lymphocyte of the memory set generated on the learning process realized by the CLONALG.

The learning updating only occurs when there is the need to reinforce the NSA knowledge, i.e., when fewer than 70% of the detectors do not satisfy the affinity criterion, matching no antigen. Then, this detector is substituted by a new detector generated in the learning process, which is called the updating knowledge.

The process in Fig. 4 is repeated until every antigen (test set) is analyzed.

7. Results and applications

This section presents the obtained results of the proposed method in the simulated test system. The algorithm was developed using MATLAB® (MATLAB 2011). Every test and simulation was performed using a PC Intel Core 2 Duo 1.9 GHz with 2 GB of RAM (Random-Access Memory). The proposed algorithm used a real distribution system with 134 buses (LaPSEE 2011).

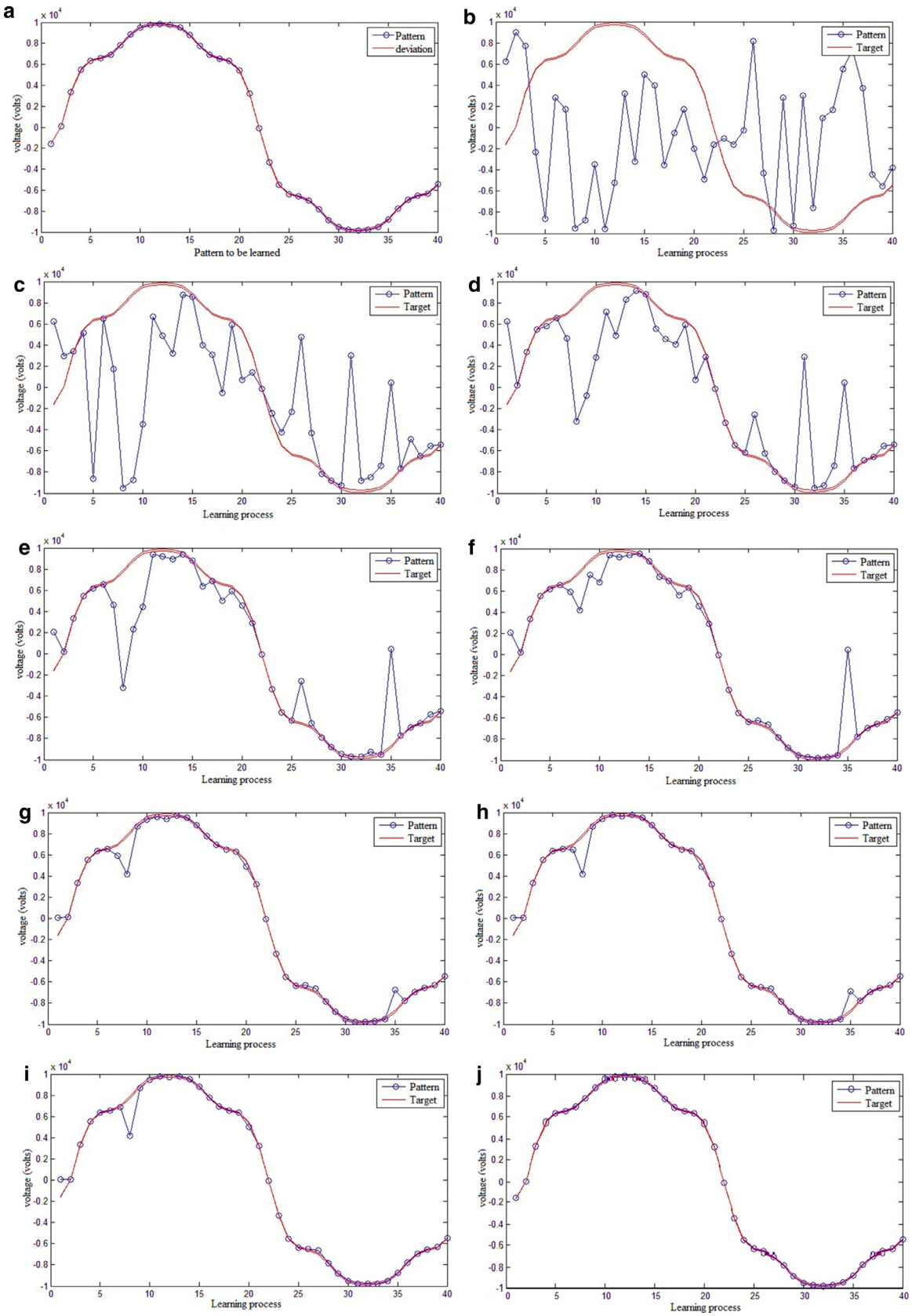

Fig. 6. Learning process of the CLONALG.

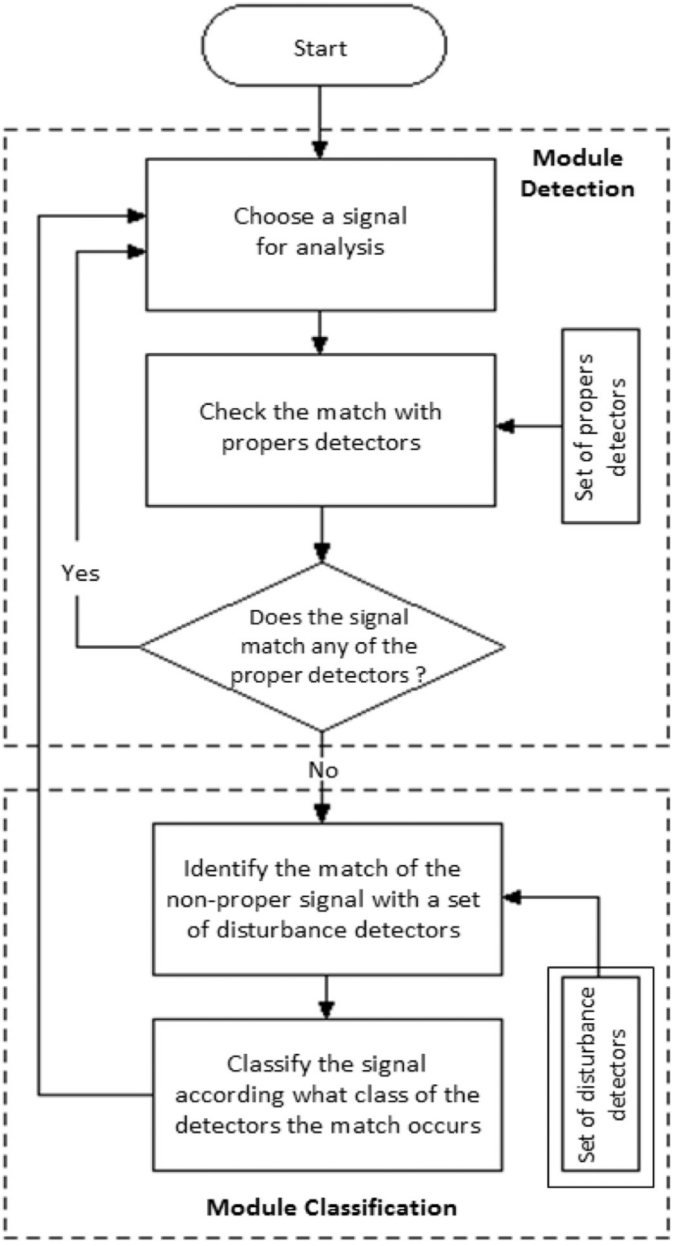

Fig. 7. Monitoring Module of the NSA.

The proposed method was evaluated to verify the efficiency, precision and robustness in the voltage disturbance diagnosis using the conventional algorithm (without continuous training), proposed and explained in Lima et al. (2013), and the algorithm with continuous training. The metrics adopted to the efficiency, precision and robustness are: ability to learn unknown patterns, percentage of correct answers and computational time.

Table 2 presents the parameters in the tests, which were obtained using an empirical test process.

Three patterns were excluded during the censor phase in both systems (with and without continuous training) to evaluate the performance of the novelty-detection, continuous-learning and knowledge-updating modules of the proposed algorithm. The excluded patterns were: swell, harmonic and oscillatory transient. They were randomly chosen like any disturbance, and the algorithms began the monitoring process online without knowing these patterns.

Table 2
Parameters.

Parameters	NSA without continuous training	NSA with continuous training
TAf	66.66%	66.66%
N	–	25
M	–	9
R	–	16
n	–	5
β	–	0.3
ρ	–	4
d	–	0

Table 3
Results of the system without continuous training.

Disturbances	134 buses		
	Tested patterns	Matching patterns	Matching (%)
Swell	72	0	0.00
Sag	72	70	97.22
Outage	48	48	100.00
Harmonic	144	0	0.00
Swell with harmonic	96	96	100.00
Sag with harmonic	96	95	98.95
Oscillatory transient	96	0	0.00
Total	624	309	49.51

Table 4
Results of the continuous-training system.

Disturbances	134 buses		
	Tested patterns	Matching patterns	Matching (%)
Swell	72	71	98.61
Sag	72	72	100.00
Outage	48	48	100.00
Harmonic	144	140	97.22
Swell with harmonic	96	96	100.00
Sag with harmonic	96	96	100.00
Oscillatory transient	96	95	98.95
Total	624	618	99.03

Table 3 shows the results with the previously mentioned patterns and configurations for the conventional diagnosis, i.e., the diagnosis without continuous training.

The conventional diagnosis system cannot identify the excluded patterns from the censor process, i.e., it cannot identify these types of patterns.

Table 4 presents the results of the diagnosis process with continuous training. The configurations and parameters are identical to those used for the conventional NSA algorithm.

Table 4 shows that the continuous-training system can learn the unknown disturbances and execute the diagnosis in another analysis. In addition, the updating knowledge module contributes to increase the matching rate for some known disturbances because the system updates the detector set of the NSA in the online monitoring process, which reinforces and improves the knowledge.

Tables 3 and 4 show an improvement in diagnosing the disturbances, i.e., the system with continuous training can evolve with the experience and diagnose the disturbances that were not previously diagnosed. The excluded patterns from the censor process were not recognized by the system without continuous training with 0% matches. After using the continuous training, the patterns are identified with a matching rate of approximately 98.95% for the swell, 97.22% for the harmonic and 98.95% for the oscillatory transient.

In conclusion, it is possible to control the quality of the generated lymphocytes in the memory set of the learning process with the CLONALG. To finish the learning process, the lymphocytes of

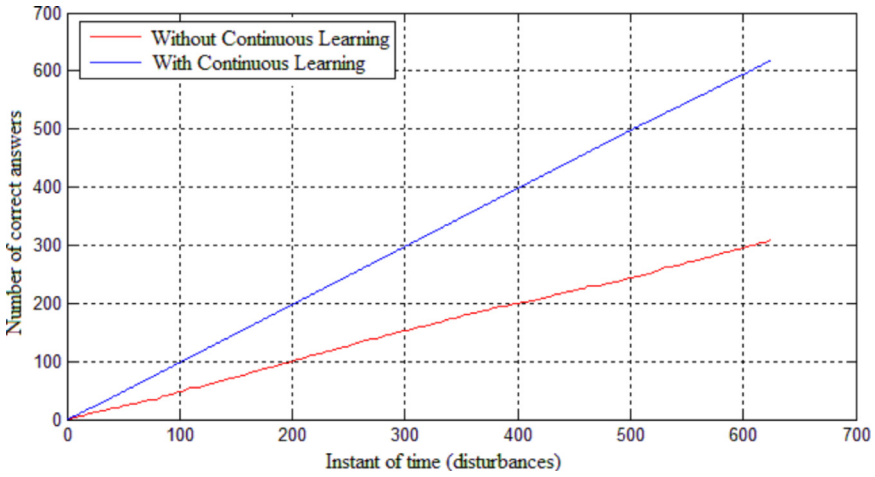

Fig. 8. Graph of the efficiency with time.

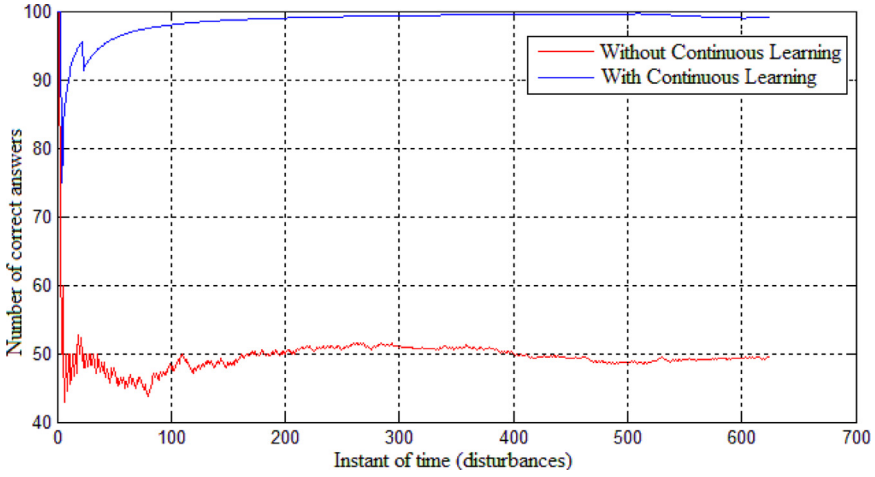

Fig. 9. Graph of the precision with time.

the memory set and the antigens (pattern to be learned) must have an affinity of at least 92%. With this criterion, it is possible to ensure that the memory lymphocytes have high affinity values and provide efficiency in the diagnosis process.

Figs. 8 and 9 show graphs of the obtained results with and without continuous training to analyze the matches and the precision in a given time. The instant of time is the occurrence of a failure, i.e., the quantity of failures that the system analyzes until the considered instant.

Fig. 8 shows a linear relationship between the instant of time (number of failures) and the number of matches until this instant of time. When the system continuously learns, the linear relationship tends to have an angular coefficient of approximately 1 and a linear coefficient of approximately 0 for each analyzed instant, i.e., an inclination of approximately 45º represents an optimal precision. For the system without continuous training, the linear relationship presents different coefficients with an inclination of approximately 22º, which confirms that after the continuous training, the system improves the performance with time.

Finally, Fig. 9 shows the precision of the system considering the analyzed instants of time (failures) until the moment. The system without continuous training has a low error precision of approximately 50% until the end. The system with continuous training has almost no error and maintains a precision of approximately 100% during the entire time.

7.1. Comparative study

Table 5 presents a comparative study that uses matching according to the proposed method and the principal references that are available in the literature. This study presented methods with continuous learning and methods without continuous learning.

The proposed method presents a matching index and a precision that is similar to other works that are available in the literature. It should be emphasised that the use of the continuous learning changes the sensitivity of the voltage disturbance diagnosis and can detect and classify any abnormality; therefore, good results are presented.

7.2. Comments about the algorithm

After carrying out the tests and obtaining the results, we identified the strengths and weaknesses of the proposed algorithm; these are presented in this section.

Strengths:

- The proposed algorithm presented excellent performance, presenting a good accuracy rate (99.03%) when compared to the best works available in the literature.
- The algorithm require low processing time.
- This algorithm has the capacity to identify unknown patterns (unknown disturbances).

Table 5
Comparative Study.

Reference	Electrical system/theoretical model	Method used	Total match (%)
(Uyar et al., 2008)	(Abdel-Galil et al., 2004)	Multilayer Perceptron (Levenberg-Marquardt) e *wavelet*	96.21
(Zhang et al., 2011)	Not specified	Decision tree and Fourier transform	99.00
(Lima et al., 2013)	134-bus	Negative Selection Algorithm	99.11
(Oleskovicz et al., 2009)	138/13,8 kV	Multilayer Perceptron e *wavelet*	99.31
(Decanini, Tonelli-Neto, Malange, & Minussi, 2011)	134-buss	Fuzzy ARTMAP neural network and *wavelet* transform	99.66
(Lima, Lotufo, & Minussi, 2015)	134-buss	Negative Selection Algorithm with *wavelet* transform	100.00
(Lima et al., 2015)	134-buss	Modified Negative Selection Algorithm	100.00
(Barros et al., 2015)	134-bus	Modified Euclidean ARTMAP neural network with continuous learning	90.32
This work	134-bus	Immunologic algorithm with continuous learning	99.03

- This method has the capacity to continuously improve/reinforce the knowledge and permanently learn without reinitializing the learning process.
- This method is able to follow innovations and modernizations of the electrical system, which include new types of prominent failures, learn with experience and constantly evolve.

Weaknesses:

- The algorithm proposed have several parameters that must be set in advance.

8. Conclusion

This work presents a new approach for voltage disturbance diagnosis with continuous training based on the BIS. Two artificial immune algorithms were used for the diagnosis and learning: NSA and CLONALG. The continuous learning algorithm is compared with the conventional algorithm, i.e., without continuous learning, proposed and explained in Lima et al. (2013). Three patterns were excluded from the censor process of both systems to evaluate the novelty detection, and the conventional process matches 49.51% of the analyzed disturbances. The algorithm with continuous learning matches 99.03% for every pattern.

The continuous-learning module can continuously learn to identify unknown disturbances. The obtained results show that the diagnosis with continuous learning performs better (in terms of efficiency, precision and robustness) than the system without the continuous learning. Therefore, the newly proposed detection strategies in this paper, which continuously learn and update knowledge, provide the diagnosis system with sufficient precision, efficiency and robustness for important applications and the capacity to follow the evolution of electrical systems such as the smart-grid environment. The main advantage of this method is the capacity to continuous learning, which is a novelty in methods of voltage disturbances diagnosis. This advantage is very important to evolution of the techniques in intelligent systems.

The results allow practical applications such as modeling and development of an efficient system for intelligent diagnosis in smart grids. Based on the accuracy index of the computational algorithm, certainly in a real network with modern relays or a robust system of acquisition data, analysis and diagnosis should provide better efficiency.

Finally, the proposed methodology when compared to other methods such as artificial neural networks presents the capacity to continuously improve/reinforce the knowledge and permanently learn without reinitializing the learning process, improving the diagnosis process.

To give sequence to this research, it is suggested the following topics: add module to perform wavelet analysis and processing of signals; investigate the possibility of investing in distribution systems environment in anticipatory character; apply the method proposed to distribution systems with non-linear loads and distributed generation.

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Electrical Demand Load Forecasting by ARIMA Regression and Artificial Neural Networks

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Abstract— **The use of hybrid techniques has been applied to solve several problems, including convergence and precision. This paper proposes a hybrid methodology using SARIMA regression models and Multilayer Perceptron Neural Network trained by Levenberg-Marquardt algorithm applied to short term load forecasting. This forecasting is a notably important task for the planning and operation of electrical power systems to anticipate when and how much generation and transmission must be available to provide the required electrical load without interruption. The results are presented for a Brazilian electrical demand energy time series.**

Keywords- SARIMA Models by Box & Jenkins, term load forecast, artificial neural network.

I. INTRODUCTION

Electrical power systems have been growing in size and complexity and it is necessary to provide alternatives to minimise the system operation and generation costs. Electrical load forecasting is a fundamental task in operation and control centres, therefore, this task must be precise to enable the system to operate with security and reliability to guarantee the energy is provide safely, economically and continuously. Some techniques that are emphasied in the literature for load forecasting [1] include: simple or multiple linear regression, exponential smoothing, state estimation, Kalman filter, and the Auto Regressive Integrated Moving Average (ARIMA) of Box & Jenkins [2]. These methods require the previous load modeling for further application. The parameters that aid the load modeling can include: meteorological conditions such as cloudy days, wind velocity, temperature or others conditions such as non-typical days (holidays, strikes, etc.) [3].

Presently, the use of intelligent techniques such, as artificial neural networks (ANN) [4] is an efficient alternative procedure for effective load forecasting. The great advantage of the ANN is the ability to handle with complex problems by training, the ability to identify and incorporate characteristics of the time series without needing a theoretical formulation and previous load modelling such as those required by statistical methods.

An interesting proposal is to conjugate statistical techniques with intelligent systems resulting in a hybrid model. There are several published using ARIMA of Box&Jenkins and artificial neural networks (Multilayer Perceptron with Back propagation algorithm) [5-9]. These works are applied to time series of several real problems.

Koutroumanidis [5] presentes a description of alternative energy sources to produce electricity. ARIMA and ANN are used, where ARIMA is responsible for generating the load parameters and ANN are used to predict the future prices of selling wood in Greece. The use of the hybrid model has obtained good results, allowing the industries to proceed in the rational planning of producing and selling wood. In the works of Khashei and Bijari [6], [7], a hybrid model with ARIMA and ANN is used to obtain a precise prediction applied to annually register the sunspot, the number of linces that are captured per year in the Mackenzie river in Canada and the dolllar/pound rate tax. The ARIMA models are used in the first phase to generate the necessary data from the historical series, and in the second phase, the ANN is used to model the data generated by ARIMA models and to predict the time series.

The works of Valenzuela [8] and Khashei [9] besides using ARIMA and ANN, also combine Fuzzy Rules and Genetic Algorithms to improve the results.

Considering the good results obtained in the literature using hybrid models, the objective of this work is to presente a hybrid methodology using ARIMA and ANN (multilayer perceptron with Levenberg-Marquardt training) to predict short term electrical loads.

At the first step of the proposed model, the ARIMA is used to model the electrical load series, providing the simulated electrical load series that will be used as input to the neural network (second step) that executes the load prediction.

The proposed methodology is tested with the electrical load historical data from 30, 60 and 90 days to predict the electrical load of the next day. To validate the proposed methodology, the results were compared with other neural network architectures [10] and with the results obtained by the classical models (ARIMA and perceptron muiltilayer neural network). In this way, it is possible to verify that the use of the hybrid model is viable and provides the best results compared to the models that are used separately.

II. TIME SERIES PREDICTION MODELS

There are several different approachs to execute time series forecasting, for for example the traditional statistical modesl such as: exponential smoothing and ARIMA of Box&Jenkins in which the future prediction values are limited by being described by a linear function of previous observations. Another class of prediction models is nonlinear, for example the artificial neural networks. The hybrid model proposed in this paper use two techniques, i.e., the linear model (ARIMA) and the nonlinear (ANN).

A. *ARIMA of Box&Jenkins models*

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The ARIMA of Box&Jenkins models are based on statistical concepts that use the correlation between observations at different times. The identification of the model structure is difficult, but there is some software [11] that executes this step automatically, avoiding the most difficult step of the analysis.

The ARIMA models assume that the future value is a linear function of past observations and random errors. The process that generates the time series is according equation (1) [12]:

$$\emptyset(B)\Delta^d Z_t = \theta(B)a_t \qquad (1)$$

of order *(p,d,q)* written as ARIMA(p, d, q), where p and q are orders of $\emptyset(B)$ and $\theta(B)$ respectivelly.

where:

Z_t : current values at time *t*;

a_t : random errors at time *t*;

$\emptyset(B)$: auto regressive operator of order p;

$= 1 - \phi_1 B - \phi_2 B^2 - \ldots - \phi_p B^p$

$\theta(B)$: moving average operator of order q;

$= 1 - \theta_1 B - \theta_2 B^2 - \ldots - \theta_q B^q$

Δ^d : series differentiation;

$= (1 - B)^d$

d : indicates number of times the series differs until it becomes stationary.

If the series is already stationary, the term d does not exist and the model is termed ARMA (p, q), according to equation (2) [12].

$$\emptyset(B)Z_t = \theta(B)a_t \qquad (2)$$

Box&Jenkins methodology [2] is most frequently used to execute time series prediction, consisting in adjusting ARIMA (p,d,q) to a data set. The construction of ARIMA models includes three steps: identification, estimation and verification [2].

The seasonal ARIMA models are an extention of ARIMA models, known as the Seasonal Auto Regressive Integrated Movin Average (SARIMA), represented by SARIMA (p, d, q)(P ,D, Q)m. They are according equation (3) [12].

$$\phi(B)\Phi(B^m)\Delta^d\Delta_m^D Z_t = \theta(B)\Theta(B^m)a_t \qquad (3)$$

where:

$\Phi(B^m)$: seasonal auto regressive operator of order P;

$= 1 - \Phi_1 B^m - 1 - \Phi_2 B^{2m} - \cdots - 1 - \Phi_P B^{Pm}$

$\Theta(B^m)$: seasonal moving average operator of order Q;

$= 1 - \Theta_1 B^m - 1 - \Theta_2 B^{m2} - \cdots - 1 - \Theta_Q B^{mQ}$

Δ_m^D : seasonal series differentiation;

$= (1 - B^m)^D$

D : number of seasonal diferences.

The identification, estimation and verification procedures of the seasonal models are identical to those without seasonality, except at the identification phase, where it is necessary to differentiate the series in relation to Δ and Δ_m to produce a stationary state, obtaining the values of d and D. The estimation and verification phases of SARIMA (p, d, q) (P,D,Q)m are similar to the ARIMA (p, d, q) [13].

B. *Artificial Neural Networks*

Artificial Neural Networks have some advantages in predicting time series due to model series with nonlinear characteristics.

The advantages of using ANN are due to the capacity of generalisation in non-stationary environments whose characteristics can change with time. The generalization of the neural networks occurs to produce adequate outputs to inputs that were not presented during the training, allowing complex and difficult problems to be solved [4].

The multilayer perceptron neural network (MLP) is an important architecture, and it is also one of the most widely used architectures that are applicable to different application problems. MLPs contain at least one hidden layer between the input and the output layers. MLP is a feed-forward network with multiple layers and supervised training [4].

One of the most important characteristics of this neural network is the ability to learn and improve the performance, thus, the neural network must be trained adjusting the weights by a training algorithm. In this study the Levenberg-Marquardt algorithm is used [14].

The weight adaptation by the Levenberg-Marquard algorithm is given by equation (4) [15]:

$$W(k+1) = W(k) - [J^T(W)\,J(W) + \mu_k I]^{-1}\,J^T(W)\,e(W) \qquad (4)$$

where:

W : synaptic weight vector;

J : Jacobian matrix;

I : identity matrix;

μ_k : Levenberg-Marquardt constant;

$$e(W) = \sum_{i=1}^{n} (y_i - y_{ei})$$

y_i : output provided by the neural network;

y_{ei} : real output.

The parameter μ_k is a training stabiliser factor that is based, on the Newton method, althougt it is faster because it, avoid long steps that can lead to a convergence error. This algorithm is considered the fastest, and it is executed with few iterations when using an adequate quantity of data. However, when this number is high this training is not adequate due to the computational effort generated by inversing matrices [14, 15].

III. Prepare Your Paper Before Styling

The proposed methodology of this article is a modification of the work presented by Abreu et. al [16]. In [16], as shown in Figure 1, a hybrid model whose objective is to obtain the prediction of electric load in the short term through the best SARIMA model has been developed .

In this article the proposed hybrid model aims at reducing the number of steps used in [16], and consequently, obtaining a good prediction of electric charges with a shorter processing time. It is important to emphasize that the difference of this work is associated with the direct process of simulated series obtained by R software [16] that is used directly in the prediction of electric load through an artificial neural network, a multi-layer perceptron, with Levenberg-Marquardt algorithm.

The methodology developed in this work consists of realising short term load forecasting using the ARIMA of the Box&Jenkins model and the Multilayer perceptron neural network trained by the Levenberg-Marquardt algorithm. The load time series contain hourly data and is considered with seasonality presenting correlations in different times, i.e., correlation between loads at night must be high and are likely greater than the correlation between loads during the morning of the same day.

Before initiating the first step, the real data are normalised to avoid saturation of the neural network. The normalisation is given as follows:

$$CN = (C \times 1{,}2) / Vmax(C)$$

where:

CN = normalised load;

C = real load;

$Vmax\ (C)$ = maximum value of the real load.

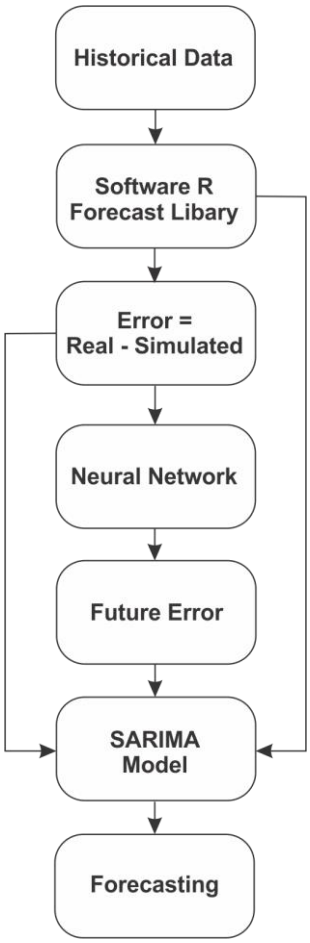

Fig. 1. Modelo Híbrido desenvolvido por Abreu [16]

After being normalised, the load data are applied to R software by forecast library [17, 18] to determine the simulated and predicted series of the best SARIMA model. The simulated series is used as the input of the neural network. Figure 2 below shows the structure.

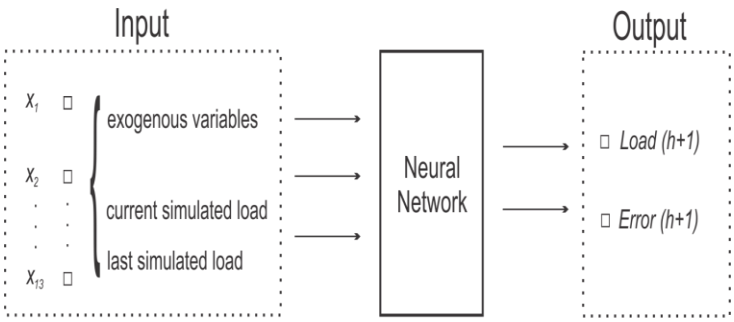

Fig. 2. Neural Network structure

The neural network specified in Figure 2 is composed of three layers, where the input layer has 13 neurons, the hidden layer has 27 neurons and the output layers contain 2 neurons.

The set containing the input and output vector is defined respectively by equations (5) and (6):

$$X_h = [\, te \;\; L(h-3) \;\; L(h-2) \;\; L(h-1) \;\; L(h)]^T , \quad X \in R^b \qquad (5)$$

$$Y = [\, C(h+1) \quad E(h+1)] , \; Y \in R^2 \qquad (6)$$

where:

b = dimension of vector X;

$L(h-v)$ = simulated load v hours previous of current hour h;

$C(h+1)$ = load value to the subsequent hour to current hour h;

$E(h+1)$ = error of the subsequent hour to current hour h;

At the input layer, vector te representes the nine inputs of the neural network composed by binary data corresponding to the exogenous variables: holidays, day of the week and hour of the day. The last four input $L(h-3)\; L(h-2)\; L(h-1)\; L(h)$, correspond to the simulated loads obtained by SARIMA at time $h-3, h-2, h-1$ and h. The output of the neural network is the predicted load and error 24 hours in advance.

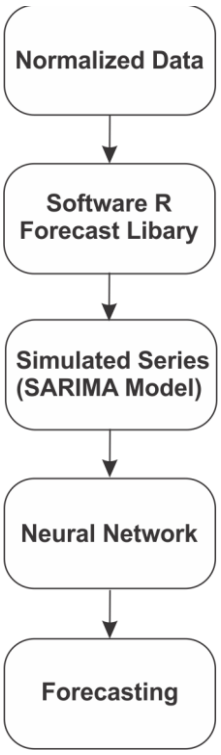

Fig. 3. Flowchart of the proposed model

The load prediction is executed by adding the outputs of the neural network according to equation (7). The proposed methodology is shown in Figure 4.

$$Prediction = \; L(h+1) + E(h+1) \qquad (7)$$

The training and test phases of the neural network were executed using the MATLAB software with the Neural Network toolbox [19].

IV. Application and Results

The proposed model aims to predict short term load forecasting 24 hours in advance.

To test the efficiency of the hybrid model, three different applications were executed according to the quantity of training days. The predited day is the 24 hours of the day following the last day of the historical training.

TABLE I. Periods used in the three applications

Application	Period	Number of vectors
1	31 days	744
2	61 days	1464
3	84 days	2208

To evaluate the precision of the results, the MAPE (mean absolute perceptual error) and the maximum prediction error are calculated. The real load values are compared with the estimated values calculated by the hybrid model. These errors are calculated according to equations (8) and (9), respectively [20].

$$MAPE = \frac{1}{NT}\sum_{h=1}^{NT}\{|C(h) - \underline{C}(h)|/C(h)\} \times 100\% \qquad (8)$$

$$Max\ error(\%) = máx\{|C(h) - \underline{C}(h)|/C(h)\} \times 100\% \qquad (9)$$

where:

$C(h)$: real load referred to hour h;

$\underline{C}(h)$: estimated load by hybrid model refered to hour h;

NT : total quantity of hours.

The obtained results with the hybrid model are compared with the following architectures: multilayer perceptron with descent gradient and momentum, multilayer perceptron with Levenberg-Marquardt training [10], and also with SARIMA models obtained for each application.

Tables 2, 3 and 4 show the results obtained analysing the MAPE and maximum error performance for applications 1, 2 and 3, respectively.

Figures 4, 5 and 6 present the load prediction curves 24 hours in advance.

TABLE II. MAPE AND MAXIMUM ERROR FOR APPLICATION 1

Models	MAPE (%)	Maximum Error (%)
Hybrid Model	1,353	1,726
SARIMA	9,096	18,623
Perceptron trained by descent gradient with momentum	1,998	4,845
Perceptron with Levenberg-Marquardt training	1,178	3,560

TABLE III. MAPE AND MAXIMUM ERROR FOR APPLICATION 2

Models	MAPE (%)	Maximum Error (%)
Hybrid Model	1,187	1,286
SARIMA	7,774	16,955
Perceptron trained by descent gradient with momentum	2,093	5,022
Perceptron with Levenberg-Marquardt training	1,490	4,267

TABLE IV. MAPE AND MAXIMUM ERROR FOR APPLICATION 3

Models	MAPE (%)	Maximum Error (%)
Hybrid Model	1,093	0,913
SARIMA	8,747	18,630
Perceptron trained by descent gradiente with momentum	2,313	7,088
Perceptron with Levenberg-Marquardt training	1,698	4,687

Analysis of Tables II, III and IV shows that the hybrid model is superior to the other neural networks and the SARIMA model for the three applications proposed. Considering the economical generation point of view, results with less than 5% precision present a significant reduction in the generation costs [21]. Thus, decreasing this precision is economical and the proposed model obtains results below 1%.

As the quantity of vectors increase, better results are obtained,which is a fundamental property of the ANN, which assures that, as more data are used for training better results are obtained at the diagnosis (test phase).

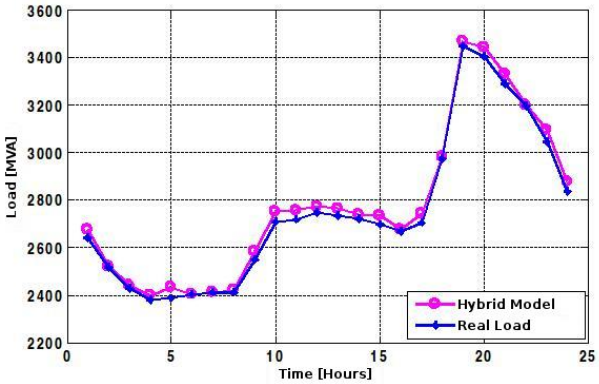

Fig. 4. Predicted and real load by hybrid model for application 1.

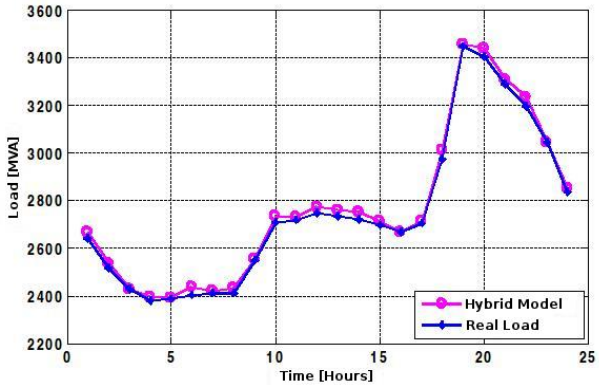

Fig. 5. Predicted and real load by hybrid model for application 2.

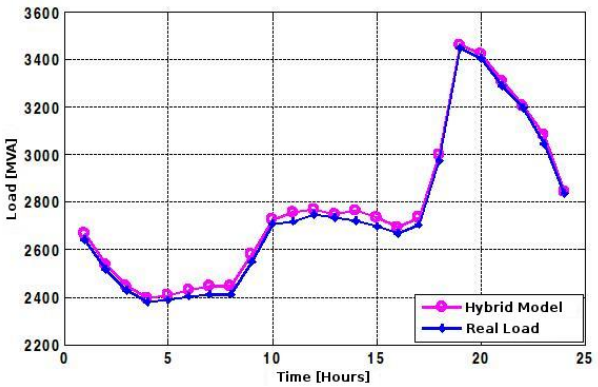

Fig. 6. Predicted and real load by hybrid model for application 2.

Analysing the results described in Tables 2 to 4, it is observed that the hybrid model has the least maximum error when compared to the three other applications, while the

MAPEs are all below 1,4%. It is emphasised that the results of the hybrid model are superior to those of the ARIMA models for the three applications, and they are superior to the Perceptron multilayer with the Levenberg-Marquardt algorithm for applications 2 and 3. This is due to the fundamental property of artificial neural networks that assures that as the quantity of data increases, the better the quality of the obtained results becomes at the diagnosis phase (test phase).

It is observed that the curves obtained by the proposed model follow the pattern of the real data for the three applications, indicating a good precision of the hybrid model.

V. CONCLUSIONS

This work developed a hybrid model using ARIMA of Box&Jenkins and artificial neural networks with the multilayer perceptron trained by Levenberg-Marquardt algorithm aiming to improve the results of short term load forecasting.

The ARIMA models are used to address the linear portion and the ANN for the nonlinear portion of the time series to execute the forecasting.

To evaluate the performance, three tests were executed containing different periods of historical data. The short term load forecasting is executed considering data from 24 hours in advance with MAPEs less than 1.34% and maximum errors less than 1.8%. It is observed that the statistical model with the intelligent system has the best results when compared to the models that are used separately.

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ARTMAP-FUZZY: SISTEMA INTELIGENTE APLICADO NA DETECÇÃO DE FALHAS DE UM TRATOR

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RESUMO

Este trabalho apresenta o desenvolvimento de um sistema inteligente que, por meio de uma rede neural artificial, atua na detecção de falhas estruturais em um trator. Simulou-se o trator por meio de um modelo numérico, construído através de equações diferenciais, o qual gera dados conforme se altera a velocidade e a distância entre as saliências no solo. Para a análise, identificação e caracterização dos dados simulados computacionalmente, foi codificado um sistema que utiliza conceitos da Teoria da Ressonância Adaptativa presentes na rede neural do tipo ARTMAP-Fuzzy. O principal objetivo deste sistema é inspecionar a estrutura do trator contribuindo para sua melhor conservação, indicando se ela se encontra em condições normais ou em situação de falha estrutural. Os resultados obtidos na aplicação da rede neural ao problema especificado mostraram-se ser eficientes e precisos.

Palavras-chave: redes neurais artificiais; ARTMAP-Fuzzy; detecção de falhas; trator

ARTMAP-FUZZY: INTELLIGENT SYSTEM APPLIED TO FAULT DETECTION IN A TRACTOR

ABSTRACT

This paper presents the development of an intelligent system that, by an artificial neural network, works in the detection of structural flaws in a tractor. It was simulated a tractor by means of a numerical model built by differential equations, which generates data as to change the speed and the distance between the protrusions on the ground. For the analysis, identification and characterization of computationally simulated data was encoded a system that utilizes concepts of Adaptive Resonance Theory present in the ARTMAP-fuzzy neural network. The main objective of this system is to inspect the tractor structure contributing to its better conservation, indicating whether it is in normal conditions or structural failure situation. The results obtained in the application of neural network to the specified problem proved to be efficient and accurate.

Keywords: artificial neural network; ARTMAP–Fuzzy; failure detection; tractor

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INTRODUÇÃO

A agricultura é uma das principais atividades econômicas da sociedade moderna e, como em todas as áreas, a informatização tem sido incorporada nos processos produtivos ao longo dos anos. A incorporação das tecnologias possibilitou a expansão da área cultivada e o aumento da produtividade.

Um crescimento ainda maior da produção agrícola depende, entre outros fatores, do nível de tecnologia em tratores usados na agricultura. Os tratores, para atenderem as exigências de custos e de produtividade, devem permitir a seus operadores a condição de exigir destas máquinas o máximo desempenho nas tarefas do campo. Sob certas condições do traçado e velocidade, o trator pode apresentar instabilidade e provocar vibrações que são prejudiciais à estrutura do trator ou, em casos mais extremos, o seu tombamento (KHOURY et al, 2004).

Desta forma é necessário desenvolver maneiras eficientes de monitorar a estrutura, visando detectar falhas em estados iniciais, intervir na sua propagação e, consequentemente, impedir o seu desgaste ou até mesmo o seu tombamento. Esta atual e importante linha de pesquisa é denominada Monitoramento da Integridade de Estruturas (SHM – *Structural Health Monitoring*).

Conforme destacado por Hall (1999), um SHM deve satisfazer os seguintes requisitos: aquisição e processamento de dados; validação e análise de sinais; identificação e caracterização de falhas; interpretação de mudanças adversas em uma estrutura; auxiliar a tomada de decisões.

Toda estrutura passa por uma avaliação de testes de integridade estrutural. Neste caso são realizadas avaliações não destrutivas (NDE – *Nondestructive Evaluation*), visando estimar o grau de segurança e confiabilidade da estrutura.

Nos últimos anos, os sistemas de monitoramento de integridade estrutural vêm sofrendo uma evolução natural, devido ao avanço das tecnologias de sensores, materiais inteligentes, técnicas eficientes de processamento de dados e sinais, e principalmente a crescente demanda de se empregar tais ferramentas em tempo real.

Neste contexto, uma possível solução, para se desenvolver um SHM moderno e eficiente é a utilização de sensores inteligentes e/ou técnicas de computação inteligentes (redes neurais artificiais, lógica fuzzy, etc.), que viabilizam habilidades como a extração de conhecimentos e informações de processos complexos, facilitam a análise de sinais e proporcionam a automatização da tomada de decisões nos SHMs.

Neste trabalho, apresenta-se a proposta de uma metodologia para o desenvolvimento do SHM para solucionar o problema baseado em técnicas de computação inteligente (CI). Neste sentido, emprega-se uma rede neural artificial (RNA) ARTMAP-*Fuzzy* (CARPENTER et al., 1992) para realizar a análise e monitoramento da integridade do trator, devido as suas características de aprendizagem e reconhecimento de padrões, e por apresentarem bom desempenho em outros tipos de problemas de reconhecimento de padrões e diagnóstico, conforme destacado pelos autores em Lima et al. (2013) e Tonelli-Neto (2012).

MATERIAL E MÉTODO

Primeiramente faz-se uma abordagem no conceito da rede neural ARTMAP-*Fuzzy* e seu algoritmo de treinamento.

A rede neural ARTMAP-Fuzzy e seu algoritmo

A rede neural artificial ARTMAP-*Fuzzy* possui uma arquitetura estável e

plástica, que garante uma vantagem com relação a outros tipos de redes neurais artificiais. A rede ARTMAP-*Fuzzy* permite a inclusão de um módulo de treinamento continuado, o qual habilita a extração de conhecimento sem a necessidade de reiniciar o processo de treinamento quando novos padrões são apresentados, ao contrário do que acontece com a maioria das redes neurais (CARPENTER et al., 1992). A RNA ARTMAP-*Fuzzy* trata-se de um sistema de aprendizagem supervisionado composta de um par de módulos ART (CARPENTER e GROSSBERG, 1987), ARTa-*Fuzzy* e ARTb-*Fuzzy*, interconectados por um módulo de memória associativa inter-ART. Esta arquitetura de rede neural incorpora a teoria *fuzzy*, o operador lógico AND *fuzzy* (^) habilitando o sistema neural de aprendizagem, em resposta aos padrões de entradas binários e analógicos, durante os intervalos [0 1] (CARPENTER et al., 1992).

Martins (2010) afirma que um mecanismo interno chamado *match-tracking* é responsável pelo processo auto regulatório da rede neural, no qual maximiza a generalização e minimiza o erro. Quando a rede neural faz uma predição errada, através de uma conexão associativa instruída, o parâmetro de monitoramento do módulo ARTa-*Fuzzy* é incrementado no mínimo necessário para corrigir o erro no módulo ARTa-*Fuzzy*. Segundo Lopes (2005), a figura 1 ilustra a arquitetura da rede neural ARTMAP-*Fuzzy*.

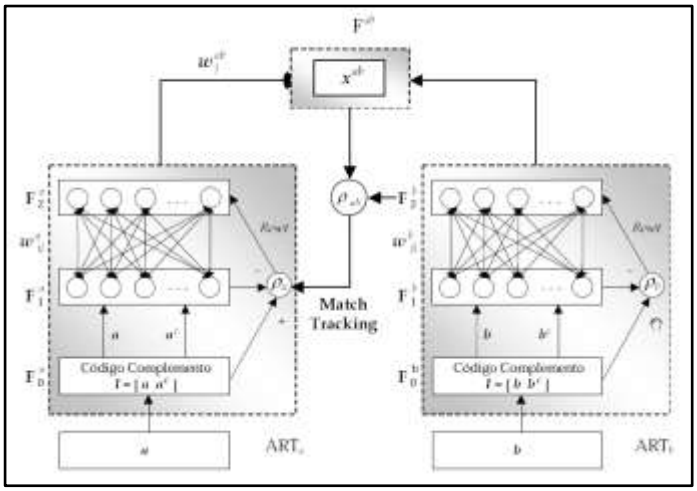

Figura 1. Arquitetura da Rede Neural Artificial ARTMAP-*Fuzzy*.

De acordo com Carpenter et al. (1992), a arquitetura ARTMAP-*Fuzzy* possui três parâmetros principais para o desenvolvimento, chamados:

- Parâmetro de escolha α (α > 0): opera na categoria de seleção;
- Taxa de treinamento β (β ϵ [0 1]): Controla a velocidade de adaptação da rede;
- Parâmetro de vigilância (ρa, ρb e ρab ϵ [0 1]): controla a ressonância da rede, nomeado os parâmetros responsáveis pelo número de categorias criadas.

Se ρ possui um valor alto, a rede neural torna-se mais seletiva reduzindo sua capacidade de generalização. Se ρ possui um valor pequeno, é reduzido o número de categorias criadas, maximizando a capacidade de generalização da rede ARTMAP-*Fuzzy*.

De acordo com Lopes (2005), a rede neural ARTMAP-*Fuzzy* consiste em dois módulos ART *Fuzzy*, o ARTa e o ARTb. E são apresentadas em seguida as etapas de como o algoritmo é executado durante a fase de treinamento.

Etapa 1: obtém-se os dados de entrada para o módulo ARTa e ARTb.

Etapa 2: efetua-se a normalização dos vetores A e B, respectivamente. Em (1) é

possível observar a normalização do vetor de entrada do módulo ARTa.

$$\bar{a} = \frac{a}{|a|} \quad (1)$$

Etapa 3: inicialmente todos os pesos da rede neural possuem o valor igual a 1, indicando que não existe categoria ativa.

$$W_j^a = W_k^b = W_j^{ab} = 1 \quad (2)$$

Etapa 4: faz-se a leitura dos parâmetros $\alpha,\ \beta,\ \rho_a, \rho_b, \rho_{ab},\ \varepsilon$.

Etapa 5: efetua-se a codificação do complemento, preservando a amplitude da informação.

$$\overline{a_i^c} = 1 - \bar{a}_i \quad (3)$$

Etapa 6: então o vetor de entrada no campo F_0 será um vetor 2M-dimensional.

$$I = [\bar{a} \quad \overline{a^c}] = \left[\overline{a_1} \quad \ldots \quad \bar{a}_M \quad \bar{a}_1^c \quad \ldots \quad \bar{a}_M^c\right] \quad (4)$$

Etapa 7: obtém-se valores os quais poderão ser escolhidos como categoria.

$$T_j = \frac{|I \wedge W_j|}{\alpha + |W_j|} \quad (5)$$

Etapa 8: escolhe-se a categoria do módulo ARTb.

$$T_k^b = \max\left\{T_k^b : k = 1, \ldots, N_b\right\} \quad (6)$$

Etapa 9: neste momento ocorre o teste de vigilância. Se houver ressonância, parte-se para a etapa 11, caso contrário, etapa 10.

$$\frac{\left|I_s \wedge W_k^b\right|}{I_s} \geq \rho_b \quad (7)$$

Etapa 10: reset. Após resetado volta-se para a etapa 8.

$$T_k^b = 0 \quad (8)$$

Etapa 11: adapta-se os pesos do módulo ARTb.

$$W_k^{novo} = \beta\left(I_s \wedge W_k^{velho}\right) + (1-\beta)W_k^{velho} \quad (9)$$

Etapa 12: atualiza-se o vetor atividade Y_b.

Etapa 13: escolhe-se a categoria do módulo ARTa.

$$T_j^a = \max\left\{T_j^a : j = 1, \ldots, N_a\right\} \quad (10)$$

Etapa 14: faz-se o teste de vigilância. Se houver ressonância, parte-se para a etapa 16, caso contrário, etapa 15.

$$\frac{\left|I_e \wedge W_j^a\right|}{I_e} \geq \rho_a \quad (11)$$

Etapa 15: reset. Após resetado volta-se para a etapa 13.

$$T_j^a = 0 \quad (12)$$

Etapa 16: realiza-se o *match tracking*. Se houver ressonância, parte-se para a etapa 19, caso contrário, etapa 17.

$$\frac{\left|Y^b \wedge W_j^{ab}\right|}{Y^b} \geq \rho_{ab} \quad (13)$$

Etapa 17: reset.

$$T_j^a = 0 \quad (14)$$

Etapa 18: acresce-se o parâmetro de vigilância, e volta-se para a etapa 13.

$$\rho_a = \frac{\left|I_e \wedge W_j^a\right|}{I_e} + \varepsilon \quad (15)$$

Etapa 19: atualiza-se o vetor atividade de F_2.

Etapa 20: adapta-se os pesos do módulo ARTa.

$$W_j^{novo} = \beta\left(I_e \wedge W_j^{velho}\right) + (1-\beta)W_j^{velho} \quad (16)$$

Etapa 21: adapta-se os pesos Inter-ART. Treinamento concluído.

Desenvolvimento e simulação computacional

O principal objetivo deste artigo consiste no desenvolvimento de um sistema inteligente para o monitoramento de falhas estruturais em um trator. Os dados são obtidos através de simulações realizadas em um modelo numérico, construído através de equações

diferenciais. O modelo do trator está ilustrado na figura 2 (SAKAI, 2001).

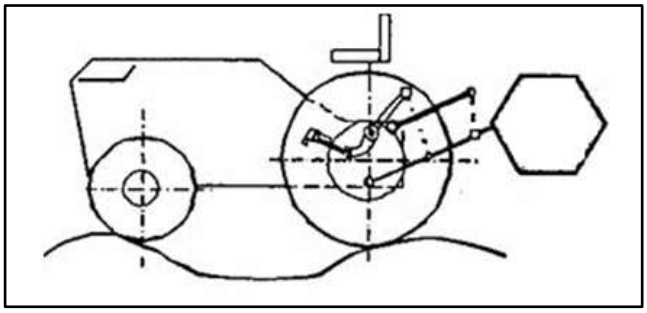

Figura 2. Modelo do trator.

Durante o trabalho no campo, as rodas traseiras e dianteiras de um trator podem perder o contato com o chão. Este fenômeno pode ocorrer de 3 modos.

- Somente as rodas dianteiras perdem contato com o chão (o trator empina a frente);
- Somente as rodas traseiras perdem contato com o chão (o trator empina a traseira);
- E as rodas traseiras e dianteiras perdem contato com o chão ao mesmo tempo (o trator dá um salto).

Diante disto, um modelo matemático simplificado em duas dimensões (horizontal e vertical) que descreve a dinâmica de um trator durante o salto é mostrado pela figura 3.

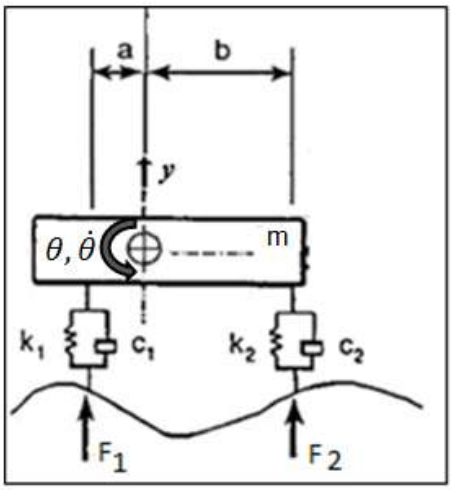

Figura 3. Modelo dinâmico.

A força de equilíbrio na direção vertical e do momento do salto é equacionada por:

$$m\ddot{y} + F_1(y,\dot{y},\theta,\dot{\theta}) + F_2(y,\dot{y},\theta,\dot{\theta}) + mg = 0 \quad (17)$$

$$J\ddot{\theta} + aF_1(y,\dot{y},\theta,\dot{\theta}) - bF_2(y,\dot{y},\theta,\dot{\theta}) = 0 \quad (18)$$

Sendo:

J = momento angular de inércia do centro de massa;

$F_1(\cdot)$ e $F_2(\cdot)$ = cargas dinâmicas nos pneus dianteiro e traseiro do trator, respectivamente, dadas por:

$$F_1(y,\dot{y},\theta,\dot{\theta}) = k_1(y + a\theta - h_1(t)) + c_1(\dot{y} + a\dot{\theta} - \dot{h}_1(t)) \quad (19)$$

$$F_2(y,\dot{y},\theta,\dot{\theta}) = k_2(y - b\theta - h_2(t)) + c_2(\dot{y} - b\dot{\theta} - \dot{h}_2(t)) \quad (20)$$

Nas equações (19) e (20), temos respectivamente $h_1(t)$ e $h_2(t)$ as quais

são excitações periódicas que simulam os desníveis da superfície: $h_1(t) = h_0 sen(\omega t)$ e $h_2(t) = h_0 sen(\omega t - \phi)$ onde, $\omega = \frac{2\pi v}{L}$ e $\phi = \frac{2\pi W_B}{L}$.

Observe que quando o trator:

- Empina a frente $F_1(\cdot) = 0$. Logo: $k_1 = c_1 = 0$;
- Empina a traseira $F_2(\cdot) = 0$. Logo: $k_2 = c_2 = 0$;
- Dá um salto $F_1(\cdot) = F_2(\cdot) = 0$. Logo: $k_1 = c_1 = k_2 = c_2 = 0$.

Para realizar as simulações das condições normais (base-line) e anormais (falha estrutural), no trator altera-se a distância entre as saliências consecutivas (L=0,6m, L=0,4m, L=0,3m, L=0,2m e L=0,15m) e variam-se as velocidades (v=1m/s até 10m/s) para cada L. Sabe-se que o trator funciona em condições normais com L=0,6m e a variação de v=1m/s a 3m/s, e sendo as outras consideradas anormais, como L=0,15m e velocidade v=6m/s que, sob certas condições do traçado e velocidade, o trator poderá apresentar instabilidade nas vibrações prejudicando o controle por parte do operador.

Funcionamento do sistema

O sistema inteligente de monitoramento e diagnóstico de falhas proposto neste artigo segue o fluxograma proposto por Lima et al. (2014) ilustrada na figura 4.

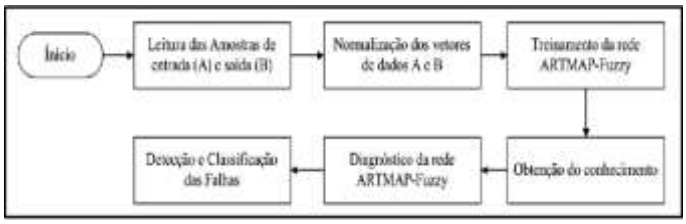

Figura 4. Funcionamento do sistema inteligente.

A fase de aquisição é um modelo numérico de um trator já mencionado anteriormente. O conjunto de entrada da rede ARTMAP-*Fuzzy* é composto por sinais de resposta em frequência.

A fase de treinamento da rede neural ARTMAP-*Fuzzy* é realizada de forma off-line e é importante para ajustar os pesos e criar as categorias que serão utilizadas no diagnóstico dos dados. Nesta fase a rede é treinada com um conjunto de dados gerando uma quantidade de categorias igual ao número de sinais de entrada.

A fase de diagnóstico é realizada de forma online. Nesta fase um novo conjunto de sinais diferentes do conjunto de sinais utilizados na fase de treinamento deve ser apresentada a rede, e cada sinal será analisado individualmente. Todos os sinais são comparados com o conhecimento (categorias criadas na fase de treinamento) obtido pela rede neural ARTMAP-*Fuzzy* e assim, quando houver um casamento entre os padrões classifica-se em categoria o sinal apresentado a rede. Obtida a categoria correspondente a um determinado padrão é, portanto, identificado a intensidade (porcentagem de falha) e a localização da falha na estrutura. E para exemplificar os sinais analisados pelo sistema, pode-se observar um sinal normal através da figura 5 e um sinal anormal através da figura 6.

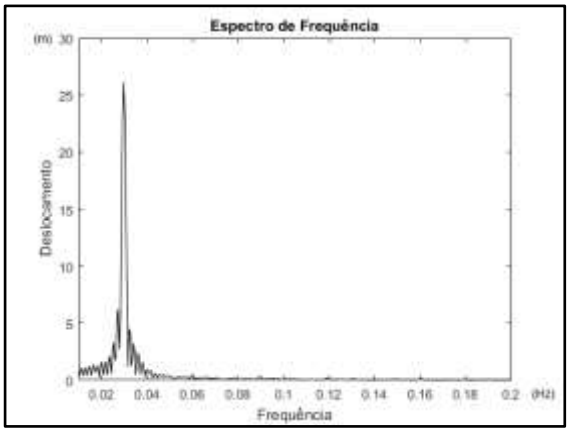

Figura 5. Representação de um sinal normal.

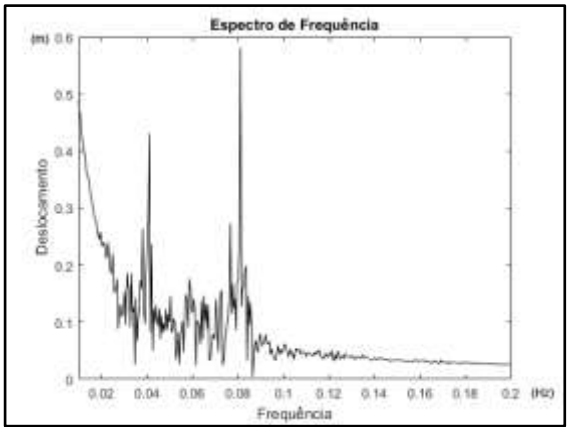

Figura 6. Representação de um sinal anormal.

Cenários propostos para análise do sistema

Para testar o sistema inteligente, foram propostos 5 cenários diferentes para treinamento e diagnóstico, representados a seguir por tabelas, onde L é a distância em metros entre as saliências, e estão dispostas em cinco linhas. Disposto em colunas estão as velocidades, de 1 a 10 m/s. Dentro das tabelas, a letra T significa que os dados foram utilizados para treinamento e a letra D para diagnóstico. Os cenários podem ser observados através das tabelas 1, 2, 3, 4 e 5. Sabe-se que são condições normais apenas os sinais onde L=0,6m e velocidade igual a 1, 2 ou 3m/s. As demais situações, são todas de falha estrutural, ou seja, 47 situações.

Tabela 1. Cenário 1.

m/s	1	2	3	4	5	6	7	8	9	10
L=0,6m	T	T	T	T	T	T	T	D	D	D
L=0,4m	T	T	T	T	T	T	T	D	D	D
L=0,3m	T	T	T	T	T	T	T	D	D	D
L=0,2m	T	T	T	T	T	T	T	D	D	D
L=0,15m	T	T	T	T	T	T	T	D	D	D

Tabela 2. Cenário 2.

m/s	1	2	3	4	5	6	7	8	9	10
L=0,6m	T	T	T	T	T	D	T	D	T	D
L=0,4m	T	T	T	T	T	D	T	D	T	D
L=0,3m	T	T	T	T	T	D	T	D	T	D
L=0,2m	T	T	T	T	T	D	T	D	T	D
L=0,15m	T	T	T	T	T	D	T	D	T	D

Tabela 3. Cenário 3.

m/s	1	2	3	4	5	6	7	8	9	10
L=0,6m	T	T	T	T	D	T	T	T	T	T
L=0,4m	D	D	D	D	T	T	T	T	T	T
L=0,3m	T	T	D	D	D	D	T	T	T	T
L=0,2m	T	D	T	T	T	T	D	D	D	T
L=0,15m	D	T	T	T	T	T	T	T	T	D

Tabela 4. Cenário 4.

m/s	1	2	3	4	5	6	7	8	9	10
L=0,6m	T	T	T	T	T	T	T	T	T	T
L=0,4m	D	T	D	T	D	T	D	T	D	T
L=0,3m	T	D	T	D	T	D	T	D	T	D
L=0,2m	D	T	D	T	D	T	D	T	D	T
L=0,15m	T	T	T	T	T	T	T	T	T	T

Tabela 5. Cenário 5.

m/s	1	2	3	4	5	6	7	8	9	10
L=0,6m	T	D	T	D	T	D	T	D	T	D
L=0,4m	T	T	T	T	T	T	T	T	T	T
L=0,3m	D	T	D	T	D	T	D	T	D	T
L=0,2m	T	T	T	T	T	T	T	T	T	T
L=0,15m	T	D	T	D	T	D	T	D	T	D

Após diversas execuções, a melhor configuração dos parâmetros independente do cenário aplicado, pode ser observada na tabela 6.

Tabela 6. Valores dos parâmetros utilizados pela rede ARTMAP-*Fuzzy*.

Parâmetros	Valores
α	0,1
β	1
ρ_a	0,45
ρ_b	1
ρ_{ab}	1
ξ	0,0001

É importante destacar que foi possível estabelecer um valor para ρ_a relativamente baixo, devido os sinais apresentarem grande diferença entre condição normal e com falha, conforme visto anteriormente através da figura 6.

O algoritmo foi desenvolvido em MATLAB®, utilizando-se a versão 6.5, e os testes realizados em um PC Intel Core 2 Duo 2.4 GHz, 6GB de memória RAM, e sistema operacional Windows 8 Professional 64 bits,

RESULTADOS E DISCUSSÃO

Para o treinamento da rede neural foi utilizado um conjunto de dados com 35 sinais. A quantidade de sinais utilizadas para o treinamento, representa 70% dos dados disponíveis. Com objetivo de avaliar a rede neural na fase de diagnóstico foi usado um conjunto de dados com 15 sinais. Os dados utilizados na fase de diagnóstico representam 30% dos dados disponíveis. A base de dados simulada apresenta 47 padrões distintos de falhas e apenas 3 em condições normais.

Na tabela 7 apresentam-se os resultados obtidos pelo sistema SHM aplicado aos dados do trator. O resultado é o mesmo para os cinco cenários propostos.

Tabela 7. Resultados obtidos aplicando rede ARTMAP-*Fuzzy*.

	Fase de Treinamento	Fase de Diagnóstico
Amostras Utilizadas	35	15
Classificações Corretas	35	15
Classificações Erradas	0	0
Acerto (*%*)	100	100
Tempo (*ms*)	300	90

Os resultados apresentados na tabela 7 foram obtidos a partir da melhor configuração do sistema, sendo possível somente após uma fase experimental de testes e ajustes de parâmetros. O índice de acerto igual a 100% demonstra que o sistema SHM proposto apresenta um ótimo desempenho.

CONCLUSÃO

O sistema de diagnóstico de falhas em estruturas proposto neste estudo apresentou ótimos resultados, alcançando um índice de acerto de 100% quando executado com a sua melhor configuração. A rede neural ARTMAP-*Fuzzy* mostrou-se apta para ser utilizada como um sistema de tempo real, pois a fase de treinamento que é mais demorada, pode ser realizada de forma off-line, enquanto que a fase de diagnóstico obteve um tempo de resposta de 6 milissegundos por sinal analisado, tempo este aceitável para um sistema de tomada de decisão online. Portanto, conclui-se que o sistema para monitoramento de falhas em estruturas proposto a partir de uma rede neural artificial ARTMAP-*Fuzzy* é confiável e preciso quando utilizado com objetivo de manter a integridade de estruturas mecânicas

Um dos motivos da resistência a tração da fibra de bambu ser maior que a da cana pode ser entendido utilizando os conhecimentos de Rudall (2007), pois através dos resultados da microtomografia da fibra de bambu, nota-se que a quantidade de xilema, que tem função

estrutural, é maior que a da fibra da cana, e que, assim também, a concentração de lignina nesta região possa ser maior que em outras áreas da planta.

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Artificial Immune Systems Applied in Data Management Solutions to the Problem of Restoration of Electrical Distribution Systems

A. M. Cossi and M. L. M. Lopes

***Abstract*— This paper proposes a method based on Artificial Immune Systems for data management solutions provided by a model used to train power grid and find possible solutions to the problem of restoration of electricity distribution systems. The method aims to manage the supply of recurring defects solutions in the power grid through information stored in a database; the network training for non-recurring defects and storage of data in the database; choosing solutions in the database to start the network training process in the search for new solutions. To train the network and find new solutions to the restoration problem, we use a model that has as one solution technique Tabu Search algorithm. To test the efficiency of the proposed method, we present results of tests and discussions carried out in a medium voltage distribution system.**

***Keywords*— Data management, artificial immune systems, distribution systems, restoration.**

I. INTRODUÇÃO

AS FERRAMENTAS desenvolvidas para resolver o Problema de Restauração de Sistemas de Distribuição de Energia Elétrica (PRSDEE) devem ser eficientes para proporcionar uma resposta rápida para a restauração do sistema evitando assim que as empresas distribuidoras de energia elétrica deixem de cumprir com as metas de qualidade impostas pelos órgãos reguladores. Neste contexto, este trabalho busca contribuir com essas ferramentas através de um método baseado em Sistemas Imunológicos Artificiais (SIA), cujo objetivo é de gerenciar os dados de soluções de PRSDEE's e tentar promover um restabelecimento do sistema de forma mais rápida.

O método proposto consiste em gerenciar: 1- o fornecimento de soluções de PRSDEE's recorrentes através de informações armazenadas na memória de um banco de dados; 2- o treinamento da rede para PRSDEE's não recorrentes e memorização dos dados das soluções na memória do banco de dados; 3- o fornecimento de soluções existentes na memória do banco de dados que sejam de boa qualidade para iniciar processos de treinamento de redes elétricas na busca de novas soluções para resolver o PRSDEE. O fato de iniciar o processo de treinamento da rede elétrica na busca de soluções de PRSDEE's através de uma solução de boa qualidade, pode reduzir significativamente o espaço de busca de soluções e consequentemente baixar o tempo de processamento, gerando assim uma resposta mais rápida para o PRSDEE.

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O método baseado em SIA funciona através do reconhecimento de cadeias de padrões pré-determinadas pelo sistema imunológico entre as células com anomalias e suas respectivas cura. Neste caso, para cada anomalia ocorrida na célula, a qual é identificada através de um antígeno, é verificada a existência ou não de anticorpos (cura) para o antígeno apresentado pela anomalia nessa célula. Assim, o método poderá fornecer anticorpos para anomalias já conhecidas em determinadas células ou criar anticorpos para anomalias desconhecidas. O funcionamento da técnica SIA é detalhada em [1] e [2].

Na literatura especializada existem diversos trabalhos que abordam o tema sobre PRSDEE's. Dentre eles podemos destacar os que utilizam como técnica de solução os métodos heurísticos [3] e metaheurísticos: algoritmos genéticos [4] e [5], tabu search [6], sistemas imunológicos artificiais [7], dentre outros. No entanto, não existem trabalhos que tratam especificamente do problema de gerenciamento de dados de soluções de problemas ligados a sistemas de distribuição de energia elétrica utilizando SIA.

O método proposto pode ser utilizado para treinar a rede considerando diversas situações de PRSDEE's e armazenar as soluções na memória do banco de dados para que possam ser utilizadas em casos de defeitos na rede de distribuição, obtendo assim uma resposta rápida para o PRSDEE. Além disso, o método pode ser adaptado e utilizado em conjunto com outras ferramentas utilizadas para resolver problemas diversos ligados a sistemas de distribuição de energia elétrica. Para treinar a rede de distribuição na busca de soluções e resolver o PRSDEE utiliza-se um modelo de restauração que possui como técnica de solução um algoritmo tabu search (TS). Ressalta-se que não é objeto deste trabalho apresentar o modelo de treinamento da rede elétrica utilizando o algoritmo TS.

Para testar a eficiência do método proposto de gerenciamento de dados de soluções para o PRSDEE, apresentam-se resultados e discussões de testes realizados em um sistema de distribuição de média tensão, cujos dados podem ser encontrados em [8].

O trabalho está organizado da seguinte forma: na Seção 2 apresenta-se em detalhes o método proposto baseado em SIA; na Seção 3 apresentam-se os resultados e comentários dos testes; e na Seção 4 as conclusões finais.

II. MÉTODO BASEADO EM SIA

O método baseado em SIA desenvolvido para o gerenciamento de dados é chamado de MGD-SIA (Método de Gerenciamento de Dados Baseado em Sistemas Imunológicos Artificiais). No PRSDEE o MGD-SIA identifica um sistema de distribuição como sendo uma célula. Um defeito no sistema

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de distribuição é então conhecido como uma anomalia nessa célula. Uma anomalia na célula é então identificada pelo defeito ocorrido em uma determinada seção do sistema de distribuição, a qual é reconhecido pelo MGD-SIA através de um agente patológico chamado de antígeno (identificação da seção onde o defeito ocorreu). Uma possível solução para o defeito ocorrido no sistema de distribuição (anomalia na célula) é reconhecida pelo MGD-SIA como sendo um anticorpo com afinidade total ou parcial ao antígeno da anomalia. Tal anticorpo seria então uma solução para o problema da restauração total ou parcial do sistema de distribuição após a ocorrência do defeito, ou seja, seria a cura para a anomalia na célula. A Fig. 1 ilustra de forma geral o método proposto.

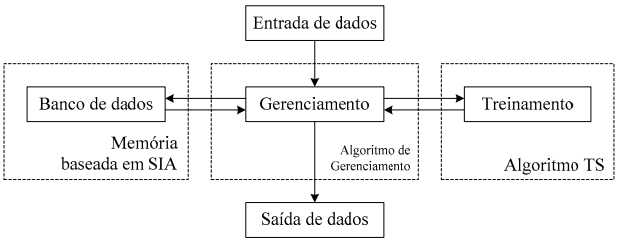

Figura. 1. Esquema geral do método baseado em SIA (MGD-SIA).

De acordo com a Fig. 1, temos:

- Entrada de dados: são os dados da célula e da anomalia (antígeno) ocorrida na célula, ou seja, são os dados do sistema elétrico, do defeito e das seções da rede afetadas pelo defeito;
- Gerenciamento: trata-se de um algoritmo de gerenciamento de dados. Este algoritmo realiza todo o processo de análise, comparação e escolha dos dados, bem como a comunicação entre a entrada de dados, o banco de dados, o treinamento e a saída de dados;
- Treinamento: algoritmo TS desenvolvido para avaliar o defeito e encontrar possíveis curas (anticorpos) para anomalias desconhecidas (antígeno) ocorrida na célula, ou seja, algoritmo TS desenvolvido para encontrar as possíveis soluções para o PRSDEE;
- Banco de dados: armazena os dados de células que já foram analisadas e de seus respectivos antígenos e anticorpos através de cadeias de padrões e suas respectivas topologias de rede, ou seja, armazena dados de sistemas elétricos que já foram estudados e dos defeitos e suas respectivas soluções.
- Saída de dados: são os padrões de anticorpos para o antígeno e suas respectivas topologias de rede, ou seja, são as possíveis soluções para o PRSDEE.

A seguir, apresenta-se de forma detalhada o MGD-SIA.

2.1 Algoritmo de Gerenciamento do MGD-SIA

O algoritmo de gerenciamento do MGD-SIA funciona da seguinte maneira:

1. Detecção de anomalia na célula;
2. Identificação, localização e classificação do agente patológico (antígeno);
3. Verificar se a célula já foi analisada e se o antígeno é conhecido:
   3.1. Se a célula nunca foi analisada, ir ao passo 4.1 (resposta primária).
   3.2. Se a célula já foi analisada e o antígeno não for conhecido, ir ao passo 4.2 (resposta primária).
   3.3. Se a célula já foi analisada e o antígeno for conhecido, ir ao passo 5 (resposta secundária).
4. Resposta primária:
   4.1. Se a célula nunca foi analisada, encontrar a cura através da produção de anticorpos com afinidade total ou parcial ao antígeno através do treinamento da rede utilizando o algoritmo TS, da seguinte maneira:
      4.1.1. Guardar a nova célula no conjunto de células analisadas;
      4.1.2. Guardar o novo antígeno no conjunto de antígenos ligados à célula;
      4.1.3. Produzir padrões de anticorpos através do treinamento da rede sem utilizar uma solução inicial conhecida;
      4.1.4. Escolher o anticorpo com maior afinidade ao antígeno (melhor função de adaptação);
      4.1.5. Guardar o anticorpo no conjunto de anticorpos ligados à célula;
      4.1.6. Adotar a cadeia de padrões do anticorpo como a cura para a anomalia;
      4.1.7. Ir ao passo 6.
   4.2. Se a célula já foi analisada, porém o antígeno não é conhecido, encontrar a cura através da produção de anticorpos com afinidade total ou parcial ao antígeno através do treinamento da rede utilizando o algoritmo TS, da seguinte maneira:
      4.2.1. Guardar o novo antígeno no conjunto de antígenos ligados à célula;
      4.2.2. Verificar a possibilidade de iniciar o treinamento da rede através de algum anticorpo, pertencente ao conjunto de anticorpos, com alguma afinidade parcial ao antígeno (iniciar o treinamento da rede através de uma solução inicial de boa qualidade), da seguinte maneira:
         i. Calcular a taxa de afinidade T_{Af} do antígeno;
         ii. Apresentar o antígeno ao conjunto de anticorpos. Neste caso calcula-se a afinidade A_{fi} de cada anticorpo em relação ao antígeno;
         iii. Se um ou mais anticorpos apresentarem uma A_{fi} maior que T_{Af}, este(s) anticorpo(s) possui(em) alguma afinidade ao antígeno. Neste caso, o anticorpo que possuir a maior afinidade será a solução inicial do algoritmo TS. Ir ao passo 4.2.3. Caso contrário, ir ao passo 4.2.4 e iniciar o treinamento da rede normalmente (sem solução inicial previamente conhecida).
      4.2.3. Produzir anticorpos através do treinamento da rede utilizando uma solução inicial conhecida e escolher aquele com maior afinidade ao

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antígeno (melhor função de adaptação). Após, ir ao passo 4.2.5.

4.2.4. Produzir anticorpos através do treinamento da rede sem utilizar uma solução inicial conhecida e escolher aquele com maior afinidade ao antígeno (melhor função de adaptação). Após, ir ao passo 4.2.5.

4.2.5. Guardar o anticorpo no conjunto de anticorpos ligados à célula;

4.2.6. Adotar a cadeia de padrões do anticorpo como a cura para a anomalia;

4.2.7. Ir ao passo 6.

5. Resposta secundária: se a célula e o antígeno forem conhecidos, encontrar a cura através de um anticorpo com afinidade total ou parcial ao antígeno, da seguinte maneira:
   5.1. Apresentar o antígeno a todos os anticorpos do conjunto de anticorpos;
   5.2. Verificar qual é o anticorpo correspondente ao antígeno (mesma cadeia de padrões);
   5.3. Apresentar a cadeia de padrões do anticorpo como cura para a anomalia;
   5.4. Ir ao passo 6.
6. Gerar a saída de dados, podendo ser uma das soluções:
   6.1. Solução 1: cadeia de padrões do anticorpo com maior afinidade (total ou parcial) ao antígeno gerado pela resposta primária;
   6.2. Solução 2: cadeia de padrões do anticorpo com maior afinidade (total ou parcial) ao antígeno gerado pela resposta secundária;
   6.3. Solução 3: nenhum anticorpo. Neste caso não foi encontrada nenhuma cura para a anomalia na célula. O sistema não será restaurado.

A Fig. 2 ilustra de forma resumida o algoritmo de gerenciamento do MGD-SIA descrito anteriormente.

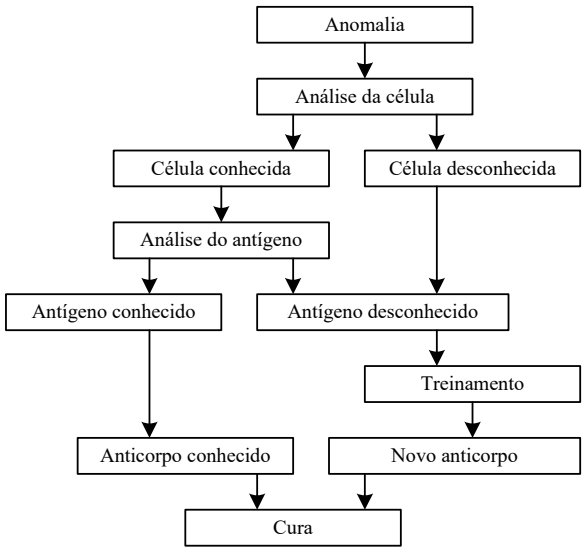

Figura. 2. Resumo do algoritmo de gerenciamento do MGD-SIA.

2.2 *Memória do* MGD-SIA

O esquema de memória do MGD-SIA consiste em armazenar dados de células estudadas após a ocorrência de uma anomalia nessa célula. Neste caso, são armazenados na memória do banco de dados o conjunto de células e os seus respectivos padrões de antígenos (dados da rede elétrica com defeito) e anticorpos (dados da rede elétrica após o restabelecimento da rede), bem como todos os demais dados da rede elétrica. A busca por soluções na memória do banco de dados consiste em analisar a afinidade entre os padrões de antígeno e anticorpos. Esse esquema de memória é detalhado a seguir.

A. Conjunto de células conhecidas (LC)

A célula representa um sistema elétrico, e é conhecida por um nome de identificação conforme a quantidade de seções. Cada seção é considerada como sendo uma região da rede de distribuição composta por várias barras de carga (consumidores).

O conjunto de células LC é formado por células conhecidas:

- $LC = \{C_{12}, C_{34}, C_{120}, ..., C_{ns}\}$

sendo: C_{12} é o sistema S12 (sistema de 12 seções); C_{34} é o sistema S34 (sistema de 34 seções); C_{120} é o sistema S120 (sistema de 120 seções), C_{ns} é o sistema S_{ns} (sistema de *ns* seções).

B. Conjunto de antígenos ligados à célula (LAg)

Um antígeno representa a anomalia (agente patológico) na célula. Assim, um defeito na seção *k* da célula C_j (rede elétrica) teria o antígeno AgC_j^k. Portanto, o antígeno é identificado pelo número da seção em que a anomalia acorreu na célula. Outro defeito em outra seção da célula seria outro antígeno ligado à mesma célula e assim por diante.

O conjunto de antígenos $LAgC_j$ é formado pelos antígenos ligados a célula sob análise:

- $LAgC_j = \{AgC_j^1, AgC_j^2, ..., AgC_j^k, ..., AgC_j^n\}$
- $AgC_j^k = \{S1C_j^k, S2C_j^k, ..., SiC_j^k, ..., SnC_j^k\}$

sendo: $LAgC_j$ é o conjunto formado por diferentes antígenos representando diferentes anomalias ocorridas na célula C_j; AgC_j^k é o antígeno que representa a anomalia ocorrida na seção *k* da célula C_j, e é formado por uma cadeia de padrões composto por: seção do defeito (seção da anomalia - S_a), seções que foram desenergizadas pelo defeito (seções com padrões não próprios – S_{np}) e seções que não foram desenergizadas (seções com padrões próprios – S_p); SiC_j^k é a seção *i* da cadeia de padrões do antígeno AgC_j^k.

C. Conjunto de anticorpos ligados à célula conhecida (LAb)

Um anticorpo representa a cura para a anomalia na célula. O anticorpo deve ter afinidade total ou parcial ao antígeno. Assim, um defeito na seção *k* da célula C_j teria o antígeno AgC_j^k. Portanto, o anticorpo com afinidade ao antígeno AgC_j^k seria AbC_j^k.

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O conjunto de anticorpos $LAbC_j$ é formado pelos anticorpos com alguma afinidade a seus respectivos antígenos ligados a célula sob análise:

- $LAbC_j = \{AbC_j^1, AbC_j^2, ..., AbC_j^k, ..., AbC_j^n\}$
- $AbC_j^k = \{S1C_j^k, S2C_j^k, ..., SiC_j^k, ..., SnC_j^k\}$

sendo: $LAbC_j$ é o conjunto de anticorpos com alguma afinidade aos seus respectivos antígenos para diferentes anomalias ocorridas célula C_j; AbC_j^k é o anticorpo com afinidade ao antígeno da anomalia ocorrida na seção *k* da célula C_j, e é representado por uma cadeia de padrões composta pelas seções S_a, S_{np} (seções restauradas ou não pelo algoritmo TS) e S_p; SiC_j^k é uma seção *i* da cadeia de padrões do anticorpo AbC_j^k.

Se a cadeia de padrões do anticorpo AgC_j^k apresentar biC_j^k=0 na posição correspondente da seção *i* ou para qualquer seção do anticorpo, significa que esta seção não foi religada pelo processo de restauração.

2.3 Cadeia de padrões

As Fig. 3 e 4 ilustram o casamento entre as cadeias de padrões de antígeno e anticorpo. Os vetores são formados pelas seções S_a, S_{np} e S_p, e pelo vetor afinidade entre essas cadeias.

No MGD-SIA, os padrões (antígeno e anticorpo) a serem analisados são os padrões próprios e não-próprios. Neste caso, a seção do defeito referente ao antígeno analisado não entra na análise de afinidade entre as cadeias, apesar de fazer parte da cadeia de padrões.

A. Padrões com afinidade total

Para haver afinidade total entre os padrões analisados do anticorpo e antígeno todas as posições do vetor afinidade deve possuir 1 (um). Neste caso, para o antígeno AgC_j^k existe um anticorpo AbC_j^k, ou seja, existe o casamento total entre as cadeias de padrões do antígeno e anticorpo. A Fig. 3 ilustra um exemplo de afinidade total do antígeno AgC_j^k com o anticorpo AgC_j^k.

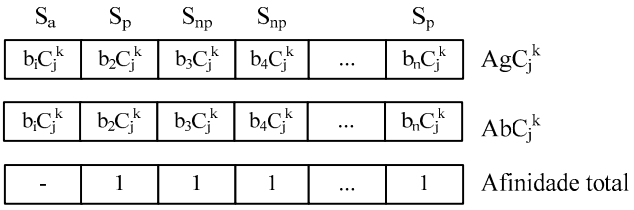

Figura 3. Comparação entre cadeia de padrões de antígeno e anticorpo com afinidade total.

B. Padrões com afinidade parcial

Caso alguma posição do vetor afinidade possua o valor 0 (zero), significa que não existe afinidade total do anticorpo ao antígeno, ou seja, para o antígeno AgC_j^k não existe um anticorpo AbC_j^k com afinidade total. Neste caso, poderá haver ou não um anticorpo uma afinidade parcial ao antígeno. A Fig. 4 ilustra um exemplo de afinidade parcial do antígeno AgC_j^k com algum anticorpo da lista de anticorpos, como por exemplo o anticorpo AgC_j^g.

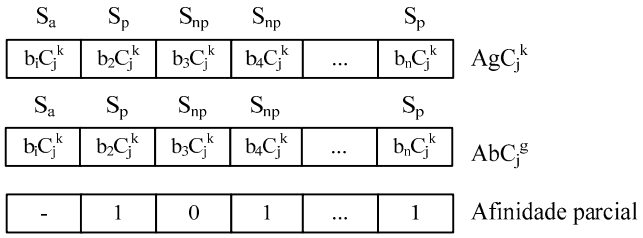

Figura 4. Comparação entre cadeia de padrões de antígeno e anticorpo com afinidade parcial.

2.4 Critério de afinidade

A. Cura com afinidade total ou parcial

A cura com afinidade total ou parcial entre anticorpo e antígeno somente poderá existir para células conhecidas. Neste caso, para uma anomalia na seção *k* da célula C_j tem-se o antígeno AgC_j^k. Se existir um anticorpo AbC_j^k na memória do banco de dados, significa que este anticorpo é uma cura total para a anomalia. No entanto, caso não exista um anticorpo ligado ao antígeno na memória do banco de dados, a cura poderá ser encontrada pelo treinamento da rede elétrica através do algoritmo TS. Caso, o algoritmo TS não encontre um anticorpo com afinidade total ao antígeno, este poderá encontrar anticorpos com afinidade parcial ao antígeno. Assim, anticorpos com afinidade parcial ao antígeno existentes na memória do banco de dados, bem como encontrados no treinamento da rede pelo algoritmo TS, podem ser curas parciais para anomalias em células.

B. Determinação de afinidade parcial

O anticorpo com afinidade parcial ao antígeno poderá ser utilizado da seguinte forma:

- Como solução inicial do algoritmo TS: caso um anticorpo possua alguma afinidade parcial ao antígeno, sua cadeia de padrões poderá ser usada como solução inicial para o treinamento da rede através do algoritmo TS na busca por soluções de boa qualidade para o PRSDEE.
- Como solução final para o PRSDEE: caso não seja encontrado nenhum anticorpo com afinidade total ao antígeno, a cadeia de padrões do anticorpo com a maior afinidade parcial ao antígeno existente no banco de dados ou obtido através do algoritmo TS poderá ser usada como solução final para o PRSDEE.

No MGD-SIA, para avaliar a afinidade parcial entre anticorpo e antígeno, utiliza-se o conceito de casamento parcial e afinidade proposto em [9]. A taxa de afinidade representa o grau de semelhança necessária para ocorrer o casamento entre as duas cadeias em análise, definida como:

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$$T_{Af} = \left(\frac{A_n}{A_t} \right) * 100 \quad (1)$$

sendo: *TAf* é a taxa de afinidade; A_n é o número de cadeias normais no problema (cadeias próprias); *At* é o número total de cadeias no problema (cadeias próprias e não próprias).

A afinidade entre os dois padrões é dada por:

$$A_{ft} = \frac{\sum_{i=1}^{L} PC}{L} * 100 \quad (2)$$

sendo: *Aft* é a porcentagem de afinidade entre os padrões analisados; *L* é a quantidade total de posições; *PC* é a posição casada; $\sum_{i=1}^{L} PC$ é o somatório (quantidade) de posições casadas.

De acordo com o trabalho apresentado em [9], se *Aft* for maior ou igual a *TAf* (*Aft* ≥ *TAf*) ocorre o casamento entre os dois padrões, ou seja, os dois são considerados semelhantes. Caso *Aft* seja menor que *TAf* (*Aft* < *TAf*) o anticorpo não reconhece o antígeno, portanto não existe casamento entre os padrões. No entanto, no PRSDEE, para haver algum grau de afinidade entre os padrões do anticorpo e antígeno, *Aft* deverá ser maior que *TAf* (*Aft* > *TAf*). Neste caso, se *Aft* for menor ou igual *TAf* (*Aft* ≤ *TAf*) não existe casamento parcial entre os padrões.

III. Testes e Resultados

Para testar a eficiência do MGD-SIA, apresentam-se resultados de testes para um defeito (anomalia) em um sistema de distribuição (célula) de média tensão, considerando que a célula é conhecida, ou seja, já foi analisada, porém com anomalia desconhecida, ou seja, não possui antígeno (defeito) e nem anticorpo com afinidade total (solução) na memória do banco de dados. Como a anomalia não possui anticorpos na memória do banco de dados, será necessário efetuar o treinamento da rede para a obtenção de anticorpos (soluções) com afinidade total ou parcial ao antígeno (defeito) da anomalia. Além disso, como a célula já foi analisada, possui anticorpos na memória do banco de dados que poderão servir como solução inicial (anticorpos com afinidade parcial ao antígeno) para o treinamento da rede através do algoritmo TS. A solução final para o PRSDEE deverá possuir uma cadeia de padrões de anticorpo com afinidade total ao antígeno, ou mesmo parcial que seja melhor que a solução inicial utilizada pelo algoritmo TS.

Considere então uma anomalia na seção 3 da célula C_{53}. Trata-se de uma célula existente no banco de dados, porém com antígeno desconhecido, contendo 53 seções (50 seções de carga e 3 seções representando as subestações). Os dados da célula C_{53} encontram-se em [8].

A. Dados inicias

O conjunto de células LC do banco de dados é dado por:

- $LC = \{C_{12}, C_{53}, C_{120}, ..., C_{ns}\}$
- C_{53}: S53 (sistema de 53 seções)

Para a célula C_{53}, temos o seguinte conjunto de antígenos:

- $LAgC_{53} = \{AgC_{53}^{2}, AgC_{53}^{7}, AgC_{53}^{8}, AgC_{53}^{11}\}$

Para a anomalia na seção 3 da célula C_{53} temos:

- Antígeno da anomalia: AgC_{53}^{3}
- $AgC_{53}^{3} \notin LAgC_{53}$

B. Anomalia na célula

A Fig. 5 ilustra a célula C_{53} (rede em operação). Considere então um defeito (anomalia) na seção 3 ($S3C_{53}^{3}$), e a Fig. 6 mostra a cadeia de padrões não próprios do antígeno (AgC_{53}^{3}) para esta anomalia. Com o defeito na seção 3, as seções 4, 5, 6, 7, 8, 26, 27 e 28 ficaram desenergizadas. Para esta anomalia, a cadeia de padrões do antígeno AgC_{53}^{3} é composto pelas seções S_a, S_p e S_{np}:

- $AgC_{53}^{3}(S_a) = \{S3C_{53}^{3}\} = \{3\}$
- $AgC_{53}^{3}(S_{np}) = \{S4C_{53}^{3}, S5C_{53}^{3}, S6C_{53}^{3}, S7C_{53}^{3}, S8C_{53}^{3}, S26C_{53}^{3}, S27C_{53}^{3}, S28C_{53}^{3}\} = \{4, 5, 6, 7, 8, 26, 27, 28\}$
- $AgC_{53}^{3}(S_p)$ = todas as demais 45 seções do sistema.

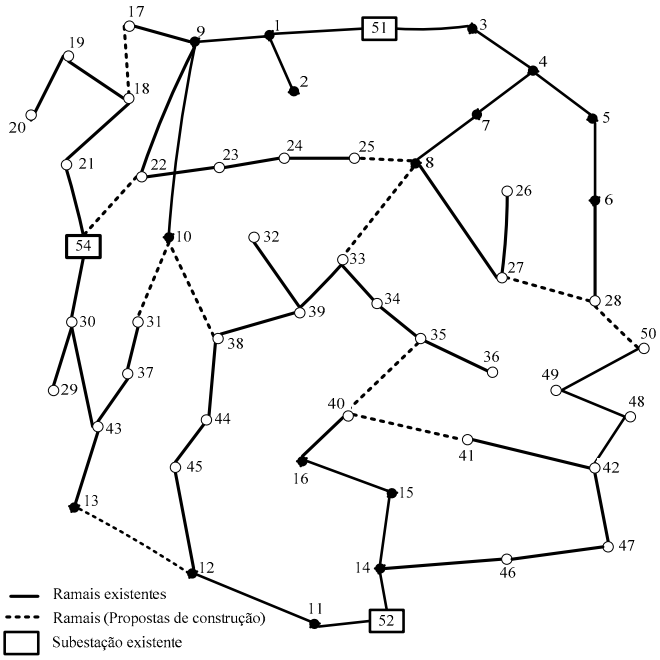

Figura 5. Célula C_{53} em operação antes de ocorrer o defeito na seção 3.

S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}
4	5	6	7	8	26	27	28

Figura 6. Cadeia de padrões não próprios (S_{np}) do antígeno AgC_{53}^{3}.

C. Solução inicial do treinamento

Após checar a lista de antígenos da célula C_{53}, o MGD-SIA encontrou o anticorpo AbC_{53}^{7} com afinidade parcial ao antígeno AgC_{53}^{3}. Este anticorpo será usado para iniciar o algoritmo TS. Trata-se de um anticorpo gerado para a anomalia na seção 7 da célula C_{53}. Para esta anomalia, a

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cadeia de padrões do antígeno AgC_{53}^{7} (Fig. 7) é composto pelas seções S_a, S_p e S_{np}:

- $AgC_{53}^{7}(S_a) = \{7\}$
- $AgC_{53}^{7}(S_{np}) = \{8, 26, 27\}$
- $AgC_{53}^{7}(S_p)$ = todas as demais 50 seções do sistema.

S_{np}	S_{np}	S_{np}
8	26	27

Figura 7. Cadeia de padrões não próprios do antígeno AgC_{53}^{7}.

A afinidade parcial entre o anticorpo AbC_{53}^{7} e o antígeno AgC_{53}^{3} foi checada através das equações 1 e 2, sendo:

$$T_{Af} = \left(\frac{45}{53}\right) * 100 = 84{,}9\% \quad (1)$$

$$A_{ft} = \frac{50}{53} * 100 = 94{,}3\ \% \quad (2)$$

Neste caso temos que *Aft > TAf*. Portanto existe afinidade parcial entre o anticorpo AbC_{53}^{7} e o antígeno AgC_{53}^{3}. Assim, este anticorpo será usado como solução inicial para o treinamento da rede através do algoritmo TS.

D. Cura para a célula

Após treinar a rede através do algoritmo TS, a cura encontrada para a anomalia na seção 3 da célula C_{53}, é dada então pelo anticorpo AbC_{53}^{3}. A Fig. 8 ilustra a célula C_{53} (sistema em operação) após a cura para a anomalia na seção 3 (anticorpo AbC_{53}^{3}).

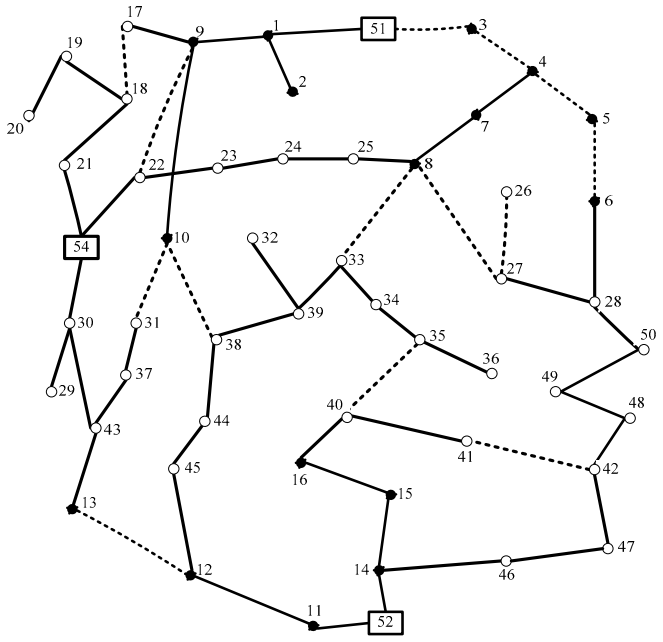

Figura 8. Cura para a anomalia na seção 3 da célula C_{53}.

Para o antígeno AgC_{53}^{3}, temos a seguinte cadeia de padrões do anticorpo AbC_{53}^{31}:

- AbC_{53}^{3} = {4, 0, 6, 7, 8, 0, 27, 28}

A Fig. 9 ilustra a afinidade entre os padrões do antígeno AgC_{53}^{3} e anticorpo AbC_{53}^{31}.

S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	S_{np}	
4	5	6	7	8	26	27	28	AgC_{53}^{3}
4	0	6	7	8	0	27	28	AbC_{53}^{3}
1	0	1	1	1	0	1	1	Afinidade parcial

Figura 9. Afinidade parcial ente o antígeno AgC_{53}^{3} e anticorpo AbC_{53}^{3}.

De acordo com a Fig. 9, percebe-se que além da seção onde ocorreu o defeito, as seções 5 e 26 não foram restauradas pelo algoritmo TS, pois violou uma das restrições impostas ao problema de restauração. A solução encontrada conseguiu restaurar as seções 4, 6, 7, 8, 27 e 28. Neste caso, tem-se uma solução com um anticorpo com afinidade parcial ao antígeno.

O tempo de processamento de treinamento da rede através do algoritmo TS partindo-se de uma solução inicial conhecida foi de 0,257 segundos. A mesma solução encontrada pelo algoritmo TS sem utilizar uma solução inicial de boa qualidade, obteve um tempo de processamento de 0,343 segundos.

IV. Conclusões

De acordo com os resultados, a ferramenta desenvolvida foi capaz de gerenciar as soluções (solução inicial existente e solução final) entre o banco de dados e o treinamento de rede. Além disso, o fato de utilizar uma solução inicial existente no banco de dados no treinamento da rede para encontrar uma solução final de boa qualidade reduziu o tempo de processamento. No entanto, este fato deve ser melhor visto em sistemas de grande porte.

Para anomalias em células que nunca foram analisadas, não existem dados na memória do banco de dados referente a essa célula. Neste caso, o algoritmo MGD-SIA deverá identificar que a célula é desconhecida e treinar a rede através do algoritmo TS para tentar resolver o problema de PRSDEE. Os resultados serão armazenados no banco de dados e poderão ser utilizados na resolução desta ou de outras anomalias na mesma célula. No entanto, para anomalias em células já analisadas e com antígenos conhecidos, o algoritmo MGD-SIA identificará que a célula e o antígeno são conhecidos e encontrará o anticorpo (solução) com afinidade ao antígeno no banco de dados. Neste caso, o tempo de resposta na busca de soluções para o PRSDEE será rápido.

O algoritmo MGD-SIA pode ser utilizado para que o operador do sistema elétrico treine situações de defeito em toda a rede e armazene os dados das soluções na memória do banco de dados para que possam ser usados para resolver problemas futuros. Ressalta-se, que o método de gerenciamento e memória de dados desenvolvido para o PRSDEE através da abordagem imunológica é simples e eficiente, e que pode ser adaptado para ser utilizado para

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outras ferramentas com aplicações diversas na área de sistemas elétricos de potência.

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CHAOS CONTROL VIA STATE-DEPENDENT RICCATI EQUATION METHOD APPLIED A NON-IDEAL VIBRATION ABSORBER COUPLED TO A NONLINEAR OSCILLATOR

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Abstract: The non-ideal vibration absorber system coupled to a nonlinear oscillator is proposed in the paper. This system is demonstrated with a chaotic behavior and we present the control technical for the chaos stabilization. The simulation results show the effectiveness of the control strategy.

AMS Subject Classification: 93C10
Key Words: chaos control, state-dependent Riccati equation, non-ideal system, vibration

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1. Introduction

In recent decades the control of structural vibration has become increasingly more interest from engineers and researchers because of the fact that these vibrations are unwanted and may cause noise emission, premature wear and fatigue failure of components. Thus there is great interest in reducing vibrations in the system through the use of control techniques [5, 6, 7].

The structural control is a technology for protection of structures which promotes a change in the stiffness and damping properties of the structure by adding external devices or by the action of external forces. Vibration absorbers have been applied to the control (reduction) of vibrations in structures, and the absorbers are simple devices that, when properly connected to a structure, are able to promote the reduction of its vibrations in an effective manner and consequently in many cases, reduction of noise levels, with the advantage of not requiring high costs for its implementation.

The tuned mass damper(TMD)is one popular device for minimize vibrations of mechanical structures. Frahm [1] was the first to introduce the TMD concept and considered a linear attachment composed of a mass and a spring coupled to a conservative linear oscillator(LO). Viguie and Kerschen proposed in [3] the tuning of a nonlinear vibration absorber coupled to an essentially nonlinear oscillator. The dynamic model of non-ideal power source can be defined for a beam excited by an unbalanced direct current (DC) motor was proposed by Kononenko [2]. This paper proposes the control of the vibration absorber coupled to a nonlinear oscillator model,and for its development we organize the paper this way. In Section 2, we present the mathematical model. In Section 3, the control design control project is propose for the minimize vibration and in Section 4, the conclusions are presented.

2. Non-ideal Model

Figure 1 shows the non-ideal and nonlinear absorber coupled to nonlinear oscillator proposed, which consist of a nonlinear oscillator composed of a mass m_1 and a cubic stiffness k_{nl1} and the absorber with a mass m_2 and a nonlinear stiffness k_{nl2}. c_1 and c_2 are the weak damping to induce energy dissipation [3].

The angular position z is non-ideal excitation response, a, b are motor torque constants, r is the distance from unbalanced mass to rotation center of the DC motor and d is related to moment of inertia of the system. All parameters are dimensionless constant positives. The term $\ddot{z}$ is due interaction

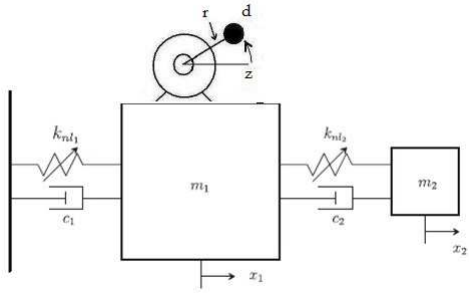


Figure 1: Non-ideal and nonlinear absorber coupled to nonlinear oscillator.

between the dynamical system and an energy source. The parameter a is the applied torque constant and depends on initial conditions and b is resistive net torque and has no influence of initial conditions and is considered as internal damping of the DC Motor.

The equations that describe the device illustrate in Figure 1 are:

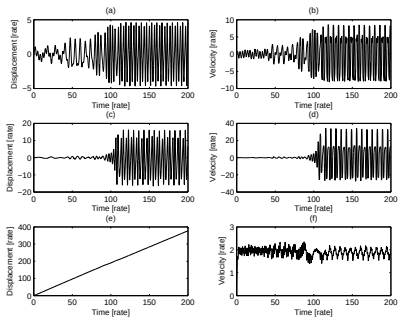
$$\begin{aligned} m_1\ddot{x}_1 + c_1\dot{x_1} - c_2(\dot{x_1} - \dot{x_2}) + k_{nl1}x_1^3 - k_{nl2} \\ (x_1 - x_2)^3 = d(z^2 cos\dot{z} + \ddot{z}sinz), \\ m_2\ddot{x}_2 + c_2(\dot{x_2} - \dot{x_1})k_{nl2}(x_2 - x_1)^3 = 0, \\ \ddot{z} + bz - r\ddot{q}sin\ddot{z} + a = 0. \end{aligned} \tag{1}$$

Rewriting the equations of the dynamical system, in state form, making $y_1 = x_1$, $y_2 = \dot{x_1}$,$y_3 = x_2$,$y_4 = \dot{x_2}$, $y_5 = z$ and $y_6 = \dot{z}$ the governing equations may be written as being:

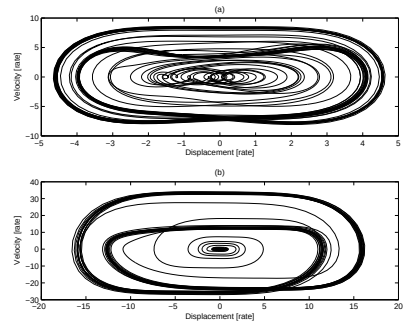
$$\begin{aligned} \dot{y}_1 &= y_2, \\ \dot{y}_2 &= \frac{-c_1y_2 + c_2(y_2 - y_4) - k_{nl1}y_1^3 - k_{nl2}(y_1 - y_3)^3}{m_1 - dr(sin^2y_5)} \\ &\quad + \frac{dy_6^2 cos(y_5) + dsin(y_5)(a - by_6)}{m_1 - dr(sin^2y_5)}, \\ \dot{y}_3 &= y_4, \\ \dot{y}_4 &= -\frac{c_2}{m_2}(y_4 - y_2) - k_{nl2}(y_3 - y_4)^3, \\ \dot{y}_5 &= y_6, \end{aligned} \tag{2}$$

$$\dot{y}_6 = \frac{(m_1 - drsin^2 y_5)(a - by_6) + rsiny_5(-c_1 y_2 + c_2(y_2 - y_4)}{m_1 - dr(sin^2(y_5))} + \frac{(-k_{nl1}y_1^3 - k_{nl2}(y1 - y3)^3 + dy_6^2 cosy_5 + dsiny_5(a - by_6))}{m_1 - dr(sin^2(y_5))}.$$

Here, x_1 and x_2 are the velocity and the displacement of the tuned mass damper, x_3 and x_4 are the velocity and displacement of the nonlinear oscillator and x_5 and x_6 are the angular position of non-ideal excitation response. The Figure 2 illustrate the dynamics behavior of the adopted dynamics model, by using numerical values, for the chosen parameters $m_1 = 1kg, m_2 = 0.05kg, c_1 = 0.002Ns/m, c_2 = 0.002Ns/m, k_1 = 0.5N/m$ and $k_2 = 0.025N/m$ [3] by using numerical values to the tuned mass damper coupled to a linear oscillator and $a = 5Nm$, $b = 1.5Nm$, $r = 0.3m$ and $d = 0.2kgm^2$ to the non-ideal excitation [8]. The initial conditions are $x_1(0.1) = 0$, $x_2(0) = 0$, $x_3(0) = 0$, $x_4(0) = 0$, $x_5(0.5) = 0$ and $x_6(0) = 0$



(a) Time History.



(b) Phase Portrait

Figure 2: Dynamical System

The eigenvalues of the system (2) indicates that the system is unstable, with $\lambda_1 = -1.6016$, $\lambda_2 = -0.5312$, $\lambda_3 = -0.0394$, $\lambda_4 = -0.0003$, $\lambda_5 = 0.3143 + 0.3586i$ and $\lambda_6 = 0.3143 - 0.3586i$.

Figure 3 illustrates the Lyapunov exponents, $\lambda_1 = 0.30392$; $\lambda_2 = 0.30538$; $\lambda_3 = -0.02534$; $\lambda_4 = -0.012883$; $\lambda_5 = -0.53184$ and $\lambda_6 = -1.5832$, demonstrating the presence of the chaos with the two Lyapunov exponent positive [4]. The total time for Lyapunov exponents computation was the $\Delta\tau = 150,000$ with time-step of $\tau = 0.5$.

Aiming to minimize vibrations and reduce the oscillatory motion caused in the system in the following section proposes the application of State-Dependent Riccati Equation (SDRE) to reduce this chaotic motion, see Figure 2b, to a

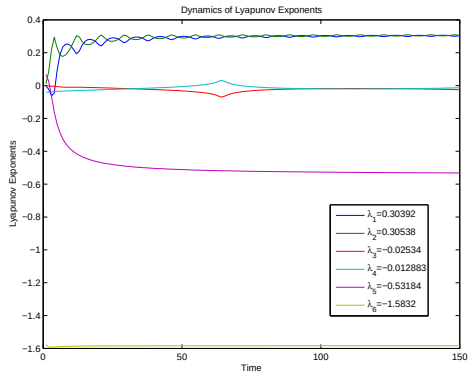


Figure 3: Dynamic of Lyapunov expoent

small stable orbit.

3. Optimal Control Design

The control objective is to stabilize the system with chaotic behavior in the small orbit periodic near the system origin. The Control Design via State-Dependent Riccati Equation (SDRE) approach for obtaining a suboptimal solution of the control problem has the following procedure [9] and [10]:

1. Represent the model in state-space form. Use direct parametrization to bring the nonlinear dynamics $\dot{x} = f(x) + g(x)$ to the state-dependent coefficient (SDC) form, as follows:

$$\begin{aligned} \dot{x} &= A(x)x + B(x)u \\ y &= C(x)x \end{aligned} \tag{3}$$

   where, $f(x) = A(x)x$ and $g(x) = B(x)$, $A(x) \in \Re^{nxn}$ is the dynamic matrix, $B(x) \in \Re^{nxm}$ is the input matrix, $C(x) \in \Re^{sxn}$ is the output matrix, $x \in \Re^n$ is the state vector $u \in \Re^m$ is the control law, $y \in \Re^s$ is the output vector.

2. Define the initial conditions $x(0) = x_0$, and choose the coefficients of positive definite weighting matrices $Q(x)$ and $R(x)$, which determine the relative importance of state $x(t)$ and control effort $u(t)$, respectively.

3. Solve the state-dependent Riccati equation given by:

$$A^T(x)P(x) + P(x)A(x) - P(x)B(x)R^{-1}(x)B^T(x)P(x) + Q(x) = 0 \tag{4}$$

4. Construct the nonlinear feedback control via:

$$u = -R^{-1}(x)B^T(x)P(x)x \tag{5}$$

The control law (5) is calculated so that the performance index given by, $J = \frac{1}{2}\int_{t_0}^{\infty}[x^TQ(x)x + u^TR(x)u]dt$ is minimized.

In the multivariable case, there always exists an infinite number of SDC parameterizations. Therefore, the choice of the matrix $A(x)$ isn't unique [9].

The pair $\{A(x), B(x)\}$ is a controllable parametrization of the nonlinear system in a region Ω if $\{A(x), B(x)\}$ is pointwise controllable in the linear sense for all $x \in \Omega$. Therefore, the choice of $A(x)$ must be such that the state-dependent controllability matrix $[B(x) \quad A(x)B(x) \quad ... \quad A^{n-1}(x)B(x)]$ has full rank [10].

The SDRE technique has been used to control various systems [11, 5]. Details about the technique SDRE can be found in [9] and [10].

Applying the above procedure in the nonlinear system (2), the state dependent coefficients are given by:

$$\begin{aligned} A_{(1,1)} &= 0 \\ A_{(1,2)} &= 1 \\ A_{(1,3)} &= 0 \\ A_{(1,4)} &= 0 \\ A_{(1,5)} &= 0 \\ A_{(1,6)} &= 0 \end{aligned} \tag{6}$$

$$\begin{aligned} A_{(2,1)} &= -(3(x_1 - x_3)^2)/400 - (21x_1^2)/20 \\ A_{(2,2)} &= -1/250 \\ A_{(2,3)} &= (3 * (x_1 - x_3)^2)/400 \\ A_{(2,4)} &= 1/500 \\ A_{(2,5)} &= (cos(x_5) * ((3x_6)/2 - 29/10))/(2 * (sin(x_5)^2/4 - 1)) \\ &\quad -(x_6^2 sin(x_5))/2 - (cos(x_5)sin(x_5)^2 \\ &\quad ((3 * x_6)/2 - 29/10))/(4(sin(x_5)^2/4 - 1)^2) \\ A_{(2,6)} &= x_6 cos(x_5) + (3sin(x_5))/(4(sin(x_5)^2/4 - 1)) \end{aligned} \tag{7}$$

$$A_{(3,1)} = 0$$

$$\begin{aligned} A_{(3,2)} &= 0 \\ A_{(3,3)} &= 0 \\ A_{(3,4)} &= 1 \\ A_{(3,5)} &= 0 \\ A_{(3,6)} &= 0 \end{aligned} \tag{8}$$

$$\begin{aligned} A_{(4,1)} &= (3(x_1 - x_3)^2)/20 \\ A_{(4,2)} &= 1/25 \\ A_{(4,3)} &= -(3(x1 - x_3)^2)/20 \\ A_{(4,4)} &= -1/25 \\ A_{(4,5)} &= 0 \\ A_{(4,6)} &= 0 \end{aligned} \tag{9}$$

$$\begin{aligned} A_{(5,1)} &= 0 \\ A_{(5,2)} &= 0 \\ A_{(5,3)} &= 0 \\ A_{(5,4)} &= 0 \\ A_{(5,5)} &= 0 \\ A_{(5,6)} &= 1 \end{aligned} \tag{10}$$

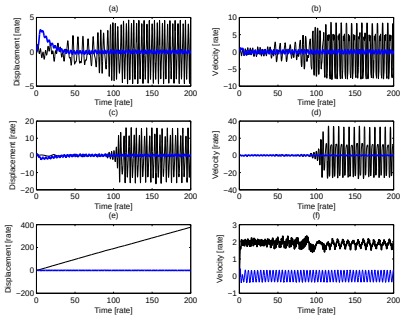
$$\begin{aligned} A_{(6,1)} &= (sin(x_5) * (k_1 + k_2))/(2(sin(x_5)^2/4 - 1)) \\ A_{(6,2)} &= 0 \\ A_{(6,3)} &= -(k_2 sin(x_5))/(2(sin(x_5)^2/4 - 1)) \\ A_{(6,4)} &= 0 \\ A_{(6,5)} &= (cos(x_5)sin(x_5)(sin(x_5^2)/4 - (sin(x_5)(k_1x_1 + k_2(x_1 - x_3)))/2 \\ &\quad +(sin(x_5)^2/4 - 1)((3x_6)/2 - 29/10) + \\ &\quad (x_6^2 cos(x_5)sin(x_5))/4))/(2(sin(x_5)^2/4 - 1)^2) \\ &\quad -((x_5 cos(x_5^2))/2 - (cos(x_5)(k_1x_1 + k_2(x_1 - x_3)))/2 \\ &\quad +(x_6^2 cos(x_5)^2)/4 - (x_6^2 sin(x_5)^2)/4 \\ &\quad +(cos(x_5)sin(x_5) * ((3x_6)/2 - 29/10))/2)/(sin(x_5)^2/4 - 1) \\ A_{(6,6)} &= -((3sin(x_5)^2)/8 + (x_6 cos(x_5)sin(x_5))/2 - 3/2)/(sin(x_5)^2/4 - 1) \end{aligned} \tag{11}$$

$$B(x) = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \tag{12}$$

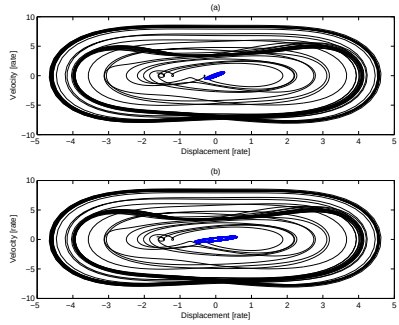
Assuming $C(x)$, $Q(x)$ and $R(x)$ are constant matrices, we have:

$$y = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad R = \begin{bmatrix} 1 \end{bmatrix} \tag{13}$$

The responses of the controlled system are shown in Figs. 4a and 4b .



(a) Control Time History.



(b) Control Phase Portrait

Figure 4: Control Dynamical System

4. Conclusion

In this paper the authors propose a model of a non-ideal vibration absorber. In section 2 the mathematical modeling is presented and computer simulations demonstrating the chaotic behavior of the system is shown in Figure 2 and 3.

We proposed the use of an SDRE control strategy to reduce the chaotic movement of this system to a stable orbit. The Figures 4 illustrated the effectiveness of the control strategy taken to these problems.

Acknowledgment

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Electrical Load Forecasting using a Hybrid Model with Artificial Neural Networks

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Abstract—**Electrical load forecasting is a very important research field in the planning and operation of power systems. To provide electrical energy safely and reliably without interruptions, the forecasting must be as precise as possible. The use of hybrid techniques has been recently extended to time series predictions with good results. This work develops a hybrid prediction model for short term load forecasting using the ARIMA (autoregressive integrated moving average) of Box and Jenkins and artificial neural networks trained by the Levenberg-Marquardt algorithm. The results were tested with data from a Brazilian company over three periods and compared with other methods considering the same time series. The proposed model has a high performance for the three applications.**

Keywords—component;ARIMA models by Box and Jenkins, short term load forecasting, artificial neural network, Levenberg-Marquardt training formattin)

I. INTRODUCTION

Load forecasting is very important for ensuring that electrical companies can provide energy securely, reliably and economically without interruptions. Advanced knowledge of the load allows the planning of the electrical system expansion, the load flow, economical operation, security and the control analysis of the electrical systems. The following techniques for load forecasting are found in the literature [1]: simple or multiple linear regression, exponential smoothing, state estimation, Kalman filters, and the ARIMA (autoregressive integrated moving average) of Box and Jenkins [2]. These methods apply the previous modeling of the load. The meteorological conditions, such as the nebulosity, the wind speed, the temperature variations and several other factors, such as nonconventional days (holidays, strikes, etc.), are used to aid the load modeling; therefore, the prediction improves with more available information [2]. A necessary condition to realise the electrical load forecasting is to obtain information from the past codified in data and to assume that the behavior of the past will be repeated in the future [3].

Methods based on artificial intelligence have shown efficiency when applied to several real life problems. Currently, artificial neural networks (ANN) [4] have been used to realize electrical load forecasting due to their ability to learn and generalize. The generalization is the capacity to produce adequate outputs for inputs that are not presented during the training, and, therefore, solve complex problems and difficult to address. Several works usevarious neural network architectures to execute electrical load forecasting [3], [5-7].

A very interesting proposal found in the literature is to use the ARIMA models with ANN exploring the advantages of linear and nonlinear modeling, providing a robust high-quality prediction model [8]. The works using the hybrid methodology with ARIMA and ANN explores time series applied to several real problems [9-13]. For example, the work of Koutroumanidis [11] describes the contribution of renewable energy sources to produce electricity, where the ARIMA is responsible for generating the load parameters and ANN to predict the price of selling wood in Greece. The hybrid model provides good results allowing a rational use of wood production and the associated market. The works of Khashei and Bijari [14], [15], consider the following time series: the solar spots by year, the number of people that are arrested per year at the Mackenzie River and the dollar/pound exchange rate. The objective is to obtain a precise prediction using the hybrid model with the ARIMA and the ANN, where the ARIMA models are used in the first phase to generate the necessary data from historical time series and forward the data to a neural network to model the data generated from the ARIMA and to predict the time series future. Each of these works divides the time series into a linear part and a nonlinear part, where the ARIMA considers the linear part and the ANN the nonlinear part of the ARIMA errors. Certain works using ARIMA and ANN combine the use of fuzzy rules [14] and genetic algorithms to improve the results [15].

This work proposed the development of a hybrid model to short-term load forecasting using ARIMA models and ANN, specifically an MLP (multi layer perceptron) trained by the Levenberg-Marquardt algorithm. The hybrid model aims to overcome the use of ARIMA and ANN individually, once the ARIMA models address the linear part of the time series and the ANN the nonlinear. Although other works use hybrid models with ARIMA and ANN [8-13], the hybrid methodology developed in this work is different, once the ANN predicts a random error that will be used in the ARIMA model to predict the electrical load. This technique has never been applied to load demand forecasting. Therefore, this work innovates in this research area. The demand series is seasonal and, thus uses the seasonal ARIMA called SARIMA. In this work the SARIMA model provides the simulated series of the real load, i.e., the

model works with the linear part of the time series. The residuals are obtained by the difference of the real load and the simulated load, which are treated by the neural network (that will predict the residuals), and afterwards are substituted on the SARIMA model to effectuate the electrical load prediction. The advantage of the neural network is that it works with the nonlinear part without using a mathematical model.

The proposed model is tested with 30, 60 and 90 days of the historical load, with the aim of predicting the load for the next day. The results were compared with other works that use the same data of the same electrical company, and the proposed model performs better.

II. HYBRID METHODOLOGY

A. ARIMA models of Box and Jenkins

The ARIMA models from Box and Jenkins were the most important models for time series prediction in several areas for at least 50 years. These methods only depend on the series data to specify the best model [1].

The ARIMA model is denoted by ARIMA (p,d,q). The future value of one variable is assumed to be a linear function of several past values and random errors. The process that generates the time series is in accordance with equation (1) [16]:

$$\emptyset(B)\Delta^d Z_t = \theta(B)a_t \qquad (1)$$

Z_t : real values at period t;

a_t : random errors at period t;

$\emptyset(B) = 1 - \phi_1 B - \phi_2 B^2 - \cdots - \phi_p B^p$: order p auto-regressive operator;

$\theta(B) = 1 - \theta_1 B - \theta_2 B^2 - \ldots - \theta_q B^q$: order q moving average operator;

$\Delta^d = (1 - B)^d$: series differentiation;

d : number of times that the series were differentiate to become stationary.

If the series is already stationary, the term d does not exist, and the model becomes an ARMA (p,q) model, which is represented by equation (2) [16].

$$\emptyset(B)Z_t = \theta(B)a_t \qquad (2)$$

When a time series presents seasonality the seasonal ARIMA models are used which are known as seasonal autoregressive Integrated Moving Average (SARIMA) and represented by SARIMA $(p, d, q)(P, D, Q)_m$. The seasonality indicates the repetition of a pattern in a time period. The representation is according equation (3) [17].

$$\phi(B)\Phi(B^m)\Delta^d \Delta_m^D Z_t = \theta(B)\Theta(B^m)a_t \qquad (3)$$

where:

$\Phi(B^m) = 1 - \Phi_1 B^m - 1 - \Phi_2 B^{2m} - \cdots - 1 - \Phi_p B^{Pm}$: order P seasonal auto regressive operator;

$\Theta(B^m) = 1 - \Theta_1 B^m - 1 - \Theta_2 B^{m2} - \cdots - 1 - \Theta_Q B^{mQ}$: order Q seasonal moving average operator;

$\Delta_m^D = (1 - B^m)^D$: differentiation of the seasonal series;

D : number of seasonal differences.

The time series prediction by Box and Jenkins [1] consists of adjusting the ARIMA(p,d,q) model to a data set. The building of this methodology is approached in the identification, estimation and verification cycle.

B. Artificial Neural Netwoks

Artificial neural networks are one of the most important models for prediction nonlinear time series. The advantages of using ANN has the advantage of being able to address nonlinear problems without using a mathematical model, and the capacity in generalizing can address non- stationary characteristics. The advantages of ANN encourage researchers to use it in time series prediction.

The MLP neural network is an important one because it is easy to implement. The MLP neural network is a feedforward architecture having at least one hidden layer between the input and output layers with supervised training [4].

One of the principal characteristics of the ANN is the capacity to learn. The ANN must be trained by an algorithm and the one chosen is the Levenberg-Marquardt which uses second order partial derivate to adjust the weights [5].

The learning rule is based on equation (4) [5].

$$W(k+1) = W(k) - [J^T(W)J(W) + \mu_k I]^{-1} J^T(W)e(W) \qquad (4)$$

where:

W : synaptic weight vector;

J : Jacobian matrix;

I : identity matrix;

μ_k : Levenberg-Marquardt constant;

$$e(W) = \sum_{i=1}^{n} (y_i - y_{ei})$$

y_i : output given by the network;

y_{ei} : real output.

The Levenberg-Marquardt constant μ_k is used as a stabilisation factor, which adjusts the approach to the Newton method by avoiding steps that can lead to convergence error. This algorithm is considered the fastest one and requires only a few iterations when using a moderate number of parameters. If the quantity of these parameters is large, the method is not adequate due to the computational effort required for matrix inversion [18].

III. DEVELOPMENT OF THE PROPOSED METHODOLOGY

A time series is considered as nonlinear function of past observations and random errors. Thus, the first step is to use the autoregressive integrated moving average models to identify the characteristics of the electrical load time series and estimate the parameters and coefficients to ultimately find the best SARIMA model of the load series. Therefore the historical data (electrical load time series) are applied into the software R by the forecast library [19, 20] to obtain the best ARIMA model.

Figure describes the first step of the hybrid methodology.

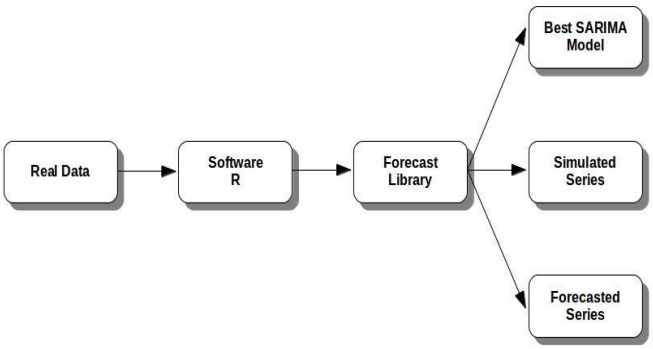

Figure 1. Flowchart of software R.

The software R provides the parameters and the coefficients of the load series and consequently the best model.

By the simulated series of the SARIMA model, the error E_h can be found by to equation (5).

$$E_h = R_h - S_h \tag{5}$$

where R_h are the real load values and S_h the simulated loads, both at hour h.

The next step of the hybrid methodology is to use the error E_h as the input of the neural network presented in Fig.2.

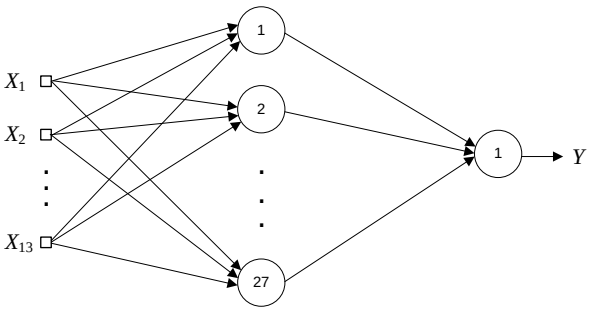

Fig. 2.- Neural network structure.

The set containing the input and output vectors of the neural network are defined as follows:

$$X_h = \left[\, te \;\; E_{(h-3)}E_{(h-2)}E_{(h-1)}E_{(h)}\right]^T, \;\; X \in R^b \tag{6}$$

$$Y = \left[E_{(h+1)}\right], \;\; Y \in R^1 \tag{7}$$

where:

b : dimension of vector X;

$E_{(h-v)}$: error v hours previous to the current time$h$;

$E_{(h+1)}$: error corresponding to the subsequent hour to the current h;

te : time vector corresponding to the historical data (holidays, day of the week, hours) in binary (-1,+1).

The neural network used in this work contains three layers, with 13 neurons at the input layer, 27 neurons at the hidden layer and one neuron at the output. It is worth mentioning that the numbers of neurons that make up each layer are chosen according to the system characteristics. According to our research group experiences, in the presented heuristic-based method, the quantity of the neurons in the hidden layer, (QNHL), is greater than or equal to the maximum of the quantity of neurons in the input layer (QNIL), and the quantity of neurons in the output layer (QNOL), equation (8). Therefore, this is the criterion that is used to determine the quantity of neurons in the hidden layer. For example the quantity of neurons in the input layer is QNIL = 13, and in order to obtain a solution with a higher quality $(2 \cdot \mathrm{QNIL} + 1)$ criterion has been used that results 27 neurons in the hidden layer."

$$QNHL \geq \max\{QNIL, QNOL\} \tag{8}$$

At the input layer, the vector te represents nine inputs composed of binary data corresponding to exogenous variables such as: holidays, days of the week and hour of the day respectively. The last four values correspond to the errors $E_{(h-3)}E_{(h-2)}E_{(h-1)}E_{(h)}$, obtained by the difference between the real data series and the simulated series, according to equation (5). The goal of the neural network is to predict the error 24h in advance.

Adding the predicted error E_{h+1} to error E_h the vector $\hat{E}_h$ is obtained represented by equation (9).

$$\hat{E}_h = \left[E_{(h-v)} \ldots E_{(h-1)}E_{(h)}E_{(h+1)}E_{(h+2)} \ldots E_{(h+23)}\right]^T \tag{9}$$

The error $\hat{\mathrm{E}}_\mathrm{h}$ is substituted in the residuals $\mathrm{a_t}$ of equation (3), providing equation (10), which predicts the electrical load. Thus, the load forecasting is realized by the SARIMA model adapted from the residuals from the ANN.

$$\phi(B)\Phi(B^m)\Delta^d\Delta_m^D Z_t = \theta(B)\Theta(B^m)\hat{E}_h \tag{10}$$

The training and test phases of the neural network were executed using the MATLAB software with the Neural Network toolbox [20]. MATLAB was also used to realize the load forecasting after the substitution of the error $\hat{\mathrm{E}}_\mathrm{h}$ into the SARIMA. Figure 3 presents a flowchart of the proposed methodology.

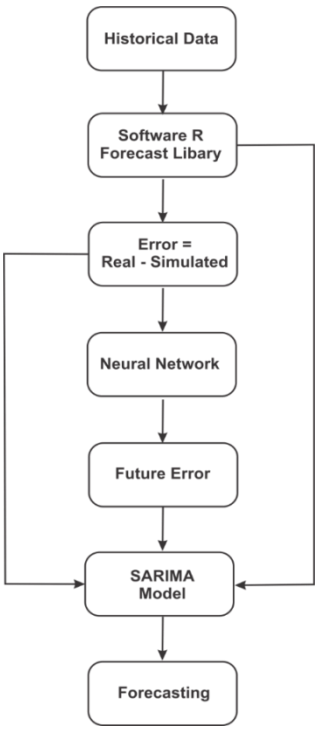

Fig. 3.- Flowchart of the proposed model.

IV. APPLICATION OF THE PROPOSED MODEL IN LOAD FORECASTING

This section presents the results obtained by the hybrid model for short-term load forecasting, i.e. 24 h in advance. To test the efficiency, three different applications are realized according to the quantity of historical data used for training, and compared with the paper by Nose Filho [22] that uses the same data with the same period of training for different neural network architectures and with the SARIMA itself for each application.

Application 1 considers 31 days composed by 744 vectors, with a vector corresponding to each hour. The 744 vectors are applied into the software R, vielding the model SARIMA $(2,0,0)(2,0,2)_{24}$. Equation (11) describes the coefficients of SARIMA model for application 1.

$$(1 - 1{,}3641.B + 0{,}3879.B^2)(1 - 0{,}2462.B^{24} - 0{,}7523.B^{48})Z_t = (1 + 0{,}0971.B^{24} - 0{,}7523.B^{48})a_t \quad (11)$$

Application 2 considers 61 days, composed by 1464 vectors and the software R provides the SARIMA $(2,0,2)(2,0,2)_{24}$. Equation (12) shows the coefficients for this application.

$$(1 - 1{,}6914.B + 0{,}7166.B^2)(1 - 0{,}5885.B^{24} - 0{,}4105.B^{48})Z_t = (1 - 0.3069 \cdot B - 0{,}2185.B^{48})(1 - 0{,}4444.B^{24} - 0{,}4060.B^{48})a_t \quad (12)$$

Application 3 considers 84 days and 2208 vectors and the model SARIMA $(3,0,1)(2,0,2)_{24}$ which are described in equation (13).

$$(1 - 0{,}3999.B - 0{,}8949.B^2 + 0{,}3808.B^3)(1 - 0{,}7955.B^{24} - 0{,}2037.B^{48})Z_t = (1 - 0{,}9779.B)(1 - 0{,}5533.B^{24} - 0{,}2837.B^{48})a_t \quad (13)$$

The SARIMA models are generated, and the errors (equation (9)) which will be used on the training and diagnosis of the ANN, are obtained. Then, the models (ANN) will predict 24 errors in advance.

Table 1 shows the parameters and specifications for the neural network.Several tests were performed, until it appears the parameters that best fit the system.

TABLE I. NEURAL NETWORK PARAMETERS

Parameters	***Application 1***	***Application 2***	***Application 3***
Quantity of layers	3	3	3
Quantity of neurons per layer	13-27-1	13-27-1	13-27-1
Goal	0,001	0,001	0,001
Maximum number of iterations	39	45	67
Minimum gradient	10-10	10-10	10-10
Levenberg-Marquardt coefficients	0,12	0,15	0,20
Decreasing Levenberg-Marquardt coefficient	0,12	0,15	0,20
Increasing Levenberg-Marquardt coefficient	10	10	10
Maximum Levenberg-Marquardt coefficient	1010	1010	1010

Figures 4, 5, and 6, present the vector errors, $\widehat{E}_h$, obtained from the difference between the real load series, the simulated load series (SARIMA), which are depicted in blue lines, while the predicted error of the ANN are depicted in red line. The error vector $\widehat{E}_h$ is substituted on the residuals of the SARIMA model, a_t, which are represented by equations (11), (12), and (13) to obtain the short-term load forecasting, Z_t.

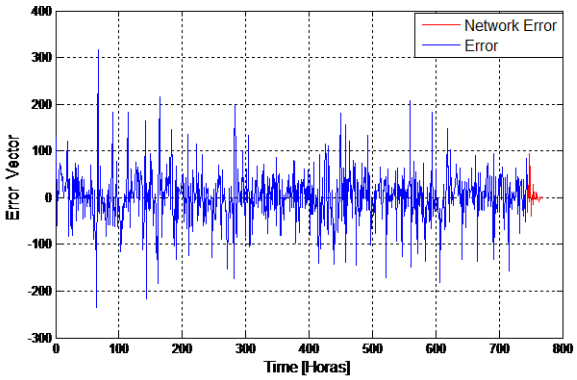

Fig. 4.- Error Vector (application 1).

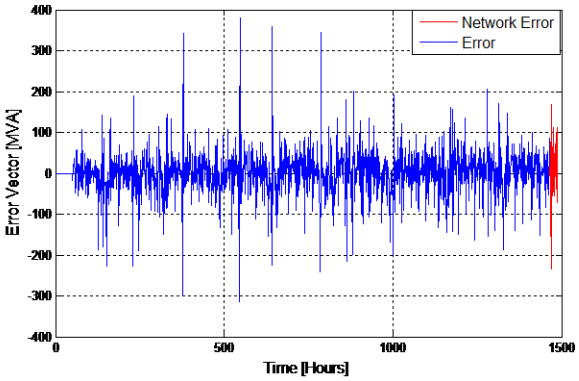

Fig. 5.- Error Vector (application 2).

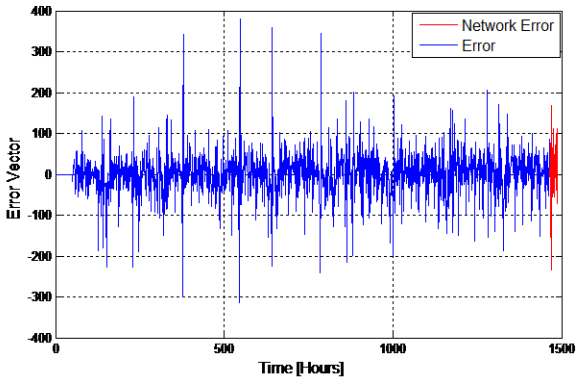

Fig. 6.- Error Vector (application 3).

The literature usually creates the analysis for the results of load forecasting using the mean absolute percentage error (MAPE) and the maximum percentage error [3], [6], which are calculated according to equations (14) and (15) [23]:

$$MAPE = \frac{1}{NT}\sum_{h=1}^{N}\left\{\frac{\left|L(h) - \underline{L}(h)\right|}{L(h)}\right\} \times 100\% \qquad (14)$$

$$Max\ Error\ (\%) = máx\left\{\frac{\left|L(h) - \underline{L}(h)\right|}{L(h)}\right\} \times 100\% \qquad (15)$$

where:

L(h): real load referred to by hour h;

$\underline{L}$(h): load estimated by the hybrid model referred to by hour h;

NT: total quantity of hours.

The obtained results are compared with the different architectures and trainings of ANN, such as MLP with descent gradient with momentum, MLP with Levenberg-Marquardt, the adaptive neural fuzzy inference system (ANFIS) and the general regression neural network (GRNN) from reference [22], well as with SARIMA itself considering the MAPE and maximum error for evaluation. The results are shown in Tables 2, 3, and 4 for applications 1, 2 and 3 respectively.

TABLE II. MAPE AND MAXIMUM ERROR FOR APPLICATION 1

Models	***MAPE***	***Max Error(%)***
Hybrid Model	0.908	1.016
SARIMA	9.096	18.623
ANN perceptron trained by descent gradient with moment [22]	1.998	4.845
ANN Perceptron with Levenberg-Marquardt training [22]	1.178	3.560
ANFIS [22]	1.090	2.861
GRNN [22]	1.003	2.410

TABLE III. MAPE AND MAXIMUM ERROR FOR APPLICATION 2

Models	***MAPE***	***Max Error(%)***
Hybrid Model	0.892	0.954
SARIMA	7.774	16.955
ANN perceptron trained by descent gradient with moment [22]	2.093	5.022
ANN Perceptron with Levenberg-Marquardt training [22]	1.490	4.267
ANFIS [22]	1.403	3.848
GRNN [22]	1.118	3.650

TABLE IV. MAPE AND MAXIMUM ERROR FOR APPLICATION 2

Models	***MAPE***	***Max Error(%)***
Hybrid Model	0.634	0.875
SARIMA	8.747	18.630
ANN perceptron trained by descent gradient with moment [22]	2.313	7.088
ANN Perceptron with Levenberg-Marquardt training [22]	1.698	4.687
ANFIS [22]	0.973	3.757
GRNN [22]	1.288	4.246

Analysis of Tables 2, 3 and 4 shows that the hybrid model is superior to the other neural networks and the SARIMA model for the three applications proposed. Considering the economical generation point of view, results with less than 5% precision present a significant reduction in the generation costs [24]. Thus, decreasing this precision is economical and the proposed model obtains results below 1%.

As the quantity of vectors increase, better results are obtained,which is a fundamental property of the ANN, which

assures that, as more data are used for training better results are obtained at the diagnosis (test phase).

Figures 7, 8 and 9 present the loads 24 h in advance.

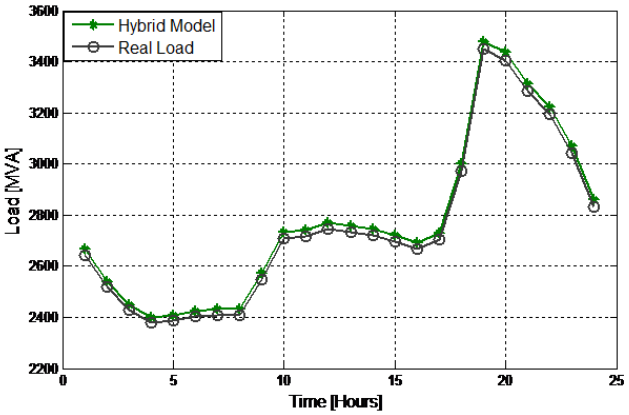

Fig. 7.- Load forecasted by the hybrid model: Application 1.

As shows in Figures 7-9, the curves obtained by the hybrid model can follow the prediction curves for the three applications proposed, demonstratingthe good precision of the proposed model.

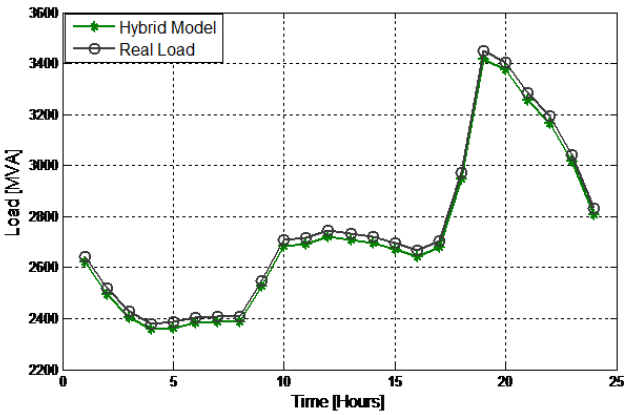

Fig. 8.- Load forecasted by the hybrid model: Application 2.

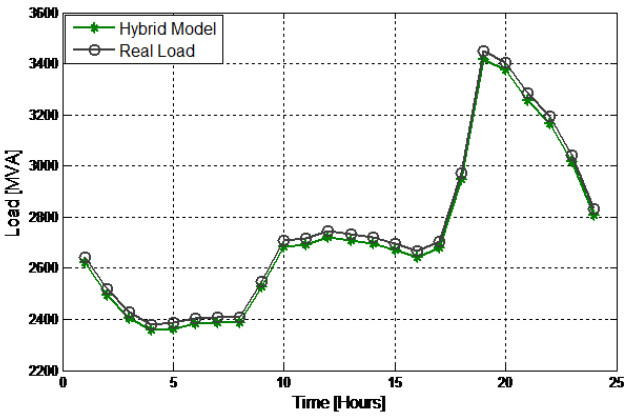

Fig. 9.- Load forecasted by the hybrid model: Application 3.

V. CONCLUSION

This study developed a hybrid methodology for short term electrical load forecasting using the ARIMA of Box and Jenkins and a neural network trained with the Levenberg-Marquardt algorithm. The ARIMA models generate the load parameters (the best model) and realize the desired prediction. This prediction is executed after the ANN realizes the error prediction, which is used with the error determined by the difference between the real data and the simulated data substituted into the ARIMA model to finally calculate the prediction. Three tests were performed for different periods by considering the historical data of a Brazilian electrical company. The short- term load forecasting considers 24 h in advance, where the proposed model performs well with a MAPE of less than 1 % and maximum errors of less than 1.1%. Thus, the proposed model can realize predictions, including non-conventional days such as weekends, reliably and with high performance. The prediction error decreased, and better results were found by increasing the data set.

The short-term load forecasting proposed in this work is efficient and the results are important considering other models found in the literature.

ACKNOWLEDGMENT

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Artificial Immune Systems Applied to the Analysis of Structural Integrity of a Building

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Keywords: Analysis of Structural Integrity, Building, Artificial Immune Systems, Negative Selection Algorithm.

Abstract. This paper presents the application of artificial immune systems for analysis of the structural integrity of a building. Inspired by a biological process, it uses the negative selection algorithm to perform the identification and characterization of structural failure. This paper presents the application of artificial immune systems for analysis of the structural integrity of a building. Inspired by a biological process, it uses the negative selection algorithm to perform the identification and characterization of structural failure. This methodology can assist professionals in the inspection of mechanical and civil structures, to identify and characterize flaws, in order to perform preventative maintenance to ensure the integrity of the structure and decision-making. In order to evaluate the methodology was made modeling a two-story building and several situations were simulated (base-line condition and improper conditions), yielding a database of signs, which were used as input data for the negative selection algorithm. The results obtained by the present method efficiency, robustness and accuracy.

Introduction

Although the technological advances in the development of metallic materials and construction, has seen a large number of newly constructed edifications had shortcomings anyway. This is due to the inappropriate use of materials, lack of care in implementing adaptations and even when of its using, all this coupled with the lack of maintenance, has created extra expenses to condominiums in repairs of faults that could be avoided [10].

Faults can represent low or no dynamic performance of the structure in relation to its durability and stability when exposed to abnormal conditions such as, for example, natural events (earthquakes, tremors, etc.).. According to [1] structures, mostly, have more severe failures in reinforced concrete structures. The integrity risk is connected to the structure of the concrete reinforcement, which consists of steel, and usually fails as corrosion or cracks.

Because of this, it is necessary to develop efficient ways to inspect such structures, to identify and characterize flaws, making it possible to make quick decisions to perform preventive maintenance to ensure the integrity of the structure and to prevent disasters.

In literature found themselves few studies that use methodologies in order to analyze of the structural integrity of buildings or similar structures, aiming to make decisions. Following are presented the main works.

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In [10] the authors present a methodology for structural analysis of buildings using a non-destructive test (NDT). In [3] proposes a fuzzy system for the detection of structural flaws using curvatures of the vibration mode shapes. In work [12] proposed a model using wavelet transform to assess structural integrity of bridges across the vibration signals. In work [11] the authors propose an experimental method to conduct structural analysis of buildings. The authors analyze experimentally a shed where failures are performed on metal columns support. In [13] proposed a multinodal genetic algorithm to diagnose damage in a steel truss bridge.

In this paper, we present a method for analyzing the integrity of buildings based on artificial immune systems. From acquisition and signal processing, it applies the negative selection algorithm (NSA) [7] to identify and characterize flaws. In the purpose of evaluating this method, we used a data base of simulated signals from a numerical model of a two-storey building, which in this case is the building structure. The building was modeled by differential equations and simulated by Matlab ® [9]. We chose to use the artificial immune systems (AIS) because the natural learning and pattern recognition and present a good performance and efficiency in other similar problems of diagnosis, as outlined in [8].

This article is organized as follows: Section 2 presents the negative selection algorithm. Modeling and simulations are shown in section 3. Section 4 presents the proposed methodology. The applications and results are given in Section 5 and, finally, section 6 presents the conclusion.

Negative Selection Algorithm (NSA)

The negative selection algorithm (NSA) [7] is a technique based on pattern recognition process exercised by the biological immune system, being developed as a computational model. ANS proposed in [7] for detecting changes systems, is based on negative selection of T cells within the thymus. This process is the breakdown of the cells between proper and non-proper taken by the organism. The algorithm runs in two stages as illustrated in Figure 1 [6]:

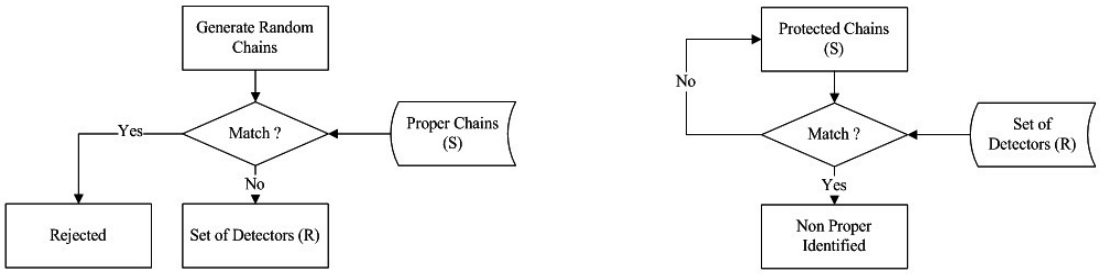


Fig. 1: Flowcharts NSA.

The censor phase of the ASN basically consists in generating a set of detectors. The detectors are analogous to type T cells matured able to recognize the pathogens. The monitoring phase consists in monitoring a system to identify a change in the behavior of the same, and sort this change using the set of detectors created in censor phase [5].

Matching Criterion

To evaluate the affinity between the chains and affirm that they are similar, we use the matching criterion. The matching criterion can be perfect or partial [8]. The perfect match occurs when the two chains being analyzed, have the same values in all positions, namely, the two chains must necessarily be equal. The marriage partial there is no need for all positions of the chains have the same value. The marriage partial there is no need for all positions of the chains have the same value. At the marriage partial, only a number of positions between the chains must have the same value to affirm marriage, and this amount was previously defined, and known as the affinity rate.

The affinity rate is defined by the following relationship [2]:

$$TAf = \left(\frac{An}{At}\right) * 100 \quad (1)$$

being:

TAf : afinity rate;

An : number of chains in normal problem (chains propers);

At : total number of chains in problem (chains propers and not propers).

The equation (1) proposes a statistical relationship between all samples of the problem to the calculation of the affinity rate. In (2) represents a way of quantifying the overall affinity between patterns [10]:

$$Af_T = \frac{\sum_{i=1}^{L} Vc}{L} * 100 \quad (2)$$

sendo:

Af_T : % affinity between patterns analyzed;

L : total number of variables;

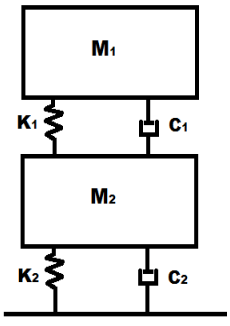
Vc : variables matching;

$\sum_{i=1}^{L} Vc$: sum (amount) of variables matching.

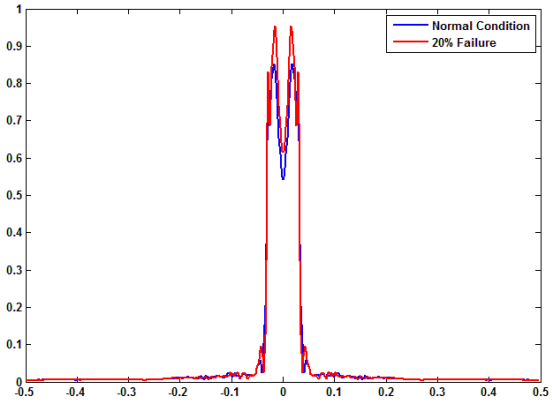
If $Af_T >= TAf$ matching occurs between the two patterns (are considered similar).

Modeling and Simulations

In this section a mathematical model was developed using differential equations of a two storey building with two degrees of freedom each. The building is adimensional model and consists of two masses (M_1 e M_2), and the stiffness coefficients (k_1 e k_2) and elasticity (c_1 e c_2), see [3]. The model is illustrated in Figure 2 (a).



(a)



(b)

Fig. 2 – (a) Structure model, (b) Frequency response.

We simulated 1400 signs in the structure, being 200 signs in normal condition, i.e., without flaw (condition base-line) and 1200 signals with structural flaws, being 200 signs for each type of failure. The signals faults were simulated with additions of 5%, 10%, 15%, 20%, 25% and 30% in mass M_1 of the structure. The structure was excited by the signal (S) as (3) [4]:

$$S = 1 * e^{-5\cos(t)} \quad (3)$$

After exciting the structure obtains the frequency response of the structure using the Fourier Fast Transform (FFT) [4]. The frequency response is obtained from the signals of the velocity and the displacement of the structure excited. Fig. 1 (b) illustrate two signals are captured in the simulations, which has a signal with a 20% failure (red) and a signal in normal condition (blue).

Proposed Methodology

The System Analysis of Structural Integrity of Buildings (SASIB) presented in this section uses based on the artificial immune systems [6], especially the negative selection algorithm [7]. This method is divided into three modules, namely: data acquisition, censor phase and monitoring phase of NSA.

System Analysis of Structural Integrity of Buildings (SASIB)

The flowchart of the operation of SASIB is shown in Figure 3.

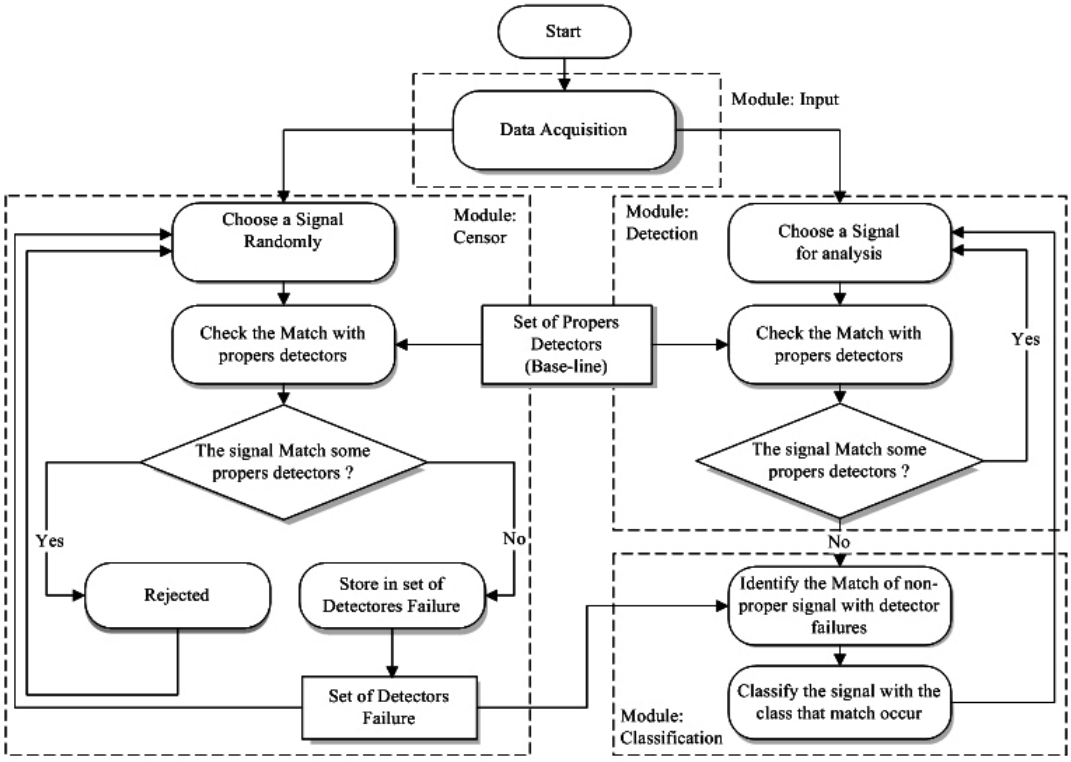


Fig. 3: Flowcharts SASIB.

The data acquisition module comprises the experimental part for excite and capture the signals in the structure as sensors, actuators, accelerometers, piezoelectric materials, etc. This module obtains the signals frequency response that will be used as input to the NSA.

In censor module is generated the set of failure detectors that will be used in the monitoring data. This process is done offline. The fault detector array comprises signals from different types of faults into the structure, which are randomly selected.

The monitoring module is composed of two stages: detection and classification of fault. The signs at faults are detected by evaluating and verifying matching criterion to the detector set (base-line signals), so does the detection of signals in fault discrimination based on self / non-self. If the signal is proper is classified the normal condition of the structure. If the signal is a signal non proper (signal failure) makes an evaluation of this signal compared to the set of failure detectors, thus if there is a marriage of the signal with a class of detectors, the signal is classified as a class of detector activated. The monitoring of the signals is done *online*.

Applications and Results

In this section we present the results obtained with the application of SASIB in the database being simulated. All tests were performed using a PC Intel Core 2 Duo 1.9 GHz, 2 GB of RAM, and operating system Windows 7 Ultimate 32-bit. The algorithm was developed in MATLAB ® [9].

In tests the objective was to verify the proposed methodology in the analysis process and monitoring of structural integrity, checking the efficiency, accuracy and the computational time of the system for different configurations of the set of detectors NSA, ie, information that represents the knowledge of the system. Used as a parameter an affinity rate (*TAf*) with a fixed value of 85.5%.

We generated four sets of detectors (*I, II, III* and *IV*) using 15%, 20%, 25% and 30% of samples of each type of structural failure, totaling 210, 280, 350 and 420 detectors respectively.Each type of standard were defined 30, 40, 50 and 60 detectors for each set of detectors. The database simulated has seven patterns faults in the building, totaling seven distinct classes

Table 1 shows the results of the system analysis of structural integrity of buildings when applied to simulated data.

Table 1 – Results obtained by the method.

Diagnostic	Set of detectors			
	I	*II*	*III*	*IV*
Samples tested	1400	1400	1400	1400
Ratings correct	1378	1386	1392	1400
Ratings wrongs	22	14	8	0
Hit (%)	98,42	99,00%	99,42%	100%
Time (ms)	189,8	193,5	183,8	190,4

Table 1 shows the results of a crossover test was conducted 20 times to the system for each set of detectors in order to ensure the accuracy of the results. You can see that the system analysis of the structural integrity of buildings presents a good performance (success rate of 100% in the best configuration), and the number of detectors directly influence the recognition of failures. You can see that the system analysis of the structural integrity of buildings presents a good performance (success rate of 100% in the best configuration), and the number of detectors directly influence the recognition of failures. Thus, it is suggested to use 30% of the data base information to generate the array of detectors NSA in order to provide robustness to the system. The time shown in Table 2 represents the average time spent to analyze and process each signal.

We can conclude that the more knowledge the NSA gets in your censor phase, the more efficient the process of recognition and classification.

Conclusion

In this paper we present a novel approach to develop a system of analysis of the structural integrity of buildings, in which we used an algorithm artificial immune negative selection, which showed excellent results, obtaining a success rate of 100% for the best configuration the system.The generation of detectors (censor phase) demand greater computational time, however not affect the algorithm as it is performed off-line. The monitoring system, from signal acquisition is done rapidly with time less than 200 milliseconds, this allows the application of SASIB on problems that need to make decisions in real time. We conclude that the SASIB proposed in this paper is fairly efficient, reliable, robust and accurate to perform the analysis the structural integrity of buildings. It is emphasized that the artificial immune algorithm presented facility used in the detection and classification of structural flaws, this is due to their ability to analyze point-to-point. This paper contributes to this line of research, presenting a new method to analyze failures in structures using techniques based on intelligent systems.

Acknowledgments

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A Comparison of Methodologies for Intelligent Computing Used to Integrity Analysis of a Structure Aeronautic

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Keywords: Analysis of Structural Integrity, Aeronautical Structures, Artificial Immune Systems, Artificial Neural Networks.

Abstract. This article presents the application and comparison of two techniques for intelligent computing to perform the analysis of the structural integrity of an aircraft structure. In this context, a ARTMAP-Fuzzy neural network and immunological negative selection algorithm are used in the identification and characterization of structural failure. The main application of these methodologies is to assist in the inspection of aircraft structures aiming at detecting and characterize flaws and decision making. To evaluate the methodology was performed modeling and simulation of signals from a numerical model using an aluminum beam. We performed a comparative analysis of methodologies, proving the efficiency of intelligent methods in the analysis of structural integrity. The results obtained by the method show efficiency, robustness and accuracy. To Evaluate the methodology was Performed modeling and simulation of signals from the numerical model using an aluminum beam. We performed a comparative analysis of methodologies, proving the efficiency of intelligent methods in the analysis of structural integrity. The results Obtained by the methods show efficiency, robustness and accuracy.

Introduction

Despite the technological advancement and the development of new materials and aircraft structures, many investments are still employed in structural analysis, with the aim of ensure economic, security and verify the integrity of the structure.

In aircraft structural failure can alter the dynamic behavior of the structure, causing a significant variation in the spatial parameters of the structure, reducing the stiffness and structural mass, and also an increased damping in most cases. Failures can occur as a consequence of various factors, such as component wear, cracks, loosening of bolted connections, or simply the combination of these elements. With the variation of the spatial parameters, the structure can present variations dynamic parameters, such as response functions, resonance frequencies, damping ratio and eigen modes of vibration of the structure [10].

In the current literature we found some studies that use methodologies to conduct analysis of structural integrity in this way, stand out the following work: in the article [6] used the method of the electromechanical impedance for monitoring aerospace structures with active piezoelectric sensors. A system for identifying and locating damage in an airplane wing was proposed in [9]. In [11] was proposing a multinodal genetic algorithm to diagnose damage in a steel truss bridge.

In this paper, we present a comparison between two methodologies for monitoring and analysis of aeronautical structures. With the acquisition and processing of signals, applies a ARTMAP-Fuzzy neural network [1] negative selection immune algorithm (NSA) [5] to identify and characterize flaws. To evaluate the methodologies used a data base of simulated signals from a

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numerical model of a beam of aluminum, which emulates a wing structure of an aircraft. The beam was modeled using the finite element method and simulated in Matlab® [8].

Negative Selection Algorithm (NSA)

The negative selection algorithm (NSA) has been proposed by [5] to detect changes in system state, and is based on negative selection of T cells within the thymus. This algorithm is the discrimination proper and non-proper cells. The algorithm can be described in the following steps [4]:

1) Censor Phase
   a. Define the proper set of chains (S) to be protected;
   b. Generate random chains and evaluate the affinity (Match) between each of them and the proper chains. If the affinity is higher than a stipulated threshold, reject the chain. Otherwise, store it in a set of detectors (R);
2) Monitoring
   a. Be the set of chains to be protected (protected chains), evaluate the affinity between each of them and the set of detectors. If the affinity is higher than a predetermined threshold, then a non-proper element is identified;

The censor phase of NSA basically consists in generating a set of detectors. The detectors are analogous to T type cells matured able to recognize pathogenic agents. The monitoring phase consists in monitoring a system to identify a change in the behavior of the same, and thus classify this change using the set of detectors created in censor phase [7]. In this study we used the partial matching criterion. The rate of affinity represents the degree of similarity needed for the marriage occurs between the two standards and is defined by the following relationship [3]:

$$TAf = \left(\frac{An}{At} \right) * 100 \quad (1)$$

being: TAf the afinity rate, An the number of chains in normal problem and At the total number of chains in problem. The equation (1) proposes a statistical relationship between all samples of the problem to the calculation of the affinity rate. In (2) represents a way of quantifying the overall affinity between patterns [7]:

$$Af_T = \frac{\sum_{i=1}^{L} Vc}{L} * 100 \quad (2)$$

Where in Af_T the % affinity between patterns analyzed, L the total number of variables and Vc the variables matching. If $Af_T >= TAf$ matching occurs between the two patterns (are considered similar).

ARTMAP-*Fuzzy* Artificial Neural Network

The ARTMAP-*Fuzzy* artificial neural network (ANN) has an architecture based on adaptive resonance theory, ie, belongs to the ART family (*Adaptive Resonance Theory*) [2]. This network corresponds to a supervised learning system consists of a pair of modules of adaptive resonance theory, ART_a-*Fuzzy* and ART_b-*Fuzzy*,which receive and classify the inputs (I_a) and desired outputs (I_b) respectively, and the inter-*ART* associative memory module checks the matching of inputs and outputs with the existing categories [1].

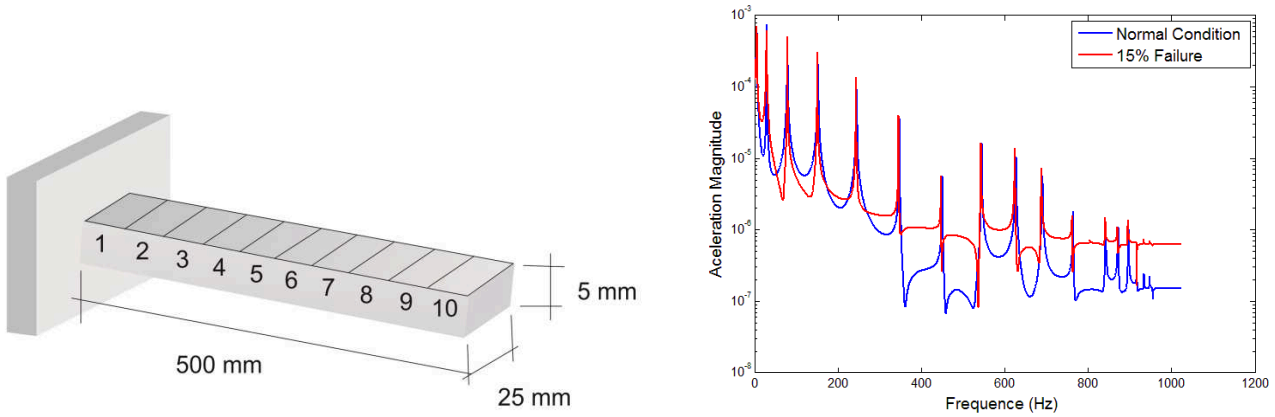
The ARTMAP-*Fuzzy* architecture has three key parameters for its development, being [1]: Parameter choice α ($\alpha > 0$): Operates in the category selection, Rate Training β ($\beta \in [0\ 1]$): Controls

the speed of adaptation of the network and Parameter monitoring (ρ_a, ρ_b e ρ_{ab} ϵ [0 1]): Controls the resonant network, namely the parameter responsible for the number of categories created.

If ρ has a large value, the neural network becomes more selective reducing its generalization capability. If ρ has a small value, it reduces the number of categories created, increasing the generalization capability of the ARTMAP-*Fuzzy* network.

Modeling and Simulations

To evaluate the methods proposed in this work, we used the finite element method to model an aluminum beam clamped-free condition in discretized with 10 finite elements with two degrees of freedom each. The material properties used are: modulus of elasticity, E = 700 GPa, density, γ = 2710 kg/m^3. The dimensions are 500*mm* long, 25*mm* wide and 5*mm* thick. Fig. 1 (a) illustrates the patterned beam.



(a) (b)

Fig. 1: (a) Beam modeled, (b) Simulated signal.

From the beam model were performed several simulations with different percentages of wear and locations of faults. The generated database consists signal captured by an accelerometer fixed to the beam. In all simulations the beam was excited in the 3rd degree of freedom (finite element two) and the signal was captured on the 19th degree of freedom (finite element ten). Thus, were simulated 540 signals in the structure, 400 without wear (base-line condition) and 140 signs with wear (structural failure). The signals at failure were simulated in wear levels 1, 5, 10, 15, 20, 25 and 30%, begging each failure has been positioned at two different locations in finite element 3 and 5 with 10 simulated signals at each location. Fig. 1 (b) presents two signals that had been captured in the simulations, a 15% failure and another under normal conditions.

Proposed Methodology

Systems integrity analysis of aeronautical structures proposed in this work are composed of three main modules: data acquisition, obtaining knowledge (training or censor) and the diagnosis, as shown in Figure 2:

ARTMAP-*Fuzzy* Artificial Neural Network . The data acquisition module comprises the experimental part for capturing the signals on the mechanical structure, such as sensors, actuators, accelerometers, etc. In the training module and the weights are adjusted created the categories (knowledge) that will be used in the diagnosis of the data by ARTMAP-*Fuzzy* neural network. In this phase the network is trained using a data set is generated and a number of categories equals the number of input signals. The diagnostic module signals the test set are presented to the network, and each signal is compared with the knowledge (categories created in the training phase) obtained by ARTMAP-*Fuzzy* neural network and so when there is a marriage between the patterns classified category in the displayed signal the ARTMAP-*Fuzzy* network. Obtained category corresponding to a given pattern is thus identified and classified failure in the structure.

Negative Selection Algorithm (NSA). The data acquisition module is the same ARTMAP *Fuzzy* neural network. The censor module the NSA is used to generate a set of fault detectors to be used in the diagnosis data. In the case of NSA the signals at faults are detected through evaluation and verifying marriage to the set of proper detectors (base-line signals) thus carries out the detection of signals in fail discrimination on the basis proper and non-proper. If the signal identified is proper the same is instantly classified as the normal condition of the structure**.** Otherwise, the signal identified for a non-proper (signal failure) carried out an evaluation of this signal, comparing it with the set of failure detectors, thus to occur wedding sign with a class of detectors, the signal is classified as a class of detector activated.

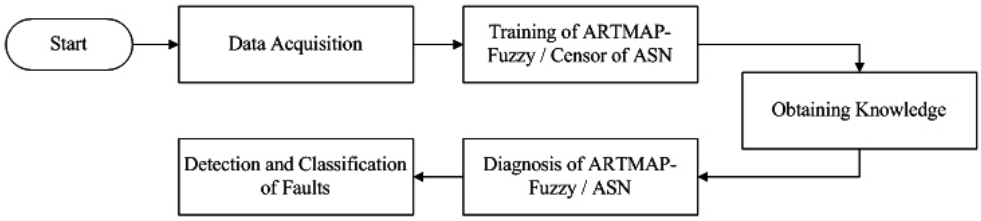


Fig 2: Functioning of the methodology.

Applications and Results

In this section we present the results obtained with the application the methods in the database simulated. All tests were performed using a PC Intel Core 2 Duo 1.9 GHz, 2 GB of RAM, operating system Windows 7 Ultimate 32 bits. The algorithm was developed in Matlab® [8].

For the ARTMAP-*Fuzzy* neural network the parameters used in the training process and diagnosis were: $\alpha = 0{,}1$, $\beta = 1$, $\rho_a = 0{,}99$, $\rho_b = 0{,}99$ e $\rho_{ab} = 1$. For the NSA was used as a parameter an affinity rate with a fixed value of 74% (calculated with equation (1)). To evaluate the ARTMAP-Fuzzy neural network the database wass divided in two parts (70% training and 30% test). To evaluate the NSA, were generated three sets of detectors (I, II and III) using 10%, 20% and 30% of available signals, totaling 54, 108 and 162 detectors, respectively. The database has seven different patterns of faults. The signals obtained by the frequency acquisition module representing the input signals of both methods. Table 1 shows the results obtained with the two systems when applied to the data set .

Table 1 - Results obtained by the method.

Diagnosis	NSA			ARTMAP-Fuzzy	
	I	*I*	*II*	*Training*	*Test*
Samples tested	540	540	540	398	142
Ratings correct	533	538	540	398	142
Ratings wrongs	7	2	0	0	0
Accuracy (%)	98.70	98,62	100.00	100.00	100.00
Time (ms)	163,5	162,10	158,00	735,23	146,25

The parameters used in the methods were obtained empirically by testing. The result was obtained by cross-reference test, performed twenty times to ensure accuracy of results. Both systems identified and classified 100% of the signals failed in its best configuration. We note that for the ARTMAP-Fuzzy neural network the more knowledge obtained in training, the more efficient the process of diagnosis. As fo the NSA, the greater the number of detectors, better diagnosis phase tracking. By comparing the proposed methods we obtain that both feature precision, robustness and efficiency, therefore, have identified and classified all faults. In relation to the processing time is greater than ARTMAP-Fuzzy network, however, both can be applied in real time, facilitating decision-making.

It is noteworthy that the two methods discussed are inspired by completely different theories, however, have accuracy, efficiency and good performance. Thus, either of the two methods can be applied to the analysis of aeronautical structures, which present good results.

Conclusion

In this paper, we propose two new approaches based on intelligent computing for analysis of aircraft structures, in this context we used an ARTMAP-Fuzzy artificial neural network and NSA, that showed good results, achieving 100% accuracy for the best configuration of both methods. Both the training and the censor phase, require more computational time, however are executed *off-line*, not causing harm to the system. The diagnostic data is performed with time less than 160 milliseconds, what allows methods to be used in real time. Thus, we conclude that the intelligent computing provides efficiency, reliability, robustness and accuracy in the analysis of aircraft structural integrity. This work contributes to the research line, presenting a comparison of two efficient methods employing intelligent techniques.

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A Comparison of Methodologies for Intelligent Computing Used to Integrity Analysis of a Structure Aeronautic

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Comparative evaluation of several methods for Canine Visceral Leishmaniasis diagnosis

Avaliação comparativa entre vários métodos para o diagnóstico da Leishmaniose Visceral Canina

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Abstract

The aim of the present study was to evaluate the serological methods using ELISA with recombinant-rK39 (ELISA-rK-39) and soluble extract-SE (ELISA-SE) antigens, the indirect fluorescence antibody test (IFAT) in comparison to an immunochromatography rapid diagnostic test (RDT-rK39) and with a direct parasitological exam (PA) for Canine Visceral Leishmaniasis (CVL) diagnosis. The results showed that 89% (60/67) of the dogs were positive for at least one serological diagnostic test. ELISA-SE was the test that detected anti-*Leishmania* antibodies in the serum of the highest number of dogs (71.6%) followed by ELISA-rK39 (65.7%), IFAT (65.7%) and RDT-rK39 (55.2%). PA detected the lowest numbers (40.3%) of positive dogs. In relation to the total of examined dogs, the Kappa indexes ($p \leq 0.05$) showed a good agreement between ELISA-SE and IFAT (88.1%; k = 0.7237), and it was also observed in the comparison of RDT-rK39 with ELISA-SE (83.6%, k= 0.6561), IFAT (83.5%, k= 0.6605) and PA (85.0%, k= 0.7074). A bad agreement was detected in any association of ELISA-rk39 with the other tests in either symptomatic or asymptomatic animals. ELISA as well as RDT using recombinant antigenic protein (rK39) were the methods that detected the lowest prevalence rates (33.3%) of CVL in asymptomatic dogs. In conclusion, only one test does not adequately identify dogs with CVL and it is necessary the association of two or more diagnostic tests. Because of the good agreement indexes of RDT-rK39 when evaluated with ELISA-SE, IFAT and PA it was suggested as a complementary method to be used in association with either ELISA-SE or IFAT, particularly in the symptomatic dogs. Furthermore, new studies are recommended in order to improve the sensitivity of tests mainly for asymptomatic dogs.

Keywords: CVL, diagnosis, dogs, *Leishmania infantum*, rK39, RDT.

Resumo

O objetivo do presente estudo foi avaliar os métodos sorológicos usando ELISA (Ensaio Imunoenzimático Indireto) com o antígeno recombinante rK39 (ELISA-rK39) e o antígeno extrato solúvel bruto (ELISA-ES) e a RIFI (Reação de Imunofluorescência Indireta) em comparação com o método imunocromatográfico rápido (RDT-rK39) e o parasitológico direto (PA), para o diagnóstico da Leishmaniose Visceral Canina (LVC) em cães de Ilha Solteira, São Paulo, Brasil. Os resultados mostraram que 89% (60/67) dos cães foram positivos por pelo menos um teste diagnóstico sorológico (RIFI, ELISA-ES, ELISA-rk39 ou RDT-rK39) e somente 40,3% (27/67) foram positivos pelo PA. O ELISA-ES foi o teste que detectou anticorpos anti-*Leishmania* em maior número de cães (71,6%) seguido por ELISA-rK39, RIFI (65,7%) e por RDT-rK39 (55,2%). No total de cães analisados (assintomáticos e sintomáticos), o índice Kappa de concordância ($p \leq 0,05$) foi considerado de boa concordância entre ELISA-ES e IFAT (88,1%; k= 0,7237) e entre RDT-rK39 com ELISA-ES (83,6%, k= 0,6561), RIFI (83,5%, k= 0,6605) e PA (85,0%, k= 0,7074). O índice de concordância ruim foi observado em qualquer associação de ELISA-rk39 com todos os outros testes nos animais sintomáticos e nos assintomáticos. Tanto o ELISA como o RDT com proteínas recombinantes (rK39) detectaram a

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menor porcentagem de cães assintomáticos (33,3%) em relação aos outros testes sorológicos. Em conclusão, somente um método diagnóstico não foi suficiente para identificar todos os cães positivos com LVC, principalmente os assintomáticos e por isso foi necessário a associação de dois ou mais métodos. Em função da boa concordância do teste RDT-rK39 com ELISA-ES, RIFI e PA, o mesmo foi sugerido como um teste complementar ao ELISA-ES ou RIFI para o diagnóstico da LVC, principalmente dos cães sintomáticos. No entanto, novos estudos são recomendados para melhorar a sensibilidade dos testes principalmente para cães assintomáticos.

Palavras-chave: LVC, diagnóstico cães, *Leishmania infantum*, rK39, RDT.

Introduction

Canine Visceral Leishmaniasis (CVL) is a disease caused by Kinetoplastida protozoa that belongs to the genus *Leishmania* (ROSS, 1903). *Leishmania infantum* (syn. *L. chagasi*) is the causative agent of visceral leishmaniasis in the New World, with endemic regions extending from southern USA to northern Argentina, including Brazil (KUHLS et al., 2011).

The main route for parasite transmission to human and animals is through the sandfly's bite, *Lutzomyia* (*L.*) *longipalpis* (Diptera: Psychodidae: Phlebotominae) (PORROZZI et al., 2006). Thus, the canine skin is the body system that most commonly presents signs of the CVL, and the presence of *Leishmania* parasites in dog skin reinforces the important role played by domesticated canid in the *L. infantum*'s life cycle (QUEIROZ et al., 2010).

Appropriate diagnostic tests are important in epidemiological studies and control programs, and for that reason the association of techniques may increase the positivity and contribute to the control of this canine disease (ASSIS et al., 2010; QUEIROZ et al., 2010). The detection of infected dogs can be made by serological analysis using an indirect fluorescence antibody test (IFAT or ELISA) or by the demonstration of the protozoan in parasitological methods (e. g., direct microscopic examination of amastigotes in smear). However, the sensitivity and the specificity of serological tests depend on the antigen, the protocol used in the laboratory, the time of infection and the serum conversion (COSTA et al., 1991; REITHINGER et al., 2002; FERREIRA et al., 2007); furthermore, the direct parasitological test is a very specific method, but the sensitivity is low and depends on the biological material and the parasitic burden (GENARO, 1993).

Specificity and sensitivity of the ELISA based immunoassay can be improved by using recombinant technology (BURNS-JUNIOR et al., 1993; MOHAPATRA et al., 2010), which drives the expression and purification of diagnostically relevant proteins in large amounts (SUNDAR; RAI et al., 2002).

In the last two decades, several *Leishmania* antigens have been genetically and antigenically characterized. Recombinant K39 antigen (rK39) is a 39-amino-acid repetitive immunodominant B-cell epitope of the 230 kDa kinesin related protein of *L. chagasi* (BURNS-JUNIOR et al., 1993; BADARÓ et al., 1996). The rK39 ELISA has proven to be suitable for detecting human and canine visceral leishmaniasis (VL) with the host producing specific antibodies against replicating *Leishmania* (QU et al., 1994; BADARÓ et al., 1996). Although, sera from patients and dogs with acute VL strongly recognized rK39, patients with asymptomatic or self-healing infections had low or undetectable levels of anti-rK39 antibodies (BADARÓ et al., 1996).

In another study, the performances of ELISA assays with different antigen preparations, such as *Leishmania amazonensis* or *Leishmania chagasi* lysates and the recombinant antigens rK-39 and rK-26, were compared to detect anti-*Leishmania* antibodies in dogs from a visceral leishmaniasis-endemic area in Brazil. In this study, the sensitivities and specificities of crude and recombinant antigens were similar and remarkably high (96-100%) identifying different clinical phases of canine leishmaniasis (ROSÁRIO et al., 2005).

Later, Mohapatra et al. (2010) designed a comparative evaluation of two recombinant hydrophilic antigens, rK9 and rK26 of *L. chagasi* along with rK39 and a crude soluble antigen (derived from *Leishmania donovani* promastigote) for the serodiagnosis of Indian visceral leishmaniasis patients by quantitative ELISA. The rK39 antigen was observed to be the most suitable antigen with 100% and 96% of sensitivity and specificity, respectively, in comparison to rK26, rK9 and and crude antigens.

The *Leishmania* rK39 Antibody Diagnostic Rapid Test (RDT) is a qualitative immunochromatography assay for detecting antibodies against *L. donovani* complex, that includes the *L. donovani* and *L. infantum* species, in whole blood, serum or plasma, but the diagnostic performance has been reported to be variable (QUINNELL et al., 2013). According to Quinnell et al. (2013) the sensitivity of this RDT method in infected dogs was relatively low (ranging from 46% to 78%), suggesting that rK39 RDTs may be useful in a veterinary clinical setting, but is too low for operational control programs.

More recently, an immunochromatography assay (Dual-Path Platform - DPP® CVL kit) was developed to detect antibodies against the rK26/rK39 fusion protein. Grimaldi Jr et al. (2012) investigated the evaluation of this method for CVL diagnosis and indicated that this kit displays high specificity (96%) but low sensitivity (47%) to identify parasite-positive dogs without signs of CVL. However, the test's sensitivity was significantly higher (98%) in disease cases, indicating that this convenient test may be useful to identify the most infectious dogs. Thus, this kit is unlikely to be sensitive enough for detecting asymptomatic canine carriers of *L. infantum*, but it may be useful for confirming clinically suspected cases due to its high specificity in symptomatic dogs (GRIMALDI JR et al., 2012).

As the euthanasia of positive dogs after serological exams is one of the most important CVL control measure, the use of appropriated diagnostic methods is decisive for the success of the control programs (REITHINGER et al., 2002); however, the efficacy of diagnostic methods is a challenge and more studies about this topic still necessary. Thus, the aim of the present study was to evaluate the serological methods using ELISA with recombinant-rK39 (ELISA-rK-39) and soluble extract-SE (ELISA-SE) antigens and an indirect fluorescence antibody test

(IFAT) in comparison with the immunochromatography assay, a rapid diagnostic test (RDT-rK39) and a direct parasitological exam (PA) for CVL diagnosis in asymptomatic and symptomatic dogs from Ilha Solteira, São Paulo, Brazil.

Materials and Methods

Animals and local of study

The dogs of the present study belong to the Animal Protector Association shelter from Ilha Solteira, located in the northwestern region of the state of São Paulo, Brazil. According to Noronha Junior et al. (2007), Ilha Solteira is considered endemic for CVL, where the reported prevalence rates were 14% in the urban areas (PAULAN et al., 2012) and 37.7% in the rural areas (PAULAN et al., 2013).

The study was approved by the Animal Ethics Committee of São Paulo State University at Jaboticabal, under the Protocol no. 015863/09. Sixty seven dogs from different breeds and ages, were clinically examined and divided into two groups according to their clinical leishmaniasis signs: asymptomatic - when they did not present any clinical sign (n=12); or symptomatic - when they presented at least one clinical sign (n=55). The main clinical signs were onychogryphosis, lymphadenomegaly, weight lost, skin lesions, apathy, opaque bristles and/or alopecia, conjunctivitis and anemia.

Parasitological exams

For parasitological exams (PA), smears of popliteal lymph node aspirates were fixed in methanol, stained with Giemsa (4%) for 20 minutes and were directly examined under a light microscope (1000× magnification) for the detection of *Leishmania* amastigotes. At least 200 microscopic fields were examined for each slide.

Indirect Fluorescence Antibody Test (IFAT)

The detection of antibodies against *L. infantum* by IFAT was performed according to the procedure of Oliveira et al. (2008) as briefly described: *Leishmania* parasites were isolated from the bone marrow of naturally infected dogs and maintained on a RPMI-1640 medium at 25 °C. For IFAT procedure, *Leishmania* promastigotes maintained in culture medium were used for antigen preparation. In duplicate, the serum dilutions were used, starting at 1:40 and placed over the antigen on the slides at 37 °C for 30 minutes, and then subjected to three washes in PBS (0.01M, pH 7.4) by immersion, five minutes each. After drying at room temperature, the slides were incubated with anti-dog IgG serum conjugated for fluorescein isothiocyanate (KPL, cat nº 02-19-02, USA), diluted at 1:30 in PBS containing 1mg% of Evan's Blue. The slides were washed again in PBS, covered with buffered glycerin and a cover slip, and then examined under a fluorescent microscope (Olympus, BX-FLA, Japan). In all the experiments, reference sera were included as negative and positive controls. For positive serum samples, parasites displayed a bright-green peripheral stain with a dull fluorescence of the cytoplasm and were considered positive serum samples at 1:40 or more (cut-off point).

Enzime-Linked Immunosorbent Assay (ELISA)

Soluble extract antigens of *L. infantum* (SE)

An indirect ELISA method (OLIVEIRA et al., 2008) was used for *Leishmania* antibodies. 100 µl of crude soluble extract of *L. infantum* antigen (ELISA-SE) was diluted in a sodium carbonate-bicarbonate buffer (0.05M, pH 9.6) and was added to each ELISA microplate, sealed and overnight incubated at 4 °C in a humid chamber. Antigen protein concentration was determined at 5 µg/ml. After coating the plates with antigens, the plates were blocked for one hour at 37 °C with 6% powered milk diluted in 0.05 M carbonate/bicarbonate buffer at pH 9.6. After wash (3 min each), 100 µl of diluted dog serum (1/400) in PBS-Tween-20 plus 5% normal rabbit serum was added in duplicates to the plate and incubated in a humid chamber at 37 °C for 60 min. Following, a 100 µl of alkaline phosphatase conjugate rabbit anti-dog IgG whole molecule (Sigma, cat. nº A 0793, USA) at 1:8000 dilution plus 0,5% normal rabbit serum was added to each well and incubated at 37 °C for 60 min in a humid chamber. After the washings, 100 µl of P-nitrophenyl phosphate (Sigma, cat. nº N-9389, USA) at 1 mg/ml in diethanolamine buffer, pH 9.8, was added and incubated for 45 min at room temperature. The reaction was stopped by adding 50 µl of 3M sodium hydroxide and the plates were read at 406 nm wavelengths using a microplate reader (Dynex Technologies, USA). To control inter-plate variation, positive and negative controls were included in each plate.

The ELISA system calibration was done according to Oliveira et al. (2008), using positive reference sera collected from *Leishmania* symptomatic dogs and negative reference sera from *Leishmania* negative dogs. The negative reference sera were samples collected from dogs living in a non-endemic areas. The enzymatic activity of each serum sample was determined by calculating the sample (S) for the positive (P), rate (S/P) at each dilution, considering positive and negative serum references. The cut off point was determined to be ELISA level (EL ≥ 3).

Recombinant K39 (rK39) antigen

For ELISA using rK39 of *L. infantum* (ELISA-rK39), the antigenic protein concentration was determined at 200 ng/mL. After coating the plates with antigens, the plates were blocked for one hour at 37 °C with 5% powered milk diluted in 0.05 M carbonate/bicarbonate buffer at pH 9.6. Dog sera were diluted at 1/50 in PBS-Tween-20. Following the incubation with dog sera, 100 µl aliquot of alkaline phosphatase conjugate rabbit anti-dog IgG whole molecule (Sigma, cat nº A 0793, USA) was diluted at 1:4000, plus 0.5% normal rabbit serum added to each well, followed by incubation at 37 °C for 60 min in a humid chamber. The reaction was detected by the substrate P-nitrophenyl phosphate (Sigma, N-9389, USA).

The calibration of the ELISA system and the enzymatic activity of each serum sample were determined as described above in 4.1. item. The cutoff point was determined to be EL ≥ 3.

Sensit Leishmania rK39 Antibody Rapid Diagnostic Test (RDT-rK39)

The Sensit *Leishmania* rK39 Antibody Diagnostic Rapid Test (RDT-rK39) is a qualitative immunochromatography assay for detecting antibodies to *Leismania donovani* complex that includes the *L. donovani* and *L. infantum* species in whole blood, serum or plasma. The RDT-rK39 kit obtained from Ubio Biotechnology Systems Pvt Ltd, Cochin Kerala, India, was used according to the manufacturer's instructions. The appearance of a red line in the test area, even a faint one, was interpreted as positive, following manufacturer's recommendations; all tested samples were classified as either positive or negative. All reactions were performed by the same person, who was not blind to the results of the reference tests.

Statistical analysis

The results of diagnostic methods were submitted for statistical analysis using Fortran Power station (version 4) program (University of Economics and Business. Austria), In order to evaluate the concurrence index among diagnostic methods, the Kappa index of agreement (at the 95% confidence interval) was used. According to Landis and Koch (1977), $k < 0.4$ is consider bad; $0.41 \geq \kappa \leq 0.60$ is accept as moderate agreement; $0.61 \geq \kappa \leq 0.80$ is a good agreement, and $\kappa > 0.8$ is an excellent agreement.

The chi-square test with a significance level of 5% was calculated using the R system (R Development Core Team, 2013) in order to assess the associations between the positivity of asymptomatic and symptomatic dogs in each diagnostic method (RDT, ELISA-SE, ELISA-rK39, IFAT and PA) for CVL.

Results and Discussion

In order to conduct this study, 67 dogs of different breeds and ages, from the animal shelter were examined according to their clinical status and by several laboratorial tests for CVL diagnosis. Thus, according to their clinical signs, 55/67 (82.1%) dogs were classified as symptomatic and the most frequent clinical signs were onycogryphosis (35/55, 63.6%), alopecia (33/55, 60%), emaciation (30/55, 54.5%), hypertrophic lymph nodes (18/55, 32.7%) and corneal opacity (15/55, 27.3%). In contrast, 12/67 (17.9%) dogs were classified as asymptomatic.

One of the greatest challenges for the disease control strategy in dogs is regarding about the appropriate diagnosis. It is clear that the precise CVL diagnosis is complex and due the fact that the infected dogs have multiple clinical aspects and no specific clinical pathological alterations, reinforce the need for a combination of two or more laboratorial methods for canine diagnosis (ASSIS et al., 2010; QUEIROZ et al., 2010; PALTRINIERI et al., 2010). Furthermore, the diagnostic performance has been reported to be variable, with some disagreements about the results of specificity and sensibility of methods in different studies (BADARÓ et al., 1983; REITHINGER et al., 2002; GOMES et al., 2008). Thus, the sensitivity and specificity of diagnostic methods are dependent of the type of antigen (e.g., crude or recombinant, parasite species, promastigotes or amastigotes, etc), the time of infection and changes to the standard experimental protocol (BADARÓ et al., 1983). We evaluate the serological methods using ELISA with two different types of antigens (crude soluble extract and rK39) and IFAT in comparison with RDT-rK39 and PA, and the results showed that 89% (60/67) of the dogs were positive for at least one serological diagnostic test, but none of the methods tested in this study was able to detect all positive dogs when individually evaluated, and for that reason it was necessary a combination of two or more methods.

The PA is a very specific method for CVL diagnosis, since it detects the parasites in the tissues, including biopsies or/and aspirates of bone marrow, lymph nodes, spleen, liver or skin (ALVAR et al., 2004; MOREIRA et al., 2007; TASCA et al., 2009; ASSIS et al., 2010; QUEIROZ et al., 2010). However, the sensitivity of this method is low, depending on biologic material, parasitic load and animal's clinical status (GENARO, 1993). In the present study, only 23/55 (41.8%) of symptomatic dogs had *Leishmania* amastigotes in samples of popliteal lymph node aspirates, and it was the lowest prevalence rate of positivism in comparison with ELISA, IFAT or RDT, which the difference was statistical significant ($p \leq 0.05$) (Table 1). For that reason, serological exams are currently used for mass-screening surveys instead of parasitological exams.

However, it is important to mention that all positive dogs by ELISA-SE (Figure 1a), IFAT (Figure 1c) or RDT-rK39 (Figure 1d), were also simultaneously positive by PA (40.3%) reinforcing the low sensitivity, but the high specificity of PA as a diagnostic method for CVL diagnosis (GENARO, 1993; BRASIL, 2003). Moreover, the Kappa indexes (Table 2) showed a moderate

Table 1. Numbers and percentages (Nº/%) of positive dogs for Canine Visceral Leishmaniosis (CVL) by serological (ELISA-SE, ELISA-rk39, IFAT and RDT) and parasitological direct exams (PA) in asymptomatic (A) and symptomatic (S) dogs, Ilha Solteira, São Paulo, Brazil, 2013.

	Numbers / percentages of positive dogs by CVL				
Groups (N)	**Elisa-SE**	**Elisa-rk39**	**IFAT**	**RDT**	**PA**
A (N=12)	7/58.3^{aA}	4/33.3^{aA}	6/50.0^{aA}	4/33.3^{aA}	4/33.3^{aA}
S (N=55)	41/74.5^{aA}	40/72.7^{aB}	38/69.1^{aA}	33/60.0^{aA}	23/41.8^{bA}
T (N=67)	**48/71.6**	**44/65.7**	**44/65.7**	**37/55.2**	**27/40.3**

(a/b) Proportions with the same small letters in lines are not significantly different (P < 0.5). (A/B) Proportions with the same capital letters in columns are not significantly different (P < 0.5). Chi-square test.

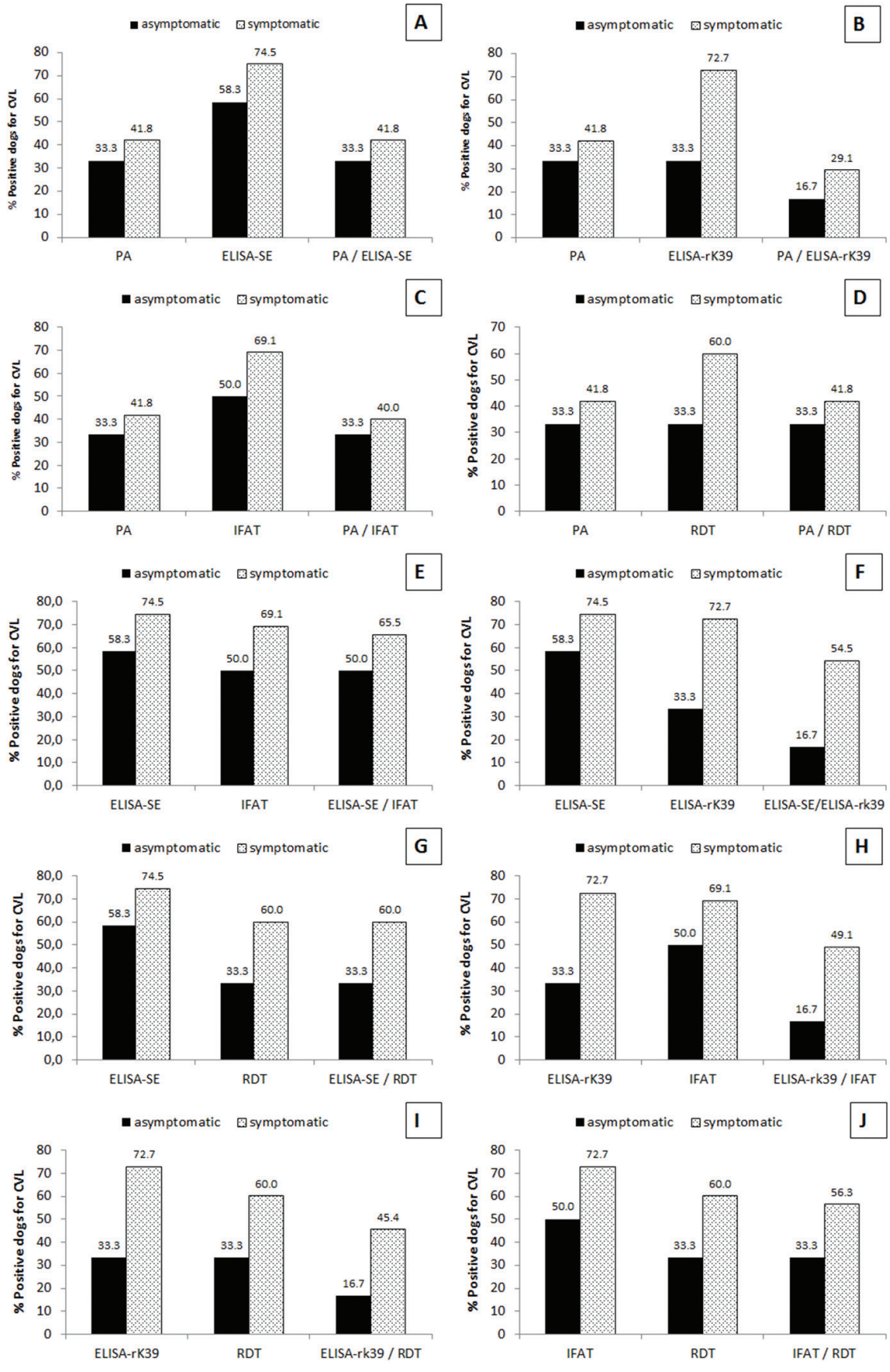

Figure 1. Results of positive dogs for Canine Visceral Leishmaniosis (CVL) compared two by two in symptomatic (n=55) and asymptomatic dogs (n=12) Ilha Solteira São Paulo, Brazil, 2013.

Table 2. Comparative analysis of diagnostic methods for Canine Visceral Leishmaniosis (CVL) in dogs classified in asymptomatic (A) and symptomatic (S). Ilha Solteira, São Paulo, Brazil, 2013.

Concordances between diagnostic methods Number of dogs / Percentage				
Techniques	**A (N= 12)**	**S (N= 55)**	**Kappa Index (κ)**	**Total (N= 67)**
Elisa-rK39 x Elisa-SE	05/41.6	34/61.8	0.0330	39/58.2
κ	–0.1053	0.0170	Bad	
Elisa-rK39 x IFAT	06/50.0	31/56.4	0.0069	37/55.2
κ	0,0000	–0.0560	Bad	
Elisa-rK39 x PA	08/66.7	24/43.6	0.0151	32/47.8
κ	0.2500	–0.0492	Bad	
Elisa-rK39 x RDT-rK39	08/66.7	32/58.1	0.1667	40/59.7
κ	0.2500	0.0800	Bad	
Elisa-SE x IFAT	11/91.7	48/87.3	0.7237	59/88.1
κ	0.8333	0.6867	Good	
Elisa-SE x PA	09/75.0	37/67.3	0.4217	46/68.7
κ	0.5263	0.3941	Moderate	
Elisa-SE x RDT-rK39	09/75.0	47/85.4	0.6561	56/83.6
κ	0.5263	0.6774	Good	
IFAT x PA	10/83.3	38/691	0.4654	48/71.6
κ	0.6667	04182	Moderate	
IFAT x RDT-rK39	10/83.3	46/83.6	0.6605	56/83.5
κ	0.6667	0.6457	Good	
RDT-rK39 x PA	12/100.0	45/81.8	0.7074	57/85.0
κ	1.000	0.6479	Good	

Elisa-rK39 (ELISA recombinant K39 antigen); Elisa-SE (ELISA crude extract soluble antigen); IFAT (indirect fluorescent antibody test); RDT-rK39 (Rapid Diagnostic Test–rK39 antigen); PA (Parasitological direct test); (κ) = Kappa Index; A= asymptomatic; S= symptomatic.

agreement between PA x ELISA-SE (68.7%, k= 0.4217) and between PA × IFAT (71.6%, k= 0.4654), but a good agreement between PA × RDT-rK39 was observed for symptomatic animals (81.8%, k= 0.6579).

According to the tests, the positivism rates for CVL were the following: 71.6% (48/67), 65.7% (44/67), 65.7% (44/67) and 55.2% (37/67) by ELISA-SE, ELISA-rk39, IFAT and RDT-rK39, respectively. Furthermore, the results showed that the serum prevalence rates were higher in symptomatic rather than in asymptomatic dogs (Table 1) in accordance to the results of Assis et al. (2010) and Queiroz et al. (2010), who compared different diagnostic tests in a group of CVL positive dogs.

When ELISA-SE and IFAT were individually or simultaneously analyzed (Figure 1e), the data showed similar numbers of positive dogs in both exams, not only symptomatic but also asymptomatic dogs. The Kappa indexes of agreement (Table 2) between both serological methods (ELISA-SE × IFAT) were considered good in the symptomatic (87.3%, k= 0.6867) and excellent in the asymptomatic group (91.7%, k= 0.8333). Similarly, Assis et al. (2010) observed also a good concordance between ELISA-SE and IFAT, indicating that either ELISA or IFAT can be recommended for CVL diagnosis during epidemiological surveys.

Moreover similar results were not observed by ELISA-rK39 when individually or simultaneously compared with PA (Figure 1b), ELISA-SE (Figure 1f), IFAT (Figure 1h) or RDT-rK39 (Figure 1i). Firstly, because ELISA-rK39 reacted with much lower numbers of asymptomatic (33.3%) in comparison with symptomatic dogs (72.7%) with statistically significant difference (Table 1). Secondly, because the Kappa indexes of ELISA-rK39 indicated a bad agreement in the association with ELISA-SE (58.2%, k= 0.0330), IFAT (55.2%, k= 0.0069), RDT-rK39 (59.7%, k= 0.1667) and PA (47.8%, k= 0.0151) independently of the clinical signs of the dogs (Table 2). Similar results were found by Zijlstra et al. (2001) reporting low sensitivity of the rK-39 antigen when a strip test was used in comparison with direct agglutination (DAT) that was 67% and 90%, respectively, with an agreement of only 69% between both diagnostic tests.

More recently, immunochromatography assays have been developed (e. g., rapid diagnostic tests - RDTs) in order to detect antibodies against one or more types of recombinant proteins of the parasite. In this work, we also analyzed one RDT kit containing one recombinant antigen, (RDT-rK39). In comparison with serological and parasitological exams, all positive dogs diagnosed by this kit were confirmed by ELISA-SE (Figure 1g), ELISA-rK39 (Figure 1i) or IFAT (Figure 1j); even though, this kit detected lower numbers of positive dogs (55.2%) than ELISA-SE (71.6%), ELISA-rK39 (65.7%) and IFAT (65.7%). Furthermore, our studies showed that the Kappa indexes of agreement of RDT-rk39 were good when compared to ELISA-SE (83.6%, k= 0.6561), IFAT (83.5%, k= 0.6605) and PA (85.0%, k= 0.7074), particularly in symptomatic dogs (Table 2). These good agreement indexes suggest that RDT may be another option to be used as a complementary method in association with either ELISA or IFAT or PA for CVL diagnosis.

Moreover, this study showed that RDT-rK39 and ELISA-rK39 detected lower numbers of positive asymptomatic dogs (33.3%) than symptomatic (60%-72.7%) demonstrating the poor efficacy of recombinant protein for the detection of infection in asymptomatic dogs. Similarly, Quinnell et al. (2013) reported higher sensitivity (99.7%) of recombinant K39 protein when used in the RDT only in dogs with clinical disease, but not in asymptomatic dogs. In other studies, the immunochromatographic tests using protein k39 as antigen, detected sensitivities ranging from 91.5% to 96.7% in symptomatic dogs (METTLER et al., 2005; LIMA et al., 2010) and only 52.9% in asymptomatic dogs (METTLER et al., 2005). According to the clinical progression of the disease, from asymptomatic to symptomatic, higher specific antibodies tend to increase with the infection's development (DEPLAZES et al., 1995), which may explain the low concentration of anti-*Leishmania* antibodies in asymptomatic dogs, while crude soluble extract (SE) may identify different clinical phases of canine leishmaniasis.

The heterogeneous humoral immune response that develops in dogs infected by *L. infantum* (DEPLAZES et al., 1995; SOLANO-GALLEGO et al., 2001) likely involves multiple antigens that are differentially recognized in different animals and different stages of disease in the same host (FALQUETO et al., 2009). These results suggest that low sensitivity to detect asymptomatic infections may due to the use of a single defined antigen. Thus, the development of RDTs using multiple defined antigens (rK39) may increase the sensitivity in asymptomatic dogs.

In Brazil, the Public Health Services recommend two serological diagnostic tests for epidemiological surveys and CVL control: the RDT (detecting antibodies against rK26 and rK39 antigens) for screening and ELISA as a confirmatory test (BRASIL, 2013). Nonetheless, recent tests of a RDT incorporating both rK26 and rK39 showed low sensitivity (47%) to detect asymptomatic canine infections (GRIMALDI JR et al., 2012). In accordance with Quinnell et al. (2013), RDT is a reasonable method for confirmation of infection in suspected clinical cases, but due the low sensitivity to detect asymptomatic infected dogs it is not indicated for large-scale epidemiological studies and operational control programs, which many dogs could be misdiagnosed as false negatives, leading to a failed control campaign. However, in this paper, RDT-rK39 presented good indexes of concordance with ELISA-SE, IFAT and PA, and when used as a screening test, could permit a faster removal of infected symptomatic reservoirs from the environment, stopping the transmission of parasites to susceptible uninfected dogs.

The results of this comparative evaluation among several diagnostic methods reinforce that only one test does not adequately identify the dogs with CVL and for that reason it is necessary a combination of different methods. All tests used in this study showed poor efficacy for detecting infection in asymptomatic dogs, particularly with recombinant protein K39. Therefore, RDT-recombinant antigenic proteins should be used only as a complementary method in association with either ELISA or IFAT, because of the good concordance among these methods, particularly in the symptomatic dogs. Furthermore, additional studies are recommended for the development of new technologies to increase the sensitivity of diagnostic methods mainly for asymptomatic dogs.

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Analysis of Structural Integrity of a Building Using an Artificial Neural Network ARTMAP-Fuzzy-Wavelet

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Keywords: Artificial Neural Network, ARTMAP-Fuzzy-Wavelet, Analysis of Structural Integrity, Building Structure.

Abstract. This paper presents an ARTMAP-Fuzzy-Wavelet artificial neural network to perform the analysis of the structural integrity of a building. The combination of Fuzzy ARTMAP neural network, wavelets transform to generate a tool that performs the identification and characterization of structural failure. This method is applied as a support tool for professionals in the inspection of mechanical and building structures to identify and characterize flaws in order to carry out preventive maintenance to ensure the integrity of the structure and decision making. In order to validate the methodology perform mathematical modeling of a building of two walk, and from this model were simulated different situations (base-line condition and improper conditions), yielding a database of signals that serve as input ARTMAP-Fuzzy-Wavelet neural network. The results obtained by ARTMAP-Fuzzy-Wavelet shown efficiency and robustness.

Introduction

The analysis of structural integrity is an important tool to help engineers and technicians to identify faults in structures such as buildings. Technological advancement has provided the development of new metallic materials and construction. however, there is a large amount of newly constructed buildings presenting failures. This is due to the inappropriate use of materials, lack of care in the execution, and even adaptations, all this added to the lack of maintenance, has generated a concern for engineers, encountering buildings with fewer than five years of age with several flaws that could be avoided [7].

In building structures failures may represent low or no dynamic performance of the structure in relation to the stability and durability when exposed to abnormal conditions such as, for example, natural events (earthquakes, tremors, etc.). According to [1] the building structures the most part show greater severity of failures in reinforced concrete structures. The risk the integrity of the structure is connected to the concrete armor, which is made of steel, and usually has flaws as corrosion and cracks. Thus it is necessary to develop efficient ways to inspect such structures, to identify and characterize failures, so that be possible make quick decisions to perform preventive maintenance to ensure the integrity of the structure and prevent disasters.

In literature found themselves few studies that use methodologies to perform the analysis of the structural integrity of buildings, buildings or similar structures, aiming to make decisions. Following is presented the main works.

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In [7] presents a methodology for structural analysis of buildings using a nondestructive testing (NDT). In [4] it proposed a fuzzy system to detect structural faults using curvatures of mode shapes of vibration. In the work [13] a model using wavelet transform to evaluate structural integrity of bridges across the vibration signals has been proposed. In the article [12] the authors proposed an experimental method to perform structural analysis of buildings. The authors experimentally analyze a shed making holes in metal support columns. In [14] proposed a multimodal genetic algorithm to diagnose damage in a steel truss bridge. In [11] presents an ARTMAP-Fuzzy neural network to monitor the structural integrity of a two storey building. In the article [8] the authors propose a methodology based on artificial immune systems for the analysis of structural integrity of buildings, the authors validated method with a numerical example of a building.
In this paper, we present a method to perform the analysis of the structural integrity of buildings using an ARTMAP-Fuzzy-Wavelet artificial neural network. From the acquisition and processing of the signals, applies an ARTMAP-Fuzzy-Wavelet neural network to identify and characterize failures. To evaluate this method, we used a database of simulated signals from a numerical model of a two-story building, which in this case is the structure. The building modeled by differential equations and simulated in Matlab [9].
The combination of wavelet transform with ARTMAP-Fuzzy neural network provides greater accuracy in fault diagnosis, because the decompose signals at the levels of resolution it is possible to easily identify abnormalities, because of the level of detail displayed in the decomposition process.

ARTMAP-Fuzzy Artificial Neural Network

The ARTMAP-Fuzzy artificial neural network corresponds to a supervised learning system comprised of a pair of modules Adaptive Resonance Theory [3], ART_a-*Fuzzy* and ART_b-*Fuzzy*, interconnected by inter-*ART* associative memory module. This neural network architecture incorporates fuzzy set theory, the AND *fuzzy* operator (^) enabling the learning of the neural system in response to binary input patterns and analog, belonging to the interval [1], [2].
A internal mechanism called *match-tracking* is responsible for self-regulating process of the neural network, in which maximize generalization and minimize the error. When the neural network makes a prediction wrong, through an associative connection instructed, the monitoring parameter of ART_a-*Fuzzy* module is incremented in minimum amount necessary to correct the error in the ART_b-*Fuzzy* module.
The ARTMAP-*Fuzzy* architecture has three main parameters for the development, namely [2]:

- Parameter choice α ($\alpha > 0$): Operates in the category selection;
- Rate Training β ($\beta \in [0\ 1]$): Controls the speed of adaptation of the network;
- Parameter monitoring (ρ_a, ρ_b e ρ_{ab} $\in [0\ 1]$): Controls the resonant network, namely the parameter responsible for the number of categories created.

If ρ has a large value, the neural network becomes more selective reducing its generalizability. If ρ has a small value, it reduces the number of categories created, increasing the generalization capability of the ARTMAP-*Fuzzy* network.
The selection of the category is performed as (2), J is active index of F_2 [2].

$$T_j = \frac{|I \wedge w_j|}{\alpha + |w_J|} \tag{1}$$

$$J = \arg\max T_j \tag{2}$$

Resonance occurs if the vigilance test (3) is satisfied for the active index J [2].

$$\frac{|I \wedge w_j|}{I} \geq \rho \tag{3}$$

If vigilance test (3) is not satisfied, the *reset* occurs and the choice function continues until a new category J satisfies (3). After the process of resonance and the modules ART_a and ART_b should check the match tracking of categories in both modules, this step is taken by Inter-*ART* module.

The Inter-*ART* module verifies the Match Tracking of information between *ART* modules. This process is done by the *Match Tracking* described in (4), where y^b is the vector of activity module ART_b [2].

$$\frac{\left|y^b \wedge w_{JK}^{ab}\right|}{y^b} \geq \rho_{ab} \tag{4}$$

If the test *Match Tracking* (4) is not satisfied, a new category of ART_a module should be chosen and introduced in the training process until this criterion is satisfied.

After making the resonance processes for modules ART_a and ART_b and the *Match Tracking* for the Inter-*ART* module is made adaptation of the synaptic weights w_J^a, w_K^b and w_{JK}^{ab} as (5), J and K are activer index [2].

$$\begin{aligned} w_J^{a+1} &= \beta(I \wedge w_J^a) + (1-\beta)w_J^a \\ w_J^{b+1} &= \beta(I \wedge w_K^b) + (1-\beta)w_K^b \\ w_{JK}^{ab+1} &= \begin{cases} 1 \rightarrow J = j, K = k \\ 0 \rightarrow J \neq j, K \neq k \end{cases} \end{aligned} \tag{5}$$

Discrete Wavelet Transform

The wavelet functions are mathematical transforms able to decompose functions, allowing rewriting these functions more detailed, i.e. with a global vision. Thus, it is possible to differentiate local characteristics of a signal in different sizes (resolutions) and, analyze all the signals by translations. As the most of wavelets has compact support, they are useful in analyzing non stationary signals. There are several wavelet families. This work considers the orthonormal family functions and the Daubechies discrete family [5] due to have faster computational algorithms [10].

Define a signal $y[t] = (y_0, \ldots, y_{n-1}, y_n)$ representing a discrete vector then it can be represented by a wavelet series as follows [10]:

$$y[t] = \sum_{k=0}^{N_J} c_{J,k}\phi_{J,k}(t) + \sum_{j=J}^{1}\sum_{l=0}^{N_j} d_{j,k}\upsilon_{j,k}(t), \nabla t \in [0, N_0] \tag{6}$$

where: J represents the resolution level, $N_j = (N/2) - 1$ represents the quantity of points in each new vector obtained by transformation, $\phi_{j,k}(t)$ and $\upsilon_{j,k}(t)$ are the wavelet and scale functions that execute the transformation; j is the scale (dilation) and k the position (translation).

The discrete wavelet transform (DWT) when applied directly to a signal to generate a set of coefficients is calculated by several entrances into a G filter (low pass) and H filter (high pass), or known as resolution levels. The filters G and H are vectors with constants already calculated that provide an orthogonal base related to the scale and wavelet functions respectively. This process IF known as Mallat Pyramidal algorithm [10] and is shown in Fig. 1 (a).

At Fig. 1 (a), c_0 corresponds to the original discrete signal ($c_0 = y[t]$), H and G represent the low pass and high pass filters respectively. The parameters d_1, d_2 and d_3 are the wavelet coefficients or detail in each resolution level and c_3 are the scale coefficients or approximation at the last level of the transform. These coefficients are obtained by convolution of the constants at filters (7) and (8), [10]:

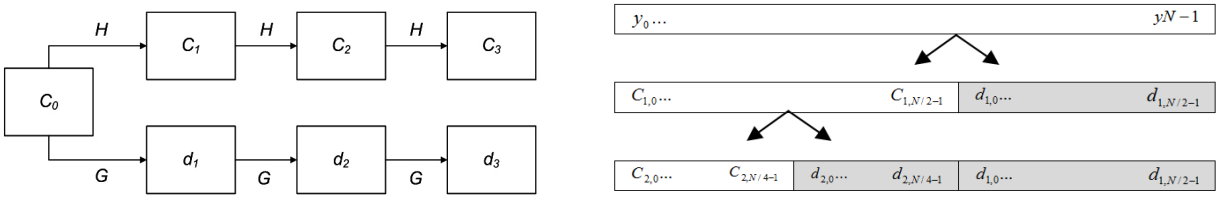
$$C_{j+1,k} = \sum_{l=0}^{D-1} h_l C_{j,2k+l} \tag{7}$$

$$d_{j+1,k} = \sum_{l=0}^{D-1} g_l C_{j,2k+l} \tag{8}$$

where: $k = [0, \ldots, (N/2^j) - 1]$ and D the quantity of constants of the filter. Thus, the coefficients $c_{j,k}$ represent the average local media and the wavelet coefficients $d_{j,k}$ represent the complementary information or the details that run away from the average media. Therefore, the transform coefficients ordered by scale (j) and position (k) are represented as follows [10]:

$$\psi = \left((C_{J,k})_{k=0}^{N_J}, \left((d_{j,k})_{k=0}^{N_J}\right)_{j=J}^{1}\right) \tag{9}$$

such that ψ is the finite representation in terms of the coefficients of the signal decomposition in equation (6). Fig. 1 (b) shows the decomposition process of a signal in two resolution levels. Observe that in each transformation level the size of the vectors is reduced by half ($N/2^J$).

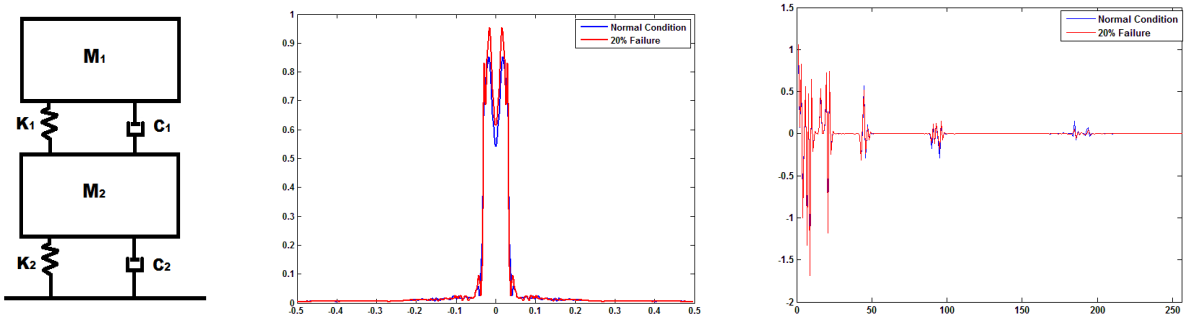


(a) (b)

Fig. 1 Pyramidal algorithm for DWT

Modeling and Simulations

In this section, a mathematical model was developed using differential equations of a two storey building with two degrees of freedom each. The building is adimensional model and consists of two masses (M_1 and M_2), and the stiffness coefficients (k_1 and k_2) and elasticity (c_1 and c_2), see [6, 11]. The model is illustrated in Fig. 2 (a).



(a) (b) (c)

Fig. 2 (a) Structure model, (b) Frequency response, (c) Wavelet domain

We simulated 700 signs in the structure, being 100 signs in normal condition, i.e., without flaw (condition base-line) and 600 signals with structural flaws, being 100 signs for each type of failure. The signals faults were simulated with additions of 5%, 10%, 15%, 20%, 25% and 30% in mass M_1 of the structure. The structure was excited by the signal (S) as [6]:

$$S = 1 * e^{-5\cos(t)} \tag{10}$$

After exciting the structure obtains the frequency response of the structure using the Fourier Fast Transform (FFT) [6]. The frequency response is obtained from the signals of the velocity and the displacement of the structure excited. Fig. 2 (b) illustrate two signals are captured in the simulations, which has a signal with a 20% failure (red) and a signal in normal condition (blue). Following applies wavelet transform to obtain the signals shown in Fig. 2 (c). The data set is formed by signals processed by the wavelet transform, in the wavelet domain.

Proposed Methodology

The system of analysis of structures integrity proposed in this paper consists of four main modules: data acquisition, wavelet decomposition, training and diagnosis of ARTMAP-*Fuzzy* neural network. The data acquisition module is composed of the experimental equipment to capture signals in the building structure, such as sensors, accelerometers, etc. The signals represent frequency response, resulting from the FFT (Fast Fourier Transform) obtained in the input module. Further, the signals are processed by Wavelet decomposition module at three levels of resolution. The set of signals obtained in the wavelet domain are used as inputs to the ARTMAP-Fuzzy neural network.

In the training module are configured weights and created the categories (knowledge) that will be used in the diagnosis of the data by ARTMAP-*Fuzzy* neural network. In this step the network is trained using a data set which generated a number of categories equals the number of input signals. This process is done *offline*. The diagnostic module a new set of signals different from the training set should be presented to the network, and each signal will be analyzed individually. All signals are compared with the knowledge (categories created in the training phase) obtained by ARTMAP-*Fuzzy* neural network and so when there is a matching criterion between the patterns are classified as category the displayed signal ARTMAP-*Fuzzy* network. Found category corresponding to a given pattern is, therefore, identified the fault in the structure. The flowchart of the operation of the method of analysis is shown in Figure 3.

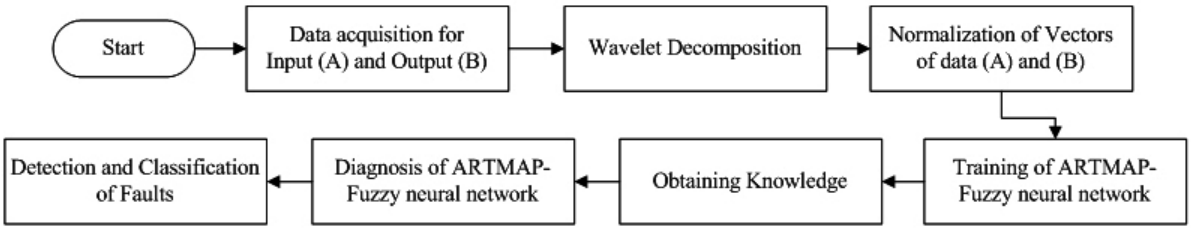


Fig. 3 Proposed Methodology

Applications and Results

In this section we present the results obtained with the application of ARTMAP-Fuzzy-Wavelet neural network in the database. The algorithm was developed in MATLAB® [9].

In order to evaluate the methodology analyzed the efficiency, accuracy and computational time in the analysis of failures in the structure. The parameters of the ARTMAP-Fuzzy-Wavelet neural network used in the process of training and diagnosis were $\alpha = 0{,}1$, $\beta = 0.9$, $\rho_a = 0{,}99$, $\rho_b = 1$ and $\rho_{ab} = 1$.

In the training phase we used a data set with 490 signs, 70 signs of the structure under normal conditions (base-line) and 420 signs of structure failure. This data set corresponds to 70% of the available data. To evaluate ARTMAP-Fuzzy-Wavelet neural network in the diagnosis phase was used a data set with 210 signals, 30 signals of the structure without fail and 180 signal failure of the structure. These data used in the tests represent 30% of the available data. The database simulated has six different patterns faults. The signals obtained by Wavelet decomposition represent the input signals to the training and diagnosis of ARTMAP-Fuzzy-Wavelet neural network. Table 1 shows the results obtained by the system failure diagnosis in structures when applied to the data set.

Table 1 Results obtained by the method

	Training Phase	Diagnostic Phase
Samples Used	490	210
Ratings Correct	490	210
Ratings Wrongs	0	0
Accuracy (%)	100,00	100,00
Time (*m*s)	783,7	186,34

To obtain these results, the system has undergone a testing phase and parameter settings. The results were obtained with the best configuration of ARTMAP-Fuzzy-Wavelet neural network. The result was obtained by cross-reference test, in which the system was run 15 times to ensure accuracy of results. We note that the analysis of the structural integrity shows a good performance, with a success rate of 100%. It is noteworthy that the wavelet module provided a more detailed analysis of the signals, so that the failures were easily identified.

Conclusion

In this article a new approach has been proposed in order to analyze the structural integrity of buildings, in which we used an ARTMAP-Fuzzy-Wavelet artificial neural network who presented excellent results, obtaining a hit rate of 100% for the best system configuration. The training phase is the one that requires more computational time; however is executed *off-line* causing no harm to the algorithm. Already the diagnostic phase of the system, from the acquisition of signals and wavelet processing is performed rapidly below 200 milliseconds, which entitles the system to be a tool used in real time. It is noteworthy that the wavelet module allows perform the analysis of the signals in a different domain, in which faults are easily identified due to the coefficients of detail of the multi-resolution analysis. Finally, it is concluded that the ARTMAP-Fuzzy-Wavelet artificial neural network proposed is very efficient, reliable and robust in the integrity analysis of building structures.

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Monitoring and Fault Identification in Aeronautical Structures Using an ARTMAP-Fuzzy-Wavelet Artificial Neural Network

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Keywords: Monitoring and Fault Identification, Aeronautical Structures, Intelligence Computation, ARTMAP-Fuzzy-Wavelet.

Abstract. This paper presents a methodology to perform the monitoring and identification of flaws in aircraft structures using an ARTMAP-Fuzzy-Wavelet artificial neural network**.** This technique is used in the detection and characterization of structural failure. The main application of this method is to assist in the inspection of aircraft structures in order to identify and characterize failures as well as decision-making, in order to avoid accidents or air crashes**.** In order to evaluate this method, the modeling and simulation of signals from a numerical model of an aluminum beam was performed. The results obtained by the method are satisfactory compared to literature.

Introduction

The monitoring and fault identification is a widely employed tool in the aircraft industry, with the aim of ensuring the integrity of structures and avoid disasters. A system for monitoring and fault identification (SHM) carries out tasks such as: acquisition and data processing, validation and analysis, detection, characterization and interpretation of adverse changes in a structure in order to assist decision-making in real time [7]. On aircraft, primarily commercial, structural failures can occur due of various factors, such as component wear, cracks, loosening of bolted joints, or simply the combination of these elements. Such failures in most cases, regardless of origin or intensity, cause a substantial change in the parameters of the structure, causing a reduced structural rigidity, and reduced mass also increased damping so that the dynamic behavior of the structure is changed [14].

Given this problem, several solutions have been proposed, such as ultrasonic inspection, X-radiography, acoustic emission testing, among other traditional techniques of SHM. Such techniques may even have good instrumental equipment and are well formulated, however, cannot meet increasing needs in industry, especially when the structures are in motion [5]. Therefore, to develop a solution for SHMS modern to structures dynamic such as aircraft is the use of smart materials, efficient techniques for processing data and signals, and the use of smart computing techniques.

In this regard, the literature, there are many works available that utilize smart materials and systems in SHM, which feature robustness, accuracy and good performance. Following we present the most relevant papers.

In [8], the authors presented the application of a genetic algorithm in conjunction with a multi-layer perceptron neural network with backpropagation to realize the detection and location of faults in a numerical model of a beam. In the work [6] we used the electromechanical impedance to monitor the aerospace structures coupled with active piezoelectric sensors. In [3] it proposed a fuzzy

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system to detect structural faults using curvatures of mode shapes of vibration. A system for identifying and locating damages in an aircraft wing utilizing a probabilistic neural network proposed in [13]. In [15] it was proposed the multimodal genetic algorithm to diagnose damages in a steel truss bridge. The work [10] proposed an ARTMAP-*Fuzzy* neural network for monitoring the integrity of mechanical structures (beams). In [9] it proposed an immune negative selection algorithm to diagnose faults in aircraft structures.

This paper presents a new approach for monitoring and fault identification in aircraft structures using an ARTMAP-Fuzzy-Wavelet neural network. From acquisition and processing of the signals, applies an ARTMAP-Fuzzy-Wavelet neural network to identify and characterize structural failure. To evaluate this method, we used a database of simulated signals from a numerical model of a beam of aluminum, which in this case is the structure of the wing of an aircraft. The beam was modeled by finite elements and simulated using Matlab [11]. The combination of wavelet transform with ARTMAP-Fuzzy neural network enables greater precision when diagnosing faults because the signals are decomposed into several levels of resolution, so that the signals are analyzed in detail, facilitating the identification of failures.

Artificial Neural Network Artmap-*Fuzzy*

The ARTMAP-*Fuzzy* artificial neural network, has an architecture based on adaptive resonance theory, ie, belongs to the family ART [1]. This network corresponds to a supervised learning system consists of a pair of modules of adaptive resonance theory, ART_a-*Fuzzy* and ART_b-*Fuzzy*, which receive and classify the inputs (I_a) and desired outputs (I_b) respectively, and the inter-ART associative memory module verifies the matching criterion of the inputs and outputs to the existing categories. Fig. 1 illustrates the architecture of ARTMAP-*Fuzzy* neural network [2].

The basic parameters of the ARTMAP-*Fuzzy* neural network are [2]:

- Parameter choice α ($\alpha > 0$): Operates in the category selection;
- Rate Training β ($\beta \in [0\ 1]$): Controls the speed of adaptation of the network;
- Parameter monitoring (ρ_a, ρ_b e $\rho_{ab} \in [0\ 1]$): Controls the resonant network, namely the parameter responsible for the number of categories created.

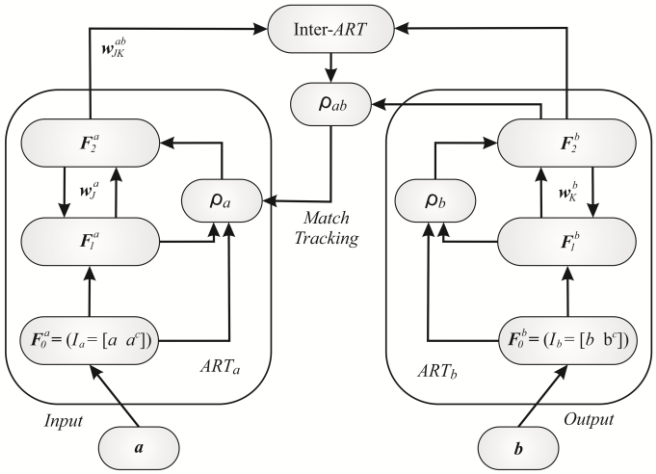

Fig. 1 ARTMAP-Fuzzy neural network [2].

Discrete Wavelet Transform

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where: J represents the resolution level, $N_j = (N/2) - 1$ represents the quantity of points in each new vector obtained by transformation, $\phi_{j,k}(t)$ and $\upsilon_{j,k}(t)$ are the wavelet and scale functions that execute the transformation; j is the scale (dilation) and k the position (translation).

The discrete wavelet transform (DWT) when applied directly to a signal to generate a set of coefficients is calculated by several entrances into a G filter (low pass) and H filter (high pass), or known as resolution levels. The filters G and H are vectors with constants already calculated that provide an orthogonal base related to the scale and wavelet functions respectively. This process IF known as Mallat Pyramidal algorithm [12] and is shown in Fig. 2 (a).

At Fig. 2 (a), C_0 corresponds to the original discrete signal ($C_0 = y[t]$), H and G represent the low pass and high pass filters respectively. The parameters d_1, d_2 and d_3 are the wavelet coefficients or detail in each resolution level and C_3 are the scale coefficients or approximation at the last level of the transform. These coefficients are obtained by convolution of the constants at filters (2) and (3), [12]:

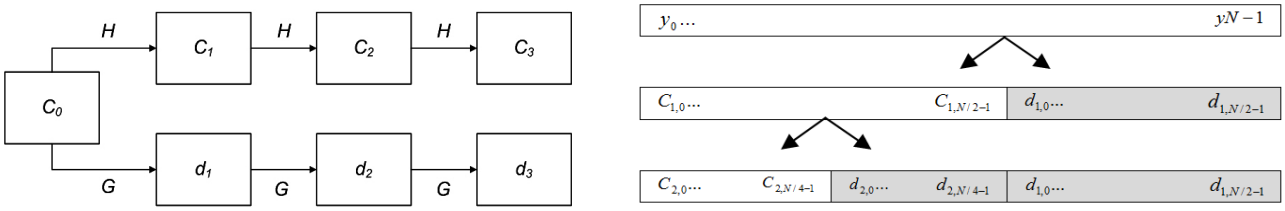
$$C_{j+1,k} = \sum_{l=0}^{D-1} h_l C_{j,2k+l} \quad (2)$$

$$d_{j+1,k} = \sum_{l=0}^{D-1} g_l C_{j,2k+l} \quad (3)$$

where: $k = [0, \ldots., (N/2^j) - 1]$ and D the quantity of constants of the filter. Thus, the coefficients $C_{j,k}$ represent the average local media and the wavelet coefficients $d_{j,k}$ represent the complementary information or the details that run away from the average media. Therefore, the transform coefficients ordered by scale (j) and position (k) are represented as follows [12]:

$$\psi = \left((C_{J,k})_{k=0}^{N_J}, \left((d_{j,k})_{k=0}^{N_J}\right)_{j=J}^{1}\right) \quad (4)$$

such that ψ is the finite representation in terms of the coefficients of the signal decomposition in equation (1). Fig. 1 (b) shows the decomposition process of a signal in two resolution levels. Observe that in each transformation level the size of the vectors is reduced by half ($N/2^J$).



(a) (b)

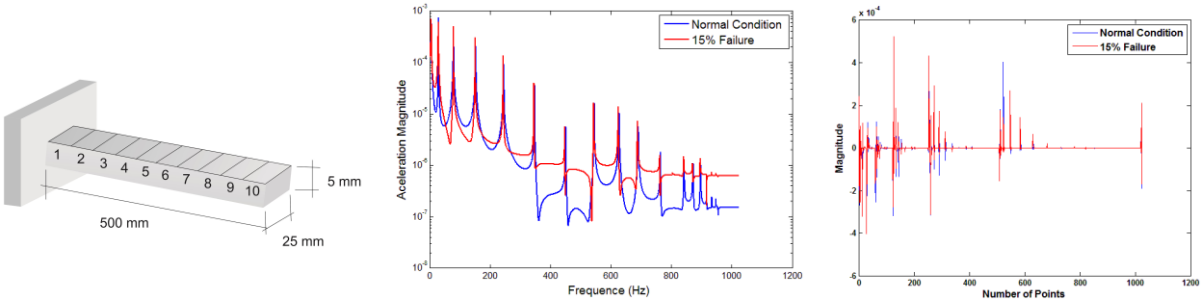
Fig. 2 Pyramidal algorithm for DWT [12]

Modeling and Simulations

The proposed model to evaluate the methodology, obtained by finite element method, was an aluminum beam in the cantilever-free condition discretized with 10 finite elements with 2 degrees of liberty each. The material properties used are the modulus of elasticity (E = 700 GPa) and the density (γ = 2710 kg/m^3). The dimensions are 500*mm* long, 25*mm* wide and 5*mm* thick. Fig. 3 (a) illustrates the patterned beam [9].

From the beam model were performed several simulations with different percentages of wear and locations of faults. The database consists generated signal captured by an accelerometer attached to the beam. In all simulations, the beam was excited in the 3rd degree of freedom (finite element 2) and the signal was captured on the 19th degree of freedom (finite element 10). Thus, were simulated 850 signals in the structure, 500 without wear (base-line condition) and 350 signs with wear (structural failure). The signals at failure were simulated in wear levels 1, 5, 10, 15, 20, 25 and 30%. For each

level of wear failure was placed in two locations (finite elements 3 and 5). Table 1 shows the number of simulations for each level of failure. Fig. 3 (b) presents two signals that had been captured in the simulations, the 15% failure and another under ordinary conditions. Following applies wavelet transform to obtain the signals shown in Fig. 3 (c). The data set is formed by signals processed by the wavelet transform, in the wavelet domain.



(a) (b) (C)

Fig. 3 (a) Beam modeled, (b) Simulated signal, (c) Wavelet domain.

Table 1 Settings of the simulations

Wear Level	0%	1%	5%	10%	15%	20%	25%	30%	Total
Number of simulations	500	50	50	50	50	50	50	50	850

Proposed Methodology

The system of monitoring and fault diagnosis is proposed in this paper consists of four main modules: data acquisition, the wavelet decomposition, training and diagnosis of ARTMAP-*Fuzzy* neural network.

The data acquisition module is composed of the experimental apparatus for capturing the signals on the mechanical structure, such as sensors, actuators, accelerometers, etc. The signals are representing the frequency response obtained by the data acquisition module.

Further, the signals are processed by Wavelet decomposition module at three levels of resolution. The set of signals obtained in the wavelet domain are used as inputs to the ARTMAP-*Fuzzy* neural network.

The module of training the ARTMAP-*Fuzzy* neural network is performed *off-line* and serves to adjust the weights and create categories (knowledge) to be used in diagnostic data. In this module, the network is trained using a data set is generated and a number of categories equals the number of input signals.

The diagnostic module is performed *online*. In this module, a new set of signals from different set of signals used in the training phase the network should be presented, and each signal will be analyzed individually. All signals are compared with the knowledge (categories created in the training phase) obtained by the ARTMAP-*Fuzzy* neural network-so when there is a marriage between the defaults is classified in the category signal presented to ARTMAP-*Fuzzy* network. Obtained the corresponding category a certain pattern is therefore identified intensity (percentage of failure) and the location of the fault in the structure. The flowchart of the functioning of the monitoring and fault diagnosis system is presented in Fig. 4.

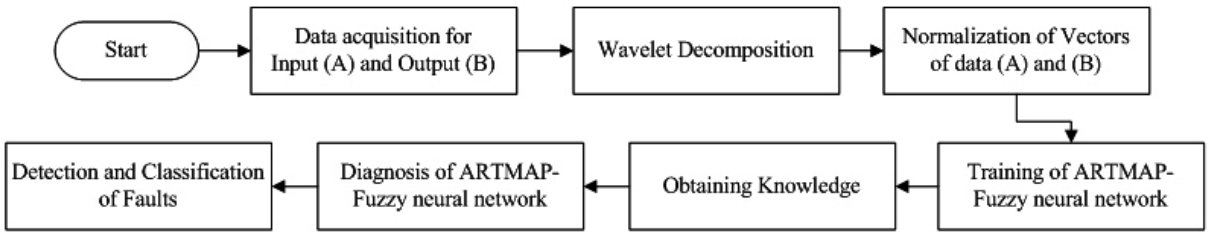


Fig. 4 Proposed Methodology

Applications and Results

In this section, we present the results obtained with the application of ARTMAP-Fuzzy-Wavelet neural network in the database. All tests were performed using a PC Intel Core 2 Duo 1.9 GHz, 2GB of RAM, operating system Windows 7 Ultimate 32 bits. The algorithm was developed in MATLAB® [11].

To test the ARTMAP-Fuzzy-Wavelet neural network was proposed analyzed the efficiency, accuracy and computation time in fault diagnosis process on the simulations performed. The parameters of the fuzzy-ARTMAP neural network-wavelet utilized in the training and diagnosis process were $\alpha = 0{,}2$, $\beta = 0.9$, $\rho_a = 0{,}8$, $\rho_b = 1$ and $\rho_{ab} = 1$.

In the training phase we used a data set with 595 signs, 350 signs of the structure under normal conditions (base-line) and 245 signs of structure failure. This data set corresponds to 70% of the available data. To evaluate ARTMAP-Fuzzy-Wavelet neural network in the diagnosis phase was used a data set with 2550 signals, 150 signals of the structure without fail and 105 signal failure of the structure. These data used in the tests represent 30% of the available data. The database simulated has seven different patterns faults. The signals obtained by Wavelet decomposition represent the input signals to the training and diagnosis of ARTMAP-Fuzzy-Wavelet neural network. Table 2 shows the results obtained by the system failure diagnosis in structures when applied to the data set.

Table 2 Results obtained by the method

	Training Phase	Diagnostic Phase
Samples Used	595	255
Ratings Correct	595	255
Ratings Wrongs	0	0
Accuracy (%)	100,00	100,00
Time (*m*s)	915,82	194,63

To obtain these results, the system has undergone a testing phase and parameter settings. The results were obtained with the best configuration of ARTMAP-Fuzzy-Wavelet neural network. The result was obtained by cross-reference test, in which the system was run 15 times to ensure accuracy of results. We note that the analysis of the structural integrity shows a good performance, with a success rate of 100%. It is noteworthy that the wavelet module provided a more detailed analysis of the signals, so that the failures were easily identified.

Conclusion

In this article, a new approach has been proposed in order to analyze the structural integrity of aeronautical structures, in which we used an ARTMAP-Fuzzy-Wavelet artificial neural network who presented excellent results, obtaining a hit rate of 100% for the best system configuration. The training phase is the one that requires more computational time; however is executed *off-line* causing no harm to the algorithm. Already the diagnostic phase of the system, from the acquisition of signals and wavelet processing is performed rapidly below 200 milliseconds, which entitles the system to be a tool used in real time. It is noteworthy that the wavelet module allows perform the analysis of the signals in a different domain, in which faults are easily identified due to the coefficients of detail of the multi-resolution analysis. Finally, it is concluded that the ARTMAP-Fuzzy-Wavelet artificial neural network proposed is very efficient, reliable and robust in the integrity analysis of aeronautical structures.

Acknowledgements

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Research Article

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Artificial Immune Systems with Negative Selection Applied To Clinical Diagnosis of Breast Cancer Samples

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Abstract

This paper uses an artificial immune systems with negative selection applied for diagnosing breast cancer samples. Taking as basis a biological process, the negative selection principle. This process is used to discriminate the samples, attaining a classification for benign or malignant cases. The main application of the method is assist professionals in the breast cancer diagnostic process, thereby providing decision-making agility, efficient treatment planning, reliability and the necessary intervention to save lives. To evaluate this method, the Wisconsin Breast Cancer Diagnosis database was used. This is an actual breast cancer database. The results obtained using the method (99.77% of accuracy in the better configuration of methodology), when compared with the specialized literature, show accuracy, robustness and efficiency in the breast cancer diagnostic process.

Keywords

Breast Cancer Samples; Clinical Diagnosis; Artificial Immune Systems; Negative Selection Algorithm.

Introduction

Cancer is a chronic disease that affects millions of people worldwide. In view of an aging population and longer life expectancy, higher rates of cancer incidences have been observed, especially breast cancer, which is one of the cancers with the highest incidence rates. This type of cancer occurs mainly in medium and low developing countries where the prevalence of cancer has become a public health problem, which requires greater vigilance, prevention behaviors, awareness policies, and effective treatments. According to the National Cancer Institute (INCA), a survey conducted in early 2014showed that breast cancer represents 22.5% of the cases affecting women in 2013, in other words, of all existing types of cancer, breast cancer is the one that affects women most [1].

In the Brazil, the breast cancer death rate is very high, probably because the disease is diagnosed at advanced stages. The statistics conducted by the World Health Organization (WHO) indicate that approximately 61% of the cases of women fighting breast cancer will result in death [2].

To reduce these rates, it is vital to conduct awareness and prevention campaigns, and especially conduct breast cancer diagnosis at an early stage. The correct diagnosis at an early stage of the disease enables quick planning decisions and evidently treatment efficiency. However, to properly diagnose cancer is a complex and difficult process because it involves many variables. The correct diagnosis requires a very experienced professional, and especially an accurate classification of the clinical stage of the tumor (cancer stage).

The traditional classification systems used are detailed and complex, and often difficult to use, posing limitations to the pathologists, and a slower decision making process [3]. Thus, it is vital to develop integrated systems that are capacitated to work with processing techniques and data analysis, and which, when combined with the professionals' experience, provide the necessary assistance to carry out the diagnosis and plan the treatment.

In this regard, the application of Artificial Intelligence (AI) could be a possible solution to the problems in the diagnosis process. The intelligent methods can extract information and knowledge from complex problems, and moreover they are easy to apply. There are several methods based on this concept, which are currently being used to assist professionals to perform the diagnosis of diseases, especially professionals with little experience. These methods provide security, reliability and agility in the diagnosis.

In the specialized literature some articles aimed at the diagnosis of breast cancer stand out. In reference [4], the authors present a method for diagnosing cancer samples using fuzzy logic and genetic algorithm. In the Neuro-Fuzzy Network (ANFIS) was used for the diagnosis of breast cancer; the results were considered satisfactory (96.30%) [5,6]. Article proposes a hybrid system which uses a neural network coupled to a specialized fuzzy system [7]. The system showed good performance when applied to the Wisconsin database [8].In a method using an ANFIS is presented for diagnosing breast cancer samples [9]. Polat K et al., states that an Artificial Immune Recognition System (AIRS) is presented to classify breast cancer samples [10]. This method is based on the immune system metaphors, and seeks to reproduce mechanisms inspired by immunology, such as competition for resources, clonal selection, affinity maturation and the formation of memory cells. The resource allocation mechanism is based on the fuzzy logic. Karabatak M, Ince MC and Avci E proposed a specialized system for diagnosing breast cancer [11].In reference12 the authors propose a new approach using a hierarchical fuzzy neural network for the diagnosis [12]. In reference 13 a TSK-type fuzzy model is presented for diagnosing breast cancer [13]. In reference 14 the authors submit a method based on the clonal selection algorithm, which adds a modification by using radial-based model of partial least squares to assist in the antibodies generation process [14]. This modification allows transferring the sample from an original stage to a final stage by a linear regression process. The key to a good classification refers to the width of the center of the radial base. The authors conducted a series of tests and obtained good results (99.58%).However, depending on the value given to the radial-base width, the classification process behaves in a disorderly manner.

AIRS is a promising technique in the field of AI. Its design is inspired by the Biological Immune Systems. It aims to computationally reproduce its main characteristics, properties and abilities [15].

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AIRS is a suitable tool for the diagnosis of diseases, a consequence of its ability to detect changes in behavior patterns [16]. AIRS allows including new disease patterns in the diagnostic process without having to reboot the memory system, that is, it enables continuous learning, which means that the system can be more efficient as new patterns are available.

This article deals with a method for diagnosing breast cancer based on artificial immune systems. From the samples of cancerous lesions, the negative selection algorithm (NSA) is implemented to differentiate between benign and non-benign samples (where there is evidence of malignancy). The samples classified as benign do not pose risks, i.e., they are not harmful to the organism. Those classified as non-benign are samples that need greater attention because they exhibit evidence of malignancy. The data analysis is performed by comparing the previously created detectors and samples, evaluating their affinity. If the affinity among the samples exceeds a predetermined threshold set by the professional, a match is found, and the sample is classified.

To evaluate the performance of the method, tests using a real database were performed, substantially explored in the literature. The database used was the Wisconsin Breast Cancer Diagnosis (WBCD) [8]. This method afforded satisfactory results, with a high generalization and reliability capacity, and low computational effort. When compared against the other articles in the specialized literature, the results showed that the method is better than the other techniques. The method presented proved to be accurate, with an average success rate of 99.77% in the diagnoses performed.

This article is organized as follows. Section 2 presents the database used in the tests. Section 3 presents the basic concepts of the biological immune system. The negative selection algorithm is discussed in Section 4. The description of the proposed methodology is shown in Section 5, 6 and 7 shows the results obtained in this article and also the conclusion, respectively.

Methods

Wisconsin Breast Cancer Diagnosis (WBCD) Database

The WBCD database was created by William H. Wolberg, a Doctor at the hospital of the University of Wisconsin-Madison, Wisconsin, United States [8].Between 1989 and 1991 Dr. Wolberg received several cases of breast tumors to be analyzed. In the analyses, the tumors were diagnosed as benign and malignant. Based on the referenced information, a database was assembled with 9 instances representing characteristics of the tumor and of course the classifications for these instances, which totaled 10 variables [17, 18].

The characteristics stored in the database are as follows:

- cell mass thickness (CT);
- cell size uniformity (CS);
- cell shape uniformity (CH);
- marginal accession (AD);
- epithelial cell size (EP);
- empty nucleus(BN);
- bland chromatin (CO);
- normal nucleolus (NN);
- mitosis (MM);
- classification (“benign” or “malignant”);

Table 1: Samples of Wisconsin Breast Cancer Diagnosis data [24].

ID	CT	CS	CH	AD	EP	BN	CO	NN	MM	CLASS
30	3	1	1	1	1	1	2	1	1	2
47	4	1	1	3	2	1	3	1	1	2
69	5	1	3	1	2	1	2	1	1	2
93	2	1	1	1	2	1	3	1	1	2
135	4	1	1	1	2	1	2	1	1	2
157	4	1	1	1	2	1	3	2	1	2
186	7	5	10	10	10	10	4	10	3	4
247	9	10	10	1	10	8	3	3	1	4
277	8	10	10	10	8	10	10	7	3	4
343	5	3	3	1	3	3	3	3	3	4
400	8	5	6	2	3	10	6	6	1	4
532	6	10	10	10	4	10	7	10	1	4

This database has 699 samples, of which 65% represent benign tumors and 35% represent malignant tumors [19]. The table shows a small sample of the data contained in this base. In this problem, class 2 corresponds to a normal pattern (“benign”) and class 4 corresponds to an abnormal pattern (“malign”) (Table 1).

Biological Immune System (BIS)

BIS is the body’s main defense against various infection agents that invade the biological body. In this case, the immune system must act instantly, effectively responding against the invading pathogens, identifying them in order to protect the human body from the eminent threat. There are two types of responses: the response by the innate immune system and the response by the adaptive system.

The innate immune system is the first line of defense with a very fast response, characterized by the phagocytic cells (granulocytes, macrophages, etc.),responsible for ingesting foreign particles in the organism, and other types of defenses such as physical (skin) and chemical barriers. The adaptive immune system is in the second level, which is capable of recognizing microorganisms such as viruses, bacteria, fungi, protozoa, helminthes and certain types of worms. The adaptive immune system is called immunological memory. It is capable of recording information from the infectious pathogens, at the initial detection, in order to expedite a response to this same type of agent that may reoccur in the future [16].

Basic Defense Mechanisms: The biological body, especially human, is protected by various cells and molecules that work in harmony, directing responses to foreign substances to the body, the so called antigens. Figure 1 shows this complex process in a simplified manner.

In step (I), the process is initiated when a pathogen (infectious agent) is ingested by a specialized antigen-presenting cell (APCs). At this stage, the pathogens are digested, fragmented in antigenic peptides. In step (II), fragments of peptides bind to MHC (major histo compatibility complex) and appear on the surface of the APC molecule. Next, in step (III), T cells, which have receptor molecules on their surface, are able to recognize different MCH/peptides antigens processed by the APC, that is, the recognition causes a status of activation. The third step is the self/nonself-discrimination carried out by the body, differentiating the cells of the infectious agents [15, 16].

Step (IV), is where the system is than active for having recognized an MHC/peptide antigen; the T cells divide and secrete

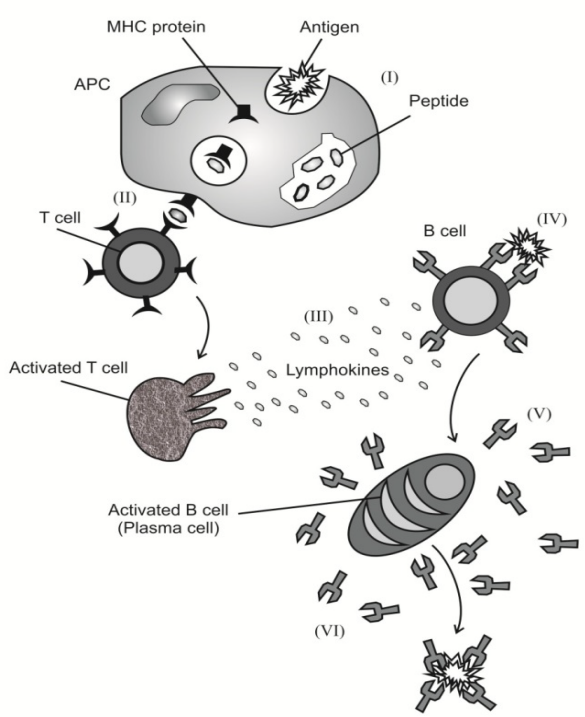

Figure 1: Simplified diagram of the recognition and activation mechanisms of BIS [16].

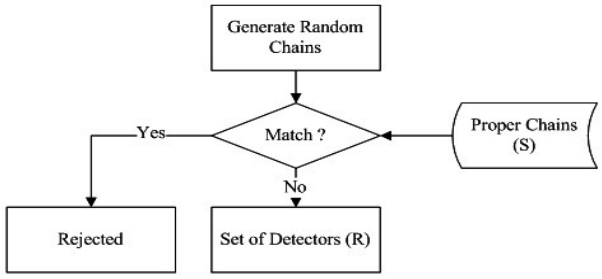

Figure 2: – Flowchart of NSA censor phase.

chemical signals (lymphokine), whose signal to other components of the immune system informs that an antigen was found. After this signaling, in step (V) the B cells which have unique receptor molecules specificity on their surface, are able to recognize the free antigens in the body, without the need for ingestion and digestion of the presenting cells (APC), and are thereby activated. When activated, the B cells, in step (VI), divide and turn into plasmocytes, which secrete high levels of antibodies.

In step (VII), these generated antibodies bond to the antigens found. Thus, the pathogen is neutralized, hence destroying the threat. Some of the T and B cells transform into memory cells, which remain circulating in the system and ensuring a quick and efficient response to a future exposure of the same type of infectious pathogen strain. It should be noted that the entire process is carried out with the cooperation of the set of forming cells of the immune system, each

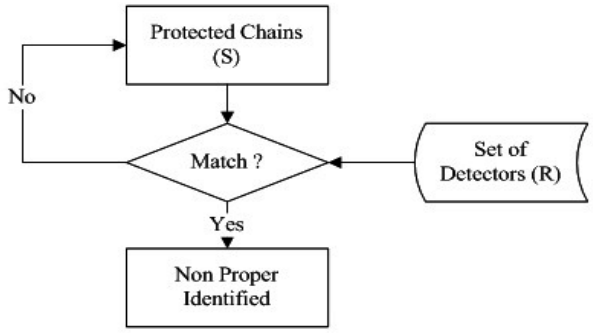

Figure 3: Flowchart of NSA Monitoring phase.

one responsible for a relatively simple function, and the set overall performs an extremely complex process [15, 16].

Negative Selection Algorithm (NSA)

NSA is a technique based on the pattern recognition process performed by the biological immune system, which is developed as a computational model. The ANS proposed by Forrest S, Perelson A, Allen L and Cherukuri R for detecting changes in the systems, is based on the negative selection of T cells in the thymus [20]. This process is the discrimination of the cells between self and non-self-performed by the organism. The algorithm is executed in two steps as shown below:

1. Censor

a) Define the set of chains (S) to be protected;

b) Generate random chains and evaluate the affinity (Match) between each of them and the chains themselves. If the affinity exceeds a threshold set, reject the chain. Otherwise, store them in a set of detectors (R).

2. Monitoring

a) Given the set of chains to be protected (protected chains), evaluate the affinity between each of them and the set of detectors. If the affinity exceeds a predetermined threshold, then a non-self-element is identified.

Figures 2 and 3 illustrate the flowcharts of the censor and monitoring phase of the negative selection algorithm (Figure 2 and 3).

In the censor step of NSA the detectors are initially defined, which represent a normal condition of the body, known as self-chains(S). The objective of this phase is to generate a set of detector patterns (R), with the ability to recognize any non-self-pattern, in the data monitoring stage. Then, with the reading of the data, the chains are randomly chosen and the affinity is verified by comparing these chains with the set of self-chains(S).Assuming that the affinity exceeds a predetermined threshold, the chain is rejected. Otherwise, this chain is accepted in the set of detectors(R), and will be used to conduct the classifications during the data monitoring. The detectors are similar to the matured T cells type capable of recognizing pathogens, which is to detect practically any not-self element, that is, a change or error in the data to be monitored [22].

In the monitoring phase, the data is monitored in order to identify changes in the behavior of the samples and then classify these

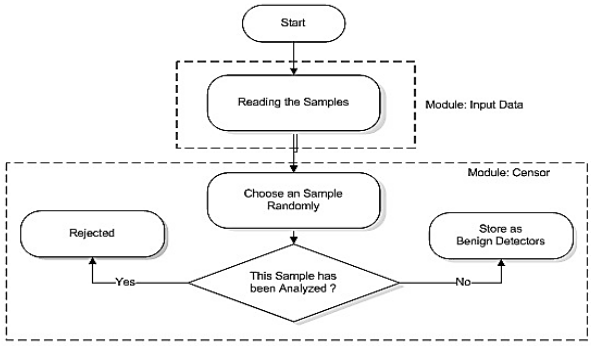

Figure 4: Flowchart of NSA Censor phase.

changes using the set of detectors created during the censor phase. Thus, by analyzing the protected chains (S) and comparing them with the set of detectors (R) the affinity between each of the chains is evaluated. If the affinity exceeds a certain threshold, then the non-self-element is detected and classified. It should be noted that the censor and monitoring phases are performed offline and in real-time, respectively [16, 22].

The result shown at the final execution of the NSA is a chain classification among the proper (normal) and non-proper (abnormal) answers.

Some observations can be made about the negative selection algorithm, highlighted as follows [23, 24]:

■ the size of the set of detectors does not necessarily increase with the number of chains to be protected;

■ the probability of detecting anomalies increases exponentially with the number of independent detection algorithms;

■ detection is symmetric;

■ There is an exponential cost to generate the detectors with regards to the size of the set of chains to be protected (self), because the random generation of chains of the R set results in there pleated generation of several chains;

Matching Criterion: To evaluate the affinity between the chains and state that they are similar, a criterion known as a match is used. The match can be perfect or partial [22].

The perfect match is when the two chains under analysis have the same values n all positions, that is, the two chains must necessarily be equal. However in the partial match there is no need for all positions of the chains to have the same value. In the partial match, only a number of positions between the chains must have the same value to affirm the match, with this amount being previously set. This quantity is known as the rate of affinity.

In this article, we chose to use the partial match proposed in reference 25, in which the rate of affinity represents the degree of similarity that must occur between the two chains under analysis for the match to be confirmed [25].

The rate of affinity is defined by the following equation [25]:

$$Af = \left(\frac{An}{At}\right) * 100 \quad (1)$$

Where:

Af: Affinity rate;

An : Number of normal samples;

At : Total number of samples.

Proposed Method

The breast cancer diagnostic system presented in this section is based on the artificial immune systems, especially in the negative selection algorithm, which was presented in section 3 of this article.

The negative selection algorithm is a computational method that seeks to reproduce the negative selection process performed by matured T cells within the human body. In summary, this process performs a self/nonself-discrimination, which represents the main diagnosis mechanism of the human body, which is responsible for identifying unknown pathogens by the immune system (viruses, bacteria, fungi, etc.).

Based on this algorithm, the diagnostic system proposed in this paper performs the self/non-self-discrimination, a benign and non-self-tumor in which no evidence of malignancy was identified.

The proposed method consists of two phases: censor and data monitoring. In the censor phase, a data census is carried out in order to transfer knowledge, creating chains of detectors to identify abnormalities (non-self) in the monitoring process. In the monitoring phase the data is analyzed and compared with the detectors created during the censor phase, in order to present a diagnosis through the self/non-self-discrimination. The following are the censor and monitoring stages of the breast cancer diagnosis system.

Censor: At this stage, the detectors are generated to be used by the AIRS during the monitoring process. Due to the fact that the problem in question has two classes of patterns, the algorithm only needs to have knowledge of the normal samples (self), so that based on this information the discrimination is performed for what is self (benign) and what is non-self (malignant). Thus, the censor phase for this system has a single step. In this step the self-pattern is defined (benign) through the routine shown in the figure, which illustrates how the process to generate the self-detectors is performed (Figure 4).

In this article we use the partial match criterion proposed by Bradley DW and Tyrrell AM [25].The rate of affinity is defined in equation (1).Using this equation, the affinity rate is estimated for this problem. This calculation is defined using equation (2), which has a total of 699 samples, of which 444 samples are normal, that is, without cancer:

$$Af = \left(\frac{444}{699}\right) * 100 = 63.52\% \quad (2)$$

The affinity ratio is of 63.52%, and that means that to confirm a match between two cancer samples there must be an affinity/similarity of at least 63.52% of the samples. The rate of casual affinity has a fixed value based on the calculation shown in equation (2), that is; it is based on the statistical sample. However, in this article the results with other rates of affinity values will be presented to provide a better assessment regarding the quality and performance of the system in the diagnostic process.

At the final of the censor phase, the NSA obtains the proper detector set as a result, which is going to be used at the monitoring phase of the samples, and therefore realizing the detection of the

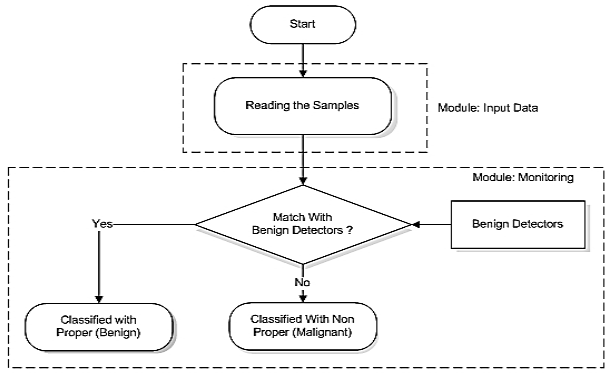

Figure 5: Flowchart of NSA Monitoring Phase.

abnormalities (malignancy samples).

Monitoring: The monitoring phase is divided into two modules, which are responsible for the reading of the data and the self and nonself-discrimination. Figure 5 illustrates the flowcharts of the data monitoring phase.

At this stage, the reading of the samples to be analyzed is performed. These samples are compared to the previously assigned benign detectors (self) in the censor phase. With the comparison of the sample using the set of detectors, a match between the sample and the set of detectors is verified. If the rate of affinity is satisfied, or if there is similarity between the samples analyzed and considered a match, then the sample is diagnosed as "benign" for having its own characteristics. Otherwise, there is no match among the samples being analyzed, thus the sample is diagnosed as "malignant" for not having characteristics known by the system, in other words, the sample is unknown. This process is repeated for each sample, and that is how the data monitoring is performed.

At the monitoring phase, the NSA obtains as a result the classification (diagnosis) of the analyzed cancer samples. The system, can present two types of diagnosis after realizing the monitoring: the classification as proper (the sample is classified as benign) and the classification as non-proper (the sample classified as malignancy).

Results

In this section we present the results obtained with the proposed method implemented based on the tests. All tests were performed using a PC Intel Core 2 Duo 1.9 GHz, 2 GB of RAM, and operating system Windows 7 Ultimate 32-bit. The algorithm was developed in MATLAB [26].

The test base used to demonstrate the performance of the proposed algorithm was the Wisconsin Breast Cancer Diagnosis Database presented in section 2 [8].

As previously reported, the WBCD base has 10 attributes. However, an improved amount of attributes to perform the diagnosis was used in this article. To choose the attributes to be used, a selection process based on the calculation of the standard deviation of the samples was done. The five attributes that showed the lowest levels of standard deviation were chosen. When the standard deviation is low, it means that the data are more homogeneous. When the standard deviation is high, it means that the data is varied. This choice was used in order to provide the system greater reliability. The standard deviation was calculated by the following equation:

$$S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

The variables chosen that exhibit the lowest standard deviation are: the cell mass thickness (CT), cell size uniformity(CS), the cell shape uniformity (CH), the empty nucleolus (BN), and normal nucleolus (NN).During the separation of these variables it was observed that some data are not usable. The table presents the data and characteristics of the WBCD database (Table 2).

Table 2: Data on the WBCD base.

Database	UCI Wisconsin Breast Cancer Data
Type	Classification
Number of Data	699
Number of usable data	683
"benign" class data	444
"malignant" class data	239
Number of Attributes	10

Table 3: Results for Test I.

Diagnostic	Samples tested	Benign	Malign	Execution time (*ms*)	Accuracy (%)	Error (%)
Detector set I	683	413	270	190,00	93,01%	6,99%
Detector set II	683	434	249	70,00	97,74%	2,26%
Detector set III	683	443	240	50,00	99,77%	0,23%

According to what is presented at table, the data base of WBCD contains 699 samples, which 459 are benign and 240 are malignancy, however only 683 are useful, resulting in 444 benign and 239 malignancy (Table 2).

Several tests were carried out to evaluate the performance of the diagnostic system. Next, the tests and their results are described.

Test I

In test I, the goal is to execute the proposed method with unmodified normal parameters and settings.

For test I, the affinity rate with a fixed value calculated in the equation 2 was used, which is of 63.52%. For this test, in the censor stage, three sets of detector patterns were generated and the test was performed on each one of them. Sets I, II and III have 5, 10 and 20 detector patterns, respectively. The detectors generated use 1.12%, 2.25% and 4.50% of the benign samples, which have a total of 444 samples. The below table shows the results for this test (Table 3).

Test I did not show good performance (accuracy level above 93%) of the diagnostic system, or that the number of benign detectors directly influences the final diagnosis. Up to 30% of the information in the database is commonly used to generate detectors, and in this case the proposal is to use up to 5% in order to provide robustness to the diagnosis. As the number of detectors increases, the diagnosis becomes more accurate, and this is because the amount of knowledge provides higher efficiency to the diagnosis.

Test II

In test II, the objective is to verify the sensitivity of the proposed method with modifications in the rate of affinity.

For test II, four affinity rates were used, with values of 40%, 60%,

Table 4: Results for Test II.

Diagnostic	Samples tested	Benign	Malign	Execution time (millisecond)	Accuracy (%)	Error (%)
Rate I	683	509	174	60,00	85,37%	14,63%
Rate II	683	444	239	70,00	100,00%	0,00%
Rate III	683	396	287	65,00	89,18%	10,82%
Rate IV	683	46	637	50,00	10,36%	89,64%

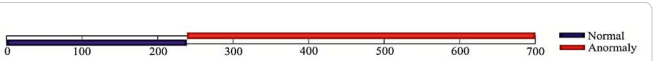

Figure 6: Data arrangement of test I.

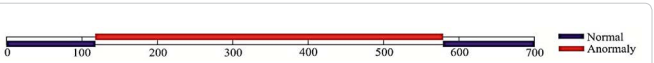

Figure 7: Data arrangement of test II.

80% and 90%. For this test, the set of pattern detectors I was used, illustrated in Test I. The below table shows the results for this test (Table 4).

Test II shows that the rate of affinity of the samples directly influences the final diagnosis. A statistical calculation is performed using equation 1 by relating the number of normal and total samples of the database. When a variation is performed to a lower rate of affinity, a very good performance is observed, which may result in 100% of accuracy. However, as the rate of affinity decreases, the match criterion is less precise, resulting in non-self-samples to be classified as self. However, when a higher rate of affinity is executed, the performance greatly worsens, and this is because the match criterion is too rigorous, in order to validate samples that have almost a perfect match, which rarely happens. Given the reduced size of the set of detectors, many wrong diagnoses can occur. To solve this problem a greater amount of benign detectors would have to be used, more than 30% of the information in the database.

It should be noted that the rate of affinity must be precisely calculated for the system to perform well in the diagnostic process, and is commonly calculated as expressed in equation 1. But this requirement is not a prerequisite during the calculation, there is no need to use equation 1 to calculate the affinity rate. This rate can change the problem is to find the value that provides accuracy to the diagnostic.

Test III

In test III, the objective is to verify the sensitivity of the proposed method in relation to the position of the samples in the set of tests.

For test III, the configuration that showed the best performance for safety and reliability during the diagnosis was used. The rate of affinity is set to a value of 63.52% and a set of detectors with 20 samples is used. Three sets of tests were prepared, and in the first one all the normal data is grouped, followed by the abnormal data. In the second test the normal data are grouped interspersed with the abnormal data, and in the third test the data are randomly grouped. The below figures illustrate the data arrangement for tests 1 and 2 (Figure 6, 7).

The below table shows the results for these tests (Table 5)

Test III shows that the diagnostic system is very sensitive regarding the data arrangement. This does not affect its performance.

Table 5: Results for Test III.

Diagnostic	Samples tested	Benign	Malign	Execution time (*ms*)	Accuracy (%)	Error (%)
Test 1	683	443	240	70,00	99,77%	0,23%
Test 2	683	440	243	75,00	99,09%	0,91%
Random	683	441	242	50,00	99,32%	0,68%

Table 6: Comparative study.

Reference	Database	Technique used	Accuracy (%)
[27]	WBCD	*Back propagation*	95,16%
[5]	WBCD	*ANFIS*	96,30%
[28]	WBCD	*Kohonen*	96,70%
[7]	WBCD	*Fuzzy*	96,71%
[4]	WBCD	*Fuzzy-genetic*	97,07%
[7]	WBCD	*ILFN and Fuzzy*	98,13%
[10]	WBCD	*SAI-Fuzzy*	98,51%
[14]	WBCD	*AIRS and radial basis network*	99,58%
This paper	WBCD	*Negative Selection Algorithm*	99,77%

Analyses of Results

The results obtained for the tests performed in this article are satisfactory (greater accuracy rate) and prove that the negative selection algorithm is effective in the diagnostic process. The parameters used, as well as the number of detectors, directly influence its performance. Thus, a trial test phase should be performed to find the correct parameters. It should be highlighted that for this article only 5% of the information samples was used to generate the benign detectors, which were incidentally generated with 30% of the information. Of a total of 444, the benign samples were randomly chosen, even samples to be defined as detectors, which is a very small amount of information. This shows that the method is robust and effective in the diagnostic process. The execution time is relatively small, which provides a rapid diagnosis.

The table 6 presents a comparative study between the proposed method and the main methods available in the specialized literature (Table 6).

The table 6 shows that the proposed method, in this work, presents a higher success rate than other techniques (Table 6).

The proposed method presents the best results, allowing the use to solve this kind of problem (Table 6). This way, some advantages are presented as follows showing that this method is better than other techniques, such as:

- The method does not present a training phase where is necessary to reinitialize the learning during the execution of the algorithm;
- The method uses only 5% of the available information to generate the set of detectors;
- The processing time is very low;
- The method allows including a continuous learning module, making possible to the system acquires knowledge during long time improving and learning new patterns not known.

Conclusion

This paper presents a breast cancer diagnostic method based on artificial immune systems, in particular the negative selection algorithm. The main stages and characteristics of the NSA and its application in the proposed problem were described. As the system's

input data, the algorithm needs only five attributes of breast cancer samples. The proposed algorithm showed excellent results, achieving a success rate of 99.77% of accuracy in all samples tested.

The detector generation phase is the one that demands greater computational time, but as it is executed off-line, it does not compromise the algorithm. The monitoring phase of the system, with the data reading, is performed rapidly with a time less than 100 milliseconds, which accredits the algorithm as a tool used in real time, given the urgency for quick decision making.

It is observed that with a minimal amount of information, that is, with a greatly reduced set of detectors, the method was able to perform a diagnosis with considerable accuracy and safety. This emphasizes that the method is robust and efficient in diagnosis using the minimal knowledge.

Therefore, it is concluded that artificial immune systems, based on the negative selection algorithm, obtained satisfactory performance in the tests conducted, and the method proved to be quite reliable, secure and robust for breast cancer diagnosis. Following this study, the focus will be on other strategies of immune systems, in order to render the application more competitive (efficiency, reliability and with greatly reduced processing time, among other requirements).

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Artificial Immune Systems with Negative Selection Applied to Health Monitoring of Aeronautical Structures

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Keywords: Health Monitoring; Aeronautical structures; Artificial Immune Systems; Negative Selection Algorithm.

Abstract. In this paper we present a system for aircraft structural health monitoring based on artificial immune systems with negative selection. Inspired by a biological process, the principle of discrimination proper/non-proper, identifies and characterizes the signs of structural failure. The main application of this method is to assist in the inspection of aircraft structures, to detect and characterize flaws and decision making in order to avoid disasters. We proposed a model of an aluminum beam to perform the tests of the method. The results obtained by this method are excellent, showing robustness and accuracy.

Introduction

In recent years the aerospace industry began to make various investments in monitoring systems for aircraft structural integrity in order to ensure safety and economy. In general, System Health Monitoring of Structures (SHM) has the ability to perform data acquisition and processing, validation and analysis, in order to identify, characterize and interpret adverse changes in a structure, with the goal of assist decision making in real time [8].

Structural flaws in aircraft may occur as a result of various factors, such as component wear, cracks, loosening of bolted connections, or simply the combination of these elements. In most cases, regardless of origin or intensity, structural failure causes a sensitive variation in the spatial parameters of the structure, providing structural stiffness reduction, mass reduction and also an increased damping, modifying the dynamic behavior of the structure. The spatial variation of the parameters affects the main dynamic parameters, such as response functions, resonance frequencies, damping ratio and eigenmodes of the structure [14].

All aeronautical structure before being put to use, goes through an evaluation and testing of structural integrity. In this case are evaluated nondestructive (NDE), in order to estimate the degree of safety and reliability of the structure. There are several techniques to make assessments and non-destructive estimate the degree of safety and reliability of the structure, such as ultrasonic inspection, X-radiography, acoustic emission testing, among other traditional techniques. Such techniques have a good instrumental implementation and are well formulated for integrity assessment; however, these techniques fail to fully satisfy the monitoring of structural integrity, further being in motion [6]. Thus, the SHM see having a natural evolution due to advancement of technologies, sensors, smart materials and processing data signals as well as the increasing demand for real-time structures, such as aircraft. The use of smart sensors in conjunction with intelligent

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computing techniques become possible alternatives for developing SHM in dynamic structures such as aircraft, in order to perform analysis and decision making.

In the current literature are available many works that use materials and intelligent systems in SHM, which feature robustness, accuracy and good performance. Following we present the relevant work.

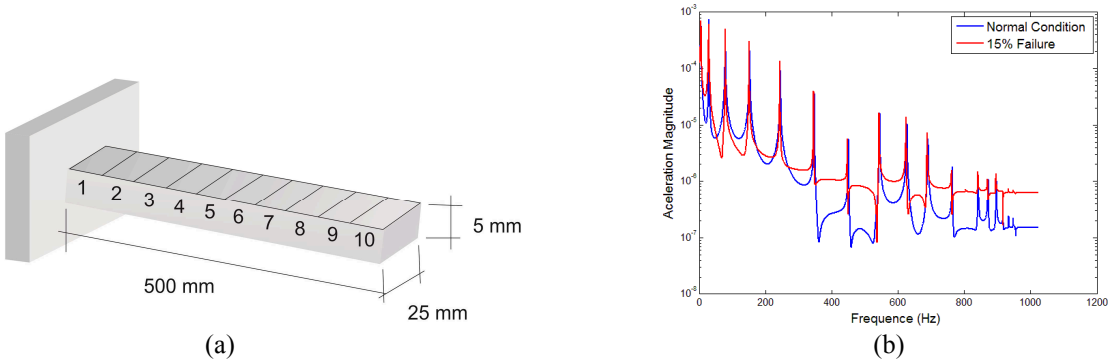
In [9], the authors describe the use of a genetic algorithm in conjunction with a multilayer neural network to perform the detection and localization of faults in a numerical model of a beam. In work [14] the authors used a multilayer neural network trained via backpropagation algorithm for detection and characterization of flaws in beams CFRP (carbon fiber reinforced). In [7] was used in the method of the electromechanical impedance for monitoring aerospace structures coupled with active piezoelectric sensors. In [2] propose a fuzzy system for detecting structural faults using curvatures of mode shapes. In the work [13] proposed a model using wavelet transform to assess structural integrity of bridges across the vibration signals. A system for identifying and locating damage in an airplane wing was proposed in [12]. In [15] proposed a genetic algorithm multinodal to diagnose damage in a steel truss bridge.

This work, we present a SHM based on artificial immune systems to monitor the health of aeronautical structures. From acquisition and signal processing, it applies the negative selection algorithm (NSA) [5] to identify and characterize the flaws. In order to evaluate the SHM aeronautical used a data base of simulated signals from a numerical model of a beam of aluminum, which in this case represents the structure of the wing of an aircraft. The beam was modeled using the finite element method and simulated in Matlab® [11]. In this work we chose to use the artificial immune systems (AIS) because of its natural learning and pattern recognition and present a good performance and efficiency in other types of problems, as shown in [10].

This article is organized as follows: Section 2 presents the modeling and simulations. The description of negative selection algorithm (NSA) is in section 3. Section 4 presents the proposed methodology. The applications and results are presented in Section 5 and finally, Section 6 concludes this work.

Modeling and Simulations

The model of the aluminum beam was developed by the finite element method in the cantilever-free condition discretized with 10 finite elements with 2 degrees of liberty each. The material properties used are the modulus of elasticity (E = 700 GPa) and the density ($\gamma = 2710$ kg/m^3). The dimensions are 500*mm* long, 25*mm* wide and 5*mm* thick. Fig. 1 (a) illustrates the patterned beam.



(a) (b)

Fig. 1 (a) Beam modeled, (b) Simulated signal.

Through beam modeled by finite element analysis was performed several simulations with different percentages of wear and locations of failures. The database is formed that was generated signal captured by an accelerometer attached to the beam. In all simulations the beam was excited in the 3rd degree of liberty and the signal was captured on the 19th degree of liberty. Thus, were simulated 1500 signals in the structure, 400 signals being in normal condition, i.e. without wear (base-line condition) and 1100 signs with wear (structural failure). The signals at failure were

simulated in wear levels 1, 5, 10, 15, 20, 25, 30, 35, 40, 45 and 50%. At each level of wear we stand failure in all finite elements. Fig. 1 (b) presents two signals that had been captured in the simulations, a 15% failure and another under normal conditions.

Table 1 (a) shows the number of simulations for each wear level. Table 1 (b) shows the number of simulations for each location of the failure in each wear levels, that is, for each level of failure ten simulations were generated at each location.

Table 1 – Settings of the simulations.

Wear Level	Number of simulations
0%	400
1%	100
5%	100
10%	100
15%	100
20%	100
25%	100
30%	100
35%	100
40%	100
45%	100
50%	100
Total	1500

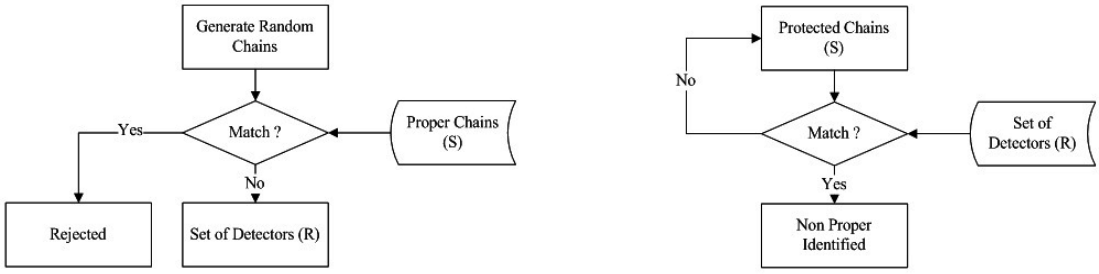
(a)

Location	Number of simulations
1	10
2	10
3	10
4	10
5	10
6	10
7	10
8	10
9	10
10	10
Total	100

(b)

Negative Selection Algorithm (NSA)

The negative selection algorithm (NSA) was proposed by [5] to detect changes in system state, and is based on negative selection of T lymphocytes in the thymus. This process works with discrimination proper cells and non-proper. The algorithm runs in two phases, as illustrated in the flowcharts in Fig. 2 [4]:



(a) (b)

Fig. 2 Flowcharts of NSA.

The censor phase of the ASN basically consists in generating a set of detectors. The detectors are similar to the maturated T cells that can recognise pathogenic agents. The monitoring phase consists in monitoring a system to identify a change in behavior, thus, this phase classifies the change using the detector set that was created in the censor phase [3].

Matching Criterion

The matching criterion is used to determine the affinity between the chains and affirm that are similar. The matching can be perfect or partial [10]. In this paper we use the partial matching. This criterion, a number of points between patterns must have the same value to confirm the matching, being the value (affinity rate) previously defined. The affinity rate represents the degree of similarity required for the matching occurs between the two patterns. The affinity rate is defined by the following relationship [1]:

$$TAf = \left(\frac{An}{At}\right) * 100 \tag{1}$$

being:

TAf : afinity rate;

An : quantity of chains in normal problem (chains propers);

At : total number of chains in problem (chains propers and non-propers).

The equation (1) proposes a statistical relationship between all samples of the problem to the calculation of the affinity rate. In (2) represents the method to quantify the total affinity with analysed patterns [10]:

$$Af_T = \frac{\sum_{i=1}^{L} Vc}{L} * 100 \tag{2}$$

where:

Af_T :% affinity with patterns analyzed;

L : total quantity of variables;

Vc : matched variables;

$\sum_{i=1}^{L} Vc$:sum (quantity) of the matched variables.

If $Af_T \geq TAf$, the matching with the patterns occurs (are considered equal/similar).

Proposed Methodology

The system health monitoring of aircraft structures presented in this section is based on artificial immune systems [4], especially in the negative selection algorithm [5]. The proposed method is divided into three modules: data acquisition, censor phase and monitoring phase of NSA.

Following shows the monitoring system of structural integrity.

Structural Integrity Monitoring System (SIMS)

The SIMS is consists of three main modules, with data acquisition, censor and monitoring. The flowchart of the operation of SIMS is given in Fig. 3.

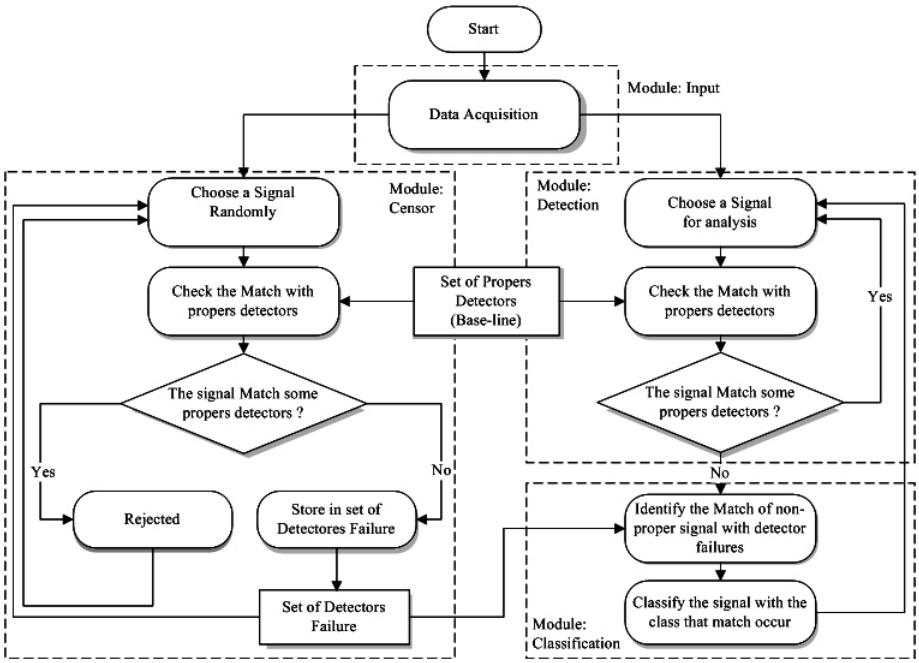


Fig. 3 Flowcharts SIMS.

The data acquisition module comprises the experimental part for capturing the signals in the structure as sensors, actuators, accelerometers, piezoelectric materials, etc.

The sensing module is carried *off-line* and used to generate the set of failure detectors that will be used in the monitoring data. In this module, the fault detector set is comprised of signals of different levels of failure structure at different locations, which are randomly selected.

The monitoring module is composed of two stages: detection and classification of failure. The signs at failure are detected by evaluating and verifying matching criterion to the detector set (base-line signals), so does the detection of signals in flaws discrimination based on propoer/non-proper. If the signal is proper is classified the normal condition of the structure. If the signal is a signal non-proper (signal failure) makes an evaluation of this signal compared to the set of failure detectors, thus if there is a match of the signal with a class of detectors, the signal is classified as a class of detector activated. The monitoring of the signals is executed *online*.

Applications and Results

In this section we present the results obtained with the application of the SIMS data base simulated. All tests were performed using a PC Intel Core 2 Duo 1.9 GHz, 2 GB of RAM, operating system Windows 7 Ultimate 32 bits. The algorithm was developed in Matlab® [11].

In tests in order to verify the proposed methodology, checking the efficiency, accuracy and computation time for different configurations of the detector set, ie, information that is used as knowledge. Was used as a parameter a afinity rate with a fixed value of 73.33% (calculated with equation (1)). We generated three sets of detectors (I, II and III) using 10%, 20% and 30% of standard samples each, totaling 110, 220 and 330 detectors. So each type were defined pattern 1, 2 and 3 detectors for each set. The simulated database includes 110 patterns of failure.

In Table 2 presents the results obtained by the system health monitoring of structures when applied to data of the aluminum beam.

Table 2 - Results obtained by the method.

Diagnosis	Set of detectors		
	I	*II*	*III*
Samples tested	1500	1500	1500
Ratings correct	1482	1495	1500
Ratings wrongs	18	5	0
Accuracy (%)	98,88	99,66%	100%
Time (ms)	160,5	163,4	162,8

The values in Table 2 represent the results of the test performed. Aiming to ensure the accuracy of the results is executed the system (twenty times) for each set of detectors is possible to see that the system of monitoring the integrity of structures presents a good performance (success accuracy of 100% in the best configuration) and the number of detectors influence the recognition of failure. It is suggested to use 30% of the database information to generate the set of detectors in order to provide robustness to the system. The more knowledge, the more efficient the process of recognition and classification of SIMS.

Conclusions

In this paper we propose a new approach to develop a SIMS. In this context has employed an artificial immune algorithm of negative selection that showed excellent results, obtaining a success accuracy of 100% for the best system configuration. The censor phase is that demand greater computational time, however is performed off-line is not causing harm to the algorithm. The phase monitoring system, from the signal acquisition is performed rapidly with time less than 170 milliseconds, which makes efficient SIMS a tool to be used in real time. Thus, it is concluded that the proposed SIMS based on artificial immune systems is very efficient, reliable, robust and accurate to perform the monitoring of the integrity of aircraft structures. This work contributes to the research area in SIMS, presenting a new method to monitor aircraft structures using intelligent techniques.

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Analysis of Structural Integrity Using an ARTMAP-Fuzzy Artificial Neural Network

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Keywords: Analysis of Structural Integrity, Building Structure, Artificial Neural Networks, Adaptive Resonance Theory.

Abstract. This paper presents the application of artificial neural networks in the analysis of the structural integrity of a building. The main objective is to apply an artificial neural network based on adaptive resonance theory, called ARTMAP-*Fuzzy* neural network and apply it to the identification and characterization of structural failure. This methodology can help professionals in the inspection of structures, to identify and characterize flaws in order to conduct preventative maintenance to ensure the integrity of the structure and decision-making. In order to validate the methodology was modeled a building of two walk, and from this model were simulated various situations (base-line condition and improper conditions), resulting in a database of signs, which were used as input data for ARTMAP-*Fuzzy* network. The results show efficiency, robustness and accuracy.

Introduction

Despite the technological advances in the development of metallic materials and construction, one can observe a large number of new buildings had shortcomings. The inappropriate use of materials, together with the lack of care in implementing adaptations when using them, all this added to the lack of maintenance, has created extra expenses to condominiums up less than five years old have to consume financial resources in repairs failures could be avoided [5].

In literature there are some studies which use methodologies for performing analysis of the structural integrity of buildings or constructions, as in [5], where the authors present a methodology for structural analysis using a non-destructive testing (NDT). In [6] it is a structural analysis of buildings after the collapses of the metal columns.

This paper presents a method for analysis and monitoring of the structural integrity of a building using an ARTMAP-Fuzzy artificial neural network. From the acquisition and signal processing, we apply an ARTMAP-Fuzzy neural network [1] to identify and characterize the flaws. In order to evaluate this method, we used a database of simulated signals from a numerical model of a two-story building, which in this case represents the structure. The building was modeled by differential equations, and was simulated by Matlab® software [3, 4].

The ARTMAP-Fuzzy neural network has a stable architecture and plastic, which ensures an advantage over other types of artificial neural networks. The ARTMAP-Fuzzy neural network allows the inclusion of a training module continued, which enables the extraction of knowledge without the need to restart the training process when new patterns are presented, unlike what happens with most neural networks. This feature makes it possible to use a reduced set of patterns in the training phase, during the analysis (application), the extraction of knowledge is continuous, ie, it is a system that seeks to improve over time [1].

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ARTMAP-*Fuzzy* Artificial Neural Network

The ARTMAP-*Fuzzy* artificial neural network corresponds to a supervised learning system comprised of a pair of modules Adaptive Resonance Theory [2], ART_a-*Fuzzy* and ART_b-*Fuzzy*, interconnected by inter-*ART* associative memory module. This neural network architecture incorporates fuzzy set theory, the AND *fuzzy* operator (^) enabling the learning of the neural system in response to binary input patterns and analog, belonging to the interval [0 1], [1].

A internal mechanism called *match-tracking* is responsible for self-regulating process of the neural network, in which maximize generalization and minimize the error. When the neural network makes a prediction wrong, through an associative connection instructed, the monitoring parameter of ART_a-*Fuzzy* module is incremented in minimum amount necessary to correct the error in the ART_b-*Fuzzy* module.

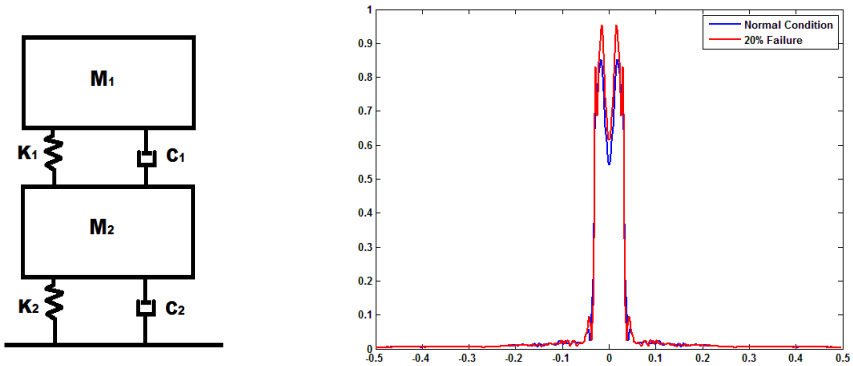
The ARTMAP-*Fuzzy* architecture has three main parameters for the development, namely [1]:

- Parameter choice α ($\alpha > 0$): Operates in the category selection;
- Rate Training β ($\beta \in [0\ 1]$): Controls the speed of adaptation of the network;
- Parameter monitoring (ρ_a, ρ_b e $\rho_{ab} \in [0\ 1]$): Controls the resonant network, namely the parameter responsible for the number of categories created.

IF ρ has a large value, the neural network becomes more selective reducing its generalizability. If ρ has a small value, it reduces the number of categories created, increasing the generalization capability of the ARTMAP-*Fuzzy* network.

Modeling and Simulations

In this section a mathematical model was developed using differential equations of a two storey building with two degrees of freedom each. The building is adimensional model and consists of two masses (M_1 e M_2), and the stiffness coefficients (k_1 e k_2) and elasticity (c_1 e c_2), see [3]. The model is illustrated in Figure 1 (a).



(a) (b)

Fig. 1 – (a) Structure model, (b) Frequency response.

We simulated 700 signs in the structure, being 100 signs in normal condition, i.e., without flaw (condition base-line) and 600 signals with structural flaws, being 100 signs for each type of failure. The signals faults were simulated with additions of 5%, 10%, 15%, 20%, 25% and 30% in mass M_1 of the structure. The structure was excited by the signal (S) as [3]:

$$S = 1 * e^{-5\cos(t)} \tag{1}$$

After exciting the structure obtains the frequency response of the structure using the Fourier Fast Transform (FFT) [3]. The frequency response is obtained from the signals of the velocity and the displacement of the structure excited. Fig. 1 (b) illustrate two signals are captured in the simulations, which has a signal with a 20% failure (red) and a signal in normal condition (blue).

Proposed Methodology

The system analysis and health monitoring of structures proposed in this paper consists of three main modules: data acquisition, training and diagnosis of ARTMAP-*Fuzzy* neural network.

The data acquisition module is composed of the experimental equipment to capture signals in the building structure, such as sensors, accelerometers, etc. The input set of ARTMAP-*Fuzzy* network consists of signals frequency response, resulting from the FFT (Fast Fourier Transform).

In the training module are configured weights and created the categories (knowledge) that will be used in the diagnosis of the data by ARTMAP-*Fuzzy* neural network. In this step the network is trained using a data set which generated a number of categories equals the number of input signals. This process is done *offline*. The diagnostic module a new set of signals different from the training set should be presented to the network, and each signal will be analyzed individually. All signals are compared with the knowledge (categories created in the training phase) obtained by ARTMAP-*Fuzzy* neural network and so when there is a matching criterion between the patterns are classified as category the displayed signal ARTMAP-*Fuzzy* network. Found category corresponding to a given pattern is, therefore, identified the fault in the structure. The flowchart of the operation of the method of analysis is shown in Figure 2.

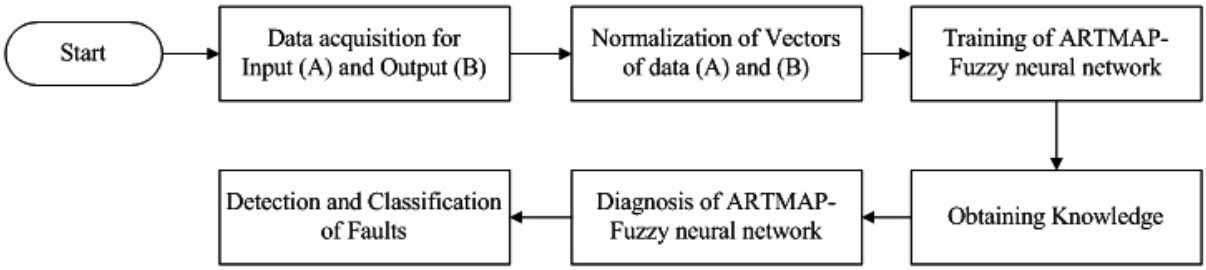


Fig. 2: Proposed Methodology.

Applications and Results

In this section we present the results obtained with the application of ARTMAP-Fuzzy neural network in the database. All tests were done using a PC Intel Core 2 Duo 1.9 GHz, 2 GB of RAM, and operating system Windows 7 Ultimate 32-bit. The algorithm was developed in MATLAB ® [4].

In order to evaluate the methodology analyzed the efficiency, accuracy and computational time in the analysis of failures in the structure. The parameters of the ARTMAP-*Fuzzy* neural network used in the process of training and diagnosis were $\alpha = 0{,}1$, $\beta = 1$, $\rho_a = 0{,}99$, $\rho_b = 0{,}99$ and $\rho_{ab} = 1$.

In the training phase we used a data set with 490 signs, 70 signs of the structure under normal conditions (base-line) and 420 signs of structure failure. This data set corresponds to 70% of the available data. To evaluate ARTMAP-*Fuzzy* neural network in the diagnosis phase was used a data set with 210 signals, 30 signals of the structure without fail and 180 signal failure of the structure.
These data used in the tests represent 30% of the available data. The database simulated has six different patterns faults. The response signals at frequencies obtained by FFT in the simulations represent the input signals to the training and diagnosis of ARTMAP-*Fuzzy* neural network. Table 1 shows the results obtained by the system failure diagnosis in structures when applied to the data set.

Table 1 - Results obtained by the method.

	Training Phase	Diagnostic Phase
Samples Used	490	210
Ratings Correct	490	210
Ratings Wrongs	0	0
Accuracy (%)	100,00	100,00
Time (*ms*)	645,1	139,00

To obtain these results, the system has undergone a testing phase and parameter settings. The results were obtained with the best configuration of ARTMAP-*Fuzzy* neural network. The result was obtained by cross-reference test, in which the system was run 20 times to ensure accuracy of results. We note that the analysis of the structural integrity shows a good performance, with a success rate of 100%. The more knowledge gained in training, the more efficient the process of diagnosis

Conclusion

In this paper we propose a new approach to develop a system of structural analysis of buildings, in this context we used an ARTMAP-Fuzzy artificial neural network which showed good results, achieving a success rate of 100% for the best system configuration. The training phase is that demand greater computational time, however is performed *off-line* is not causing harm to the algorithm. The diagnostic phase of the system, from the signal acquisition is performed rapidly with time less than 140 milliseconds, which makes the system to be a tool used in real time.Thus, we conclude that the proposed system based on ARTMAP-Fuzzy network is very efficient, reliable, robust and accurate to perform the analysis and monitoring of the integrity of structures buildings. We emphasize that this work presents a new method to monitor buildings using intelligent techniques.

Acknowledgments

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A NOTE OF OPTIMAL LINEAR CONTROL TECHNICAL APPLIED IN THE CHAOS STABILIZATION IN THE MATHIEU-VAN DER POL AUTONOMOUS OSCILLATOR

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Abstract: This paper presents the application of the Linear Optimal Control Technical to control the chaotic movement of a hyperchaotic system with three positive Lyapunov exponents. The Mathieu - van der Pol autonomous system is demonstrated with a chaotic behavior and we present the control technical for the chaos stabilization. The simulation results show the effectiveness of the control strategy.

AMS Subject Classification: 34H10, 49N05
Key Words: control design, Tri-Chaos, nonlinear dynamics

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1. Introduction

A new four order system – Mathieu-van der Pol autonomous system is proposed by [1]. In this work, the authors present a new hyperchaotic system with three positive Lyapunov exponents (called Tri-Chaos), as well as the implementation of an electronic circuit.

Chaos control problems consist of attempts to stabilize a chaotic system to an equilibrium point, a periodic orbit, or more general, about a given reference trajectory. In the last years, a significant interest in control of the nonlinear systems, exhibiting chaotic behavior, has been observed and many of the techniques were discussed in the literature recently, see [2]-[11]. The aim of this paper is to propose the application of the Linear Optimal Control [11] to control the chaotic movement of a hyperchaotic system with three positive Lyapunov exponents proposed by Shih-Yu et al. [1].

The paper is organized as follows: in Section 2 we demonstrate the mathematical model and show the nonlinear dynamics of the model. In Section 3, we discuss a control design problem for non-linear model. In Section 4, we give the concluding remarks. Finally, we list out the bibliographic references used in this paper.

2. Non-Linear Model

The work [1] introduced a new hiperchaotic system with three positive Lyapunov expoents, also tri-chaos call. Via linear coupling, Mathieu, and van der Pol systems are coupled with each other and then become a new four order system-Mathieu-van der Pol autonomous system. The authors propose a new chaotic system with three Lyapunov exponents, and the governing equations of system, Mathieu-van der Pol autonomous system, are:

$$\begin{aligned} \dot{x_1} &= x_2, \\ \dot{x_2} &= -(a+bx_3)x_1 - (a+bx_3){x_1}^3 - cx_2 + dx_3, \\ \dot{x_3} &= x_4, \\ \dot{x_4} &= -ex_3 + f(1-{x_3}^2)x_4 + gx_1, \end{aligned} \tag{1}$$

where x_1, x_2, x_3 and x_4 are four states of the system, a, b, c, d, e, f and g are the parameters of the Mathieu-van der Pol system, [1].

We choose the chaotic behavior for analyzing when the parameters $a = 91.7$, $b = 5.023$, $c = 0.01$, $d = 91$, $e = 87.001$, $f = 18$ and $g = 9.5072$, see [1].

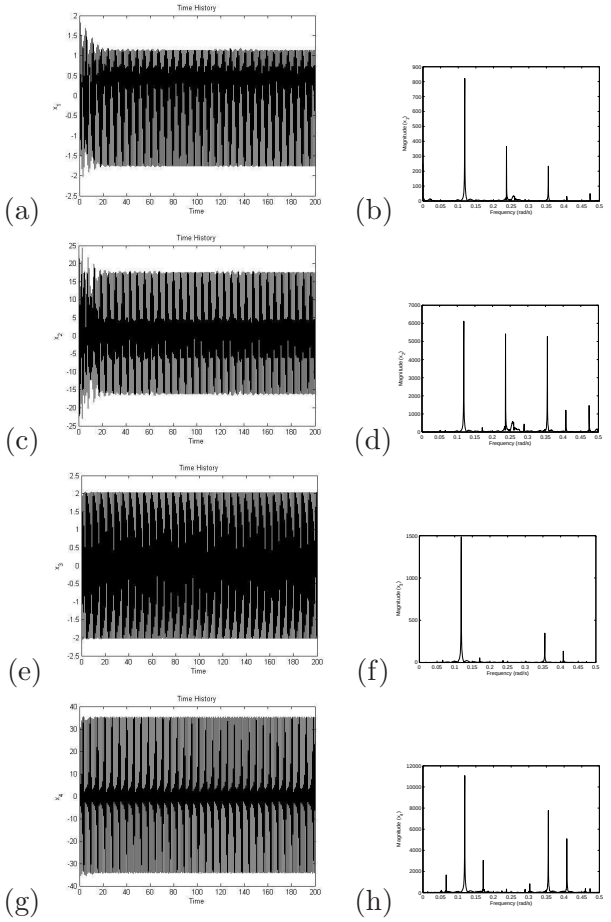
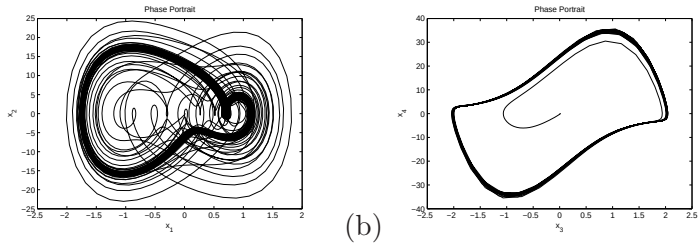


Figure 1: (a)Dynamical behavior of the time history (x_1), (b) FFT for (x_1), (c) Dynamical behavior of the time history (x_2), (d) FFT for (x_2), (e) Dynamical behavior of the time history (x_3), (f) FFT for (x_3), (g) Dynamical behavior of the time history (x_4), and (h) FFT for (x_4).

Figures 1 show the dynamical behavior of the time history and FFTs for states of the system, respectively.

Figures 2 show the portrait phase for the chaotic behavior.

Figure 3 illustrates the four Lyapunov exponents, $\lambda_1 = 12.126062$; $\lambda_2 = 7.7712$; $\lambda_3 = 7.7814$ and $\lambda_4 = -9.6868$, demonstrating the presence of the Tri-Chaos with the three Lyapunov exponent positive. The total time for Lyapunov



(a) (b)

Figure 2: Phase Portrait. (a) x_1 and x_2 projections and (b) x_3 and x_4 projections

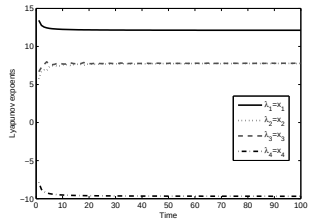


Figure 3: Dynamics of Lyapunov exponents for system

exponents computation was $\Delta\tau = 100,000$ with time-step of $\tau = 0.5$.

3. Control Design

In this section we develop an optimal linear control design for the systems considered, i.e. the Mathieu-van der Pol autonomous system, reducing the chaotic movement to a stable orbit.

Next, we present a summary of the theory of the used methodology. Due to the simplicity in configuration and implementation, the linear state feedback control is especially attractive, see e.g. [11]. It is used before on some mathematical models, as in [6]-[10], undeserved others. We remarked that this approach is analytical, without dropping any non-linear term.

Let the nonlinear governing equations of motion (1) be rewritten in a state form:

$$\dot{x} = Ax + g(x) + U. \tag{2}$$

If one considers a vector function $\tilde{x}$, that characterizes the desired trajectory, and takes the control U vector consisting of two parts: $\tilde{u}$ being the feed forward

and u_f is a linear feedback, in such a way it gives

$$u_f = Bu, \tag{3}$$

where B is a constant matrix. Next, by taking the deviation of the trajectory of system (2) to the desired one $y = x - \tilde{x}$.

It is optimal, in order to transfer the non-linear system (3) from any initial to final state $y(t_f) = 0$, minimizing the functional $\tilde{J} = \frac{1}{2}\int_0^\infty (y^T \tilde{Q} y + u^t R u)dt$, where the symmetric matrix $P(t)$ is evaluated through the solution of the matrix Ricatti differential equation

$$PA + A^T P - PBR^{-1}B^T P + Q = 0, \tag{4}$$

satisfying the final condition $P(t_f) = 0$.

In addition, with the feedback control (3), there exists a neighborhood $\Gamma_0 \subset \Gamma$,$\Gamma \subset \Re^n$, of the origin such that if $x_0 \in \Gamma_0$, the solution $x(t) = 0$, $t \geq 0$ of the controlled system (4) is locally asymptotically stable, and $J_{min} = x_0^T P(0) x_0$. Finally, if $\Gamma = \Re^n$, then the solution $y(t) = 0$,$t > 0$ of the controlled system (4) is globally asymptotically stable.

Using the theorem from [12], the dynamic error y can be minimized ($y \to 0$).

3.1. Theorem Linear Optimal Control

If there are matrices Q and R, positive definite, Q symmetric, and such that the matrix

$$\tilde{Q} = Q - G^T(x, \tilde{x})P - PG(x, \tilde{x}) \tag{5}$$

is positive definite for the limited matrix G, then the linear feedback control

$$u = -R^{-1}B^T P_y \tag{6}$$

is optimal, in order to drive the non-linear system (3) of any initial state to the terminal state

$$y(\infty) = 0, \tag{7}$$

minimizing the functional

$$J = \int_0^\infty (y^T \tilde{Q} y + u^T R u)dt, \tag{8}$$

where the symmetric matrix P is calculated from the nonlinear Riccati equation:

$$PA + A^T P - PBR^{-1}B^T P + Q = 0. \tag{9}$$

Next, we will apply this methodology to model (1).

3.2. Application of the Linear Optimal Control to the System

The equations (1) describe the system:

$$\begin{aligned} \dot{x_1} &= x_2 + U, \\ \dot{x_2} &= -(a + bx_3)x_1 - (a + bx_3){x_1}^3 - cx_2 + dx_3, \\ \dot{x_3} &= x_4, \\ \dot{x_4} &= -ex_3 + f(1 - {x_3}^2)x_4 + gx_1, \end{aligned} \tag{10}$$

where the function of control U is defined in the equation (2).

We will obtain $B = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$, $y = \begin{pmatrix} x_1 - \tilde{x}_1 \\ x_2 - \tilde{x}_2 \\ x_3 - \tilde{x}_3 \\ x_4 - \tilde{x}_4 \end{pmatrix}$, $\tilde{x} = \begin{pmatrix} 0.01sin(\pi t) \\ 0.01sin(\pi t) \\ 0.01sin(\pi t) \\ 0.01sin(\pi t) \end{pmatrix}$, $Q = I_4$, $A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ -92.7777 & -0.01 & 90.9497 & 0 \\ 0 & 0 & 0 & 0 \\ 9.5071 & 0 & -87.0046 & 17.9982 \end{pmatrix}$, where the controllability matrix R of the system to the pair $[A, B]$ is obtained by $R = [B|AB|A^2B|A^3B] \neq 0$.

Thus, $R = \begin{pmatrix} 1 \end{pmatrix}$.

Then the matrix $P(t)$ is done by

$$P = \begin{pmatrix} 0.4349 & -0.0018 & -0.0448 & -0.0135 \\ -0.0018 & 0.0043 & 0.0110 & -0.0056 \\ -0.4482 & 0.0110 & 1.1717 & -0.1709 \\ -0.0135 & -0.0056 & -0.1709 & 0.0578 \end{pmatrix}$$

and (an optimal control)

$$u = -2.8743x_1 + 0.7939x_2 + 56.3594x_3 - 13.2351x_4.$$

The trajectories of the system, without and with control may be seen through Figures 4. According to the optimal control verification [12], the function (4) is numerically calculated across $L(t) = y^T \tilde{Q} y$, where $L(t)$ is defined positive and it is show in Figure 5.

4. Conclusions

In this work the presence of chaotic behavior is verified in the proposed system. It is mathematically modeled by means of a set of four ordinary differential equations.

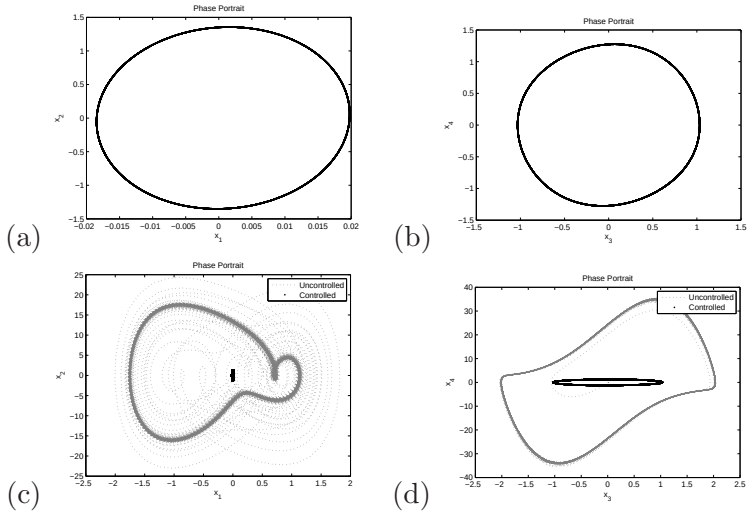


Figure 4: Phase Portrait. (a) x_1 and x_2 controlled projections, (b) x_3 and x_4 controlled projections, (c) x_1 and x_2 uncontrolled and controlled projections, (d) x_3 and x_4 uncontrolled and controlled projections.

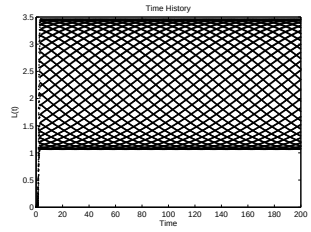


Figure 5: Controlled dynamical behavior of the time history (time 0-200).

We also propose to use an optimal linear feedback control strategy, applied to mathematical model proposed. This kind of control strategy is reducing the chaotic movement of the system to a small stable orbit.

Figures 4 and 5 illustrate the effectiveness of the control strategy to the Mathieu-van der Pol autonomous system.

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Neural network based on adaptive resonance theory with continuous training for multi-configuration transient stability analysis of electric power systems

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ABSTRACT

This work presents a methodology to analyze electric power systems transient stability for first swing using a neural network based on adaptive resonance theory (ART) architecture, called Euclidean ARTMAP neural network. The ART architectures present plasticity and stability characteristics, which are very important for the training and to execute the analysis in a fast way. The Euclidean ARTMAP version provides more accurate and faster solutions, when compared to the fuzzy ARTMAP configuration. Three steps are necessary for the network working, training, analysis and continuous training. The training step requires much effort (processing) while the analysis is effectuated almost without computational effort. The proposed network allows approaching several topologies of the electric system at the same time; therefore it is an alternative for real time transient stability of electric power systems. To illustrate the proposed neural network an application is presented for a multi-machine electric power systems composed of 10 synchronous machines, 45 buses and 73 transmission lines.

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1. Introduction

Stability analysis (angle and voltage) is a very important investigation procedure in electric power systems. The main objective is to analyse and execute strategies that could guarantee the energy supplying with quality, and furthermore avoid or at least minimize the energy supplying interruptions to the users. The stability associated to the angles is the transient stability, consisting in evaluating the effects proceeding from pertubations that can cause undesirable oscillations on the angles of the synchronous machines [1,2]. This analysis is effectuated by solving nonlinear differential equations which describe the dynamics of the system (oscillation equation of the synchronous machine) and afterwards, analyising and evaluating the angular position of each synchronous machine during the time, known as simulation [3]. The techniques used for simulation are precise and do not present restrictions to the model employed. Nevertheless, they need much CPU due to complex calculus evolved. Generally, it also provides only the qualitative information, i.e., the system is stable or unstable for a particular contingency. This information is not always useful to the operation, mainly in unstable cases, where preventive procedures are necessary (generation reallocation, load-shedding, etc.) which are quantified in term of the knowledge of how much the system is unstable (quantitative information).

An alternative procedure consists in obtaining the analysis without solving the differential equations. The direct method of Lyapunov (DML) is one of the procedures and the results are considered adequate when using principally the classical model [1,2].

Taking into account the quantity of deffects to be analysed, the complexity and the dimension of the modern power systems, the simulation as well as the Lyapunov direct method are alternatives that are not adequate to applications in real time. New approaches based on artificial intelligence specially the neural networks [4], are found on the literature to overcome the difficulties. Therefore, this work aims to investigate an application of artificial neural networks (ANN) on transient stability analysis (TSA) of electric power systems. The contingencies used, in this paper, are balanced three-phase short-circuits with a transmission line tripped. This proposal can consider several topologies of the electric network, that is a more plausible scenario of the operation point, when compared to a unique (fixed) topology as commonly adopted on the literature. Therefore, it is a new procedure for neural networks applications compared to those available nowadays.

TSA with neural networks in generally uses feedforward neural networks with the training based on backpropagation algorithm [4,5]. This technique is efficient according to the specialized literature and is a banchmark in precision. Nevertheless the processing time is high. It is emphasized that the specialized literaure con-

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tains few references of stability analysis with neural networks, e.g., [6–13], among others.

This way the use of Euclidean ARTMAP neural network is investigated [14]. This network pertains to a set of proposals (ART, ART1, ART2 and ARTMAP) [15,16] appropriated to classify binary and/or analogic patterns, with a supervised or non-supervised training. ARTMAP neural network is an architecture based on supervised training for multi-input and multi-output mappings. The ART (adaptive resonance theory) family [15] has the characteristics of stability (capacity of learning by adjusting the weights) and plasticity (capacity of continuous learning with the inclusion of new patterns without loosing the memory in relation to previous patterns). The Euclidean ARTMAP neural network was proposed on the literature as an alternative to overcome some operational difficulties like those found on neural networks pertaining to the ART family. The input data are necessarily not normalized and codified as the ARTMAP neural network, reducing the dimension of the input vector and the execution time of training and analysis. Firstly the geometric representation of the categories of the neural network was with hyper rectangles, and now is represented by circles, improving the precision of the results and sometimes the best way to group the data. Another advantage of employing this new neural network is the possibility of using the continuous training. With this procedure the neural network can be trained off-line. The training considers a set of data generated by a computational program, the teacher, and always updated based on data from the electrical systems or obtained by simulation. Neural networks of the ART family, considering the plasticity and the low execution time (training, continuous training and analysis) are therefore compatible with on line applications and are used to implement this procedure. To illustrate the proposed method results are presented for a multi-machine system.

2. System model

Considering an electrical power system composed of *ng* synchronous machines, the dynamic behavior of the *i*th machine can be described by the following equation [2]:

$$M_i \frac{d^2\theta_i}{dt^2} - g_i(\theta) = 0, \quad i \in NG \tag{1}$$

where

$$g_i(\theta) = Pm_i - Pe_i - M_i \frac{PCOA}{MT}; \tag{2}$$

$M_i = 2H_i/\omega_s$; H_i is the inertia constant (s); ω_s is the synchronous velocity (elect. rad./s) $= 2\pi f^0$; f_0 is the nominal frequency of the system (Hz); θ_l is the rotor angle of the *i*th synchronous machine referred to the center of the angles (CA) $= \delta_i - \delta^0$; δ_i rotor angle of the *i*th synchronous machine measured in relation to a machine rotating in synchronous velocity (elect. rad.); $\delta^0 = \sum_{j \in NG} M_j \delta_j$; Pm_i is the input mechanical power (pu); Pe_i is the output mechanical power (pu); *PCOA* is the accelerating power of the center of angles (CA) $\sum_{j \in NG}(Pm_j - Pe_j)$; $MT = \sum_{j \in NG} M_j$; $NG \triangleq \{1, 2, \ldots, ng\}$ = index set of the machines that compose the system; *ng* is the number of synchronous machines.

3. Transient stability analysis

TSA of electrical power systems considering a contingency of index *r* can be effectuated using the security margin criterion (*M*) [17]:

$$M_r = \frac{Ecrit_r - Ee_r}{Ecrit_r} \tag{3}$$

where $Ecrit_r$ is the total critical energy of the system; Ee_r is the total energy of the system evaluated on the time of the elimination of the fault (*te*).

Critical energy (*Ecrit*), as well as critical time (*tcrit*), can be determined by PEBS (potential energy boundary surface) method [2], or another procedure that presents a similar result, mainly in precision. For the classical model, PEBS method shows the same results as the simulation [17]. The total energy related to the system defined by equation (1) is given by [17]:

$$E(\theta, \omega) = Ec(\omega) + Ep(\theta) \tag{4}$$

where

$$Ec(\omega) = \frac{1}{2}\sum_{i \in N} M_i \omega_i^2 \quad \text{(kinetics energy)} \tag{5}$$

$$Ep(\theta) = -\sum_{i \in N} \int_{\theta_i^p}^{\theta_i} g_i(\theta)\, d\theta_i \quad \text{(potential energy)} \tag{6}$$

Therefore, transient stability for the *r*th contingency is evaluated by security margin on the following way [17]:

- If $M_r \geq 0$, system is stable;
- If $M_r < 0$, system is unstable.

The terms stable and unstable are referred to the concept of first oscillation transient stability.

4. ARTMAP neural network based on euclidean distance

ARTMAP neural network [15] is adequate to solve several complex problems, as for example in electric power systems: security analysis, transient stability analysis, load forecasting, etc. [1,6–13,16,17]. The importance of employing neural networks is due to the stability and plasticity characteristics, which offer stable learning, multiple generalization and fast convergence. Nevertheless the ART family neural networks present some operational difficulties as little sensitivity to the network parameters (vigilance parameters, etc.) and the precision of the analysis. The effects of the parametric sensitivity have been solved or attenuated by employing new conceptions of training and ART neural networks architecture. Thus, several proposals are presented on the specialized literature aiming to solve the imprecision problem. Generally the imprecision are due to the mechanisms of choosing the categories and the vigilance tests that need to be improved.

Therefore, this section presents the Euclidean neural network architecture that is the model chosen in this work to solve the TSA problem. It is a conception that aims to improve the quality of the solutions (more precision without affecting the time of calculus necessary to effectuate the training and other routines), when compared to the conventional ARTMAP neural network.

The fuzzy ARTMAP neural network uses hyper rectangles to geometrically represent the categories, which synthesize the input patterns [15]. The use of hyper rectangles to represent the categories induces to the L_1 norm, which is used on the calculus of the activation function and, also on the match function of the fuzzy ARTMAP neural network. Meanwhile, in many cases hyper rectangles not necessarily can be the best way to group the input data, neither the geometry based on L_1 norm leads to optimal classification results as compared to other geometries.

Considering what was mentioned above, it is interesting to find different types of geometries to efficiently represent the categories of the descendent ART neural networks. The first investigation with this conception is related to the Gaussian ARTMAP developed by [18], which adopted the L_2 norm and the distances defined in function of the Mahalanobis distance [19].

It is emphasized that the descendent ART neural networks, based on the Euclidean distance is a particular case of the Mahalanobis ART neural networks. Mahalanobis conception does not preserve the plasticity characteristics, and thus, the Euclidean distance is chosen.

The traditional fuzzy ARTMAP neural network needs normalization and complementary codification of the input patterns, which doubles the dimension of the network introducing a considerable operational cost. Therefore, ART and ARTMAP neural networks based on Euclidean distance [14] and Mahalanobis distance [19], were developed without needing the normalization and complementary codification of the input vectors. In this case, the input vector dimension is reduced to the half, when compared to the dimension of a conventional fuzzy ARTMAP algorithm.

ARTMAP neural architecture based on Euclidean distance uses the Euclidean distance on the weight adaptation and *match* function of the ART and ARTMAP neural networks [14].

$$T_j = (\boldsymbol{a} - \boldsymbol{w}_j)(\boldsymbol{a} - \boldsymbol{w}_j)^T \tag{7}$$

$$FM_J = \frac{T_J}{\max(\boldsymbol{a}\boldsymbol{a}^T,\ \boldsymbol{w}_J\boldsymbol{w}_J^T)} \tag{8}$$

where $\boldsymbol{w}_j$ is the jth vector weight; $\boldsymbol{a}$ is the input vector; β is the training rate; T_j is the jth choice function; T_0 is the empty activation neuron (recognize category without the presence of an input pattern).

All these parameters are associated to ART/ARTMAP neural network training according to the references [15].

The choice of the candidate categories satisfies the minimum activation and the resonance occurs when the match function of the chosen category finds the vigilance criterion, being different from the ART and ARTMAP neural networks. Thus:

$$T_J = \min\ \{T_j : j = 1, \ldots, M\} \tag{9}$$

$$FM_J < \rho \tag{10}$$

The main resonance condition is now lower vigilance and more quantity of output nodes. The weight adaptation remains the same as the traditional ARTMAP architecture:

$$\boldsymbol{w}_J^{(k+1)} = \beta(\boldsymbol{a}) + (1-\beta)\boldsymbol{w}_J^{(k)} \tag{11}$$

The training algorithm is essentially the same as the classical ARTMAP neural network. One difference is on the vigilance parameter, which is reduced to the minimum value of the activation function minus a very small positive value in case the vigilance criterion fails [14]. The Euclidean distance decides the classification, i.e., the pattern whose Euclidean distance with the model is minimum is the correct prediction. The geometric representation of the models, which are represented by the mean of the examples, of the ARTMAP neural architecture based on Euclidean distance, are circles with the centers on the mean of the examples with radius ρ (vigilance parameter).

A mechanism of updating the recognition categories is included on the prediction phase as an innovation. The criterion consists in defining a minimum quantity of elements demanded for each category generated on the training phase ($NC_{\min}$). During the presentation of an input pattern on the prediction phase, the predicted class is evaluated and, if it has a quantity of patterns inferior to the predefined one, then a new pattern is included to this class by updating the weight vectors, increasing in one the quantity of vectors pertaining to that category, and thus successively for every pattern presented to the network. This mechanism allows the categories being constantly updated, improving the region of definition and consequently the precision of the results.

During the presentation of the input vectors on the training and prediction phase, an empty neuron is included to the network, which represents a recognition category without input patterns and activation level T_0 (value of the activation function). This idea originally belongs to Vakil-Baghmished and Pavesic [20] for a fuzzy ARTMAP neural network, decreasing the time spent on the training phase. This procedure establishes a minimum value for the activation function ($0 \le T_0 \le 1$), decreasing the candidates to winner and consequentely decreasing the quantity of vectors during the competition.

The idea in this work is to determine a maximum distance that a vector can have with a determined recognition category, once the ARTMAP neural network based on the Euclidean distance suggests the choice of a candidate category with minimum activation. This procedure reduces the quantity of neurons candidates to winner (T_j), to represent the input vector (recognition category), and also reduces the search time for this neuron. This maximum value T_0 is initially determined experimentally. If neuron T_0 is chosen ($0 \le T_0 \le 1$) during the training to represent the present input vector, then a new category is created. T_0 is maintained on the prediction phase and if it is chosen as a winner category then for that category there is no prediction. A more detailed analysis for this vector is suggested to verify the possibility to include on the network as new information (continuous training). Considering the tests effectuated, it is observed that there is no need to arbitrate the parameter T_0 with hardness; it is enough that it is contained on the indicated interval.

5. Transient stability analysis: proposed solution using euclidean ARTMAP neural network

This section presents a proposal for TSA based on Euclidean ARTMAP neural network. This proposal contains the training, continuous training and analysis phases. It is a mechanism where the training is effectuated using data generated by simulation and the continuous training is permanently effectuated while the analysis is occurring. The information comes from a TSA computational program containing the electrical network data, some predefined contingencies (three-phase short-circuit with transmission line tripped) and different topologies that provide the associated security margin. Afterwards the information used on the training is pre-processed and chosen to become the input/output stimulus.

It is emphasized that considering different topologies for the analysis is a different proposal from those found on the literature. It is a device seeking a more real procedure for the electrical power system, i.e., the topology is not fixed.

5.1. Input stimulus

The proposed neural structure determines the security margin considering three-phase short-circuits with outage of a transmission line for TSA of electrical power systems. The input ($\boldsymbol{X}$) and output ($\boldsymbol{Y}$) stimulus for the training phase are defined using a hybrid computational program [17], employing the simulation concept and the Lyapunov direct method (energy method). Thus, the input pattern vectors of the neural network (input of the ARTa module) are defined according to [11]:

$$\boldsymbol{X}_r \triangleq [\boldsymbol{P}\,\boldsymbol{Q}\,\boldsymbol{\Lambda}_r] \tag{12}$$

where $\boldsymbol{P} = [P_1\, P_2 \cdots P_{\mathrm{ns}}]$; $\boldsymbol{Q} = [Q_1\, Q_2 \cdots Q_{\mathrm{ns}}]$; P_i is the active electrical power of the ith bus of the system; Q_i is the reactive electrical power of ith bus of the system; $\boldsymbol{\Lambda}_r$ is the vector containing the contingency index and information about the topology of the system represented in binary code [0,+1]

The pattern $\boldsymbol{X}$ is a n-dimentional vector:

$$n = 2ns + nc \tag{13}$$

where *ns* is the number of the buses of the systems; *nc* is the number of bits corresponding to the number (index) of the contingency and configuration of the system.

Considering the classical model, only active and reactive power is used to evaluate the transient stability, corresponding to the analysis with the security margin criterion. The network topology and other parameters (inertia constant, transient reactance, etc.) are considered constant. The ideal is to use only the causal variables (active and reactive power) and the parameters expressing changes on the electrical network (contingency representation). It is emphasized that angles and velocities are elements resulted from the process, i.e., they are in function of the power (active and reactive) and the type and circumstance of the fault.

5.2. Output stimulus

The stimulus to be applied to the ARTb module on the Euclidean ARTMAP neural network is the security margin corresponding to the input pattern vectors (ARTa module) described by equation (12). The strategy proposed in this work is the use of intervals of security margin where the contingencies are, according to what was proposed in reference [17]:

$$\vartheta = \frac{1 - M^{A}}{1 - M^{B}} - 1 \tag{14}$$

where ϑ is the effort of security margin displacement; M^{A} is the value of initial security margin of the interval; M^{B} is the value of final security margin of the interval.

Therefore, the interval in terms of security margin concept is defined as [11]:

$$IMS \triangleq \left(M_k^{A},\ M_k^{B}\right) \tag{15}$$

where *IMS* is the interval considered; *k* is the index that represents the *k*th interval.

This conception is used in this work to define the intervals associated to the contingencies as shown in Table 1, considering $\vartheta = 0.35$ and the security margins belonging to the interval $(-3;\ 1)$.

A security margin equal to -3 is defined as an inferior limit of reference, and values lesser than -3 (more negative) are very unstable contingencies, leading to generator unities outages and sometimes to an undesirable blackout, if the preventive tasks are not correctly effectuated. Therefore, the interval $(M \leq -3)$ groups the contingencies that produce great instability on the system. For contingencies associated to security margins near the unity, the system behavior is very strong.

Then, the output is defined as:

$$\boldsymbol{Y} \triangleq [\boldsymbol{IMS}_k] \tag{16}$$

where $\boldsymbol{IMS}_k$ is the vector corresponding to the *k*th interval of the security margin associated to the contingency represented in binary code.

Table 1
Binary representaiton of the fuzzy ARTMAP neural network.

Number	Security margin (*M*)	Security margin intervals (*IMS*)
1	−3	≤−3
2	−1.9630	(−3; −1.9630]
3	−1.1948	(−1.9630; −1.1948]
4	−0.6258	(−1.1948; −0.6258]
5	−0.2043	(−0.6258; −0.2043]
6	0.10790	(−0.2043; 0.10790]
7	0.33920	(0.10790; 0.33920]
8	0.51050	(0.33920; 0.51050]
9	0.63740	(0.51050; 0.63740]
10	0.73140	≥0.63740

5.3. Schematic representation

The basic scheme of solving the TSA problem by neural networks is presented. Fig. 1 shows the principal routines composing the methodology for the problem. The neural network corresponds to a version of the ART neural family (Euclidean ARTMAP) with a supervised training. It is an approach for applications in real time. Neural networks contain three fundamental phases: training, continuous training and analysis. The training phase requires great quantity of processing time, while the analysis is effectuated almost without computational effort. The continuous training is incorporated during the analysis by executing several routines: data acquisition, processing and weight adjustment in a similar way for the neural network training. It is a procedure that permanently maintains the knowledge extraction of the problem to be approached (transient stability analysis), from the training effectuated considering the data generated by computational simulation with the results coming from the operational center. Therefore, they are modules of filters (consider or reject the information) and memory (filing, updating (reinforcement) and diagnosis). This step is different from the initial training, considering that the operation maintains great number of data and results that can be useful or not. Thus, it is necessary to develop adequate routines for processing, filing and using this information. The continuous training module can be improved to consider more elaborated ones (e.g., the Park complete model). By simulation it is not yet possible to generate an index for more elaborated models, but by neural networks is possible to generate a hybrid index combining the simulation and some method based on the energy function method. This device will be studied in future researches. Developing such procedure certainly will produce significant results for the electrical power systems operation, i.e., there exists the possibility of effectively execute the analysis with quality in real time. Next section presents a scheme for the continuous training.

5.4. Continuous training execution algorithm

The continuous training incorporated on the Euclidean ARTMAP neural network was based on the concepts of *Novelty Detection* (ND) [21] and *Anomaly Detection* (AD) [22] that includes a new pattern on the memory of the neural network. The concept of ND is similar to the AD one. The difference with them is that the ND referrers to finding new events, i.e., not to detect anomalies but novelties. These events cause some damages (material prejudices, etc.). Prevention is very important to avoid accidents, or at least minimize the consequences.

This concept is used to develop the routine for activate the inclusion of new patterns from the *match tracking* mechanism of the Euclidean ARTMAP neural network. This strategy is always used for simulated data and principally for information available on the data of electrical enterprises.

The procedure adopted in this work aims to developing a mechanism of defining the moment where is necessary to effectuate the continuous training and principally where the data are available to include or reject these data on the weight matrices of the neural network. This procedure is described as follows.

5.4.1. Criteria for including or rejecting the data

The adaptive execution of the ART neural network family allows that new patterns may be permanently included to the memory, improving the performance and with a more efficient execution. It is necessary a mechanism that selects the data that are very important to the network memory to such routine works efficiently. Since an empty neuron T_0, has been introduced on the Euclidean ARTMAP neural network, as described in Section 4, the examination of the new pattern is effectuated, which was allocated in this category

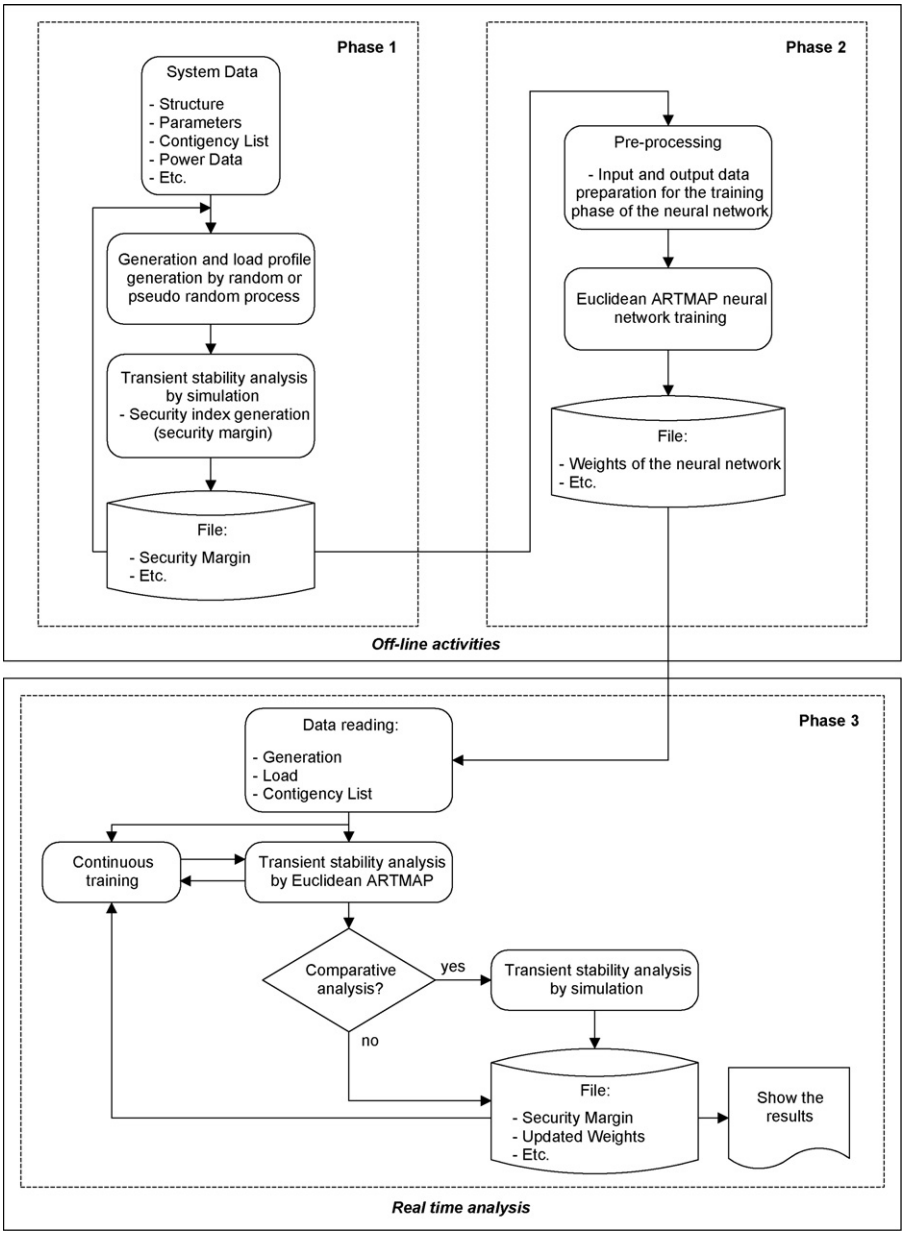

Fig. 1. Flowchart of the TSA by neural networks.

(T_0) during the analysis phase. If the T_0 neuron was chosen as the winner category, then a new category is created (composed of this unique pattern) and such a pattern is indexed as undetermined, and furthermore an analysis is necessary to know if this pattern is really important to be added to the memory or not. The importance consists in presenting the subsequent patterns to the neural network. If other patterns are indexed to the new category, then the category is considered as a novelty, from a predefined number ($N_{\min}$), and the weight vector become part of the network memory. Otherwise, the undetermined category is eliminated from the process after including a certain number of vectors on the network ($N_{\max}$) without $N_{\min}$ be attained. Several undetermined categories can be active at the same time, if T_0 is selected in each presentation of the subsequent patterns. These categories are eliminated or included on the network according to the predefined $N_{\min}$ and $N_{\max}$.

Sometimes it is necessary to erase memory of the neural network. After effectuating the training and the continuous training some categories can become inactive without prediction as no pattern was presented to the network. These categories can be eliminated excluding the corresponding weight vector and if it is interesting to include again later, this is effectuated with the continuous training.

The continuous training is also effectuated from the updating criterion of the recognition categories, as defined on Section 5.

The following steps describe the execution of the continuous training module, if it is settled during the analysis phase.

Step 1. Presentation of the input vector to the Euclidean ARTMAP neural network.
Step 2. If the winner category (chosen category to allocate the present pattern) has activation T_0, i.e., no existent class is similar to the present pattern to allocate, then a new category is created, indexed as undetermined, and follows the presentation of a new pattern to the network (Step 1).
Step 3. If the winner category, T_J, is one of the existent categories, the prediction is effectuated. If the minimum quantity of

this category (NC_{min}) is not yet attained, then a new pattern is added to this class. Otherwise, follows the presentation of a new pattern (Step 1).

Step 4. If the winner category is defined as undetermined, then add the present pattern to the category and verify if N_{min} is attained. If N_{min} is attained, then the category is a novelty, and finally new information is added to the memory of the network. Otherwise, follows with a new presentation of a new pattern (Step 1).

Step 5. Eliminate the categories of the process indexed as undetermined which minimum number of examples (N_{min}) is not attained after a maximum number of vectors (N_{max}) have been presented to the network.

Considering the steps described above, the principal parameters evolved on the process can be resumed as follows:

- Nc_{min}: minimum number of elements that a category, formed on the training phase, must have to be efficient to realize a good prediction.
- N_{min}: minimum number of elements that a category indexed as undetermined must have to be defined as novelty.
- N_{max}: maximum number that a category indexed as undetermined, without N_{min} must wait to be rejected as new information and eliminated of the process.

It is observed that the continuous training process is effectuated as new data were available (acquisition from the electrical Power system operation point and/or simulations) to be added to the neural network weight matrices. These new data are included or not according to the criterion fixed on the algorithm. Therefore, there is no mathematical equation to express the continuous training, but a heuristic procedure [22].

6. Results

Results are presented for the TSA of electrical power systems by Euclidean ARTMAP neural network [14], considering a test system that corresponds to a possible configuration of South Brazilian electric power system, composed of 10 synchronous machines, 45 buses and 73 transmission lines, as shown in Fig. 2.

Results obtained by neural networks are shown and compared to those produced by Hybrid Simulation (HS) which is a referee in terms of precision considering the classical model [17] and by Fuzzy ARTMAP neural network. Three-phase short-circuits with outage of transmission line and fault elimination time equal to nine cycles (0.15 s) is considered. Neural network training is effectuated considering a set of generation and load profiles and respective security margins, associated to a set of contingencies. Each profile corresponds to a generation dispatch, in relation to the base-case, effectuated in a random way to attend the demand, also in a random way for each bus of the system. The generation and load variation are calculated considering 80% (minimum) and 130% (maximum) of the base-case (100%). Therefore, each profile is generated considering a percent variation around the nominal state (base-case) and a random number for the generation of the random sequence. Thus, for equal percents, different random numbers generate different generation dispatches and different load profiles. This procedure generates an adequate set of patterns for the training phase.

Different generation and load profiles (obtained with different random numbers) are considered when compared to the data used on the training phase. The set of contingencies is shown in Table 2. Five contingencies are randomly chosen to illustrate the proposed methodology. However this number can be increased according to the needs of the user, without any problem related to the formulation. Neural network training is effectuated with a set containing about 1500 generation and load profiles (around 100 for each contingency for each configuration of the system) and the corresponding security margins (for the five contingencies). Three different configurations are considered: (1) system operating with every transmission lines; (2) system operating with the outage of the line between nodes 16 and 18; (3) system operating with the outage of the line between nodes 18 and 19. This procedure illustrates the possibility of effectuating the analysis considering several power profiles, as well as several topologies, that could be the more plausible operational situation.

The contingency analysis is effectuated after realizing the Euclidean ARTMAP neural network training. Comparative results with the pattern results (generated by executing the HS program [17], microcomputer version program) and the results obtained by fuzzy ARTMAP neural network and Euclidean ARTMAP neural network are presented. The results generated with the HS program are used as reference, considering the classical model, due to the precision. It is emphasized that any method can be employed as a teacher for transient stability analysis, since the chosen methodology contains resources that generates a stability index and not only information about if the system is stable or unstable, as for example in conventional simulation [3].

Table 3 presents the predictions provided by the Euclidean ARTMAP neural network (intervals *IMS*'s). This table also contains the results obtained by simulation (reference results) and fuzzy ARTMAP neural network for comparison. Column 1 of Table 3 shows the number of the fault (considering the five contingencies adopted) and the respective configuration (considering three different topologies). The perceptual and the seed used on the training phase are shown on columns 2 and 3 respectively. 150 analyses were effectuated considering the same contingencies used on the training phase, with different generation and load profiles (obtained with different seeds).

It is emphasized that the error on the diagnosis with the three methodologies is proportional to the difference with the showed classes respectively. For example, if for some analysed contingencies the order of the classes observed is 3, 1 and 2, by HS program, fuzzy ARTMAP, Euclidean ARTMAP, respectively, the analysis by ARTMAP neural network presents a larger error (consequntely with less precision) in comparison with the Euclidean ARTMAP. Therefore, the distance with the correct class is 2 and 1 respectively. Nevertheless, a detail must be observed, i.e. if the class is on the right or on the left side of the correct class, concluding that results on the left side are preferable due to be a more critical result.

Euclidean ARTMAP neural network training is effectuated with the following parameters that were experimentally determined:

- initial vigilance parameter of ARTa network ($\overline{\rho}_a$) = 0.97;
- vigilance parameter of ARTb network (ρ_b) = 0.20;
- vigilance parameter of inter-ART parameter (ρ_{ab}) = 0.95;
- training parameter β = 0.01;
- number of intervals (*IMS*) = 10.

Table 2
Set of contingencies.

Fault number	Fault		
	Short-circuit bus	Line tripped	
		Initial bus	Final bus
1	17	16	17
2	41	41	42
3	43	43	44
4	44	18	44
5	44	41	44

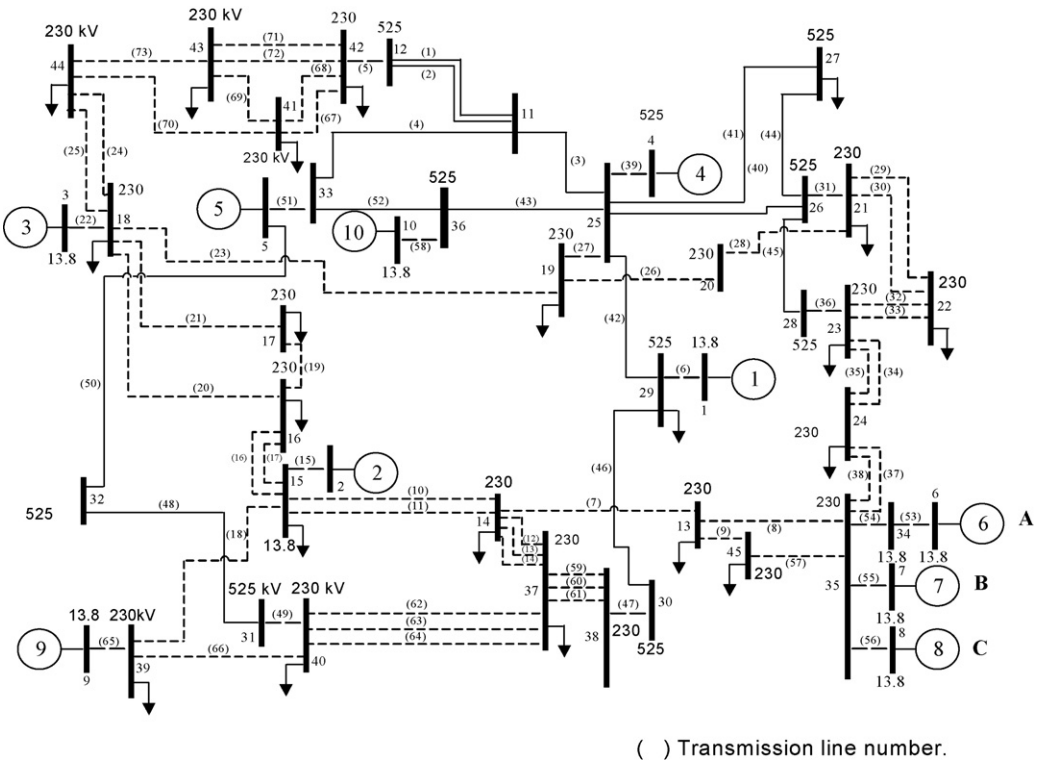

Fig. 2. Electric power system.

Table 3 contains 61 transient stability analyses which results are not equal to those obtained by simulation (reference). The results in eminence are referred to the improved results, compared with those previously predicted with the fuzzy ARTMAP neural network. This improvement is based on the security margin interval with the reference method (HS). The results in bold are those where there is no coincidence with the Euclidean ARTMAP neural network, which previously were considered better by fuzzy ARTMAP neural network. It is concluded that the Euclidean ARTMAP neural network provides better solutions in precision. In some cases there is no coincidence with the predictions with the Euclidean ARTMAP neural network and the simulation (considered as a reference in terms of precision). Probabily this is due to the reduced quantitiy of examples used on the training, principally in some particular examples. For example, the cases pertaining to classes 1,2 and 3 are in the majority predicted in class 4, meaning that those classes (1,2 and 3) do not have enough training data to represent the category on ART module. Precision increases as increases the number of stimulus of training. This problem can be overcome when the continuous training is executed. The results in shadow refer to the fuzzy ARTMAP neural network where the precision is improved, although the Euclidean ARTMAP neural network (in bold) does not confirm some predictions. Even so, the correct results increased in relation to the errors, as observed.

The parameters used in these experiences ($\overline{\rho}_a$, ρ_b, ρ_{ab}, β and *IMS*) can be changed to increase the quality (precision) of the solutions, clearly based on the experience with this kind of problem. It is emphasized that the parameter *IMS* represents the quantity of the intervals used. In the case of the user desires a great number of intervals, it is necessary to adopt a great number of *bits*. It is observed that the representation in Table 1 contains 4 *bits*.

6.1. Results considering the continuous training

Considering the continuous training, as described on Section 5.4, tests were effectuated using the same data of the analysis phase, which results are on the previous section. The continuous training considers the data that are being presented to the network and if necessary adding these data to the memory to improve the region that defines the existent categories or creating new categories considering the new information. Table 4 presents these results, comparing with the results given by the Euclidean ARTMAP neural network without including the continuous training.

The adopted parameters are:

- $NC_{min} = 100$;
- $N_{min} = 5$;
- $N_{max} = 1000$.

It is emphasized that in Tables 3 and 4 the correct results, for comparison, are those presented at column 4, i.e., the HS column.

For executing the continuous training 150 patterns were used, the same as those for the transient stability analysis, which results are in Table 3. Table 4 presents 23 transient stability analyses that were changed. These results are referred to the data that were improved with the continuous training compared with the results of the Euclidean ARTMAP neural network without the continuous training. The other 38 patterns were maintained.

Observing the results presented in Table 4, the continuous training provides better results, i.e., some patterns have the security margin interval coincident with the reference result and others are very close to the correct interval, once it is allowed to include patterns to the network even during the analysis phase. This inclusion improves the definition of the recognition categories formed during the training phase, improving the quality of the results continuously. The patterns can also be recognized as novelty, creating a new category and allowing that the process recognizes a new situation. For example, the process that should have predictions on classes 1, 2 and 3 could better differentiate the categories and execute a better prediction, with some errors

Table 3
Results of the TSA obtained by simulation (reference results), Fuzzy ARTMAP neural network and by Euclidean ARTMAP neural network.

Fault number/configuration	Profile generation/load		Security margin interval (*IMS*)		
	Base-case (%)	Seed used	HS	Euclidean ARTMAP	Euclidean ARTMAP with CT
1/1	130	427	3	4	4
1/1	120	143	3	5	4
1/1	**115**	**440**	**5**	**5**	**4**
1/1	90	793	9	8	8
1/1	80	771	9	8	8
2/1	130	8	4	5	4
2/1	125	14	6	5	6
2/1	110	259	6	7	6
2/1	90	468	9	8	8
2/1	85	473	9	8	8
2/1	80	277	9	8	8
3/1	130	841	5	4	4
3/1	120	785	8	7	7
3/1	110	844	7	6	7
3/1	105	911	7	6	7
3/1	95	680	9	8	8
4/1	125	539	4	5	4
4/1	85	728	8	7	7
4/1	80	573	9	8	8
5/1	130	121	3	4	4
5/1	125	93	2	4	4
5/1	120	138	3	4	3
1/2	120	914	3	4	4
1/2	115	962	3	4	4
1/2	85	493	5	6	5
2/2	130	399	3	4	4
2/2	125	803	3	4	4
2/2	95	276	9	8	8
2/2	85	25	5	6	5
2/2	80	380	8	7	7
3/2	**130**	**748**	**3**	**3**	**4**
3/2	**125**	**799**	**3**	**3**	**4**
3/2	95	852	7	6	7
3/2	90	810	7	6	7
4/2	130	505	3	4	4
4/2	120	763	3	5	4
4/2	115	903	4	5	4
4/2	80	756	8	7	7
5/2	130	641	3	4	4
5/2	120	790	5	4	5
5/2	110	981	3	4	4
5/2	80	720	9	8	8
1/3	130	544	1	4	4
1/3	125	816	1	4	4
1/3	120	886	1	4	4
1/3	115	934	2	4	4
1/3	110	924	2	4	4
1/3	105	938	3	4	4
1/3	95	892	3	4	3
2/3	130	395	1	4	4
2/3	115	171	1	4	4
3/3	**80**	**522**	**9**	**9**	**8**
4/3	130	296	3	4	4
4/3	**115**	**474**	**3**	**3**	**4**
4/3	90	443	3	4	4
5/3	130	24	1	4	4
5/3	125	483	1	4	4
5/3	120	409	1	4	4
5/3	115	848	1	4	4
5/3	110	794	2	4	4
5/3	105	548	2	4	4

that can be overcome as new patterns are presented to the network on the analysis process and incorporated to the memory to the Euclidean ARTMAP, according to what was described on the algorithm shown in Fig. 1. The results are also given by executing the continuous training with new simulations obtained from a teaching program, e.g., the HS program [17]. The greatest errors occur for the contingencies pertaining to the interval 1 (supercritical contingencies), which probably are due to the few quantity of examples available. Nevertheless, as more patterns are available for executing the continuous training, the Euclidean ARTMAP neural network will improve the performance and become more expert.

The continuous training is a very important routine when changes are detected on the system and thus allowing the improvement of the existent categories and executing predictions with more precision.

Table 4
Results of the transient stability analysis, obtained by simulation (reference results) and by Euclidean ARTMAP considering the continuous training (CT).

Fault number/configuration	Profile generation/load		Security margin interval (*IMS*)		
	Base-case (%)	Seed used	HS	Euclidean ARTMAP	Euclidean ARTMAP with CT
2/1	85	473	9	8	9
2/1	80	277	9	8	9
3/1	120	785	8	7	8
4/1	85	728	8	7	8
5/1	130	121	3	4	3
1/2	120	914	3	4	3
1/2	115	962	3	4	3
3/2	130	748	3	4	3
3/2	125	799	3	4	3
4/2	80	756	8	7	8
5/2	110	981	3	4	3
1/3	125	816	1	4	2
1/3	120	886	1	4	2
1/3	110	924	2	4	3
2/3	130	395	1	4	3
3/3	80	522	9	8	9
4/3	130	296	3	4	3
4/3	115	474	3	4	3
5/3	130	24	1	4	3
5/3	125	483	1	4	3
5/3	120	409	1	4	2
5/3	115	848	1	4	1
5/3	110	794	2	4	2

7. Conclusions

This work developed a procedure for multi-machine electrical power systems TSA using Euclidean ARTMAP neural network. Results were obtained considering an electrical power system corresponding to the south Brazilian region composed of 45 buses, 73 trasmission lines and 10 synchronous machines. The input stimuli of the neural network are the active and reactive power vectors (analog data) and data corresponding to the contingencies and configuration of the system (binary data). The outputs are the security margins of the system represented by intervals between the maximum and the minimum values. Thus, it is possible to represent the outputs in binary code (binary output). The training was executed considering five contingencies (three-phase short-circuit with outage of a transmission line) and three possible configurations of the system. This procedure allows considering several topologies of the electrical network, which is a new proposal not yet, available on the literature. The contingencies are arbitrated as incidents in several points of the electrical network (5 places). Loading levels are randomly arbitrated considering a universe between 80 and 130% of the total load of the system. For the prediction phase, a scheme that includes new patterns to the memory of the neural network was implemented, thus improving the definition region of the recognition categories and consequently the precision of the results. The obtained results are satisfactory, i.e., the proposed objectives were attained, which were to investigate the application of neural networks for executing diagnosis of electrical power systems transient stability. The Euclidean ARTMAP neural network provides very fast solutions (<1 second with a Pentium IV of 1.8 MHz microcomputer, considering the votation strategy with five repetition training). The solution quality is considered adequate. Concluding, the Euclidean ARTMAP neural network gives very fast answers (there is compatibility with applications on line) and the precision is dependent of implementing an adequate training scheme, with a very well elaborated quantity of input/output data (sufficient quantity and quality). The principal reference of the ARTMAP neural network is the characteristics of plasticity. This property allows conceiving mechanisms that include the continuous training (training reinforcement) without destructing the knowledge extracted previously (in most of the neural networks existent on the specialized literature the training reinforcement needs to reinitialize the process destructing the knowledge previously acquired). Therefore, the neural network is continuously improving the performance to provide more precise solutions according to what was described earlier.

The proposed neural network will have a precision increment when the continuous training is intensified. This is the principal proposal of this paper, i.e., a neural network proposal that achieves the improvement as the time passes, including new topologies of the electrical network. This is the principal contribution of this paper when compared to the specialized literature. Therefore, the electrical power system operation can incorporate a new procedure (methodology) to analyze the transient stability with an adequate reliability (precision). Afterwards, from the analysis effectuated considering the topologies provided, the neural network will search the improvement (incremental knowledge acquisition) without overspending the computational costs, and therefore the analysis can be realized with a computational time compatible for applications in real time.

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Multinodal Load Forecasting in Power Electric Systems using a Neural Network with Radial Basis Function

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Keywords: Multinodal Forecast of Electric Load, Artificial Neural Networks, Backpropagation Algorithm, Radial Basis Function.

Abstract. In this paper we present the results of the use of a methodology for multinodal load forecasting through an artificial neural network-type Multilayer Perceptron, making use of radial basis functions as activation function and the Backpropagation algorithm, as an algorithm to train the network. This methodology allows you to make the prediction at various points in power system, considering different types of consumers (residential, commercial, industrial) of the electric grid, is applied to the problem short-term electric load forecasting (24 hours ahead). We use a database (Centralised Dataset - CDS) provided by the Electricity Commission de New Zealand to this work.

Introduction

The load forecasting in electric power systems is of great importance because the most studies (power flow, economic dispatch, expansion planning, buying and selling energy, etc.) can only be effected if there was a reliable estimate of the load to be served.

The literature has several statistical methods that are used for electric load forecasting. The inconvenience of these methods is the need for prior modeling of the load, so that it can be applied later. A direct application that has shown great efficiency is the use of models with intelligent techniques, particularly, the Artificial Neural Network (ANN).

The problem electric load forecasting through ANNs, refers to a solution based on historical data considering the training load, weather information, time, etc. However, load forecasting becomes more complex when it comes to multinodal forecasting, i.e., when estimates of the loads are required at various points of the grid, e.g., substations, transformers, specific areas in distribution systems involving residential, commercial and industrial. It is noteworthy that in the literature dealing with the many load forecasting as a problem of load forecasting global.

Design for multinodal Load forecasting

In this paper, we propose a system for load forecasting multinodal through a scheme whose nodal loads are estimated from knowledge of participation factors of the global burden (global forecast)

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Control Design Applied to a Micro Electro Mechanical System: MEMS Comb Drive

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Abstract and Figures

This paper presents the linear optimal control technique for reducing the chaotic movement of the micro-electro-mechanical Comb Drive system to a small periodic orbit. We analyze the non-linear dynamics in a micro-electro-mechanical Comb Drive and demonstrated that this model has a chaotic behavior. Chaos control problems consist of attempts to stabilize a chaotic system to an equilibrium point, a periodic orbit, or more general, about a given reference trajectory. This technique is applied in analyzes the nonlinear dynamics in an MEMS Comb drive. The simulation results show the identification by linear optimal control is very effective.

SEM image of the chaotic MEM… | a presents the time history for … | (a) Time History for 1 x ; (b) Tim…

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Control Design Applied to a Micro Electro Mechanical System: MEMS Comb drive

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Keywords: Micro electro Mechanical System, Linear Optimal Control, Chaos, Control

Abstract. This paper presents the linear optimal control technique for reducing the chaotic movement of the micro-electro-mechanical Comb Drive system to a small periodic orbit. We analyze the non-linear dynamics in a micro-electro-mechanical Comb Drive and demonstrated that this model has a chaotic behavior. Chaos control problems consist of attempts to stabilize a chaotic system to an equilibrium point, a periodic orbit, or more general, about a given reference trajectory. This technique is applied in analyzes the nonlinear dynamics in an MEMS Comb drive. The simulation results show the identification by linear optimal control is very effective.

Introduction

The field of micro machining is forcing itself a profound redefinition of the nature and to attributes of electronic devices. The technologies of micro-electro mechanical systems (MEMS) have many applications in recent years, for example in wireless communications devices, biological and chemical sensing, force sensing and scanning probe microscopes [1,2,3,4]. This technology allows motion to be incorporated into the function of micro scale devices. However, the design of such mechanical systems may be quite challenging due to nonlinear effects that may strongly affect the dynamics.

Analyses of nonlinear effects have great importance for micro-electro mechanical systems (MEMS). Nonlinear systems have been shown to exhibit chaotic dynamics, as well a periodic oscillations. A chaotic system is a nonlinear deterministic system, which it is very sensitive to small perturbations on its initial conditions, and its long time behavior it is unpredictable.

Chaos control problems consists of attempts to stabilizes a chaotic system to an equilibrium point, a periodic orbit, or more general, about a given reference trajectory. In the last years, a significant interest in control of the nonlinear systems, exhibiting chaotic behavior, has been observed and many of the techniques were discussed in the literature [5,6] and so on, recently

Micro Electro Mechanical System

The micro-electro-mechanical oscillator, illustrated in Fig. 1, consists of the noninterdigitated comb drive actuators (AC and DC), suspending flexures (K), and a shuttle mass (M) [7].

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Fig. 1. SEM image of the chaotic MEM oscillator [7].

The two sets of noninterdigitated comb drives are used for periodic excitation (AC) and for DC tuning (DC) [7,8].

The equation of motion for micro-electro-mechanical Comb drive is

$$m\ddot{x} + c\dot{x} + F_r(x) + F_{es}(x,t) = 0 \quad (1)$$

where, m represents the mass of the shuttle and c damping coefficient. $F_r(x) = \left(k_1 x + k_3 x^3\right)$ is the elastic restoring force provided by the springs, with k_1 and k_3 are stiffness coefficients.

$$F_{es}(x,t) = \left(r_{10} x + r_{30} x^3\right) V_0^2 + \left(r_{1A} x + r_{3A} x^3\right) V(t)^2 \quad (2)$$

is the electrostatic restoring, with the parameters (r_{10} ; r_{30}) and (r_{1A} ; r_{3A}) are electrostatic stiffness coefficients, corresponding to the tuning comb drive and excitation comb drive, respectively; $V(t) = V_A\ 1 + \cos wt$ is signal applied to the AC comb drive (decouples parametric and harmonic excitation) [9] with amplitude V_A. Finally, V_0 is the voltage amplitude to the DC comb drive [10].

Including a dissipative force due to damping in the system, the no dimensional equation of motion is

$$z'' + \alpha z' + \beta z + \delta z^3 + \gamma(1 + \cos\Omega\tau)z + \eta(1 + \cos\Omega\tau)z^3 = 0 \quad (3)$$

where $w_0 = k_1/m$, $\tau = w_0 t$ (scale time), $\Omega = w/w_0$ (scale driving frequency), $z = x/x_0$ (scale displacement, where x_0 is a characteristic length): $\beta = 1 + r_{10}V_0^2/k_1$; $\delta = x_0^2\left(k_3 + r_{30}V_0^2\right)/k_1$; $\gamma = r_{1A}V_A^2/k_1$; $\eta = x_0^2 r_{3A}V_A^2/k_1$ and $\alpha = c/w_0 m$.

Rewriting the dynamical system (3), in the space-state form, through the transformations $x_1 = z$ and $x_2 = \dot{z}$, we will obtain:

$$\begin{cases} \dot{x}_1 = x_2 \\ \dot{x}_2 = -\alpha x_2 - [\beta + \gamma(1 + \cos\Omega\tau)]x_1 - [\delta + \eta(1 + \cos\Omega\tau)]x_1^3 \end{cases} \quad (4)$$

or in the matrix form

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -\beta - \gamma(1 + \cos\Omega\tau) & -\alpha \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - [\delta + \eta(1 + \cos\Omega\tau)] \cdot x_1^3 \quad (5)$$

The parameters for the chaotic system, which was fabricated using standard silicon on insulator processing [11,12] device shown in Fig. 1, are

r_{10}= -4.7e-3 [μN/(V^2 μm)], r_{30}= 1.8e-4 [μN/(V^2 μm^3)],
r_{1A}= 1.1e-3 [μN/(V^2 μm)], r_{30}= -1.4e-4 [μN/(V^2 μm^3)],
k_1= 9.6 [μN/(V^2μm)], k_3= 7.3 [μN/(V^2 μm^3)],
m=1.6e-9 [kg], V_0=46.5 [V], V_A=15 [V], x_0 = 1,
w/2π=2000 Hz.

Figure 2a presents the time history for x_1 and Figure 2b shows the time history for x_2.

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a) b)

c) d)

Fig. 2. (a) Time History for x_1; (b) Time History for x_2; (c) Stability Diagram for Micro-Electro-Mechanical System; and (d) Phase Plane of the chaotic oscillator..

The behavior showing in the figure 2, demonstrate a chaotic movement of MEMS Comb drive with positive Lyapunov exponent (λ=1.2). Figure 2c the stability diagram

Linear Optimal Control

We consider the nonlinear system

$$\dot{x} = Ax + g(x) \tag{6}$$

where $x \in R^n$ is vector of states, $A \in R^{n \times n}$ is constant matrix, and g(x) is vector, whose elements are continuous functions.

In several problems of engineering, physics, economy, ecology, etc., the objective is to choose a control law U that moves the system of the disturbed regime to a desired one, either an equilibrium fixed point or a periodic or non periodic orbit.

Let's consider a vector function $\tilde{x}$ that characterizes the desired trajectory. The controlled system is

$$\dot{x} = Ax + g(x) + U \tag{7}$$

where $U \in R^m$ is a control vector that consists of two parts

$$U = \tilde{u} + u_t \tag{8}$$

The part $\tilde{u}$ is the feed forward control that may be expressed as:

$$\tilde{u} = \dot{\tilde{x}} - A\tilde{x} - g(\tilde{x}) \tag{9}$$

and the control u_f is a linear feedback and it may be expressed as:

$$u_t = Bu \tag{10}$$

where $B \in R^{m \times m}$ is constant matrix.

Defining

$$y = x - \tilde{x} \tag{11}$$

as a variation of the trajectory of the system (7) of the trajectory desired, and admitting (8)-(10), arrive at the variation equation :

$$\dot{y} = Ay + g(x) - g(\tilde{x}) + Bu \tag{12}$$

The non-linear part of the system (11) can be written as

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where $G(x,\tilde{x})$ is a limited matrix, whose elements depend on x and $\tilde{x}$. Admitting (11), the system (9) has the following form:

$$\dot{y} = Ay + G(x,\tilde{x})y + Bu \tag{14}$$

Then we will use the theorem done by Rafikov and Balthazar [13,14,15].

Linear Optimal Control for the Comb Drive System.

The equations (4) describing the Comb Drive system controlled are:

$$\begin{cases} \dot{x}_1 = x_2 + U \\ \dot{x}_2 = -\alpha x_2 - [\beta + \gamma(1+\cos\Omega\tau)]x_1 - [\delta + \eta(1+\cos\Omega\tau)]x_1^3 \end{cases} \tag{15}$$

where the function of control U is determinate by (7).

The matrix A is

$$A = \begin{bmatrix} 0 & 1 \\ -8.5412 & 0 \end{bmatrix} \tag{16}$$

$$\text{and } B = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \tag{17}$$

Verifying the controllability of the system [16], by using $R = [B \mid AB]$, where R is controllability matrix for the pair of matrices [A, B] done by

$$R = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \tag{18}$$

Then, its matrix R has rank 2, because $\det(R) = -1$.

Through that information, we decided to take our unstable system for that stable fixed point, calculating like this the optimal linear control in that point, as it proceeds.

Where $y = \begin{bmatrix} x_1 - \tilde{x}_1 \\ x_2 - \tilde{x}_2 \end{bmatrix}$, $\tilde{x} = \begin{bmatrix} 0.3 \\ 0.02 \end{bmatrix}$ and the matrix $Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is a semi definitive positive matrix.

Obtaining

$$P = \begin{bmatrix} 4.1728 & 0.1164 \\ 0.1164 & 0.4825 \end{bmatrix} \tag{19}$$

and solving the algebraic equation of Riccati (19) the function of optimal control u has the following form:

$$u = 0.0584x_1 + 0.3417x_2 \tag{20}$$

The trajectories of the system without control and controlled system can be shown in the Fig. 3, what it proves the effectiveness of the used technique.

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Figure 3 (a). Time History Uncontrolled and Controlled for x_1 ; (b) Time History Uncontrolled and Controlled for x_2 ; (c) Phase Plane Uncontrolled and Controlled for the System, and (d) Time History Controlled for x_1 and x_2 ;

Conclusion

In this work, we proposed the optimal linear control technique applied to MEMS Comb drive.

A Comb drive system is modeling using the Micro-electro-mechanical system. In comparing of numerical results of controlled system with the non controlled system (Figure 3) we can verify that control orbit generated by Linear Optimal Control (Figure 3) has small diameter.

The technique control reducing the chaotic movement of the Comb drives systems to a small orbit periodic. The Figure 3, illustrate the effectiveness of the control technique to this problem.

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... This is substantially different from the classical magneto-elastic buckled beam, where a shaker is needed to provide a sinusoidal forcing signal only in the horizontal direction [Moon, 2004]. The peculiarity of the system is, then, to exploit a noise-like mechanical, internally generated signal, a further confirmation to the role of noisy signals in the generation of regular chaotic oscillations [Harada et al., 1996;Chotorlishvili et al., 2011;Siewe Siewe et al., 2010;Steur, 2007; Bonato Altran et al., 2011] . ...

A New Chaotic Electro-Mechanical Oscillator

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Análise de Estabilidade de Tensão de Sistemas Elétricos Usando uma Rede Neural Artmap *Fuzzy*

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Resumo. Apresenta-se, neste trabalho, uma nova metodologia para o diagnóstico da estabilidade de tensão estática de sistemas elétricos de potência. Esta metodologia refere-se a um sistema neural de inferência baseado em uma arquitetura neural ARTMAP *fuzzy*, cujo treinamento é realizado a partir de uma base de dados gerada, via simulação, usando-se um programa computacional: cálculo de fluxo de potência, margem de segurança e montagem da base de dados (que constitui os estímulos de entrada / saída da rede neural). Este sistema se destaca por apresentar resultados precisos com alta rapidez de resposta, o que permite, aos usuários, trabalhar com mais flexibilidade em ambiente em que modificações estruturais são requeridas (situação real de operação dos sistemas), se comparadas às demais redes neurais. Como forma de ilustrar a estrutura neural proposta, apresenta-se um aplicação considerando-se um sistema elétrico de potência composto por 45 barras, 72 linhas de transmissão e 10 máquinas síncronas.

Palavras-chave. Sistemas elétricos de potência, estabilidade de tensão, teoria da ressonância adaptativa.

1. Introdução

A estabilidade de tensão refere-se à investigação sobre o comportamento do perfil de tensão nodal do sistema, em destaque, a observação e identificação da deficiência de suporte de tensão em decorrência do aumento da potência consumida.

A estabilidade de tensão pode ser abordada sob dois pontos de vista: estático e dinâmico. O caráter dinâmico requer um tratamento como um fenômeno que pode ser modelado por um conjunto de equações diferenciais ordinárias não-lineares [8]. Trata-se de uma análise complexa e sofisticada, principalmente quando se aborda sistemas de grande porte. Uma forma mais simples, porém igualmente importante da estabilidade de tensão, refere-se à observação do comportamento das tensões nodais, considerando-se o aumento gradativo do perfil do carregamento do sistema, ou seja, a análise qualitativa do ponto de observação. Neste caso, a análise pode ser tratada como um problema linear. As inferências sobre o sistema são baseadas na análise do comportamento do ponto de equilíbrio, mais especificamente, a partir

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da análise da matriz jacobiana (J^0) das equações de potência nodais do sistema, constituindo um problema de fluxo de potência, formulado via método de Newton-Raphson [1, 8]. Deste modo, a partir da análise da matriz jacobiana J^0, pode-se inferir sobre a estabilidade do sistema. Esta análise pode ser realizada, por exemplo, usando a decomposição de autovalores e de autovetores da matriz jacobiana, vetor tangente, entre outras técnicas. O comportamento do sistema, para pequenas perturbações, pode ser analisado através dos autovalores da matriz J^0. Se todos os autovalores de J^0 forem reais e positivos, pode-se concluir que o sistema é estável para pequenas perturbações. Contudo, a instabilidade se caracteriza por haver, pelo menos, um autovalor com a parte real negativa. O ponto limítrofe ocorre quando a matriz jacobiana J^0 torna-se singular (há a ocorrência de, pelo menos, 1 autovalor nulo). Neste caso, o determinante da matriz jacobiana J^0 torna-se, também, nulo. Esta tem sido a forma abordada na maioria das referências bibliográficas pertinentes. Neste artigo, propõe-se o desenvolvimento de um sistema neural para execução do diagnóstico da estabilidade (de caráter estático) de tensão de sistemas de energia [1, 8]. O funcionamento das redes neurais é constituído por duas fases fundamentais: treinamento e análise. A fase de treinamento requer uma grande quantidade de processamento para sua realização, enquanto que a fase de análise é efetivada, praticamente, sem esforço computacional. Esta é, portanto, a principal justificativa para o uso de redes neurais para a resolução de problemas complexos que exigem soluções rápidas, como é o caso de aplicações em tempo real. A fase de treinamento da rede neural é realizada, na maioria das vezes, através da técnica de retropropagação [6], que é reconhecidamente muito lenta e, em muitos casos, não converge, principalmente quando se trabalha com grandes bancos de dados. Novas estruturas neurais têm sido abordadas, visando resolver esta questão do excessivo tempo de processamento para a realização do treinamento. Neste contexto, destacam-se as redes neurais da família ART (*Adaptive Resonance Theory*) [3], que apresentam as características de estabilidade (capacidade de aprender através do ajuste dos seus pesos) e plasticidade (capacidade de continuar a aprender com a inclusão de novos padrões, sem perder a memória relacionada aos padrões anteriores). Assim, neste artigo apresenta-se um sistema neural para a execução do diagnóstico da estabilidade estática de tensão. A rede neural usada é uma arquitetura ART-descendente [3], ou seja, uma rede ARTMAP *fuzzy* com inclusão de melhorias, visando torná-la, principalmente, mais precisa, se comparada à formulação original [3]. Esta rede neural possui, como critério de treinamento, um índice (margem) de segurança baseado na análise de sensibilidade da função determinante da matriz J^0, calculada empregando-se conceitos da álgebra de Kronecker [5]. Visando testar o desempenho da metologia proposta, apresenta-se uma aplicação considerando-se um sistema elétrico de potência composto por 10 barramentos, 72 linhas de transmissão e 10 máquinas síncronas.

2. Metodologia Proposta

A seguir, apresenta-se o esquema proposto (sistema neural) para realizar a análise de estabilidade de tensão de sistemas elétricos de potência. O funcionamento das redes

neurais é basicamente, em duas fases principais: (1) treinamento ou aprendizado; (2) testes e diagnósticos. A rede neural usada é uma arquitetura ARTMAP *fuzzy* [3]. A fase de treinamento é realizada usando um "professor" representado por um simulador (programa computacional *Simul* [4] implantado no Laborátorio Computacional do Departamento de Engenharia Elétrica - Unesp - Campus de Ilha Solteira) que realiza cálculo de redes elétricas: matrizes de rede, fluxo de potência e análise de estabilidade transitória. Este programa foi adaptado para executar, também, os cálculos associados à estabilidade estática de tensão, conforme o critério apresentado na Seção 3, e a base de dados, constituída por pares de padrões de entrada e de saída da rede neural, para a realização do treinamento neural. O fluxograma, da referida metodologia, é mostrado na Figura 1.

A parte do fluxograma, com linhas tracejadas, refere-se ao módulo que executa o treinamento continuado. Ou seja, sempre que houver disponibilidade de dados de entrada para a execução da análise de estabilidade de tensão pela rede neural (bloco 1), se julgado como sendo um vetor (P, Q, Z) relevante, usa-se este vetor e a saída ($\mathcal{M}$) associada para serem inseridos na base de dados, via adaptação dos pesos. O vetor Z contém as informações binárias que codificam a topologia da rede elétrica, os dados das contingências, etc.

3. Critério de Análise da Estabilidade Estática de Tensão

A análise de estabilidade estática de sistemas de energia consiste na investigação comportamental do modelo correspondente à solução das equações linearizadas do fluxo de potência [8], executando-se a barra de referência [1]:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = [J^0] \begin{bmatrix} \Delta\theta \\ \Delta V \end{bmatrix}$$

sendo: ΔP o vetor das potências ativas das barras PV e QP; ΔQ o vetor das potências reativas das barras PQ; $\Delta\theta$ o vetor dos ângulos nas barras PV e PQ; ΔV o vetor das tensões nodais das barras PQ e; $J^0 \underline{\Delta}$ matriz jacobiana $= \begin{bmatrix} H & N \\ M & L \end{bmatrix}$ em que *H, N, M* e *L* são as submatrizes de J^0 correspondente a $\partial P/\partial\theta$, $\partial P/\partial V$, $\partial Q/\partial Q\theta$ e $\partial Q/\partial V$, respectivamente.

Neste artigo, pretende-se, então, investigar a questão da estabilidade estática de tensão, através da análise de sensibilidade, usando como função o determinante da matriz J^0. Ressalta-se que o cálculo do determinante de uma matriz pode ser realizado de forma bastante rápida e eficiente [7], se comparado com os principais métodos comumente disponíveis na literatura, ainda que esta função não seja utilizada neste trabalho e sim a sua derivada. Desta forma, através de expansão por série de Taylor, aproximação de primeira ordem, pode-se estimar a variação do determinante da matriz J^0 ($\det(J^0)$), como mostra-se a seguir.

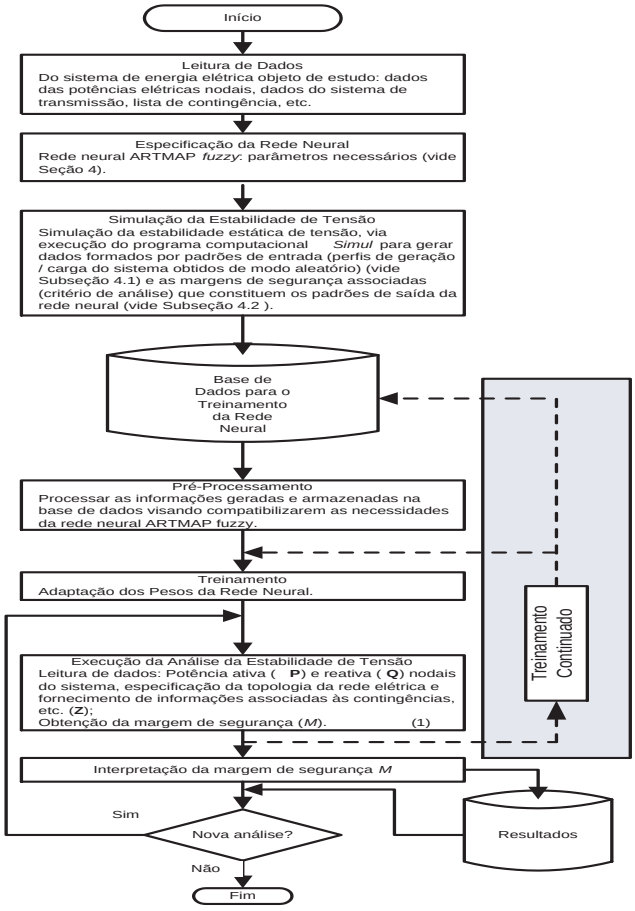

Figura 1: Fluxograma do sistema neural proposto para realizar análise de estabilidade de tensão de sistemas elétricos de potência.

Considere [5]:

$$\begin{aligned} f(J) &= f(J^0) + \epsilon Tr\left\{D\left(J^0\right)^T\right\} + \phi\left(\epsilon\right) \\ &= f(J^0) + \epsilon\left\langle\psi(D(J^0)), \psi(\Delta J)\right\rangle + \phi(\epsilon), \quad \epsilon \in \mathbb{R}. \end{aligned} \tag{3.1}$$

sendo: $J = J^0 + \epsilon\Delta J$; $\langle u, v\rangle$ o produto interno entre u e v; ∇ o gradiente de uma função escalar; $D(J^0)\underline{\Delta}\psi^{-1}(\nabla_\psi f(J^0)) = \partial f(J)/\partial J|_0$; $\nabla_\psi f(J^0) = \psi(D(J^0))$; Tr o traço de uma matriz; ; $\Delta J = \begin{bmatrix} \Delta J_H & \Delta J_N \\ \Delta J_M & \Delta J_L \end{bmatrix}$; ϵ = parâmetro de perturbação e, $\psi(\Delta J)\ \underline{\Delta}\ [\Delta J_{11} \ldots \Delta J_{p1} \ldots \Delta J_{1p} \ldots \Delta J_{pp}]^T$.

O determinante da matriz J pode ser expresso por [5]

$$\det(J) = \det\left\{J^0\left(I + \epsilon(J^0)^{-1}\Delta J\right)\right\},\ J^0 \in \mathbb{R}^{p\times p},\ (J^0)^{-1}\exists \tag{3.2}$$

sendo I a matriz identidade $(p \times p)$.

A equação (3.2) pode ser desenvolvida do seguinte modo:

$$\begin{aligned} \det(J) &= det(J^0)det(I + \epsilon(J^0))^{-1}\Delta J \\ &= det(J^0) + \epsilon Tr\left(det(J^0)(J^0)^{-1}\Delta J\right) + \phi(\epsilon) \end{aligned} \tag{3.3}$$

Como $det(J^0)(J^0)^{-1} = adj(J^0)$, então, a equação (3.6), utilizando-se as equações (3.1-3.3), pode ser expressa por:

$$det(J) = det(J^0) + \epsilon \left\langle \psi(adj(J^0)^T), \psi(\Delta J) \right\rangle + \phi(\epsilon)$$

sendo adj(J^0) a matriz adjunta de J^0 , *i.e.*, adj(J^0) = cof$((J^0)^T)$.

A partir da equação (3.2), conclui-se que a sensibilidade da função $det(J^0)$ pode ser expressa por :

$$\partial det(J)/\partial J = \psi(adj(J)^T)$$

Dividindo-se a equação (3.2) por $det(J^0)$, obtém-se:

$$\tau = 1 + \Delta\tau + \phi(\epsilon) \tag{3.4}$$

sendo : τ a função determinante normalizada, ou seja $\tau \ \underline{\Delta} \frac{\det(J)}{\det(J^0)}$, $\psi(\epsilon) = \frac{\phi(\epsilon)}{\det(J^0)}$, e $\Delta\tau = \epsilon \left\langle \psi(((J^0))^{-1})^T, \psi(\Delta J) \right\rangle$.

Também, a partir da equação (3.4), conclui-se que:

$$\left(\partial det\left(J^0\right)/\partial J|_{J=J^0}\right) = \psi(((J^0)^{-1})^T) \tag{3.5}$$

A equação (3.5) representa a sensibilidade da função determinante normalizada. Assim, fazendo-se $a = \psi\{((J^0)^{-1})^T\}$ (vetor) e $C\underline{\Delta}\psi[(J^0)^{-1}]^T = \begin{bmatrix} C_H & C_N \\ C_M & C_L \end{bmatrix}$ obtém-se a seguinte representação para a que é a sensibilidade da função determinante normalizada, na forma vetorial:

$$a = [C_{11} \cdots C_{p1} \cdots C_{1p} \cdots C_{pp}]^T .$$

A variação linear da função determinante normalizada $\Delta\tau$ pode ser expressa por:

$$\Delta\tau = \epsilon \sum_{i=1}^{p} \sum_{j=1}^{p} C_{ij}\Delta J_{ij}$$

Deve-se observar que a matriz ΔJ é uma estrutura esparsa, semelhante à esparsidade da matriz Ybarra. Assim, poucos são os elementos de ΔJ que são diferentes de zero.

Desconsiderando-se os termos não-lineares $\phi(\epsilon)$, a partir da equação (3.4), fazendo-se $\tau = 0$ e $\epsilon = 1$, obtém-se:

$$\langle a, \psi(\Delta J) \rangle = -1. \tag{3.6}$$

A equação (3.6) estabelece a relação entre os vetores $\psi\{((J^0)^{-1})^T\}$ e $\psi(\Delta)$ tais que reproduzem, considerando-se o modelo linear, um valor nulo para a função $\det(J)$, ou seja, uma condição limite para a estabilidade de tensão. Deste modo, a partir desta equação, pode-se estimar a distância euclidiana que corresponde à margem de segurança ou a distância em que o ponto de operação do sistema encontra-se em relação à condição limítrofe da estabilidade estática:

$$\mathcal{M} = \sqrt{\frac{1}{\langle a, a\rangle}} \tag{3.7}$$

correspondente a uma solução da equação (3.6) para $\psi(\Delta J) = -\dfrac{a}{\langle a, a\rangle}$.

A margem de segurança $\mathcal{M}$, calculada através da equação (3.7), corresponde a uma solução de mínima norma.

A escolha da representação dada pela equação (3.4), tem sido importante, isto porque permite realizar as análises levando-se em conta a estrutura real do sistema (sem redução).

Definição 3.1. *Autovalor crítico* ($\lambda_{min}(J^0)$) *é definido como sendo o menor autovalor positivo da matriz jacobiana* J^0.

O autovalor crítico (definição 3.1) corresponde ao autovalor de interesse para análise da estabilidade de tensão, tendo em vista que é aquele que indica a máxima aproximação do colapso de tensão e, portanto, apresenta maior possibilidade de se tornar nulo à medida que houver o aumento da demanda de energia elétrica.

A partir da álgebra matricial, pode-se chegar ao seguinte resultado [5, 2]:

$$\lambda_{min}(J^0) \geq \mathcal{M} \geq 0.$$

Deste modo, a margem de segurança $\mathcal{M}$ representa uma estimativa pessimista (favorável à segurança do sistema) do parâmetro $\lambda_{min}(J^0)$. A estimativa de $\lambda_{min}(J^0)$ pode ser calculada, de forma alternativa, aplicando-se a norma matricial inversa de J^0. Este resultado é similar ao apresentado na referência [8]. O índice $\mathcal{M}$ será, então, usado como critério de inferência da estabilidade estática da tensão de sistemas elétricos de potência, via redes neurais, como formulado nesta seção.

4. Rede Neural ARTMAP *Fuzzy*

A rede neural ARTMAP *Fuzzy* [3] é formada por 2 módulos ART *Fuzzy* interligados através de um mecanismo chamado inter-ART, como será descrito adiante. Assim, primeiramente descreve-se o módulo ART *Fuzzy*. Para uma melhor fundamentação operacional das redes neurais ART-descendentes, os vetores usados são representados por linhas e não por colunas, como habitualmente é adotado na literatura especializada.

Os dados de entrada são denotados por vetor $a = [a_1 \ldots a_R]$ R-dimensional, e é normalizado com o intuito de evitar a proliferação de categorias. Assim, $\overline{a} = \dfrac{a}{|a|}$ sendo $\overline{a}$ o vetor de entrada normalizada e $|a| = \sum a_i$.

O vetor de entrada será um vetor $2R$-dimensional, denotado por:

$$E = [\overline{a}\ \overline{a}^c] = [\overline{a}_1 \cdots \overline{a}_R\ \overline{a}_1^c \cdots \overline{a}_R^c]$$

em que $\overline{a}^c = \overline{a}_i^c = 1 - \overline{a}_i^c$.

O vetor de atividade de F_2 é simbolizado por $y = [y_1 \cdots y_S]$ sendo S o número de categorias criadas em F_2. Deste modo, tem-se:

$$y = \begin{cases} 1, & \text{se o nó } J \text{ de } F_2 \text{ é ativo} \\ 0, & \text{caso contrário} \end{cases}$$

Os parâmetros utilizados no processamento da rede ART *fuzzy* são: parâmetro de escolha ($\eta > 0$), taxa de treinamento ($\beta \in [0,1]$) e parâmetro de vigilância ($\rho \in [0,1]$). Inicialmente todos os pesos possuem valor igual a 1, indicando assim que não existe nenhuma categoria ativa.

A rede neural ARTMAP [3] é uma rede mais sofisticada, que emprega o treinamento supervisionado, *i.e.*, baseado em estímulos de entrada-saída. É composta por dois módulos ART *fuzzy*; ART_a e ART_b interconectados através de módulo de memória associativa inter-ART, F^{ab}, que possui um mecanismo auto-regulador denominado *match tracking*, que tem a incumbência de maximizar a generalização e minimizar o erro da rede.

O vetor de entrada da rede ART_a é representado pelo vetor $a = [a_1 \ldots a_{R_a}]$, R_a-dimensional, e o vetor de entrada da rede ART_b (que corresponde à saída desejada, referente ao padrão de entrada da rede ART_a) é representado pelo vetor $b = [b_1 \ldots b_{R_b}]$, R_b-dimensional [3].

Os parâmetros usados na rede ARTMAP *fuzzy* são os mesmos usados na rede ART *fuzzy*. A diferença da rede neural ART *fuzzy* é o parâmetro de vigilância do módulo inter-ART, ρ_{ab} ($\rho_{ab} \in [0,1]$).

A rede ARTMAP realiza o processamento de duas redes ART (ART_a e ART_b), sendo que, após a ressonância ser confirmada em cada rede, tem-se C categoria ativa para a rede ART_a e K categorias ativas para a rede ART_b.

Através do processo *match tracking* é verificado se a categoria ativa no ART_a corresponde ao vetor de saída desejado apresentado no ART_b. O critério de vigilância é dado por [3]:

$$\frac{|y^b \wedge w_{Ck}^{ab}|}{|y^b|} \geq \rho_{ab} \tag{4.1}$$

em que y^b é o vetor saída do ART_b (padrão de atividade F_2^b).

Se (4.1) não for satisfeito, é realizado um incremento mínimo no parâmetro de vigilância ART_a, suficiente apenas para excluir a atual categoria e selecionar outra categoria que se tornará ativa e entrará novamente no processo até que (4.1) seja obedecida. Com a ressonância confirmada, os pesos dos módulos ART_a e ART_b, são atualizados utilizando o mesmo critério da rede neural ART *fuzzy*. A adaptação para o módulo inter-ART é efetuada da seguinte forma: $w_{Ck}^{ab} = 1$ para $k = K$ e $w_{Ck}^{ab} = 0$ para $k \neq K$.

4.1. Estímulo de Entrada da Rede Neural ARTMAP *Fuzzy*

Os vetores padrões de entrada da rede neural são definidos como [4]:

$$X = [P\ Q\ Z]$$

sendo: X o vetor padrão de entrada da rede neural; $P = [P_1\ P_2\ \cdots P_n]$; $Q = [Q_1\ Q_2\ \cdots Q_n]$; P_i a potência ativa da i-ésima barra do sistema; Q_i a potência reativa da i-ésima barra do sistema; Z o vetor contendo as informações em código binário e, n o número de barras do sistema.

Para a realização do treinamento - extração do conhecimento baseado em estímulos de entrada / saída - deve-se proceder a apresentação de um conjunto de dados, no caso, $X = [P\ Q\ Z]$ (entrada) e Y (saída), constituindo um conjunto de pares de treinamento. Trata-se de geração por vetores P e Q, para a geração e para a carga do sistema por um procedimento de distribuição aleatória da geração (despacho aleatório para atender a demanda) e, também, de distribuição aleatória da carga (demanda aleatória) [4]. O vetor Z é formado por informações (em código binário), representando a topologia da rede elétrica, os dados das contingências, etc.

Para se gerar um grande espectro de variação da demanda, o critério a ser usado refere-se à distribuição aleatória da demanda e, conseqüentemente, da geração para atender a demanda, tomando-se variações percentuais da carga e da geração em torno do caso base (considerando como sendo o perfil de geração / carga de 100%).

4.2. Estímulo de Saída da Rede Neural ARTMAP *Fuzzy*

Os estímulos de saída compreendem, neste artigo, os valores da margem de segurança ($\mathcal{M}$), calculados conforme descrito na Seção 3. Ou seja, os padrões de treinamento correspondem aos parâmetros: $X_j = [P_j\ \ Q_j\ \ Z_j]$ (entradas); $Y_j = [\underline{\mathcal{M}_\mathrm{j}}]$ (saidas); $j = 1,\ 2,\ \cdots,\ np$, sendo $\underline{\mathcal{M}_\mathrm{j}}$ a $j-$ésima margem de segurança representada por classes de severidade; np o número de pares de padrões para a fase de treinamento.

A saída $\underline{\mathcal{M}_\mathrm{j}}$ é codificada por classes (em código binário), sendo que cada classe expressa o grau de severidade.

5. Aplicação

Apresentam-se, a seguir, os resultados obtidos através do uso do método proposto, considerando-se um sistema composto por 45 barras, 72 linhas de transmissão e 10 máquinas síncronas. Os dados das máquinas síncronas e do sistema de transmissão pode ser visto na Figura 2.

O treinamento da rede neural foi realizado considerando-se um conjunto de 700 perfis de geração / carga e respectivas margens de segurança $\mathcal{M}$. Cada perfil corresponde a um redespacho de geração em relação ao caso base, realizado de forma aleatória para atender a demanda, também fixada de modo aleatório em cada barra. O universo de variação de carga encontra-se correspondido entre 65 e 135%, em relação à carga total do sistema. Portanto, cada perfil é gerado considerando-se um percentual de variação em torno do estado nominal (caso base) e determinada semente para o processo de geração de seqüências aleatórias. Assim, para um mesmo percentual, diferentes sementes poderão gerar diferentes despachos de geração de diferentes perfis de carga. Realizado o treinamento da rede neural, os testes (análise da estabilidade de tensão) podem ser efetivados, cujos resultados são mostrados na Tabela 1. Os resultados da análise são apresentados na forma de classes de 1 a 10.

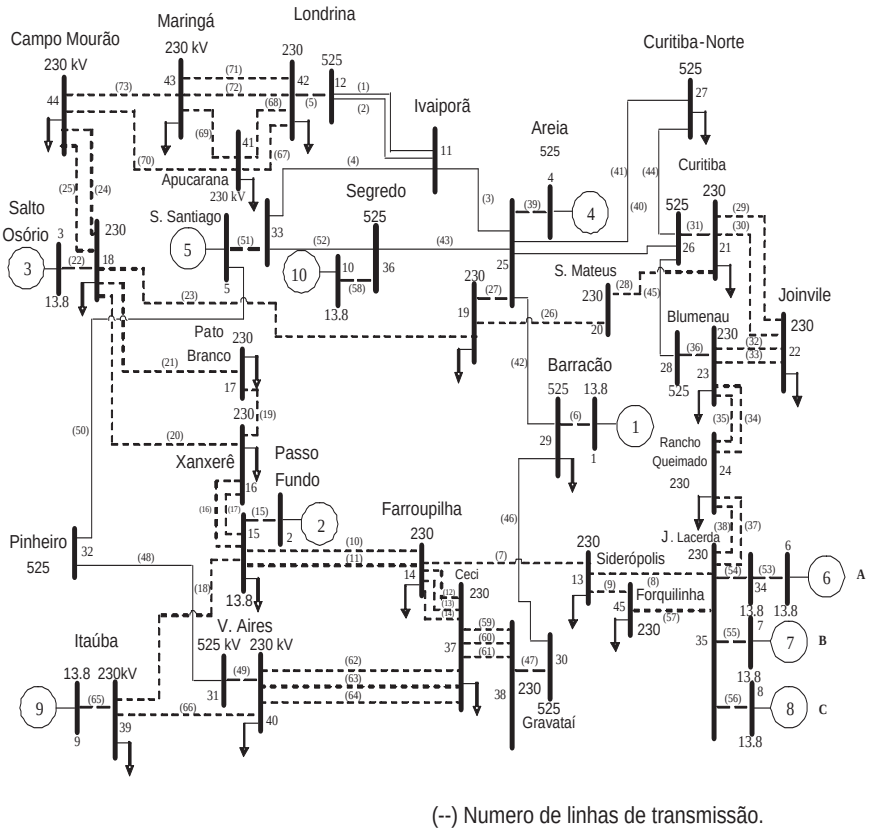

Figura 2: Representação do sistema teste.

Estas classes são adotadas visando constituir as saídas em código binário, *i.e.*, um modo mais adequado para se usar as redes neurais da família ART. Assim, adotou-se a seguinte definição das classes $1, 2, \ldots, 10$ (representação com 4 *bits*) correspondem a $0 \leq \mathcal{M} \leq 0.1,\ 0.1 \leq \mathcal{M} \leq 0.2,\ \ldots,\ 0,9 \leq \mathcal{M} \leq 1.0$, respectivamente. Com estas definições, pode-se inferir sobre as "distâncias" que cada perfil de geração / carga se encontra em relação à fronteira da estabilidade de tensão. Tais intervalos, associados às classes, podem ser definidos, aumentando-os ou diminuindo-os de acordo com o interesse do usuário. Neste caso, devendo-se ajustar o número de *bits* em função da maior ou da menor discretização usada. Ressalta-se que os acertos do sistema neural são superiores a 85%, considerando-se 700 simulações realizadas. Este percentual, certamente irá aumentar, à medida em que novos padrões forem incorporados à base de dados, através do processo referente ao treinamento continuado.

Tabela 1: Comparação dos métodos.

		Margem $\mathcal{M}$				Margem $\mathcal{M}$	
%	Semente	*Simul*	Artmap Fuzzy	%	Semente	*Simul*	Artmap Fuzzy
-32.5	123	.5994	6	17.5	123	.6722	7
-27.5	123	.6354	7	17.5	6691	.6183	7
-22.5	614	.7096	8	22.5	145	.6055	6
-12.5	31	.6551	8	22.5	221	.6528	7
-7.5	123	.7003	7	22.5	377	.5662	6
-2.5	7116	.7134	8	27.5	7116	.4349	4
7.5	31	.7089	8	27.5	8555	.5840	6
12.5	74	.6694	7	32.5	123	.5873	6

6. Conclusão

Foi apresentada uma nova metodologia para análise de estabilidade de tensão estática de sitemas elétricos de potência multinodal. Trata-se de uma metodologia usando uma rede neural ARTMAP *fuzzy*, cujo treinamento é realizado a partir de uma base de dados gerada via simulação (realizada usando o programa computacional *Simul* [4]); cálculo de fluxo de potência, bem como outras grandezas que foram implementadas visando atender a necessidade do estudo proposto. Tendo em vista que a estratégia de obtenção desta base de dados (definição de perfil de geração e de carga de modo aleatório) permitiu estabelecer um espectro de variação de modos operativos do sistema que "simule"estados operativos reais, a arquitetura ARTMAP *fuzzy*, tal como as demais redes da família ART, apresenta a característica de plasticidade que admite implementar o treinamento continuado, o que se diferencia, de forma vantajosa, em relação às tradicionais redes neurais comumente usadas na literatura especializada.

Abstract. This work presents a new methodology for the analysis of the static voltage stability in electric power systems. It is a neural inference system based on *fuzzy* ARTMAP architecture. This system presents accurate results with very fast response, being able to realize real time analysis. Besides, it allows the users work with flexibility where structural modifications are required (real situation in system operation), when compared to tradicional neural networks used in specialized literature. To illustrate the proposed neural network an apllication is presented considering an eletrical power system composed by 45 buses, 72 transmission line and 10 syncrhonous machines.

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Sensitivity analysis by neural networks applied to power systems transient stability

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Abstract

This work presents a procedure for transient stability analysis and preventive control of electric power systems, which is formulated by a multilayer feedforward neural network. The neural network training is realized by using the back-propagation algorithm with fuzzy controller and adaptation of the inclination and translation parameters of the nonlinear function. These procedures provide a faster convergence and more precise results, if compared to the traditional back-propagation algorithm. The adaptation of the training rate is effectuated by using the information of the global error and global error variation. After finishing the training, the neural network is capable of estimating the security margin and the sensitivity analysis. Considering this information, it is possible to develop a method for the realization of the security correction (preventive control) for levels considered appropriate to the system, based on generation reallocation and load shedding. An application for a multimachine power system is presented to illustrate the proposed methodology.
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Keywords: Sensitivity analysis; Preventive control; Transient stability; Neural networks; Back-propagation

1. Introduction

Sensitivity analysis [1] is a very important tool to solve several problems present in many areas of human knowledge: engineering, mathematics, physics, economy, medicine, biology, etc., especially when nonlinearities are involved. Therefore, it is possible to infer about the behavior of the system face to parametric variations without the need of solving a problem that involves great complexity, described by a set of nonlinear differential and algebraic equations. The comportamental conclusions are extracted from the calculation of the derivative function under analysis. This is the problem studied in this work, analysis of transient stability for electric energy systems and specifically the preventive control problem [2].

Transient stability analysis is one of the main studies used in electric power systems (EPS). It is a procedure to evaluate the effects caused by perturbations which origin great deviations on the angles of the synchronous machines, e.g., short-circuit, outage/input of electric equipment. In this case, the model of the system is described by a set of nonlinear algebraic and differential equations [3]. Due to unstable cases and/or equipment capability violations, it is necessary to adopt procedures that can lead the system to a secure state, known as security control. The methods for dynamical preventive control have been proposed recently and the publication available in the literature is not abundant, e.g., [2–6].

This work presents a methodology based on neural networks [7–9] to analyze the transient stability – considering short-circuit faults with transmission line outages – and, principally to provide the sensitivity analysis of EPS, that represent the necessary instruments for implementing the preventive control. Neural networks are important resources to deal with the preventive control problem, due to the training be an off-line activity and the analysis be concluded with minimal computational effort (basically the calculus with the input and output of the neural network), being useful for applications in real time. It is important to emphasize that the sensitivity calculus is carried out without computational effort. On the other hand, to obtain the sensitivity model by conventional procedures, require a large quantity of complex matrix calculation, spending much time, principally for applications in huge systems.

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The neural network used is a feedforward multilayer with training by back-propagation (BP) algorithm [10]. The BP algorithm training rate is adjusted by a fuzzy controller [11–14], which monitories the global error and the global error variation during the training as well the adaptive process [15,16]. It is an optimal mechanism that reduces the convergence time and improves the precision of the results, as observed in ref. [15]. The variables used on the training are causal variables of a problem of transient stability analysis (active and reactive nodal electric power which are the input neural network stimulus) and the security margins (output neural network stimulus) generated using the potential energy boundary surface (PEBS) iterative method [17], microcomputer version. The security margin expressed in function of total energy can be interpreted as being a measure of the distance in relation to the condition of the instability of the system. The sensitivity model is referred to the relation with the security margin and the nodal electric power. Thus, the generation reallocation and load shedding necessary for obtaining a secure state of the system can be evaluated, that is, a security level considered adequate for transient stability. It is a tool that can either aid the prevention or, at least, minimize the effects of a blackout [18–20]. An application considering a multimachine system is presented for testing the proposed methodology.

2. System model

Considering an electrical power system composed of *ns* synchronous machines, the dynamical behavior of the *i*th machine, related to Center of Angles (CA), is described by the following differential equation (classical model) [12,21]:

$$M_i\ddot{\theta}_i - P_i(\theta) = 0, \quad i \in N \tag{1}$$

where

$$P_i(\theta) = \mathrm{Pm}_i - \mathrm{Pe}_i - \frac{M_i \mathrm{PCOA}}{MT}; \tag{2}$$

M_i is $2H_i/\omega_s$; $\omega_s \triangleq$ synchronous speed of the rotor, $2\pi f_0$; H_i is the inertia constant (s); f_0 is the nominal frequency of system (Hz); $\theta_i \triangleq$ rotor angle of *i*th synchronous machine related to CA (electrical radians), $\delta_i - \delta_0$; δ_i is the rotor angle of *i*th synchronous machine in relation to synchronously rotating reference frame (in electrical radians); δ_0 is $\sum_{j \in N} M_j\delta_j$; Pm_i is the mechanical power of input (pu); Pe_i is the electrical power of output (pu); PCOA $\triangleq$ accelerating power of CA, $\sum_{j \in N}(\mathrm{Pm}_j - \mathrm{Pe}_j)$; MT is $\sum_{j \in N} M_j$; $N \triangleq$ index set of synchronous machines that comprises the system, $\{1, 2, \ldots, ns\}$; *ns* is the number of electrical synchronous machines.

3. Transient stability analysis

The transient stability analysis of EPS, considering a contingency of index *r*, is effectuated using the security margin criterion [12,17,21]:

$$\mathcal{M}_r = \frac{E_{\mathrm{crit}_r} - E_{\mathrm{e}_r}}{E_{\mathrm{crit}_r}} \tag{3}$$

where E_{crit_r} is the total critical energy of the system; E_{e_r} is the total energy of the system evaluated on the instant of the fault elimination (te).

The critical energy (E_{crit}) and the critical time (t_{crit}), is determined by the iterative PEBS method [4,16], or another procedure that presents a similar result, principally in relation to precision. The total energy related to system (1), is given by [12,17,21]:

$$E(\theta, \omega) = E_{\mathrm{c}}(\omega) + E_{\mathrm{p}}(\theta) \tag{4}$$

where

$$E_{\mathrm{c}}(\omega) = \text{kinetic energy} = \frac{1}{2}\sum_{i \in N} M_i w_i^2; \tag{5}$$

$$E_{\mathrm{p}}(\theta) = \text{potential energy} = -\sum_{i \in N} \int_{\theta_i^p}^{\theta_i} P_i(\theta)\, \mathrm{d}\theta_i. \tag{6}$$

Then, the transient stability for the *r*th contingency is evaluated by the security margin on the following way [17]:

- if $\mathcal{M}_r \geq 0$, the system is considered *stable*, for transient stability;
- if $\mathcal{M}_r < 0$, the system is considered *unstable*, for transient stability.

4. Dynamic preventive control

Considering a list composed of *S* contingencies, the security margin of the system must satisfy the following relation [4]:

$$\mathcal{M} \geq M_{\min} \tag{7}$$

where $M_{\min}$ is the minimum limit of the security margin of the system ($M_{\min} > 0$); $\mathcal{M} \triangleq \min(\mathcal{M}_r, r = 1, 2, \ldots, S)$.

The control actions must cause modifications on the security margins and the following relations must be satisfied [19,4]:

$$\mathcal{M}_r = (\mathcal{M}_r^0 + \Delta\mathcal{M}_r) \geq M_{\min}, \quad r = 1, 2, \ldots, S \tag{8}$$

where $\mathcal{M}_r$ is the security margin referred to the *r*th contingency.

The necessary changing ($\Delta\mathcal{M}_r$) to correct the security margin – in terms of a vector $\boldsymbol{X}$ – is estimated by the sensitivity theory, of first order, according to [19]:

$$\Delta\mathcal{M}_r \cong \left[\frac{\partial\mathcal{M}_r}{\partial \boldsymbol{X}}\right]^{\mathrm{T}} \Delta\boldsymbol{X} \tag{9}$$

or

$$\Delta t_{\mathrm{crit}_r} \cong \left[\frac{\partial t_{\mathrm{crit}_r}}{\partial \boldsymbol{X}}\right]^{\mathrm{T}} \Delta\boldsymbol{X} \tag{10}$$

where $\partial\mathcal{M}_r/\partial\boldsymbol{X}$ is the sensitivity of the security margin in relation to the vector $\boldsymbol{X}$; $\partial t_{\mathrm{crit}}/\partial\boldsymbol{X}$ is the sensitivity of the critical time in relation to the vector $\boldsymbol{X}$; $\Delta\boldsymbol{X}$ is the vector corresponding to the changing on the components of vector $\boldsymbol{X}$.

Equations (9) and (10) represent the sensitivity models considering the interest variables: security margin and critical time, respectively.

The vector $\boldsymbol{X}$, in this work, is represented by the nodal active power. The sensitivity $\partial \mathcal{M}_r/\partial \boldsymbol{X}$ is developed in Section 6 by neural networks.

5. Neural network structure

This section presents a multilayer feedforward neural network, whose training is realized by a back-propagation algorithm, which is a systematic procedure of adaptation of the neural network weights based on the descendent gradient [15].

The ith output element (neuron) [6] is a linear combination of the element inputs x_j that are connected to the element i by the weight w_{ij}:

$$\vartheta_i = \sum_j w_{ij} x_j \tag{11}$$

Each element can have a bias w_0 plus an extra constant input $x_0 = +1$. The linear output ϑ_i is finally converted in a nonlinear function, as a sigmoid, relay [8,10], etc. The relay functions are appropriated for binary systems, while the sigmoid functions can be employed for both continuous and binary systems.

6. Neural network training using the adaptive process

The back-propagation algorithm (BP) is realized on the following way:

(1) Present to the network a pattern $\boldsymbol{X}$, that gives an output $\boldsymbol{Y}$.
(2) Calculus of the error (difference between the desired value and the output) for each output.
(3) Determination of the back-propagated error by the network associated to the partial derivative of the quadratic error of each element related to the weights.
(4) Adjustment of the weights of each element.
(5) A new pattern is presented to the network and the process is repeated until the convergence, i.e. (|error| ≤ predefined tolerance).

Random numbers is adopted for the initial weights, considering the interval {0,1} [10]. The back-propagation BP algorithm consists of adapting the weights in such way that the quadratic error be minimized. The summation of the instantaneous quadratic error of each neuron of the last layer (network output) is given by:

$$\varepsilon^2 = \sum_{i=1}^{no} \varepsilon_i^2 \tag{12}$$

where ε_i is the $d_i - y_i$; d_i is the desired output of the ith element of the last layer; y_i is the output of the ith element of the last layer; no is the quantity of neurons of the last layer.

Considering the ith neuron of the network and using the descent gradient method [10], the adjustment of the weights is formulated by [10]:

$$\boldsymbol{V}_i(r+1) = \boldsymbol{V}_i(r) + \theta_i(r) \tag{13}$$

where $\theta_i(r)$ is $-\gamma[\nabla_i(r)]$; γ is the control of the stability parameter or training rate; $\nabla_i(r)$ is the gradient of the quadratic error related to the weights of the neuron i; $\boldsymbol{V}_i \triangleq$ vector containing the weights of the neuron i, $[w_{0i}\ w_{1i}\ w_{2i} \ldots w_{ni}]^{\mathrm{T}}$.

The adopted direction in ref. (13) for minimizing the objective function of the quadratic error corresponds to the opposite direction of the gradient. The parameter γ determines the size of the vector $\theta_i(r)$. The nonlinear function used is the sigmoid function defined by [7,10]:

$$y_i = \frac{1 - \mathrm{e}^{(-\lambda\vartheta_i + \rho)}}{1 + \mathrm{e}^{(-\lambda\vartheta_i + \rho)}} \quad \text{for } y_i \in [-1, +1] \tag{14}$$

or

$$y_i = \frac{1}{1 + \mathrm{e}^{(-\lambda\vartheta_i + \rho)}} \quad \text{for } y_i \in [0, +1] \tag{15}$$

where λ is the constant that determines the inclination of the function y_i; ρ is the constant that determines the translation of the function y_i.

The gradient $\nabla_i(r)$ is represented by:

$$\nabla_i(r) = \frac{\partial \varepsilon_i^2}{\partial V_i(r)} = 2\varepsilon_i \frac{\partial \varepsilon_i}{\partial V_i(r)} \tag{16}$$

Differentiating the error in relation to the vector V_i, follows:

$$\frac{\partial \varepsilon}{\partial V_i} = -\frac{\partial y_i}{\partial V_i} = \frac{\partial y_i}{\partial \vartheta_i}\frac{\partial \vartheta_i}{\partial V_i} = \sigma_i \frac{\partial \vartheta_i}{\partial V_i} \tag{17}$$

where $\sigma_i \triangleq$ derivative of the sigmoid function, related to ϑ_i;

$$= \frac{1}{2}\lambda(1 - y_i^2), \tag{18}$$

for sigmoid function in Eq. (14);

$$= \lambda y_i(1 - y_i), \tag{19}$$

for sigmoid function in Eq. (15).

Then

$$\frac{\partial \vartheta_i}{\partial V_i} = \boldsymbol{X}_i \tag{20}$$

where $\boldsymbol{X}_i \triangleq$ pattern vector, $[x_{0i}\ x_{1i}\ x_{2i} \ldots x_{ni}]^{\mathrm{T}}$.

Therefore, by using the gradient descent method, the following schema of the adaptation of the weights is obtained [15,22,10]:

$$\boldsymbol{V}_i(r+1) = \boldsymbol{V}_i(r) + 2\gamma\beta_i \boldsymbol{X}_i \tag{21}$$

If the ith element is in the last layer, then:

$$\beta_i = \sigma_i \varepsilon_i \tag{22}$$

If the ith element is in the other layers, then:

$$\beta_i = \sigma_i \sum_{j \in Q(i)} w_{ij}\beta_j \tag{23}$$

where $Q(i)$ is the set of the index that are in the next layer to the layer of the ith element and are interconnected to the ith element.

The parameter γ used as a stability control of the iterative process is dependent on λ [22]. For convenience, the parameter γ (training rate) is redefined as follows [13]:

$$\gamma = \frac{\gamma^*}{\lambda} \tag{24}$$

Substituting (24) in (21), the dependence on the amplitude σ related to λ is canceled. The amplitude σ is maintained constant for every λ. This alternative is important, because λ is only acting on the left and right tails of σ. Then (21) is written as follows [13]:

$$\boldsymbol{V}_i(r+1) = \boldsymbol{V}_i(r) + \left\{ 2\gamma^* \frac{\beta_i}{\lambda} \right\} \boldsymbol{X}_i \tag{25}$$

The BP algorithm is considered on the technical literature as a benchmark in precision, although the convergence is very slow. Therefore, it is proposed to adjust the convergence rate γ^* during the convergence process aiming at reducing the execution training time. The adjustment of γ^* is executed by the procedure based on the fuzzy controller.

Similar to the procedure of the adjustment of the weights effectuated by the descent gradient method, the adaptation of the inclination and translation parameters of the postsynaptic function is realized.

The general form of the postsynaptic functions used for adjusting the neural network is given by [16]:

$$y_i = f(\lambda, \rho, \vartheta, w) \tag{26}$$

The inclination and translation parameters contain learning rates denoted by γ^λ and γ^ρ, respectively.

Fig. 1 illustrates the neural architecture with the adjustments of the inclination and translation parameters of the postsynaptic function, i.e., the sigmoid function [16].

The adjustment of the inclination parameter of the sigmoid function is given by [15]:

$$\boldsymbol{\lambda}_i(r+1) = \boldsymbol{\lambda}_i(r) + \theta_i^\lambda(r) \tag{27}$$

where $\theta_i^\lambda(r)$ is $-\gamma^\lambda[\nabla_i^\lambda(r)]$; $\nabla_i^\lambda(r)$ is the gradient of the quadratic error related to the inclination of the neuron i; $\boldsymbol{\lambda}_i \triangleq$ vector containing the inclinations of the neuron i, $[\lambda_{0i}\ \lambda_{1i}\ \lambda_{2i} \ldots \lambda_{ni}]^{\mathrm{T}}$.

The gradient ∇_i^λ is represented by [15]:

$$\nabla_i^\lambda(r) = \frac{\partial \varepsilon_i^2}{\partial \lambda_i(r)} = 2\varepsilon_i \frac{\partial \varepsilon_i}{\partial \lambda_i} = -2\varepsilon_i \frac{\partial y_i}{\partial \lambda_i} = -2\varepsilon_i \sigma_i^\lambda \tag{28}$$

where $\sigma_i^\lambda \triangleq$ derivative of the sigmoid function, related to λ_i.

The general rule that defines the adaptation of the inclination parameter of the sigmoid function is given by the following way:

$$\lambda_i(r+1) = \lambda_i(r) + 2\gamma^\lambda \beta_i^\lambda \tag{29}$$

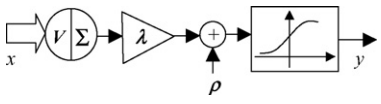

Fig. 1. Neural architecture for adjusting the inclination and translation parameters of the postsynaptic function.

The rule that defines the calculus of β_i^λ follows the next two Eqs. (30) and (31), according to:

(a) if it is in the last layer:

$$\beta_i^\lambda = \sigma_i^\lambda \varepsilon_i \tag{30}$$

(b) if it is in the other layers:

$$\beta_i^\lambda = \sigma_i^\lambda \sum_{j \in Q(i)} w_{ij} \beta_j^\lambda \tag{31}$$

Similar to the procedure described above, the adjustment of the translation parameter of the sigmoid function is given by:

$$\boldsymbol{\rho}_i(r+1) = \boldsymbol{\rho}_i(r) + \theta_i^\rho(r) \tag{32}$$

where $\theta_i^\rho(r)$ is $-\gamma^\rho[\nabla_i^\rho(r)]$; $\nabla_i^\rho(r)$ is the gradient of the quadratic error related to the displacement of the neuron i; $\boldsymbol{\rho}_i \triangleq$ vector containing the displacements of the neuron i, $[\rho_{0i}\ \rho_{1i}\ \rho_{2i} \ldots \rho_{ni}]^{\mathrm{T}}$.

The gradient ∇_i^ρ is calculated on the following way:

$$\nabla_i^\rho(r) = \frac{\partial \varepsilon_i^2}{\partial \rho_i(r)} = 2\varepsilon_i \frac{\partial \varepsilon_i}{\partial \rho_i} = -2\varepsilon_i \frac{\partial y_i}{\partial \rho_i} = -2\varepsilon_i \sigma_i^\rho \tag{33}$$

where $\sigma_i^\rho \triangleq$ derivative of the sigmoid function, related to ρ_i.

The adaptation rule of the translation parameter of the sigmoid function is shown by equation [15]:

$$\rho_i(r+1) = \rho_i(r) + 2\gamma^\rho \beta_i^\rho \tag{34}$$

The rule that defines the calculus of β_i^ρ follows the Eqs. (35) and (36), according to:

(a) if it is in the last layer:

$$\beta_i^\rho = \sigma_i^\rho \varepsilon_i \tag{35}$$

(b) if it is in the other layers:

$$\beta_i^\rho = \sigma_i^\rho \sum_{j \in Q(i)} w_{ij} \beta_j^\rho \tag{36}$$

The adaptation rule of the inclination and translation parameters of the network is calculated in an iterative form for every ith neuron. Therefore, the adaptive neural network always gives an output and the network is faster than the conventional ones.

7. Sensitivity analysis by neural networks

Supplying a pattern $\boldsymbol{X} \in \Re^n$ to the network, which gives an output $\boldsymbol{Y} \in \Re^m$ initializes the BP algorithm. In the sequence, an error in each output (the difference between the desired value and the output) is calculated. Next step is to determine the error propagated in inverse way by the network associated to the partial derivative of the quadratic error of each element related to the weights, and finally to adjust the weights in each element. Then, a new pattern is presented, and the pattern must be repeated until convergence ($|\text{error}| \leq$ predefined tolerance).

After concluding this step, the training mechanism does not actuate, including the fuzzy controller. Thus, the network is able to generalize, i.e., applying any input pattern vector, propagating the signal on the straight sense (input to output), an evaluation of the analysis (diagnosis) results on the output, providing a mapping, $\boldsymbol{X} \rightarrow \boldsymbol{Y} = \boldsymbol{f}(\boldsymbol{X})$, $\boldsymbol{X} \in \Re^n$ and $\boldsymbol{Y} \in \Re^m$.

The derivatives of the output variables (sensitivity analysis) in relation to the input vector components are estimated using a neural network structure trained according to the following description. The sensitivity analysis, by neural networks, is used in this work to obtain $\partial \mathcal{M}_r / \partial \boldsymbol{X}$ (problem defined by Eq. (9)). For the solution of the preventive control problem, the generation reallocation and load shedding (according to the formulation described in Section 4) is adopted where the vector $\boldsymbol{X}$ corresponds to the nodal active power ($\boldsymbol{P}$).

Thus, consider $\boldsymbol{X}^k$ and $\boldsymbol{Y}^k$ as being the kth pair of input and output vector of the neural network. The feedforward neural network shown in Fig. 2 is also considered. It is the representation of a network composed of three layers, where the variables on the principal points of the network and the weight matrices are defined. The input and output layers have n and m neurons, respectively, where n is the dimension of input vector $\boldsymbol{X}^k$ and m is the dimension of output vector $\boldsymbol{Y}^k$.

The partial derivative of y_p^k (pth component of vector $\boldsymbol{Y}^k$) in relation to x_j^k (jth component of the input vector $\boldsymbol{X}^k$) is desired to obtain. Then, to obtain these partial derivatives, is necessary to obtain the intermediate partial derivatives (on the output of the neurons of the hidden layers) of the neural network. Thus, the calculus of partial derivatives of z_i^k (ith component of output vector $\boldsymbol{z}^k$), in relation to x_j^k, is obtained in the following way [8]:

$$\frac{\partial z_i^k}{\partial x_j^k} = w_i^k \frac{\lambda}{2}(1 - \{z_i^k\}^2) \tag{37}$$

for sigmoid function (14), or

$$\frac{\partial z_i^k}{\partial x_j^k} = w_i^k \lambda z_i^k (1 - z_i^k) \tag{38}$$

for sigmoid function (15), where x_j^k is the jth component of input vector $\boldsymbol{X}^k$; $\boldsymbol{X}^k = [x_1^k\, x_2^k \ldots x_n^k]^{\mathrm{T}}$; k is the index referred to the kth pattern vector.

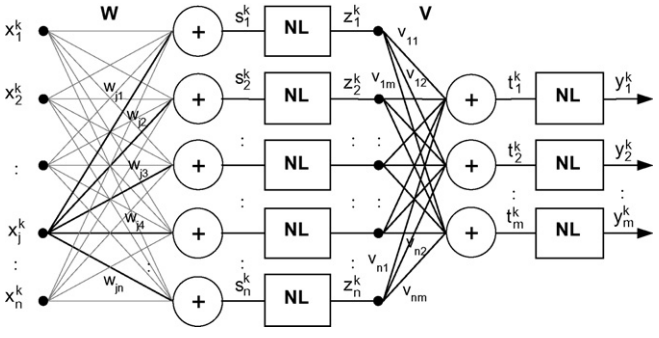

Fig. 2. Feedforward neural network.

The pth intermediate output (input of the sigmoid function) of the output layer of the neural network is expressed by:

$$t_p^k = \{\boldsymbol{z}^k\}^{\mathrm{T}} \boldsymbol{v}_p, \qquad p = 1, 2, \ldots, m \tag{39}$$

where $\boldsymbol{z}^k = [z_1^k\, z_2^k \ldots z_n^k]^{\mathrm{T}}$; $\boldsymbol{v}_p = [v_{1p}\, v_{2p} \ldots v_{np}]^{\mathrm{T}}$.

Thus,

$$\frac{\partial y_p^k}{\partial x_j^k} = \left\{ \frac{\partial y_p^k(\lambda, t_p^k)}{\partial t_p^k} \right\} \frac{\partial t_p^k}{\partial x_j^k} \tag{40}$$

as

$$\frac{\partial t_p^k}{\partial x_j^k} = \{\boldsymbol{v}_p^k\}^{\mathrm{T}} \frac{\partial \boldsymbol{z}^k}{\partial x_j^k} \tag{41}$$

where

$$\frac{\partial \boldsymbol{z}^k}{\partial x_j^k} = \left[\frac{\partial z_1^k}{\partial x_j^k} \frac{\partial z_2^k}{\partial x_j^k} \cdots \frac{\partial z_n^k}{\partial x_j^k} \right]^{\mathrm{T}} \tag{42}$$

Then, substituting Eqs. (14) or (15) in Eq. (40), follows:

$$\frac{\partial y_p^k}{\partial x_j^k} = \frac{\lambda^2}{4(1 - \{y_p^k\}^2)} \{\boldsymbol{v}_p\}^T \boldsymbol{b}_j^k \tag{43}$$

for sigmoid function (14), or

$$\frac{\partial y_p^k}{\partial x_j^k} = \lambda^2 y_p^k (1 - y_p^k) \{\boldsymbol{v}_p\}^{\mathrm{T}} \boldsymbol{b}_j^k \tag{44}$$

for sigmoid function (15), where

$$\boldsymbol{b}_j^k = [(1 - \{z_1^k\}^2) w_{j1} (1 - \{z_2^k\}^2) w_{j2} \ldots (1 - \{z_n^k\}^2) w_{jn}]^{\mathrm{T}} \tag{45}$$

for function (14), or

$$\boldsymbol{b}_j^k = [z_1^k (1 - z_1^k) w_{j1}\, z_{2k} (1 - z_1^k) w_{j2} \ldots z_n^k (1 - z_n^k) w_{jn}]^{\mathrm{T}} \tag{46}$$

for function (15).

8. Application

The application is conducted by using a system composed of 10 synchronous machines, 73 transmission lines and 45 busses, based on a configuration of the South Brazilian System, whose online diagram is shown in Fig. 3, and the other data are in ref. [4].

Three-phase short-circuits eliminated in 0.15 s (9 cycles for a 60 Hz operation), by tripping a transmission line are analyzed.

The neural network training was effectuated considering a set of 50 generation and load profiles and respective security margin. Each profile corresponds to a generation reallocation in relation to a base case in a random way to attend the demand, also fixed in a random way in each bus. The universe of the load variation is between 95% and 112.5%, in relation to the nominal load of the system. Therefore, each profile is generated considering a variation percentile around the nominal state (base

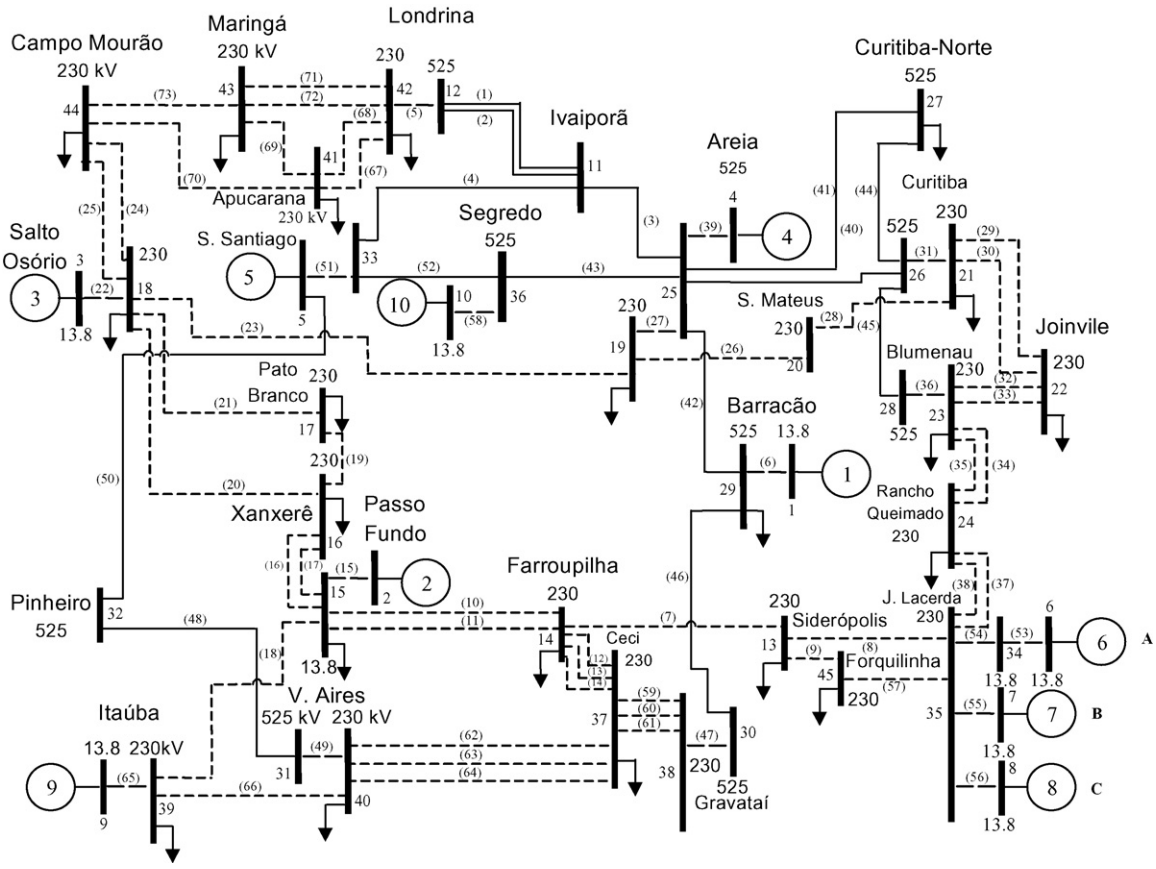

Fig. 3. South-Brazilian System used on the simulations.

case—100%) and a respective random number to the random sequence generation process. Thus, to a same percentile, different generation random numbers generate different generation dispatches of different load profiles. This proceeding generates an adequate set of patterns to the training phase. The results are shown in Table 1.

Table 2 shows the results of the transient stability analysis for 15 contingencies randomly selected. The generation/load profile is chosen in a pseudo random form (112.5% of the base case and random number equal to 2520). It represents a possible level of generation to attend a determined demand. It is a hypothetical

Table 1
Results of the transient stability analysis for the initial generation and load profile (base case—100%)

Contingency number	Data of the defect		Security margin ($\mathcal{M}$)
	Bus under short-circuit	Transmission line removed	
1	17	16–17	0.8470
2	15	15–16	0.8250
3	36	25–36	0.9161
4	29	25–29	0.8056
5	15	14–15	0.8416
6	33	11–33	0.8630
7	29	29–30	0.8481
8	16	16–17	0.8447
9	16	16–18	0.8300
10	25	25–27	0.9440
11	30	29–30	0.9800
12	39	39–40	0.2089
13	13	13–45	0.9320
14	25	25–29	0.9503
15	25	25–26	0.9444

Table 2
Results of the transient stability analysis considering a generation and load profile corresponding to a pseudo-random distribution of 12.5% in relation to the base case and a random number equal to 2520

Contingency number	Data of the defect		Security margin ($\mathcal{M}$)
	Bus under short-circuit	Transmission line removed	
1	17	16–17	0.8222
2	15	15–16	0.7262
3	36	25–36	0.9016
4	29	25–29	0.7824
5	15	14–15	0.7494
6	33	11–33	0.8829
7	29	29–30	0.8704
8	16	16–17	0.8311
9	16	16–18	0.8123
10	25	25–27	0.9449
11	30	29–30	0.9750
12	39	39–40	−0.8188
13	13	13–45	0.4829
14	25	25–29	0.9421
15	25	25–26	0.9486

operation profile, where is realized the security evaluation of the system, and in this work consists of the transient stability analysis considering a group of contingencies.

Contingence 12 is the most critical (least security margin $\mathcal{M}_{12}^0 = -0.8188$). Taking into account the objective of this work that is the generation of the sensitivity model (by neural networks) with the security margin and the power vector (generation and load), i.e., the relation with these two variables is defined. The precision of this sensitivity model can be verified executing variations, for example, on the generation profile of the system and "measure" the variation produced on the security margin.

Therefore, to correct the security margin associated to this contingence, and meet the restriction defined by Eq. (8): $\mathcal{M}_{12} = \mathcal{M}_{12}^0 + \Delta\mathcal{M}_{12} \geq M_{\min}$. $M_{\min} = 0.3$ is adopted [4]. This contingence corresponds to a three-phase short circuit on bus 39, with outage of the transmission line between busses 39 and 40.

It is emphasized that, the neural network can generate, transient stability restrictions (Eq. (8)) for a set of interest contingencies, e.g., for every contingency that presents security margin less than a predefined level. These restrictions can be aggregated to other restrictions of the system (power balance, physical limits of the electric equipment, etc.) and the preventive control problem solved by using some other optimization method.

For solving the preventive control, the generation reallocation and the load shedding is adopted. The neural network for the referred contingence produces a sensitivity coefficient of best performance (most absolute value) referred to the generation and load. For stabilizing the system, a load shedding on bus 40 and a reduction of generation on bus 9 is necessary. Table 3 shows the data related to carry out the security correction.

Suppose that the security margin of $\mathcal{M}_{12}^0 = -0.8188$ is desired to correct for $\mathcal{M}_i \geq M_{\min} = 0.3$. Then, the control action (generation reallocation/load shedding) gives a security margin displacement of $\Delta\mathcal{M} = 1.1188$. This goal is attained by sensibility analysis through a generation reallocation on bus 9 and a load shedding on bus 40, corresponding to 0.8 pu. The final security margin is 0.330, which is a good approximation of $\mathcal{M}_{12} \geq M_{\min}$.

Table 3
Analyzed results before and after the control action (generation reallocation and load shedding)

State	Item	Value
Initial	Initial security margin ($\mathcal{M}^0$)	−0.8188
Goal	Goal security margin ($M_{\min}$)	0.3
Security margin displacement	Necessary security margin variation ($\mathcal{M}^0 - M_{\min}$)	1.1188
Identification of the more sensible busses	More sensible generation bus	9
	More sensible load bus	40
Sensitivity analysis	Sensitivity coefficient of bus 9 ($\partial\mathcal{M}_r/\partial P_9$)	−1.1435
	Sensitivity coefficient of bus 40 ($\partial\mathcal{M}_r/\partial P_{40}$)	0.255
Control action	Generation reallocation at bus 9	−0.8
	Load shedding at bus 40	0.8
Final	Final security margin ($\mathcal{M}^f$)	0.330

Table 4
Transient stability study considering a generation and load profile corresponding to a pseudo-random distribution of 12.5% in relation to the base case, random number equal to 2520, active generation reduction on bus 9 and load shedding on bus 40 equal to 0.8 pu.

Contingency number	Data of the defect		Security margin ($\mathcal{M}$)
	Bus under short-circuit	Transmission Line removed	
1	17	16–17	0.8063
2	15	15–16	0.7169
3	36	25–36	0.8895
4	29	25–29	0.7679
5	15	14–15	0.7406
6	33	11–33	0.8304
7	29	29–30	0.8718
8	16	16–17	0.8148
9	16	16–18	0.7934
10	25	25–27	0.9413
11	30	29–30	0.9800
12	39	39–40	0.3301
13	13	13–45	0.4719
14	25	25–29	0.9342
15	25	25–26	0.9470

Table 4 shows the results after generation reallocation and load shedding.

Therefore, the generation reallocation produces a security margin correction of −0.8188 to 0.330, i.e., $\mathcal{M}_2 \geq M_{\min}$. The other contingencies also attend this requirement. The objectives are attained, i.e., $\mathcal{M}_i \geq M_{\min}$, $i = 1, 2, \ldots, 7$.

The neural network training is implemented using four different neural networks, according to Tables 5 and 6. They correspond to conventional back-propagation training, fuzzy back-propagation, conventional back-propagation training with adaptation of sigmoid function (translation and inclination) and

Table 5
Parameters of the neural networks used

	BP conventional	BP fuzzy
Number of pattern vectors	50	50
Number of layers	3	3
Number of neurons per layer	(43,55,1)	(43,55,1)
Sigmoid function	$\frac{1-\exp(-\lambda s_i)}{1+\exp(-\lambda s_i)}$	$\frac{1-\exp(-\lambda s_i)}{1+\exp(-\lambda s_i)}$
Training rate (γ^0)	0.07	0.07
Inclination of sigmoid function (λ^0)	0.3	0.3
Momentum term	No	No
Fuzzy controller inclusion	No	Yes
Exponential decline	No	0.75
Inclusion of the adaptation mechanism for λ and ρ	No	No
Training rate of the inclination of the sigmoid function	No	No
Training rate of the translation of the sigmoid function	No	No
Training time (s)	7379.70	6898.63
Execution time of the generation and analysis of the sensitivity model (s)	0.015	0.015

Table 6
Parameters of the neural networks used

	BP conventional with adaptation of inclination and translation of the sigmoid function	BP fuzzy with adaptation of inclination and translation of the sigmoid function
Number of pattern vectors	50	50
Number of layers	3	3
Number of neurons per layer	(43,55,1)	(43,55,1)
Sigmoid function	$\frac{1-\exp(-\lambda s_i)}{1+\exp(-\lambda s_i)}$	$\frac{1-\exp(-\lambda s_i)}{1+\exp(-\lambda s_i)}$
Training rate (γ^0)	0.07	0.07
Inclination of sigmoid function (λ^0)	0.3	0.3
Momentum term	No	No
Inclusion of the fuzzy controller	No	Yes
Exponential decline	No	0.65
Inclusion of the adaptation mechanism for λ and ρ	Yes	Yes
Training rate of inclination of sigmoid function	0.007	0.007
Training rate of translation of sigmoid function	0.1	0.1
Training time (s)	635.80	616.52
Execution time of generation and execution of the sensitivity analysis model (s)	0.015	0.015

fuzzy back-propagation with adaptation of sigmoid function (translation and inclination).

The four different neural networks give the same solution, and the differences are on the training time. Considering the back-propagation training, sometimes the convergence is very slow and could not converge, and the adaptations like fuzzy back-propagation or the inclination and translation of the sigmoid function always provide a solution with less time, and also less number of iterations during the training according to Tables 5 and 6.

9. Conclusion

A procedure for analyzing the transient stability and the preventive control of electric energy systems formulated using the sensitivity analysis generated by feedforward neural networks is proposed in this work. The neural network training is realized using the back-propagation algorithm with a fuzzy controller [13], associated with an adaptive process [15]. This algorithm gives a faster convergence and more accurate results [15], when compared to the traditional back-propagation algorithm, by adjusting the training rate and using the information of the error and the global error variation. The adaptation of the translation and inclination parameters also provides a guarantee of obtaining a solution by increasing the space of search and avoiding the paralysis process. Once the training is carried out, an off-line procedure, the neural network is able to estimate the security margin and the sensitivity analysis (a procedure effectuated almost on-line, and without computational effort). With these information, it is possible to develop a procedure for realizing the security correction (preventive control) for levels considered adequate for the system: in this work corresponds to $\mathcal{M} \geq M_{\min} = 0.3$.

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Electric load forecasting using a fuzzy ART&ARTMAP neural network

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Abstract

This work presents a neural network based on the ART architecture (adaptive resonance theory), named fuzzy ART&ARTMAP neural network, applied to the electric load-forecasting problem. The neural networks based on the ART architecture have two fundamental characteristics that are extremely important for the network performance (stability and plasticity), which allow the implementation of continuous training. The fuzzy ART&ARTMAP neural network aims to reduce the imprecision of the forecasting results by a mechanism that separate the analog and binary data, processing them separately. Therefore, this represents a reduction on the processing time and improved quality of the results, when compared to the Back-Propagation neural network, and better to the classical forecasting techniques (ARIMA of Box and Jenkins methods). Finished the training, the fuzzy ART&ARTMAP neural network is capable to forecast electrical loads 24 h in advance. To validate the methodology, data from a Brazilian electric company is used.
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Keywords: Adaptive resonance theory; Electric load forecasting; Electric power systems; Neural networks; Fuzzy logic; Fuzzy ART&ARTMAP neural network

1. Introduction

Expansion planning, load flow, economic operation, security analysis and electric energy systems control are some of studies that effectively depend on the behaviour of the load profile [1], i.e., forecasting of future information of a time series based on passed values. In the literature, there are many methods to effectuate load forecasting, for example, multiple or simple linear regression, exponential smoothing, state estimation, Kalman filter, ARIMA of Box&Jenkins methods [1–3]. All of these methods need a previous modelling to be applied a posteriori. It is necessary to know some information as for example: cloudy days, wind velocity, sudden temperature variation and non-conventional day effects (holidays, strikes, etc.), to model the load.

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After modelling the load, and using this information, the algorithm is initialised to take the results.

The use of neural networks is nowadays an alternative method that is becoming an efficient technique to solve the electric load-forecasting problem.

There are researches that treat this problem using the Back-Propagation algorithm [4,5]. This algorithm is considered on the specialised literature a benchmark in precision. However, the convergence is slow, although there are some adaptations to improve the performance [4,6]. Therefore, the idea is to use a neural network that combines good results with a faster processing. The ART neural networks are architectures based on adaptive resonance theory, having two fundamental characteristics for the network processing: the stability and the plasticity (capacity of learning new patterns without loosing the memory or the patterns already trained) [7]. These two characteristics provide to the network a stable learning and a faster convergence [8]. The objective of this work is to develop another neural network based on the ART architecture, called fuzzy ART&ARTMAP neural network [9], and apply it to the electric load-forecasting problem [10–12].

The fuzzy ART&ARTMAP neural network is composed of two modules: (1) fuzzy ART and (2) fuzzy ARTMAP. The ART network is architecture based on a supervised training for multi dimensional mapping (several inputs and several outputs), composed of two ART modules connected to an inter-ART module [7]. The fuzzy ART module classifies the input vectors in categories, composed of analog data that are converted to binary values by an active/binary code converter. The input set of fuzzy ARTMAP module is composed of binary data (that have already been converted in binary), plus the binary information referred to time (month, day of the week, holiday, and hour, etc). The fuzzy ARTMAP output module is composed of electric load values referred to the subsequent hour, and are processed in classes/areas and codified in binary. This way, this network aims to solve, or at least reduce the imprecision of the forecasting results, by a mechanism that separate the analog and binary data, and processing them separately. The fuzzy ARTMAP module receives only binary data, allowing applications in large-scale systems. It is important to emphasise that this architecture preserves the plasticity, allowing the implementation of continuous training, thus becoming an important tool for electric load forecasting.

To illustrate the proposed method, an example is presented considering historical data from a Brazilian electric company.

It is emphasised that, in this work, for convenience the vectors are represented by rows and not by columns (as usually).

2. ART (Adaptive Resonance Theory)

In 1976, Grossberg introduced the adaptive resonance theory [13,14], concept as a cognitive human information process, through a self-organisation, effectuating stable category recognition and answering an arbitrary sequence of input patterns [7,15–17]. This kind of network has a self-organisation and a self-stabilisation that allows solving the stability–plasticity dilemma. Thus, the ART network is capable to assimilate new things while maintaining those already instructed [15].

The ART network structure is composed of three layers as shown on Fig. 1 [18]:

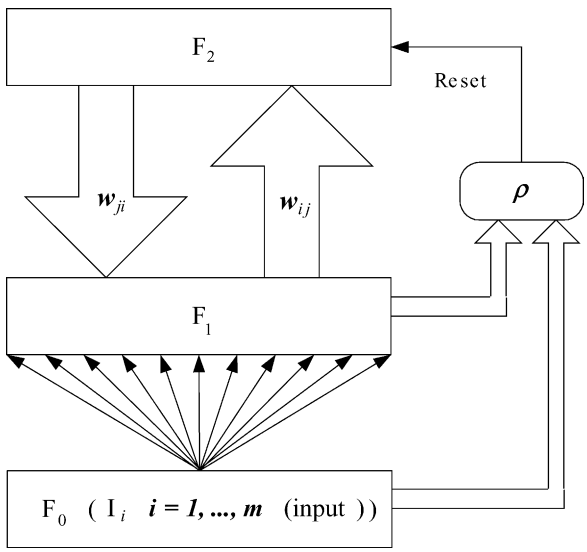

Fig. 1. ART network structure.

- F_0: input layer (receives and stores the new input patterns);
- F_1: comparing layer (intensification of the difference and elimination of the noises of the input patterns);
- F_2: recognition layer (store the categories (clusters)).

There are two essential memories to execute the ART neural network, according to [14]. They are:

1. LTM (long term memory): weight connection between the layers:
   $F_1 \rightarrow F_2$ (bottom-up);
   $F_2 \rightarrow F_1$ (top-down);
2. STM (short term memory): activity patterns developed on the layers F_1 and F_2.

The performance of the ART network is effectuated as follows [19]:

1. An input pattern is presented to the network;
2. If the input pattern is similar to patterns already stored on the ART network LTM $\rightarrow$ the input pattern pertains to a category already formed;
3. If the input pattern is not similar to any pattern already stored on the ART network LTM $\rightarrow$ the input pattern does not pertain to any category already formed, and the network will form a new category, which will store the new input pattern.

The vigilance parameter is responsible for identifying this similarity [19].

2.1. Fuzzy ART

This architecture has a non-supervised training and uses structure calculus based on fuzzy logic.

The fuzzy ART network can be considered as a modification of ART1 network. Input patterns of fuzzy ART are binary data as well as analog data, but the ART1 network uses only binary data.

2.2. Fuzzy ART algorithm

The principal steps of the fuzzy ART algorithm are resumed on the following procedures.

2.2.1. Input data

The input pattern vector of the F_0 layer is represented by the vector $\mathbf{a}$, $\mathbf{a} = [\mathbf{a}_1 \ \ldots \ \mathbf{a}_M]$, M-dimensional. This vector is normalised to avoid category proliferation, according to the equation:

$$\bar{\boldsymbol{a}} = \frac{\boldsymbol{a}}{|\boldsymbol{a}|} \tag{1}$$

where:

$$|\boldsymbol{a}| \equiv \sum_i a_i$$

and $\bar{\boldsymbol{a}}$ normalised input vector.

Afterwards, it is effectuated the codification of the complement, preserving the amplitude information. The complement codification is represented as Eq. (2) [18]:

$$\bar{a}_i^c = 1 - \bar{a}_i \tag{2}$$

where $\bar{a}^c$ is complement of the normalised vector.

Then, the input vector of the F_0 layer, is a $2M$-dimensional vector denoted by Eq. (3) [11,18]:

$$\boldsymbol{I} = [\bar{\boldsymbol{a}}\, \bar{\boldsymbol{a}}^c] \equiv [\bar{a}_1 \cdots \bar{a}_M \cdots \bar{a}_1^c \cdots \bar{a}_M^c] \tag{3}$$

The activity vector of F_1 is indicated by $\mathbf{x} = [x_1 \ldots x_M]$, according to Eq. (4) described below:

$$x = \begin{cases} I & \text{if } F_2 \text{ is inactive} \\ I \wedge w_J & \text{if the } J\text{th node of } F_2 \text{ is active} \end{cases} \tag{4}$$

The activity vector of F_2 is represented by $\mathbf{y} = [y_1 \ \ldots \ y_N]$, where N is the number of the categories created in F_2. Thus, according to [20]:

$$\begin{aligned} y_J &= 1 \\ y_j &= 0 \quad \text{for } j \neq J. \end{aligned} \tag{5}$$

2.2.2. Parameters

There are three fundamental parameters for the performance and learning of the fuzzy ART network according to [14,21]:

1. Chosen parameter ($\alpha > 0$): acts on the category selection;
2. Training rate ($\beta \in [0,1]$): controls the velocity of the network adaptation;
3. Vigilance parameter ($\rho \in [0,1]$): controls the network resonance. The vigilance parameter is the responsible of the number of the categories formed

in F_2. If ρ is very large, it produces a good classification, providing the generation of several categories (the network has less errors but has less capacity of generalisation). If ρ is very small, it will generate few categories (the network will have more capacity of generalisation, but more possibility of making mistakes).

2.2.3. Weight initialisation

Initially every weight has the same value equal to 1, i.e.:

$$w_{j1}(0) = \cdots = w_{jM}(0) = 1 \tag{6}$$

showing that there is no active category.

2.2.4. Choice category

Given the input vector **I** in F_1, for each node j in F_2, i.e., for each category, the choice function T_j is determined by Eq. (7) [18,19]:

$$T_j = \frac{|\boldsymbol{I} \wedge w_j|}{\alpha + |w_j|} \tag{7}$$

where, $\wedge$ is the fuzzy operator AND, defined by Eq. (8):

$$(\boldsymbol{p} \wedge \boldsymbol{q}) \equiv \min\ (p_i, q_i) \tag{8}$$

The choice is realised when at least one of the nodes in F_2, become active. The category is chosen as the active node J according to [18,21]:

$$J = \underset{j \in 1,\ldots,N}{\arg\max}\ T_j \tag{9}$$

If there is more than one active category, the chosen category is that one with the smaller index.

2.2.5. Resonance or reset

Resonance occurs if the vigilance criterion Eq. (10) is satisfied:

$$\frac{|\boldsymbol{I} \wedge w_J|}{|\boldsymbol{I}|} \geq \rho \tag{10}$$

If Eq. (10) is not satisfied, then the reset occurs. On the reset, the node J of F_2 is excluded from the search process of the active category given by Eq. (9). This is, T_J assumes a value equal to zero and it is chosen a new category by Eq. (9) to realise the resonance process. This procedure is executed until the network finds a category that satisfies Eq. (10).

2.2.6. Learning

After the input vector **I** has completed the resonance state, it follows the learning process, where occurs the modification of the weight vector given by Eq. (11) [19,20]:

$$w_J^{\text{new}} = \beta(I \wedge w_J^{\text{old}}) + (1 - \beta) w_J^{\text{old}} \tag{11}$$

where J is active category.

Two types of training dependent of the value of β, are possible:

1. Fast training: $\beta = 1$, permits the system to adapt faster;
2. Slow training: $0 < \beta < 1$, obliges the weights of the system to adapt slowly.

2.3. Fuzzy ARTMAP

The fuzzy ARTMAP architecture is an auto-organised learning system [22,23]. This kind of network has supervised training and pertains to the ART family, i.e., the structure is based on the adaptive resonance theory and is similar to the fuzzy ART network, employing calculus based on the fuzzy logic.

This network is composed of two fuzzy ART modules: ART_a and ART_b, interconnected by an inter-ART by an associative memory module as illustrated on Fig. 2 [7]. The inter-ART module has a self-regulator mechanism named match tracking, whose objective is to maximise the generalisation and minimise the network error [24].

The F_2^a layer is connected to the inter-ART module by the weights w_{jk}^{ab}, assuming initially a value equal to one ($w_{jk}^{ab} = 1$), showing that there is no association with the ART_a and ART_b modules [20].

2.4. Fuzzy ARTMAP algorithm

The steps of the fuzzy ARTMAP algorithm are summarised below.

2.4.1. Input data

The input pattern vector of the ART_a network is represented by the vector $\boldsymbol{a} = [a_1 \ldots a_{M_a}]$, M_a-dimensional, and the input pattern vector of the ART_b network (corresponding to the desired output referred

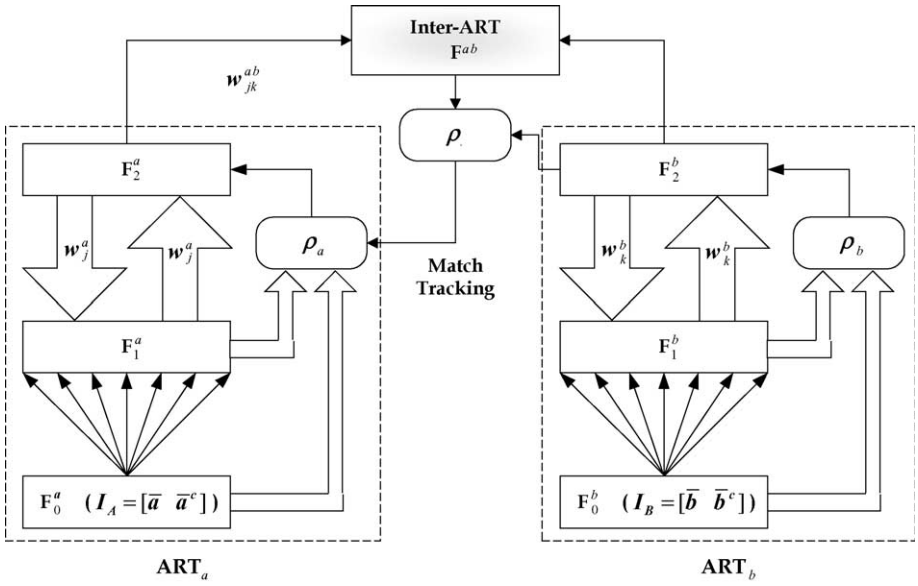

Fig. 2. Structure of the ARTMAP network.

to the input pattern of ART_a network) is represented by vector $\boldsymbol{b} = [b_1 \ldots b_M]$, M_b-dimensional [22].

2.4.2. Parameters

The parameters used on the fuzzy ARTMAP network are the same as those of the fuzzy ART network, and receive a pattern for each specific module. The difference of this network is correspondent to the parameters ρ_{ab}, $\rho_{ab} \in [0,1]$, representing the vigilance parameter referred to the inter-ART module.

2.4.3. Algorithm structure

The ARTMAP network effectuates the processing of two ART networks (ART_a and ART_b), and after the resonance is confirmed in each network, one has:

- J, active category for the ART_a network;
- K, active category for the ART_b network.

The next step is to verify using the match tracking, if the active category on ART_a corresponds to the desired output vector presented on ART_b. The vigilance criterion is given by Eq. (12):

$$\frac{\left|y^{\mathrm{b}} \wedge w^{\mathrm{ab}}_{\mathrm{Jk}}\right|}{\left|y^{\mathrm{b}}\right|} \geq \rho_{ab} \tag{12}$$

where y^{b} is ART_b output vector (activity pattern in F^b_2).

If Eq. (12) is not satisfied, it is effectuated a little increment on the vigilance parameter ART_a, only enough to exclude that category and select another category which will be active and reintroduced on the process until Eq. (12) be satisfied.

2.4.4. Learning

After occurring the resonance by the vigilance criterion Eq. (12), it is effectuated the weight adaptation. The adaptation of the ART_a and ART_b module weights, is the same as that of fuzzy ART network, and the adaptation for the inter-ART module is effectuated as follows [20]:

$$w^{\mathrm{ab}}_{\mathrm{JK}} = 1$$

$$w^{\mathrm{ab}}_{\mathrm{Jk}} = 0 \quad \text{for } k \neq K$$

3. Fuzzy ART&ARTMAP neural network

Neural networks based on the resonance adaptive theory have architecture characteristics, which are extremely important for systems that need immediate answers. Therefore, this is the great interest of using this kind of network, exploring the characteristics of

stability and plasticity, particularities of the ART family networks. One of the few disadvantages of the ART networks can be the great sensibility in relation to the parameters and the complexity of analysing the results obtained by the network.

This two kind of networks has a great flexibility, which allows idealising new architectures (applied to specific problems). The architectures are based on the adaptive resonance theory to solve the problems found on the ART architecture in relation to the category chosen criterion and the vigilance test, which cause imprecision on the ART network.

This way, the objective of this work is to develop an architecture based on the adaptive resonance theory, being capable to solve or at least to reduce the imprecision of load forecasting results, by a mechanism that separate the analog and binary data, and processing separately.

The proposed network is called fuzzy ART&ARTMAP and is composed of two modules:

1. Fuzzy ART;
2. Fuzzy ARTMAP.

The function of ART module is to classify the input vectors in categories composed of analog data (electrical load values), which are converted in binary values by an analog/binary converter (A/B). The ARTMAP module input data is composed of these binary data plus a set of other binary information referred to: month, day of the week, holiday, hour, etc.

The output of ARTMAP module is constituted of electrical load values referred to the subsequent hour. These load values are normalised by a maximum pattern load value, pertaining to the interval between 0 and 1. This interval [0,1] is partitioned in 127 classes, and each class is divided in 15 areas. The normalised load values are identified inside the classes and the areas, therefore a respective area and class represent each load value. Then, the values are converted in binary by an A/B converter and used as an output set of the ARTMAP module.

The ARTMAP module receives only binary data, allowing applications in large systems. This, represents a reduction on the processing time and a quality of the results, when compared to the Back-Propagation neural network and better to the classical forecasting methods (ARIMA of Box&Jenkins, and others) [1].

It is emphasised that this conception preserves the characteristic of plasticity, therefore allowing the implementation of a continuous training that consists of a very important tool for electrical power system operation, in special for electrical load forecasting. Fig. 3 illustrates the fuzzy ART&ARTMAP neural network.

4. Electrical load forecasting

Short term electrical load forecasting (daily forecasting) is executed as follows:

- The output of a determined instant is used as input for the subsequent instant.
- The network input pattern data is divided in two parts:
  1. The first part is the input pattern of the ART module and is composed of historic data (hour) predefined in an interval, for example in a month. In a determined hour h, the input is defined as the load values extracted from the historic data (current hour, 1, 2 and 3 hours before the current one).
  2. The second part corresponds to the input pattern data, which with the binary data form the ART module from the input set of the ARTMAP module. These data are referred to the time (month, day of the week, holiday or not, hour, etc.), all them in a binary form. The results can be improved including other variables like: temperature, humidity, wind velocity, cloudy day, etc., nevertheless it is not used in this work.

The network output corresponds to the load value referred to the hour $(h + 1)$, which is processed in classes and areas and afterwards codified in binary code.

The input patterns of the network are represented as follows:

$$P^{\text{an}}(h) = [L(h-p)\cdots L(h-2)L(h-1)L(h)], \quad P^{\text{an}} \in R^m$$

$$\Lambda^{\text{bin}}(h) = [t], \quad \Lambda^{\text{bin}} \in R^n$$

where $\mathbf{P}^{\text{an}}(h)$ is analog input data; $\boldsymbol{\Lambda}^{\text{bin}}(h)$, binary input data; m, dimension of the $\mathbf{P}^{\text{an}}$ vector; n, dimension of

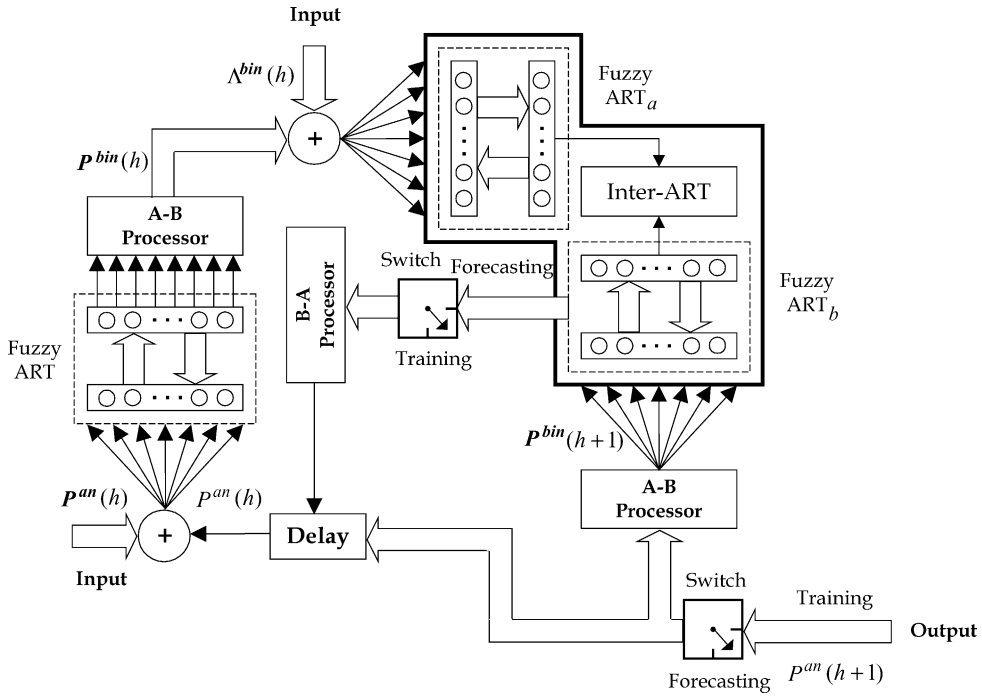

Fig. 3. Fuzzy ART&ARTMAP network.

the $\boldsymbol{\Lambda}^{\text{bin}}$ vector; $L(h-p)$, load value, p hours before the current hour h; $\mathbf{t}$, time vector referred to the historic data (month, day of the week, holiday, hour, etc.).

The vector $\mathbf{P}^{\text{an}}(h)$ gets in the fuzzy ART module, which classifies in categories. In the sequence, these categories are converted in binary providing the vector $\mathbf{P}^{\text{bin}}(h)$.

The load values representing the network output pattern are normalised according to Eq. (13), where $L_n(h+1)$ pertains to the interval [0,1].

$$L_n(h+1) = \frac{L(h+1)}{\text{CMAX}} \tag{13}$$

where, $L_n(h+1)$, normalised electrical load referred to the hour subsequent to the current hour h; $L(h+1)$, electrical load value corresponding to the hour subsequent to the current hour h; CMAX, maximum load value permitted.

The interval [0,1] is discretized in 127 classes, and each class is divided in 15 areas. This avoids the network output to provide load values alike. Then, the network can identify the load value as close as possible to the reality. This way, one class, and one area represent each $L_n(h+1)$, respectively:

$$\boldsymbol{L}_n^{\text{bin}}(h+1) = [\underbrace{101\ldots0}_{\boldsymbol{C}^{\text{bin}}}\underbrace{001\ldots1}_{\boldsymbol{A}^{\text{bin}}}]$$

where is $\mathbf{C}^{\text{bin}}$, vector of classes representing $L_n(h+1)$; $\mathbf{A}^{\text{bin}}$, vector of areas representing $L_n(h+1)$; $\boldsymbol{L}_n^{\text{bin}}(h+1)$, binary representation of $L_n(h+1)$.

Each class and area is converted in binary code. This way, the input and output vectors that compose the fuzzy ART&ARTMAP are illustrated on the following way:

$$\boldsymbol{X}(h) = [\boldsymbol{P}^{\text{bin}}(h)\Lambda^{\text{bin}}(h)], \quad \boldsymbol{Y} \in R^{\text{mb}+n}$$

$$\boldsymbol{Y}(h) = L_n^{\text{bin}}(h+1), \quad \boldsymbol{Y} \in R^{\text{cb}+\text{ab}}$$

where mb is dimension of vector $\mathbf{P}^{\text{bin}}$; cb, dimension of vector $\mathbf{C}^{\text{bin}}$; ab, dimension of vector $\mathbf{A}^{\text{bin}}$.

Electrical loads $L(h-p), \ldots, L(h-1)$ represent the feedback loop with a delay on the output. The data used on the training referred to the time (for each vector) are day of the week and hour of the day.

Table 1
Specification of the fuzzy ART&ARTMAP neural network

Item	Value
Number of pattern vectors	504
Training rate, β	1.00
Choice parameter, α	1.30
Vigilance parameter (ART_a), ρ_a	0.94
Vigilance parameter (ART_b), ρ_b	0.99
Vigilance parameter (inter-ART), ρ_{ab}	1.00

Considering the input vector representation which pertains to the fuzzy ART&ARTMAP network such that: $\mathbf{X}(h)$, dimension 20, $[\mathbf{P}^{bin}(h)\ \boldsymbol{\Lambda}^{bin}(h)]$, with $\mathbf{P}^{bin}(h)$ with dimension 12 and $\boldsymbol{\Lambda}^{bin}(h)$ dimension 8, and $\mathbf{Y}(h)$ dimension equal to 11.

The historical data are the electrical load and were obtained from a Brazilian electric company. These data contain hour load from the year 1998, and it is considered the non-typical days (holidays) and special days (Saturday and Sundays) and days from a typical week. The historic data used to effectuate the training of the fuzzy ART&ARTMAP network are between 8 and 28 July, 1998.

Therefore, a vector with 504 inputs and respective outputs represents the data set. Table 1 shows the principal parameters used on the training and diagnosis of the developed neural network.

Fig. 4 shows the load forecasting results (real data and fuzzy ART&ARTMAP neural network) referred to 29 July, 1998. The mean absolute percentile error

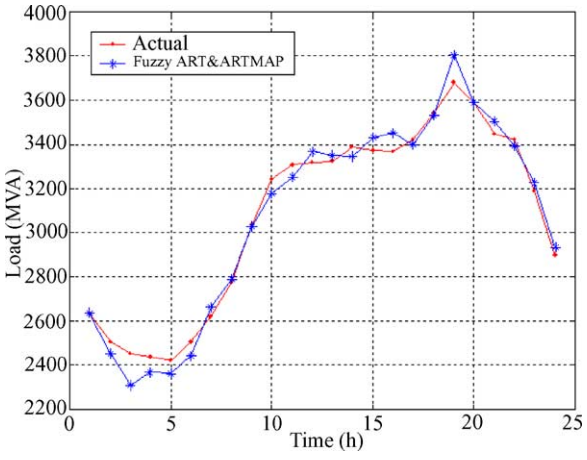

Fig. 4. Load forecasting results.

Table 2
Comparative results

Item	Conventional Back-Propagation	Fuzzy ART&ARTMAP
Processing time (s)	1165.98	2.04
Gain GT	–	571.56
MAPE (%)	1.80	1.62
MAX (%)	6.20	5.68

(MAPE) (14) [25] and the maximum error (MAX) Eq. (15) are defined to improve the analysis of the results for the daily forecasting. This way, they are shown below:

$$\mathrm{MAPE} = \frac{1}{N_h}\left\{\sum_{i=1}^{N}\frac{(|L(h) - \underline{L}(h)|)}{L(h)}\right\} \times 100 \qquad (14)$$

$$\mathrm{MAX}(\%) = \max\left\{\frac{|L(h) - \underline{L}(h)|}{L(h)}\right\} \times 100 \qquad (15)$$

where $L(h)$ is real value of the load referred to the hour h; $\underline{L}(h)$, estimated value of the load referred to the hour h; N_h, total number of hours.

The execution time gain (processing) Eq. (16) is defined to analyse the performance of the developed methodology, considering the necessary time to effectuate the training and the diagnosis, on the following way:

$$\mathrm{GT} = \frac{\mathrm{TBP}}{\mathrm{TFA}} \qquad (16)$$

where TBP is Back-Propagation conventional execution time (s); TFA, fuzzy ART&ARTMAP execution time (s).

The program was elaborated in FORTRAN and processed in a Pentium 4 (1.7 GHz and 256 MB of RAM memory). The processing time is referred only to the execution of the algorithm, excluding the operations of data reading and data output. Comparative results are shown on Table 2.

5. Conclusion

It is developed a procedure for electrical load forecasting by a fuzzy ART&ARTMAP neural network. The network is an innovation among the ART

architecture existent on the literature. In this network all the processing is based on binary data, which facilitate the application in large systems and permitting the network processing on-line. The presented results use data from a Brazilian electrical company. The load forecasting is executed considering one day ahead. The ART&ARTMAP network is advantageous face to the Back-Propagation network considering the processing time [4,6]. The mean absolute percentile error (MAPE) and the maximum daily forecasting error are 1.62 and 5.68%, respectively. Concluding, the formulation presented in this work provide the following advantages:

- Reduction on the processing time;
- The plasticity characteristic is preserved;
- Application in large scale systems;
- Implementation in continuous systems;
- Results with better quality.

Acknowledgments

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Electrical load forecasting formulation by a fast neural network

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The objective of this work is to develop a methodology for electric load forecasting based on a neural network. Here, backpropagation algorithm is used with an adaptive process that based on fuzzy logic and using a decaying exponential function to avoid instability in the convergence process. This methodology results in fast training, when compared to the conventional formulation of backpropagation algorithm. The results are presented using data from a Brazilian Electric Company, and shows a very good performance for the proposal objective.

Keywords: Load forecasting, short term, neural networks, backpropagation, fuzzy logic

1. INTRODUCTION

Expansion planning, load-flow, economic operation and unit commitment, security analysis and control of electric energy systems are effectively dependent on the previous behavior of load profile (O'Donovan [5]), i.e. forecasting of future information of a time series based on past values. Recently with the arrival of competitive power markets, load forecasting importance has strongly been considered (Bunn [3], Doulai [6], Wang [19]). In technical literature many methods are implemented for short term load forecasting: simple or multiple linear regression, exponential smoothing, state estimation, Kalman filter, ARIMA of Box and Jenkins (Almeida *et al.* [1]). All these methods, need a previous load modeling for further application. To model the load it is necessary to know some information like: cloudy days, wind speed, temperature variations and effects of non-conventional days (holidays, strikes, etc).

Having obtained the load using these informations, an appropriated algorithm is used to get the results. The use of neural networks nowadays is a very efficient method mainly to the special days case, (Hippert *et al.*, [7]). In the specialized literature there are available several references that deal with load forecasting problems by neural networks (Charytoniuk and Chen [4], Doulai and Cahill [6], Hippert *et al.* [7], Kodogianis and Anagnostakis [11] and [12], Liang and Cheng [13], Osowski and Siwek [15], Srinivasan *et al.* [16], Tamimi and Egbert [17], Wang at al. [19], among others). Nevertheless, the objective of this work is to develop a methodology to electric load forecasting using an artificial neural network (ANN) (Kartalopoulos [10]) with training by backpropagation (BP) (Werbos [20]). The BP algorithm is considered, in technical literature, as a benchmark in precision. However the convergence using BP algorithm is quite slowly. The objective proposal of this work is to adjust the training rate γ during the convergence process, to reduce the execution time. The γ adjustment is done using a fuzzy controller. We also use a decaying exponential function that establishes a priority in the regulator actuation in the initial training time and avoid instability in the convergence process.

It must be emphasized that, in this work, we intend to

develop a methodology to short term electric load forecasting based on feedforward neural networks where the training is done by backpropagation algorithm (backpropagation neural network), that is efficient in precision and speed, when compared with the conventional backpropagation training. The comparison with backpropagation neural networks and traditional prediction methods (Box-Jenkins, etc.) have been done in (Almeida *et al.* [1], Charytoniuk and Chen [3], Tamimi and Egbert [17]), where neural networks show some advantages such that; it is not necessary to select a prediction method, great robustness to noises, etc. Therefore, it is understood that the neural networks present a superior or equal performance compared to the traditional statistical load forecasting methods. Furthermore it is a very promising technique and a good alternative to the applications in temporal series predictions area.

To illustrate the proposed method, an example is presented considering a historical data from a Brazilian Electric Company.

2. NEURAL NETWORK STRUCTURE

The *i*th output element (neuron) (Widrow and Lehr [21]) is a linear combination of the element inputs x_j that are connected to the element *i* by the bias w_{ij}:

$$\vartheta_i = \sum_j x_j \tag{1}$$

Each element can have a bias w_0 fed by an extra constant input $x_0 = +1$.

The linear output ϑ_i is finally converted in a non-linearity as a sigmoid, or a relay (Kartalopoulos [6], Widrow and Lehr [21]), etc.

3. NEURAL NETWORK TRAINING

The BP training is initialized by presenting a pattern $\boldsymbol{X}$ to the network that gives an output $\boldsymbol{Y}$. Following the calculated error in each output which is the difference between the desired value and the output. The next step is to determine the back propagated error by the network associated to the partial derivative of the quadratic error of each element related to the weights, and finally the weights of each element are adjusted. Then a new pattern is presented, and the process must be repeated until convergence is reached, where | error | ≤ a chosen tolerance. The initial weights are usually adopted as random numbers (Widrow and Lehr [21]). The BP algorithm consists in adapting the weights such that the network quadratic error is minimized. The sum of the instantaneous quadratic error of each neuron of the last layer (network output) is given by (Widrow and Lehr [17]):

$$\varepsilon^2 = \sum_{i=1}^{no} \varepsilon_i^2 \tag{2}$$

where

$\varepsilon_i = d_i - y_i$;
d_i = desired output of the *i*th element of the last layer ;
γ_i = output of the *i*th element of the last layer;
no = number of neurons of the last layer.

Considering the *i*th network neuron and using the descent gradient (Kartalopoulos [6], Widrow and Lehr [21]) method, the weight adjustments can be formulated by (Widrow and Lehr [21]):

$$\boldsymbol{V}_i (r+1) = \boldsymbol{V}_i (r) + \theta_i (r) \tag{3}$$

where

$\theta_i (r)$ = $\gamma [\nabla_i (r)]$;
γ = stability control parameter or training rate;
$\nabla_i (r)$ = quadratic error gradient related to neuron *i* weights;
$\boldsymbol{V}_i$ = vector containing neuron *i* weights = $[w_{0i}\ w_{1i}\ w_{2i} \ldots\ w_{ni}]^T$.

The adopted direction in equation (3) to minimize objective function of the quadratic error corresponds to the gradient opposite direction. The γ parameter determines the vector length $\theta_i(r)$. Providing that this work deals with load forecasting where the values are always positive, the nonlinear function to be used is the sigmoid function defined by Kartalopoulos [10], Widrow and Lehr [21] which is varying between 0 and +1 is given as:

$$y_i = 1 / \{(1 + \exp(-\lambda \vartheta_i)\} \tag{4}$$

where

λ = constant that determines the function y_i slope.

The BP algorithm can be found in the literature in several ways to become computationally faster. A very interesting formulation is the BP with momentum (Widrow and Lehr [21]).

Where the gradient is calculated as from equation (3), then by considering the sigmoid function defined in equation (4) and the momentum term, the following adaptation weight scheme is obtained (Minussi and Silveira [14], Widrow and Lehr [21]):

$$\boldsymbol{V}_{ij} (r+1) = \boldsymbol{V}_{ij} (r) + \Delta_{\boldsymbol{V}ij} (r) \tag{5}$$

where

$\Delta_{\boldsymbol{V}ij} (r)$ = $2\, y\, (1 - \eta)\, \beta_j\, x_i + \eta\ \Delta_{\boldsymbol{V}ij}$ (r– 1); (6)
$\boldsymbol{V}_{ij}$ = weight corresponding to the connection with the *i*th and the *j*th neuron;
η = momentum constant ($0 \le \eta \le 1$) (Widrow and Lehr [21]).

If the *j*th element is in the last layer, then:

$$\beta_j = \sigma_j\, \varepsilon_j \tag{7}$$

where

σ_j $\underline{\Delta}$ sigmoid function derivative, given by equation (4), related to:

$$\vartheta_i = \lambda y_j (1 - y_j) \tag{8}$$

If the *j*th element is in another layer, we have:

$$\beta_j = \sigma_j \sum_{k \in S(j)} w_{jk} \beta_k \tag{9}$$

where

$S(j)$ = set of the element subtitles that are in the next layer to the *j*th element layer and are interconnected to the *j*th element.

The γ parameter which is used as a stability control of the iterative process, depends of λ (Minussi and Silveira [14]).

The network weights are randomly initialized considering the interval {0,1}. By convenience, the parameter γ (training rate) can be redefined as:

$$\gamma = \gamma^* / (\lambda) \tag{10}$$

Substituting equation (10) into equation (6), the amplitude dependency of σ related to λ. is cancelled. The σ amplitude will be maintained constant to every λ. This alternative is important considering that λ will only actuate in the left and right tails of σ. Thus, equation (6) can be written as:

$$\Delta_{Vij}(r) = \{ 2 \gamma^* (1 - \eta) \beta_j / \lambda \} x_i + \eta (V_{ij} (r - 1) \tag{11}$$

The BP training can be done by two ways.

- *Procedure 1*
  Adjusting the network weights (considering all layers), converging each pattern until the input pattern set is completed. The process will be repeated until total convergence occurs, i.e. the quadratic error will be less than a predefined tolerance taking into account the given pattern.
- *Procedure 2*
  It is the same as the first one, but doing only one iteration per pattern (weight adjusting).

4. A FUZZY LOGIC APPROACH

This work proposes an approach to adjust the training rate γ^* during the convergence process which results a reduction of the training time execution. The γ^* adjustment is done by a proceeding based on a fuzzy controller. The basic idea of the methodology consists of determining the system state, defined as the global error εg and the global error variation $\Delta\varepsilon g$, objecting a control structure that leads the error to be zero in a reduced iteration number, when compared to the conventional procedures. In this work, the control is formulated using the fuzzy logic concepts. The global error εg and its variation $\Delta\varepsilon g$ are the system state components, and $\Delta\gamma^*$ is the control action that must be done in the system. Initially, the global error is defined as:

$$\varepsilon g = \sum_{j=1}^{np} \sum_{j=1}^{np} \varepsilon_i^2 \tag{12}$$

where

εg = global error of the neural network;
np = number of the network pattern vectors.

The global error corresponds to the calculation of all output errors, output neurons, considering all network pattern vectors. The training must be executed using procedure 2, one iteration per pattern. The global error is calculated in each iteration and the parameter γ^*, is adjusted by increasing $\Delta\gamma^*$ determined by fuzzy logic. The system state and the control action are defined as:

$$E^q = [\varepsilon g \ \Delta\varepsilon g^q \ ] \ T \tag{13}$$
$$u^q = \Delta\gamma^{*q} \tag{14}$$

where q = the index that gives the current iteration.

For a very big input pattern $\boldsymbol{X}$, εg and $\Delta\varepsilon g$ can saturate. Then, the adaptive control is done using an exponential decreasing function applied to the fuzzy controller response. In such a way, the adaptive controller is given by:

$$\Delta\gamma^{*q} = \exp(-\alpha q) \ \Delta\psi^q \tag{15}$$

where

α = an arbitrary positive number;
$\Delta\psi^q$ = the increase proceeding from the fuzzy controller at the instant q.

This parameter will be used to adjust the network weight set referred to the subsequent iteration.

The process is repeated until the training is completed. This is a very simple procedure whose control system requires an additional effort, providing that the controller has two input variables and only one output. This is an improvement to that presented in Reference. [2], i.e. by using the same variables εg and $\Delta\varepsilon g$ to do the control. However, this work introduces the following contributions; 1) improvements of BP algorithm are gained; and 2) the proposed fuzzy controller is a new one, where a set of rules and the use of an exponential decreasing function applied to the controller response.

Definition. Consider a set of objects *B*. Then, a fuzzy set *A* in *B* is defined as being a set of coordinated pairs (Terano *et al*. [18]):

$$A = \{(b, \mu_A(b)) \mid b \in B\} \tag{16}$$

where

$\mu_A(b)$ = membership function of *b* in *A*.

The membership function $\mu_A(b)$ shows the degree of how *b* is in *A*.

The fuzzy control is basically formed by three parts: *fuzzification* that converts real variables in linguistic variables; *inference* that consists of the manipulation of rules using *if-then* statements, and fuzzy operations (AND, OR, NOT, etc.) and *defuzzification* that converts the results obtained (linguistic variables) in real variables, which

form the control action. The fuzzy membership functions can have three different forms, such as: triangular, trapezoidal and gaussian, according to the preference/experience of the designer. The most common defuzzification method is the centroid (Terano *et al*. [18]) that finds the inertial center of the fuzzy set solution. We have the following equation that consider a discrete fuzzy set (Terano *et al*. [18]):

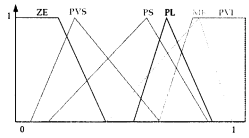
$$u = \left(\sum_{i=1}^{nr} \mu_i \, a_i \right) / \left(\sum_{i=1}^{nr} \mu_i \right) \qquad (17)$$

where

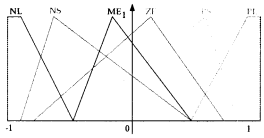
μ_i = value of the membership function;
a_i = value of the set that contains a membership value;
nr = number of fuzzy rules.

The u value calculated by equation (17) corresponds to the inertial center projection of the figure defined by the rule set over the control variable.

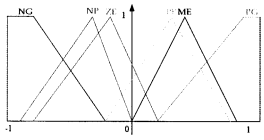
Each state variable must be represented between 3 and 7 fuzzy sets. The control variable must also be represented by the same number of fuzzy sets. The εg variable must be normalized considering as a schedule factor the first global error generated by the network, i.e. $q = 0$. With this representation, the variation interval will be between 0 and +1. If the adaptation heuristic is accordingly coincident the process convergence will be an exponential decreasing. The $\Delta\varepsilon g$ variable will vary between –1 and +1. If the convergence process is exponential decreasing, the $\Delta\varepsilon g$ values will always be negative. In this case, although the $\Delta\varepsilon g$ schedule is between –1 and +1, it must be employed in the rule set, an accurate adjustment between – 1 and 0. In the other interval (0, +1], the adjustment can be more relaxed. The membership functions of the fuzzy sets for several variables used in the neural network BP fuzzy controlled training are shown in Figure 1.



(a)Membership function for the global error εg.



(b)Membership function for the global error variation $\Delta\varepsilon g$.



(c)Membership function for the control action $\Delta\psi$.

Figure 1 Membership functions for the variables εg, Δεg and Δα. Where:
NL = negative large
NS = negative small
ZE= near to zero
PVS = positive very small
ME = medium
PL = positive large
PVL= positive very large
lu = variation limit of Aw

In fuzzy controller the rules are modified as a decision table form. Each input represents the fuzzy value of the variable $\Delta\psi$ given the global error values εg and the global error variation $\Delta\varepsilon g$. The parameter γ^* should be arbitrated in function of λ (sigmoid function slope). The variations γ^* also follow the same procedure. Then, the *lu* parameter must be adjusted to attend each problem specifications and in function of λ.

Table 1 shows the fuzzy rules set in a total of 30 rules. The number of rules can be increased to improve the network performance during the training. This Table is done based on the experience of the user according to references (John [8], Kaehler [9] and Terano *et al.* [18]).

5. LOAD FORECASTING FORMULATION

Short term load forecasting (daily forecasting) is done according to the following: implementation of a recurrence in the output of a determined instant will be used as an input on the subsequent instant. There are considered the hourly historic data in a predefined interval, e.g. monthly. The input network to an hour h is defined as the load val-

Table 1 Fuzzy controller rules

1	if	εg	is	*ZE*	and	$\Delta\varepsilon g$	is *NL*,	then	$\Delta\psi$	is *ZE*
2	if	εg	is	*ZE*	and	$\Delta\varepsilon g$	is *NS*,	then	$\Delta\psi$	is *NL*
3	if	εg	is	*ZE*	and	$\Delta\varepsilon g$	is *ZE*,	then	$\Delta\psi$	is *PS*
4	if	εg	is	*ZE*	and	$\Delta\varepsilon g$	is *PS*,	then	$\Delta\psi$	is *PL*
5	if	εg	is	*ZE*	and	$\Delta\varepsilon g$	is *ME*,	then	$\Delta\psi$	is *ZE*
6	if	εg	is	*PVS*	and	$\Delta\varepsilon g$	is *NL*,	then	$\Delta\psi$	is *ZE*
7	if	εg	is	*PVS*	and	$\Delta\varepsilon g$	is *NS*,	then	$\Delta\psi$	is *ZE*
8	if	εg	is	*PVS*	and	$\Delta\varepsilon g$	is *ZE*,	then	$\Delta\psi$	is *NL*
9	if	εg	is	PVS	and	$\Delta\varepsilon g$	is *PS*,	then	$\Delta\psi$	is *ME*
10	if	εg	is	*PVS*	and	$\Delta\varepsilon g$	is *PL*,	then	$\Delta\psi$	is *ZE*
11	if	εg	is	*PS*	and	$\Delta\varepsilon g$	is *NL*,	then	$\Delta\psi$	is *NS*
12	if	εg	is	*PS*	and	$\Delta\varepsilon g$	is *NS*,	then	$\Delta\psi$	is *ME*
13	if	εg	is	*PS*	and	$\Delta\varepsilon g$	is *ZE*,	then	$\Delta\psi$	is NL
14	if	εg	is	*PS*	and	$\Delta\varepsilon g$	is *ME*,	then	$\Delta\psi$	is *ZE*
15	if	εg	is	*PS*	and	$\Delta\varepsilon g$	is *PL*,	then	$\Delta\psi$	is *NL*
16	if	εg	is	*ME*	and	$\Delta\varepsilon g$	is *NL*,	then	$\Delta\psi$	is *PL*
17	if	εg	is	*ME*	and	$\Delta\varepsilon g$	is *NS*,	then	$\Delta\psi$	is *NL*
18	if	εg	is	*ME*	and	$\Delta\varepsilon g$	is *PS*,	then	$\Delta\psi$	is *PL*
19	if	εg	is	*ME*	and	$\Delta\varepsilon g$	is *ME*,	then	$\Delta\psi$	is *PS*
20	if	εg	is	*ME*	and	$\Delta\varepsilon g$	is *PL*,	then	$\Delta\psi$	is *ME*
21	if	εg	is	*PL*	and	$\Delta\varepsilon g$	is *NS*,	then	$\Delta\psi$	is *ME*
22	if	εg	is	*PL*	and	$\Delta\varepsilon g$	is *ZE*,	then	$\Delta\psi$	is *PS*
23	if	εg	is	*PL*	and	$\Delta\varepsilon g$	is *PS*,	then	$\Delta\psi$	is *NS*
24	if	εg	is	*PL*	and	$\Delta\varepsilon g$	is *ME*,	then	$\Delta\psi$	is *ZE*
25	if	εg	is	*PL*	and	$\Delta\varepsilon g$	is *PL*,	then	$\Delta\psi$	is *ME*
26	if	εg	is	*PVL*	and	$\Delta\varepsilon g$	is *NL*,	then	$\Delta\psi$	is *PL*
27	if	εg	is	*PVL*	and	$\Delta\varepsilon g$	is *ZE*,	then	$\Delta\psi$	is *NS*
28	if	εg	is	*PVL*	and	$\Delta\varepsilon g$	is *PS*,	then	$\Delta\psi$	is *ZE*
29	if	εg	is	*PVL*	and	$\Delta\varepsilon g$	is *ME*,	then	$\Delta\psi$	is *NL*
30	if	εg	is	*PVL*	and	$\Delta\varepsilon g$	is *PL*,	then	$\Delta\psi$	is *ZE*

ues, extracted from the historical data in four instants (current value, one, two and three hours before), temperature, etc., and data referred to time (month, day of the week, holiday and hour, etc.). The output network corresponds to the load value referred to the hour (h + 1). The output/input set is defined considering this strategy until all time series interval time is completed. This scheme can be modified to improve the results by introducing other variables (nebulous day, etc.). Then the input and output vectors are respectively defined as following:

$$\boldsymbol{X}(h) = [\, \boldsymbol{t}^T\, L(h-3)\; L(h-2)\; L(h-1)\; L(h)]^T\,,\; \boldsymbol{X} \in R^m \quad (18)$$

$$\boldsymbol{Y}(h) = [\; \mathrm{L}(h+1)\;]\;,\; \boldsymbol{Y} \in R^l \quad (19)$$

where:

m = dimension of vector $\boldsymbol{X}$;

$L(h-\mathrm{p})$ = the load value at the p hours before the current hour h;

$L(h+1)$ = electric load value corresponding to the subsequent hour to current hour h;

t = time vector referred to the historic data (month, day of the week, holiday, hour, etc) represented in a similar way as the binary code (– 1,+1).

The adoption of a binary representation is preferable in relation to (0, +1) representation, considering that the network input component "0" does not modify the weights. Using this approach the (–1, +1) representation gives a faster convergence and consequently being more efficient.

The electric loads $L(h-3)$, ... , $L(h-1)$ represent the feedback link, with a delay in the output. Then, this network is a recurrent one, and is considered a sigmoid function defined by equation (4). In the BP algorithm, the adaptation of the weights that connect to each neuron is carried out using, basically, the error propagated in the inverse direction, multiplied by $\sigma(\vartheta)$ (partial derivative of y with respect to ϑ) and the input in the referred neuron. In this way, the weights have effective adjustment only for values of ϑ situated in the *central body* of $\sigma(\vartheta)$ function. As the weights become significant, high or low a deceleration of adjustments occurs, having a tendency to a complete paralysis, exactly where the ends of both *right* and *left* tails of $\sigma(\vartheta)$ are met together. In view of this, in the specific case of the problem of electric load forecasting determination, it is proposed the use of sigmoid function relatively small λ (inferior to 1), providing longer tails (Minussi and Silveira [14]). This will permit a less restrictive choice of the network weights, if compared to the adopted in the bibliography. In such a way this reduces the possibility of occurrence of paralysis and increases the speed of the BP algorithm convergence (Minussi and Silveira [14]).

The training data used in this work are (for each vector): the time data, month day, week day (if it is a holiday or not), day hour, the current hour loads, and load values considering the previous three hours. The future load (one hour in advance) is the network output. There are not considered the temperature data because they are not available. Considering a binary representation, the vector **t** will have a dimension of 8, that joining the load data of 12 components (m = 12). The historical data are electric loads and are obtained from a Brazilian Electricity Company. These data contain hourly loads of the year 1998, that related to non typical days (holidays), special days (Saturdays and Sundays) and days from a typical week.

In our experience we consider two different artificial neural networks designated by neural network 1 and neural network 2. Neural network 1 is composed by the historic data between June 17th, 1998 and July 28th, 1998. Therefore, there are 1008 input/output vectors. Table 2 shows the principal parameters referred to the used neural network and the training while Figure 2 shows the results of load forecasting (by conventional BP and fuzzy controlled BP) referred to the defined day July 29th, 1998. Neural network 2 formed by the historic data between May 2th, 1998 and July 25th, 1998 is composed by 2016 input/output vectors. In Table 2 it is verified that the principal parameters referred to the used neural network and the training while Figure 3 shows the results of load forecasting (by conventional BP and fuzzy controlled BP) referred to the defined day July 26th, 1998.

For a precision analysis, we define the mean absolute percentage error (*MAPE*) (Srinivasan et al. [16]) and the maximum error of the daily forecasting, comparing the real load values with the estimated values by neural network in the following way:

$$MAPE = \frac{1}{N}\left\{\sum_{i=1}^{N} (|\, L(h) - \underline{L}(h)|\,) / L(h)\right\} \times 100 \quad (20)$$

$$Maximum\ error\ (\%) = max\{|\, L(h) - \underline{L}(h)\,| / L(h)\} \times 10 \quad (21)$$

Table 2 Neural network specification

Item	Neural Network 1	Neural Network 2
Pattern vectors number	1008	2016
Layers number	3	3
Neurons number per layer	12-30-1	12-30-1
Tolerance	0.025	0.030
Training rate γ*	5.5	6.0
Moment term η	0.9	0.9
Inclination of sigmoid function λ	0.3	0.3
Parameter α	0.81	0.4

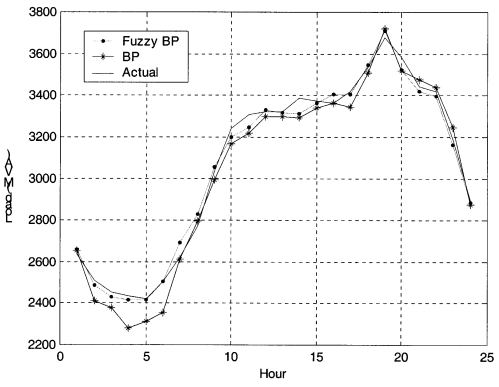


Figure 2 Load forecasting results to neural network 1

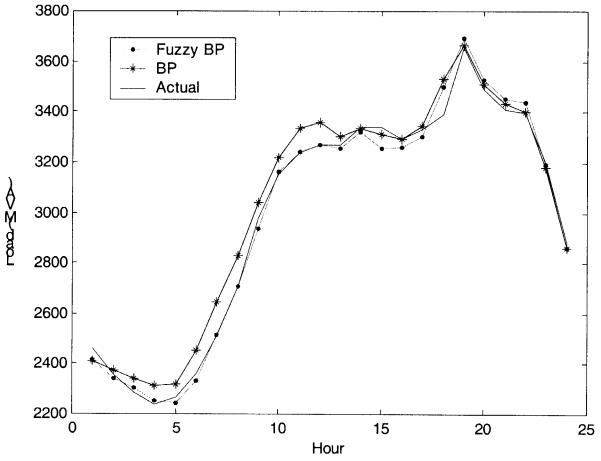


Figure 3 Load forecasting results to neural network 2

where

$L(h)$ = real load value referred to hour h;
$\underline{L}(h)$ = estimated load value referred to hour h;
N = total number of hours.

Some other forecasting are done by considering several load sets. In all cases, it was observed that the fuzzy controlled backpropagation neural network shows a better performance, either in precision or in speed, when compared to the results obtained from a conventional backpropagation, according to Tables 3 and 4. It is also observed that, for obtaining a similar forecasting, using the traditional backpropagation, it is necessary to reduce the tolerance of the iterative process.

6. CONCLUSION

A methodology for electric load forecasting is developed using a neural network The training is done based on a fuzzy controller BP algorithm. The presented results consider the historical data of a Brazilian Electricity Company. The short term load forecasting is done considering 24 hours in advance. It can be verified using these results, that the proposed formulation presents a superior performance, regarding the precision and speed, when compared to conventional BP algorithm. The observed mean absolute percentage error (*MAPE*) and the maximum error of the daily forecasting was 0.91% and 2.71%, respectively. Using the traditional backpropagation, the observed errors were respectively, 1.97% and 6.36% for neural network 1. The MAPE and the Maximum Error obtained on neural network 2 were 0.94% and 3.20%, respectively, while on conventional backpropagation these errors were approximately around 1.85% and 5.52%, respectively.

Table 3 Comparitive results to application 1

Item	Conventional BP	BP with fuzzy controller
Number of cycles	2452	1932
Processing time (s)	221.95	179.12
MAPE (%)	1.97	0.91
Maximum error (%)	6.36	2.71

Table 4 Comparitive results to application 2

Item	Conventional BP	BP with fuzzy controller
Number of cycles	1351	1455
Processing time (s)	98.49	103.49
MAPE (%)	1.85	0.94
Maximum error (%)	5.52	3.20

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